Diagnostic trouble code list



NOTE! This information is only available in English.

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100001	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 1
ABS	C100025	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 1
ABS	C100029	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 1
ABS	C10002F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100031	If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set.	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 1
ABS	C100038	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 1
ABS	C100054	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 1
ABS	C100101	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100125	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 2
ABS	C100129	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 2
ABS	C10012F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 2
ABS	C100131	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100138	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 2
ABS	C100154	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 2
ABS	C100201	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 3
ABS	C100225	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100229	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 3
ABS	C10022F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 3
ABS	C100231	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 3
ABS	C100238	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100254	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 3
ABS	C100301	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 4
ABS	C100325	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 4
ABS	C100329	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C10032F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 4
ABS	C100331	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 4
ABS	C100338	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 4
ABS	C100354	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 4
ABS	C100B01	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100B25	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 5
ABS	C100B29	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h).	Wheel Speed Sensor 5
ABS	C100B2F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed"	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h).	Wheel Speed Sensor 5
ABS	C100B31	If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set.	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100B38	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	"ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed"	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 5
ABS	C100B54	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 5
ABS	C100C01	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	"ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed"	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 6
ABS	C100C25	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100C29	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h).	Wheel Speed Sensor 6
ABS	C100C2F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h).	Wheel Speed Sensor 6
ABS	C100C31	a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set.	No Signal	Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 6
ABS	C100C38	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 6
ABS	C100C54	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001011	-Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Front ABS Inlet Valve
ABS	C001012	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS function: Function is not operable, predefined or latest normally calculated values used as outputs EBD function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Front ABS Inlet Valve
ABS	C001013	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Front ABS Inlet Valve
ABS	C001111	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001112	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Front ABS Outlet Valve
ABS	C001113	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Front ABS Outlet Valve
ABS	C001411	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Front ABS Inlet Valve
ABS	C001412	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function:Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Front ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001413	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Front ABS Inlet Valve
ABS	C001511	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Front ABS Outlet Valve
ABS	C001512	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function:Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Front ABS Outlet Valve
ABS	C001513	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100A95	The error is set if the wheelspeed difference of an axle during curve driving is not plausible compared the other axles (curve driving above 15km/h necessary).	Incorrect Assembly	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	The wheel speed sensor cables are exchanged at the ECU connector (the left side sensor cable is connected to the right side sensor connector and the right side sensor cable is connected to the left side sensor connector)	Diagnostic Service 14. No additional action is needed.	Check the installation order of the wheel speed sensors of all axles.	Wheel Speed Sensor Configuration
ABS	C001811	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Left Rear ABS Inlet Valve
ABS	C001812	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Rear ABS Inlet Valve
ABS	C001813	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Rear ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001911	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Rear ABS Outlet Valve
ABS	C001912	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to outlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Rear ABS Outlet Valve
ABS	C001913	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Rear ABS Outlet Valve
ABS	C001C11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Rear ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001C12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to Inlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Rear ABS Inlet Valve
ABS	C001C13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Rear ABS Inlet Valve
ABS	C001D11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Rear ABS Outlet Valve
ABS	C001D12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to outlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Rear ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001D13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Rear ABS Outlet Valve
ABS	C10044A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Left Front ABS Valve Module
ABS	C10054A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Right Front ABS Valve Module
ABS	C10064A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Left Rear ABS Valve Module
ABS	C20074A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Right Rear ABS Valve Module
ABS	C200068	The error is set if not all necessary data is available for EBD-function	Event Information	ABS Function: Full operation, all calculations are able to run EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	brake pedal demand signal not available	Diagnostic Service 14. No additional action is needed.	Check brake demand pressure sensor signal components: check cable, sensor and signal range (full brake application) or permanent brake application signal (offset error)	Electric Brake Force Distribution

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C200163	Duration of ABS activity is too long or ABS brake pressure demands are low for an implausible long time. Detection time (3s-10min) depends on driving situations.	Circuit / Component Protection Time-Out	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	fault of at least one wheel speed sensor or its mounting, damages of at least one tone wheel, noisy wheel speed signal, wrong data in dataset, differences in tyre sizes, puncture of a tyre, continuous brake force on a wheel (dragging wheel, without brake pressure).	Diagnostic Service 14. No additional action is needed.	Check tyre, check pneumatic and mechanical parts of wheel brakes. Check wheel speed signal components: check cable, sensor and tone wheel. Check if configuration stored in the ECU corresponds to wheel rolling circumference and tone-wheel teeth number. If those do not conform, then correct the configuration values or change the part of the vehicle.	Anti-Lock Brake System
ABS	C100E11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Additional ABS Outlet Valve
ABS	C100E12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to outlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Additional ABS Outlet Valve
ABS	C100E13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	Function: Restricted Limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Additional ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C101011	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Restricted Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Additional ABS Outlet Valve
ABS	C101012	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to outlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Additional ABS Outlet Valve
ABS	C101013	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Additional ABS Outlet Valve
ABS	C100F11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Restricted Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Additional ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100F12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to Inlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Additional ABS Inlet Valve
ABS	C100F13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Restricted Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Additional ABS Inlet Valve
ABS	C100D11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Left Additional ABS Inlet Valve
ABS	C100D12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Additional ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100D13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Additional ABS Inlet Valve
ABS	C10124A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Right Additional ABS Valve Module
ABS	C10114A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Left Additional ABS Valve Module
ABS	U001088	The error is set if Backbone CAN is in bus off status.	Bus off	ABS Function: Full operation, all calculations are able to run EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damaged Backbone CAN network wiring and/or connectors (e.g. short circuit), electro-magnetic disturbance, damaged ECU, other Backbone CAN ECU's	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors and wiring nodes between other backbone ECU's and central ECU. Check for sufficient power supply. If no fault found, check if no disturbing improper devices are mounted. If error still active check with other backbone ECU's. If error still active check with other central ECU.	CAN Communication Backbone 1 Net
ABS	U300006	ABS is not fully operational or not available at all	Algorithm Based Failures	ABS Function:Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Full operation, all calculations are able to run	Configuration check interrupted or not finished internally	Diagnostic Service 14. No additional action is needed.	Reset system. Check warning lamp during driving. If failure still active, replace ECU.	Electronic control unit
ABS	U300041	Checksum error detected in the application software	General Checksum Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	previously interrupted flash download, or damage of the ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	U300044	The error is set if RAM error in central ECU is detected	Data Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage of the central ECU (defect RAM cel)	Diagnostic Service 14. No additional action is needed.	Replace central ECU.	Electronic control unit
ABS	U300045	Error detected in the application software memory	Program Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	damage of the central ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Replace central ECU.	Electronic control unit
ABS	U300046	Error is set if incompatibilities between EEPROM dataset and drive software, hardware or bootloader versions are found, if there are unplausible data in the dataset, or if EEPROM has a failure	Calibration / Parameter Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	ROM and EEPROM do not match or invalid parameter	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace ECU.	Electronic control unit
ABS	U300049	By several monitoring functions	Internal Electronic Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Too high battery voltage or damage of the vehicle's power supply system or damage in the wiring or connectors for the ECU supply or damage in the wiring or damaged dataset or damaged software or damaged ECU.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Electronic control unit
ABS	U30004A	More components installed then configured	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a component connected without being configured.	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	U30004B	Temperature monitoring is carried out by reading the hardware status information of the ECU. Temperature threshold is 100°C +/- 5°C	Over Temperature	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Ambient temperature too high or damage of the central ECU	Diagnostic Service 14. No additional action is needed.	"Check ECU location and suroundings for high temperatures. - If fault is active, replace it. "	Electronic control unit
ABS	U300094	By several monitoring functions.	Unexpected Operation	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damaged dataset or damaged software or damaged ECU, also too weak brakes can be the reason for this fault.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
ABS	U300316	Failure is set if supply voltage of the ECU is out of range. (<20V)	Circuit Voltage Below Threshold	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Battery is not charged enough or damage of the vehicle's power supply system or damage in the wiring or connectors for the ECU supply or damage of the ECU	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
ABS	U300317	Failure is set if supply voltage of the ECU is out of range. (>32V)	Circuit Voltage Above Threshold	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage of the vehicle's power supply system or damage of the ECU	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system.	Battery Power
ABS	C100001	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100025	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 1
ABS	C100029	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 1
ABS	C10002F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 1
ABS	C100031	If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set.	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100038	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 1
ABS	C100054	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 1
ABS	C100101	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 2
ABS	C100125	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100129	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 2
ABS	C10012F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 2
ABS	C100131	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 2
ABS	C100138	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100154	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 2
ABS	C100201	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 3
ABS	C100225	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 3
ABS	C100229	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C10022F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 ... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 3
ABS	C100231	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 3
ABS	C100238	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 3
ABS	C100254	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 3
ABS	C100301	Hardware fault is detected by wheel speed input circuit of the ECU.	General Electrical Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Damage of the wheel speed sensor wiring or connector, wheel speed sensor or its wiring shorted to ground or supply	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from ECU. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 .... 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range"	Wheel Speed Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100325	If the minimum measurable wheel-speed is above its limit value, then fault is set. (Limit speed is 10 [km/h])	Signal Shape / Waveform Failure	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, toggling tone wheel, damaged tone wheel (missing or destroyed tone wheel teeth)	Diagnostic Service 14. No additional action is needed.	"Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting."	Wheel Speed Sensor 4
ABS	C100329	ABS activation frequency on wheel is implausible	Signal invalid	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 4
ABS	C10032F	Plausibility test for the wheel speed sensor signal frequency (frequency jumps)	Signal Erratic	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Large airgap, dirty tone wheel, missing or destroyed tone wheel teeth, toggling tone wheel, production-failures of tone wheel, damaged wheel speed sensor, wheel speed sensor mounting	Diagnostic Service 14. No additional action is needed.	"Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting (Check ""Klämhylsa""). Check the cables and connectors between ECU and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 … 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). "	Wheel Speed Sensor 4
ABS	C100331	"If a not plausible drop of the wheel speed signal happens (or signal is frequently jumping between above 11[km/h] and zero), error is set. "	No Signal	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	"Damage of cables or connectors (Short circuit between H and L) Too large airgap between tone wheel and wheel speed sensor, damage of wheel speed sensor mounting, damaged or wrong tone wheel, damaged wheel speed sensor or ECU electronic circuit"	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors between ECU and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between ECU GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from ECU. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h)."	Wheel Speed Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C100338	The error is set if the current wheel speed differs (>25%) from the average of the other wheels for a long time.	Signal Frequency Incorrect	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	tyre with wrong circumference is mounted, wrong tone wheel, puncture of the tyre, continuous brake force on a wheel without brake activated by ABS. Damage in the wiring and/or connectors between WSS and ECU. Undetected fault of the wheel speed sensor, too big airgap (dropped out wheel speed sensor)	Diagnostic Service 14. No additional action is needed.	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel Speed Sensor 4
ABS	C100354	The deviation between the learned wheel circumferences and the EOL programmed values are too large (deviation > 25%).	Missing Calibration	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Limited values used as outputs or limited functionality, some calculations not executed	Tyre with wrong circumference is mounted on one of the axels, wrong calibration of C3 -signal, tachograph not calibrated, wrong parameter for wheel cirumference, number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed.	"Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check"	Wheel Speed Sensor 4
ABS	C001011	-Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Front ABS Inlet Valve
ABS	C001012	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS function: Function is not operable, predefined or latest normally calculated values used as outputs EBD function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Front ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001013	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Front ABS Inlet Valve
ABS	C001111	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Front ABS Outlet Valve
ABS	C001112	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Front ABS Outlet Valve
ABS	C001113	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001411	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Front ABS Inlet Valve
ABS	C001412	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function:Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Front ABS Inlet Valve
ABS	C001413	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Front ABS Inlet Valve
ABS	C001511	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001512	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function:Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Front ABS Outlet Valve
ABS	C001513	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Full operation, all calculations are able to run	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Front ABS Outlet Valve
ABS	C100A95	The error is set if the wheelspeed difference of an axle during curve driving is not plausible compared the other axles (curve driving above 15km/h necessary).	Incorrect Assembly	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	The wheel speed sensor cables are exchanged at the ECU connector (the left side sensor cable is connected to the right side sensor connector and the right side sensor cable is connected to the left side sensor connector)	Diagnostic Service 14. No additional action is needed.	Check the installation order of the wheel speed sensors of all axles.	Wheel Speed Sensor Configuration
ABS	C001811	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Rear ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001812	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Rear ABS Inlet Valve
ABS	C001813	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Rear ABS Inlet Valve
ABS	C001911	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Left Rear ABS Outlet Valve
ABS	C001912	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Left Rear ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001913	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed. EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Left Rear ABS Outlet Valve
ABS	C001C11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Rear ABS Inlet Valve
ABS	C001C12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to Inlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Rear ABS Inlet Valve
ABS	C001C13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Restricted/limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Rear ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C001D11	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance between the output and the vehicle ground is less then 5 Ohm.	Circuit Short To Ground	ABS Function: RestrictedLimited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to ground on high side) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness."	Right Rear ABS Outlet Valve
ABS	C001D12	Fault will be detected by activation of test pulse and reading back hardware status information. If the output is higher then 3 V for the duration of the detection time.	Circuit Short To Battery	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (low resistance to voltage source above 1V) and/or connectors between PCV and ECU, or damaged ECU	Diagnostic Service 14. No additional action is needed.	"Check the cables and connectors to outlet valve of PCV Right for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible."	Right Rear ABS Outlet Valve
ABS	C001D13	Fault will be detected by activation of test pulse and reading back hardware status information. Resistance at output pin is higher than 1 kOhm.	Circuit Open	ABS Function: Limited values used as outputs or limited functionality, some calculations not executed EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage in the wiring (interruption) and/or connectors between PCV and ECU, damaged PCV or ECU	Diagnostic Service 14. No additional action is needed.	"Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness."	Right Rear ABS Outlet Valve
ABS	C10044A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Left Front ABS Valve Module
ABS	C10054A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Right Front ABS Valve Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	C10064A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Left Rear ABS Valve Module
ABS	C20074A	ECU is configured for one axle valve but 2 valves are connected	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a valve connected without being configured	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle (2s2m or 2s1m on this axle). If the 2s2m is correct, correct dataset of ECU	Right Rear ABS Valve Module
ABS	C200068	The error is set if not all necessary data is available for EBD-function	Event Information	ABS Function: Full operation, all calculations are able to run EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	brake pedal demand signal not available	Diagnostic Service 14. No additional action is needed.	Check brake demand pressure sensor signal components: check cable, sensor and signal range (full brake application) or permanent brake application signal (offset error)	Electric Brake Force Distribution
ABS	C200163	Duration of ABS activity is too long or ABS brake pressure demands are low for an implausible long time. Detection time (3s-10min) depends on driving situations.	Circuit / Component Protection Time-Out	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	fault of at least one wheel speed sensor or its mounting, damages of at least one tone wheel, noisy wheel speed signal, wrong data in dataset, differences in tyre sizes, puncture of a tyre, continuous brake force on a wheel (dragging wheel, without brake pressure).	Diagnostic Service 14. No additional action is needed.	Check tyre, check pneumatic and mechanical parts of wheel brakes. Check wheel speed signal components: check cable, sensor and tone wheel. Check if configuration stored in the ECU corresponds to wheel rolling circumference and tone-wheel teeth number. If those do not conform, then correct the configuration values or change the part of the vehicle.	Anti-Lock Brake System
ABS	U001088	The error is set if Backbone CAN is in bus off status.	Bus off	ABS Function: Full operation, all calculations are able to run EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damaged Backbone CAN network wiring and/or connectors (e.g. short circuit), electro-magnetic disturbance, damaged ECU, other Backbone CAN ECU's	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors and wiring nodes between other backbone ECU's and central ECU. Check for sufficient power supply. If no fault found, check if no disturbing improper devices are mounted. If error still active check with other backbone ECU's. If error still active check with other central ECU.	CAN Communication Backbone 1 Net
ABS	U300006	ABS is not fully operational or not available at all	Algorithm Based Failures	ABS Function:Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Full operation, all calculations are able to run	Configuration check interrupted or not finished internally	Diagnostic Service 14. No additional action is needed.	Reset system. Check warning lamp during driving. If failure still active, replace ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	U300041	Checksum error detected in the application software	General Checksum Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	previously interrupted flash download, or damage of the ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace ECU.	Electronic control unit
ABS	U300044	The error is set if RAM error in central ECU is detected	Data Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage of the central ECU (defect RAM cel)	Diagnostic Service 14. No additional action is needed.	Replace central ECU.	Electronic control unit
ABS	U300045	Error detected in the application software memory	Program Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	damage of the central ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Replace central ECU.	Electronic control unit
ABS	U300046	Error is set if incompatibilities between EEPROM dataset and drive software, hardware or bootloader versions are found, if there are unplausible data in the dataset, or if EEPROM has a failure	Calibration / Parameter Memory Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	ROM and EEPROM do not match or invalid parameter	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace ECU.	Electronic control unit
ABS	U300049	By several monitoring functions	Internal Electronic Failure	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Too high battery voltage or damage of the vehicle's power supply system or damage in the wiring or connectors for the ECU supply or damage in the wiring or damaged dataset or damaged software or damaged ECU.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ABS	U30004A	More components installed then configured	Incorrect Component Installed	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	There is a component connected without being configured.	Diagnostic Service 14. No additional action is needed.	Check the configuration of the vehicle	Electronic control unit
ABS	U30004B	Temperature monitoring is carried out by reading the hardware status information of the ECU. Temperature threshold is 100°C +/- 5°C	Over Temperature	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Ambient temperature too high or damage of the central ECU	Diagnostic Service 14. No additional action is needed.	"Check ECU location and suroundings for high temperatures. - If fault is active, replace it. "	Electronic control unit
ABS	U300094	By several monitoring functions.	Unexpected Operation	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damaged dataset or damaged software or damaged ECU, also too weak brakes can be the reason for this fault.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
ABS	U300316	Failure is set if supply voltage of the ECU is out of range. (<20V)	Circuit Voltage Below Threshold	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Battery is not charged enough or damage of the vehicle's power supply system or damage in the wiring or connectors for the ECU supply or damage of the ECU	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
ABS	U300317	Failure is set if supply voltage of the ECU is out of range. (>32V)	Circuit Voltage Above Threshold	ABS Function: Function is not operable, predefined or latest normally calculated values used as outputs EBD Function: Function is not operable, predefined or latest normally calculated values used as outputs	Damage of the vehicle's power supply system or damage of the ECU	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system.	Battery Power
ACM	U001088		Bus off	.	.	Diagnostic Service 14. No additional action is needed.	.	CAN Communication Backbone 1 Net
ACM	U200988		Bus off	.	.	Diagnostic Service 14. No additional action is needed.	.	CAN Communication Powertrain Net
ACM	U010000	Lost communication with ECM	No Sub Type Information	.	.	Diagnostic Service 14. No additional action is needed.	.	Lost Communication with EMS
ACM	P054415	Measured voltage outside the limits specified by the sensor.	Circuit Short To Battery or Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Exhaust Gas Temperature Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P205A13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted AdBlue tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed.	Check: AdBlue tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
ACM	P060747		Watchdog / Safety µC Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed.	.	Control Module Performance
ACM	P06B100	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness	Diagnostic Service 14. No additional action is needed.	Check: Harness	Sensor Power Supply A Circuit Low
ACM	P06B200	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness	Diagnostic Service 14. No additional action is needed.	Check: Harness	Sensor Power Supply A Circuit High
ACM	P06B400	Measured voltage below limit.	No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed.	Check: Harness	Sensor Power Supply B Circuit Low
ACM	P06B500	Measured voltage above limit.	No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed.	Check: Harness	Sensor Power Supply B Circuit High
ACM	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	Download issue Software download application	Diagnostic Service 14. No additional action is needed.	Check: Software download application	Electronic control unit
ACM	P204F94		Unexpected Operation	.	.	Diagnostic Service 14. No additional action is needed.	.	SCR System Performance Dosing Failure
ACM	P20C000	Measured voltage above limit.	No Sub Type Information	.	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed.	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Hose Heater 2 Short Circuit High
ACM	P20BF00	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Hose Heater 2 Short Circuit Low
ACM	P20BD13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Heater "B" Control
ACM	P208E00		No Sub Type Information	No urea dosing	dosing valve injector	Diagnostic Service 14. No additional action is needed.	Check: dosing valve injector	Aftertreatment Reagent Dosing Valve Clogged
ACM	P06E817	Measured voltage above limit.	Circuit Voltage Above Threshold	.	Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check: Short circuit to battery	Sensor Power Supply "C" Circuit High
ACM	P06E718	Measured voltage below limit.	Circuit Current Below Threshold	.	Short circuited to ground voltage	Diagnostic Service 14. No additional action is needed.	Check: Short circuited to ground voltage	Sensor Power Supply "C" Circuit Low
ACM	P107B13	Measured voltage outside the limits specified by the sensor.	Circuit Open	Limited retarder torque	Connector not properly mounted Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Retarder Water Temperature Sensor
ACM	P107C13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Retarder Oil Temperature Sensor
ACM	P107D07		Mechanical Failures	The retarder does not work	.	Diagnostic Service 14. No additional action is needed.	.	Retarder Valve
ACM	P107D13	Measured voltage outside the limits specified by the sensor.	Circuit Open	The retarder does not work	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Retarder Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P107E13	Measured voltage outside the limits specified by the sensor.	Circuit Open	The retarder does not work	Connector not properly mounted Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Retarder Air Pressure Sensor
ACM	P202D7A		Fluid Leak or Seal Failure	No urea dosing	ACM dosing module: Dosing valve Nozzle clogged Pressure line clogged Pressure line pipe leakage, ACM dosing module: Dosing valve, ACM pump module: Vent valve, ACM pump module: Internal leakage	Diagnostic Service 14. No additional action is needed.	Check: Pressure line pipe leakage, ACM dosing module: Dosing valve, ACM pump module: Vent valve, ACM pump module: Internal leakage	Aftertreatment Reagent Pressure Leakage
ACM	P204A13	Measured voltage outside the limits specified by the sensor.	Circuit Open	No urea dosing	Connector not properly mounted ACM Pump module: AdBlue pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed.	Check: ACM Pump module: AdBlue pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
ACM	P205C00	Measured voltage below limit.	No Sub Type Information	.	AdBlue tank temp sensor Cables Connectors AdBlue tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed.	Check: AdBlue tank temp sensor, Cables, Connectors	Aftertreatment Reagent Tank Temperature Short Circuit Low
ACM	P203D00	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed.	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
ACM	P208A13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Pump "A" Control
ACM	P054500	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Exhaust Gas Temperature Short Circuit Low
ACM	P20B297		Component or System Operation Obstructed or Blocked	.	Cable Connectors AdBlue tank heating valve Cable, Connectors, AdBlue tank heating valve	Diagnostic Service 14. No additional action is needed.	Check: Cable, Connectors, AdBlue tank heating valve	Reductant Heater Coolant Control Valve Performance/Stuck Open
ACM	P20B400	Measured voltage above limit.	No Sub Type Information	.	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed.	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Reductant Heater Coolant Control Valve Circuit High
ACM	P20B300	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Reductant Heater Coolant Control Valve Circuit Low
ACM	P20B113	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Heater Coolant Control Valve Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P204800	Measured voltage below limit.	No Sub Type Information	No urea dosing	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Dosing Valve Short Circuit Low
ACM	P204713	Measured voltage outside the limits specified by the sensor.	Circuit Open	No urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors, ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed.	Check: Cable / Connectors, ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
ACM	P20C400	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Hose Heater 3 Short Circuit High
ACM	P20C300	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
ACM	P20C113	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Heater "C" Control
ACM	P20BC00	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Hose Heater 1 Short Circuit High
ACM	P20BB00	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
ACM	P20B913	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Heater "A" Control
ACM	P20A300	Measured voltage above limit.	No Sub Type Information	.	Internal electrical failure in valve within the ACM Valve within the ACM	Diagnostic Service 14. No additional action is needed.	Check: Valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit High
ACM	P20A200	Measured voltage below limit.	No Sub Type Information	.	Internal electrical failure in valve within the ACM Valve within the ACM	Diagnostic Service 14. No additional action is needed.	Check: Valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
ACM	P20A013	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Reductant Purge Control Valve
ACM	P20A107		Mechanical Failures	.	Internal electrical failure in valve within the ACM Valve within the ACM	Diagnostic Service 14. No additional action is needed.	Check: Valve within the ACM	Aftertreatment Reagent Direction Valve Mechanical Problem
ACM	P065900	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	ACM Actuator High Supply 1 Short Circuit High
ACM	P065800	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	ACM Actuator High Supply 1 Short Circuit Low
ACM	P208D00	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Pump Control Short Circut High
ACM	P208C00	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Aftertreatment Reagent Pump Control Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P203A13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Reductant Level Sensor "A"
ACM	P056300		No Sub Type Information	.	.	Diagnostic Service 14. No additional action is needed.	.	ECU Battery Potential Above Range
ACM	P204C00	Measured voltage below limit.	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: AdBlue pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed.	Check: ACM Pump module: AdBlue pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
ACM	U300A11	Measured voltage below limit.	Circuit Short To Ground	.	Short circuited to ground voltage	Diagnostic Service 14. No additional action is needed.	Check: Short circuited to ground voltage	Key Position
ACM	U300A12	Measured voltage above limit.	Circuit Short To Battery	.	Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check: Short circuit to battery	Key Position
ACM	P20E892		Performance or Incorrect Operation	.	Empty AdBlue tank AdBlue filter clogged AdBlue inlet pipe leakage/blocked AdBlue tank ventilation ACM Pump module: Pump Empty AdBlue tank, AdBlue filter clogged, AdBlue inlet pipe leakage/blocked, AdBlue tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed.	Check: Empty AdBlue tank, AdBlue filter clogged, AdBlue inlet pipe leakage/blocked, AdBlue tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low
ACM	P21CA00		No Sub Type Information	.	.	Diagnostic Service 14. No additional action is needed.	.	Reductant Control Module Supply Voltage
ACM	P10AD13	Measured voltage outside the limits specified by the sensor.	Circuit Open	.	Connector not properly mounted	Diagnostic Service 14. No additional action is needed.	Check: Connector not properly mounted	Reductant Pump "A" Control Low Side
ACM	P10AE00	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Reductant Pump "A" Control Low Side Circuit Low
ACM	P10AF00	Measured voltage above limit.	No Sub Type Information	.	Short circuit to battery Harness Actuator	Diagnostic Service 14. No additional action is needed.	Check: Harness Actuator	Reductant Pump "A" Control Low Side Circuit High
ACM	P10CE97		Component or System Operation Obstructed or Blocked	.	.	Diagnostic Service 14. No additional action is needed.	.	Reductant Return No Flow Detected
ACM	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed.	.	Control Module
ACM	P24FF00		No Sub Type Information	.	.	Diagnostic Service 14. No additional action is needed.	.	Reductant Temperature Too High
ACM	U1FFF68	The factory mode is active through P1HCF	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
ACM	U012200	Lost Communication With Body Control Module	No Sub Type Information	.	.	Diagnostic Service 14. No additional action is needed.	.	Lost Communication With Vehicle Dynamics Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P109A00	Temperature above normal operating conditions	No Sub Type Information	The hydraulic retarder temperature symbol is sent The retarder does not work	Too high water temperature.	Diagnostic Service 14. No additional action is needed.	Check: Too high water temperature.	Retarder Water Temperature Moderately High
ACM	P109C00	Measured voltage below limit.	No Sub Type Information	Limited retarder torque	Short circuited to ground voltage Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Retarder Water Temperature Sensor Circuit Low
ACM	P109D00	Temperature above normal operating conditions	No Sub Type Information	Limited retarder torque	Too high retarder oil temperature	Diagnostic Service 14. No additional action is needed.	Check: Too high retarder oil temperature	Retarder Oil Temperature High
ACM	P109F00	Measured voltage below limit.	No Sub Type Information	.	Short circuited to ground voltage Harness Sensor	Diagnostic Service 14. No additional action is needed.	Check: Harness Sensor	Retarder Oil Temperature Sensor Circuit Low
ACM	P10A300		No Sub Type Information	The retarder valve is closed The retarder does not work	.	Diagnostic Service 14. No additional action is needed.	.	Retarder Air Pressure High
ACM	P10A464		Signal Plausibility Failure	Limited retarder torque	.	Diagnostic Service 14. No additional action is needed.	.	Retarder Air Pressure Sensor Range/Performance
ACM	P10A500	Measured voltage above limit.	No Sub Type Information	The retarder does not work	Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check: Short circuit to battery	Retarder Air Pressure Sensor Circuit High
ACM	P10A600	Measured voltage above limit.	No Sub Type Information	The retarder does not work	Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check: Short circuit to battery	Retarder Valve Circuit High
ACM	P10A700	Measured voltage below limit.	No Sub Type Information	The retarder does not work	Short circuited to ground voltage	Diagnostic Service 14. No additional action is needed.	Check: Short circuited to ground voltage	Retarder Valve Circuit Low
Alarm	U005588	CAN Security Subnet is off.	Bus off	Lack of CAN communication causes: - Alarm can not be armed or disarmed - sensors which are delivered via CAN can not trigger an alarm - no feedback to CIOM about current Alarm ECU state - no feedback to CIOM about last alarm reason - no feedback to CIOM about active DTCs in Alarm ECU (fault event gateway) - alarm is not triggered due to lack of CAN communication	CAN_H is shorted to GND CAN_L is shorted to UBATT	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors.	CAN Communication Security Net
Alarm	U300041	Parameters checksum not valid.	General Checksum Failure	Boodloader is informed about this fault and then application will be never started after reset.	AECU configuration parameters corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice	Electronic control unit
Alarm	U300044	AECU volatile memory error	Data Memory Failure	SW does not specify any functional behavior when this DTC is set.	AECU corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice.	Electronic control unit
Alarm	U300045	AECU non-volatile memory error	Program Memory Failure	Boodloader is informed about this fault and then application will be never started after reset.	AECU corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Alarm	U300046	AECU EEPROM is corrupted.	Calibration / Parameter Memory Failure	SW does not specify any functional behavior when this DTC is set.	AECU corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice.	Electronic control unit
Alarm	U300049	AECU internal hardware fault - CPU clock failure	Internal Electronic Failure	SW does not specify any functional behavior when this DTC is set.	AECU corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice.	Electronic control unit
Alarm	U300054	Routine for Ultrasonic/Inclination Sensors calibration has to be started. If routine ended successful then DTC will be PASSED, when routine failed then DTC will be FAILED. Until routine starts the DTC will be NOT TESTED.	Missing Calibration	Any defects regarding Calibration for Ultrasonic/Inclination sensors can cause invalid work of Ultrasonic/Inclination sensors: - Ultrasonic/Inclination sensors can be more or less sensitive and alarm can be false generated or at all - Ultrasonic/Inclination sensors can not be activated. So no alarm due to: movement detection or inclination	Ultrasonic/Inclination Sensors Calibration Routine not completed - or wrong calibration data - or lost communication with sensor - or calibration verification not passed.	Diagnostic Service 14. No additional action is needed.	Perform the calibration of MDD sensors (launch the routines R1AIA & R1AIB).	Electronic control unit
Alarm	U300094	AECU will be reset due to watchdog timer expiration	Unexpected Operation	SW does not specify any functional behavior when this DTC is set.	SW Failure.	Diagnostic Service 14. No additional action is needed.	Replace the component and send it to backoffice.	Electronic control unit
Alarm	U300316	AECU power supply <17V	Circuit Voltage Below Threshold	It can cause: - power supply for MDD and siren can not be maintained. It can cause that alarm is triggered or MDD and siren can not be armed - power supply for uController is too low and Alarm ECU works unstable. False alarm possible.	Battery level is below 17V.	Diagnostic Service 14. No additional action is needed.	Inspect the battery Inspect the harness	Battery Power
Alarm	U300317	AECU power supply > 31V	Circuit Voltage Above Threshold	It can cause: - alarm ECU hardware can be damged	Alternator/external charger voltage above 31V	Diagnostic Service 14. No additional action is needed.	Inspect the battery Inspect the harness	Battery Power
Alarm	B150697	Panic button stays in pushed state longer as defined in PanicCabButtonPulseTimeout parameter.	Component or System Operation Obstructed or Blocked	Presence of defect inhibits the Panic Alarm button.	Panic button hardware failure.	Diagnostic Service 14. No additional action is needed.	Inspect the battery Inspect the harness	Panic button
Alarm	B150249	Siren internal fault	Internal Electronic Failure	Siren won't ring.	Siren corrupted.	Diagnostic Service 14. No additional action is needed.	Replace the siren and send it to backoffice.	Siren Module
Alarm	B150268	Siren and AECU not correctly paired.	Event Information	Not possible to leave service mode.	Interruption of the procedure or Siren corrupted. Or Siren missing.	Diagnostic Service 14. No additional action is needed.	Check siren connection else Replace the siren and send it to backoffice.	Siren Module
Alarm	B150287	Siren doesn't communicate with Alarm ECU.	Missing Message	Siren won't ring.	Connector between the truck wiring harness and the Siren detached.	Diagnostic Service 14. No additional action is needed.	Inspect the siren, the connector and the harness.	Siren Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Alarm	B150349	Ultrasonic/Inclination Sensors internal fault	Internal Electronic Failure	Data from Ultrasonic/Inclination Sensors are invalid and might not detect movement in cab, or generate false alarm.	1. internal eeprom in ultrasonic or inclination sensor 2. bad result of ultrasonic selftest (selftest is run just before ultrasonic is being armed) 3. bad result of inclination sensor (selftest is run just before inclination is being armed) Two possible causes for defect 2 & 3: - missing microphone or sounder connected to ultrasonic - bad position of inclination sensor. Sensor shall lie horizontally, not tilted, not aside	Diagnostic Service 14. No additional action is needed.	First check MDD position Then check microphone & sounder If no issue found, replace the MDD and send it to backoffice.	Motion Detection Module
Alarm	B150387	Ultrasonic/Inclination Sensors don't communicated with Alarm ECU.	Missing Message	Ultrasonic/Inclination Sensors can't trigger alarm: movement detection or inclination detection won't work.	Connector between wiring harness of the truck and the Ultrasonic/Inclination Sensors detached.	Diagnostic Service 14. No additional action is needed.	Inspect the MDD, the connector and the harness.	Motion Detection Module
Alarm	U114111	Communication network between AECU, Ultrasonic/Inclination Sensors, Siren have been shorted to ground.	Circuit Short To Ground	If the Alarm System is in Service Mode, the Siren doesn't ring. Else, the siren is triggered and rings as if the system was hacked Ultrasonic/Inclination Sensors don't trigger the Alarm system.	Wiring harness faulty	Diagnostic Service 14. No additional action is needed.	- Inspect the Siren, the MDD - Inspect the harness and connectors.	Alarm Private Net
Alarm	U114112	Communication network between AECU, Ultrasonic/Inclination Sensors, Siren have been shorted to Ubatt.	Circuit Short To Battery	If the Alarm System is in Service Mode, the Siren doesn't ring. Else, the siren is triggered and rings as if the system was hacked Ultrasonic/Inclination Sensors don't trigger the Alarm system.	Wiring harness faulty	Diagnostic Service 14. No additional action is needed.	- Inspect the Siren, the MDD - Inspect the harness and connectors.	Alarm Private Net
Alarm	U114186	Communication network between AECU, Ultrasonic/Inclination Sensors, Siren not valid.	Signal Invalid	Data from Ultrasonic/Inclination Sensors, or siren are invalid and it might cause: - generate false alarm	1. Wiring harness faulty 2. EMC problem	Diagnostic Service 14. No additional action is needed.	- inspect the MDD and the siren - replace the component and send it to backoffice	Alarm Private Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Alarm	B150411	AECU 12V Power supply output to Ultrasonic/Inclination Sensors and Siren has been shorted to ground.	Circuit Short To Ground	When defect is present then 12V (Power supply for Siren and Ultrasonic/Inclination Sensors) is switched off. After that, every 1sec there is an attempt to switch on 12V. Those attempts are performed as long as short is present. Presence of short 12V at arming siren or Ultrasonic/Inclination Sensors causes that siren and Ultrasonic/Inclination can not be armed. When presence of short 12V takes place when siren is already armed than alarm is triggered. Siren starts sounding using backup battery. In that case siren backup battery must be already charged.	Wiring harness faulty	Diagnostic Service 14. No additional action is needed.	- inspect the MDD and the siren - inspect the harness	Alarm Private Net Modules Supply Circuit
Alarm	B150417	AECU 12V Power supply output to Ultrasonic/Inclination Sensors and Siren >12V	Circuit Voltage Above Threshold	When defect is present then 12V (Power supply for Siren and Ultrasonic/Inclination Sensors) is switched off. After that, every 1sec there is an attempt to switch on 12V. Those attempts are performed as long as short is present. Presence of short 12V at arming siren or Ultrasonic/Inclination Sensors causes that siren and Ultrasonic/Inclination Sensors can not be armed. When presence of short 12V takes place when siren is already armed than alarm is triggered. Siren starts sounding using backup battery. In that case siren backup battery must be already charged.	Wiring harness faulty	Diagnostic Service 14. No additional action is needed.	- inspect the MDD and the siren - inspect the harness	Alarm Private Net Modules Supply Circuit
Alarm	B150511	AECU output to LED has been shorted to ground.	Circuit Short To Ground	LED is permanently OFF and it doesn't reflect current Alarm ECU state (e.g. UNSET- no blink LED, SET- blink LED 1Hz etc).	Wiring harness faulty	Diagnostic Service 14. No additional action is needed.	- inspect the LED - inspect the harness	LED Power Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Alarm	U1FFF68	Factory mode active.	Event Information	Not possible to leave service mode. Alarm is not armed.	P1C54 set to true.	Diagnostic Service 14. No additional action is needed.	Exit Factory Mode	Factory Mode Active
TECU	P107177		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107179		Mechanical Linkage Failure			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107192		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107194		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107196		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107197		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P106D17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106D18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106D19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106C17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106C18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106C19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106B17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106B18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106B19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106A17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106A18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106A19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106719		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)
TECU	P106618		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)
TECU	P106619		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106517		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P106518		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P106519		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P105017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105054		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105422		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105818		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105821		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105716		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105721		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105729		Signal invalid			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105A17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105A18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105A19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105B17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105B18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105B19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105C17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)
TECU	P105C18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)
TECU	P105C19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)
TECU	P105D17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)
TECU	P105D18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105D19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)
TECU	P105E17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E54		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105F17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F54		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P107538		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107571		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107574		Actuator Slipping			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107575		Emergency Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107577		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107592		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107594		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107471		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107477		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107492		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107494		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107496		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107497		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107371		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107377		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107392		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107394		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107396		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107397		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107277		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107292		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107294		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107297		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P105916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105918		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105922		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105992		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105998		Component or System Over Temperature			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P106154		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Slipping Point Position
TECU	P106216		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Position Sensor Supply (SEPoC5V)
TECU	P106218		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Position Sensor Supply (SEPoC5V)
TECU	P106329		Signal invalid			Diagnostic Service 14. No additional action is needed.		Gearbox Inclination Sensor (SEIG)
TECU	P106817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Fast Clutch Valves Low Side Circuit (VAF-)
TECU	P106819		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Fast Clutch Valves Low Side Circuit (VAF-)
TECU	P106917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Slow Clutch Valves Low Side Circuit (VAS-)
TECU	P106919		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Slow Clutch Valves Low Side Circuit (VAS-)
TECU	P107017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107694		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gearbox Brake (Countershaft Brake / Input Shaft Brake)
TECU	P106417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)
TECU	P106418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106419		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)
TECU	U000188		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 2 Net
TECU	U001088		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 1 Net
TECU	P105117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105118		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105119		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105154		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105217		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105218		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105229		Signal invalid			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105254		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105262		Signal Compare Failure			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105378		Alignment or Adjustment Incorrect			Diagnostic Service 14. No additional action is needed.		Clutch Plate Wear
TECU	P105594		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Brake Switch Status
TECU	U114100		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With HMIIOM
TECU	U010000		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication with EMS
TECU	U014000		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With VMCU
TECU	P060100		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Internal Control Module Memory Check Sum Error
TECU	U114286		Signal Invalid			Diagnostic Service 14. No additional action is needed.		Wheel-Based Vehicle Speed
TECU	U300017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300044		Data Memory Failure			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300046		Calibration / Parameter Memory Failure			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U30004B		Over Temperature			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Battery Power
TECU	U300317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Battery Power
TECU	U007387		Missing Message			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet
TECU	U007388		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet
TECU	U012900		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With Brake System Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U030055		Not Configured			Diagnostic Service 14. No additional action is needed.		Program Memory EEPROM Development SW
TECU	P107B86		Signal Invalid			Diagnostic Service 14. No additional action is needed.		Accelerator Pedal Position / Brake Switch Signal Performance
TECU	U1FFF68		Event Information			Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	P10AE81		Invalid Serial Data Received			Diagnostic Service 14. No additional action is needed.		Engaged Gear in Neutral
TECU	P107177		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107179		Mechanical Linkage Failure			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107192		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107194		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107196		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P107197		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Split Engagment System
TECU	P106D17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106D18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106D19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 3 Valve (VAG3)
TECU	P106C17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106C18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106C19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 2 Valve (VAG2)
TECU	P106B17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106B18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106B19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear Reverse Valve (VAGR)
TECU	P106A17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106A18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106A19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gear 1 Valve (VAG1)
TECU	P106717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106719		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Disengagement Valve (VASD)
TECU	P106617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106618		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)
TECU	P106619		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Disengagement Valve (VAFD)
TECU	P106517		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P106518		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P106519		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Slow Engagement Valve (VASE)
TECU	P105017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105054		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Range Position Sensor (SEPoR)
TECU	P105417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105422		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Tank Air Pressure Sensor (SEPrSA)
TECU	P105816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105818		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105821		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105716		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105721		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105729		Signal invalid			Diagnostic Service 14. No additional action is needed.		Main Shaft Speed Sensor (SESM)
TECU	P105A17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105A18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105A19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		High Range Valve (VAHR)
TECU	P105B17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105B18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105B19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Low Range Valve (VALR)
TECU	P105C17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)
TECU	P105C18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105C19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Direct Split Valve (VADS)
TECU	P105D17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)
TECU	P105D18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)
TECU	P105D19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Indirect Split Valve (VAIDS)
TECU	P105E17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105E54		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Shifter 1 Position Sensor (SEPo1R)
TECU	P105F17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P105F54		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Shifter 2 Position Sensor (SEPo23)
TECU	P107538		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107571		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107574		Actuator Slipping			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107575		Emergency Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107577		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107592		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107594		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Clutch System
TECU	P107471		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107477		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107492		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107494		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107496		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107497		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Gears 2 and 3 Engagement System
TECU	P107371		Actuator Stuck			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107377		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107392		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107394		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107396		Component Internal Failure			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107397		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Gears 1 and R Engagement System
TECU	P107277		Commanded Position Not Reachable			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107292		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107294		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P107297		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Range Engagement System
TECU	P105916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105918		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105922		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105992		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105998		Component or System Over Temperature			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P106154		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Slipping Point Position
TECU	P106216		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Position Sensor Supply (SEPoC5V)
TECU	P106218		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Position Sensor Supply (SEPoC5V)
TECU	P106329		Signal invalid			Diagnostic Service 14. No additional action is needed.		Gearbox Inclination Sensor (SEIG)
TECU	P106817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Fast Clutch Valves Low Side Circuit (VAF-)
TECU	P106819		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Fast Clutch Valves Low Side Circuit (VAF-)
TECU	P106917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Slow Clutch Valves Low Side Circuit (VAS-)
TECU	P106919		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Slow Clutch Valves Low Side Circuit (VAS-)
TECU	P107017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Brake Valve (VAGB)
TECU	P107694		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Gearbox Brake (Countershaft Brake / Input Shaft Brake)
TECU	P106417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)
TECU	P106419		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Fast Engagement Valve (VAFE)
TECU	U000188		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 2 Net
TECU	U001088		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 1 Net
TECU	P105117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105118		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105119		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105154		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Split Position Sensor (SEPoS)
TECU	P105217		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105218		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105229		Signal invalid			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105254		Missing Calibration			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105262		Signal Compare Failure			Diagnostic Service 14. No additional action is needed.		Clutch Cylinder Position
TECU	P105378		Alignment or Adjustment Incorrect			Diagnostic Service 14. No additional action is needed.		Clutch Plate Wear
TECU	P105594		Unexpected Operation			Diagnostic Service 14. No additional action is needed.		Brake Switch Status
TECU	U114100		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With HMIIOM
TECU	U010000		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication with EMS
TECU	U014000		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With VMCU
TECU	P060100		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Internal Control Module Memory Check Sum Error
TECU	U114286		Signal Invalid			Diagnostic Service 14. No additional action is needed.		Wheel-Based Vehicle Speed
TECU	U300017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300044		Data Memory Failure			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300046		Calibration / Parameter Memory Failure			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U30004B		Over Temperature			Diagnostic Service 14. No additional action is needed.		Electronic control unit
TECU	U300316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Battery Power
TECU	U300317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Battery Power
TECU	U007387		Missing Message			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet
TECU	U007388		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U012900		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Lost Communication With Brake System Control Module
TECU	U030055		Not Configured			Diagnostic Service 14. No additional action is needed.		Program Memory EEPROM Development SW
TECU	P107B86		Signal Invalid			Diagnostic Service 14. No additional action is needed.		Accelerator Pedal Position / Brake Switch Signal Performance
TECU	U1FFF68		Event Information			Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	P10AE81		Invalid Serial Data Received			Diagnostic Service 14. No additional action is needed.		Engaged Gear in Neutral
TECU	P107177	-The indirect split gear can not engage	Commanded Position Not Reachable	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU Dragging clutch Broken synchronizer	Diagnostic Service 14. No additional action is needed.	GCU, Synchronizer, Driven plate, CCA	Split Engagment System
TECU	P107179	-The split cylinder position leaves the neutral position -The split cylinder valves are inactive	Mechanical Linkage Failure	None	Valve, air leakage, calibration of split	Diagnostic Service 14. No additional action is needed.	Sensor or gear position detent	Split Engagment System
TECU	P107192	-The neutral split gear can not engage	Performance or Incorrect Operation	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU	Diagnostic Service 14. No additional action is needed.	GCU	Split Engagment System
TECU	P107194	-The split cylinder position leaves the indirect position -The split cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Worn out engagement teeth or parts in GCU	Diagnostic Service 14. No additional action is needed.	Engaging sleeve, ring, spline, position detent GCU	Split Engagment System
TECU	P107196	-The split cylinder position leaves the direct position -The split cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Worn out engagement teeth or parts in GCU	Diagnostic Service 14. No additional action is needed.	Engaging sleeve, ring, spline, position detent GCU	Split Engagment System
TECU	P107197	-The direct split gear can not engage	Component or System Operation Obstructed or Blocked	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU Dragging clutch Broken synchronizer	Diagnostic Service 14. No additional action is needed.	GCU, Synchronizer, Driven plate, CCA	Split Engagment System
TECU	P106D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106A18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106717	1. The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106718	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106719	The low-side drive is short circuit to Gnd. Also set for all VAS low sides when VAS has a weak SCL	Circuit Current Above Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106617	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106618	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106619	The low-side drive is short circuit to Gnd. Also set for all VAF low sides when VAF has a weak SCL	Circuit Current Above Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106517	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P106518	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P106519	The low-side drive is short circuit to Gnd. Also set for all VAS low sides when VASHS has a weak SCL	Circuit Current Above Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P105017	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105018	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105019	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105054	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Range calibration not done after software download	Diagnostic Service 14. No additional action is needed.	Calibrate range	Range Position Sensor (SEPoR)
TECU	P105417	The sensor signal is above normal range	Circuit Voltage Above Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105418	The sensor signal is below normal range	Circuit Current Below Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105422	1. Sensor signal is in normal range 2. Pressure is above 10.0 [bar]	Signal Amplitude Greater Than Maximum	None	High air pressure or fault on pressure sensor	Diagnostic Service 14. No additional action is needed.	Pressure sensor	Tank Air Pressure Sensor (SEPrSA)
TECU	P105816	The sensor voltage signal is below normal range when enabled and in normal or below normal range when disabled.	Circuit Voltage Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105817	The sensor voltage signal is above normal range	Circuit Voltage Above Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105818	The sensor voltage signal is below normal range when sensor is enabled and above normal range when disabled.	Circuit Current Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105821	1. The sensor speed signal is within normal range and, 2. The counter shaft speed differs from both the main shaft speed (SESM) and the vehicle speed received from the vehicle ecu when the counter shaft is rotating	Signal Amplitude Less Than Minimum	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105716	ASIC reports short circuit Gnd.	Circuit Voltage Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105717	ASIC reports short circuit Ubatt or supply is greater than 12 V.	Circuit Voltage Above Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105718	ASIC reports open circuit.	Circuit Current Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105721	1. Sensor signal is within normal range and 2. The mainshaft speed differs from both the value of the countershaft speed and the vehicle speed recieved from the vehicle ECU	Signal Amplitude Less Than Minimum	Slow shifting Harsh shifting	Fault on mainshaft speed sensor or teeth wheel	Diagnostic Service 14. No additional action is needed.	Sensor or teeth wheel	Main Shaft Speed Sensor (SESM)
TECU	P105729	The sensor indicates wrong travelling direction	Signal invalid	Slow shifting Harsh start Starting gear D/Rev does not engage - when driving/moving	Mainshaft speed sensor	Diagnostic Service 14. No additional action is needed.	Sensor	Main Shaft Speed Sensor (SESM)
TECU	P105A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve	High Range Valve (VAHR)
TECU	P105A18	The low side drive is Open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)
TECU	P105A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)
TECU	P105D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)
TECU	P105D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105E17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate mainbox	Shifter 1 Position Sensor (SEPo1R)
TECU	P105F17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105F18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)
TECU	P105F19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)
TECU	P105F54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate mainbox	Shifter 2 Position Sensor (SEPo23)
TECU	P107538	Normal control signal exceeds its upper limit, clutch does not reach set point without extra activation. - Measured after 5s has elapsed from the point of time when control mode was last changed, any of the valves (VAFE, VASE, VAFD, VASD) has been used continuously more than an adjustable mean level of activation during an adjustable time - Clutch position is outside tolerance limits to any disengaged set point for more than an adjustable time. - The engine is running and,	Signal Frequency Incorrect	Harsh start Harsh shifting Slow shifting	Air leakage in clutch system	Diagnostic Service 14. No additional action is needed.	CVU or CCA	Clutch System
TECU	P107554	The clutch has not been calibrated after replacement.	Missing Calibration	Starting gear D/Rev does not engage	Not calibrated after clutch plate replacement	Diagnostic Service 14. No additional action is needed.	Calibrate clutch system	Clutch System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107571	The clutch can not transfer a specific torque without slipping	Actuator Stuck	Engine derate	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch System
TECU	P107572	It takes too long time to reach engage set point.	Actuator Stuck Open	Harsh start	Air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	CVU	Clutch System
TECU	P107574	The clutch transfers too much torque in disengaged position.	Actuator Slipping	Unexpected acceleration	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch System
TECU	P107575	1. The clutch is not in disengaged position, and maximum level of activation requested for more than an adjustable time limit. 2. Engine is running 3. (A gear is engaged OR Gearbox is in neutral and vehicle is moving) 4. Normal clutch control exceeded	Emergency Position Not Reachable	Starting gear D/Rev does not engage Engine shuts down unexpectedly	Air leakage or fault on CVU or CCA	Diagnostic Service 14. No additional action is needed.	CVU, CCA or air hoses	Clutch System
TECU	P107577	1. The clutch does not disengage/engage properly when commanded 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low airpressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above 10 dgC	Commanded Position Not Reachable	Slow shifting	Fault on CVU, CCA, clutch system or sensor	Diagnostic Service 14. No additional action is needed.	Sensor, CVU or CCA	Clutch System
TECU	P107586	Vehicle starts to move but the clutch position indicates open position and input shaft speed is equal to engine speed	Signal Invalid	Starting gear D/Rev does not engage Engine shuts down unexpectedly	Cylinder connection failure	Diagnostic Service 14. No additional action is needed.		Clutch System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107592	1. The clutch engages when not commanded and, 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low air pressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above a specific limit	Performance or Incorrect Operation	Slow shifting	Air leakage in CCA or in air hose or position sensor	Diagnostic Service 14. No additional action is needed.	CVU, CCA, air hose	Clutch System
TECU	P107594	1. The clutch disengages when not commanded and, 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low air pressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above a specific limit	Unexpected Operation	Slow shifting	Air leakage in clutch system, valves in CVU	Diagnostic Service 14. No additional action is needed.	Valve	Clutch System
TECU	P107471	-The cylinder for gear 2/3 can not reach neutral position	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107477	-The cylinder for gear 2/3 can not reach engaged position for gear 3.	Commanded Position Not Reachable	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107492	-The cylinder for gear 2/3 leaves the engaged postion for gear 3. - The 2/3 cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107494	-The cylinder for gear 2/3 leaves the neutral position. - The 2/3 cylinder valves are inactive	Unexpected Operation	Short loss of torque	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107496	-The cylinder for gear 2/3 leaves the engaged postion for gear 2. - The 2/3 cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107497	-The cylinder for gear 2/3 can not reach engaged position for gear 2.	Component or System Operation Obstructed or Blocked	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107371	-The cylinder for gear 1/R can not reach neutral position	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107377	-The cylinder for gear 1/R can not reach engaged position for reverse gear.	Commanded Position Not Reachable	Blocked out gears - Only forward gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107392	-The cylinder for gear 1/R leaves the engaged position for reverse reverse. -The 1/R cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107394	-The cylinder for gear 1/R leaves the neutral position -The 1/R cylinder valves are inactive	Unexpected Operation	Short loss of torque	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107396	-The cylinder for gear 1/R leaves the engaged position for gear 1. -The 1/R cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107397	-The cylinder for gear 1/R can not reach engaged position for gear 1.	Component or System Operation Obstructed or Blocked	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107277	-The low range gear can not engage	Commanded Position Not Reachable	Blocked out gears - Only high range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P107292	-The range cylinder leaves the high range position -The range cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107294	-The range cylinder leaves the low range position -The range cylinder valves are inactive	Unexpected Operation	Short loss of torque	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System
TECU	P107297	-The high range gear can not engage	Component or System Operation Obstructed or Blocked	Blocked out gears - Only low range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P105916	The sensor signal is below normal range	Circuit Voltage Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105918	The sensor signal is above normal range	Circuit Current Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105922	1.The sensor signal is in normal range 2. Temerature is above 100dgC during 18000s	Signal Amplitude Greater Than Maximum	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105992	1. The sensor signal is in normal range 2. Temperature above 120dgC during 600s	Performance or Incorrect Operation	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105998	1. The sensor signal is in normal range 2. Temperature is aove 140dgC during 30s	Component or System Over Temperature	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P106154	1. The CS in NVRAM is not correct, or 2. The slippoint has not been calibrated	Missing Calibration	Starting Gear D/Rev does not engage	Slippoint not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate slip point	Clutch Cylinder Slipping Point Position
TECU	P106216	Voltage is above normal range (5,5 V)	Circuit Voltage Below Threshold	Harsh start Slow shifting	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch Position Sensor Supply (SEPoC5V)
TECU	P106218	Voltage is below normal range (4,5 V)	Circuit Current Below Threshold	Harsh start Slow shifting	Fault on sensor or harness	Key off/key on required after one minute to reactivate diagnosis.	Sensor or harness	Clutch Position Sensor Supply (SEPoC5V)
TECU	P106329	More than half of the readings from the inclination sensor are faulty or doesn't change for 10 s.	Signal invalid	Wrong starter gear Unsuitable gear selected	Electrical fault on inclination sensor	Key off/key on required to reactivate diagnosis.	TECU	Gearbox Inclination Sensor (SEIG)
TECU	P107017	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107018	The low side drive is open circuit	Circuit Current Below Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107019	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107694	The gearbox brake does not brake when the valve is activated	Unexpected Operation	Slow shifting - When shifting to start gear	Air leakage, worn out disc or diaphragm in AMT-PS Air leakage, worn out discs in AMT-E	Diagnostic Service 14. No additional action is needed.	Brake	Gearbox Brake (Countershaft Brake / Input Shaft Brake)
TECU	P106417	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	P106418	The low-side drive is open circuit	Circuit Current Below Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	P106419	The low-side drive is short circuit to Gnd. Also set for all VAF low sides when VAFHS has a weak SCL	Circuit Current Above Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	U000188	TEA2+: CAN transiever signals "bus-off" on Backbone2	Bus off	Gearshifting inoperative - Manual gear shifting not possible Transmission fails to secure for crank	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 2 Net
TECU	U001088	This FMI shall be set if the CAN transiever signals "bus-off"	Bus off	Harsh shifting Gear shifts only at certain engine speeds Transmission fails to secure for crank AMT-PS: No powershift	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 1 Net
TECU	P105117	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)
TECU	P105118	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105119	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)
TECU	P105154	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Split calibration not done after software download	Diagnostic Service 14. No additional action is needed.	Calibrate split	Split Position Sensor (SEPoS)
TECU	P105217	Sensor signal is short circuit to Ubatt or sensor supply, when at least one of the sensor signals is above normal range (above 4,5 V) and the other is within normal range (0,5 V to 4,5 V).	Circuit Voltage Above Threshold	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105218	Open circuit on any of the sensor signals is detected when one of the sensor signals is within normal range (0,5V to 4,5 V) and the other sensor signal is below normal range. This includes either sensor signal being short circuit to Gnd.	Circuit Current Below Threshold	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105229	Each sensor signal is either below or above normal (0,5 to 4,5V) when: 1. Status of the sensor ASIC reports internal fault on the sensor, or 2. Open circuit on Gnd pin or, 3. Open circuit on Supply pin	Signal invalid	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105254	1. The CS in NVRAM is not correct 2. The cylinder has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Clutch stroke calibration not done after software download or after exchanged clutch plate	Diagnostic Service 14. No additional action is needed.	Calibrate clutch stroke	Clutch Cylinder Position

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105262	The sum of the clutch positions indicated by the sensor signal and the inverted sensor signal is too small (below 4,5V) and both signals is within normal range (0,5V to 4,5 V).	Signal Compare Failure	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Cylinder Position
TECU	P105378	The clutch wear is more than or equal to "Service due position"	Alignment or Adjustment Incorrect	None	Clutch worn	Diagnostic Service 14. No additional action is needed.	Clutch disc	Clutch Plate Wear
TECU	P105594	The gearlever is moved from neutral without prior application of the service brake and vehicle speed below 5 km/h	Unexpected Operation	Starting gear D/Rev does not engage	Brake pedal or parking brake (VTC only) not pressed when moving gear lever to Automatic, Manual or Reverse	Diagnostic Service 14. No additional action is needed.	None	Brake Switch Status
TECU	U114100	The FMI shall be set if any of the signals GearLeverPosition (BackBone1) or GearLeverPositionBackup (BackBone2) from HMIIOM is not received within 500ms.	No Sub Type Information	Gear shifting inoperative Gear shifting inoperative - Manual gear shifting not possible	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With HMIIOM
TECU	U010000	The FMI shall be set if the signal EngineSpeed from EMS is not received within 120ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication with EMS
TECU	U014000	The FMI shall be set if the signal AcceleratorPedalPosition1 from VMCU not received within 60 ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With VMCU
TECU	P060100	Calculated signature does not match signature provided during software download	No Sub Type Information	Engine derate	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Internal Control Module Memory Check Sum Error

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U114286	The FMI shall be set if any of the signals FrontAxleSpeed, FrAxleLeftWheelRelativeSpeed, FrAxleRightWheelRelativeSpeed, RrAxle1LeftWheelRelativeSpeed, RrAxle1RightWheelRelativeSpeed from EBS are incorrect and IDTC_EBS_MS is not set	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Wheel-Based Vehicle Speed
TECU	U300017	The voltage level for DIEE-pin is high	Circuit Voltage Above Threshold	Transmission fails to secure for crank	Fault on flexfoil or on the connector on the gcu	Diagnostic Service 14. No additional action is needed.	Flexfoil or connector on GCU	Electronic control unit
TECU	U300044	Flash CS error or program code missing	Data Memory Failure	Transmission fails to secure for crank	No software	Program ECU with SW	Software download	Electronic control unit
TECU	U300046	Flash CS dataset error or program code missing	Calibration / Parameter Memory Failure	Transmission fails to secure for crank	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Electronic control unit
TECU	U30004B	temp above 125dgC	Over Temperature	None	Too high temperature for pcb	Diagnostic Service 14. No additional action is needed.	None	Electronic control unit
TECU	U300316	24V system: below 17V 12V system: below 9V	Circuit Voltage Below Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	U300317	24V system: above 36V 12V system: above 19V	Circuit Voltage Above Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	U007387	The FMI shall be set if the signal EngineSpeed_VP24 from EECU is not recieved within 100ms	Missing Message	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	U007388	The FMI shall be set if the CAN transciever signals "busOFF"	Bus off	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	P109317	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109319	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109417	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit
TECU	P109419	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P109517	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109519	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109617	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P109619	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P10A017	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Slow Valves High Side Circuit (VASHS)
TECU	P10A019	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh start Harsh shifting	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Slow Valves High Side Circuit (VASHS)
TECU	P10A117	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Fast Valves High Side Circuit (VAFHS)
TECU	P10A119	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Slow shifting Harsh shifting	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Fast Valves High Side Circuit (VAFHS)
TECU	U012900	The FMI shall be set if the signal FrontAxleSpeed from EBS is not recieved within 240ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With Brake System Control Module
TECU	U030055	Test software is installed	Not Configured	None	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software dataset	Program Memory EEPROM Development SW
TECU	U030081	Wrong dataset for gearbox type OR incorrect structural checksum on the dataset.	Invalid Serial Data Received	Unsuitable gear selected Unexpected acceleration Harsh shifting	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software download	Program Memory EEPROM Development SW
TECU	P107B86	The FMI shall be set if any of the signals AcceleratorPedalPosition1 or BrakeSwitch from the VMCU are incorrect.	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Accelerator Pedal Position / Brake Switch Signal Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U1FFF68	Parameter Factory Mode is true	Event Information	None	Software in Factory Mode	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	P10AE81	Constant value for countershaft and mainshaft speed ratio when gearbox should be in neutral.	Invalid Serial Data Received	Gear shifting inoperative	Oil level, broken rod or fork	Diagnostic Service 14. No additional action is needed.	Engagement system	Engaged Gear in Neutral
TECU	P107177	-The indirect split gear can not engage	Commanded Position Not Reachable	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU Dragging clutch Broken synchronizer	Diagnostic Service 14. No additional action is needed.	GCU, Synchronizer, Driven plate, CCA	Split Engagment System
TECU	P107179	-The split cylinder position leaves the neutral position -The split cylinder valves are inactive	Mechanical Linkage Failure	None	Valve, air leakage, calibration of split	Diagnostic Service 14. No additional action is needed.	Sensor or gear position detent	Split Engagment System
TECU	P107192	-The neutral split gear can not engage	Performance or Incorrect Operation	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU	Diagnostic Service 14. No additional action is needed.	GCU	Split Engagment System
TECU	P107194	-The split cylinder position leaves the indirect position -The split cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Worn out engagement teeth or parts in GCU	Diagnostic Service 14. No additional action is needed.	Engaging sleeve, ring, spline, position detent GCU	Split Engagment System
TECU	P107196	-The split cylinder position leaves the direct position -The split cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Worn out engagement teeth or parts in GCU	Diagnostic Service 14. No additional action is needed.	Engaging sleeve, ring, spline, position detent GCU	Split Engagment System
TECU	P107197	-The direct split gear can not engage	Component or System Operation Obstructed or Blocked	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Air leakage, defect valve GCU Dragging clutch Broken synchronizer	Diagnostic Service 14. No additional action is needed.	GCU, Synchronizer, Driven plate, CCA	Split Engagment System
TECU	P106D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106A18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106717	1. The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106718	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106719	The low-side drive is short circuit to Gnd. Also set for all VAS low sides when VAS has a weak SCL	Circuit Current Above Threshold	Harsh start Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Disengagement Valve (VASD)
TECU	P106617	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106618	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106619	The low-side drive is short circuit to Gnd. Also set for all VAF low sides when VAF has a weak SCL	Circuit Current Above Threshold	Harsh shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Disengagement Valve (VAFD)
TECU	P106517	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P106518	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P106519	The low-side drive is short circuit to Gnd. Also set for all VAS low sides when VASHS has a weak SCL	Circuit Current Above Threshold	Harsh start	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Slow Engagement Valve (VASE)
TECU	P105017	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105018	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105019	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105054	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Range calibration not done after software download	Diagnostic Service 14. No additional action is needed.	Calibrate range	Range Position Sensor (SEPoR)
TECU	P105417	The sensor signal is above normal range	Circuit Voltage Above Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105418	The sensor signal is below normal range	Circuit Current Below Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105422	1. Sensor signal is in normal range 2. Pressure is above 10.0 [bar]	Signal Amplitude Greater Than Maximum	None	High air pressure or fault on pressure sensor	Diagnostic Service 14. No additional action is needed.	Pressure sensor	Tank Air Pressure Sensor (SEPrSA)
TECU	P105816	The sensor voltage signal is below normal range when enabled and in normal or below normal range when disabled.	Circuit Voltage Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105817	The sensor voltage signal is above normal range	Circuit Voltage Above Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105818	The sensor voltage signal is below normal range when sensor is enabled and above normal range when disabled.	Circuit Current Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105821	1. The sensor speed signal is within normal range and, 2. The counter shaft speed differs from both the main shaft speed (SESM) and the vehicle speed received from the vehicle ecu when the counter shaft is rotating	Signal Amplitude Less Than Minimum	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor (SESC)
TECU	P105716	ASIC reports short circuit Gnd.	Circuit Voltage Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105717	ASIC reports short circuit Ubatt or supply is greater than 12 V.	Circuit Voltage Above Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105718	ASIC reports open circuit.	Circuit Current Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105721	1. Sensor signal is within normal range and 2. The mainshaft speed differs from both the value of the countershaft speed and the vehicle speed recieved from the vehicle ECU	Signal Amplitude Less Than Minimum	Slow shifting Harsh shifting	Fault on mainshaft speed sensor or teeth wheel	Diagnostic Service 14. No additional action is needed.	Sensor or teeth wheel	Main Shaft Speed Sensor (SESM)
TECU	P105729	The sensor indicates wrong travelling direction	Signal invalid	Slow shifting Harsh start Starting gear D/Rev does not engage - when driving/moving	Mainshaft speed sensor	Diagnostic Service 14. No additional action is needed.	Sensor	Main Shaft Speed Sensor (SESM)
TECU	P105A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve	High Range Valve (VAHR)
TECU	P105A18	The low side drive is Open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)
TECU	P105A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Direct Split Valve (VADS)
TECU	P105D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)
TECU	P105D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)
TECU	P105D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Indirect Split Valve (VAIDS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105E17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate mainbox	Shifter 1 Position Sensor (SEPo1R)
TECU	P105F17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105F18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)
TECU	P105F19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 2 Position Sensor (SEPo23)
TECU	P105F54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate mainbox	Shifter 2 Position Sensor (SEPo23)
TECU	P107538	Normal control signal exceeds its upper limit, clutch does not reach set point without extra activation. - Measured after 5s has elapsed from the point of time when control mode was last changed, any of the valves (VAFE, VASE, VAFD, VASD) has been used continuously more than an adjustable mean level of activation during an adjustable time - Clutch position is outside tolerance limits to any disengaged set point for more than an adjustable time. - The engine is running and,	Signal Frequency Incorrect	Harsh start Harsh shifting Slow shifting	Air leakage in clutch system	Diagnostic Service 14. No additional action is needed.	CVU or CCA	Clutch System
TECU	P107554	The clutch has not been calibrated after replacement.	Missing Calibration	Starting gear D/Rev does not engage	Not calibrated after clutch plate replacement	Diagnostic Service 14. No additional action is needed.	Calibrate clutch system	Clutch System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107571	The clutch can not transfer a specific torque without slipping	Actuator Stuck	Engine derate	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch System
TECU	P107572	It takes too long time to reach engage set point.	Actuator Stuck Open	Harsh start	Air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	CVU	Clutch System
TECU	P107574	The clutch transfers too much torque in disengaged position.	Actuator Slipping	Unexpected acceleration	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch System
TECU	P107575	1. The clutch is not in disengaged position, and maximum level of activation requested for more than an adjustable time limit. 2. Engine is running 3. (A gear is engaged OR Gearbox is in neutral and vehicle is moving) 4. Normal clutch control exceeded	Emergency Position Not Reachable	Starting gear D/Rev does not engage Engine shuts down unexpectedly	Air leakage or fault on CVU or CCA	Diagnostic Service 14. No additional action is needed.	CVU, CCA or air hoses	Clutch System
TECU	P107577	1. The clutch does not disengage/engage properly when commanded 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low airpressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above 10 dgC	Commanded Position Not Reachable	Slow shifting	Fault on CVU, CCA, clutch system or sensor	Diagnostic Service 14. No additional action is needed.	Sensor, CVU or CCA	Clutch System
TECU	P107586	Vehicle starts to move but the clutch position indicates open position and input shaft speed is equal to engine speed	Signal Invalid	Starting gear D/Rev does not engage Engine shuts down unexpectedly	Cylinder connection failure	Diagnostic Service 14. No additional action is needed.		Clutch System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107592	1. The clutch engages when not commanded and, 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low air pressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above a specific limit	Performance or Incorrect Operation	Slow shifting	Air leakage in CCA or in air hose or position sensor	Diagnostic Service 14. No additional action is needed.	CVU, CCA, air hose	Clutch System
TECU	P107594	1. The clutch disengages when not commanded and, 2. There is no active faultcode on the SEPoC and, 3. There is no active faultcode for low air pressure and, 4. There is no active faultcode on any of the clutch cylinder valves and, 5. The PCB temperature is above a specific limit	Unexpected Operation	Slow shifting	Air leakage in clutch system, valves in CVU	Diagnostic Service 14. No additional action is needed.	Valve	Clutch System
TECU	P107471	-The cylinder for gear 2/3 can not reach neutral position	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107477	-The cylinder for gear 2/3 can not reach engaged position for gear 3.	Commanded Position Not Reachable	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107492	-The cylinder for gear 2/3 leaves the engaged postion for gear 3. - The 2/3 cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107494	-The cylinder for gear 2/3 leaves the neutral position. - The 2/3 cylinder valves are inactive	Unexpected Operation	Short loss of torque	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107496	-The cylinder for gear 2/3 leaves the engaged postion for gear 2. - The 2/3 cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107497	-The cylinder for gear 2/3 can not reach engaged position for gear 2.	Component or System Operation Obstructed or Blocked	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 2 and 3 Engagement System
TECU	P107371	-The cylinder for gear 1/R can not reach neutral position	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107377	-The cylinder for gear 1/R can not reach engaged position for reverse gear.	Commanded Position Not Reachable	Blocked out gears - Only forward gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107392	-The cylinder for gear 1/R leaves the engaged position for reverse reverse. -The 1/R cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107394	-The cylinder for gear 1/R leaves the neutral position -The 1/R cylinder valves are inactive	Unexpected Operation	Short loss of torque	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107396	-The cylinder for gear 1/R leaves the engaged position for gear 1. -The 1/R cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107397	-The cylinder for gear 1/R can not reach engaged position for gear 1.	Component or System Operation Obstructed or Blocked	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107277	-The low range gear can not engage	Commanded Position Not Reachable	Blocked out gears - Only high range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P107292	-The range cylinder leaves the high range position -The range cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107294	-The range cylinder leaves the low range position -The range cylinder valves are inactive	Unexpected Operation	Short loss of torque	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System
TECU	P107297	-The high range gear can not engage	Component or System Operation Obstructed or Blocked	Blocked out gears - Only low range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P105916	The sensor signal is below normal range	Circuit Voltage Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105918	The sensor signal is above normal range	Circuit Current Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105922	1.The sensor signal is in normal range 2. Temerature is above 100dgC during 18000s	Signal Amplitude Greater Than Maximum	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105992	1. The sensor signal is in normal range 2. Temperature above 120dgC during 600s	Performance or Incorrect Operation	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105998	1. The sensor signal is in normal range 2. Temperature is aove 140dgC during 30s	Component or System Over Temperature	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P106154	1. The CS in NVRAM is not correct, or 2. The slippoint has not been calibrated	Missing Calibration	Starting Gear D/Rev does not engage	Slippoint not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate slip point	Clutch Cylinder Slipping Point Position
TECU	P106216	Voltage is above normal range (5,5 V)	Circuit Voltage Below Threshold	Harsh start Slow shifting	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch Position Sensor Supply (SEPoC5V)
TECU	P106218	Voltage is below normal range (4,5 V)	Circuit Current Below Threshold	Harsh start Slow shifting	Fault on sensor or harness	Key off/key on required after one minute to reactivate diagnosis.	Sensor or harness	Clutch Position Sensor Supply (SEPoC5V)
TECU	P106329	More than half of the readings from the inclination sensor are faulty or doesn't change for 10 s.	Signal invalid	Wrong starter gear Unsuitable gear selected	Electrical fault on inclination sensor	Key off/key on required to reactivate diagnosis.	TECU	Gearbox Inclination Sensor (SEIG)
TECU	P107017	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107018	The low side drive is open circuit	Circuit Current Below Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107019	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107694	The gearbox brake does not brake when the valve is activated	Unexpected Operation	Slow shifting - When shifting to start gear	Air leakage, worn out disc or diaphragm in AMT-PS Air leakage, worn out discs in AMT-E	Diagnostic Service 14. No additional action is needed.	Brake	Gearbox Brake (Countershaft Brake / Input Shaft Brake)
TECU	P106417	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	P106418	The low-side drive is open circuit	Circuit Current Below Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	P106419	The low-side drive is short circuit to Gnd. Also set for all VAF low sides when VAFHS has a weak SCL	Circuit Current Above Threshold	Slow shifting	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or CVU	Clutch Fast Engagement Valve (VAFE)
TECU	U000188	TEA2+: CAN transiever signals "bus-off" on Backbone2	Bus off	Gearshifting inoperative - Manual gear shifting not possible Transmission fails to secure for crank	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 2 Net
TECU	U001088	This FMI shall be set if the CAN transiever signals "bus-off"	Bus off	Harsh shifting Gear shifts only at certain engine speeds Transmission fails to secure for crank AMT-PS: No powershift	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 1 Net
TECU	P105117	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)
TECU	P105118	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105119	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Split Position Sensor (SEPoS)
TECU	P105154	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Split calibration not done after software download	Diagnostic Service 14. No additional action is needed.	Calibrate split	Split Position Sensor (SEPoS)
TECU	P105217	Sensor signal is short circuit to Ubatt or sensor supply, when at least one of the sensor signals is above normal range (above 4,5 V) and the other is within normal range (0,5 V to 4,5 V).	Circuit Voltage Above Threshold	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105218	Open circuit on any of the sensor signals is detected when one of the sensor signals is within normal range (0,5V to 4,5 V) and the other sensor signal is below normal range. This includes either sensor signal being short circuit to Gnd.	Circuit Current Below Threshold	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105229	Each sensor signal is either below or above normal (0,5 to 4,5V) when: 1. Status of the sensor ASIC reports internal fault on the sensor, or 2. Open circuit on Gnd pin or, 3. Open circuit on Supply pin	Signal invalid	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch Cylinder Position
TECU	P105254	1. The CS in NVRAM is not correct 2. The cylinder has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Clutch stroke calibration not done after software download or after exchanged clutch plate	Diagnostic Service 14. No additional action is needed.	Calibrate clutch stroke	Clutch Cylinder Position

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105262	The sum of the clutch positions indicated by the sensor signal and the inverted sensor signal is too small (below 4,5V) and both signals is within normal range (0,5V to 4,5 V).	Signal Compare Failure	Harsh start Slow shifting	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Cylinder Position
TECU	P105378	The clutch wear is more than or equal to "Service due position"	Alignment or Adjustment Incorrect	None	Clutch worn	Diagnostic Service 14. No additional action is needed.	Clutch disc	Clutch Plate Wear
TECU	P105594	The gearlever is moved from neutral without prior application of the service brake and vehicle speed below 5 km/h	Unexpected Operation	Starting gear D/Rev does not engage	Brake pedal or parking brake (VTC only) not pressed when moving gear lever to Automatic, Manual or Reverse	Diagnostic Service 14. No additional action is needed.	None	Brake Switch Status
TECU	U114100	The FMI shall be set if any of the signals GearLeverPosition (BackBone1) or GearLeverPositionBackup (BackBone2) from HMIIOM is not received within 500ms.	No Sub Type Information	Gear shifting inoperative Gear shifting inoperative - Manual gear shifting not possible	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With HMIIOM
TECU	U010000	The FMI shall be set if the signal EngineSpeed from EMS is not received within 120ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication with EMS
TECU	U014000	The FMI shall be set if the signal AcceleratorPedalPosition1 from VMCU not received within 60 ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With VMCU
TECU	P060100	Calculated signature does not match signature provided during software download	No Sub Type Information	Engine derate	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Internal Control Module Memory Check Sum Error

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U114286	The FMI shall be set if any of the signals FrontAxleSpeed, FrAxleLeftWheelRelativeSpeed, FrAxleRightWheelRelativeSpeed, RrAxle1LeftWheelRelativeSpeed, RrAxle1RightWheelRelativeSpeed from EBS are incorrect and IDTC_EBS_MS is not set	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Wheel-Based Vehicle Speed
TECU	U300017	The voltage level for DIEE-pin is high	Circuit Voltage Above Threshold	Transmission fails to secure for crank	Fault on flexfoil or on the connector on the gcu	Diagnostic Service 14. No additional action is needed.	Flexfoil or connector on GCU	Electronic control unit
TECU	U300044	Flash CS error or program code missing	Data Memory Failure	Transmission fails to secure for crank	No software	Program ECU with SW	Software download	Electronic control unit
TECU	U300046	Flash CS dataset error or program code missing	Calibration / Parameter Memory Failure	Transmission fails to secure for crank	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Electronic control unit
TECU	U30004B	temp above 125dgC	Over Temperature	None	Too high temperature for pcb	Diagnostic Service 14. No additional action is needed.	None	Electronic control unit
TECU	U300316	24V system: below 17V 12V system: below 9V	Circuit Voltage Below Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	U300317	24V system: above 36V 12V system: above 19V	Circuit Voltage Above Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	U007387	The FMI shall be set if the signal EngineSpeed_VP24 from EECU is not recieved within 100ms	Missing Message	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	U007388	The FMI shall be set if the CAN transciever signals "busOFF"	Bus off	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	P109317	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109319	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109417	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit
TECU	P109419	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P109517	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109519	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109617	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P109619	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P109D16	Voltage above 5,5 V	Circuit Voltage Below Threshold		None	Key off/key on required after one minute to reactivate diagnosis.	None	Ultra Crawler Position Sensor Supply Circuit (SEPoUC)
TECU	P109D18	Voltage is below 4,5 V	Circuit Current Below Threshold		None	Key off/key on required after one minute to reactivate diagnosis.	None	Ultra Crawler Position Sensor Supply Circuit (SEPoUC)
TECU	P10A017	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Slow Valves High Side Circuit (VASHS)
TECU	P10A019	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh start Harsh shifting	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Slow Valves High Side Circuit (VASHS)
TECU	P10A117	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Fast Valves High Side Circuit (VAFHS)
TECU	P10A119	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Slow shifting Harsh shifting	Fault on harness or CVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch Fast Valves High Side Circuit (VAFHS)
TECU	U012900	The FMI shall be set if the signal FrontAxleSpeed from EBS is not recieved within 240ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With Brake System Control Module
TECU	U030055	Test software is installed	Not Configured	None	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software dataset	Program Memory EEPROM Development SW
TECU	U030081	Wrong dataset for gearbox type OR incorrect structural checksum on the dataset.	Invalid Serial Data Received	Unsuitable gear selected Unexpected acceleration Harsh shifting	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software download	Program Memory EEPROM Development SW

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107B86	The FMI shall be set if any of the signals AcceleratorPedalPosition1 or BrakeSwitch from the VMCU are incorrect.	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Accelerator Pedal Position / Brake Switch Signal Performance
TECU	U1FFF68	Parameter Factory Mode is true	Event Information	None	Software in Factory Mode	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	P10AE81	Constant value for countershaft and mainshaft speed ratio when gearbox should be in neutral.	Invalid Serial Data Received	Gear shifting inoperative	Oil level, broken rod or fork	Diagnostic Service 14. No additional action is needed.	Engagement system	Engaged Gear in Neutral
TECU	P10C417	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Crawler gears not available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Valve (VAUC)
TECU	P10C418	The low side drive is Open circuit	Circuit Current Below Threshold	Blocked out gears - Crawler gears not available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Valve (VAUC)
TECU	P10C419	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Crawler gears not available Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Valve (VAUC)
TECU	P10C517	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Neutral Valve (VAUCN)
TECU	P10C518	The low side drive is Open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Neutral Valve (VAUCN)
TECU	P10C519	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Ultra Crawler Neutral Valve (VAUCN)
TECU	P10C617	Sensor signal is short circuit to Ubatt or sensor supply, when at least one of the sensor signals is above normal range (above 4,5 V) and the other is within normal range (0,5 V to 4,5 V).	Circuit Voltage Above Threshold	Slow gearchanges - When shifting to crawler gear(s)	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Ultra Crawler Position Sensor (SEPoUC)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10C618	Open circuit on any of the sensor signals is detected when one of the sensor signals is within normal range (0,5V to 4,5 V) and the other sensor signal is below normal range. This includes either sensor signal being short circuit to Gnd.	Circuit Current Below Threshold	Slow gearchanges - When shifting to crawler gear(s)	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Ultra Crawler Position Sensor (SEPoUC)
TECU	P10C629	Each sensor signal is either below or above normal (0,5 to 4,5V) when: 1. Status of the sensor ASIC reports internal fault on the sensor, or 2. Open circuit on Gnd pin or, 3. Open circuit on Supply pin	Signal invalid	Slow gearchanges - When shifting to crawler gear(s)	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Ultra Crawler Position Sensor (SEPoUC)
TECU	P10C654	1. The CS in NVRAM is not correct 2. The cylinder has not been calibrated	Missing Calibration	Starting gear D/Rev does not engage	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate gearbox	Ultra Crawler Position Sensor (SEPoUC)
TECU	P10C662	The sum of the clutch positions indicated by the sensor signal and the inverted sensor signal is too small (below 4,5V) and both signals is within normal range (0,5V to 4,5 V).	Signal Compare Failure	Slow gearchanges - When shifting to crawler gear(s)	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Ultra Crawler Position Sensor (SEPoUC)
TECU	P10C771	-The cylinder for ultra crawler gear can not reach engaged position.	Actuator Stuck	Blocked out gears - Crawler gear(s) not available Automatic Gear shifting only possible at certain engine speed		Diagnostic Service 14. No additional action is needed.		Ultra Crawler Gear Engagement System
TECU	P10C794	-The cylinder for ultra crawler gear leaves engaged position -The ultra crawler gear cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently		Diagnostic Service 14. No additional action is needed.		Ultra Crawler Gear Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10C796	-The cylinder for ultra crawler gear leaves neutral position -The ultra crawler gear cylinder valves are inactive	Component Internal Failure	Short loss of torque		Diagnostic Service 14. No additional action is needed.		Ultra Crawler Gear Engagement System
TECU	P10C797	-The cylinder for ultra crawler gear can not reach neutral position.	Component or System Operation Obstructed or Blocked	Blocked out gears - Only crawler gear(s) available Automatic Gear shifting only possible at certain engine speed		Diagnostic Service 14. No additional action is needed.		Ultra Crawler Gear Engagement System
TECU	P10CA17	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Ultra Crawler Valve High Side Circuit (VAUCHS)
TECU	P10CA19	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Crawler gear(s) not available Automatic Gear shifting only possible at certain engine speed	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Ultra Crawler Valve High Side Circuit (VAUCHS)
TECU	P106D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 5, 6, 11, and 12 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 3 Valve (VAG3)
TECU	P106C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears -Gear 3, 4, 9 and 10 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P106C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 2 Valve (VAG2)
TECU	P106B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Reverse gears not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear Reverse Valve (VAGR)
TECU	P106A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106A18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P106A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Gear 1, 2, 7 and 8 not available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gear 1 Valve (VAG1)
TECU	P105017	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105018	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105019	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Range Position Sensor (SEPoR)
TECU	P105054	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Range calibration not done after software download	Diagnostic Service 14. No additional action is needed.	Calibrate range	Range Position Sensor (SEPoR)
TECU	P105417	The sensor signal is above normal range	Circuit Voltage Above Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105418	The sensor signal is below normal range	Circuit Current Below Threshold	No Pop up Indicating: Transmission low air pressure	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Tank Air Pressure Sensor (SEPrSA)
TECU	P105422	1. Sensor signal is in normal range 2. Pressure is above 10.0 [bar]	Signal Amplitude Greater Than Maximum	None	High air pressure or fault on pressure sensor	Diagnostic Service 14. No additional action is needed.	Pressure sensor	Tank Air Pressure Sensor (SEPrSA)
TECU	P105716	ASIC reports short circuit Gnd.	Circuit Voltage Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105717	ASIC reports short circuit Ubatt or supply is greater than 12 V.	Circuit Voltage Above Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105718	ASIC reports open circuit.	Circuit Current Below Threshold	Slow shifting Harsh shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Main Shaft Speed Sensor (SESM)
TECU	P105721	1. Sensor signal is within normal range and 2. The mainshaft speed differs from both the value of the countershaft speed and the vehicle speed recieved from the vehicle ECU	Signal Amplitude Less Than Minimum	Slow shifting Harsh shifting	Fault on mainshaft speed sensor or teeth wheel	Diagnostic Service 14. No additional action is needed.	Sensor or teeth wheel	Main Shaft Speed Sensor (SESM)
TECU	P105729	The sensor indicates wrong travelling direction	Signal invalid	Slow shifting Harsh start Starting gear D/Rev does not engage - when driving/moving	Mainshaft speed sensor	Diagnostic Service 14. No additional action is needed.	Sensor	Main Shaft Speed Sensor (SESM)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105A17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve	High Range Valve (VAHR)
TECU	P105A18	The low side drive is Open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to high range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)
TECU	P105A19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	High Range Valve (VAHR)
TECU	P105B17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears - Not possible to shift to low range Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105B19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Low Range Valve (VALR)
TECU	P105E17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105E18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Shifter 1 Position Sensor (SEPo1R)
TECU	P105E54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate mainbox	Shifter 1 Position Sensor (SEPo1R)
TECU	P107554	The clutch has not been calibrated after replacement.	Missing Calibration	Starting gear D/Rev does not engage	Not calibrated after clutch plate replacement	Diagnostic Service 14. No additional action is needed.	Calibrate clutch system	Clutch System
TECU	P107574	The clutch transfers too much torque in disengaged position.	Actuator Slipping	Unexpected acceleration	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch System
TECU	P107371	-The cylinder for gear 1/R can not reach neutral position	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107377	-The cylinder for gear 1/R can not reach engaged position for reverse gear.	Commanded Position Not Reachable	Blocked out gears - Only forward gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107392	-The cylinder for gear 1/R leaves the engaged position for reverse reverse. -The 1/R cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107394	-The cylinder for gear 1/R leaves the neutral position -The 1/R cylinder valves are inactive	Unexpected Operation	Short loss of torque	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107396	-The cylinder for gear 1/R leaves the engaged position for gear 1. -The 1/R cylinder valves are inactive	Component Internal Failure	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	calibrate gear, engagement teeth	Gears 1 and R Engagement System
TECU	P107397	-The cylinder for gear 1/R can not reach engaged position for gear 1.	Component or System Operation Obstructed or Blocked	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gears 1 and R Engagement System
TECU	P107277	-The low range gear can not engage	Commanded Position Not Reachable	Blocked out gears - Only high range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P107292	-The range cylinder leaves the high range position -The range cylinder valves are inactive	Performance or Incorrect Operation	Gear disengage inadvertently	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System
TECU	P107294	-The range cylinder leaves the low range position -The range cylinder valves are inactive	Unexpected Operation	Short loss of torque	Worn engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement teeth Position detent	Range Engagement System
TECU	P107297	-The high range gear can not engage	Component or System Operation Obstructed or Blocked	Blocked out gears - Only low range gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Synchronizer, air leakage or piston rod	Diagnostic Service 14. No additional action is needed.	GCU, synchronizer	Range Engagement System
TECU	P105916	The sensor signal is below normal range	Circuit Voltage Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105918	The sensor signal is above normal range	Circuit Current Below Threshold	None	Sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105922	1.The sensor signal is in normal range 2. Temerature is above 100dgC during 18000s	Signal Amplitude Greater Than Maximum	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105992	1. The sensor signal is in normal range 2. Temperature above 120dgC during 600s	Performance or Incorrect Operation	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105998	1. The sensor signal is in normal range 2. Temperature is aove 140dgC during 30s	Component or System Over Temperature	None	Insufficient cooling Insufficient lubrication, low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Cooler Oil pump sugsil blocked channels. Check oli level, gearbox leakage,	Gearbox Oil Temperature Sensor (SETeGO)
TECU	P106097	High clutch temperature (estimated not measured)	Component or System Operation Obstructed or Blocked	No powershift	High clutch load	Diagnostic Service 14. No additional action is needed.		Clutch Plate Load
TECU	P106154	1. The CS in NVRAM is not correct, or 2. The slippoint has not been calibrated	Missing Calibration	Starting Gear D/Rev does not engage	Slippoint not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate slip point	Clutch Cylinder Slipping Point Position
TECU	P106329	More than half of the readings from the inclination sensor are faulty or doesn't change for 10 s.	Signal invalid	Wrong starter gear Unsuitable gear selected	Electrical fault on inclination sensor	Key off/key on required to reactivate diagnosis.	TECU	Gearbox Inclination Sensor (SEIG)
TECU	P107017	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107018	The low side drive is open circuit	Circuit Current Below Threshold	Slow shifting - When shifting to start gear	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107019	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed	Fault on actuator or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Gearbox Brake Valve (VAGB)
TECU	P107694	The gearbox brake does not brake when the valve is activated	Unexpected Operation	Slow shifting - When shifting to start gear	Air leakage, worn out disc or diaphragm in AMT-PS Air leakage, worn out discs in AMT-E	Diagnostic Service 14. No additional action is needed.	Brake	Gearbox Brake (Countershaft Brake / Input Shaft Brake)
TECU	U000188	TEA2+: CAN transiever signals "bus-off" on Backbone2	Bus off	Gearshifting inoperative - Manual gear shifting not possible Transmission fails to secure for crank	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 2 Net
TECU	U001088	This FMI shall be set if the CAN transiever signals "bus-off"	Bus off	Harsh shifting Gear shifts only at certain engine speeds Transmission fails to secure for crank AMT-PS: No powershift	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Backbone 1 Net
TECU	P105378	The clutch wear is more than or equal to "Service due position"	Alignment or Adjustment Incorrect	None	Clutch worn	Diagnostic Service 14. No additional action is needed.	Clutch disc	Clutch Plate Wear
TECU	P105594	The gearlever is moved from neutral without prior application of the service brake and vehicle speed below 5 km/h	Unexpected Operation	Starting gear D/Rev does not engage	Brake pedal or parking brake (VTC only) not pressed when moving gear lever to Automatic, Manual or Reverse	Diagnostic Service 14. No additional action is needed.	None	Brake Switch Status

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U114100	The FMI shall be set if any of the signals GearLeverPosition (BackBone1) or GearLeverPositionBackup (BackBone2) from HMIIOM is not received within 500ms.	No Sub Type Information	Gear shifting inoperative Gear shifting inoperative - Manual gear shifting not possible	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With HMIIOM
TECU	U010000	The FMI shall be set if the signal EngineSpeed from EMS is not received within 120ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication with EMS
TECU	U014000	The FMI shall be set if the signal AcceleratorPedalPosition1 from VMCU not received within 60 ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With VMCU
TECU	P060100	Calculated signature does not match signature provided during software download	No Sub Type Information	Engine derate	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Internal Control Module Memory Check Sum Error
TECU	U114286	The FMI shall be set if any of the signals FrontAxleSpeed, FrAxleLeftWheelRelativeSpeed, FrAxleRightWheelRelativeSpeed, RrAxle1LeftWheelRelativeSpeed, RrAxle1RightWheelRelativeSpeed from EBS are incorrect and IDTC_EBS_MS is not set	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Wheel-Based Vehicle Speed
TECU	U300017	The voltage level for DIEE-pin is high	Circuit Voltage Above Threshold	Transmission fails to secure for crank	Fault on flexfoil or on the connector on the gcu	Diagnostic Service 14. No additional action is needed.	Flexfoil or connector on GCU	Electronic control unit
TECU	U300044	Flash CS error or program code missing	Data Memory Failure	Transmission fails to secure for crank	No software	Program ECU with SW	Software download	Electronic control unit
TECU	U300046	Flash CS dataset error or program code missing	Calibration / Parameter Memory Failure	Transmission fails to secure for crank	Bad software	Diagnostic Service 14. No additional action is needed.	Software download	Electronic control unit
TECU	U30004B	temp above 125dgC	Over Temperature	None	Too high temperature for pcb	Diagnostic Service 14. No additional action is needed.	None	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U300316	24V system: below 17V 12V system: below 9V	Circuit Voltage Below Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	U300317	24V system: above 36V 12V system: above 19V	Circuit Voltage Above Threshold	Gear shifting inoperative	Fault on generator or external charger	Diagnostic Service 14. No additional action is needed.	Generator or external charger	Battery Power
TECU	P108917	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Engagement Valve
TECU	P108918	The low-side drive is open circuit	Circuit Current Below Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Engagement Valve
TECU	P108919	The low-side drive is short circuit to Gnd or other active low side. Also set for all VAF2 low sides when VAF2HS has a weak SCL.	Circuit Current Above Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Engagement Valve
TECU	P108A17	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Disengagement Valve
TECU	P108A18	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Disengagement Valve
TECU	P108A19	The low-side drive is short circuit to Gnd or other active low side. Also set for all VAS2 low sides when VAS2HS has a weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Disengagement Valve
TECU	P108B17	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Disengagement Valve
TECU	P108B18	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Disengagement Valve
TECU	P108B19	The low-side drive is short circuit to Gnd or other active low side. Also set for all VAF2 low sides when VAF2HS has a weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Disengagement Valve
TECU	P108C17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Second Gear Valve
TECU	P108C18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Second Gear Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P108C19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Second Gear Valve
TECU	P108D17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Third Gear Valve
TECU	P108D18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Third Gear Valve
TECU	P108D19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed No powershift	Fault on valve or flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Valve or flexfoil	Neutral and Third Gear Valve
TECU	P108E17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Rear Valve
TECU	P108E18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Rear Valve
TECU	P108E19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Rear Valve
TECU	P108F17	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Front Valve
TECU	P108F18	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Front Valve
TECU	P108F19	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Synchronizer Front Valve
TECU	P109017	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Rear Neutral Valve
TECU	P109018	The low side drive is open circuit	Circuit Current Below Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Rear Neutral Valve
TECU	P109019	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Rear Neutral Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P109117	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front Neutral Valve
TECU	P109118	The low side drive is open circuit	Circuit Current Below Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front Neutral Valve
TECU	P109119	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front Neutral Valve
TECU	P109217	The low side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front and Countershaft Rear Valve (VACFCR)
TECU	P109218	The low side drive is open circuit	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front and Countershaft Rear Valve (VACFCR)
TECU	P109219	The low side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front and Countershaft Rear Valve (VACFCR)
TECU	U007387	The FMI shall be set if the signal EngineSpeed_VP24 from EECU is not recieved within 100ms	Missing Message	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	U007388	The FMI shall be set if the CAN transciever signals "busOFF"	Bus off	Slow shifting	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	CAN Communication Powertrain Subnet
TECU	P107C17	Signal voltage is above normal range	Circuit Voltage Above Threshold	No powershift	Fault on sensor, flexfoil or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Pressure Sensor
TECU	P107C18	The sensor signal is below normal range	Circuit Current Below Threshold	No powershift	Fault on sensor, flexfoil or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Pressure Sensor
TECU	P107C62	1. The position sensor and pressure sensor differ. 2. The pressure sensor does not correspond to the valve activation.	Signal Compare Failure	No powershift	Fault on pressure sensor	Diagnostic Service 14. No additional action is needed.	Pressure sensor or clutch system	Clutch 1 Pressure Sensor
TECU	P107D17	Signal voltage is above normal range	Circuit Voltage Above Threshold	No powershift	Fault on sensor, flexfoil or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Pressure Sensor
TECU	P107D18	The sensor signal is below normal range	Circuit Current Below Threshold	No powershift	Fault on sensor, flexfoil or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Pressure Sensor
TECU	P107D62	1. The position sensor and pressure sensor differ. 2. The pressure sensor does not correspond to the valve activation.	Signal Compare Failure	No powershift	Fault on pressure sensor	Diagnostic Service 14. No additional action is needed.	Pressure sensor or clutch system	Clutch 2 Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107E16	The sensor voltage signal is below normal range when enabled and in normal or below normal range when disabled.	Circuit Voltage Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 2
TECU	P107E17	The sensor voltage signal is above normal range	Circuit Voltage Above Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 2
TECU	P107E18	The sensor voltage signal is below normal range when sensor is enabled and above normal range when disabled.	Circuit Current Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 2
TECU	P107E21	1. The sensor signal is within normal range and, 2. The counter shaft speed differs from both the main shaft speed (SESM) and the vehicle speed recieved from the vehicle ecu when the counter shaft is rotating	Signal Amplitude Less Than Minimum	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 2
TECU	P107F17	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Second Gear Position Sensor
TECU	P107F18	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Second Gear Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P107F19	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Second Gear Position Sensor
TECU	P107F54	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate gear 2	Second Gear Position Sensor
TECU	P108017	Either sensor pin is short circuit to high potential. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC and passive Aout is higher than 0,5 V.	Circuit Voltage Above Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Third Gear Position Sensor
TECU	P108018	Open circuit between the pins where the sensor should be connected. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is higher than 0,5 V and passive Aout is lower than 0,5 V.	Circuit Current Below Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Third Gear Position Sensor
TECU	P108019	Sensor pin is short circuit to ground. Set by SW when no signal pulses detected from ASIC and active Aout from ASIC is below 0,5V.	Circuit Current Above Threshold	Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Third Gear Position Sensor
TECU	P108054	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate gear 3	Third Gear Position Sensor
TECU	P108129	The DCVU signals an error on the SePoCF sensor. Could be SCL, OC or SCH.	Signal invalid	Transmission fails to secure for crank	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	DCVU	Countershaft Front Position Sensor
TECU	P108154	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate counter shaft front sleeve	Countershaft Front Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P108229	The DCVU signals an error on the SePoCR sensor. Could be SCL, OC or SCH.	Signal invalid	Transmission fails to secure for crank	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	DCVU	Countershaft Rear Position Sensor
TECU	P108254	The Checksum of the NVRAM is not correct, or the sensor has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate counter shaft rear sleeve	Countershaft Rear Position Sensor
TECU	P108349	DCVU reports a fatal internal error.	Internal Electronic Failure	No powershift	Fault on DCVU	Diagnostic Service 14. No additional action is needed.	DCVU	Dual Clutch Valve Unit
TECU	P108386	DCVU does not receive status message from TECU	Signal Invalid	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Dual Clutch Valve Unit
TECU	P108387	TECU does not receive correct messages from the DCVU	Missing Message	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Dual Clutch Valve Unit
TECU	P108393	The DCVU signals an error on the 5V feed to the DCVU.	No Operation	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Dual Clutch Valve Unit
TECU	P108402	1. The position sensor and the pressure sensor differ. 2. The position sensor does not correspond to the valve activation.	General Signal Failure	No powershift	Fault on position sensor	Diagnostic Service 14. No additional action is needed.	Position sensor or clutch system	Clutch 2 Position Sensor
TECU	P108417	Sensor signal is short circuit to Ubatt or sensor supply, when at least one of the sensor signals is above normal range (above 4,5 V) and the other is within normal range (0,5 V to 4,5 V).	Circuit Voltage Above Threshold	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Position Sensor
TECU	P108418	Open circuit on any of the sensor signals is detected when one of the sensor signals is within normal range (0,5V to 4,5 V) and the other sensor signal is below normal range. This includes either sensor signal being short circuit to Gnd.	Circuit Current Below Threshold	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P108429	Each sensor signal is either below or above normal (0,5 to 4,5V) when: 1. Status of the sensor ASIC reports internal fault on the sensor, or 2. Open circuit on Gnd pin or, 3. Open circuit on Supply pin	Signal invalid	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Position Sensor
TECU	P108454	1. The CS in NVRAM is not correct 2. The cylinder has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate clutch system	Clutch 2 Position Sensor
TECU	P108462	The sum of the clutch positions indicated by the sensor signal and the inverted sensor signal is too small (below 4,5V) and both signals is within normal range (0,5V to 4,5 V).	Signal Compare Failure	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 2 Position Sensor
TECU	P108516	Voltage above 5.5 V	Circuit Voltage Below Threshold	Harsh start Slow shifting No powershift	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch 1 Position Sensor and Clutch 2 Pressure Sensor Supply Circuit
TECU	P108518	Voltage below 4.5 V	Circuit Current Below Threshold	Harsh start Slow shifting No powershift	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch 1 Position Sensor and Clutch 2 Pressure Sensor Supply Circuit
TECU	P108616	Voltage above 5.5 V	Circuit Voltage Below Threshold	Harsh start Slow shifting No powershift	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch 2 Position Sensor and Clutch 1 Pressure Sensor Supply Circuit
TECU	P108618	Voltage below 4.5 V	Circuit Current Below Threshold	Harsh start Slow shifting No powershift	Fault on harness	Key off/key on required after one minute to reactivate diagnosis.	Harness	Clutch 2 Position Sensor and Clutch 1 Pressure Sensor Supply Circuit
TECU	P108717	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Fast Disengagement Valve 2
TECU	P108718	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Fast Disengagement Valve 2
TECU	P108719	The low-side drive is short circuit to Gnd or other active low side. Also set for all VAF2 low sides when VAF2HS has a weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Fast Disengagement Valve 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P108817	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Engagement Valve
TECU	P108818	The low-side drive is open circuit	Circuit Current Below Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Engagement Valve
TECU	P108819	The low-side drive is short circuit to Gnd or other active low side. Also set for all VAS2 low sides when VAS2HS has a weak SCL.	Circuit Current Above Threshold	No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Engagement Valve
TECU	P109317	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109319	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Gear 1 Valve High Side Circuit (VAG1HS)
TECU	P109417	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit
TECU	P109419	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Reverse Gear Valve High Side Circuit
TECU	P109517	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109519	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 1 High Side Circuit (VAIN1HS)
TECU	P109617	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P109619	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears - Very limited number of gears available Automatic Gear shifting only possible at certain engine speed AMT-PS: No powershift	Fault on flexfoil	Diagnostic Service 14. No additional action is needed.	Flexfoil	Internal Valve 2 High Side Circuit (VAIN2HS)
TECU	P109717	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Slow Valves High Side Circuit
TECU	P109719	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh start Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Slow Valves High Side Circuit
TECU	P109817	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Fast Valves High Side Circuit (VAF1HS)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P109819	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 1 Fast Valves High Side Circuit (VAF1HS)
TECU	P109917	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Valves High Side Circuit (VAS2HS)
TECU	P109919	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Slow Valves High Side Circuit (VAS2HS)
TECU	P109A17	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Valves High Side Circuit (VAF2HS)
TECU	P109A19	The high side drive is short circuit to Gnd. Also set for all HS + LS when weak SCL.	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Clutch 2 Fast Valves High Side Circuit (VAF2HS)
TECU	P109B17	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Countershaft Extension Valve High Side Circuit (VACEXHS)
TECU	P109B19	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Countershaft Extension Valve High Side Circuit (VACEXHS)
TECU	P109C17	The high side drive is short circuit to Ubatt	Circuit Voltage Above Threshold	None	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Countershaft Front and Countershaft Rear Valve High Side Circuit
TECU	P109C19	The high side drive is short circuit to Gnd	Circuit Current Above Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Diagnostic Service 14. No additional action is needed.	Harness	Countershaft Front and Countershaft Rear Valve High Side Circuit
TECU	P109E16	Voltage above 5,5 V	Circuit Voltage Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front and Countershaft Rear Sensor Supply Circuit
TECU	P109E18	Voltage is below 4,5 V	Circuit Current Below Threshold	Blocked out gears Automatic Gear shifting only possible at certain engine speed No powershift	Fault on harness or DCVU	Key off/key on required after one minute to reactivate diagnosis.	Harness	Countershaft Front and Countershaft Rear Sensor Supply Circuit
TECU	P109F16	Voltage above 11 V	Circuit Voltage Below Threshold	Gear shifting inoperative - Only gear shifting at stand still	Fault on flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Flexfoil	Mainshaft and Countershaft Sensor Supply Circuit
TECU	P109F18	Voltage is below 7 V	Circuit Current Below Threshold	Gear shifting inoperative - Only gear shifting at stand still	Fault on flexfoil	Key off/key on required after one minute to reactivate diagnosis.	Flexfoil	Mainshaft and Countershaft Sensor Supply Circuit
TECU	P10A217	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Disengagement Valve 1 (VAFD11)
TECU	P10A218	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Disengagement Valve 1 (VAFD11)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10A219	The low-side drive is short circuit to Gnd. Also set for all VAF1 low sides when VAF1 has a weak SCL	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Disengagement Valve 1 (VAFD11)
TECU	P10A317	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Disengagement Valve (VASD1)
TECU	P10A318	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Disengagement Valve (VASD1)
TECU	P10A319	The low-side drive is short circuit to Gnd. Also set for all VAS1 low sides when VAS1 has a weak SCL	Circuit Current Above Threshold	Harsh shifting No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Disengagement Valve (VASD1)
TECU	P10A417	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	Harsh start No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Engagement Valve (VASE1)
TECU	P10A418	The low-side drive is open circuit	Circuit Current Below Threshold	Harsh start No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Engagement Valve (VASE1)
TECU	P10A419	The low-side drive is short circuit to Gnd. Also set for all VAS1 low sides when VAS1HS has a weak SCL	Circuit Current Above Threshold	Harsh start No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Slow Engagement Valve (VASE1)
TECU	P10A517	The low-side drive is short circuited to Ubatt	Circuit Voltage Above Threshold	No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Engagement Valve (VAFE1)
TECU	P10A518	The low-side drive is open circuit	Circuit Current Below Threshold	No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Engagement Valve (VAFE1)
TECU	P10A519	The low-side drive is short circuit to Gnd. Also set for all VAF1 low sides when VAF1HS has a weak SCL	Circuit Current Above Threshold	No powershift	Fault on actuator or harness	Diagnostic Service 14. No additional action is needed.	Harness or DCVU	Clutch 1 Fast Engagement Valve (VAFE1)
TECU	P10A602	1. The position sensor and pressure sensor differ. 2. The position sensor does not correspond to the valve activation.	General Signal Failure	No powershift	Fault on position sensor	Diagnostic Service 14. No additional action is needed.	Position sensor or clutch system	Clutch 1 Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10A617	Sensor signal is short circuit to Ubatt or sensor supply, when at least one of the sensor signals is above normal range (above 4,5 V) and the other is within normal range (0,5 V to 4,5 V).	Circuit Voltage Above Threshold	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Position Sensor
TECU	P10A618	Open circuit on any of the sensor signals is detected when one of the sensor signals is within normal range (0,5V to 4,5 V) and the other sensor signal is below normal range. This includes either sensor signal being short circuit to Gnd.	Circuit Current Below Threshold	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Position Sensor
TECU	P10A629	Each sensor signal is either below or above normal (0,5 to 4,5V) when: 1. Status of the sensor ASIC reports internal fault on the sensor, or 2. Open circuit on Gnd pin or, 3. Open circuit on Supply pin	Signal invalid	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Position Sensor
TECU	P10A654	1. The CS in NVRAM is not correct 2. The cylinder has not been calibrated	Missing Calibration	Transmission fails to secure for crank	Not calibrated after software download	Diagnostic Service 14. No additional action is needed.	Calibrate clutch system	Clutch 1 Position Sensor
TECU	P10A662	The sum of the clutch positions indicated by the sensor signal and the inverted sensor signal is too small (below 4,5V) and both signals is within normal range (0,5V to 4,5 V).	Signal Compare Failure	Harsh start Slow shifting No powershift	Fault on sensor or harness	Diagnostic Service 14. No additional action is needed.	Sensor or harness	Clutch 1 Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10A716	The sensor voltage signal is below normal range when enabled and in normal or below normal range when disabled.	Circuit Voltage Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 1
TECU	P10A717	The sensor voltage signal is above normal range	Circuit Voltage Above Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 1
TECU	P10A718	The sensor voltage signal is below normal range when sensor is enabled and above normal range when disabled.	Circuit Current Below Threshold	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 1
TECU	P10A721	1. The sensor signal is within normal range and, 2. The counter shaft speed differs from both the main shaft speed (SESM) and the vehicle speed recieved from the vehicle ecu when the counter shaft is rotating	Signal Amplitude Less Than Minimum	Harsh starts Starting gear D/Rev does not engage - when driving/moving Slow shifting No powershift	Fault on sensor or flexfoil	Diagnostic Service 14. No additional action is needed.	Sensor or flexfoil	Countershaft Speed Sensor 1
TECU	U012900	The FMI shall be set if the signal FrontAxleSpeed from EBS is not recieved within 240ms on BackBone1	No Sub Type Information	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Lost Communication With Brake System Control Module
TECU	U030055	Test software is installed	Not Configured	None	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software dataset	Program Memory EEPROM Development SW
TECU	U030081	Wrong dataset for gearbox type OR incorrect structural checksum on the dataset.	Invalid Serial Data Received	Unsuitable gear selected Unexpected acceleration Harsh shifting	Wrong dataset	Diagnostic Service 14. No additional action is needed.	Software download	Program Memory EEPROM Development SW
TECU	P107B86	The FMI shall be set if any of the signals AcceleratorPedalPosition1 or BrakeSwitch from the VMCU are incorrect.	Signal Invalid	Blocked out gears Automatic Gear shifting only possible at certain engine speed Harsh start	Fault on harness	Diagnostic Service 14. No additional action is needed.	Harness	Accelerator Pedal Position / Brake Switch Signal Performance
TECU	U1FFF68	Parameter Factory Mode is true	Event Information	None	Software in Factory Mode	Diagnostic Service 14. No additional action is needed.		Factory Mode Active

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10AA71	-The cylinder for gear 2 can not reach engaged position.	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gear 2 Engagement System
TECU	P10AA94	-The cylinder for gear 2 leaves engaged position. -The gear 2 cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gear 2 Engagement System
TECU	P10AA96	-The cylinder for gear 2 leaves neutral position. -The gear 2 cylinder valves are inactive	Component Internal Failure	None	Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gear 2 Engagement System
TECU	P10AA97	-The cylinder for gear 2 can not reach neutral position.	Component or System Operation Obstructed or Blocked	Blocked out gears - Only gear 3,4,9 and 10 available Automatic Gear shifting only possible at certain engine speed No powershift	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gear 2 Engagement System
TECU	P10AB71	-The cylinder for gear 3 can not reach engaged position.	Actuator Stuck	Blocked out gears Automatic Gear shifting only possible at certain engine speed	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gear 3 Engagement System
TECU	P10AB94	-The cylinder for gear 3 leaves engaged position. -The gear 3 cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Engagement teeth, Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gear 3 Engagement System
TECU	P10AB96	-The cylinder for gear 3 leaves neutral position. -The gear 3 cylinder valves are inactive	Component Internal Failure	None	Gearbox calibration, Position detent GCU	Diagnostic Service 14. No additional action is needed.	GCU	Gear 3 Engagement System
TECU	P10AB97	-The cylinder for gear 3 can not reach neutral position.	Component or System Operation Obstructed or Blocked	Blocked out gears - Only gear 5,6,11 and 12 available Automatic Gear shifting only possible at certain engine speed No powershift	Gearbox calibration, Air leakage	Diagnostic Service 14. No additional action is needed.	GCU	Gear 3 Engagement System
TECU	P10AC71	-The cylinder for countershaft front gear can not reach engaged position.	Actuator Stuck	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed No powershift	Air leakage, engagement teeth	Diagnostic Service 14. No additional action is needed.	DCVU, engagement sleeve/gear	CF Engagement System
TECU	P10AC94	-The cylinder for countershaft front gear leaves engaged position -The countershaft front gear cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Position detent, worn out engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement sleeve or gear, Detent	CF Engagement System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10AC96	-The cylinder for countershaft front gear leaves neutral position -The countershaft front gear cylinder valves are inactive	Component Internal Failure	Short loss of torque	Position detent	Diagnostic Service 14. No additional action is needed.	Detent	CF Engagement System
TECU	P10AC97	-The cylinder for countershaft front gear can not reach neutral position.	Component or System Operation Obstructed or Blocked	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed No powershift	Air leakage	Diagnostic Service 14. No additional action is needed.	DCVU	CF Engagement System
TECU	P10AD71	-The cylinder for countershaft rear gear can not reach engaged position.	Actuator Stuck	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed No powershift	Air leakage, engagement teeth	Diagnostic Service 14. No additional action is needed.	DCVU, engagement sleeve/gear	CR Engagement System
TECU	P10AD94	-The cylinder for countershaft rear gear leaves engaged position -The countershaft rear gear cylinder valves are inactive	Unexpected Operation	Gear disengage inadvertently	Position detent, worn out engagement teeth	Diagnostic Service 14. No additional action is needed.	Engagement sleeve or gear, Detent	CR Engagement System
TECU	P10AD96	-The cylinder for countershaft rear gear leaves neutral position -The countershaft rear gear cylinder valves are inactive	Component Internal Failure	None	Position detent	Diagnostic Service 14. No additional action is needed.	Detent	CR Engagement System
TECU	P10AD97	-The cylinder for countershaft rear gear can not reach neutral position.	Component or System Operation Obstructed or Blocked	Blocked out gears - Only every second gear available Automatic Gear shifting only possible at certain engine speed No powershift	Air leakage	Diagnostic Service 14. No additional action is needed.	DCVU	CR Engagement System
TECU	P10AE81	Constant value for countershaft and mainshaft speed ratio when gearbox should be in neutral.	Invalid Serial Data Received	Gear shifting inoperative	Oil level, broken rod or fork	Diagnostic Service 14. No additional action is needed.	Engagement system	Engaged Gear in Neutral
TECU	P10AF92	Sync phase too long	Performance or Incorrect Operation	No powershift	Air leakage, worn out friction foil	Diagnostic Service 14. No additional action is needed.	DCVU, synch. mech. friction foil.	Sync
TECU	P10AF94	Sync phase too long	Unexpected Operation	No powershift	Air leakage, worn out friction foil	Diagnostic Service 14. No additional action is needed.	DCVU, synch. mech. friction foil.	Sync
TECU	P10A871	The clutch can not transfer a specific torque without slipping	Actuator Stuck	Engine derate	Fault on clutch system or calibration	Diagnostic Service 14. No additional action is needed.	Clutch cylinder, CCA, driven plate or calibrate clutch slip point	Clutch 1 System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10A872	It takes too long time to reach engage set point.	Actuator Stuck Open	Blocked out gears - Only even gears available Automatic Gear shifting only possible at certain engine speed No powershift	Severe air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU/ DCCA/ Clutch	Clutch 1 System
TECU	P10A873	It takes too long time to reach disengage set point or disengage setpoint is unintentionally left.	Actuator Stuck Closed	Blocked out gears - Only even gears available Automatic Gear shifting only possible at certain engine speed No powershift	Severe air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU/ DCCA/ Clutch	Clutch 1 System
TECU	P10A875	The clutch cannot disengage even with extra activation and there is a gear engaged on clutch 1 and the engine is running.	Emergency Position Not Reachable	Engine shuts down unexpectedly	Air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU/ DCCA/ Clutch	Clutch 1 System
TECU	P10A877	The clutch movement when only VASE is active is too small.	Commanded Position Not Reachable	Harsh start Harsh shifting No powershift	Mechanical fault on slow engagement valve (VASE)	Diagnostic Service 14. No additional action is needed.	DCVU	Clutch 1 System
TECU	P10A892	Measured after 5s has elapsed from when control inputs were last changed, any of the valves (VAFD, VASD) has been used more than the upper limit for mean level of activation.	Performance or Incorrect Operation	Blocked out gears - Only even gears available Automatic Gear shifting only possible at certain engine speed No powershift	Minor air leakage	Diagnostic Service 14. No additional action is needed.	DCVU	Clutch 1 System
TECU	P10A894	Measured after 5 s has elapsed from when control inputs were last changed, any of the valves (VAFE, VASE) has been used more than the upper limit for mean level of activation.	Unexpected Operation	Blocked out gears - Only even gears available Automatic Gear shifting only possible at certain engine speed No powershift	Minor air leakage	Diagnostic Service 14. No additional action is needed.	DCVU / DCCA	Clutch 1 System
TECU	P10A971	Clutch can not transfer a specific torque without slipping	Actuator Stuck	Engine derate	Worn out driven plate	Diagnostic Service 14. No additional action is needed.	Clutch	Clutch 2 System
TECU	P10A972	It takes too long time to reach engage set point.	Actuator Stuck Open	Blocked out gears - Only uneven gears available Automatic Gear shifting only possible at certain engine speed No powershift	Severe air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU/ DCCA/ Clutch	Clutch 2 System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10A973	It takes too long time to reach disengage set point or disengage setpoint is unintentionally left.	Actuator Stuck Closed	Blocked out gears - Only uneven gears available Automatic Gear shifting only possible at certain engine speed No powershift	Severe air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU/ DCCA/ Clutch	Clutch 2 System
TECU	P10A975	The clutch cannot disengage even with extra activation and there is a gear engaged on clutch 1 and the engine is running.	Emergency Position Not Reachable	Engine shuts down unexpectedly	Air leakage or mechanical error	Diagnostic Service 14. No additional action is needed.	DCVU / DCCA/ Clutch	Clutch 2 System
TECU	P10A977	The clutch movement when only VASE is active is too small.	Commanded Position Not Reachable	Harsh start Harsh shifting No powershift	Mechanical fault on slow engagement valve (VASE)	Diagnostic Service 14. No additional action is needed.	DCVU	Clutch 2 System
TECU	P10A986	Vehicle starts to move but the clutch position indicates open position and input shaft speed is equal to engine speed	Signal Invalid	Starting gear D/Rev does not engage Engine shuts down unexpectedly	Cylinder connection failure	Diagnostic Service 14. No additional action is needed.		Clutch 2 System
TECU	P10A992	Measured after 5s has elapsed from when control inputs were last changed, any of the valves (VAFD, VASD) has been used more than the upper limit for mean level of activation.	Performance or Incorrect Operation	Blocked out gears - Only uneven gears available Automatic Gear shifting only possible at certain engine speed No powershift	Minor air leakage	Diagnostic Service 14. No additional action is needed.	DCVU / DCCA	Clutch 2 System
TECU	P10A994	Measured after 5 s has elapsed from when control inputs were last changed, any of the valves (VAFE, VASE) has been used more than the upper limit for mean level of activation.	Unexpected Operation	Blocked out gears - Only uneven gears available Automatic Gear shifting only possible at certain engine speed No powershift	Minor air leakage	Diagnostic Service 14. No additional action is needed.	DCVU	Clutch 2 System
Audio	U114163	Button physically continuously pressed for 2 minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	EJECT button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		EJECT button electrical switch
Audio	U114263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	TA button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		INFORMATION button electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114363	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	RADIO button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		RADIO button electrical switch
Audio	U114463	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	CD button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		CD button electrical switch
Audio	U114563	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	POWER button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		POWER button electrical switch
Audio	U114663	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	NAVIGATION button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		NAVIGATION button electrical switch
Audio	U114863	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 1 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 1 button electrical switch
Audio	U114963	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 2 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 2 button electrical switch
Audio	U115263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK DOWN button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK DOWN button electrical switch
Audio	U114A63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 3 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 3 button electrical switch
Audio	U114B63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 4 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 4 button electrical switch
Audio	U114C63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 5 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 5 button electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114D63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 6 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 6 button. electrical switch
Audio	U114E63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	MENU button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		MENU button electrical switch
Audio	U115163	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK UP button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK UP button electrical switch
Audio	U004611	CAN Bus Line physical short to GND	Circuit Short To Ground	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device connected to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004612	CAN Bus Line physical short to Battery	Circuit Short To Battery	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004688	CAN HW unable to connect to bus	Bus off	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus - CAN hardware issue resulting in Bus off	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U300041	Incorrect software signature. Corrupted memory.	General Checksum Failure	Unit will not start main application.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300044	RAM memory error	Data Memory Failure	Ram memory error. Unexpected behaviour. Power cycle might solve the problem	- Checksum calculation reports error	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300045	Corrupted memory	Program Memory Failure	Can only be reported from the SBL.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300046	Checksum error for calibration memory, (DST).	Calibration / Parameter Memory Failure	Unexpected behaviour.	- Power loss during software download - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300094	Faulty operation	Unexpected Operation	Unexpected behaviour.	- Checksum error. - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300316	Battery Voltage < 10 V. The radio will performe a controlled shut down.	Circuit Voltage Below Threshold	When reaching threshold level, (~10V), audio unit will turn off amplifier. When reaching ~7V radio will start a controlled shutdown.	- Open fuse - DCDC not working	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	U300317	Battery Voltage > 16 V.	Circuit Voltage Above Threshold	When reaching threshold level, (~16V), DTC will be set.	- DCDC not working correctly.	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	B100211	Speaker output short circuited to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100212	Speaker output short circuited to V-batt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B100213	Speaker is in open circuit.	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100411	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100412	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100413	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100613	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100811	Mono input is shorted to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problems - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100812	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100912	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition	- Harness problem - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input - circuit
Audio	B100E13	AUX1 Not detected	Circuit Open	AUX1 is not connected or is not connected correctly.	- Radio connector not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 detection circuit
Audio	B100F12	AUX1 Common Short to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition.	- Attached device not correctly installed - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 common circuit
Audio	B101011	AUX1 Left is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101012	AUX1 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problems	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101111	AUX1 Right is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101112	AUX1 Right is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101613	Speaker is in open circuit	Circuit Open	Shorted channels will be muted until recovery from error condition	- No speaker connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U1FFF68	- ECU in factory mode.	Event Information		Parameter set to "1" when delivering truck.	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
Audio	B100011	Pin short circuited to power ground	Circuit Short To Ground	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100012	Pin short circuited to power supply	Circuit Short To Battery	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100013	Microphone input, open circuit	Circuit Open	Fucntionality of microphone is not supported until error recovery	- Microphone not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100A12	AUX2 Left is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Left Channel Circuit
Audio	B100B12	AUX2 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Right Channel Circuit
Audio	B100C12	AUX2 Common is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Common Circuit
Audio	B100D13	AUX2 detection circuit open	Circuit Open	AUX2 is not connected or is not connected correctly	- Audio connector not connected	Diagnostic Service 14. No additional action is needed.		AUX 2 Detection Circuit
Audio	B101408	USB connection error (H/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	B101419	Power consumption from USB device > 500 mA	Circuit Current Above Threshold	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken. - Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	U115600	Unable to authenticate an iPod	No Sub Type Information	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported	Diagnostic Service 14. No additional action is needed.		USB, Ipod Communication
Audio	U11594B	CD mechanism overheat > 70 degree C.	Over Temperature	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit overheated	Diagnostic Service 14. No additional action is needed.		CD Mechanism Temperature
Audio	B101808	USB communication error (S/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Communication
Audio	U115707	Internal failure inside the CD mechanism of the Audio unit. (i.e mechanical , electrical).	Mechanical Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U115768	Incorrect loaded CD	Event Information	If a load failure occurs, the CD shall move to the ejected position. The CD shal remian in the ejected position and the user shall be notified that there is no CD present.	- Disc inserted upside down. - Disc broken - CD stucked	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115877	CD mechanism unable to read CD	Commanded Position Not Reachable	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Disc format not supported - Audio broken - Heavy vibration prevent the Audio unit to focus on media	Diagnostic Service 14. No additional action is needed.		CD Mechanism Optical Focus
Audio	U115A49	Internal failure in Audio unit. CD mechanism communication problem.	Internal Electronic Failure	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Communication
Audio	B101321	No statations detected when scanning the frequency band	Signal Amplitude Less Than Minimum	Tuner wont be able to tune to stations.	- No radio reception - Antenna not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Antenna circuit
Audio	U114163	Button physically continuously pressed for 2 minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	EJECT button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		EJECT button electrical switch
Audio	U114263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	TA button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		INFORMATION button electrical switch
Audio	U114363	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	RADIO button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		RADIO button electrical switch
Audio	U114463	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	CD button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		CD button electrical switch
Audio	U114563	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	POWER button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		POWER button electrical switch
Audio	U114663	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	NAVIGATION button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		NAVIGATION button electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114863	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 1 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 1 button electrical switch
Audio	U114963	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 2 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 2 button electrical switch
Audio	U115263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK DOWN button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK DOWN button electrical switch
Audio	U114A63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 3 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 3 button electrical switch
Audio	U114B63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 4 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 4 button electrical switch
Audio	U114C63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 5 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 5 button electrical switch
Audio	U114D63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 6 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 6 button. electrical switch
Audio	U114E63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	MENU button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		MENU button electrical switch
Audio	U115163	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK UP button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK UP button electrical switch
Audio	U004611	CAN Bus Line physical short to GND	Circuit Short To Ground	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device connected to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004612	CAN Bus Line physical short to Battery	Circuit Short To Battery	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U004688	CAN HW unable to connect to bus	Bus off	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus - CAN hardware issue resulting in Bus off	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U300041	Incorrect software signature. Corrupted memory.	General Checksum Failure	Unit will not start main application.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300044	RAM memory error	Data Memory Failure	Ram memory error. Unexpected behaviour. Power cycle might solve the problem	- Checksum calculation reports error	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300045	Corrupted memory	Program Memory Failure	Can only be reported from the SBL.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300046	Checksum error for calibration memory, (DST).	Calibration / Parameter Memory Failure	Unexpected behaviour.	- Power loss during software download - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300094	Faulty operation	Unexpected Operation	Unexpected behaviour.	- Checksum error. - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300316	Battery Voltage < 10 V. The radio will performe a controlled shut down.	Circuit Voltage Below Threshold	When reaching threshold level, (~10V), audio unit will turn off amplifier. When reaching ~7V radio will start a controlled shutdown.	- Open fuse - DCDC not working	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	U300317	Battery Voltage > 16 V.	Circuit Voltage Above Threshold	When reaching threshold level, (~16V), DTC will be set.	- DCDC not working correctly.	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	B100211	Speaker output short circuited to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100212	Speaker output short circuited to V-batt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100213	Speaker is in open circuit.	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100411	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100412	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100413	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100613	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100811	Mono input is shorted to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problems - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input + circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B100812	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100912	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition	- Harness problem - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input - circuit
Audio	B100E13	AUX1 Not detected	Circuit Open	AUX1 is not connected or is not connected correctly.	- Radio connector not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 detection circuit
Audio	B100F12	AUX1 Common Short to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition.	- Attached device not correctly installed - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 common circuit
Audio	B101011	AUX1 Left is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101012	AUX1 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problems	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101111	AUX1 Right is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101112	AUX1 Right is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101613	Speaker is in open circuit	Circuit Open	Shorted channels will be muted until recovery from error condition	- No speaker connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	U1FFF68	- ECU in factory mode.	Event Information		Parameter set to "1" when delivering truck.	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
Audio	B100011	Pin short circuited to power ground	Circuit Short To Ground	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100012	Pin short circuited to power supply	Circuit Short To Battery	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100013	Microphone input, open circuit	Circuit Open	Fucntionality of microphone is not supported until error recovery	- Microphone not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100A12	AUX2 Left is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Left Channel Circuit
Audio	B100B12	AUX2 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Right Channel Circuit
Audio	B100C12	AUX2 Common is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Common Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B100D13	AUX2 detection circuit open	Circuit Open	AUX2 is not connected or is not connected correctly	- Audio connector not connected	Diagnostic Service 14. No additional action is needed.		AUX 2 Detection Circuit
Audio	B101408	USB connection error (H/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	B101419	Power consumption from USB device > 500 mA	Circuit Current Above Threshold	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken. - Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	U115600	Unable to authenticate an iPod	No Sub Type Information	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported	Diagnostic Service 14. No additional action is needed.		USB, Ipod Communication
Audio	U11594B	CD mechanism overheat > 70 degree C.	Over Temperature	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit overheated	Diagnostic Service 14. No additional action is needed.		CD Mechanism Temperature
Audio	B101808	USB communication error (S/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Communication
Audio	U115707	Internal failure inside the CD mechanism of the Audio unit. (i.e mechanical , electrical).	Mechanical Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115768	Incorrect loaded CD	Event Information	If a load failure occurs, the CD shall move to the ejected position. The CD shal remian in the ejected position and the user shall be notified that there is no CD present.	- Disc inserted upside down. - Disc broken - CD stucked	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115877	CD mechanism unable to read CD	Commanded Position Not Reachable	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Disc format not supported - Audio broken - Heavy vibration prevent the Audio unit to focus on media	Diagnostic Service 14. No additional action is needed.		CD Mechanism Optical Focus
Audio	U115A49	Internal failure in Audio unit. CD mechanism communication problem.	Internal Electronic Failure	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Communication
Audio	B101321	No statations detected when scanning the frequency band	Signal Amplitude Less Than Minimum	Tuner wont be able to tune to stations.	- No radio reception - Antenna not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Antenna circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114163	Button physically continuously pressed for 2 minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	EJECT button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		EJECT button electrical switch
Audio	U114263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	TA button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		INFORMATION button electrical switch
Audio	U114363	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	RADIO button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		RADIO button electrical switch
Audio	U114463	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	CD button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		CD button electrical switch
Audio	U114563	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	POWER button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		POWER button electrical switch
Audio	U114663	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	NAVIGATION button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		NAVIGATION button electrical switch
Audio	U114863	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 1 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 1 button electrical switch
Audio	U114963	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 2 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 2 button electrical switch
Audio	U115263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK DOWN button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK DOWN button electrical switch
Audio	U114A63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 3 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 3 button electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114B63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 4 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 4 button electrical switch
Audio	U114C63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 5 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 5 button electrical switch
Audio	U114D63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 6 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 6 button. electrical switch
Audio	U114E63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	MENU button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		MENU button electrical switch
Audio	U115163	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK UP button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK UP button electrical switch
Audio	U004611	CAN Bus Line physical short to GND	Circuit Short To Ground	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device connected to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004612	CAN Bus Line physical short to Battery	Circuit Short To Battery	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004688	CAN HW unable to connect to bus	Bus off	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus - CAN hardware issue resulting in Bus off	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U300041	Incorrect software signature. Corrupted memory.	General Checksum Failure	Unit will not start main application.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300044	RAM memory error	Data Memory Failure	Ram memory error. Unexpected behaviour. Power cycle might solve the problem	- Checksum calculation reports error	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300045	Corrupted memory	Program Memory Failure	Can only be reported from the SBL.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300046	Checksum error for calibration memory, (DST).	Calibration / Parameter Memory Failure	Unexpected behaviour.	- Power loss during software download - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300094	Faulty operation	Unexpected Operation	Unexpected behaviour.	- Checksum error. - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U300316	Battery Voltage < 10 V. The radio will performe a controlled shut down.	Circuit Voltage Below Threshold	When reaching threshold level, (~10V), audio unit will turn off amplifier. When reaching ~7V radio will start a controlled shutdown.	- Open fuse - DCDC not working	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	U300317	Battery Voltage > 16 V.	Circuit Voltage Above Threshold	When reaching threshold level, (~16V), DTC will be set.	- DCDC not working correctly.	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	B100211	Speaker output short circuited to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100212	Speaker output short circuited to V-batt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100213	Speaker is in open circuit.	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100411	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100412	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100413	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100613	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100811	Mono input is shorted to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problems - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100812	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100912	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition	- Harness problem - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input - circuit
Audio	B100E13	AUX1 Not detected	Circuit Open	AUX1 is not connected or is not connected correctly.	- Radio connector not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 detection circuit
Audio	B100F12	AUX1 Common Short to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition.	- Attached device not correctly installed - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 common circuit
Audio	B101011	AUX1 Left is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101012	AUX1 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problems	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B101111	AUX1 Right is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101112	AUX1 Right is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101613	Speaker is in open circuit	Circuit Open	Shorted channels will be muted until recovery from error condition	- No speaker connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	U1FFF68	- ECU in factory mode.	Event Information		Parameter set to "1" when delivering truck.	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
Audio	B100011	Pin short circuited to power ground	Circuit Short To Ground	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100012	Pin short circuited to power supply	Circuit Short To Battery	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100013	Microphone input, open circuit	Circuit Open	Fucntionality of microphone is not supported until error recovery	- Microphone not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100A12	AUX2 Left is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Left Channel Circuit
Audio	B100B12	AUX2 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Right Channel Circuit
Audio	B100C12	AUX2 Common is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Common Circuit
Audio	B100D13	AUX2 detection circuit open	Circuit Open	AUX2 is not connected or is not connected correctly	- Audio connector not connected	Diagnostic Service 14. No additional action is needed.		AUX 2 Detection Circuit
Audio	B101408	USB connection error (H/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	B101419	Power consumption from USB device > 500 mA	Circuit Current Above Threshold	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken. - Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	U115600	Unable to authenticate an iPod	No Sub Type Information	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported	Diagnostic Service 14. No additional action is needed.		USB, Ipod Communication

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U11594B	CD mechanism overheat > 70 degree C.	Over Temperature	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit overheated	Diagnostic Service 14. No additional action is needed.		CD Mechanism Temperature
Audio	B101808	USB communication error (S/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Communication
Audio	U115707	Internal failure inside the CD mechanism of the Audio unit. (i.e mechanical , electrical).	Mechanical Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115768	Incorrect loaded CD	Event Information	If a load failure occurs, the CD shall move to the ejected position. The CD shal remian in the ejected position and the user shall be notified that there is no CD present.	- Disc inserted upside down. - Disc broken - CD stucked	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115877	CD mechanism unable to read CD	Commanded Position Not Reachable	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Disc format not supported - Audio broken - Heavy vibration prevent the Audio unit to focus on media	Diagnostic Service 14. No additional action is needed.		CD Mechanism Optical Focus
Audio	U115A49	Internal failure in Audio unit. CD mechanism communication problem.	Internal Electronic Failure	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Communication
Audio	B101321	No statations detected when scanning the frequency band	Signal Amplitude Less Than Minimum	Tuner wont be able to tune to stations.	- No radio reception - Antenna not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Antenna circuit
Audio	U114163	Button physically continuously pressed for 2 minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	EJECT button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		EJECT button electrical switch
Audio	U114263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	TA button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		INFORMATION button electrical switch
Audio	U114363	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	RADIO button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		RADIO button electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114463	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	CD button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		CD button electrical switch
Audio	U114563	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	POWER button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		POWER button electrical switch
Audio	U114663	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	NAVIGATION button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		NAVIGATION button electrical switch
Audio	U114863	Button physically continuously pressed for one minute due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 1 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 1 button electrical switch
Audio	U114963	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 2 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 2 button electrical switch
Audio	U115263	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK DOWN button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK DOWN button electrical switch
Audio	U114A63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 3 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 3 button electrical switch
Audio	U114B63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 4 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 4 button electrical switch
Audio	U114C63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 5 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 5 button electrical switch
Audio	U114D63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	PRESET 6 button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		PRESET 6 button. electrical switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U114E63	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	MENU button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		MENU button electrical switch
Audio	U115163	Button physically continuously pressed for two minutes due to user or mechanical issue.	Circuit / Component Protection Time-Out	Fucntionality of button is ignored until error recovery.	SEEK UP button stucked. Continuous detection of keypress	Diagnostic Service 14. No additional action is needed.		SEEK UP button electrical switch
Audio	U004611	CAN Bus Line physical short to GND	Circuit Short To Ground	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device connected to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004612	CAN Bus Line physical short to Battery	Circuit Short To Battery	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U004688	CAN HW unable to connect to bus	Bus off	Enter Can Bus off state and awaits recovery conditions.	- Harness problem - Broken device attached to CAN bus - CAN hardware issue resulting in Bus off	Diagnostic Service 14. No additional action is needed.		CAN Communication Infotainment Net
Audio	U300041	Incorrect software signature. Corrupted memory.	General Checksum Failure	Unit will not start main application.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300044	RAM memory error	Data Memory Failure	Ram memory error. Unexpected behaviour. Power cycle might solve the problem	- Checksum calculation reports error	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300045	Corrupted memory	Program Memory Failure	Can only be reported from the SBL.	- Power loss during software donwload - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300046	Checksum error for calibration memory, (DST).	Calibration / Parameter Memory Failure	Unexpected behaviour.	- Power loss during software download - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300094	Faulty operation	Unexpected Operation	Unexpected behaviour.	- Checksum error. - Broken unit	Diagnostic Service 14. No additional action is needed.		Electronic control unit
Audio	U300316	Battery Voltage < 10 V. The radio will performe a controlled shut down.	Circuit Voltage Below Threshold	When reaching threshold level, (~10V), audio unit will turn off amplifier. When reaching ~7V radio will start a controlled shutdown.	- Open fuse - DCDC not working	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	U300317	Battery Voltage > 16 V.	Circuit Voltage Above Threshold	When reaching threshold level, (~16V), DTC will be set.	- DCDC not working correctly.	Diagnostic Service 14. No additional action is needed.		Battery Power
Audio	B100211	Speaker output short circuited to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100212	Speaker output short circuited to V-batt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100213	Speaker is in open circuit.	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RF+ circuit
Audio	B100411	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B100412	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100413	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LF+ circuit
Audio	B100611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100613	Speaker is in open circuit	Circuit Open	Shorted channel will be muted until recovery from error condition.	- Speaker not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output LR+ Circuit
Audio	B100811	Mono input is shorted to GND	Circuit Short To Ground	Shorted channel will be muted until recovery from error condition.	- Harness problems - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100812	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition.	- Harness problem	Diagnostic Service 14. No additional action is needed.		Mono input + circuit
Audio	B100912	Mono input is shorted to Vbatt	Circuit Short To Battery	Mono channel will be muted until recovery from error condition	- Harness problem - The attached device is broken	Diagnostic Service 14. No additional action is needed.		Mono input - circuit
Audio	B100E13	AUX1 Not detected	Circuit Open	AUX1 is not connected or is not connected correctly.	- Radio connector not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 detection circuit
Audio	B100F12	AUX1 Common Short to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition.	- Attached device not correctly installed - Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 common circuit
Audio	B101011	AUX1 Left is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101012	AUX1 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problems	Diagnostic Service 14. No additional action is needed.		AUX 1 left channel circuit
Audio	B101111	AUX1 Right is shorted to GND	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Attached device is not correctly installed - Harness problems - Mono plug used, (TRS jack)	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101112	AUX1 Right is shorted to Vbatt	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 1 right channel circuit
Audio	B101611	Speaker output short circuited to GND.	Circuit Short To Ground	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101612	Speaker output short circuited to Vbatt.	Circuit Short To Battery	Shorted channels will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	B101613	Speaker is in open circuit	Circuit Open	Shorted channels will be muted until recovery from error condition	- No speaker connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Speaker output RR+ circuit
Audio	U1FFF68	- ECU in factory mode.	Event Information		Parameter set to "1" when delivering truck.	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
Audio	B100011	Pin short circuited to power ground	Circuit Short To Ground	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	B100012	Pin short circuited to power supply	Circuit Short To Battery	Fucntionality of microphone is not supported until error recovery	- Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100013	Microphone input, open circuit	Circuit Open	Fucntionality of microphone is not supported until error recovery	- Microphone not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Microphone input circuit
Audio	B100A12	AUX2 Left is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Left Channel Circuit
Audio	B100B12	AUX2 Rigth is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Right Channel Circuit
Audio	B100C12	AUX2 Common is shorted to Vbatt	Circuit Short To Battery	Shorted channel will be muted until recovery from error condition	- Harness problem	Diagnostic Service 14. No additional action is needed.		AUX 2 Common Circuit
Audio	B100D13	AUX2 detection circuit open	Circuit Open	AUX2 is not connected or is not connected correctly	- Audio connector not connected	Diagnostic Service 14. No additional action is needed.		AUX 2 Detection Circuit
Audio	B101408	USB connection error (H/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	B101419	Power consumption from USB device > 500 mA	Circuit Current Above Threshold	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device broken. - Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Circuit
Audio	U115600	Unable to authenticate an iPod	No Sub Type Information	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported	Diagnostic Service 14. No additional action is needed.		USB, Ipod Communication
Audio	U11594B	CD mechanism overheat > 70 degree C.	Over Temperature	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit overheated	Diagnostic Service 14. No additional action is needed.		CD Mechanism Temperature
Audio	B101808	USB communication error (S/W detection)	Bus Signal / Message Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Attached device not supported - Harness problem	Diagnostic Service 14. No additional action is needed.		USB Communication
Audio	U115707	Internal failure inside the CD mechanism of the Audio unit. (i.e mechanical , electrical).	Mechanical Failures	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control
Audio	U115768	Incorrect loaded CD	Event Information	If a load failure occurs, the CD shall move to the ejected position. The CD shal remian in the ejected position and the user shall be notified that there is no CD present.	- Disc inserted upside down. - Disc broken - CD stucked	Diagnostic Service 14. No additional action is needed.		CD Mechanism Disc Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
Audio	U115877	CD mechanism unable to read CD	Commanded Position Not Reachable	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Disc format not supported - Audio broken - Heavy vibration prevent the Audio unit to focus on media	Diagnostic Service 14. No additional action is needed.		CD Mechanism Optical Focus
Audio	U115A49	Internal failure in Audio unit. CD mechanism communication problem.	Internal Electronic Failure	If an error is detected this shall be communicated to the HMI and to diagnostic function.	- Audio unit broken	Diagnostic Service 14. No additional action is needed.		CD Mechanism Communication
Audio	B101321	No statations detected when scanning the frequency band	Signal Amplitude Less Than Minimum	Tuner wont be able to tune to stations.	- No radio reception - Antenna not connected - Harness problem	Diagnostic Service 14. No additional action is needed.		Antenna circuit
BBM	B101012	Portpin shorted to Battery	Circuit Short To Battery	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B2:10	BBM Digital Output 16 switch
BBM	B101112	Portpin shorted to Battery	Circuit Short To Battery	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B2:11	BBM Digital Output 17 switch
BBM	B101212	Portpin shorted to Battery	Circuit Short To Battery	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B2:3	BBM Digital Output 18 switch
BBM	B100011	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:3	BBM Digital Output 1 switch
BBM	B100111	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:1	BBM Digital Output 2 switch
BBM	B100311	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:4	BBM Digital Output 3 switch
BBM	B100411	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:2	BBM Digital Output 4 switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BBM	B100511	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:21	BBM Digital Output 5 switch
BBM	B100611	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:10	BBM Digital Output 6 switch
BBM	B100711	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:22	BBM Digital Output 7 switch
BBM	B100811	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:20	BBM Digital Output 8 switch
BBM	B100815	Portpin shorted to Battery or Open	Circuit Short To Battery or Open	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:20	BBM Digital Output 8 switch
BBM	B100911	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:8	BBM Digital Output 9 switch
BBM	B100915	Portpin shorted to Battery or Open	Circuit Short To Battery or Open	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:8	BBM Digital Output 9 switch
BBM	B100A11	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:6	BBM Digital Output 10 switch
BBM	B100B11	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:9	BBM Digital Output 11 switch
BBM	B100C11	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:7	BBM Digital Output 12 switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BBM	B100D11	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:17	BBM Digital Output 13 switch
BBM	B100E11	Portpin shorted to GND	Circuit Short To Ground	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:5	BBM Digital Output 14 switch
BBM	B100F12	Portpin shorted to Battery	Circuit Short To Battery	Functional Behavior depending on connected equipment	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B2:2	BBM Digital Output 15 switch
BBM	B150211	Portpin shorted to GND	Circuit Short To Ground	External switch power feed is off	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:16	BBM Switch Power Supply Circuit
BBM	B150215	Portpin shorted to Battery or open	Circuit Short To Battery or Open	External switch power feed is off	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:16	BBM Switch Power Supply Circuit
BBM	B152B11	Voltage on A01 pin is 0 or << 3.447 V	Circuit Short To Ground	Sensor Power Supply is off.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:14	A/C Refrigerant High Pressure Sensor 5V Supply
BBM	B152B12	Voltage on A01 pin > 3.689 V	Circuit Short To Battery	Sensor Power Supply is off.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:14	A/C Refrigerant High Pressure Sensor 5V Supply
BBM	B152B16	Voltage on A01 pin is < 3.447	Circuit Voltage Below Threshold	Sensor Power Supply is off.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:14	A/C Refrigerant High Pressure Sensor 5V Supply
BBM	B152B17	Voltage on A01 pin >3.689	Circuit Voltage Above Threshold	Sensor Power Supply is off.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and B3:14	A/C Refrigerant High Pressure Sensor 5V Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BBM	B152C00	On Park Brake Release Failure	No Sub Type Information	Parking Brake release is inhibited	1. The digital input of BBM responsible for activating Remote Parking Brake Release Inhibition (function CtaBsPBReleaseInhibit_status) was active while vehicle speed was above 10km/h. 2.The digital input of BBM responsible for inhibiting Remote Parking Brake Release Inhibition (function CtaBsPBReleaseInhibCancel_stat) has been active for more than 60s (duration defined by P1KKE).	Diagnostic Service 14. No additional action is needed.	11. The digital input of BBM responsible for activating Remote Parking Brake Release Inhibition (function CtaBsPBReleaseInhibit_status) was active while vehicle speed was above 10km/h(speed defined by development parameter P1KKD). This is an assumption that there is a failure in the bodybuilder equipment because function should only be requested at standstill. WORKSHOP ACTION: Recommendation is to check the equipment feeding this BBM digital input. 2. The digital input of BBM responsible for inhibiting Remote Parking Brake Release Inhibition (function CtaBsPBReleaseInhibCancel_stat) has been active for more than 60s (duration defined by P1KKE). This is an assumption that the emergency PB release switch is stucked in position pushed. WORKSHOP ACTION: Recommendation is to check the switch and wire feeding this BBM digital input	Remote Park Brake Release Inhibition Error
BBM	U000188	On Busoff	Bus off	No Communication	Can Comm error Backbone 2	Diagnostic Service 14. No additional action is needed.	Check Backbone 2 CAN connector.	CAN Communication Backbone 2 Net
BBM	U001088	On Busoff	Bus off	No Communication	Can Comm error Backbone 1	Diagnostic Service 14. No additional action is needed.	Check Backbone 1 CAN connector.	CAN Communication Backbone 1 Net
BBM	U200688	On Busoff	Bus off	No Communication	Can Comm error Backbone Proprietary	Diagnostic Service 14. No additional action is needed.	Check BackBone proprietary Net CAN connector.	CAN Communication BB proprietary Net
BBM	U300041	Faulty SW and/or Data	General Checksum Failure	ECU shall not operate	Faulty SW and/or Data	Diagnostic Service 14. No additional action is needed.	1. check the signature value which is stored in memory. 2. checkif unexpected resets 3. Replace the BBM and check	Electronic control unit
BBM	U300044	RAM error	Data Memory Failure	ECU shall not operate	RAM error	Diagnostic Service 14. No additional action is needed.	1. checkif unexpected resets 2. Replace the BBM and check	Electronic control unit
BBM	U300045	NVRAM / ROM Error	Program Memory Failure	DTC shall be sent to cluster	NVRAM / ROM Error	Diagnostic Service 14. No additional action is needed.	1. checkif unexpected resets 2. Replace the BBM and check	Electronic control unit
BBM	U300049	ECU Internal Error	Internal Electronic Failure	DTC shall be sent to cluster	ECU Internal Error	Diagnostic Service 14. No additional action is needed.	1. checkif unexpected resets 2. Replace the BBM and check	Electronic control unit
BBM	U300094	Faulty operation	Unexpected Operation	DTC shall be sent to cluster	Faulty operation	Diagnostic Service 14. No additional action is needed.	1. checkif unexpected resets 2. Replace the BBM and check	Electronic control unit
BBM	U300316	ECU Low Voltage <10V	Circuit Voltage Below Threshold	DTC shall be sent to cluster	Low Battery Voltage	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
BBM	U300317	ECU high Voltage> 33V	Circuit Voltage Above Threshold	DTC shall be sent to cluster	High Battery Voltage	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
BBM	U1FFF68	Truck is in Factory Mode. Factory mode address parameter is set to value "1"	Event Information	DTC shall be sent to cluster	Whenever P1C54 (Shared Factory mode parameter) is set to TRUE, this DTC shall be set. Unit not configured correctly when leaving factory.	Diagnostic Service 14. No additional action is needed.	Check the Factory mode Parameter value P1C54,	Factory Mode Active
BBM	U250044	Internal Data - RAM Inconsistency detected Inconsistency detected by internal routines. In normal situations, the ECU will recover automatically, no action needed. DTC set for fault tracing.	Data Memory Failure	BBM does not work properly.	- Inproper shutdown - power cut - In rare situations HW failure - Software error	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BBM	U250045	Internal Data - ROM Inconsistency detected Inconsistency detected by internal routines. In normal situations, the ECU will recover automatically DTC set for fault tracing.	Program Memory Failure	BBM does not work properly.	- Inproper shutdown - power cut - In rare situations HW failure - Software error	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency
BBM	U114268	If DTC's reached maximum of 29	Event Information	Cluster takes care of this DTC	ECU has too many DTCs set	Diagnostic Service 14. No additional action is needed.	1. Check No of active DTCs count if more than 29	DTC Notification Max Limit Reached
BLECU	C100323	- Hall sensor level doesn't change during reference run from low to high.(ESW bit read from AMIS during reference run is not changed)	Signal Stuck Low	- AFS functionallity not available. - Neutral position is secured because of HardReference (The motor is able to detect the mechanical block of extremities.), no other movement until the error is solved. (If block detection is activated through Motor Dataset, possible to be reported also 0x97 – Component or System Operation Obstructed or Blocked)	- Headlamp stepper motor failure (hall sensor is broken; motor is blocked).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Left Bending Lights Stepper Motor Circuit
BLECU	C100324	- Hall sensor level doesn't change during reference run from high to low.(ESW bit read from AMIS during reference run is not changed)	Signal Stuck High	- AFS functionallity not available. - Neutral position is secured because of HardReference(The motor is able to detect the mechanical block of extremities.), no other movement until the error is solved. (If block detection is activated through Motor Dataset, possible to be reported also 0x97 – Component or System Operation Obstructed or Blocked).	- Headlamp stepper motor failure (hall sensor is broken; motor is blocked).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Left Bending Lights Stepper Motor Circuit
BLECU	C100363	- Stepper motor overusage warning. OverUsage counter > OverUse2stThreshold.	Circuit / Component Protection Time-Out	- AFS functionallity not available untill usage drops down. - The motors are positioned in neutral position until the usage drops below parameter OverUse1stThreshold.	- Too many movements on headlamp stepper motor.	Diagnostic Service 14. No additional action is needed.	- No action required.	Left Bending Lights Stepper Motor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	C100416	- Satellite supply voltage is out of range less than UV_KL15_12V (typical 10V).	Circuit Voltage Below Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position when the voltage drops below UV_KL15_12V-1V. - If the voltage drop is below 7V, 12 V supply is shutted down and neutral possition is not secured.	- Headlamp stepper motor power supply shorted to ground. - BLECU hardware failure (DC/DC converter failure). - Headlamp stepper motor hardware failure.	Diagnostic Service 14. No additional action is needed.	1. Inspect power supply harness between ECU and sattelites. 2.Replace the ECU. 3.If still active replace headlamp.	Bending Light Satellite Power Supply Circuit
BLECU	C100417	- Satellite supply voltage is out of range - greater than OV_KL15_12V (typical 14V).	Circuit Voltage Above Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position when the voltage goes above OV_KL15_12V +1V.	- Headlamp stepper motor power supply shorted to battery. - BLECU hardware failure (DC/DC converter failure).	Diagnostic Service 14. No additional action is needed.	1. Inspect power supply harness between ECU and sattelites. 2.Replace the ECU. 3.If still active replace headlamp.	Bending Light Satellite Power Supply Circuit
BLECU	C100013	- Electrical problem on satellite's stepper motor or stepper driver. Reported by AMIS.	Circuit Open	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (wire to stepper motor is broken; open load for motor coils).	Diagnostic Service 14. No additional action is needed.	- Stepper motor defect. Replace headlamp.	Right Bending Lights Stepper Driver Circuit
BLECU	C100016	- Electrical problem on satellite's stepper driver.Power drop on sattelite's supply line. - Reported by AMIS, supply lower than 8.5V	Circuit Voltage Below Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (undervoltage on motor driver; stepper motor driver reset).	Diagnostic Service 14. No additional action is needed.	1. Inspect power supply harness between ECU and sattelites. Measure 12V at left headlamp. When NOK test with other ECU. 2. Stepper motor defect. Replace headlamp.	Right Bending Lights Stepper Driver Circuit
BLECU	C100019	- Electrical problem on satellite's stepper motor or stepper driver. Reported by AMIS, current on coils higher than 800 mA.	Circuit Current Above Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (short circuit on motor coils).	Diagnostic Service 14. No additional action is needed.	- Stepper motor defect. Replace headlamp.	Right Bending Lights Stepper Driver Circuit
BLECU	C10004B	- Stepper motor driver shutdown due to high temperature (greater than TW + 10°C).(155degreesC +/-7degreesC). Reported by AMIS.	Over Temperature	- AFS functionallity not available untill the motor cools down - temperature drops below TW. - The motor is commanded to move to neutral position. The motor is internally shutted down for its own protection until it cooles down.Neutral position is not secured. - Motors are blocked in latest position.	- Headlamp stepper motor overheated.	Diagnostic Service 14. No additional action is needed.	- No action required.	Right Bending Lights Stepper Driver Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	C100068	- Stepper motor driver temperature warning (145degreesC +/- 7degreesC).	Event Information	- AFS functionallity not available untill the motor cools down - temperature drops below TW. - The motor is commanded to move to neutral position.	- Warning temperature threshold for headlamp stepper motor is reached.	Diagnostic Service 14. No additional action is needed.	- No action required.	Right Bending Lights Stepper Driver Circuit
BLECU	C100097	- Reported by application when the motor is moving through 0 and hall does not change.	Component or System Operation Obstructed or Blocked	- AFS functionallity not available. - The motor is commanded to move to neutral position. - The motor is blocked, neutral position is not secured. The application is waiting for Reference conditions to be fulfilled.	- Mechanical defect on headlamp stepper motor (motor stall; step loss; mechanical defect on sattelite motor).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Right Bending Lights Stepper Driver Circuit
BLECU	C100113	- Electrical problem on satellite's stepper motor or stepper driver. Reported by AMIS.	Circuit Open	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (wire to stepper motor is broken; open load for motor coils).	Diagnostic Service 14. No additional action is needed.	- Stepper motor defect. Replace headlamp.	Left Bending Lights Stepper Driver Circuit
BLECU	C100116	- Electrical problem on satellite's stepper driver.Power drop on sattelite's supply line. - Reported by AMIS, supply lower than 8.5V.	Circuit Voltage Below Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (undervoltage on motor driver; stepper motor driver reset).	Diagnostic Service 14. No additional action is needed.	1. Inspect power supply harness between ECU and sattelites. Measure 12V supply. If NOK test with other ECU. If OK -> Stepper motor defect. 2. Replace headlamp.	Left Bending Lights Stepper Driver Circuit
BLECU	C100119	- Electrical problem on satellite's stepper motor or stepper driver. Reported by AMIS, current on coils higher than 800 mA.	Circuit Current Above Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Headlamp stepper motor hardware failure (short circuit on motor coils).	Diagnostic Service 14. No additional action is needed.	- Stepper motor defect. Replace headlamp.	Left Bending Lights Stepper Driver Circuit
BLECU	C10014B	- Stepper motor driver shutdown due to high temperature (greater than TW + 10°C)(155degreesC +/-7degreesC). Reported by AMIS.	Over Temperature	- AFS functionallity not available untill the motor cools down - temperature drops below TW. - The motor is commanded to move to neutral position. The motor is internally shutted down for its own protection until it cooles down.Neutral position is not secured. - Motors are blocked in latest position.	- Headlamp stepper motor overheated.	Diagnostic Service 14. No additional action is needed.	- No action required.	Left Bending Lights Stepper Driver Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	C100168	- Stepper motor driver temperature warning (145degreesC +/- 7degreesC)	Event Information	- AFS functionallity not available untill the motor cools down - temperature drops below TW. - The motor is commanded to move to neutral position.	- Warning temperature threshold for headlamp stepper motor is reached.	Diagnostic Service 14. No additional action is needed.	- No action required.	Left Bending Lights Stepper Driver Circuit
BLECU	C100197	- Reported by application when the motor is moving through 0 and hall does not change.	Component or System Operation Obstructed or Blocked	- AFS functionallity not available. - The motor is commanded to move to neutral position. - The motor is blocked, neutral position is not secured. The application is waiting for Reference conditions to be fulfilled.	- Mechanical defect on headlamp stepper motor (motor stall; step loss; mechanical defect on sattelite motor).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Left Bending Lights Stepper Driver Circuit
BLECU	C100223	- Hall sensor level doesn't change during reference run from low to high.(ESW bit read from AMIS during reference run is not changed).	Signal Stuck Low	- AFS functionallity not available. - Neutral position is secured because of HardReference(The motor is able to detect the mechanical block of extremities.), no other movement until the error is solved. (If block detection is activated through Motor Dataset, possible to be reported also 0x97 – Component or System Operation Obstructed or Blocked).	- Headlamp stepper motor failure (hall sensor is broken; motor is blocked).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Right Bending Lights Stepper Motor Circuit
BLECU	C100224	- Hall sensor level doesn't change during reference run from high to low.(ESW bit read from AMIS during reference run is not changed).	Signal Stuck High	- AFS functionallity not available. - Neutral position is secured because of HardReference(The motor is able to detect the mechanical block of extremities.), no other movement until the error is solved. (If block detection is activated through Motor Dataset, possible to be reported also 0x97 – Component or System Operation Obstructed or Blocked).	- Headlamp stepper motor failure (hall sensor is broken; motor is blocked).	Diagnostic Service 14. No additional action is needed.	1. Check if module is blocked (maybe cable is inhibited), try to release to release it. 2. Perform reference run, use the "Move x Satellite motor" routine to find a block point by moving the motors to left/right limits and back to zero position and than check if the new Zero position is identical to the initial one , if module moveable and error still present stepper motor is defect. 3. Replace headlamp.	Right Bending Lights Stepper Motor Circuit
BLECU	C100263	- Stepper motor overusage warning. OverUsage counter > OverUse2stThreshold	Circuit / Component Protection Time-Out	- AFS functionallity not available untill usage drops down. - The motors are positioned in neutral position until the usage drops below parameter OverUse1stThreshold.	- Too many movements on headlamp stepper motor.	Diagnostic Service 14. No additional action is needed.	- No action required.	Right Bending Lights Stepper Motor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	U100287	- No communication with sattelite. - No answear from AMIS driver.	Missing Message	- AFS functionallity not available. - The motor is commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Communication between BLECU and headlamp stepper motor cannot be established (satellite failure; harness troubles; LIN wire open).	Diagnostic Service 14. No additional action is needed.	1. Check truck harness (12V, LIN from rightt to left headlamp). 2. Change headlamp.	Right Headlamp Module Data Link
BLECU	U100011	- Electrical defect on LIN harness. - Reported by SBC. - LIN shorted to ground.	Circuit Short To Ground	- AFS functionallity not available. - No LIN communication with either satellite. Neutral position is not secured. The application is waiting for this error to be solved. The motors are going to neutral position by their own.	- LIN wire between BLECU and either headlamp stepper motor is shorted to ground.	Diagnostic Service 14. No additional action is needed.	- Inspect LIN harness if is shorted with a ground wire.	LIN Advanced Lighting Net
BLECU	U100012	- Electrical defect on LIN harness. - Reported by SBC. - LIN shorted to battery.	Circuit Short To Battery	- AFS functionallity not available. - No LIN communication with either satellite. Neutral position is not secured. The application is waiting for this error to be solved. The motors are going to neutral position by their own.	- LIN wire between BLECU and either headlamp stepper motor is shorted to battery.	Diagnostic Service 14. No additional action is needed.	- Inspect LIN harness if is shorted with a power supply wire.	LIN Advanced Lighting Net
BLECU	U100049	- Electrical defect on LIN hardware. - Reported by SBC. - TXDL is clamped dominant.	Internal Electronic Failure	- AFS functionallity not available. - No LIN communication with either satellite. Neutral position is not secured. The application is waiting for this error to be solved. The motors are going to neutral position by their own.	- BLECU hardware failure (LIN transceiver).	Diagnostic Service 14. No additional action is needed.	1. Inspect LIN harness failure 2. Download a new MSW and Datasets 3. Replace the ECU.	LIN Advanced Lighting Net
BLECU	U100187	- No communication with sattelite. - No answear from AMIS driver.	Missing Message	- AFS functionallity not available. - The motor is commanded to move to neutral position . Neutral position is not secured. The application is waiting for this error to be solved.	- Communication between BLECU and headlamp stepper motor cannot be established (satellite failure; harness troubles; LIN wire open).	Diagnostic Service 14. No additional action is needed.	1. Check truck harness (12V, LIN from rightt to left headlamp). 2. Change headlamp.	Left Headlamp Module Data Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	U200487	- BLECU has problems communicating on CAN network.	Missing Message	- AFS functionallity might not be available. - Possibly no CAN communication with BLECU. - The motors are commanded to move to neutral position if the messages used as algorithm input are missing. The application is waiting for this error to be solved.	- CAN communication problems (wrong timing configuration; message(s) not received from FCIOM).	Diagnostic Service 14. No additional action is needed.	1. Inspect CAN harness failure 2. Download a new MSW and Datasets 3. Replace the ECU.	CAN Communication Advanced Lightning Net
BLECU	U200488	- BLECU does not communicate on CAN network. - Error frames present on CAN bus.	Bus off	- AFS functionallity not available. - The motors are commanded to move to neutral position . The application is waiting for this error to be solved.	- Too many errors on CAN bus. - CAN harness problems.	Diagnostic Service 14. No additional action is needed.	1. Inspect CAN harness failure 2. Download a new MSW and Datasets 3. Replace the ECU.	CAN Communication Advanced Lightning Net
BLECU	U300041	- Incorrect checksum on parameter datasets. Reported by Vector RomCheck.	General Checksum Failure	- AFS functionallity not available. - BLECU is not communicating to FCIOM. - The application in invalidated and only the boot is active, 12V supply is not activ. Neutral position is not secured.	- Incorrect datasets programmed on BLECU. - BLECU flash memory corruption.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U300044	- RAM check failed.	Data Memory Failure	- AFS functionallity not available. - The error is reported after the ECU already recover. Neutral position is not secured The application will wait next power cycle to recover.	- BLECU RAM memory corruption.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U300045	- Incorrect checksum on application. Reported by Vector RomCheck.	Program Memory Failure	- AFS functionallity not available. - BLECU is not communicating to FCIOM. - The application in invalidated and only the boot is active, 12V supply is not activ. Neutral position is not secured.	- Incorrect application programmed on BLECU. - BLECU flash memory corruption.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U300046	- EEPROM related errors.	Calibration / Parameter Memory Failure	- Autorecovery. EEPROM old data mught get lost (DTCs, ...)	- Emulated EEPROM block failure.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
BLECU	U300049	- Failure of microcontroller's oscillator or PLL not working at specified frequency.	Internal Electronic Failure	- AFS functionallity not available. - The error is reported after the ECU already recover. Neutral position is not secured The application will wait next power cycle to recover.	- BLECU's microcontroller clock failure.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U30004B	- Temperature inside BLECU reaches high values (T > TSensor3ndThreshold(approx 130 degrees Celsius).	Over Temperature	- AFS functionallity not available. - The motors are commanded to move to neutral position until the temperature drops below TSensor2ndThreshold and a new power cycle is performed.	- BLECU is overheated.	Diagnostic Service 14. No additional action is needed.	- No action required.	Electronic control unit
BLECU	U300054	- Invalid calibration or manufacturing data has been detected.	Missing Calibration	- AFS functionallity not available.12V supply is active.	- BLECU did not passed all EOL tests. - BLECU flash memory damaged.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U300094	- Unexpected microcontroller reset detected.	Unexpected Operation	- AFS functionallity not available. - The error is reported after the ECU already recover. Neutral position is not secured The application will wait next power cycle to recover.	- BLECU watchdog reset.	Diagnostic Service 14. No additional action is needed.	1. Download a new MSW and Datasets 2. Replace the ECU.	Electronic control unit
BLECU	U300316	- Supply voltage is out of range (less than UV_UBAT (typical 16V)).	Circuit Voltage Below Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . The application is waiting for this error to be solved.	- BLECU is not powered as specified.	Diagnostic Service 14. No additional action is needed.	- Check ECU supply harness and voltage levels.	Battery Power
BLECU	U300317	- Supply voltage is out of range (above OV_UBAT (typical 32V).	Circuit Voltage Above Threshold	- AFS functionallity not available. - The motors are commanded to move to neutral position . The application is waiting for this error to be solved.	- BLECU is not powered as specified.	Diagnostic Service 14. No additional action is needed.	- Check ECU supply harness and voltage levels.	Battery Power
BLECU	U100368	More than 28 DTCs are currently set.	Event Information	- Depending on the other DTSs set.	Too many errors present on ECU. Harnesses failure, power failure.	Diagnostic Service 14. No additional action is needed.	- See action for each DTC.	DTC Notification Max Limit Reached
XCIOM	P250000	If the logic is low (U<3.5V) then something is wrong and this DTC should be set.	No Sub Type Information	*Alternator is not charging	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and L-Terminal	Generator/Alternator Lamp/L-Terminal Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C103011	I > 10,336A..13,962A (T<20ºC) I > 5,906A..7,978A (T>20ºC) (I > 11,767A..15,895A Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and APM	APM redundancy power supply
XCIOM	C103012	U = 5V	Circuit Short To Battery	*In case of short circuit to battery, the output cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and APM	APM redundancy power supply
XCIOM	C10CD11	I > 2,405A..3,250A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Area Inhibit valve	Area inhibit valve circuit
XCIOM	C10CD15	I < 40mA (worst case: I < 5mA...70mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Area Inhibit valve	Area inhibit valve circuit
XCIOM	U002888	Bus off detected, CAN communication not working	Bus off	Vehicle cannot be used.	1. Wiring harness 2. ECU failure (any ECU connected to Chassis network)	Diagnostic Service 14. No additional action is needed.	Check harness Replace ECU	CAN Communication Chassis Net
XCIOM	U114149	ECU LIN hardware fault.	Internal Electronic Failure	*LIN communication error	1. Wiring harness 2. ECU HW faulty (transceiver). 3. BATS (Battery Sensor) failure 4. ECU SW error (BSW driver).	Diagnostic Service 14. No additional action is needed.	1. Check harness 2. Replace ECU	LIN11 Communication Net
XCIOM	U114186	Invalid Signal on LIN detected	Signal Invalid	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	- software error - configuration error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	LIN11 Communication Net
XCIOM	U114187	Message on LIN missing detected	Missing Message	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	- software error - configuration error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	LIN11 Communication Net
XCIOM	U114188	LIN Bus Off detected	Bus off	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	- software error - configuration error - broken lines - component broken	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check ECUs	LIN11 Communication Net
XCIOM	C10F013	Battery Sensor Open Load Detected	Circuit Open	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	- Battery Sensor Error - Harness Error	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Battery Sensor	Battery Sensor Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10F016	Battery Sensor low voltage detected	Circuit Voltage Below Threshold	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Missing.	- Battery Sensor Error - Harness Error	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Battery Sensor	Battery Sensor Module
XCIOM	C10F017	Battery Sensor high voltage detected	Circuit Voltage Above Threshold	*Sensor HW damage eventually	- Battery Sensor Error - Harness Error	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Battery Sensor	Battery Sensor Module
XCIOM	C10F045	Programm memory failure detected	Program Memory Failure	*No display of battery data	Software Error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Battery Sensor Module
XCIOM	C10F048	Supervision software failure detected	Supervision Software Failure	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	Software Error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Battery Sensor Module
XCIOM	C10F094	Unexpected operation detected	Unexpected Operation	*BatterySOH, BatteryCurrent, BatterySOHTrustLevel and BatterySOCnom = Error.	Operation Error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Battery Sensor Module
XCIOM	U30004A	- Incorrect Component Installed	Incorrect Component Installed	Fault tracing	- Incorrect Component Installed	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U30004B	The temperature has exceeded the threshold on one of the outputs.	Over Temperature	The output that is overheated will be turned OFF.	The ambient temperature is too high.	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Electronic control unit
XCIOM	C10EF11	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Drive valve	Front Wheel Drive Valve Circuit
XCIOM	C10EF12	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	* The output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Drive valve.	Front Wheel Drive Valve Circuit
XCIOM	C10EF15	I < 100mA (worst case: 88 mA < I < 115 mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Drive valve	Front Wheel Drive Valve Circuit
XCIOM	C103911	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Difflock valve	Front Wheels Diff lock valve
XCIOM	C103912	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	* The output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Difflock valve	Front Wheels Diff lock valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C103915	I < 100mA (worst case: 88 mA < I < 115 mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	"1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Wheel Difflock valve	Front Wheels Diff lock valve
XCIOM	C101264	1. The engine isn't running but the primary steering switch is still indicated as operational 2. The vehicle is at standstill but the secondary steering switch is still indicated as operational	Signal Plausibility Failure	*Indication of open circuit of one, two or three of the monitoring switches	1. Open circuit on one or two of the steering servo switches. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and the steering servo switches	Hydraulic Steering Switch Circuit
XCIOM	C101212	U=5,0V	Circuit Short To Battery	*Failing to monitor the the primary and secondary steering servo circuits.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and the steering servo switches	Hydraulic Steering Switch Circuit
XCIOM	C103A11	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and 5th wheel common valve	5th Wheel Common Valve Circuit
XCIOM	C103A15	I < 100mA (worst case: 88 mA < I < 115 mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component will be continouosly activated if low side path is activated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and 5th wheel common valve	5th Wheel Common Valve Circuit
XCIOM	C101F12	I > 1,201A..2,175A	Circuit Short To Battery	*In case of short circuit to battery, the component will be continouosly activated if low side path is activated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and 5th wheel common valve	Slideable 5th Wheel Valve Circuit
XCIOM	C10D211	I > 6,922A..9,35A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *WrknLghtChassisOutpFalr_stat = 2 (Short circuit to ground)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and 5th Wheel Light	Fifth Wheel Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10D215	I < 150mA (worst case: I < 20mA...270mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *WrknLghtChassisOutpFalr_stat = 1 (Open circuit)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and 5th Wheel Light	Fifth Wheel Light Circuit
XCIOM	C10EE7A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	1. Broken Bellows 2. Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Front Axle Left/Right Bellow Circuit
XCIOM	C11237A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	1. Broken Bellows 2. Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left Bellow Circuit
XCIOM	C11247A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	1. Broken Bellows 2. Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Right Bellow Circuit
XCIOM	C108407	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left - Right Pressure Sensor Circuit
XCIOM	C108416	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left - Right Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108417	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left - Right Pressure Sensor Circuit
XCIOM	C108516	U < P1CBY.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Pressure Sensor Supply Voltage Circuit
XCIOM	C108517	U > P1CBY.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Pressure Sensor Supply Voltage Circuit
XCIOM	C111D07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C111D16	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left Pressure Sensor Circuit
XCIOM	C111D17	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left Pressure Sensor Circuit
XCIOM	C111E07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Right Pressure Sensor Circuit
XCIOM	C111E16	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Right Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C111E17	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Sensor broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Right Pressure Sensor Circuit
XCIOM	C10CB11	I > 1.5A	Circuit Short To Ground	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Load Exhaust Circuit
XCIOM	C10CB12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Load Exhaust Circuit
XCIOM	C10CB15	I < 25mA	Circuit Short To Battery or Open	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong 6. Connector Not Connected	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Load Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10CB72	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Load Exhaust Circuit
XCIOM	C10CB73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Load Exhaust Circuit
XCIOM	C10CC11	I > 1.5A	Circuit Short To Ground	Normal software operation	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Left - Right Circuit
XCIOM	C10CC12	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Left - Right Circuit
XCIOM	C10CC15	I < 25mA	Circuit Short To Battery or Open	Normal software operation	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong 6. Connector Not Connected	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Left - Right Circuit
XCIOM	C10CC72	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 1- or 2-channel Solenoid Valve Left - Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10ED07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left - Right Channel Circuit
XCIOM	C111F11	I > 1.5A	Circuit Short To Ground	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C111F12	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C111F15	I < 25mA	Circuit Short To Battery or Open	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong 6. Connector Not Connected	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C111F72	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C112011	I > 1.5A	Circuit Short To Ground	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C112012	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C112015	I < 25mA	Circuit Short To Battery or Open	Normal software operation.	1. Broken Cable 2. Solenoid in valve block broken 3. Connector Broken 4. ECU Broken 5. Connector Stift Wrong 6. Connector Not Connected	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C112072	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C112107	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Left Channel Circuit
XCIOM	C112207	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation.	1. Solenoid mechanical stuck 2. Pipes wrongly connected 3. Pipes broken 4. Dirt in valve 5. Dirt in pneumatic connectors 6. Dirt in pipes 7. Mechanical fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Axle Right Channel Circuit
XCIOM	C108307	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 182]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left/Right Level Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108316	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 181]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left/Right Level Sensor Circuit
XCIOM	C108317	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 180]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left/Right Level Sensor Circuit
XCIOM	C108354	Level sensor is not calibrated	Missing Calibration	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 183]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Calibrate the front left/right level sensor	Air Suspension Front Left/Right Level Sensor Circuit
XCIOM	C108616	U < P1B9V.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 291]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Level Sensor Supply Voltage Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108617	U > P1B9V.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 290]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Level Sensor Supply Voltage Circuit
XCIOM	C10F816	U < P1B9V.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 389]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Front Axle Load Sensor Supply Voltage Circuit
XCIOM	C10F817	U > P1B9V.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 388]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Front Axle Load Sensor Supply Voltage Circuit
XCIOM	C10EC16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 281]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Front Axle Load Sensor Circuit
XCIOM	C10EC17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 280]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Front Axle Load Sensor Circuit
XCIOM	C10EC54	Load sensor is not calibrated.	Missing Calibration	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 378]: No load sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Calibrate the front axle load sensor	Front Axle Load Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C111B07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 174]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left Level Sensor Circuit
XCIOM	C111B16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 173]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left Level Sensor Circuit
XCIOM	C111B17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 172]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Left Level Sensor Circuit
XCIOM	C111B54	Level sensor is not calibrated	Missing Calibration	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 175]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Calibrate the front left level sensor	Air Suspension Front Left Level Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C111C07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 178]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Right Level Sensor Circuit
XCIOM	C111C16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 177]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Right Level Sensor Circuit
XCIOM	C111C17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 176]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Front Right Level Sensor Circuit
XCIOM	C111C54	Level sensor is not calibrated	Missing Calibration	Air suspension front limphome: It is only possible to control the vehicle in manual mode. However the bogie functionality is still working as in normal conditions. All rear sensor inputs are evaluated etc. Sensor inputs for the front axle are ignored and the front axle functionality is in limphome.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 179]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Calibrate the front right level sensor	Air Suspension Front Right Level Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	P206511	U < 0,15V	Circuit Short To Ground	*FuelLevel = 254 (Error) *FuelVolume = 2046 (Error)	" 1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Fuel level sensor	Fuel Level Sensor Circuit
XCIOM	P206512	U = 5V	Circuit Short To Battery	*FuelLevel = 254 (Error) *FuelVolume = 2046 (Error)	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Fuel level sensor	Fuel Level Sensor Circuit
XCIOM	P206513	4,212V < U < 5,0V	Circuit Open	*FuelLevel = 254 (Error) *FuelVolume = 2046 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Fuel level sensor	Fuel Level Sensor Circuit
XCIOM	P206516	0,15V <= U < 0,807V	Circuit Voltage Below Threshold	*FuelLevel = 254 (Error) *FuelVolume = 2046 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Fuel level sensor	Fuel Level Sensor Circuit
XCIOM	P206517	2,998V < U < 4,212V	Circuit Voltage Above Threshold	*FuelLevel = 254 (Error) *FuelVolume = 2046 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Fuel level sensor	Fuel Level Sensor Circuit
XCIOM	C103711	I > 1,293A..1,746A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and the lubrication motor	Chassis lubrication motor
XCIOM	C103715	Off state pull down resistor (10K)	Circuit Short To Battery or Open	*In case of open circuit, the output cannot be activated. *In case of short circuit to battery, the output cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and the lubrication motor	Chassis lubrication motor
XCIOM	P107938	Frequency input signal outside the range (9Hz to11Hz) OR (36Hz to 44Hz).	Signal Frequency Incorrect	*MainSwitch_Status = 2 (Error)	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Generator/Alternator	Generator/Alternator Shut Down Input
XCIOM	C103411	I > 1,263A..1,706A (I > 3,355A..4,532A Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and NOX component	NOX 1 power supply
XCIOM	C103412	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	The output cannot be deactivated. The Nox sensor will always be powered.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and NOX component	NOX 1 power supply
XCIOM	P107711	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 1 Valve Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	P107712	U = 5V	Circuit Short To Battery	*In case of short circuit to battery, the component cannot be deactivated	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 1 Valve Circuit
XCIOM	P107715	I < 100mA (worst case: 88 mA < I < 115 mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 1 Valve Circuit
XCIOM	P107811	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 2 Valve Circuit
XCIOM	P107812	U = 5V	Circuit Short To Battery	*In case of short circuit to battery, the component cannot be deactivated	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 2 Valve Circuit
XCIOM	P107815	I < 100mA (worst case: 88 mA < I < 115 mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Power Take off valve	Power Takeoff 2 Valve Circuit
XCIOM	C10CE11	I > 2,405A..3,250A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Range Inhibit valve	Range inhibit valve circuit
XCIOM	C10CE15	I < 40mA (worst case: I < 5mA...70mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Range Inhibit valve	Range inhibit valve circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	P085100	The neutral switch shall be considered to be jammed in an inactive position when this has been valid for more than the configurable time NeutralSwitchFailure DetectionTimeLimit, and the SecuredNeutralPositi on signal and the TransmissionNeutral Switch shall then both be set to "Error" until the value in signal NeutralSwitch_stat changes.	No Sub Type Information	*Neutral switch stuck in inactive position	neutral switch jammed in an inactive position	Diagnostic Service 14. No additional action is needed.	Check Neutral switch	Neutral Switch Input Circuit Low
XCIOM	P085200	"The neutral switch shall be considered to be jammed in an active position when this has been valid for more than the configurable time NeutralSwitchFailure DetectionTimeLimit, and the SecuredNeutralPositi on signal and the TransmissionNeutral Switch shall then both be set to ""Error"" until the value in signal NeutralSwitch_stat changes. "	No Sub Type Information	*Neutral switch stuck in active position	neutral switch jammed in an active position	Diagnostic Service 14. No additional action is needed.	Check Neutral switch	Neutral Switch Input Circuit High
XCIOM	C111A11	I > 1,207A..1,631A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Short stop heater	Short Stop Heater Pump Circuit
XCIOM	C111A12	U = 5V	Circuit Short To Battery	*In case of short circuit to battery, the component cannot be deactivated	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Short stop heater	Short Stop Heater Pump Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C102F11	I > 12,906A..17,433A (I > 15,379A..20,773A Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TECU	TECU power supply
XCIOM	C102F12	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated. The TECU will always be powered and thereby current consumption is increased.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TECU	TECU power supply
XCIOM	C103111	I > 0,352A...0,475A (I > 3,355A..4,532A Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TPM	TPM power supply
XCIOM	C103112	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	The output cannot be deactivated. The TPM will always be powered.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TPM.	TPM power supply
XCIOM	C103B11	I > 0,901A..1,217A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer PWM valve	Trailer PWM Valve Supply
XCIOM	C103B12	Short to Battery detected (Offstate Diagnostic)	Circuit Short To Battery	*In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer PWM valve	Trailer PWM Valve Supply
XCIOM	C103B15	I < 50mA (worst case: I < 30mA...75mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component will be continouosly activated if low side path is activated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer PWM valve	Trailer PWM Valve Supply
XCIOM	C103C12	I > 0,914A..1,655A	Circuit Short To Battery	*In case of short circuit to battery, the component will be continouosly activated if low side path is activated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer PWM valve	Trailer PWM Valve Ground
XCIOM	C101C11	I > 4,885A..6,599A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer fog lights	Trailer fog lights

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C101C15	I < 250mA (worst case: I < 140mA...335mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer fog lights	Trailer fog lights
XCIOM	C101D11	I > 12,868A..17,382A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer reverse lights	Trailer Reverse Lights Circuit
XCIOM	C101D15	I < 250mA (worst case: I < 100mA...350mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer reverse lights	Trailer Reverse Lights Circuit
XCIOM	P107A11	U < 0,15V	Circuit Short To Ground	*ManTransmiOilTemp = Error	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Gearbox Oil Temp sensor	Gearbox Oil Temperature Sensor Circuit
XCIOM	P107A12	U = 5V	Circuit Short To Battery	*ManTransmiOilTemp = Error	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Gearbox Oil Temp sensor	Gearbox Oil Temperature Sensor Circuit
XCIOM	P107A13	4,212V <= U < 5V	Circuit Open	*ManTransmiOilTemp = Error	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Gearbox Oil Temp sensor	Gearbox Oil Temperature Sensor Circuit
XCIOM	P107A16	0,15V <= U < 0,186	Circuit Voltage Below Threshold	*ManTransmiOilTemp = Error	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Gearbox Oil Temp sensor	Gearbox Oil Temperature Sensor Circuit
XCIOM	P107A17	4,191V < U < 4,212V	Circuit Voltage Above Threshold	*ManTransmiOilTemp = Error	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Gearbox Oil Temp sensor	Gearbox Oil Temperature Sensor Circuit
XCIOM	U300041	- ECU ROM checksum error.	General Checksum Failure	Check started at ECU startup	- ECU SW has been tampered - ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300044	ECU volatile memory error.	Data Memory Failure	ECU does not work properly.	1. ECU faulty (volatile memory)	Diagnostic Service 14. No additional action is needed.	If this DTC occurs frequently with similar DTC, this could indicate HW failure	Electronic control unit
XCIOM	U300045	ECU non volatile memory error.	Program Memory Failure	ECU does not work properly.	1. ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	If this DTC occurs frequently with similar DTC, this could indicate HW failure	Electronic control unit
XCIOM	U300049	ECU internal hardware fault.	Internal Electronic Failure	ECU does not work properly.	1. ECU faulty (BSW driver / transceiver). 2. Software error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300094	ECU watchdog fault.	Unexpected Operation	ECU does not work properly, potentially in FailSafe mode.	1. ECU faulty (watchdog fault) 2. Software error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	U1FFF68	Factory mode parameter P1C54 is active	Event Information	A truck delivered with Factory mode active will have unspecified behavior that might influence customer safety.	Factory mode not disabled	Diagnostic Service 14. No additional action is needed.	Set P1C54 (Shared Factory mode parameter) for the ECU to FALSE before shipping from factory.	Factory Mode Active
XCIOM	U114268	Too many DTC for Diagnostic Warning Manger (DWM) to handle.	Event Information	Max. number of DTCs reached. No more DTCs will be stored.	1. Refer to the DTC's that are active.	Diagnostic Service 14. No additional action is needed.	None, this is a notification to the driver	DTC Notification Max Limit Reached
XCIOM	U250145	ECU has detected Internal Data Fragmentation and cannot save internal status information like DTCs.	Program Memory Failure	ECU cannot save internal data like new DTC	- Inproper shutdown - power cut - In rare situations HW failure due to memory degradation - Software error	Diagnostic Service 14. No additional action is needed.	Transfer data, run RID R1AHD, restore data may fix the problem	Internal Data Storage Fragmented
XCIOM	U250245	ECU detected Internal Data Storage Discarded, possibly discarding internal data but selfhealing attempted.	Program Memory Failure	Fault tracing	- Software error - Memory degradation due to end-of-life	Diagnostic Service 14. No additional action is needed.	No action due to this DTC, the ECU has tried to recover. There are likely other DTCs that need to be handled as a consequence.	Internal Data Storage Discarded
XCIOM	C103311	I > 0,054A..0,072A (I > 9,7A - 15,8A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *Battery is not charged	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Alternator	Alternator control
XCIOM	C103312	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated. Alternator regulation is set to 100%	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Alternator	Alternator control
XCIOM	U300046	- Program memory corruption - ROM Checksum error	Calibration / Parameter Memory Failure	Fault tracing	- Memory hardware error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300716	U < 16 V	Circuit Voltage Below Threshold	ECU only partly working - some outputs are not powered	- Power supply "B" broken - Battery voltage too low - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "B"
XCIOM	U300717	U > 32 V	Circuit Voltage Above Threshold	ECU potentially blown	- Battery voltage too high - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "B"
XCIOM	U300616	U < 16 V	Circuit Voltage Below Threshold	ECU only partly working - some outputs are not powered	- Power supply "A" broken - Battery voltage too low - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "A"
XCIOM	U300617	U > 32 V	Circuit Voltage Above Threshold	ECU potentially blown	- Battery voltage too high - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "A"
XCIOM	U300093	Unexpected ECU operation.	No Operation	ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors. Too many restarts or RAM/ROM errors.	ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors. Too many restarts or RAM/ROM errors.	Diagnostic Service 14. No additional action is needed.	Powercycle ECU Reprogram SW Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	P064511	Low voltage present when not expected	Circuit Short To Ground	A/C Clutch Relay Control Circuit cannot be activated	Wiring harness, CCM pin A12 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & A/C Clutch Relay	A/C Clutch Relay Control Circuit
CCM	P064515	Short to battery or open circuit conditions detected	Circuit Short To Battery or Open	A/C Clutch Relay Control Circuit cannot be activated/deactivated	Wiring harness, CCM pin A12 is short to battery OR open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring on CCM & A/C Clutch Relay	A/C Clutch Relay Control Circuit
CCM	B150512	Inside Temperature Sensor Reference Voltage > 4.9V	Circuit Short To Battery	Inside Temperature Sensor Reference Voltage reporting unexpected values	Inside Temperature Sensor Reference Voltage, CCM pin B01 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged. DTC may also be set when Vbat injected at ITOS supply pin B10 when ITOS is attached.	Inside Temperature Sensor Reference Voltage
CCM	B150514	Inside Temperature Sensor reference Voltage < 0.97V	Circuit Short To Ground or Open	Inside Temperature Sensor Reference Voltage reporting unexpected values	-CCM pin B01 short to ground -CCM pin B01 open circuit -Connector damaged	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor Reference Voltage
CCM	B150612	Inside Temperature Sensor Surface Temperature Sensor voltage > 4.9V	Circuit Short To Battery	Inside Temperature Sensor Surface Temperature Sensor reporting unexpected values	Inside Temperature Sensor Surface Temperature Sensor Voltage, CCM pin B02 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor Surface Temperature Sensor
CCM	B150614	Inside Temperature Sensor Surface Temperature Sensor Voltage < 0.97V	Circuit Short To Ground or Open	Inside Temperature Sensor Surface Temperature Sensor reporting unexpected values	-CCM pin B02 short to ground -CCM pin B02 open circuit -Connector damaged	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor Surface Temperature Sensor
CCM	B150714	Inside Temperature Sensor Sun Sensor voltage < 0.97V	Circuit Short To Ground or Open	Inside Temperature Sensor Sun Sensor reporting unexpected values	-CCM pin	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor Sun Sensor
CCM	B150811	Inside Temperature Sensor supply voltage < 4.0V	Circuit Short To Ground	Inside Temperature Sensor Sun Sensor reporting unexpected values	CCM pin B10 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor 5V supply
CCM	B150812	Inside Temperature Sensor supply voltage > 6.0 V	Circuit Short To Battery	Inside Temperature Sensor general reduced functionality	CCM pin B01 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Temperatur Sensor (ITOS) properly attached and not damaged	Inside Temperature Sensor 5V supply
CCM	U114149	Climate control moduel hardware failure detected	Internal Electronic Failure	No Communication on LinBus. No communication between CCM and Slave nodes connected on LinBus	Climate control module damaged	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	LIN10 Communication Net
CCM	U114188	Bus off state detected	Bus off	No communication LIN10 General reduced/no system functionality on: - Parking Heater - Climate Control Panel - Defog Sensor	- Short to Ground on CCM pin A14 OR Connector D, pin X:4 OR X:11 OR X:17 - Connecotor damaged - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Check Defog, Parking Heater & Climate Control Panel (CCP) are connected to CCM	LIN10 Communication Net
CCM	B150D16	Climate control panel Supply voltage < 7V	Circuit Voltage Below Threshold	Climate Control Panel general reduced/no functionality	-Drained battery -Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check Climate Control Panel power supply	Climate Control Panel
CCM	B150D17	Climate control panel supply voltage > 20V	Circuit Voltage Above Threshold	Climate Control Panel general reduced/none functionality	Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Climate Control Panel
CCM	B150D44	Climate control panel internal memory volatile error	Data Memory Failure	General reduced functionality on Climate Control Panel	Climate control panel corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Control Panel HW	Climate Control Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B150D45	Climate control panel internal ROM is corrupted	Program Memory Failure	General reduced functionality on Climate Control Panel	Climate control panel corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Control Panel HW	Climate Control Panel
CCM	B150D46	Climate control panel EEPROM memory corrupted.	Calibration / Parameter Memory Failure	General reduced functionality on Climate Control Panel	Climate control panel corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2.Replace Climate Control Panel HW	Climate Control Panel
CCM	B150D49	Climate control panel hardware fault	Internal Electronic Failure	General reduced functionality on Climate Control Panel	Climate control panel corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Control Panel (CCP) HW	Climate Control Panel
CCM	B150D86	Climate control panel checksum or communication error detected	Signal Invalid	No Message received from Climate Control Panel. No Other Signal DTC can be reported.	-Communication error beteween Climate control panel and Climate control unit -Climate control panel corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Control Panel HW	Climate Control Panel
CCM	B150D87	Climate control panel malfunction	Missing Message	No Message received from Climate Control Panel No other Signal DTC can be reported	-Climate control panel is not answering for 6 frames -Climate control panel is not connected to LinBus or power, Climate control unit pin A14/A05 -Climate control panel defective -SC to GND A14	Diagnostic Service 14. No additional action is needed.	Check Climate Control Panel (CCP) is properly attached	Climate Control Panel
CCM	B150D94	Unexpected operation by Climate control panel detected	Unexpected Operation	General reduced functionality on Climate Control Panel	-Unexpected reset due to defective climate control panel -Watchdog failure -Climate control panel damaged	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Control Panel (CCP) HW	Climate Control Panel
CCM	B150E49	Wrong signal value to Climate control unit	Internal Electronic Failure	General reduced functionality of Defog Sensor	Defog sensor corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Climate Defog Sensor HW	Defog Sensor
CCM	B150E86	CCM detects signal error from Defog Sensor	Signal Invalid	No Message is received from Defog Sensor No Other Signal DTC can be reported	-Checksum error -Communication error between Climate control unit and Defog sensor -Defog sensor corrupted	Diagnostic Service 14. No additional action is needed.	Check Defog Sensor wiring not damaged & properly attached	Defog Sensor
CCM	B150E87	No response from Defog Sensor	Missing Message	No Message is received from Defog Sensor No other Signal DTC can be reported	-Defog sensor is not answering for 6 frames -Defog sensor is not connected to LinBus or power supply -Defog sensor is not connected to climate control module pin A14/A05 -SC to GND A14	Diagnostic Service 14. No additional action is needed.	Check Defog Sensor is properly attached	Defog Sensor
CCM	B150E96	Internal Defog sensor failure detected	Component Internal Failure	General reduced functionality of Defog Sensor	Defog sensor corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Defog Sensor HW	Defog Sensor
CCM	U003788	Bus off state	Bus off	CAN bus communication is disdurbed/interrupted: General reduced functionality of CCM	-Climate control unit pin A03/A02 short to ground -Open circuit on Climate control unit pin A03/A02 -Connecotor damaged -Transciever damaged	Diagnostic Service 14. No additional action is needed.	1. Check CAN-buss wires and connectors 2. Check CCM wiring & Connectors	CAN Communication Cab Net
CCM	U300041	Parameters checksum not valid.	General Checksum Failure	General reduced functionality of the ECU	ECU configuration parameters corrupted.	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	U300044	ECU volatile memory error	Data Memory Failure	General reduced functionality of the ECU	ECU corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	Electronic control unit
CCM	U300045	ECU ROM is corrupted	Program Memory Failure	General reduced functionality of the ECU	ECU corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	Electronic control unit
CCM	U300046	ECU EEPROM is corrupted.	Calibration / Parameter Memory Failure	General reduced functionality of the ECU	ECU corrupted.	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	Electronic control unit
CCM	U300049	ECU internal hardware fault - CPU clock failure	Internal Electronic Failure	CCM not operational	ECU corrupted.	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace CCM HW	Electronic control unit
CCM	U300054	stepper motor calibration completed unsuccessfully	Missing Calibration	Ventilated areas and heating/cooling might be incorrect	Started calibration not finished OR calibration not started when supposed to.	Diagnostic Service 14. No additional action is needed.	1. Restart calibration of HVAC stepper motors 2. Check stepper motors not malfunctioning	Electronic control unit
CCM	U300316	Supply voltage < 16V	Circuit Voltage Below Threshold	General reduced functionality of ECU	-Drained battery -Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check CCM Supply wiring and connectors not damaged & properly attached	Battery Power
CCM	U300317	Supply voltage > 32V	Circuit Voltage Above Threshold	General reduced functionality of ECU	Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check CCM Supply wiring and connectors not damaged & properly attached	Battery Power
CCM	U1FFF68	DTC NOT SUPPORTED IN SW: CANNOT BE SET	Event Information	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	Factory Mode Active
CCM	U024388	Bus off state	Bus off	- No communication on CAN possible - General reduced system functionality	-CCM pin A01/A04 short to ground -CCM pin A01/A04 open circuit -Connecotor damaged -Transciever damaged	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check CCM wiring & Connectors	CAN Communication Climate Subnet
CCM	P053011	A/C Refrigerant High Pressure Sensor detects sensor signal < 0.25V	Circuit Short To Ground	A/C Refrigerant High Pressure Sensor values unexpected	A/C Refrigerant High Pressure Sensor supply short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Pressure Sensor	A/C Refrigerant High Pressure Sensor Circuit
CCM	P053012	A/C Refrigerant High Pressure Sensor detects sensor signal > 4.75V	Circuit Short To Battery	A/C Refrigerant High Pressure Sensor values unexpected	A/C Refrigerant High Pressure Sensor Circuit supply short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Pressure Sensor	A/C Refrigerant High Pressure Sensor Circuit
CCM	B151E16	EACC voltage supply < 60V	Circuit Voltage Below Threshold	General reduced functionality of ECU	-Drained battery -Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check supply cabels not damaged and properly attached 2. Check power supply system	Electric AC Compressor High Voltage Supply
CCM	B151E17	EACC voltage supply > 218V	Circuit Voltage Above Threshold	General reduced functionality of ECU	Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Electric AC Compressor High Voltage Supply
CCM	B151E19	High Voltage current > 175A	Circuit Current Above Threshold	General reduced functionality of ECU	-Internal EACC shortcut to lower voltage -Current sensor (IGBT) malfunction	Diagnostic Service 14. No additional action is needed.	Check EACC supply wire is properly attached	Electric AC Compressor High Voltage Supply
CCM	B151E1D	Deviating current flow detected	Circuit Current Out of Range	General reduced functionality of ECU	Internal componenet error. Occurance not meassurable with AM Tool -Current sensor values out of range -One or more of the phases in the Electric AC Compressor motor is broken	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Check if Phases in the Electric AC Compressor motor is broken	Electric AC Compressor High Voltage Supply
CCM	B151E49	Deviating voltage detected	Internal Electronic Failure	General reduced functionality of ECU	-Voltage sensor values out of range -EACC defective	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace EACC HW	Electric AC Compressor High Voltage Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151F1C	12V power supply is out of range, 8,5V > supply voltage > 16V	Circuit Voltage Out of Range	General reduced functionality of ECU	Shortcut to lower or higher voltage	Diagnostic Service 14. No additional action is needed.	Check EACC supply wire and connetor is not damaged & properly attached	Electric AC Compressor
CCM	B151F1E	Deviating temperature or abnormal temperature detected	Circuit Resistance Out of Range	General reduced functionality of ECU	Internal ECU faliture. Occurance not meassurable with AM Tool. -Temperature sensor values out of range	Diagnostic Service 14. No additional action is needed.	1. Check PCCU environment 2. Restart SW 3. Replace PCCU HW if fault is still present when environment temperature is in nomral range	Electric AC Compressor
CCM	B151F49	Internal electronic failure EACC	Internal Electronic Failure	General reduced functionality of ECU	-Internal shortcut to lower or higher voltage -EACC defective	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace EACC HW	Electric AC Compressor
CCM	B151F4B	Temperature is to high, >125 °C	Over Temperature	General reduced functionality of ECU	-High load on EACC -High ambient temperature	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral conditions	Electric AC Compressor
CCM	B151F86	Communication problem	Signal Invalid	General reduced functionality of ECU	-Signal is not in its valid range -EACC corrupted -Parking Cooler Control Unit corrupted	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Check if PCCU or EACC has faulty HW	Electric AC Compressor
CCM	B151F97	-Excessive load -Loss of synchronism -Incorrect compressor start	Component or System Operation Obstructed or Blocked	General reduced functionality of ECU	-Blocked compressor -Low lubrication -Induced voltage was abnormal at Sensor-less Drive -Unsuccessfull shift between drive modes	Diagnostic Service 14. No additional action is needed.	Check if Blocked compressor and Low lubrication on EACC	Electric AC Compressor
CCM	B150F11	Low voltage present when not expected OR discharge temp <-40	Circuit Short To Ground	Unexpected values reported from Air Discharge Temperature Sensor	Wiring harness, CCM pin B16 Short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Air Discharge Sensor	Air Discharge Temperature Sensor
CCM	B150F15	Short to battery or open circuit conditions detected	Circuit Short To Battery or Open	Unexpected values reported from Air Discharge Temperature Sensor	Wiring harness, CCM pin B16 is short to battery Or open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Air Discharge Sensor	Air Discharge Temperature Sensor
CCM	B150911	Low voltage present when not expected	Circuit Short To Ground	Air Quality Sensor not working	Wiring harness, CCM pin A08 OR A14 short to ground	Diagnostic Service 14. No additional action is needed.	1. Check if additional DTC B151911 is set. Due to CCM HW restrictions within CCM B151911 may cause B150911 to be set. - Clear DTC with diagnostic service 14 - Restart truck - Read if B150911 is still set 2. Check wiring and connectors between CCM & Air Quality Sensor (AQS)	Air Quality Sensor 12V Supply Circuit
CCM	B150A12	Voltage present when not expected OR Voltage >12V	Circuit Short To Battery	Unexpected values reported from Air Quality Sensor	- Wiring harness, CCM pin A08 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Air Quality Sensor (AQS)	Air Quality Sensor Ground Circuit
CCM	B150A14	Short to ground or open circuit conditions detected	Circuit Short To Ground or Open	Unexpected values reported from Air Quality Sensor	Wiring harness, CCM pin A08 short to ground Or open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Air Quality Sensor (AQS)	Air Quality Sensor Ground Circuit
CCM	B150B03	CCM detects signal error from AQS Sensor.	FM (Frequency Modulated) / PWM (Pulse Width Modulated) Failures	Unexpected values reported from Air Quality Sensor	CCM pin B07 has the status - Short to Ground - Short to Supply - Open Circuit Internal error in AQS.	Diagnostic Service 14. No additional action is needed.	1. Check if there is displayed only fault code B150A14: If yes, wiring issue of supply (GND) 2. Check if there is displayed fault code B150B03 and not B150A14: No wiring issue of supply (GND), could be wiring issue of signal line or sensor itself	Air Quality Sensor Control Signal

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151811	A/C Refrigerant High Pressure Sensor detects sensor signal < 0.25V	Circuit Short To Ground	Unexpected values reported from Pressure Sensor	Wiring harness, CCM pin A13 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Pressure Sensor	A/C Refrigerant High Pressure Sensor 5V Supply
CCM	B151812	A/C Refrigerant High Pressure Sensor detects sensor signal > 4.75V	Circuit Short To Battery	Unexpected values reported from Pressure Sensor	Wiring harness, CCM pin A13 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Pressure Sensor	A/C Refrigerant High Pressure Sensor 5V Supply
CCM	B151011	Low voltage present when not expected	Circuit Short To Ground	Blower Control not working	Wiring harness, CCM pin A16 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & Blower Control	Blower Control
CCM	B151012	CCM detects voltage above normal range OR voltage present when not expected.	Circuit Short To Battery	Blower Control not working	Wiring harness, CCM pin A16 short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Blower Control	Blower Control
CCM	B151112	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Defrost & Demist Stepper Motor general reduced functionality	- Wiring harness, CCM pin A09 OR pin B17-B21 is short to battery OR -Defrost steeper motor defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Defrost & Demist Stepper Motor Circuit
CCM	B151114	- set condition 10 times error recognize - Failure will only handled if not outside of system voltage range - debouncing: 2 s => 10 x 2 s = 20 s - DTC call once at every check	Circuit Short To Ground or Open	Defrost & Demist Stepper Motor general reduced functionality	1. Wiring harness, component or ECU 2. Stepper defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Defrost & Demist Stepper Motor Circuit
CCM	B151212	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Recirculation Stepper Motor general reduced functionality	- Wiring harness, CCM pin A09 OR B11-B15 is short to battery OR - REC/OSA stepper motor is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Recirculation Stepper Motor Circuit
CCM	B151214	CCM detects Short to ground or open circuit conditions	Circuit Short To Ground or Open	Recirculation Stepper Motor general reduced functionality	- Wiring harness, CCM oin A09 OR B11-B15 is short to ground or has open circuit OR - REC/OSA steppper motor defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Recirculation Stepper Motor Circuit
CCM	B151312	CCM detects voltage above normal volatage OR Voltage present when not expected	Circuit Short To Battery	Temperature Door Stepper Motor general reduced functionality	- Wiring harness, CCM pin A06 OR B17-B21 is short to battery OR - Air mix stepper motor is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Temperature Door Stepper Motor Circuit
CCM	B151314	CCM detects Short to ground or open circuit conditions	Circuit Short To Ground or Open	Temperature Door Stepper Motor general reduced functionality	- Wiring harness, CCM oin A06 OR B17-B21 is short to ground or has open circuit OR - Air mix steppper motor defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Temperature Door Stepper Motor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151412	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Water Valve Stepper Motor general reduced functionality	- Wiring harness, CCM pin A06 OR B11-B15 is short to battery OR - Watervalve stepper motor is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Water Valve Stepper Motor Circuit
CCM	B151414	CCM detects Short to ground or open circuit conditions	Circuit Short To Ground or Open	Water Valve Stepper Motor general reduced functionality	- Wiring harness, CCM pin A06 OR B11-B15 is short to ground or has open circuit OR - Watervalve steppper motor defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Water Valve Stepper Motor Circuit
CCM	B151712	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Floor & Ventilation Stepper Motor general reduced functionality	- Wiring harness, CCM pin A09 OR B05-B09 is short to battery OR - Air distrubution stepper motor(s) is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Floor & Ventilation Stepper Motor Circuit
CCM	B151714	CCM detects Short to ground or open circuit conditions	Circuit Short To Ground or Open	Floor & Ventilation Stepper Motor general reduced functionality	- Wiring harness, CCM pin A09 OR B05-B09 is short to ground or has open circuit OR - Air distrubution steppper motor(s) is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Floor & Ventilation Stepper Motor Circuit
CCM	P053511	CCM detects voltage below normal voltage OR low voltage present when not expected	Circuit Short To Ground	A/C Evaporator Temperature Sensor unexpected values	Wiring harness, CCM pin B19 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & Evaporator Sensor	A/C Evaporator Temperature Sensor Circuit
CCM	P053515	CCM detects Short to battery or open circuit conditions	Circuit Short To Battery or Open	A/C Evaporator Temperature Sensor unexpected values	Wiring harness, CCM pinB19 is short to battery or open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring on CCM & Evaporator Sensor	A/C Evaporator Temperature Sensor Circuit
CCM	B151911	CCM detects voltage below normal voltage OR low voltage present when not expected	Circuit Short To Ground	Climate Control Panel / Defog Sensor not working	Wiring harness, CCM pin A14 is short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM, Defog sensor & CCP	Panel / Defog Sensor Power Supply Circuit
CCM	B151912	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Climate Control Panel / Defog Sensor general reduced functionality	Wiring harness, CCM pin A14 is short to battery	Diagnostic Service 14. No additional action is needed.	Check wiring on CCM, Defog sensor & CCP	Panel / Defog Sensor Power Supply Circuit
CCM	B152212	CCM detects voltage above normal range OR voltage present when not expected	Circuit Short To Battery	Air Distribution Stepper general reduced functionality	- Wiring harness, CCM pin A09 OR B05-B09 is short to battery OR - Air distrubution stepper motor is defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Air Distribution Stepper Circuit
CCM	B152214	CCM detects Short to ground or open circuit conditions	Circuit Short To Ground or Open	Air Distribution Stepper general reduced functionality	- Wiring harness, CCM pin A09 OR B05-B09 is short to ground or has open circuit OR - Air distrubution steppper motor defect	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Air Distribution Stepper Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B150C16	Parking heater supply voltage < 21.5V	Circuit Voltage Below Threshold	General reduced functionality of Parking Heater	-Drained battery -Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check Parking Heater power supply system	Parking Heater Module
CCM	B150C17	Parking heater supply voltage > 30V	Circuit Voltage Above Threshold	General reduced functionality of Parking Heater	Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Parking Heater Module
CCM	B150C49	Internal hardware failure detected	Internal Electronic Failure	General reduced functionality of Parking Heater	- Parking heater hardware is defective - Temperature sensor defective - Air inlet defective	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace Parking Heater HW	Parking Heater Module
CCM	B150C4B	Air heater above normal operation temperature	Over Temperature	General reduced functionality of Parking Heater	Heating air system (inlets, ducts, outlets) restricted or blocked.	Diagnostic Service 14. No additional action is needed.	Check Heating air system (inlets, ducts, outlets) restricted are not blocked	Parking Heater Module
CCM	B150C86	Communication error between parking heater and CCM	Signal Invalid	- No Message is received from Parking Heater - No Other Signal DTC can be reported	-CCM pin A14/A05 communication error -Parking heater corrupted	Diagnostic Service 14. No additional action is needed.	Check Parking heater wiring not damaged & properly attached	Parking Heater Module
CCM	B150C87	Parking heater not answering	Missing Message	- No Message is received from Parking Heater. - No Other Signal DTC can be reported	-Parking Heater is not answering for 6 frames -Parking heater is not connected to CCM pin A05/A14 -Connectors A05/A14 defective -SC to GND A14	Diagnostic Service 14. No additional action is needed.	Check Parking Heater is properly attached	Parking Heater Module
CCM	B150C93	Heater does not start on first and second trial	No Operation	General reduced functionality of Parking Heater	-Fuel line empty -Fuel line damaged -Combustion air supply system restricted -Exhaust system restricted -Burner worn out	Diagnostic Service 14. No additional action is needed.	Check Fuel line, Combustion air supply system, Exhaust system and if Burner worn out	Parking Heater Module
CCM	B150C97	Repeated parking heater falitures has occurred	Component or System Operation Obstructed or Blocked	General reduced functionality of Parking Heater	-Overfeeding of fuel -Heater is locked -Abnormal fuel provision -Air supply system defective -Exhaust system defective -Burner defective	Diagnostic Service 14. No additional action is needed.	Check if overfeeding fuel and if Heater is blocked	Parking Heater Module
CCM	U114311	Low voltage present when not expected	Circuit Short To Ground	Parking Heater Burner Motor not working	-Parking heater burner motor supply short to ground -Parking heater burning motor blocked	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Parking Heater Burner Motor
CCM	U114313	Open circuit conditions detected	Circuit Open	Parking Heater Burner Motor not working	-Parking heater burner motor supply short to ground -Parking heater burner motor is broken	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM and HVAC	Parking Heater Burner Motor
CCM	U114511	Low voltage present when not expected	Circuit Short To Ground	Parking Heater Dosing Pump not working	-Supply cable of dosing pump is short to ground -Parking heater dosing Pump is broken	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM and HVAC	Parking Heater Dosing Pump
CCM	U114513	Open circuit conditions detected	Circuit Open	Parking Heater Dosing Pump not working	-Supply cable of the dosing pump is open circuit -Parking heater dosing pump is broken	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM and HVAC	Parking Heater Dosing Pump

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	U114611	Low voltage present when not expected	Circuit Short To Ground	Unexpected values from Parking Heater Water Temperature Sensor	-Parkig heater temperature sensor supply cable short to ground -Parking heater temperature sensor is defective	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Parking Heater Water Temperature Sensor
CCM	U114613	Open circuit conditions detected	Circuit Open	Unexpected values from Parking Heater Water Temperature Sensor	-Temperatur sensor cable open circuit -Temperature sensor is defective	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & HVAC	Parking Heater Water Temperature Sensor
CCM	U114711	Low voltage present when not expected	Circuit Short To Ground	Unexpected values from Parking Heater Air Outlet Temperature Sensor	-Parking heater outlet sensor cable short to ground -Parking heater air outlet temperature sensor damaged	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Parking Heater Air Outlet Temperature Sensor
CCM	U114713	Open circuit conditions detected	Circuit Open	Parking Heater Air Outlet Temperature Sensor unexpected values	-Parking heater outlet temperature sensor open circuit -Parking heater air outlet temperature sensor defective.	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM and HVAC	Parking Heater Air Outlet Temperature Sensor
CCM	U114811	Low voltage present when not expected	Circuit Short To Ground	Parking Heater Flame Detector not working	-Parking heater supply cable short to ground -Parking heater flame detector damaged	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM and HVAC	Parking Heater Flame Detector
CCM	U114813	Open circuit conditions detected	Circuit Open	Parking Heater Flame Detector not working	-Parking heater flame detector open circuit -Parking heater flame detector defective	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & HVAC	Parking Heater Flame Detector
CCM	U114893	Flame interrupt during heater operation and/or no successful restart	No Operation	Parking Heater Flame Detector unexpected behaviour	Flame interrupt during heater operation, no successful restart - Fuel line empty - Fuel line damaged - Combustion air supply system restricted - Exhaust system restricted - Burner worn out	Diagnostic Service 14. No additional action is needed.	Check Fuel line, Combustion air supply system, Exhaust system and if Burner worn out	Parking Heater Flame Detector
CCM	U114411	Low voltage present when not expected	Circuit Short To Ground	Parking Heater Glow Plug not working	Supply cable of the parking heater glow plug is short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Parking Heater Glow Plug
CCM	U114413	Open circuit conditions detected	Circuit Open	Parking Heater Glow Plug not working	-Parking heater glow plug supply cable Open Circuit -Parking heater glow plug defective	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & HVAC	Parking Heater Glow Plug
CCM	B151A86	PCCU detects communication error between PCCU and ElectricalAirConditionCompressor (EACC) on LIN17	Signal Invalid	General reduced functionality of ECU	- PCCU in failmode OR corrupted - EACC in failmode OR corrupted	Diagnostic Service 14. No additional action is needed.	Check wiring between EACC & PCCU is not damaged and corectly attached	LIN17 communication bus

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151A87	PCCU missing message from ElectricalAirConditionCompressor (EACC)	Missing Message	General reduced functionality of ECU	- Communication signal disturbances - Faulty LIN driver component EACC - EACC not activated - Incorrect commande was sent (sleep command) - EACC is still in boot mode (standby)	Diagnostic Service 14. No additional action is needed.	1. Restart ECU SW 2. Check EACC is properly connected 3. Replace ECU	LIN17 communication bus
CCM	B151A88	Bus off state for LIN17 between PCCU and ElectricalAirConditionCompressor (EACC)	Bus off	No CAN communication on between PCCU and EACC General reduced system functionality	- Communication signal disturbances - Faulty LIN driver component EACC	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check EACC & PCCU properly connected	LIN17 communication bus
CCM	B151B16	PCCU supply voltage < 20V	Circuit Voltage Below Threshold	General reduced functionality of ECU	- Hardware malfunctions within the PCCU - SC to lower voltage - Corrosion in LV connector - Current leakage, LV power supply - Internal fault ElectricalAirConditionCompressor	Diagnostic Service 14. No additional action is needed.	Check PCCU Supply wiring and connectors not damaged & properly attached	Parking Cooler Control Unit
CCM	B151B17	PCCU supply voltage > 32V	Circuit Voltage Above Threshold	General reduced functionality of ECU	- Hardware malfunctions within the PCCU - Shortcut to higher voltage - Open circuit LV ground wire (lost reference to ground)	Diagnostic Service 14. No additional action is needed.	Check PCCU Supply wiring and connectors not damaged & properly attached	Parking Cooler Control Unit
CCM	B151B49	Internal HW failure detected within PCCU	Internal Electronic Failure	General reduced functionality of ECU	Hardware malfunctions within the PCCU	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace PCCU HW	Parking Cooler Control Unit
CCM	B151B4B	PCCU internal temperature > 80°C	Over Temperature	General reduced functionality of ECU	- ElectricalAirConditionCompressor on High torque load (-> high current consumption) - Faulty temperature sensor (drifting offset) - High ambient temperature - Soiled PCCU (covered with heat insulating dirt) - Bad ambient cooling performance	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Parking Cooler Control Unit
CCM	B151B64	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	Signal Plausibility Failure	DTC NOT SUPPORTED IN SW: CANNOT BE SET	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	Parking Cooler Control Unit
CCM	B151C11	PCCU detects condenser fans output SC to ground	Circuit Short To Ground	General reduced functionality of ECU	- Power supply wire to Condenser fan shorted to ground - Condensor fan motor is broken (internal short to ground) - PCCU fan power supply (internal) is shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wiring on PCCU not damaged and properly attached	Parking Cooler Control Unit Condenser Fans Output

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151C12	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	Circuit Short To Battery	DTC NOT SUPPORTED IN SW: CANNOT BE SET	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	Parking Cooler Control Unit Condenser Fans Output
CCM	B151C13	PCCU detects open circuit conditions	Circuit Open	General reduced functionality of ECU	- Power supply wire to Condenser fan broken - Ground supply wire to Condenser fan broken - Condensor fan motor is broken (internal open circuit) - PCCU fan power supply broken (internal open circuit)	Diagnostic Service 14. No additional action is needed.	Check wiring on PCCU not damaged and properly attached	Parking Cooler Control Unit Condenser Fans Output
CCM	B151C19	PCCU current > 12A	Circuit Current Above Threshold	General reduced functionality of ECU	- Mechanical obstacles - Dirty fan blades - Power supply wire to Condenser fan SC to lower voltage - Condensor fan motor is shorted to lower voltage - PCCU fan power supply (internal) is shorted to lower voltage	Diagnostic Service 14. No additional action is needed.	1. Check PCCU fan not blocked 2. Check wiring properly attached	Parking Cooler Control Unit Condenser Fans Output
CCM	B151C1C	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	Circuit Voltage Out of Range	DTC NOT SUPPORTED IN SW: CANNOT BE SET	FAULT CODE REMOVED FROM SOFTWARE: DTC CAN NEVER BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	DTC NOT SUPPORTED IN SW: CANNOT BE SET	Parking Cooler Control Unit Condenser Fans Output
CCM	U114949	PCCU detects internal HW failure	Internal Electronic Failure	General reduced functionality of ECU	PCCU LIN HW failure	Diagnostic Service 14. No additional action is needed.	1. Restart SW 2. Replace PCCU HW	LIN17 Communication Net
CCM	B151D13	PCCU detects open circuit conditions for itself OR ElectricalAirConditionCompressor (EACC)	Circuit Open	General reduced functionality of ECU	- PCCU high voltage supply wire to EACC broken - PCCU high voltage ground wire to EACC broken - PCCU high voltage supply connection broken (internal) - EACC high voltage supply connection broken (internal) - HVIL loop broken	Diagnostic Service 14. No additional action is needed.	Check wiring on PCCU not damaged and properly attached	Parking Cooler Control Unit High Voltage Output
CCM	B151D16	PCCU supply voltage < 105 V	Circuit Voltage Below Threshold	General reduced functionality of ECU	- HV supply wire to ElectricalAirConditionCompressor (EACC) shorted to lower voltage - PCCU HV supply (internal) is shorted to lower voltage - EACC HV supply (internal) is shorted to lower voltage	Diagnostic Service 14. No additional action is needed.	Check PCCU & EACC Supply wiring and connectors not damaged & properly attached	Parking Cooler Control Unit High Voltage Output
CCM	B151D17	PCCU supply voltage > 170V	Circuit Voltage Above Threshold	General reduced functionality of ECU	- No ElectricalAirConditionCompressor (EACC) load - Hardware malfunctions within the PCCU	Diagnostic Service 14. No additional action is needed.	Check PCCU Supply wiring and connectors not damaged & properly attached	Parking Cooler Control Unit High Voltage Output

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CCM	B151D19	PCCU current in High Voltage supply to ElectricalAirConditionCompressor (EACC) > 4,7A	Circuit Current Above Threshold	General reduced functionality of ECU	- High load on EACC - Leak currents - Mechanical obstacles EACC - Internal fault PCCU	Diagnostic Service 14. No additional action is needed.	Check PCCU wiring properly attached and connectors not damaged	Parking Cooler Control Unit High Voltage Output
CCM	B152019	PCCU detects to high internal current	Circuit Current Above Threshold	General reduced functionality of ECU	Fault caused by current leakage. Level event (current) of fault code set not meassurable. - High Voltage cable shafted or damaged - Shortcircuit inside High Voltage connectors (HV+ and HV-) - Internal short circuit (inside PCCU or ElectricalAir ConditionCompressor)	Diagnostic Service 14. No additional action is needed.	Check PCCU wiring properly attached and connectors not damaged	Parking Cooler Control Unit High Voltage Isolation
CCM	B152113	PCCU detects open circuit conditions on itself OR on ElectricalAirConditionCompressor (EACC) for High Voltage supply	Circuit Open	General reduced functionality of ECU	HV connectors (PCCU or EACC) disconnected HVIL wires are cut	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between PCCU & EACC	Parking Cooler Control Unit High Voltage Interlock Loop
CCM	U114211	Low voltage present when not expected	Circuit Short To Ground	Parking Heater Water Pump not working	wiring harness, CCM pin A14/A05 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors on CCM & HVAC	Parking Heater Water Pump
CCM	U114213	Open circuit conditions detected	Circuit Open	No Functionallity change at CCM side.	Wiring harness, CCM pin A14/A05 open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring and connectors between CCM & HVAC	Parking Heater Water Pump
CCM	B150311	Left Side Sun Sensor supply voltage < 0.2V	Circuit Short To Ground	Left Side Sun Sensor unexpected values	CCM pin A10 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Left Side Sun Sensor properly attached and not damaged	Left Side Sun Sensor
CCM	B150315	Short to battery or open circuit conditions detected	Circuit Short To Battery or Open	Left Side Sun Sensor unexpected values	CCM pin A10 short to battery or open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Left Side Sun Sensor properly attached and not damaged	Left Side Sun Sensor
CCM	B150411	Right side sun sensor supply voltage < 0.2V	Circuit Short To Ground	Right Side Sun Sensor unexpected values	CCM pin A07 short to ground	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Right Side Sun Sensor properly attached and not damaged	Right Side Sun Sensor
CCM	B150415	Short to battery or open circuit conditions detected	Circuit Short To Battery or Open	Right Side Sun Sensor unexpected values	CCM pin A07 short to battery or open circuit	Diagnostic Service 14. No additional action is needed.	Check wiring between CCM & Right Side Sun Sensor properly attached and not damaged	Right Side Sun Sensor
CIOM	B100111	Input stay connected to GND more than defined time (P1DWC).	Circuit Short To Ground	Auxiliary Horn function switched off until the low voltage state exit (or button release).	Harness issue, button unexpected usage: press too long.	Diagnostic Service 14. No additional action is needed.	- Inspect the switch - Inspect the harness and connectors	Auxiliary Horn Switch Circuit
CIOM	B100112	Resident voltage level in circuit when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the switch - Inspect the harness and connectors	Auxiliary Horn Switch Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150C63	Switch stay in push state more than allowed time: depends of switch.	Circuit / Component Protection Time-Out	Axle load request deactivated and button press is not processed.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	- Inspect the switch	Axle Load Distribution Button
CIOM	U114288	Communication off on LIN bus.	Bus off	ECU invalidate related LIN signals.	1. Harness issue: short circuit to GND/BAT or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors	LIN1 Communication Net
CIOM	U114388	Communication off on LIN bus.	Bus off	ECU invalidate related signals.	1. Harness issue: short circuit to GND/BAT or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors	LIN2 communication Net
CIOM	U114488	Communication off on LIN bus.	Bus off	ECU invalidate related signals.	1. Harness issue: short circuit to GND/BAT or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors	LIN3 Communication Net
CIOM	U114588	Communication off on LIN bus.	Bus off	ECU invalidate related signals.	1. Harness issue: short circuit to GND/BAT or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors	LIN4 Communication Net
CIOM	U114688	Communication off on LIN bus.	Bus off	ECU invalidate related signals.	1. Harness issue: short circuit to GND/BAT or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	Inspect the harness or connectors	LIN5 Communication Net
CIOM	U114E86	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	Inspect the Flexible Switch Panel	Flexible Switch Panel Link
CIOM	U114E87	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Flexible Switch Panel - Inspect the harness	Flexible Switch Panel Link
CIOM	U114786	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Cruise Control Free Wheel	Cruise Control Free Wheel Module Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U114787	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Cruise Control Free Wheel - Inspect the harness	Cruise Control Free Wheel Module Link
CIOM	U114886	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Exterior Light Control Panel	Exterior Light Control Panel Link
CIOM	U114887	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Exterior Light Control Panel - Inspect the harness	Exterior Light Control Panel Link
CIOM	U114986	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Interior Light Control Panel	Interior Light Control Panel Link
CIOM	U114987	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Interior Light Control Panel - Inspect the harness	Interior Light Control Panel Link
CIOM	U114A86	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Remote Control ECS Module	Remote Control ECS Module Link
CIOM	U114A87	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Remote Control ECS Module - Inspect the harness	Remote Control ECS Module Link
CIOM	U114B86	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the High Living Electronics Control Module	High Living Electronics Control Module Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U114B87	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the High Living Electronics Control Module - Inspect the harness	High Living Electronics Control Module Link
CIOM	U114C86	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Enviroment Control Module Low	Living Enviroment Control Module Low Link
CIOM	U114C87	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Enviroment Control Module Low - Inspect the harness	Living Enviroment Control Module Low Link
CIOM	U114D86	Invalid frame received from LIN slave.	Signal Invalid	ECU invalidate related signals.	"LIN slave issue. Invalid parameters setting."	Diagnostic Service 14. No additional action is needed.	- Inspect the Diff Lock Free Wheel Module	Diff Lock Free Wheel Module Link
CIOM	U114D87	LIN slave not responding.	Missing Message	ECU invalidate related signals.	1. LIN slave issue. 2. Harness issue. 3. Invalid parameters setting. Note: Set together with B100313 (CIOM 12V power supply circuit - Circuit Open) indicates the global 12V power supply outputs issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Diff Lock Free Wheel Module - Inspect the harness	Diff Lock Free Wheel Module Link
CIOM	B150216	LIN slave measures power supply voltage lower than 7.5V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Cruise Control Free Wheel Panel - Inspect the harness	Cruise Control Free Wheel Panel
CIOM	B150217	LIN slave measures power supply voltage higher than 18.5V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Cruise Control Free Wheel Panel - Inspect the harness	Cruise Control Free Wheel Panel
CIOM	B150244	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Cruise Control Free Wheel Panel	Cruise Control Free Wheel Panel
CIOM	B150245	LIN slave ROM check fail (FCI reception).	Program Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Cruise Control Free Wheel Panel	Cruise Control Free Wheel Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150246	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Cruise Control Free Wheel Panel	Cruise Control Free Wheel Panel
CIOM	B150294	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Cruise Control Free Wheel Panel	Cruise Control Free Wheel Panel
CIOM	B150316	LIN slave measures power supply voltage lower than 7.5V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Diff Lock Free Wheel Panel - Inspect the harness	Diff Lock Free Wheel Panel
CIOM	B150317	LIN slave measures power supply voltage higher than 18.5V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Diff Lock Free Wheel Panel - Inspect the harness	Diff Lock Free Wheel Panel
CIOM	B150344	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Diff Lock Free Wheel Panel	Diff Lock Free Wheel Panel
CIOM	B150345	LIN slave ROM check fail (FCI reception).	Program Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Diff Lock Free Wheel Panel	Diff Lock Free Wheel Panel
CIOM	B150346	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Diff Lock Free Wheel Panel	Diff Lock Free Wheel Panel
CIOM	B150394	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Diff Lock Free Wheel Panel	Diff Lock Free Wheel Panel
CIOM	B150516	LIN slave measures power supply voltage lower than 7.5V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Interior Light Control Panel - Inspect the harness	Interior Light Control Panel
CIOM	B150517	LIN slave measures power supply voltage higher than 18.5V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Interior Light Control Panel - Inspect the harness	Interior Light Control Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150544	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Interior Light Control Panel	Interior Light Control Panel
CIOM	B150545	LIN slave ROM check fail (FCI reception).	Program Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Interior Light Control Panel	Interior Light Control Panel
CIOM	B150546	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Interior Light Control Panel	Interior Light Control Panel
CIOM	B150549	Push button scan is not configured correctly (FCI reception).	Internal Electronic Failure	LIN slave invalidates corresponding signals.	1. LIN slave EOL configuration not correct	Diagnostic Service 14. No additional action is needed.	- Replace the Interior Light Control Panel	Interior Light Control Panel
CIOM	B150594	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Interior Light Control Panel	Interior Light Control Panel
CIOM	B150916	Supply voltage measured by LIN slave < 7V±0.5.	Circuit Voltage Below Threshold	LIN slave behaviour: - All illumination will be turned off after 5s. - All switch status will be set to "Not pressed". - Untrustable signals:None	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Flexible Switch Panel - Inspect the harness	Flexible Switch Panel
CIOM	B150917	Supply voltage measured by LIN slave > 18V±0.5.	Circuit Voltage Above Threshold	LIN slave behaviour: - All illumination will be turned off immediately. - All switch status will be set to "Not pressed". - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Flexible Switch Panel - Inspect the harness	Flexible Switch Panel
CIOM	B150944	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Flexible Switch Panel	Flexible Switch Panel
CIOM	B150979	Read switch ID error or no response from flexible switch (FCI reception).	Mechanical Linkage Failure	LIN Master invalidates the LIN input signals corresponding to position of the switch in failure(set them to Error).	1. Flexible switch removed/disconnected; 2. Flexible switch failure: - wrong data in EEPROM - noise on the IIC-Bus - SCL has a short circuit to GND or VCC - SDA has a short circuit to GND or VCC - CS has a short circuit to GND or VCC	Diagnostic Service 14. No additional action is needed.	- Inspect connection between Switches and Panels	Flexible Switch Panel
CIOM	B150616	LIN slave measures power supply voltage lower than 7.5V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Exterior Light Control Panel - Inspect the harness	Exterior Light Control Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150617	LIN slave measures power supply voltage higher than 18.5V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: - Standby state. - Untrustable signals:None.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Exterior Light Control Panel - Inspect the harness	Exterior Light Control Panel
CIOM	B150644	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Exterior Light Control Panel	Exterior Light Control Panel
CIOM	B150645	LIN slave ROM check fail (FCI reception).	Program Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Exterior Light Control Panel	Exterior Light Control Panel
CIOM	B150646	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Exterior Light Control Panel	Exterior Light Control Panel
CIOM	B150649	Push button scan is not configured correctly (FCI reception).	Internal Electronic Failure	LIN slave invalidates corresponding signals.	1. LIN slave EOL configuration not correct.	Diagnostic Service 14. No additional action is needed.	- Replace the Exterior Light Control Panel	Exterior Light Control Panel
CIOM	B150694	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Exterior Light Control Panel	Exterior Light Control Panel
CIOM	B150816	LIN slave measures power supply voltage lower than 8.0V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: the fault is reported when the system has going back from high voltage state to normal. This means that when FCI is reported the unit is function correctly. In the low voltage state the LEDs are deactivated and the LIN communication is shut down.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Environment Control Module Low - Inspect the harness	Living Environment Control Module Low
CIOM	B150817	LIN slave measures power supply voltage higher than 18.0V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: the fault is reported when the system has going back from high voltage state to normal. This means that when FCI is reported the unit is function correctly. In the high voltage state the LEDs are deactivated and the LIN communication is shut down.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Environment Control Module Low - Inspect the harness	Living Environment Control Module Low
CIOM	B150846	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Living Environment Control Module Low	Living Environment Control Module Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150894	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Living Environment Control Module Low	Living Environment Control Module Low
CIOM	B150716	LIN slave measures power supply voltage lower than 7.5V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: turn of LIN communication, backlights and display.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Environment Control Module High - Inspect the harness	Living Environment Control Module High
CIOM	B150717	LIN slave measures power supply voltage higher than 19.0V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: turn of LIN communication, backlights and display.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Living Environment Control Module High - Inspect the harness	Living Environment Control Module High
CIOM	B150744	LIN slave RAM check fail (FCI reception).	Data Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue. 2. LIN slave RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	Replace the Living Environment Control Module High	Living Environment Control Module High
CIOM	B150745	LIN slave ROM check fail (FCI reception).	Program Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	Replace the Living Environment Control Module High	Living Environment Control Module High
CIOM	B150746	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	Replace the Living Environment Control Module High	Living Environment Control Module High
CIOM	B150749	- Button stuck - Failure to communicate (I2C) with LCD display. - Fail to read button statuses (SPI bus error) - LIN communication error	Internal Electronic Failure	LIN slave: invalidates corresponding signals.	- LCD display failure; - Button interface (internal) failure; - Button stuck; - LIN communication issue;	Diagnostic Service 14. No additional action is needed.	Replace the Living Environment Control Module High	Living Environment Control Module High
CIOM	B150794	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	Replace the Living Environment Control Module High	Living Environment Control Module High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B150416	LIN slave measures power supply voltage lower than 8.0V (FCI reception).	Circuit Voltage Below Threshold	LIN slave: the fault is reported when the system has going back from high voltage state to normal. This means that when FCI is reported the unit is function correctly. In the low voltage state the LEDs are deactivated and the LIN communication is shut down.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Wired Control ECS Panel - Inspect the harness	Wired Control ECS Panel
CIOM	B150417	LIN slave measures power supply voltage higher than 18.0V (FCI reception).	Circuit Voltage Above Threshold	LIN slave: the fault is reported when the system has going back from high voltage state to normal. This means that when FCI is reported the unit is function correctly. In the high voltage state the LEDs are deactivated and the LIN communication is shut down.	1. Harness issue; 2. CIOM 12V output failure; 3. Faulty LIN slave hardware that misdetect the supply level;	Diagnostic Service 14. No additional action is needed.	- Inspect the Wired Control ECS Panel - Inspect the harness	Wired Control ECS Panel
CIOM	B150446	LIN slave flash corruption detected during writing of calibration data (FCI reception).	Calibration / Parameter Memory Failure	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Wired Control ECS Panel	Wired Control ECS Panel
CIOM	B150494	LIN slave unexpected SW operation executed; LIN slave watchdog reset performed; LIN slave overrun situation detected;	Unexpected Operation	LIN Master invalidates the LIN input signals (set them to Error).	1. LIN slave SW issue; 2. LIN slave MCU not correct;	Diagnostic Service 14. No additional action is needed.	- Replace the Wired Control ECS Panel	Wired Control ECS Panel
CIOM	B100311	Output short-circuit to GND.	Circuit Short To Ground	Switch off 12V power. Periodically check if the short circuit has been solved (5s period). Functions activation not possible through LIN components (ILCP, ELCP CCFW, DLFW, RCECS, LECM).	"1. Harness issue; 2. LIN slave internal short circuit"	Diagnostic Service 14. No additional action is needed.	- Inspect the harness - Inspect LIN modules	ECU 12V power supply circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B100312	Output short-circuit to BAT+.	Circuit Short To Battery	Switch off 12V power. Periodically check if the short circuit has been solved (5s period). Functions activation not possible through LIN components (ILCP, ELCP CCFW, DLFW, RCECS, LECM). One or more functions of ECU do not perform as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness - Inspect LIN modules	ECU 12V power supply circuit
CIOM	B100313	Open circuit conditions detected on all 4 output pins. Threshold for open circuit detection: <1mA external current at output voltages between 10V and 14V.	Circuit Open	One or more functions of ECU do not perform as expected. Functions activation not possible through LIN components (ILCP, ELCP CCFW, DLFW, FSP, RCECS, LECM).	1. Harness issue, most probably close to CIOM connectors.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	ECU 12V power supply circuit
CIOM	B100316	Feedback voltage measured on the output is lower than 11,4V threshold (configurable).	Circuit Voltage Below Threshold	One or more functions of ECU may not work as expected.	"1. Harness issue; 2. CIOM internal HW failure; 3. LIN slave internal HW failure."	Diagnostic Service 14. No additional action is needed.	- Inspect the harness - Inspect LIN modules	ECU 12V power supply circuit
CIOM	B100317	Feedback voltage measured on the output is higher 14V threshold (configurable).	Circuit Voltage Above Threshold	One or more functions of ECU may not work as expected.	"1. Harness issue; 2. CIOM internal HW failure;"	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	ECU 12V power supply circuit
CIOM	B100211	Input stay connected to GND more than defined time (P1DWB).	Circuit Short To Ground	City Horn function switched off until the low voltage state exit (or button release).	Harness issue, button unexpected usage: press too long.	Diagnostic Service 14. No additional action is needed.	- Inspect the City Horn Switch - Inspect the harness	City Horn Switch
CIOM	B100212	Resident voltage level in circuit when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the City Horn Switch - Inspect the harness	City Horn Switch
CIOM	U000188	CAN failure detected on BB2;	Bus off	Signals invalidated by com layer accordingly to com database deadline settings. Functions processed accordingly to fallback modes.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to BB2.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	CAN Communication Backbone 2 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U001088	CAN failure detected BB1.	Bus off	Signals invalidated by com layer accordingly to com database deadline settings. Functions processed accordingly to fallback modes.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to BB1.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	CAN Communication Backbone 1 Net
CIOM	U003788	CAN failure detected Cab subnet.	Bus off	Signals invalidated by com layer accordingly to com database deadline settings. Functions processed accordingly to fallback modes.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to Cab subnet.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	CAN Communication Cab Net
CIOM	U005588	CAN failure detected on Security Subnet.	Bus off	Signals invalidated by com layer accordingly to com database deadline settings. Functions processed accordingly to fallback modes.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to Security subnet.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	CAN Communication Security Net
CIOM	U200388	CAN failure detected on FMS subnet.	Bus off	Signals invalidated by com layer accordingly to com database deadline settings. Functions processed accordingly to fallback modes.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to FMS subnet.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness	CAN Communication FMS Net
CIOM	U300005	Not supported SW configuration: invalid parameters configuration settings.	System Programming Failures	General reduced functionality of the ECU. One or more functions of ECU do not perform as expected, but return automatically to normal operation after failure is removed (mainly the function concerned by failure will be switched off, e.g. not available signals sent on output).	1. SW release issue; 2. Not supported truck configuration build: - Invalid ECS functions HMI settings configured; - Invalid DiffLock function HMI settings configured; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Contact back office	Electronic control unit
CIOM	U300041	Faulty MSW or/and DST downloaded.	General Checksum Failure	MSW invalidationed: DTC reported by PBL.	1. Invalid SW downloaded; 2. Invalid HW used; SW or HW complex system issue, to be rised to back-office;	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U300044	RAM corruption detected.	Data Memory Failure	ECU reset.	1. Invalid SW processing; 2. RAM corruption (intermittent fault due to working environment); 3. RAM physical issue (continuous defect); SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U300045	MSW memory corruption.	Program Memory Failure	MSW invalidation after ECU reset.	1. SWDL invalid process (cut of power during SWDL, etc) 2. Flash memory physical defect; 3. Invalid SW operation; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U300046	Dataset memory corruption.	Calibration / Parameter Memory Failure	MSW invalidation after ECU reset.	1. Incorrect DST download/change (e.g. cut of power during change). 2. Invalid SW operation (mainly in case of parameters change processing); 3. Flash memory physical defect; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U300049	Internal electronic failure detected.	Internal Electronic Failure	One or more functions of ECU are not available/not processed in expected way.	Internal electronic failure: - µC peripheric circuit invalid state; - LIN tranceiver circuit broken; - CAN tranceiver circuit broken; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U30004A	Not supported SW configuration downloaded in HW.	Incorrect Component Installed	Keyfob/FMS functions inactive in case of invalid HW(low HW version instead of high HW version).	1. Not correct component installed or not valid SW configuration downloaded: - Remote keyless entry function activation in not supported HW version; - FMS gateway activation in not supported HW version. 2. Not correct or not supported truck configuration build. SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U300054	No at least one NAD assigment procedure performed.	Missing Calibration	Flexible switch panels not processed. No functions available through Flexible switches.	1. Invalid ECU installation procedure: Node Assignment procedure execution not performed; 2. ECU lost the RID procedure execution status (SW issue), to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Perform Node Assignment procedure	Electronic control unit
CIOM	U300055	No specific SW dataset downloaded.	Not Configured	General reduced functionality of the ECU	1. No valid SW configuration downloaded (no valid LIN schedule table selected through dataset); SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U300094	SW unexpected behaviour.	Unexpected Operation	ECU reset.	1. SW bug: - unexpected SW operation executed; - watchdog reset performed; - overrun situation detected; - insynchronized communication circuit state detected; 2. RAM corruption; 3. Flash corruption; SW or HW complex system issue, to be rised to back-office. Note: If for some reason the ECU power supply voltage (battery voltage) has exceeded 38V this DTC shall be ignored.	Diagnostic Service 14. No additional action is needed.	Replace Cab I/O Module	Electronic control unit
CIOM	U300316	Power supply voltage is lower than 16V (configurable).	Circuit Voltage Below Threshold	One or more functions of ECU do not perform as expected, but return automatically to normal operation after failure is removed. ESCL output switched low (off); Front Lid Lock Relay Control switched low (off); Inputs processing switched off; Digital output diagnostics activated (ESCL, Front Lid Outputs);	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the harness	Battery Power

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	U300317	Power supply voltage is higher than 32V (configurable).	Circuit Voltage Above Threshold	One or more functions of ECU do not perform as expected, but return automatically to normal operation after failure is removed. Immobilizer processing disabled; ESCL output switched low (off); Front Lid Lock Relay Control switched low (off); Inputs processing switched off; Digital output diagnostics activated (ESCL, Front Lid Outputs);	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the harness	Battery Power
CIOM	U1FFF68	Factory Mode is active in the ECU.	Event Information	Lock control function is working in factory mode: pre-delivery mode ignored.	P1C54 parameter is set to TRUE.	Diagnostic Service 14. No additional action is needed.	Exit Factory Mode	Factory Mode Active
CIOM	B151112	Resident voltage level when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Seat Belt Buckle Switch - Inspect the harness	Seat Belt Buckle Switch
CIOM	U114F68	Maximum number of DTC reached.	Event Information	No new DTC set until the issue solved or DTC list cleared.	1. Unexpected environment conditions; 2. ECU failure	Diagnostic Service 14. No additional action is needed.	Clear All DTC	DTC Notification Max Limit Reached
CIOM	B150A4A	Flexible switch not supported by SW: unknown switch ID, not associated with any known function.	Incorrect Component Installed	Corresponding FS ignored.	1. Flexible switch failure; 2. Not supported flexible switch installed (not valid vehicle range);	Diagnostic Service 14. No additional action is needed.	Replace faulty Flexible Switch	Unidentified switch
CIOM	B150B4A	Not supported double switch configuration present on truck.	Incorrect Component Installed	Corresponding FS ignored.	1. Double flexible switch mounting.	Diagnostic Service 14. No additional action is needed.	Remove one of two identical switches	Not supported double switch
CIOM	B151031	Flexible switch responce to flexible switch panel invalid: not possible to idenfity ID (switch ID0 reported by FSP).	No Signal	Corresponding FS ignored.	Flexible switch failure	Diagnostic Service 14. No additional action is needed.	Replace faulty Flexible Switch	Flexible Switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B151095	Flexible switch requested to be present in configuration is missing.	Incorrect Assembly	Corresponding FS ignored.	1. Configuration failure; 2. Flexible switch removed/disconnected; 3. Flexible switch panel failure. 4. Flexible switch failure.	Diagnostic Service 14. No additional action is needed.	- Inspect flexible switch panel - Inspect the harness - Inspect flexible switches connection	Flexible Switch
CIOM	B100411	Output short-circuit to GND.	Circuit Short To Ground	ECU behaviour: Current operation aborted until new request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 1 Control Circuit - Inspect the harness	Front Lid Lock Relay 1 Control Circuit
CIOM	B100412	Output short-circuit to BAT+.	Circuit Short To Battery	System effect: Latch is not possible to lock or/and to unlock.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 1 Control Circuit - Inspect the harness	Front Lid Lock Relay 1 Control Circuit
CIOM	B100413	Open circuit conditions detected on output pin. Threshold for open circuit detection: <100uA @ 0,8V.	Circuit Open	System effect: Latch is not possible to lock or/and to unlock.	1. Harness issue; 2. Control relay issue (FRC). 3. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 1 Control Circuit - Inspect the harness	Front Lid Lock Relay 1 Control Circuit
CIOM	B100416	Feedback voltage measured on the output is lower defined threshold (configurable). Threshold: <(0.1*VBAT)-0.2.	Circuit Voltage Below Threshold	System effect: Latch is not possible to lock or/and to unlock. Operation is not completed.	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 1 Control Circuit - Inspect the harness	Front Lid Lock Relay 1 Control Circuit
CIOM	B100417	Feedback voltage measured on the output is higher defined threshold (configurable). Threshold: <0.05*VBAT.	Circuit Voltage Above Threshold	System effect: Latch is not possible to lock or/and to unlock. Operation is not completed.	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 1 Control Circuit - Inspect the harness	Front Lid Lock Relay 1 Control Circuit
CIOM	B100511	Output short-circuit to GND.	Circuit Short To Ground	ECU behaviour: Current operation aborted until new request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 2 Control Circuit - Inspect the harness	Front Lid Lock Relay 2 Control Circuit
CIOM	B100512	Output short-circuit to BAT+.	Circuit Short To Battery	System effect: Latch is not possible to lock or/and to unlock.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 2 Control Circuit - Inspect the harness	Front Lid Lock Relay 2 Control Circuit
CIOM	B100513	Open circuit conditions detected on output pin. Threshold for open circuit detection: <100uA @ 0,8V.	Circuit Open	System effect: Latch is not possible to lock or/and to unlock.	1. Harness issue; 2. Control relay issue (FRC). 3. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 2 Control Circuit - Inspect the harness	Front Lid Lock Relay 2 Control Circuit
CIOM	B100516	Feedback voltage measured on the output is lower defined threshold (configurable). Threshold: <(0.1*VBAT)-0.2.	Circuit Voltage Below Threshold	System effect: Latch is not possible to lock or/and to unlock. Operation is not complited.	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 2 Control Circuit - Inspect the harness	Front Lid Lock Relay 2 Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B100517	Feedback voltage measured on the output is higher defined threshold (configurable). Threshold: <0.05*VBAT.	Circuit Voltage Above Threshold	System effect: Latch is not possible to lock or/and to unlock. Operation is not completed.	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Lock Relay 2 Control Circuit - Inspect the harness	Front Lid Lock Relay 2 Control Circuit
CIOM	B151312	Resident voltage level in circuit when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Latch Sensor - Inspect the harness	Front Lid Latch Sensor
CIOM	B150E16	Keyfob battery voltage below threshold.	Circuit Voltage Below Threshold	Keyfob button requests will be executed intermittently or not executed.	Keyfob battery voltage too low	Diagnostic Service 14. No additional action is needed.	Replace Keyfob battery	Key Fob
CIOM	B150E63	One of the keyfob buttons stay too long time in press position, more than 10s.	Circuit / Component Protection Time-Out	Keyfob goes to local save power mode.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	- Inspect Keyfob Note: - DID P1DJ8 can be used to get additional information.	Key Fob
CIOM	B151463	Switch stay in push state more than allowed time: depends of P1CUF.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Loading Level Flexible Switch	Loading Level Flexible Switch
CIOM	B151763	Switch stay in push state more than allowed time: depends of P1EXK.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Ferry Flexible Switch	Ferry Flexible Switch
CIOM	B151863	Switch stay in push state more than allowed time: depends of P1DWD.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Kneeling Flexible Switch	Kneeling Flexible Switch
CIOM	B151963	Switch stay in push state more than allowed time: depends of P1IZ2.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Loading Level Adjustment Flexible Switch	Loading Level Adjustment Flexible Switch
CIOM	B151A63	Switch stay in push state more than allowed time: depends of P1LXQ.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Front Park Brake Release Flexible Switch	Front Park Brake Release Switch
CIOM	B151512	Resident voltage level when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect Luggage Compartment Switch - Inspect the harness	Luggage Compartment Switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B100011	Output short-circuit to GND	Circuit Short To Ground	Output activation forbidden.	1. Harness issue. 2. ESCL internal failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the ESCL Logic Activation Power Supply Circuit - Inspect the harness	ESCL Logic Activation Power Supply Circuit
CIOM	B100012	Output short-circuit to BAT+.	Circuit Short To Battery	Output activation forbidden.	1. Harness issue. 2. CIOM internal failure (very low probability).	Diagnostic Service 14. No additional action is needed.	- Inspect the ESCL Logic Activation Power Supply Circuit - Inspect the harness	ESCL Logic Activation Power Supply Circuit
CIOM	B100013	Open circuit conditions detected on output pin	Circuit Open	ATTENTION: this diagnostic is disabled. DTC, output activation forbidden.	"1. Harness issue; 2. CIOM internal HW failure;"	Diagnostic Service 14. No additional action is needed.	- Inspect the ESCL Logic Activation Power Supply Circuit - Inspect the harness	ESCL Logic Activation Power Supply Circuit
CIOM	B100016	Feedback voltage measured on the output is lower defined threshold. Threshold: < Vbat - 2V.	Circuit Voltage Below Threshold	Output activation forbidden.	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the ESCL Logic Activation Power Supply Circuit - Inspect the harness	ESCL Logic Activation Power Supply Circuit
CIOM	B100017	Feedback voltage measured on the output is higher defined threshold. Threshold: 5V.	Circuit Voltage Above Threshold	Output activation forbidden	1. Harness issue; 2. CIOM internal HW failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the ESCL Logic Activation Power Supply Circuit - Inspect the harness	ESCL Logic Activation Power Supply Circuit
CIOM	B151212	Resident voltage level when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or button itself will blow when button is pressed; - if short circuit with protected output, this output will be affected when button is pressed (short to ground from output perspective);	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Swivel Seat Switch - Inspect the harness	Swivel Seat Switch
CIOM	B150F63	Switch stay in push state more than allowed time: > 90s.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	"1. Button failure. 2. Driver press the button too long time."	Diagnostic Service 14. No additional action is needed.	Inspect Traction switch	Traction switch
CIOM	U115054	No at least one Immobilizer Transponder Matching procedure performed.	Missing Calibration	Immobilizer function always active Not possible to crank	1. Invalid ECU installation procedure: Immobilizer Transponder Matching procedure not performed; 2. ECU internal failure, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	Perform Immobilizer Transponder Matching procedure	Immobilizer Transponder Data Link
CIOM	U115081	CIOM - Keyfob transponder communication failure.	Invalid Serial Data Received	PIN code strategy activation.	1. Transponder NOK; 2. CIOM internal falure.	Diagnostic Service 14. No additional action is needed.	Inspect the Immobilizer Transponder	Immobilizer Transponder Data Link
CIOM	U115088	CIOM - Immobilizer antenna link NOK.	Bus off	PIN code strategy activation.	1. Harness issue: antenna disconnected, SC to GND. 2. Antenna failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Immobilizer Transponder - Inspect the harness	Immobilizer Transponder Data Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
CIOM	B151363	Front Lid Lock Sensor has not changed the state to expected value (lock or unlock position accordingly to ongoing operation) in defined time (development configurable value).	Circuit / Component Protection Time-Out	Latch controlled maximum allowed time. Continuous application of the maximum time operation can result in premature wear of the latch.	- harness issue on frond lid lock control system - sensor failure (if latch state confirmed physically) - front lid latch broken	Diagnostic Service 14. No additional action is needed.	- Inspect the Front Lid Latch Sensor	Front Lid Latch Sensor
CIOM	B150463	Signal from LIN slave stay in active state too long time: depends of switch. See Proposed Workshop Action for more information.	Circuit / Component Protection Time-Out	Accordingly to functional requirements.	1. Button failure. 2. Driver press the button too long time.	Diagnostic Service 14. No additional action is needed.	Inspect Wired Control ECS Panel	Wired Control ECS Panel
DACU	B150593	The Collision Mitigation (CM) system shall be automatically disabled after 3 system activations	No Operation	CM disabled and driver notified if there are many interventions or disturbances in ECU. CM unstable and would have faulty operations if run in an environment with too many interventions	1. Extreme driving conditions are worst 2. Unclaibrated sensors mounted on trucks 3. Calibration of FLS/LPOS gone bad	Diagnostic Service 14. No additional action is needed.	1. Interview the driver about the events,If the driver says that the events were incorrect next step is to 2. checkFLS/LPOS/SteeringWheelAngle/yawRate calibration 3. check FLS/LPOS connector or any obstacles in sensors 4. Replace the sensor and check	Collision Mitigation System
DACU	U000188	Busoff on Can Comm Backbone 2	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Backbone 2 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Backbone 2 CAN connector.	CAN Communication Backbone 2 Net
DACU	U001088	Busoff on Can Comm Backbone 1.	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Backbone 1 CAN Can Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Backbone 1 CAN connector.	CAN Communication Backbone 1 Net
DACU	U200188	Busoff on Can Comm Active safety 1	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Active safety 1 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Active safety 1 CAN connector.	CAN Communication Active Safety Proprietary 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	U200288	Busoff on Can Comm Active safety 2	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Active safety 2 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Active safety 2 CAN connector.	CAN Communication Active Safety Proprietary 2 Net
DACU	U300041	Faulty SW and/or Data	General Checksum Failure	ECU shall not operate	ECU SW has been tampered ECU faulty (non volatile memory ROM)	Diagnostic Service 14. No additional action is needed.	1. Check the signature value which is stored in memory. 2. Check if unexpected resets 3. Replace the DACU and check	Electronic control unit
DACU	U300044	RAM error	Data Memory Failure	RAM detection shall be performed by ECU after sleep mode or power on.	ECU faulty (volatile memory RAM)	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300045	NVRAM / ROM error is detected	Program Memory Failure	Checks for NVRAM / ROM error is detection Example 1.Write to NVRAM fails 2. ECU detects a ROM memory failure	ECU faulty (non volatile memory RAM)	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300049	ECU Internal Error	Internal Electronic Failure	Checks for The MCU clock error is detection, CAN tranciever hardware error that inhibits RX/TX transmission	1. ECU faulty (BSW driver / transceiver). 2. ECU specific Software error.	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300094	Faulty operation	Unexpected Operation	ECU checks for unexpected operations like 1. unexpected reset due to null pointer reference 2. division by zero 3. Watch dog failure	1. ECU faulty Operations (watchdog fault) 2. ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors 3. Software error	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300316	ECU Low Voltage <10V	Circuit Voltage Below Threshold	ECU Not functional	1. Broken/bad power lines(Rusted). 2. Battery voltage too low	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
DACU	U300317	ECU high Voltage> 32V	Circuit Voltage Above Threshold	ECU operates Normal	1. Broken/bad power lines. 2. Battery voltage too high. 3. DACU power circuitry gone bad.	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
DACU	U1FFF68	Truck is in Factory Mode	Event Information	Truck is in Factory Mode	Truck is in Factory Mode	Diagnostic Service 14. No additional action is needed.	Check the Factory Mode parameter Value: Parameter P1C54	Factory Mode Active
DACU	U250044	Internal Data - RAM Inconsistency detected	Data Memory Failure	RAM detection shall be performed after sleep mode or power on and this DTC set if RAM check failed	1. ECU faulty (volatile memory RAM) 2. Normal operation crrupted the data in RAM	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency
DACU	U250045	Internal Data - ROM Inconsistency detected	Program Memory Failure	Checks for NVRAM / ROM error is detection Example: 1.Write to NVRAM fails 2. ECU detects a ROM memory failure	1. ECU faulty (Non volatile memory ROM) 2. Normal operation crrupted the data in ROM	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B100211	FCW LED - Short to Ground	Circuit Short To Ground	LED shall not be powered and visual warning for ACC and CM Functionality is disabled	1. LED damaged/Internally shorted/broken. 2. Wires broken. Insulation damage or its wire shorted to truck body. 3. DACU supply circuitry for LED faulty.	Diagnostic Service 14. No additional action is needed.	1. Check the FCW HUD connections and its wiring 2. Check LEDs short/Open with multimeter	FCW HUD Power Supply
DACU	B100215	FCW LED -Short to Batt or open Ck	Circuit Short To Battery or Open	LED shall not be powered Up and visual warning for ACC and CM Functionality is disabled	1. LED damaged/open internally/broken. 2. Wires broken. Insulation damage or LED wires shorted to battery voltage. 3. LED Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check the FCW HUD connections and its wiring 2. Check LEDs short/Open with multimeter	FCW HUD Power Supply
DACU	U114268	More than Maximum allowed other DTCs (29 DTC's) with WIB Bit set are active in the ECU	Event Information	This component is responsible to collect DTC information from the DEM and to send the DiagnosticFaultStatus signal to the DiagnosticWarningManager . The DiagnosticFaultStatus will be sent periodically from the application and will also be used as a heartbeat signal to monitor that the ECU is still alive.	Major ECU fault.	Diagnostic Service 14. No additional action is needed.	1. Check No of active DTCs count if more than 29	DTC Notification Max Limit Reached
DACU	B100111	FLS - Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Adaptive Cruise Contro and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100112	FLS - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100113	FLS - Open Circuit	Circuit Open	Sensor shall not be powered and Adaptive Cruise Contro and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3. FLS Connector open/FLS disconnected	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100116	FLS - Low Voltage < 10V	Circuit Voltage Below Threshold	Sensor shall not be powered and Adaptive Cruise Control and Collision Mitigation Functionalities are disabled	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B100117	FLS - High Voltage > 32V	Circuit Voltage Above Threshold	Sensor shall not be powered and Adaptive Cruise Control and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. FLS Wires broken. Insulation damage. 3. Bad connection/contacts with FLS connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B150311	Lane Change Support LED - Short to Ground	Circuit Short To Ground	LED shall not be powered and visual warning for Lane Change Support Functionality is disabled	1. LED damaged/Internally shorted/broken. 2. Wires broken. Insulation damage or its wire shorted to truck body. 3.DACU supply circuitry for LED faulty.	Diagnostic Service 14. No additional action is needed.	1. Check the LCS HUD connections and its wiring 2. Check LEDs short/Open with multimeter	LCS LED Power Supply
DACU	B150315	LCS LED - Short to Batt or open Ck	Circuit Short To Battery or Open	LED shall not be powered Up and visual warning for Lane Change Support Functionality is disabled	1. LED damaged/open internally/broken. 2. Wires broken. Insulation damage or LED wires shorted to battery voltage. 3. LED Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check the LCS HUD connections and its wiring 2. Check LEDs short/Open with multimeter	LCS LED Power Supply
DACU	B100011	Lane Positioning Sensor (LPOS) - Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100012	Lane Positioning Sensor(LPOS) - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100013	Lane Positioning Sensor(LPOS) - Open Circuit	Circuit Open	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. LPOS Connector open/LPOS disconnected 2. DACU supply circuitry for LPOS faulty	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100016	Lane Positioning Sensor(LPOS) - Low Voltage < 8.5V	Circuit Voltage Below Threshold	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Bad connection/contacts with LPOS connector	Diagnostic Service 14. No additional action is needed.	1. Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100017	Lane Positioning Sensor(LPOS) - High Voltage > 32V	Circuit Voltage Above Threshold	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. LPOS wiring Insulation damage. 3. Bad connection/contacts with LPOS connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1. Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150416	Side Sensor - Low Voltage <10V. Failure event sent	Circuit Voltage Below Threshold	If Side Sensor voltage supply out of range (abnormal low) then then Lane Change Support(LCS) System status is set to error, SideObjectDetectedT2P set to error and Report to DEM. Side Sensor voltage supply out of range (abnormal low) information is received through DiagnosticsFaultStatus_DM 1 signal	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply 3. check Side sensor for physical damage	Lane Change Support Side Sensor Module
DACU	B150417	Side Sensor - High Voltage > 14V. Failure event sent	Circuit Voltage Above Threshold	If Side Sensor voltage supply out of range (abnormal high) then Lane Change Support(LCS) System status is set to error. SideObjectDetectedT2P set to error and Report to DEM Side Sensor voltage supply out of range (abnormal high) information is received through DiagnosticsFaultStatus_DM 1 signal	1. Connected component broken. 2. Sidesensor Wires broken. Insulation damage. 3. Bad connection/contacts with Sidesensor connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply 3. check Side sensor for physical damage	Lane Change Support Side Sensor Module
DACU	B150438	Message Update Rate is not correct.	Signal Frequency Incorrect	If Side Sensor local CAN abnormal update rate detected then set SideObjectDetectedT2P as error and Report to DEM. Side Sensor local CAN abnormal update rate information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side Sensor local CAN abnormal update rate detected 2. Sidesensor is faulty 3. Sidesensor CAN lines got damaged/bad connections/connector loose	Diagnostic Service 14. No additional action is needed.	1. check SS signal frequency 2. Check its CAN and Supply connectors 3. Restart and check 4. Check its supply 5. check Side sensor for physical damage	Lane Change Support Side Sensor Module
DACU	B150454	CCP(CAN Calibration Protocol) Initialization Error with SideSensor	Missing Calibration	Lane Change Support(LCS) will Initialize the Side Sensor after powering up the sensor and read/upload parameter from sensor. If there is some mismatch in the sensor parameter and the DACU parameter, the differed parameter will be sent to sensor in a new CCP session. Gateway object detection message received from Side Sensor	1. Side Sensor CCP initialisation not completed due to communication failure 2. 3 attempts of CCP initialisation failed. 3. No proper Vehicle mode transition/short transition of vehicle mode	Diagnostic Service 14. No additional action is needed.	1. check Side sensor for physical damage 2. Check its CAN and Supply connectors 3. Restart and check 4. Check its supply	Lane Change Support Side Sensor Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150487	Communication Failure with Side Sensor, No messages recieved from Side sensor	Missing Message	If Side sensor hardware is faulty then detected or blocked then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor hardware fault information is received through DiagnosticsFaultStatus_DM 1 signal.	1. No supply to Side sensor 2. Sidesensor is faulty/damaged/connector to sensor is loose or damaged 3. 3 attempts of CCP initialisation failed so it require another Vehicle mode transition.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply	Lane Change Support Side Sensor Module
DACU	B150492	Side sensor incorrect operation	Performance or Incorrect Operation	If Side sensor software is faulty then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error. Side sensor software fault information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side sensor is faulty 2. Sidesensor CAN lines got damaged/bad connections/connector loose 3. Supply to Sidesensor abnormal so sidesensor might behaves abnormal	Diagnostic Service 14. No additional action is needed.	1. check Side sensor for physical damage 2. Check its CAN and Supply connectors	Lane Change Support Side Sensor Module
DACU	B150496	Side Sensor Internal failure	Component Internal Failure	If Side sensor hardware fault detected then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor hardware fault information is received through DiagnosticsFaultStatus_DM 1 signal.	1. Side sensor is faulty/internally shorted 2. Sidesensor CCP initialisation improper so it may require another Vehicle mode transition 3. Supply to Sidesensor abnormal so sidesensor might behaves abnormal	Diagnostic Service 14. No additional action is needed.	1. Check Side sensor 2. Check its CAN and Supply connectors	Lane Change Support Side Sensor Module
DACU	B150497	Side sensor operation obstructed or blocked.	Component or System Operation Obstructed or Blocked	If Side sensor blocked then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor blocked information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side Sensor is either Blocked or sensor failure to detect objects or obstructed(sensor faulty). 2. Sidesensor CCP initialisation improper so it may require another Vehicle mode transition 3. Improper sully to Sidesensor	Diagnostic Service 14. No additional action is needed.	1. Check SS whether it is aged or it is under working condition. 2. Check any obstacles in Sensor 3. Check its connector	Lane Change Support Side Sensor Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150498	Side sensor over temperature (> 85ºC OR < -40ºC)	Component or System Over Temperature	If Side Sensor temperature out of range then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side Sensor temperature out of range information is received through DiagnosticsFaultStatus_DM 1 signal.	1. Side sensor temperature out of range 2. Due to loose contact in its connector it is over heated 3. Supply to sidesensor is too high.	Diagnostic Service 14. No additional action is needed.	1. Check SS tempearture 2. Check sidesensor connector and its wiring 3. Check with new sensor	Lane Change Support Side Sensor Module
DACU	B150211	Side Sensor- Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Lane Change Support System status is set to disabled/error based on Vehicle Mode	1. Connected component broken. 2. Wires broken. Insulation damage. 3. DACU supply circuitry for Sidesensor faulty.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors/wirings to the sensors 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150212	SS - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for Sidesensor faulty.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors to the sensors 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150213	Side Sensor - Open Circuit	Circuit Open	Sensor shall not be powered and Lane Change Support System status is set to disabled/error based on Vehicle Mode	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Sidesensor Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors to the sensors 2. Check if sensor disconnected 3. check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150216	Side Sensor - Low Voltage <10V	Circuit Voltage Below Threshold	Sensor shall not be powered and Lane Change Support System status is set to disabled (if battery power <10) /error based on Vehicle Mode	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150217	Side Sensor - High Voltage > 14V	Circuit Voltage Above Threshold	Sensor shall not be powered and Lane Change Support System status is set to disabled (if battery power >32)/error based on Vehicle Mode	1. Connected component broken. 2. Sidesensor Wires broken. Insulation damage. 3. Bad connection/contacts with Sidesensor connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	C006368	High noise in Yaw Rate signal input to DACU	Event Information	YawRateStatus shall be set to status NOK. But the best possible Yawrate value shall be sent in the signal "YawRateCompensated"	The last 10 samples are evaluated and if the maximum and minimum values of YawRate differ more than +/- 0.2 rad/s (default) from the mean value the DTC is set.	Diagnostic Service 14. No additional action is needed.	Do nothing(this DTC just explains why the function was disabled)	Yaw Rate Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150593	The Collision Mitigation (CM) system shall be automatically disabled after 3 system activations	No Operation	CM disabled and driver notified if there are many interventions or disturbances in ECU. CM unstable and would have faulty operations if run in an environment with too many interventions	1. Extreme driving conditions are worst 2. Unclaibrated sensors mounted on trucks 3. Calibration of FLS/LPOS gone bad	Diagnostic Service 14. No additional action is needed.	1. Interview the driver about the events,If the driver says that the events were incorrect next step is to 2. checkFLS/LPOS/SteeringWheelAngle/yawRate calibration 3. check FLS/LPOS connector or any obstacles in sensors 4. Replace the sensor and check	Collision Mitigation System
DACU	B150968	CM input signal failure	Event Information	NA	failures in the input signals of CM	Diagnostic Service 14. No additional action is needed.	NA	Collision Warning With Emergency Brake, Input Signal Failure
DACU	B150A68	FCW input signal failure	Event Information	NA	failures in the input signals of FCW	Diagnostic Service 14. No additional action is needed.	NA	FCW Failure
DACU	B150882	Alive request from the watch dog is not updated	Alive / Sequence Counter Incorrect / Not Updated	NA	CollisionSituationAliveRequest value is 14 or CollisionSituationAliveRequest counter is not updated properly.	Diagnostic Service 14. No additional action is needed.	NA	Collision Status Alive Monitoring
DACU	B150B68	Stuck of these critical Input Signals Yaw rate Vehicle speed Target fusion CollisionSituation Status signal Vehicle width Equivalent wheelbase turning Acceleration radar ego speed acceleration	Event Information	Each time an indication of failure is detected an emergency brake shall not be allowed within a time window of 10s or as long as the failure exists	Critical signals behaviour is incorrect for 10 times or too long time.	Diagnostic Service 14. No additional action is needed.	NA	Safety Critical Signals Monitoring
DACU	U000188	Busoff on Can Comm Backbone 2	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Backbone 2 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Backbone 2 CAN connector.	CAN Communication Backbone 2 Net
DACU	U001088	Busoff on Can Comm Backbone 1.	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Backbone 1 CAN Can Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Backbone 1 CAN connector.	CAN Communication Backbone 1 Net
DACU	U200188	Busoff on Can Comm Active safety 1	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Active safety 1 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	1. Check Active safety 1 CAN connector. 2. Check for P1GD8 parameter(has to be true).	CAN Communication Active Safety Proprietary 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	U200288	Busoff on Can Comm Active safety 2	Bus off	No Communication and error frames generated. The FDC(Fault detection count) shall increase each time the ECU reaches bus off status and decreases when ECU leaves bus off status.	Active safety 2 CAN Comm lines short to Ground/Battery.	Diagnostic Service 14. No additional action is needed.	Check Active safety 2 CAN connector.	CAN Communication Active Safety Proprietary 2 Net
DACU	U300041	Faulty SW and/or Data	General Checksum Failure	ECU shall not operate	ECU SW has been tampered ECU faulty (non volatile memory ROM)	Diagnostic Service 14. No additional action is needed.	1. Check the signature value which is stored in memory. 2. Check if unexpected resets 3. Replace the DACU and check	Electronic control unit
DACU	U300044	RAM error	Data Memory Failure	RAM detection shall be performed by ECU after sleep mode or power on.	ECU faulty (volatile memory RAM)	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300045	NVRAM / ROM error is detected	Program Memory Failure	Checks for NVRAM / ROM error is detection Example 1.Write to NVRAM fails 2. ECU detects a ROM memory failure	ECU faulty (non volatile memory RAM)	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300049	ECU Internal Error	Internal Electronic Failure	Checks for The MCU clock error is detection, CAN tranciever hardware error that inhibits RX/TX transmission	1. ECU faulty (BSW driver / transceiver). 2. ECU specific Software error.	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300094	Faulty operation	Unexpected Operation	ECU checks for unexpected operations like 1. unexpected reset due to null pointer reference 2. division by zero 3. Watch dog failure	1. ECU faulty Operations (watchdog fault) 2. ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors 3. Software error	Diagnostic Service 14. No additional action is needed.	1. Checkif unexpected resets 2. Replace the DACU and check	Electronic control unit
DACU	U300316	ECU Low Voltage <10V	Circuit Voltage Below Threshold	ECU Not functional	1. Broken/bad power lines(Rusted). 2. Battery voltage too low	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
DACU	U300317	ECU high Voltage> 32V	Circuit Voltage Above Threshold	ECU operates Normal	1. Broken/bad power lines. 2. Battery voltage too high. 3. DACU power circuitry gone bad.	Diagnostic Service 14. No additional action is needed.	1. Check the batt voltage 2. Check the FRC connector	Battery Power
DACU	U1FFF68	Truck is in Factory Mode	Event Information	Truck is in Factory Mode	Truck is in Factory Mode	Diagnostic Service 14. No additional action is needed.	Check the Factory Mode parameter Value: Parameter P1C54	Factory Mode Active
DACU	U250044	Internal Data - RAM Inconsistency detected	Data Memory Failure	RAM detection shall be performed after sleep mode or power on and this DTC set if RAM check failed	1. ECU faulty (volatile memory RAM) 2. Normal operation crrupted the data in RAM	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency
DACU	U250045	Internal Data - ROM Inconsistency detected	Program Memory Failure	Checks for NVRAM / ROM error is detection Example: 1.Write to NVRAM fails 2. ECU detects a ROM memory failure	1. ECU faulty (Non volatile memory ROM) 2. Normal operation crrupted the data in ROM	Diagnostic Service 14. No additional action is needed.	1. Make proper shutdown to avoid this problem(Re-start the system.	Internal Data Inconsistency

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B100211	FCW LED - Short to Ground	Circuit Short To Ground	LED shall not be powered and visual warning for ACC and CM Functionality is disabled	1. LED damaged/Internally shorted/broken. 2. Wires broken. Insulation damage or its wire shorted to truck body. 3. DACU supply circuitry for LED faulty.	Diagnostic Service 14. No additional action is needed.	1. Check the FCW HUD connections and its wiring 2. Check LEDs short/Open with multimeter	FCW HUD Power Supply
DACU	B100215	FCW LED -Short to Batt or open Ck	Circuit Short To Battery or Open	LED shall not be powered Up and visual warning for ACC and CM Functionality is disabled	1. LED damaged/open internally/broken. 2. Wires broken. Insulation damage or LED wires shorted to battery voltage. 3. LED Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check the FCW HUD connections and its wiring 2. Check LEDs short/Open with multimeter	FCW HUD Power Supply
DACU	U114268	More than Maximum allowed other DTCs (29 DTC's) with WIB Bit set are active in the ECU	Event Information	This component is responsible to collect DTC information from the DEM and to send the DiagnosticFaultStatus signal to the DiagnosticWarningManager . The DiagnosticFaultStatus will be sent periodically from the application and will also be used as a heartbeat signal to monitor that the ECU is still alive.	Major ECU fault.	Diagnostic Service 14. No additional action is needed.	1. Check No of active DTCs count if more than 29	DTC Notification Max Limit Reached
DACU	B100111	FLS - Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Adaptive Cruise Contro and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100112	FLS - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100113	FLS - Open Circuit	Circuit Open	Sensor shall not be powered and Adaptive Cruise Contro and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3. FLS Connector open/FLS disconnected	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B100116	FLS - Low Voltage < 10V	Circuit Voltage Below Threshold	Sensor shall not be powered and Adaptive Cruise Control and Collision Mitigation Functionalities are disabled	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B100117	FLS - High Voltage > 32V	Circuit Voltage Above Threshold	Sensor shall not be powered and Adaptive Cruise Control and Collision Mitigation Functionalities are disabled	1. Connected component broken. 2. FLS Wires broken. Insulation damage. 3. Bad connection/contacts with FLS connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check FLS sensor for physical damage	Forward Looking Sensor Power Supply
DACU	B150C68	Stuck of Input Signal Front Axle Speed.	Event Information	Signal Vehicle Speed Compensated will signal error.	if the insignal Front Axle Speed is monitored to be not updated and the signal Vehicle Speed Compensated shall signal error.	Diagnostic Service 14. No additional action is needed.	Do nothing(this DTC just explains why the function was disabled)	Front Axle Speed Signal, Monitoring
DACU	B150311	Lane Change Support LED - Short to Ground	Circuit Short To Ground	LED shall not be powered and visual warning for Lane Change Support Functionality is disabled	1. LED damaged/Internally shorted/broken. 2. Wires broken. Insulation damage or its wire shorted to truck body. 3.DACU supply circuitry for LED faulty.	Diagnostic Service 14. No additional action is needed.	1. Check the LCS HUD connections and its wiring 2. Check LEDs short/Open with multimeter	LCS LED Power Supply
DACU	B150315	LCS LED - Short to Batt or open Ck	Circuit Short To Battery or Open	LED shall not be powered Up and visual warning for Lane Change Support Functionality is disabled	1. LED damaged/open internally/broken. 2. Wires broken. Insulation damage or LED wires shorted to battery voltage. 3. LED Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check the LCS HUD connections and its wiring 2. Check LEDs short/Open with multimeter	LCS LED Power Supply
DACU	B100011	Lane Positioning Sensor (LPOS) - Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100012	Lane Positioning Sensor(LPOS) - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for LPOS faulty.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100013	Lane Positioning Sensor(LPOS) - Open Circuit	Circuit Open	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. LPOS Connector open/LPOS disconnected 2. DACU supply circuitry for LPOS faulty	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B100016	Lane Positioning Sensor(LPOS) - Low Voltage < 8.5V	Circuit Voltage Below Threshold	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Bad connection/contacts with LPOS connector	Diagnostic Service 14. No additional action is needed.	1. Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B100017	Lane Positioning Sensor(LPOS) - High Voltage > 32V	Circuit Voltage Above Threshold	Sensor shall not be powered and Lane Keeping Support(LKS) Functionality is disabled	1. Connected component broken. 2. LPOS wiring Insulation damage. 3. Bad connection/contacts with LPOS connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1. Check Supply voltage and its connectors/wiring 2. Check LPOS sensor for physical damage	LPOS Power supply
DACU	B150416	Side Sensor - Low Voltage <10V. Failure event sent	Circuit Voltage Below Threshold	If Side Sensor voltage supply out of range (abnormal low) then then Lane Change Support(LCS) System status is set to error, SideObjectDetectedT2P set to error and Report to DEM. Side Sensor voltage supply out of range (abnormal low) information is received through DiagnosticsFaultStatus_DM 1 signal	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply 3. check Side sensor for physical damage	Lane Change Support Side Sensor Module
DACU	B150417	Side Sensor - High Voltage > 14V. Failure event sent	Circuit Voltage Above Threshold	If Side Sensor voltage supply out of range (abnormal high) then Lane Change Support(LCS) System status is set to error. SideObjectDetectedT2P set to error and Report to DEM Side Sensor voltage supply out of range (abnormal high) information is received through DiagnosticsFaultStatus_DM 1 signal	1. Connected component broken. 2. Sidesensor Wires broken. Insulation damage. 3. Bad connection/contacts with Sidesensor connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply 3. check Side sensor for physical damage	Lane Change Support Side Sensor Module
DACU	B150438	Message Update Rate is not correct.	Signal Frequency Incorrect	If Side Sensor local CAN abnormal update rate detected then set SideObjectDetectedT2P as error and Report to DEM. Side Sensor local CAN abnormal update rate information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side Sensor local CAN abnormal update rate detected 2. Sidesensor is faulty 3. Sidesensor CAN lines got damaged/bad connections/connector loose	Diagnostic Service 14. No additional action is needed.	1. check SS signal frequency 2. Check its CAN and Supply connectors 3. Restart and check 4. Check its supply 5. check Side sensor for physical damage	Lane Change Support Side Sensor Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150454	CCP(CAN Calibration Protocol) Initialization Error with SideSensor	Missing Calibration	Lane Change Support(LCS) will Initialize the Side Sensor after powering up the sensor and read/upload parameter from sensor. If there is some mismatch in the sensor parameter and the DACU parameter, the differed parameter will be sent to sensor in a new CCP session. Gateway object detection message received from Side Sensor	1. Side Sensor CCP initialisation not completed due to communication failure 2. 3 attempts of CCP initialisation failed. 3. No proper Vehicle mode transition/short transition of vehicle mode	Diagnostic Service 14. No additional action is needed.	1. check Side sensor for physical damage 2. Check its CAN and Supply connectors 3. Restart and check 4. Check its supply	Lane Change Support Side Sensor Module
DACU	B150487	Communication Failure with Side Sensor, No messages recieved from Side sensor	Missing Message	If Side sensor hardware is faulty then detected or blocked then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor hardware fault information is received through DiagnosticsFaultStatus_DM 1 signal.	1. No supply to Side sensor 2. Sidesensor is faulty/damaged/connector to sensor is loose or damaged 3. 3 attempts of CCP initialisation failed so it require another Vehicle mode transition.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors 2. Check its supply	Lane Change Support Side Sensor Module
DACU	B150492	Side sensor incorrect operation	Performance or Incorrect Operation	If Side sensor software is faulty then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error. Side sensor software fault information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side sensor is faulty 2. Sidesensor CAN lines got damaged/bad connections/connector loose 3. Supply to Sidesensor abnormal so sidesensor might behaves abnormal	Diagnostic Service 14. No additional action is needed.	1. check Side sensor for physical damage 2. Check its CAN and Supply connectors	Lane Change Support Side Sensor Module
DACU	B150496	Side Sensor Internal failure	Component Internal Failure	If Side sensor hardware fault detected then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor hardware fault information is received through DiagnosticsFaultStatus_DM 1 signal.	1. Side sensor is faulty/internally shorted 2. Sidesensor CCP initialisation improper so it may require another Vehicle mode transition 3. Supply to Sidesensor abnormal so sidesensor might behaves abnormal	Diagnostic Service 14. No additional action is needed.	1. Check Side sensor 2. Check its CAN and Supply connectors	Lane Change Support Side Sensor Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	B150497	Side sensor operation obstructed or blocked.	Component or System Operation Obstructed or Blocked	If Side sensor blocked then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side sensor blocked information is received through DiagnosticsFaultStatus_DM 1 signal	1. Side Sensor is either Blocked or sensor failure to detect objects or obstructed(sensor faulty). 2. Sidesensor CCP initialisation improper so it may require another Vehicle mode transition 3. Improper sully to Sidesensor	Diagnostic Service 14. No additional action is needed.	1. Check SS whether it is aged or it is under working condition. 2. Check any obstacles in Sensor 3. Check its connector	Lane Change Support Side Sensor Module
DACU	B150498	Side sensor over temperature (> 85ºC OR < -40ºC)	Component or System Over Temperature	If Side Sensor temperature out of range then set SideObjectDetectedT2P as error and Report to DEM. Lane Change Support(LCS) System status is set to error Side Sensor temperature out of range information is received through DiagnosticsFaultStatus_DM 1 signal.	1. Side sensor temperature out of range 2. Due to loose contact in its connector it is over heated 3. Supply to sidesensor is too high.	Diagnostic Service 14. No additional action is needed.	1. Check SS tempearture 2. Check sidesensor connector and its wiring 3. Check with new sensor	Lane Change Support Side Sensor Module
DACU	B150211	Side Sensor- Short to Ground	Circuit Short To Ground	Sensor shall not be powered and Lane Change Support System status is set to disabled/error based on Vehicle Mode	1. Connected component broken. 2. Wires broken. Insulation damage. 3. DACU supply circuitry for Sidesensor faulty.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors/wirings to the sensors 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150212	SS - Short to Batt	Circuit Short To Battery	Sensor shall be power ON even on Inactive condition	1. Connected component broken. 2. Wires broken. Insulation damage. 3.DACU supply circuitry for Sidesensor faulty.	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors to the sensors 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150213	Side Sensor - Open Circuit	Circuit Open	Sensor shall not be powered and Lane Change Support System status is set to disabled/error based on Vehicle Mode	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Sidesensor Connector open/SideSensor disconnected	Diagnostic Service 14. No additional action is needed.	1. Check its CAN and Supply connectors to the sensors 2. Check if sensor disconnected 3. check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150216	Side Sensor - Low Voltage <10V	Circuit Voltage Below Threshold	Sensor shall not be powered and Lane Change Support System status is set to disabled (if battery power <10) /error based on Vehicle Mode	1. Broken/bad power lines(Rusted) or Insulation damage 2. Connected component/its wire broken. 3. Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check Side sensor for physical damage	Lane Change Support Power Supply
DACU	B150217	Side Sensor - High Voltage > 14V	Circuit Voltage Above Threshold	Sensor shall not be powered and Lane Change Support System status is set to disabled (if battery power >32)/error based on Vehicle Mode	1. Connected component broken. 2. Sidesensor Wires broken. Insulation damage. 3. Bad connection/contacts with Sidesensor connector got shorted with battery supply	Diagnostic Service 14. No additional action is needed.	1.Check Supply voltage and its connectors/wiring 2. Check Side sensor for physical damage	Lane Change Support Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DACU	C006368	High noise in Yaw Rate signal input to DACU	Event Information	YawRateStatus shall be set to status NOK. But the best possible Yawrate value shall be sent in the signal "YawRateCompensated"	The last 10 samples are evaluated and if the maximum and minimum values of YawRate differ more than +/- 0.2 rad/s (default) from the mean value the DTC is set.	Diagnostic Service 14. No additional action is needed.	Do nothing(this DTC just explains why the function was disabled)	Yaw Rate Sensor
DM	B150214	24V Power supply lost (Fault Code Indication received from Window Antipinch).	Circuit Short To Ground or Open	Power window function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors.	Window Antipinch Module
DM	B150216	Window Antipinch measures 24V power supply voltage lower than 18,5 V (Fault Code Indication received from Window Antipinch).	Circuit Voltage Below Threshold	Power window function is not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors. - Inspect the Door seal ageing.	Window Antipinch Module
DM	B150217	Window Antippinch measures power supply voltage higher than 32V (Fault Code Indication received from Window Antipinch).	Circuit Voltage Above Threshold	Power window function is not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors. - Inspect the Door seal ageing.	Window Antipinch Module
DM	B15021C	Window Antipinch measures 12V power supply voltage lower than 7.5V or higher than 18,5V (Fault Code Indication received from Window Antipinch).	Circuit Voltage Out of Range	Power window function is not working as expected.	1. Harness issue; 2. Driver Door Module 12V output failure; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors.	Window Antipinch Module
DM	B150249	Window Antipinch internal fault (Fault Code Indication received from Window Antipinch).	Internal Electronic Failure	Power window function is not working as expected.	1. Internal uC failure detected; 2. Temperature sensor defective; 3. Hall sensor defective	Diagnostic Service 14. No additional action is needed.	- Replace the Window Antipinch Module.	Window Antipinch Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150254	Window position information is lost (Fault Code Indication received from Window Antipinch).	Missing Calibration	Power window function is not working as expected. Manual Up is working. Manual Down is working. Auto-Down is working. Auto-Up is not working.	1. During a window motor movement the 12V or 24V power supply was suddenly interrupted; 2. Internal WindowAntipinch failure (Hall sensor error, Window moving while position counting not possible etc..); 3. Block event occured during reversal; 4. Window positon is unplausible; 5. Sealing ageing (that generated decalibration) 6. Change of door and glass type information sent over LIN during Window Antipinch run time.	Diagnostic Service 14. No additional action is needed.	- Perform the calibration (Press window up button until the window raise to the top and maintain it press for 3 seconds after that). - Inspect the Door seal ageing.	Window Antipinch Module
DM	B150255	No valid configuration information received from Driver Door Module over LIN: - Glass Type - Door Type (Fault Code Indication received from Window Antipinch).	Not Configured	Power window function is not working as expected. Manual Up is working. Manual Down is working. Auto-Down is working. Auto-Up is not working.	1. Harness issue; 2. Glass and door configuration information sent over LIN not available or error; 3. Change of door and glass type information sent over LIN during Window Antipinch run time.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness and connectors. - Redownload the Door Module Software. - Contact back office.	Window Antipinch Module
DM	B150271	Internal Window Antipinch relay is stuck (Fault Code Indication received from Window Antipinch).	Actuator Stuck	Power window function is not working as expected.	1. Mechanical defect of the relay contacts.	Diagnostic Service 14. No additional action is needed.	- Replace the Window Antipinch Module.	Window Antipinch Module
DM	B150298	Window Motor is too hot according Window Antipinch overheating SW algorithm (LIN Fault Code Indication received from Window Antipinch).	Component or System Over Temperature	Window motor is inhibited by Window Antipinch. Power window function is not working as expected, but returns to normal operation once the motor has cooled down.	1. Motor thermal protection is activated because motor was actived during too long time.	Diagnostic Service 14. No additional action is needed.	- Cool down the motor during 20 minutes minimum (Do not use it). - Inspect the Door seal ageing.	Window Antipinch Module
DM	U114286	Invalid frame received from LIN slave.	Signal Invalid	Driver Door Module invalidates Window Antipinch signals.	LIN slave issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	Window Antipinch Link
DM	U114287	LIN slave not responding.	Missing Message	Driver Door Module invalidates Window Antipinch signals.	LIN slave issue. Harness issue. Invalid parameters setting.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	Window Antipinch Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U114188	Communication fault detected on the LIN bus.	Bus off	Driver Door Module invalidates Window Antipinch and Door Control Panel signals.	1. Harness issue: short circuit to GND/BAT/12V power supply or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	LIN8 Communication Net
DM	B150316	Door Control Panel measures power supply voltage lower than 7.5V (Fault Code Indication received from Door Control Panel).	Circuit Voltage Below Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue; 2. Driver Door Module 12V output failure; 3. Faulty Door Control Panel hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness and connectors.	Driver Door Control Panel
DM	B150317	Door Control Panel measures power supply voltage higher than 18.5V (Fault Code Indication received from Door Control Panel).	Circuit Voltage Above Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue; 2. Driver Door Module 12V output failure; 3. Faulty Door Control Panel hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness and connectors.	Driver Door Control Panel
DM	B150344	Door Control Panel RAM check fail (Fault Code Indication received from Door Control Panel).	Data Memory Failure	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	1. Door Control Panel slave SW issue. 2. Door Control Panel RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Driver Door Control Panel
DM	B150345	Door Control Panel ROM check fail (Fault Code Indication received from Door Control Panel).	Program Memory Failure	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	1. Door Control Panel ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Driver Door Control Panel
DM	B150346	Door Control Panel flash corruption detected during writing of calibration data (Fault Code Indication received from Door Control Panel).	Calibration / Parameter Memory Failure	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	1. Door Control Panel flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Driver Door Control Panel
DM	B150349	Door Control Panel Buttons configuration failure (Fault Code Indication received from Door Control Panel).	Internal Electronic Failure	All functions handled by Door Control Panel are not working as expected.	1. Door Control Panel End Of Line configuration not correct.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Driver Door Control Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150394	Door Control Panel unexpected SW operation executed; Door Control Panel slave watchdog reset performed; Door Control Panel slave overrun situation detected; (Fault Code Indication received from Door Control Panel).	Unexpected Operation	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	1. Door Control Panel SW issue; 2. Door Control Panel MCU not correct.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Driver Door Control Panel
DM	U114386	Invalid frame received from LIN slave.	Signal Invalid	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	LIN slave issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Driver Door Control Panel Link
DM	U114387	LIN slave not responding.	Missing Message	Driver Door Module invalidates the Door Control Panel signals. All functions handled by Door Control Panel are not working as expected.	LIN slave issue. Harness issue. Invalid parameters setting.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Driver Door Control Panel Link
DM	B100011	Output pin short circuit to GND.	Circuit Short To Ground	Turns off Door Control Panel 12V power supply; All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. DCP internal short circuit.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100012	Output pin short circuit to BAT+.	Circuit Short To Battery	Turns off Door Control Panel 12V power supply; All functions handled by Door Control Panel are not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100013	Open circuit conditions detected on output pin.	Circuit Open	All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. DCP internal failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100016	Feedback voltage measured on the output is lower than 11,4V threshold (configurable).	Circuit Voltage Below Threshold	All functions handled by Door Control Panel may not work as expected.	1. Harness issue. 2. DDM internal HW issue. 3. DCP internal HW failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100017	Feedback voltage measured on the output is higher 14V threshold (configurable).	Circuit Voltage Above Threshold	All functions handled by Door Control Panel may not work as expected.	1. Harness issue. 2. DDM internal HW issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100111	Output pin short circuit to GND.	Circuit Short To Ground	Turns off Window Antipinch 12V power supply. Power window function is not working as expected.	1. Harness issue. 2.Antipinch internal short circuit.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100112	Output pin short circuit to BAT+.	Circuit Short To Battery	Turns off Window Antipinch 12V power supply. Power window function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100113	Open circuit conditions detected on output pin.	Circuit Open	Power window function is not working as expected.	1. Harness issue. 2. Antipinch internal failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100116	Feedback voltage measured on the output is lower than 11,4V threshold (configurable).	Circuit Voltage Below Threshold	Power window function may not work as expected.	1. Harness issue. 2. DDM internal HW issue. 3. Antipinch internal HW failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100117	Feedback voltage measured on the output is higher 14V threshold (configurable).	Circuit Voltage Above Threshold	Power window function may not work as expected.	1. Harness issue. 2. DDM internal HW issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	U005588	CAN Security failure detected.	Bus off	Driver Door Module does not communicate on CAN bus. One or more functions of Driver Door Module are not working as expected.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to Security subnet.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	CAN Communication Security Net
DM	U300005	Not supported SW configuration; Invalid parameters settings;	System Programming Failures	One or more functions of Driver Door Module are not working as expected.	1. SW release issue; 2. Not supported truck configuration build; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U300041	Flash memory HW failure.	General Checksum Failure	MSW invalidation after ECU reset: DTC reported by Primary Boot Loader. Driver Door Module are not working as expected	SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U300044	RAM corruption detected.	Data Memory Failure	ECU reset.	1. Invalid SW processing; 2. RAM corruption (intermittent fault due to working environment); 3. RAM physical issue (continuous defect); SW or HW complex system issue, to be rised to back-office	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U300045	Program memory corruption.	Program Memory Failure	MSW invalidation after ECU reset. Driver Door Module are not working as expected.	1. SWDL invalid process (cut of power during SWDL, etc) 2. Flash memory physical defect; 3. Invalid SW operation; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U300046	Dataset memory corruption.	Calibration / Parameter Memory Failure	MSW invalidation after ECU reset. Driver Door Module are not working as expected	1. Incorrect dataset download/change (e.g. cut of power during change). 2. Invalid SW operation (mainly in case of parameters change processing); 3. Flash memory physical defect; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U300049	Internal electronic failure.	Internal Electronic Failure	One or more functions of Driver Door Module are not working as expected.	Internal electronic failure: - µC peripheric circuit invalid state; - LIN tranceiver circuit broken; - CAN tranceiver circuit broken; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U30004A	Power window function with antipinch activated on not valid HW. Electrical Power window function activated on not valid HW.	Incorrect Component Installed	Power window function is not working as expected.	Not correct component installed or not valid SW configuration downloaded. SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Contact back office.	Electronic control unit
DM	U300055	No specific SW dataset downloaded.	Not Configured	One or more functions of Driver Door Module are not working as expected.	No valid SW configuration downloaded: - no valid LIN schedule table selected through dataset - no valid latch motor type selected through dataset - no valid window motor type selected through dataset - no valid mirror motor type selected through dataset SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Contact back office.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U300094	SW unexpected behaviour:	Unexpected Operation	ECU reset.	1. SW bug; - unexpected SW operation executed; - watchdog reset performed; - overrun situation detected; - unsynchronized communication circuit state detected; 2. RAM corruption; 3. Flash corruption; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Driver Door Module Unit.	Electronic control unit
DM	U300316	Power supply voltage is lower than 16V (configurable).	Circuit Voltage Below Threshold	One or more functions of Driver Door Module are not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the Driver Door Module Unit. - Inspect the harness and connectors.	Battery Power
DM	U300317	Power supply voltage is higher than 32V (configurable).	Circuit Voltage Above Threshold	One or more functions of Driver Door Module are not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the Driver Door Module Unit. - Inspect the harness and connectors.	Battery Power
DM	U1FFF68	Factory Mode is active in the ECU.	Event Information	No functional impact.	P1C54 parameter is set to TRUE.	Diagnostic Service 14. No additional action is needed.	Exit Factory mode.	Factory Mode Active
DM	B100D13	Both lock & unlock signals stay in Open Circuit state longer than allowed (1500 ms configurable) after electrical lock or unlock request.	Circuit Open	if Open Circuit is on lock sensor : -for a lock command, the latch will be lock and unlock 1s later. -for an unlock command,the latch will be unlock. if Open Circuit is on unlock sensor : -for a lock command, the latch will be lock. -for an unlock command, the latch will be unlock.	1. Harness issue; 2. Latch failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch lock unlock sensors
DM	B100411	Output pin short circuit to GND	Circuit Short To Ground	Locking function is not working as expected : For Lock command, it will not be executed. For Unlock command, the latch will be unlock.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 1
DM	B100412	Output pin short circuit to BAT+	Circuit Short To Battery	Locking function is not working as expected : For Lock command, the latch will be lock For Unlock command, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 1
DM	B100511	Output pin short circuit to GND.	Circuit Short To Ground	Locking function is not working as expected : For Lock command, the latch will be lock. For Unlock command, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100512	Output pin short circuit to BAT+	Circuit Short To Battery	Locking function is not working as expected : For Lock command, it will not be executed. For Unlock command, the latch will be unlock.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 2
DM	B150513	Open circuit conditions detected on output pin.	Circuit Open	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Actuator Circuit
DM	B100D11	Both lock & unlock signals stay in SC to ground state.	Circuit Short To Ground	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch lock unlock sensors
DM	B150712	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is open. If sensor has blown, door will always be considered as closed; - if short circuit with protected output, this output will be affected when door is open (short to ground from concerned output perspective);	1. Harness issue; 2. Driver Door Module failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Open Sensor
DM	B150812	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is locked. If sensor has blown, door will always be considered as unlocked. - if short circuit with protected output, this output will be affected when door is locked (short to ground from concerned output perspective);	1. Harness issue; 2. Driver Door Module failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Lock Sensor
DM	B150912	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is unlocked. If sensor has blown, door will always be considered as locked. - if short circuit with protected output, this output will be affected when door is unlocked (short to ground from concerned output perspective);	1. Harness issue; 2. Driver Door Module failure	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Unlock Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100211	Output pin short circuit to GND.	Circuit Short To Ground	Switch off output until : 1. Next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp mounting issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100212	Output pin short circuit to BAT+.	Circuit Short To Battery	Switch off output until : 1. Next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100213	Open circuit conditions detected on output pin.	Circuit Open	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100311	Output pin short circuit to GND.	Circuit Short To Ground	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp mounting issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit
DM	B100312	Output pin short circuit to BAT+.	Circuit Short To Battery	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit
DM	B100313	Open circuit conditions detected on output pin.	Circuit Open	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit
DM	B100811	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Mirrors adjustment actuators common circuit
DM	B100812	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Mirrors adjustment actuators common circuit
DM	B100911	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100912	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100913	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100919	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100A11	Output pin short circuit to GND	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A12	Output pin short circuit to BAT+	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A13	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100B11	Output pin short circuit to GND	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B12	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B13	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100C11	Output pin short circuit to GND	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100C12	Output pin short circuit to BAT+	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100C13	Open circuit conditions detected on output pin	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100C19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100611	Output pin short circuit to GND.	Circuit Short To Ground	Power window function is not working as expected. - For window down request, window will be driven down. - For window up request, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 1
DM	B100612	Output pin short circuit to BAT+.	Circuit Short To Battery	Power window function is not working as expected. - For window down request, it will not be executed. - For window up request, window will be driven up.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 1
DM	B100711	Output pin short circuit to GND.	Circuit Short To Ground	Power window function is not working as expected. - For window down request, it will not be executed. - For window up request, window will be driven up.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 2
DM	B100712	Output pin short circuit to BAT+.	Circuit Short To Battery	Power window function is not working as expected. - For window down request, window will be driven down. - For window up request, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 2
DM	B150613	Open circuit conditions detected on output pin	Circuit Open	Power window function is not working as expected. - For window down request, it will not be executed. - For window up request, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150618	Feedback load current measured on the output is lower than defined threshold. Threshold is dynamic and based on multicriteria formula (voltage, temperature and motor type).	Circuit Current Below Threshold	Power window function is not working as expected.	1. Anormal window mechanical resistance, 2. Power motor issue (Thermal switch open, etc) 3. Door sealing ageing.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor
DM	B150619	Feedback load current measured on the output is higher than defined threshold. Threshold is dynamic and based on multicriteria formula (voltage, temperature and motor type).	Circuit Current Above Threshold	Power window function is not working as expected.	1. Anormal window mechanical resistance, 2. Power motor issue, 3. Door sealing ageing.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor
DM	U114668	Maximum number of DTC reached: 29 DTC.	Event Information	No new DTC set until the failure is removed or DTC list cleared.	1. Unexpected environment conditions; 2. ECU failure	Diagnostic Service 14. No additional action is needed.	- Clear all DTC.	DTC Notification Max Limit Reached
xBS	U014087	The error is set if messages from VMCU are not comming or missing.	Missing Message	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Failure is out of EAS system.	Diagnostic Service 14. No additional action is needed.	Failure is out of EAS system. Check ECU:s transmitting VMCU signals and connected components.	Lost Communication With VMCU
xBS	U300007	The error is set if the ECU's moisture indicator indicates too high moisture content.	Mechanical Failures	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Mechanical damage of the EAS ECU's housing or damage of the EAS ECU, connector	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU and check connector	Electronic control unit
xBS	U300041	Checksum error detetced in the application software	General Checksum Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Previously interrupted flash download, or damage of the EAS ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace EAS ECU.	Electronic control unit
xBS	U300044	The error is set if RAM error in EAS ECU is detected	Data Memory Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the EAS ECU (defect RAM cel)	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U300045	Error detected in the application software memory	Program Memory Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the EAS ECU (defect program memory)	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU.	Electronic control unit
xBS	U300046	Error is set if incompatibilities between EEPROM dataset and drive software, hardware or bootloader versions are found, if there are unplausible data in the dataset, or if EEPROM has a failure	Calibration / Parameter Memory Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Main switch (power supply of ECU) was switched off while EAS afterrun, incorrect data in the dataset, dataset inconsistent to ECU, damaged EAS ECU (EEPROM)	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace EAS ECU.	Electronic control unit
xBS	U300049	An ECU self monitoring function detected an ECU hardware failure	Internal Electronic Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of EAS ECU	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU.	Electronic control unit
xBS	U30004B	Temperature monitoring is carried out by reading the hardware status information of the ECU.	Over Temperature	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Ambient temperature too high or damage of the EAS ECU	Diagnostic Service 14. No additional action is needed.	Check ECU location and suroundings for high temperatures. - If fault is active, replace it.	Electronic control unit
xBS	U300094	An ECU plausibility monitoring function detected an ECU hardware failure	Unexpected Operation	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of EAS ECU	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU	Electronic control unit
xBS	U300096	The error is set if not safety critical components of the ECU are damaged	Component Internal Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the EAS ECU	Diagnostic Service 14. No additional action is needed.	Replace EAS ECU	Electronic control unit
xBS	U002888	The error is set if Chassis CAN is in bus off status.	Bus off	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damaged Chassis CAN network wiring and/or connectors (e.g. short circuit), electro-magnetic disturbance, damaged EAS ECU, other damaged Chassis CAN ECU's	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors and wiring nodes between other Chassis CAN ECU's and EAS ECU. Check for sufficient power supply. If no fault found, check if no disturbing improper devices are mounted. If error still active check with other chassis CAN ECU's. If error still active check with other EAS ECU.	CAN Communication Chassis Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U300316	Failure is set if supply voltage of the EAS ECU is out of range (supply voltage is equal or lower than 15V).	Circuit Voltage Below Threshold	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Battery is not charged enough or damage of the vehicle's power supply system, or damage in the wiring or connectors for the EAS ECU supply	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
xBS	U300317	Failure is set if supply voltage of the EAS ECU is out of range (supply voltage is equal or higher than 36V).	Circuit Voltage Above Threshold	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the vehicle's power supply system	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system.	Battery Power
xBS	U30031C	Failure is set if supply voltage of the EAS ECU is out of range (supply voltage is out of [19.5 ... 32]V).	Circuit Voltage Out of Range	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Battery is not charged enough or damage of the vehicle's power supply system, or damage in the wiring or connectors for the EAS ECU supply	Diagnostic Service 14. No additional action is needed.	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
xBS	C005122	The failure is reported if the sensor signal exceeds the maximum possible range of the steering wheel angle set by EoL-parameter.	Signal Amplitude Greater Than Maximum	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Sensor is calibrated or mounted wrongly or steering system does not correspond to the vehicle specification (wrong steering gearbox or steering gear arm) or wrong maximum parameter in dataset or steering wheel angle sensor damaged.	Diagnostic Service 14. No additional action is needed.	Check if sensor is calibrated correctly: with engine running, set steered wheels in straight position, read steering angle via diagnosis, check if the value of steering angle is 0 +- 10 degrees. If value is not OK, recalibrate the sensor. While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle values via diagnosis (+-360 degrees). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005127	The failure is reported, if the steering angle sensor signal changes too fast.	Signal Rate of Change Above Threshold	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Mounting or drive of the sensor defect or sensor defect.	Diagnostic Service 14. No additional action is needed.	Check the mounting and the drive of the sensor. If fault still active, replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005128	The failure is reported if the offset of the steering angle sensor is higher than 90 degrees (depending on dataset parameter) or if the fast offset adjustment of the sensor (started via diagnosis) results in a high offset (90 degrees depending on dataset parameter).	Signal Bias Level Out of Range / Zero Adjustment Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Sensor is calibrated or mounted wrongly or wrong parameter for threshold in dataset or steering angle sensor is damaged.	Diagnostic Service 14. No additional action is needed.	Check the mounting of the sensor and check if sensor is calibrated correctly (+-10 degree in straight line position). If necessary recalibrate the sensor. If fault is still active check steering system for specified parts. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C005149	The steering angle sensor has an internal supervision which monitors the operation of the sensor and indicates its status via CAN-Sensor. The EAS ECU monitors the sensor status flags and the range of the sensor values.	Internal Electronic Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Internal sensor error	Diagnostic Service 14. No additional action is needed.	Replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005154	This failure is reported if the End Of Line Calibration of the steering angle sensor is not finished. (SAS fast adjustment calibration routine has not been started).	Missing Calibration	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	ECU or SAS was replaced or initial dataset reloaded.	Diagnostic Service 14. No additional action is needed.	Recalibrate the steering angle sensor. (Start fast adjustment routine)	Steering Wheel Position Sensor
xBS	C005155	This failure is reported if a steering angle sensor is detected, however it is not configured.	Not Configured	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Installed but not configured (in ECU) component	Diagnostic Service 14. No additional action is needed.	Check if configuration matches installation. Download correct dataset.	Steering Wheel Position Sensor
xBS	C005161	Not calibrated status of the steering angle sensor is detected by the EAS ECU.	Signal Calculation Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	New steering angle sensor was mounted or steering angle sensor was reset or calibration of steering angle sensor was not successful or sensor unit is damaged or EAS ECU is damaged	Diagnostic Service 14. No additional action is needed.	Calibrate the sensor again. If calibration cannot be performed successfully, check with an other sensor. If it still not possible to calibrate the sensor, check with other ECU.	Steering Wheel Position Sensor
xBS	C005176	The failure is reported, if the sign of the steering angle sensor signal is incorrect.	Wrong Mounting Position	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Sensor is mounted wrongly or the steering angle sensor is damaged.	Diagnostic Service 14. No additional action is needed.	While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle sign values via diagnosis (+-360 degrees, sign should be negative while turning left and positive while turning right). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005193	The failure is reported if after a vehicle standstill a left and a right curve has been detected, but the steering angle sensor signal has not changed.	No Operation	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Sensor is mounted wrongly or the steering angle sensor is damaged.	Diagnostic Service 14. No additional action is needed.	Check correct mounting or damages of the sensor. If mounting is OK, then replce the sensor.	Steering Wheel Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U012687	The error is set if the EAS ECU does not receive any messages from the steering angle sensor.	Missing Message	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damaged wiring and/or connectors between steering angle sensor and EAS ECU steering angle sensor is not connected to the EAS ECU, damaged steering angle sensor, damaged EAS ECU.	Diagnostic Service 14. No additional action is needed.	Check the wiring and connectors between steering angle sensor and EAS ECU and external yaw rate sensor and EAS ECU for open circuit and physical damage (including check for chafing damage), check the power supply of the ESP sensors 8,3V - 14,5V, if error still active, replace steering angle sensor, if error still active, replace EAS ECU.	Lost Communication With Steering Angle Sensor Module
xBS	C100C01	"The 12V voltage and the current of Steering Angle sensor supply are monitored (hardware feedback signals on central ECU). The error is set if there is a parasitic voltage greater then - Above 4.5V before switching ON sensor supply - Overcurrent is detected during normal operation (200-600 mA). - External sensor sends voltage error information."	General Electrical Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damage of the Steering Angle sensor wiring or connectors, or damage of the central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and Steering Angle sensor for short-circuits. 1. Remove sensor and check for faultcodes again. 2. If still a problem with sensors removed the problem is in the harness or ECU. 3. Check wiring for shortcuts to ground or other signals and CNT (below 3 ohm). Repair if faulty 4. If harness is ok and there is still active faultcode, replace ECU. 5. If harness is ok and no active fault try to move harness to reproduce the fault.	EBS ESP Sensor Supply
xBS	C100C16	The error is set, if the Steering Angle sensor supply voltage is lower than the working limit (8.3 [V]).	Circuit Voltage Below Threshold	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damage of the Steering Angle sensor wiring or connectors, or damage of the central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and Steering Angle sensor for short-circuits. If error is still active replace central ECU.	EBS ESP Sensor Supply
xBS	C100C17	The error is set, if the Steering Angle Sensor supply voltage is higher than the working limit (14.5 [V]).	Circuit Voltage Above Threshold	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damage of the Steering Angle sensor wiring or connectors, damage of the central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and Steering Angle sensor for short-circuits to other signals higher than 12 V. If error is still active replace central ECU.	EBS ESP Sensor Supply
xBS	C100C49	Hardware test is performed and feedback is monitored.	Internal Electronic Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Damaged central ECU	Diagnostic Service 14. No additional action is needed.	Replace central ECU	EBS ESP Sensor Supply
xBS	U012987	The error is set if messages from brakes are not coming or missing.	Missing Message	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Failure is out of EAS system.	Diagnostic Service 14. No additional action is needed.	Failure is out of EAS system. Check ECU:s transmitting EBS signals and connected components.	Lost Communication With Brake System Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U116968	The error is set if the number of DTCs to be sent to diagnostics fault status message exceeds 29.	Event Information	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Over 29 different faults set	Diagnostic Service 14. No additional action is needed.	Failure is out of EAS system. Check ECU:s transmitting EBS signals and connected components.	DTC Notification Max Limit Reached
xBS	U116A68	CRC or message counter error in external Steering Angle Sensor messages.	Event Information	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Failure is out of EAS system.	Diagnostic Service 14. No additional action is needed.	Failure is out of EAS system.	Steering Angle Sensor Message Checksum
xBS	C100811	Error is set if the steering angle sensor supply can not be switched on or no valid supply voltage for the steering angle sensor is available during switched on. The supply voltage is less than 4.75V measured on the ECU pin.	Circuit Short To Ground	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between steering angle sensor and central ECU, damaged sterring angle sensor, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and rear axle steering angle sensor for short-circuits. The insulation resistance should be higher than 200kOhm. Disconnect the sensor, switch the ECU to After Market mode and measure the voltage of the sensor supply. If it's lower than 4.8V then replace the central ECU else replace rear axle steering angle sensor.	Rear Active Steering Angle Sensor
xBS	C100816	Error is set if any of the two rear sensors signal voltages is less than 0.2V.	Circuit Voltage Below Threshold	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between steering angle sensor and central ECU, damaged sterring angle sensor, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and steering angle sensor signal for short-circuits and open-circuits.The insulation resistance should be higher than 200 kOhm and the wire resistance should be less than 5 Ohm. Switch the ECU to After Market mode and measure the voltage of the sensor signal. If it's lower than 0.4V (end of line parameter) then replace rear axle steering angle sensor else replace the central ECU.	Rear Active Steering Angle Sensor
xBS	C100817	Error is set if any of the two rear sensors voltage is more than 4.8V	Circuit Voltage Above Threshold	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between steering angle sensor and central ECU, damaged sterring angle sensor, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and steering angle sensor signal for short-circuits.The insulation resistance should be higher than 200 kOhm. Switch the ECU to After Market mode and measure the voltage of the sensor signal. If it's higher than 4.6V (end of line parameter) then replace rear axle steering angle sensor else replace the central ECU.	Rear Active Steering Angle Sensor
xBS	C10081C	Error is set if sum of the rear sensors' voltage less than 4.7V or greater than 5.3V	Circuit Voltage Out of Range	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between steering angle sensor and central ECU, damaged sterring angle sensor, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and steering angle sensor signal for short-circuits and open-circuits. The insulation resistance should be higher than 200 kOhm and the wire resistance should be less than 5 Ohm. If the error is not active, then the axle must turn between the 2 end positions (by diagnostics) while the sensor voltages are measured. If the error is continuously active then switch the ECU to After Market mode and measure the voltage of both sensor signals. If sum of the signals is lower than 4.8V (end of line parameter) or higher than 5.2 then replace rear axle steering angle sensor else replace the central ECU (only if the error is continuously present and the sum of the sensor voltages measured by external tool is correct).	Rear Active Steering Angle Sensor
xBS	C100854	Error is set if the steering angle sensor offset is not calibrated.	Missing Calibration	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Sensor is not calibrated.	Diagnostic Service 14. No additional action is needed.	Calibrate sensor, if not succesful reset the sensor with the diagnostic tool, and then calibrate it again. If calibration cannot be performed successfully, check mechanical parts, specs and other active failures. If error still active replace the sensor.	Rear Active Steering Angle Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100911	The error is set if status feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between proportional valve and central ECU, damaged proportional valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kOhm. If the resistance of the proportional valve is lower than 20 Ohm replace the proportional valve. If error is still active replace the central ECU.	Rear Active Steering Proportional Valve A
xBS	C100915	The error is set if status feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between proportional valve and central ECU, damaged proportional valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and proportional valve for short-circuits and for open circuit. The insulation resistance must be higher than 200 kOhm and the wire resistance should be less than 3 Ohm. If the resistance of the proportional valve is higher than 30 Ohm replace the proportional valve. If error is still active replace the central ECU.	Rear Active Steering Proportional Valve A
xBS	C100A11	The error is set if status feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between proportional valve and central ECU, damaged proportional valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kOhm. If the resistance of the proportional valve is lower than 20 Ohm replace the proportional valve. If error is still active replace the central ECU.	Rear Active Steering Propotional Valve B
xBS	C100A15	The error is set if status feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between proportional valve and central ECU, damaged proportional valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and proportional valve for short-circuits and for open circuit. The insulation resistance must be higher than 200 kOhm and the wire resistance should be less than 3 Ohm. If the resistance of the proportional valve is higher than 30 Ohm replace the proportional valve. If error is still active replace the central ECU.	Rear Active Steering Propotional Valve B
xBS	C100B11	The error is set if status feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between cut-off valve and central ECU, damaged cut-off valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and cutoff valve for short-circuits. The insulation resistance must be higher than 200 kOhm. If the resistance of the cutoff valve is lower than 18 Ohm replace the cutoff valve. If error is still active replace the central ECU.	Rear Active Steering Cutoff Valve
xBS	C100B15	The error is set if status feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Damage of the wiring or connector between cut-off valve and central ECU, damaged cut-off valve, damaged central ECU	Diagnostic Service 14. No additional action is needed.	Check the cables and connectors between central ECU and cutoff valve for short-circuits and for open circuit. The insulation resistance must be higher than 200 kOhm and the wire resistance should be less than 3 Ohm. If the resistance of the cutoff valve is higher than 28 Ohm replace the cutoff valve. If error is still active replace the central ECU.	Rear Active Steering Cutoff Valve
xBS	C100B71	The error is set if the angle moves more than 2 degrees (end of line parameter) during a test where the cutoff valve is opened and a taget value is requested.	Actuator Stuck	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Valve is stuck in closed position	Diagnostic Service 14. No additional action is needed.	Replace hydraulic valve	Rear Active Steering Cutoff Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109B68	Fault is set if one of the external CAN input signals of RASec is not available or the signal indicates an error: - RearAxleSteeringFunctionDisable (RAS switch status) - EngineSpeed - WheelBasedVehicleSpeed (TACHO speed) - PusherAxleLoad - TagAxleLoad - LiftAxle1Position - LiftAxle2Position	Event Information	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	Disturbed CAN communication, faulty other component, faulty other ECU (outside of the EBS ECU)	Diagnostic Service 14. No additional action is needed.	Check other ECUs for related fault codes	Rear Active Steering Function
xBS	C108B02	The error is set if the difference between the calibrated vehicle speed and mean drive axle wheelspeed is higher than 20 km/h (EOL parameter)	General Signal Failure	Normal function. Highest speed of mean front axle wheel speed and vehicle speed is used	Incorrect calibration of vehicle speed, wheel speed sensor failure	Diagnostic Service 14. No additional action is needed.	Check other related fault codes. Check tyre sizes of the driven wheels. If tyre sizes are correct, check tachosignal (according to regulations). If tachosignal is not correct, check components related to tachometer.	Rear Active Steering Inter Axle Vehicle Speed
xBS	C108A01	The error is set if the compensation offset gets higher than 2 degrees (EOL parameter)	General Electrical Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Incorrect wheel alignment, rear axle angle sensor value drift	Diagnostic Service 14. No additional action is needed.	Align wheels and recalibrate	Rear Active Steering Rear Angle
xBS	C108A64	The error is set if rear steering angle is too high (EOL parameter)	Signal Plausibility Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	ECU failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Rear Active Steering Rear Angle
xBS	C108A77	The error is set if the angle is deviating too much from the target lower or higher than expected. The accepted deviation between angle and target is defined by an EOL characteristics.	Commanded Position Not Reachable	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	Any hydraulic failure in the vehicle (low hydraulic oil level, hydraulic pump) or hydraulic valve/cylinder failure	Diagnostic Service 14. No additional action is needed.	Check hydraulic oil level, check hydraulic pump, check hydraulic valve/cylinder	Rear Active Steering Rear Angle

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108964	The error is set if both proportional valves are activated at the same time	Signal Plausibility Failure	Rear axle steering will be in passive failure mode i.e. the axle will be self steered and hydraulically damped. Risk of damaging valves when reversing.	ECU failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Rear Active Steering Control Valves
xBS	C102C11	Internal valve short to ground.	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Outlet Solenoid
xBS	C102C12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Outlet Solenoid
xBS	C102C13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Outlet Solenoid
xBS	C102D11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Outlet Solenoid
xBS	C102D12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Outlet Solenoid
xBS	C102D13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Outlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102E11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Outlet Solenoid
xBS	C102E12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Outlet Solenoid
xBS	C102E13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Outlet Solenoid
xBS	C102F11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Outlet Solenoid
xBS	C102F12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Outlet Solenoid
xBS	C102F13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Outlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103011	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Outlet Solenoid
xBS	C103012	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Outlet Solenoid
xBS	C103013	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Outlet Solenoid
xBS	C103111	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Outlet Solenoid
xBS	C103112	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Outlet Solenoid
xBS	C103113	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Outlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103211	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Outlet Solenoid
xBS	C103212	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Outlet Solenoid
xBS	C103213	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Outlet Solenoid
xBS	C103311	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Outlet Solenoid
xBS	C103312	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Outlet Solenoid
xBS	C103313	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Outlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103411	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Outlet Solenoid
xBS	C103412	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Outlet Solenoid
xBS	C103413	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Outlet Solenoid
xBS	C103511	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Backup Solenoid
xBS	C103512	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Backup Solenoid
xBS	C103513	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102111	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Inlet Solenoid
xBS	C102112	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Inlet Solenoid
xBS	C102113	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Inlet Solenoid
xBS	C102211	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Inlet Solenoid
xBS	C102212	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Inlet Solenoid
xBS	C102213	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Inlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102311	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Inlet Solenoid
xBS	C102312	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Inlet Solenoid
xBS	C102313	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Inlet Solenoid
xBS	C102411	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Inlet Soleniod
xBS	C102412	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Inlet Soleniod
xBS	C102413	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Inlet Soleniod

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102511	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Inlet Solenoid
xBS	C102512	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Inlet Solenoid
xBS	C102513	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Inlet Solenoid
xBS	C102611	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Inlet Solenoid
xBS	C102612	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Inlet Solenoid
xBS	C102613	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Inlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102711	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Inlet Solenoid
xBS	C102712	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Inlet Solenoid
xBS	C102713	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Inlet Solenoid
xBS	C102811	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Inlet Solenoid
xBS	C102812	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Inlet Solenoid
xBS	C102813	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Inlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102911	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Inlet Solenoid
xBS	C102912	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Inlet Solenoid
xBS	C102913	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Inlet Solenoid
xBS	C102A11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Inlet Solenoid
xBS	C102A12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Inlet Solenoid
xBS	C102A13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Inlet Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102B11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Outlet Solenoid
xBS	C102B12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Outlet Solenoid
xBS	C102B13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 1 EPM Outlet Solenoid
xBS	C103611	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Backup Solenoid
xBS	C103612	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Backup Solenoid
xBS	C103613	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 2 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103711	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Backup Solenoid
xBS	C103712	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Backup Solenoid
xBS	C103713	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 3 EPM Backup Solenoid
xBS	C103811	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Backup Solenoid
xBS	C103812	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Backup Solenoid
xBS	C103813	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 4 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103911	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Backup Solenoid
xBS	C103912	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Backup Solenoid
xBS	C103913	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 5 EPM Backup Solenoid
xBS	C103A11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Backup Solenoid
xBS	C103A12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Backup Solenoid
xBS	C103A13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 6 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103B11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Backup Solenoid
xBS	C103B12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Backup Solenoid
xBS	C103B13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 7 EPM Backup Solenoid
xBS	C103C11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Backup Solenoid
xBS	C103C12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Backup Solenoid
xBS	C103C13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 8 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C103D11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Backup Solenoid
xBS	C103D12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Backup Solenoid
xBS	C103D13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 9 EPM Backup Solenoid
xBS	C103E11	Internal valve short to ground	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Backup Solenoid
xBS	C103E12	Internal valve short to battery	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Backup Solenoid
xBS	C103E13	Open circuit in the internal valve	Circuit Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Replace EPM, There is no meassuer to be done, the DTC is set in the EPM and send over the information to the EBS ECU. The error is also stored in the EPM.	Wheel 10 EPM Backup Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105349	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 1 EPM Pressure Sensor
xBS	C105449	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 2 EPM Pressure Sensor
xBS	C105549	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 3 EPM Pressure Sensor
xBS	C105649	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 4 EPM Pressure Sensor
xBS	C105749	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 5 EPM Pressure Sensor
xBS	C105849	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 6 EPM Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105949	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 7 EPM Pressure Sensor
xBS	C105A49	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 8 EPM Pressure Sensor
xBS	C105B49	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 9 EPM Pressure Sensor
xBS	C105C49	- Sensor signal is not plausible while not braking (offset failure) - Sensor signal is < 0,13V or too high (outside of valid signal range).	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged - Wrong piping - EPM damaged	Diagnostic Service 14. No additional action is needed	Check silencer. Check piping for wrong connections (supply and brake cylinder). If it is OK, replace EPM.	Wheel 10 EPM Pressure Sensor
xBS	C107429	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and PCVs (both sides). Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 1 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107462	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 1 EPM Pressure Control
xBS	C107471	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 1 EPM Pressure Control
xBS	C107472	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 1 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107473	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 1 EPM Pressure Control
xBS	C107529	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	"Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced."	Wheel 2 EPM Pressure Control
xBS	C107562	Difference between measured full pressure and supply pressure > 2.2 bar	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 2 EPM Pressure Control
xBS	C107571	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 2 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107572	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 2 EPM Pressure Control
xBS	C107573	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 2 EPM Pressure Control
xBS	C107629	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 3 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107662	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 3 EPM Pressure Control
xBS	C107671	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 3 EPM Pressure Control
xBS	C107672	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 3 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107673	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 3 EPM Pressure Control
xBS	C107729	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 4 EPM Pressure Control
xBS	C107762	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 4 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107771	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 4 EPM Pressure Control
xBS	C107772	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 4 EPM Pressure Control
xBS	C107773	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 4 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107829	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 5 EPM Pressure Control
xBS	C107862	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 5 EPM Pressure Control
xBS	C107871	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 5 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107872	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 5 EPM Pressure Control
xBS	C107873	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 5 EPM Pressure Control
xBS	C107929	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 6 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107962	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 6 EPM Pressure Control
xBS	C107971	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 6 EPM Pressure Control
xBS	C107972	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 6 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107973	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 6 EPM Pressure Control
xBS	C107A29	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 7 EPM Pressure Control
xBS	C107A62	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 7 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107A71	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 7 EPM Pressure Control
xBS	C107A72	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 7 EPM Pressure Control
xBS	C107A73	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 7 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107B29	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 8 EPM Pressure Control
xBS	C107B62	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 8 EPM Pressure Control
xBS	C107B71	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 8 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107B72	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 8 EPM Pressure Control
xBS	C107B73	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 8 EPM Pressure Control
xBS	C107C29	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 9 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107C62	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 9 EPM Pressure Control
xBS	C107C71	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 9 EPM Pressure Control
xBS	C107C72	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 9 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107C73	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 9 EPM Pressure Control
xBS	C109541	One of the EPMs has a checksum error	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	EPM damaged	Diagnostic Service 14. No additional action is needed	Start system with ignition Off/On check if another EPM faultcode is active, faulttrace the new DTC, Download correct drive software into the EPM ,if the original error is still active, disconnect all EPMs connect EPMs sekvential identify the faulty EPM replace faulty EPM.	Electro-Pneumatic Module
xBS	C109546	EPM sent reconfiguration request more than once in the same power-on cycle.	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - Wrong EPM HW type connected	Diagnostic Service 14. No additional action is needed	Download a new dataset, if the error is still active, Chk power supply for the correct EPM, replace EPM	Electro-Pneumatic Module
xBS	C109555	EPM is detected, but not configured.	Not Configured	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - Installed but not configured component	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Download correct/new dataset to the ECU. Check the wiring (for the correct power supply is connected to the correct EPM.) from central ECU to EPM.	Electro-Pneumatic Module
xBS	C109594	EPM Bootloader detected a drive software incompatibility error with the ECU hardware type.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Incompatible EPM software parts (ECU hardware vs. drive sw, bootloader vs. drive software) - EPM damaged	Diagnostic Service 14. No additional action is needed	Download correct drive software into the EPM; If error is still active, replace EPM	Electro-Pneumatic Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107D29	Difference between measured full pressure and supply pressure > 2.2 bar	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged or plugged piping from pressure supply to EPM - Damaged or open piping from EPM to PCV - Received supply pressure values from APM incorrect - EPM damage - PCV damage	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Control maximum pressure at the corresponding EPM via diagnostics. Read pressure value via diagnostics. Measure pressure in brake chamber externally with 2% precision. Compare the two pressure values with each other and with the defined nominal maximal braking pressure. If measured pressures are smaller than (nominal maximal braking pressure - 1bar), then EPM should be replaced. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should also be replaced.	Wheel 10 EPM Pressure Control
xBS	C107D62	Difference between right and left on on axle > 0.8 bar (Axles with 2 independent channels are monitored)	Signal Compare Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged, - Damaged pneumatic pipe (leakage) between EPM and wheel brake - EPM damaged - Damaged pneumatic pipe between EPM and supply pressure if 2x1CH is used for one axle.	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure (only for 2x1CH on same axle), and between EPM and brake chamber. Check silencer. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at either EPM on the corresponding axle via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demaned by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced. Make the same test with the other EPM on the axle.	Wheel 10 EPM Pressure Control
xBS	C107D71	- At beginning of braking the EPMs of one axle are switched in backup-mode: Backup pressure < pressure level dependent on FBM stroke. - At end of braking the EPM is switched to backup-mode: Exhaust performance is not sufficient.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication	- Silencer plugged in EPM or FBM - Pneumatic pipes between FBM and EPM open or plugged - FBM damaged (mechanic/pneumatic part) - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check backup piping between FBM and EPM. If there is no problem with pneumatic piping and silencer: Ensure that reservoirs are fully charged (until cut-off pressure). Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached. If difference between the two externally measured pressure is greater than 0.5 bar + 5% then EPM should also be replaced. Otherwise, if DTC is still active, replace FBM. Release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 10 EPM Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107D72	Difference between pressure measured by EPM and demanded pressure > 1.5 bar	Actuator Stuck Open	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Silencer plugged in EPM or FBM - EPM Damaged - FBM damaged - Residual backup pressure between FBM and EPM	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between FBM and EPM. Check silencer. Check components for possible leakages (mainly FBM). If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics to 8 bar, and then control pressure from 8 to 1 bar in 1 step. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If pressure is greater than 1.5 bar 1 s after reducing demand pressure to 1 bar, then check EPM silencer. If EPM silencer is OK then EPM should be replaced. Drive the system to backup via diagnostics. Measure backup pressure and pressure in brake chamber externally with 2% precision. Apply brake pedal until ca. 2 bar is reached then release brake pedal fast. If backup pressure is greater than 0.3 bar 1 s after releasing brake pedal then check FBM silencer. If FBM silencer is OK, then FBM should be replaced. If backup pressure is less than 0.3 bar 1 s after releasing brake pedal but pressure in the brake chamber is still greater than 0.3 bar, then check EPM silencer. If EPM silencer is OK, then EPM should be replaced.	Wheel 10 EPM Pressure Control
xBS	C107D73	- Difference between pressure measured by EPM and demanded pressure > 1.5 bar - Pressure sensor signal> 2 bar when applying full brake. - Pressure sensor signal remains < demanded pressure and supply pressure during inlet valve protection mode of the EPM pressure control.	Actuator Stuck Closed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged piping or part between EPM and wheel brake - Damaged piping or part between EPM and supply pressure - Received supply pressure values from APM incorrect - EPM damaged	Diagnostic Service 14. No additional action is needed	Check pneumatic piping between EPM and supply pressure, between EPM and brake chamber. Check piping and components for possible leakages. If not problem was seen: Ensure that reservoirs are fully charged (until cut-off pressure). Measure pressure in supply reservoir and compare with read out pressure from APM via diagsnotics. Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. If difference between pressure demanded by diagnostics and value read by diagnostics is greater than 0.5 bar + 5%, then EPM should be replaced.	Wheel 10 EPM Pressure Control
xBS	C105D11	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 1 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105D12	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 1 EPM Power Supply
xBS	C105D1B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 1 EPM Power Supply
xBS	C105D95	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on..	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 1 EPM Power Supply
xBS	C105E11	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 2 EPM Power Supply
xBS	C105E12	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 2 EPM Power Supply
xBS	C105E1B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 2 EPM Power Supply
xBS	C105E95	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 2 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105F11	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 3 EPM Power Supply
xBS	C105F12	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 3 EPM Power Supply
xBS	C105F1B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 3 EPM Power Supply
xBS	C105F95	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 3 EPM Power Supply
xBS	C106011	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 4 EPM Power Supply
xBS	C106012	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 4 EPM Power Supply
xBS	C10601B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 4 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106095	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 4 EPM Power Supply
xBS	C106111	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line	Wheel 5 EPM Power Supply
xBS	C106112	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 5 EPM Power Supply
xBS	C10611B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 5 EPM Power Supply
xBS	C106195	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 5 EPM Power Supply
xBS	C106211	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 6 EPM Power Supply
xBS	C106212	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 6 EPM Power Supply
xBS	C10621B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 6 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106295	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 6 EPM Power Supply
xBS	C106311	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 7 EPM Power Supply
xBS	C106312	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 7 EPM Power Supply
xBS	C10631B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 7 EPM Power Supply
xBS	C106395	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 7 EPM Power Supply
xBS	C106411	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 8 EPM Power Supply
xBS	C106412	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 8 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C10641B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 8 EPM Power Supply
xBS	C106495	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 8 EPM Power Supply
xBS	C106511	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 9 EPM Power Supply
xBS	C106512	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 9 EPM Power Supply
xBS	C10651B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 9 EPM Power Supply
xBS	C106595	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 9 EPM Power Supply
xBS	C106611	Supply voltage < 4V	Circuit Short To Ground	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check if generator diode is defect according to the manual of the generator. Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits. If no error found, disconnect EPM from central ECU, Wiring isolation resistances between EPM supply line and EPM ground line and between EPM supply line and vehicle ground should be above 500 Ohm. If this is not the case, disconnet supply connector from EPM, and repeat previous meauserement. If measurement is OK, then replace EPM, otherwise repair cabling. If isolation resistance with cables connected to EPM are above the limit of 500 Ohm recheck faultpresent if DTC still active then replace central ECU.	Wheel 10 EPM Power Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106612	Supply voltage > 4V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) (Exclude ShortCut between EPM supply.) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM (including check for chafing damage) for short circuits to other voltage source. Drive the system to full backup via diagnostics service, then disconect EPM supply from central ECU(Meassuer at the pin of the ECU), and measure EPM supply voltage output of central ECU. If voltage is higher than 4V then replace central ECU, otherwise repair wiring harness.	Wheel 10 EPM Power Supply
xBS	C10661B	Implausible voltage drop during valve activation	Circuit Resistance Above Threshold	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from central ECU to the corresponding EPM for too high resistance. Resistance of supply wires disconnected from both central ECU and EPM should be below 3 Ohm. Check connectors for corrosion.	Wheel 10 EPM Power Supply
xBS	C106695	Supply voltage of a switched-off EPM > 4V, while this EPM is switched off and another is switched on.	Incorrect Assembly	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) (Short curcuit between EPM supplies)	Diagnostic Service 14. No additional action is needed	Get the same failure for the other EPM from the fault storage. Check the cables and connectors from central ECU to both effected EPM:s (including check for chafing damage) for short circuits between each other, and repair wiring or connector.	Wheel 10 EPM Power Supply
xBS	C106709	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 1 EPM
xBS	C106712	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 1 EPM
xBS	C106714	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 1 EPM
xBS	C106729	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 1 EPM
xBS	C106741	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 1 EPM
xBS	C106746	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 1 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106749	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 1 EPM
xBS	C10674A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 1 EPM
xBS	C106794	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 1 EPM
xBS	C106809	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 2 EPM
xBS	C106812	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 2 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106814	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 2 EPM
xBS	C106829	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 2 EPM
xBS	C106841	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 2 EPM
xBS	C106846	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 2 EPM
xBS	C106849	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 2 EPM
xBS	C10684A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 2 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106894	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 2 EPM
xBS	C106909	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 3 EPM
xBS	C106912	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 3 EPM
xBS	C106914	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 3 EPM
xBS	C106929	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 3 EPM
xBS	C106941	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 3 EPM
xBS	C106946	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 3 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106949	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 3 EPM
xBS	C10694A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 3 EPM
xBS	C106994	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 3 EPM
xBS	C106A09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 4 EPM
xBS	C106A12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 4 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106A14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 4 EPM
xBS	C106A29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 4 EPM
xBS	C106A41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 4 EPM
xBS	C106A46	Faulty EEPROM memory or data is not consisten	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 4 EPM
xBS	C106A49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 4 EPM
xBS	C106A4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 4 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106A94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 4 EPM
xBS	C106B09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 5 EPM
xBS	C106B12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 5 EPM
xBS	C106B14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 5 EPM
xBS	C106B29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 5 EPM
xBS	C106B41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 5 EPM
xBS	C106B46	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 5 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106B49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 5 EPM
xBS	C106B4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 5 EPM
xBS	C106B94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 5 EPM
xBS	C106C09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 6 EPM
xBS	C106C12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 6 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106C14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 6 EPM
xBS	C106C29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 6 EPM
xBS	C106C41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 6 EPM
xBS	C106C46	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 6 EPM
xBS	C106C49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 6 EPM
xBS	C106C4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 6 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106C94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 6 EPM
xBS	C106D09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 7 EPM
xBS	C106D12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 7 EPM
xBS	C106D14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 7 EPM
xBS	C106D29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 7 EPM
xBS	C106D41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 7 EPM
xBS	C106D46	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 7 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106D49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 7 EPM
xBS	C106D4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 7 EPM
xBS	C106D94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, replace central ECU.	Wheel 7 EPM
xBS	C106E09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 8 EPM
xBS	C106E12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 8 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106E14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 8 EPM
xBS	C106E29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 8 EPM
xBS	C106E41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 8 EPM
xBS	C106E46	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 8 EPM
xBS	C106E49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 8 EPM
xBS	C106E4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only	Wheel 8 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106E94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 8 EPM
xBS	C106F09	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 9 EPM
xBS	C106F12	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 9 EPM
xBS	C106F14	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 9 EPM
xBS	C106F29	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 9 EPM
xBS	C106F41	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 9 EPM
xBS	C106F46	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 9 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C106F49	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 9 EPM
xBS	C106F4A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check/re-load configuration of EPM (also at other EPM:s where same DTC is reported). If after updating all EPM:s, these types of DTC:s are reported again regarding all EPM:s, then) Check the pinning of the power supply for the EPM at the ECU, check the wiring harness for wrong pinning, check/re-load configuration of central ECU. If after updating all EPM:s, these types of DTC:s are reported only regarding some of the EPM:s, replace those EPM:s	Wheel 9 EPM
xBS	C106F94	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 9 EPM
xBS	C107009	EPM Internal test indicating error	Component Failures	Driver informed by YELLOW warning and brake malfunction indication	- EBS system was not shut down properly (direct switch off with main power switch). - EPM damaged	Diagnostic Service 14. No additional action is needed	Let the EBS system shut down properly. Switch on the system again. If failure is no longer active, no further action is needed. If DTC 0x504912..0x505212 is active, check the lining wear sensor wiring/connection. If failure is still active replace EPM.	Wheel 10 EPM
xBS	C107012	ALS (Axle Load Sensor) supply > 6V.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. If error still active, disconnect the connector of axle load sensor at the EPM, measure voltage between vehicle ground and supply of the axle load sensor. If voltage is higher than 3V, then repair the wiring, otherwise replace the EPM.	Wheel 10 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107014	ALS (Axle Load Sensor) supply < 4.5V.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from axle load sensor to the corresponding EPM (including check for chafing damage) for short circuits. Check the axle load sensor for correct operation. Disconnect the connector of axle load sensor at the EPM, measure the resistance between the supply of the axle load sensor and the vehicle ground. If resistance is lower than 200kOhm, then repair wiring, if error still active, replace the EPM.	Wheel 10 EPM
xBS	C107029	Gradient of pressure sensor signal is implausible.	Signal invalid	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	Check piping (especially P1 supply pressure input); Check if supply pressure information is correct; Replace the EPM if piping and supply pressure information are correct.	Wheel 10 EPM
xBS	C107041	Invalid checksum is detected.	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 10 EPM
xBS	C107046	Faulty EEPROM memory or data is not consistent	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Mismatch between EPM software and EPM EEPROM, - EPM supply (UB) is switched off while EBS Power-down, - EPM damaged	Diagnostic Service 14. No additional action is needed	If the SWDownload is possible download the correct drive software into the EPM; If error is active or inactive replace the EPM. The faultcode will be detected in the EPM and stored in the EPM, and sent to the EBS ECU.	Wheel 10 EPM
xBS	C107049	- EPM hardware fault is detected - Special inactive DTC message is sent with wrong content (Dynamic built in test failed) - Pressure demand value sent back by the EPM does not match any of the last 6 pressure demand values from the central ECU.	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	If there is no Brake CAN communication DTC and the error is still active / inactive, replace EPM.	Wheel 10 EPM
xBS	C10704A	- ECU receives negative or incorrect response from an EPM upon the configuration check or configuration download messages - ECU detects incompatible EPM software parts.	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 10 EPM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107094	- Fault code received from the EPM does not match to any of the defined error code registered in the central ECU - EPM does not fulfill a required test task from the central ECU - Mismatches between EPM Bootloader, drive software and hardware - Difference in voltage measured by central ECU's and EPM's > 3V.	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Download software update to the EPM. If error is still active(if occurnce counter is higher then 2) and error is stored only for one EPM, replace the EPM. If error is active (if occurnce counter is higher then 2) reported for every EPM, If newer SW in Central system do SWDL, if error still occuer replace central ECU.	Wheel 10 EPM
xBS	C100411	Resistance between vehicle ground and PCV ground < 20 kOhm..	Circuit Short To Ground	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm If Values are not ok Replace PCV, Otherwise repair cabling harness.	Pressure Control Valve Ground Circuit
xBS	C100412	- Short to battery on all configured PCV outputs at the same time	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cable of the PCV ground and the cables and connectors to all installed PCV for short circuit to other wires (including check for chafing damage). Repair if faulty. If fault is still active, disconnect PCVs from the central ECU. Measure voltage on the connector between the wires of the PCVs and the vehicle ground. If the voltage is higher than 1V, repair the wiring harness. Otherwise measure the resistance between the PCV ground and the vehicle ground. If the resistance is lower than 200 kOhm, repair the wiring harness. Otherwise replace the central ECU.	Pressure Control Valve Ground Circuit
xBS	C100555	Valve (resistance to ground < 10kOhm) is detected, but not configured.	Not Configured	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - Installed but not configured component	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. If configuration matches installation, check the connector. If no fault is found with the connector, then check the wiring (including check for chafing damage), and repair if faulty. Insulation resistance of wiring (of the inlet and the outlet valve) towards vehicle and PCV ground must be higher than 200 kOhm.	Pressure Control Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100592	- Faulty ECU internal PCV test circuit - Short circuit to low voltage signal or weak short (50Ohm-1kOhm) circuit to battery.	Performance or Incorrect Operation	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check for other fault codes (also in other systems) and repair them. Erase fault codes and re-check the fault code. If error is still active, progress with the other actions. Check the wiring and connectors. Repair if faulty. Disconnect the wires from the central ECU. Measure resistance of PCV lines against the other signals of the connector. If any resistance is lower than 200 kOhm, then repair the faulty wires. Otherwise measure resistance between the ground of both PCVs and the vehicle ground. If all resistances are higher than 200 kOhm, then replace the central ECU. Else disconnect the wires also from the PCVs and measure resistance between the ground wires of both PCVs and the vehicle ground. If resistance is lower than 200 kOhm, then repair the faulty wires. If fault is still active, replace the central ECU.	Pressure Control Valve
xBS	C100616	Voltage < 4 V	Circuit Voltage Below Threshold	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM (including check for chafing damage). Disconnect central ECU and TCM. Check the cables and connectors for the pressure sensor for short cuts and interruptions, resistance of TCM supply and pressure sensor supply should be less than 3 Ohm. If higher then there is an interruption on wire. Resistance between pressure sensor supply connection and all wires of different components on the same connector and to vehicle ground should be larger than 200 kOhm. If no fault is found, then replug cable again, activate Aftermarket diagnostic mode and check the voltages between the ECU pins (Voltage of TCM supply should be in range [UB - 3V ... UB + 3V]. Voltage difference should be in range [4.5 ... 5.5] V between TCM sensor supply and sensor ground.). If the measured voltages are not in the given range replace TCM. If there a still an active fault (same fault code), replace the ECU.	Trailer Control Modulator pressure sensor supply circuit
xBS	C100617	Voltage > 6 V	Circuit Voltage Above Threshold	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM (including check for chafing damage). Activate AM mode. Disconnect TCM (on TCM side) and measure sensor supply wire harness to vehicle ground. Voltage should be below 1V. If above repair wire harness. Fault trace by resistance between pressure sensor supply connection and all wires of different components on the same connector should be larger than 200 kOhm. If the voltage is below 1V then connect TCM and measure again. If higher then 6 V replace TCM. If no fault is found, then replug cable again, activate Aftermarket diagnostic mode and check the voltages between the ECU pins (Voltage of TCM supply should be in range [UB - 3V ... UB + 3V]. Voltage difference should be in range [4.5 ... 5.5] V between TCM sensor supply and sensor ground.). If there a still an active fault (same fault code), replace the ECU. If there is no active fault code try to move the harness to reproduce the fault.	Trailer Control Modulator pressure sensor supply circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100021	Sensor signal voltage < 0.3V.	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM. Disconnect connector on TCM side and ECU side. Measure insulation resistance between ground (vehicle-GND and TCM-sensor GND) and signal sensor line. Resistance must be higher then 200kOhm. If it is lower then repair wiring harness. If it is not lower then connect both connectors set the ECU to After market mode via diagnosis and measure sensor signal line voltage (on TCM side). If voltage is higher then 0.4V then TCM is OK (then go to A). If not change TCM. A. Measure sensor signal voltage on ECU side. If it is higher then 0.4V and error is still active then replace ECU. If voltage is lower then the harness is the root cause.	Trailer Control Module Pressure Sensor
xBS	C100022	Sensor signal voltage > 4.5V.	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM. Disconnect connector on TCM side. Measure voltage between sensor signal and sensor signal ground. Voltage should be lower then 0.2V. If voltage is higher then there is a problem in the wiring harness. If voltage is lower then connect TCM and set the ECU to After market mode via diagnosis and measure again, if higher then 4.5V then replace TCM. If voltage is not higher then 4.5V, then measure on ECU side. If voltage there is lower then 4.5 V and error is still active then replace ECU.	Trailer Control Module Pressure Sensor
xBS	C100028	Sensor signal offset is out of range <0.5 or >0.85V.	Signal Bias Level Out of Range / Zero Adjustment Failure	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - Pneumatic problem - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect trailer. Set the ECU in After market mode via diagnosis. Release parking brake (SECURE VEHICLE). Disconnect connector on TCM side, measure voltage on the harness between sensor signal and sensor ground. If voltage is higher then 0.2V then repair wiring harness. If voltage is lower then connect TCM and measure pressure on yellow coupling head with all brakes released. If pressure is lower then (<0.1 bar) and the voltage on sensor signal is higher then 0.85V then replace TCM. If the pressure is higher then the 0.1 bar, then there is a pneumatic problem (TCM, parkbrake, FBM). Measure pressure on the parking brake input on the TCM (port 43), pressure should be higher than 6bar (KB to verify) all the time. If pressure is lower then check parking brake. Else check pressure on port 42 (backup) of TCM. If pressure is higher then 0.1 bar check footbrake module else replace TCM.	Trailer Control Module Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100112	Voltage > 5 V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [10..25] Ohm, resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm. If not ok then measure directly on TCM connector, alve should be in [10..25] Ohm. if not ok then replace TCM else repair wiring. Connect again on ECU side and disconnect from TCM side: Set ECU in AM mode: Measure voltage difference between valve and vehicle ground should be lower than 3 V. If not ok then repair harness. If result is ok reconnect everything delete error code, check if error code appears again (activate Inlet valve via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Inlet Valve
xBS	C100114	Voltage < 5 V	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [10..25] Ohm, resistance between actual valve connection and all wires of different components on the same connector and to vehicle ground should be larger than 200 kOhm. If not ok then measure directly on TCM connector, valve should be in [10..25] Ohm. if not ok then replace TCM else repair wiring. If result is ok reconnect everything delete error code, check if error code appears again . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Inlet Valve
xBS	C100212	Voltage > 5 V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [10..25] Ohm, resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm. If not ok then measure directly on TCM connector, alve should be in [10..25] Ohm. if not ok then replace TCM else repair wiring. Connect again on ECU side and disconnect from TCM side: Set ECU in AM mode: Measure voltage difference between valve and vehicle ground should be lower than 3 V. If not ok then repair harness. If result is ok reconnect everything delete error code, check if error code appears again (activate Outlet valve via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100214	Voltage < 5 V	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [10..25] Ohm, resistance between actual valve connection and all wires of different components on the same connector and to vehicle ground should be larger than 200 kOhm. If not ok then measure directly on TCM connector, valve should be in [10..25] Ohm. if not ok then replace TCM else repair wiring. If result is ok reconnect everything delete error code, check if error code appears again . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Outlet Valve
xBS	C100312	Voltage > 5 V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [6..17] Ohm, resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm. If not ok then measure directly on TCM connector, alve should be in [6..17] Ohm. if not ok then replace TCM else repair wiring. Connect again on ECU side and disconnect from TCM side: Set ECU in AM mode: Measure voltage difference between valve and vehicle ground should be lower than 3 V. If not ok then repair harness. If result is ok reconnect everything delete error code, check if error code appears again (activate Backup valve via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Backup Valve
xBS	C100314	Voltage < 5 V	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check all cables and connectors from the TCM to the central ECU (including check for chafing damage). With valve wiring disconned from central ECU: resistance of valve should be in [6..17] Ohm, resistance between actual valve connection and all wires of different components on the same connector and to vehicle ground should be larger than 200 kOhm. If not ok then measure directly on TCM connector, valve should be in [6..17] Ohm. if not ok then replace TCM else repair wiring. If result is ok reconnect everything delete error code, check if error code appears again . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Trailer Control Module Backup Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C001011	Resistance between the output and the vehicle ground < 5 Ohm.	Circuit Short To Ground	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Left Front ABS Inlet Valve
xBS	C001012	Voltage > 3 V	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Front ABS Inlet Valve
xBS	C001013	Resistance at output > 1 kOhm..	Circuit Open	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Front ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C00101A	Resistance between the output and the PCV ground < 5 Ohm.	Circuit Resistance Below Threshold	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Front ABS Inlet Valve
xBS	C001095	Short circuit between inlet and outlet valve	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check short circuit of all components where this type of fault is detected against each other. Disconnect central ECU and faulty PCV. Check short circuit of wire of PCV_FL_IV (highside of valve - X2/3) and wires of all other components where shorted channels were detected against each other. Insulation resistance must be higher than 200 kOhm, otherwise repair faulty wires and/or connectors. If no fault found, reconnect the valves and repeat the previous measurement. If fault is still active, replace the PCV. If result is OK, reconnect everything, delete error code, check if error code appears again (activate the PCV via diagnosis) . If error is not detected again, the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Front ABS Inlet Valve
xBS	C001111	Resistance between the output and the vehicle ground < 5 Ohm.	Circuit Short To Ground	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Left Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C001112	Voltage > 3 V	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Front ABS Outlet Valve
xBS	C001113	Resistance at output > 1 kOhm.	Circuit Open	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Left Front ABS Outlet Valve
xBS	C00111A	Resistance between the output and the PCV ground < 5 Ohm.	Circuit Resistance Below Threshold	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Left Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C001163	Valve is set exhaust > 1 min (when using I/O DID "PCV Valves", 0x1212).	Circuit / Component Protection Time-Out	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Too long valve activation by Diag-I/O-control	Diagnostic Service 14. No additional action is needed	Check if diagnostic command is shorter than 1 min.	Left Front ABS Outlet Valve
xBS	C001195	Short circuit between inlet and outlet valve	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check short circuit of all components where this type of fault is detected against each other. Disconnect central ECU and faulty PCV. Check short circuit of wire of PCV_FL_IV (highside of valve - X2/3) and wires of all other components where shorted channels were detected against each other. Insulation resistance must be higher than 200 kOhm, otherwise repair faulty wires and/or connectors. If no fault found, reconnect the valves and repeat the previous measurement. If fault is still active, replace the PCV. If result is OK, reconnect everything, delete error code, check if error code appears again (activate the PCV via diagnosis) . If error is not detected again, the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Left Front ABS Outlet Valve
xBS	C001411	Resistance between the output and the vehicle ground < 5 Ohm.	Circuit Short To Ground	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Front ABS Inlet Valve
xBS	C001412	Voltage > 3 V	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Front ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C001413	Resistance at output > 1 kOhm.	Circuit Open	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Front ABS Inlet Valve
xBS	C00141A	Resistance between the output and the PCV ground < 5 Ohm.	Circuit Resistance Below Threshold	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV	Right Front ABS Inlet Valve
xBS	C001495	Short circuit between inlet and outlet valve	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check short circuit of all components where this type of fault is detected against each other. Disconnect central ECU and faulty PCV. Check short circuit of wire of PCV_FL_IV (highside of valve - X2/3) and wires of all other components where shorted channels were detected against each other. Insulation resistance must be higher than 200 kOhm, otherwise repair faulty wires and/or connectors. If no fault found, reconnect the valves and repeat the previous measurement. If fault is still active, replace the PCV. If result is OK, reconnect everything, delete error code, check if error code appears again (activate the PCV via diagnosis) . If error is not detected again, the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Front ABS Inlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C001511	Resistance between the output and the vehicle ground < 5 Ohm.	Circuit Short To Ground	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resitance between PCV and vehicle Gnd. Resistance must be higher than 200kOhm Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Front ABS Outlet Valve
xBS	C001512	Voltage > 3 V	Circuit Short To Battery	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors to Inlet valve of PCV Left for short circuit to all other cables and pins (including check for chafing damage). With valve wiring disconned from central ECU: resistance between actual valve connection and all wires of different components on the same connector should be larger than 200 kOhm, voltage difference between valve and vehicle ground should be lower than 1 V. In case of lower resistance or higher voltage difference, repair wiring harness. If result is ok reconnect everything delete error code, check if error code appears again (activat PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Front ABS Outlet Valve
xBS	C001513	Resistance at output > 1 kOhm.	Circuit Open	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If values are not ok replace PCV, otherwise repair cabling harness.	Right Front ABS Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C00151A	Resistance between the output and the PCV ground < 5 Ohm.	Circuit Resistance Below Threshold	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - PCV damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling and connectors from ECU to PCV including chafing. Disconnect the wiring on the ECU side. 1) Measure resistance between Inlet/outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If result of measurement is not ok continue with 2) If result is ok reconnect everything delete error code, check if error code appears again (activate PCV via diagnosis) . If error is not detected again. The error is intermittent try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible. 2) Disconnect wiring from PCV Measure resistance between Inlet/Outlet Valve and PCV GND. Resistance must be between 10 and 22Ohm. If Values are not ok Replace PCV, Otherwise repair cabling harness.	Right Front ABS Outlet Valve
xBS	C001563	Valve is set exhaust > 1 min (when using I/O DID "PCV Valves", 0x1212).	Circuit / Component Protection Time-Out	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Too long valve activation by Diag-I/O-control	Diagnostic Service 14. No additional action is needed	Check if diagnostic command is shorter than 1 min.	Right Front ABS Outlet Valve
xBS	C001595	Short circuit between inlet and outlet valve	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check short circuit of all components where this type of fault is detected against each other. Disconnect central ECU and faulty PCV. Check short circuit of wire of PCV_FL_IV (highside of valve - X2/3) and wires of all other components where shorted channels were detected against each other. Insulation resistance must be higher than 200 kOhm, otherwise repair faulty wires and/or connectors. If no fault found, reconnect the valves and repeat the previous measurement. If fault is still active, replace the PCV. If result is OK, reconnect everything, delete error code, check if error code appears again (activate the PCV via diagnosis) . If error is not detected again, the error is intermittent, try to reproduce the error by moving cables and connectors. If error can not be reproduced no further action is possible.	Right Front ABS Outlet Valve
xBS	C109895	Short circuit between backup and inlet valve	Incorrect Assembly	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors (also for chafing damage) between central ECU and TCM; Disconnect connectors of the central ECU and of the TCM. Check for short circuit between the wires of the inlet and backup valves of the TCM. If resistance is lower than 200 kOhm, repair faults in the wiring; Measure resistance on the inlet and backup valve of the TCM directly on the TCM connector; If resistance is lower than 20 [Ohm], replace the TCM; Otherwise delete fault codes and try to move the harness to reproduce the fault.	Trailer Control Module Backup Valve And Inlet Valve
xBS	C109995	Short circuit between backup and outlet valve	Incorrect Assembly	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors (also for chafing damage) between central ECU and TCM; Disconnect connectors of the central ECU and of the TCM. Check for short circuit between the wires of the inlet and backup valves of the TCM. If resistance is lower than 200 kOhm, repair faults in the wiring; Measure resistance on the inlet and backup valve of the TCM directly on the TCM connector; If resistance is lower than 20 [Ohm], replace the TCM; Otherwise delete fault codes and try to move the harness to reproduce the fault.	Trailer Control Module Backup Valve And Outlet Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109A95	Short circuit between inlet and outlet valve	Incorrect Assembly	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors (also for chafing damage) between central ECU and TCM; Disconnect connectors of the central ECU and of the TCM. Check for short circuit between the wires of the inlet and backup valves of the TCM. If resistance is lower than 200 kOhm, repair faults in the wiring; Measure resistance on the inlet and backup valve of the TCM directly on the TCM connector; If resistance is lower than 20 [Ohm], replace the TCM; Otherwise delete fault codes and try to move the harness to reproduce the fault.	Trailer Control Module Inlet Valve And Outlet Valve
xBS	C100712	Voltage > 5 V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM (including check for chafing damage). Disconnect central ECU and TCM. Check the cables and connectors. Check cable and TCM resistances according to the values given in field detection. Connect the wiring to TCM: Check TCM voltages according to the values given in field 'detection'; With wiring disconnected from both central ECU and TCM: resistance of each wire should be less than 3 [Ohm], resistance between any two wires and resistance between any wire and vehicle ground should be larger than 200 [kOhm]; With wiring disconnected from central ECU: resistance between TCM supply and inlet or outlet valve should be 12..23 [Ohm], resistance between TCM supply and backup valve should be 6..16 [Ohm]; In Aftermarket mode and with wiring disconnected from central ECU: voltage between any valve and TCM supply should be lower than 5 [V].	Trailer Control Modulator
xBS	C100714	Fault will be detected by activation of test pulse and reading back hardware status information.	Circuit Short To Ground or Open	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - TCM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and TCM (including check for chafing damage). Disconnect central ECU and TCM. Check the cables and connectors. Check cable and TCM resistances with wiring disconnected from central ECU: resistance between TCM supply and inlet or outlet valve should be [10 .. 25] Ohm, resistance between TCM supply and backup valve should be [6 .. 17] Ohm. If not ok measure the same thing on directly on the TCM. If not ok, replace TCM. If ok, then repair wiring harness. Resistance between any TCM pin and vehicle ground should be larger than 40 kOhm. If not ok on wire harness to the same test on TCM. If not ok on TCM, replace TCM else repair wiring harness.	Trailer Control Modulator
xBS	C10074A	TCM configured but not populated in ECU	Incorrect Component Installed	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	Wrong ECU hardware type	Diagnostic Service 14. No additional action is needed	Check that ECU HW matches the chassi number. If not replace ECU.	Trailer Control Modulator
xBS	C100755	TCM detected, but not configured.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged wire(s) or connector(s) - Installed but not configured component	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation (check if TCM is installed and connected). If configuration matches installation, check connectors. If no fault is found with the connector, then check wiring (including check for chafing damage), and repair if faulty.	Trailer Control Modulator
xBS	C108829	Pressure sensor signal > 10 bar	Signal invalid	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Supply pressure for TCM at supply pressure input is too high (APM) - Damaged wire(s) or connector(s) - TCM damaged	Diagnostic Service 14. No additional action is needed	Check maximum supply pressure for TCM at red coupling; Maximum supply pressure must be lower than 9,5 [bar]; If maximum supply pressure is correct; Measure pressure externally with 2 % precision on the yellow coupling head with brake pedal fully pressed; Compare the measured pressure with the TCM pressure read via diagnosis. In case of a difference higher than 0,5 [bar]+5 %; Check cables and connectors (to pressure sensor ground and signal) between ECU and TCM for damages, chafing marks, dirt, dust, water in the connectors; Replace the TCM	Trailer Control Module Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108864	Trailer brake is activated > 10 min	Signal Plausibility Failure	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged park brake(pressure output or signal from APM) - Damaged piping - Damaged/ too long usage of stretch brake - Damaged/ wrongly installed FBM (residual backup pressure without FBM actuation) - TCM damaged	Diagnostic Service 14. No additional action is needed	Valve controlled trailer brake signal must have state deactivated, if valve controlled trailer brake is released. Check valve controlled trailer brake signal, and repair if faulty. Perform the needed reverse test procedure. If some other TCM related fault is reported during this procedure, follow the repair procedure of the reported fault.	Trailer Control Module Pressure Control
xBS	C108867	Implausible high pressure at end of braking and valve controlled trailer brake is activated implausibly long	Signal Incorrect After Event	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- TCM damaged - Valve controlled trailer brake signal is incorrect (it is always active)	Diagnostic Service 14. No additional action is needed	Valve controlled trailer brake signal must have state deactivated, if valve controlled trailer brake is released. Check valve controlled trailer brake signal, and repair if faulty. Perform the needed reverse test procedure. If some other TCM related fault is reported during this procedure, follow the repair procedure of the reported fault.	Trailer Control Module Pressure Control
xBS	C108871	Backup pressure does not exceed minimum pressure level dependent on FBM stroke.	Actuator Stuck	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Pneumatic pipes between FBM and TCM open or stuffed - Damage in the mechanical or pneumatic part of the FBM - Damaged parking brake(pressure output or signal from APM) - TCM damaged - Supply pressure at TCM supply pressure input is not sufficient, supply pressure circuit for trailer - Brake piping of the yellow coupling head open	Diagnostic Service 14. No additional action is needed	Check backup piping between the FBM and the TCM, repair if faulty; Check the wiring and the connectors between the central ECU and the TCM, repair if faulty; Check piping between the supply pressure and the TCM, repair if faulty; Compare supply pressure of the TCM measured by external pressure gauge (with a maximum tolerance of 2%) with the supply pressure information transmitted by the APM. If the pressure difference is higher than 1 bar + 5%, then check the APM; Set ECU into Aftermarket Diagnostic Mode; Press the brake pedal fully; Check backup pressure increase from FBM with external pressure gauge with a maximum tolerance of 2%; Compare the backup pressure with the pressure of the TCM read via diagnosis; The dynamic behaviour of both signals must be similar. In case of a deviation, piping between the FBM and the TCM must be repaired; Otherwise the TCM is defect; Replace the TCM.	Trailer Control Module Pressure Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108872	Difference between pressure measured by TCM and demanded pressure > 1.5 bar	Actuator Stuck Open	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- TCM silencer plugged - TCM damaged - Damaged park brake(pressure output or signal from APM) - Damaged/ wrongly installed FBM (residual backup pressure without FBM actuation)	Diagnostic Service 14. No additional action is needed	Disconnect trailer; Do the revers test to exclude temporary faults; Check parking brake valve by reading parking brake status from APM; Check the silencer of the TCM; Set ECU into Aftermarket Diagnostic Mode; Press the brake pedal fully and release it; Check pressure decrease with external pressure gauge with a maximum tolerance of 2%. Compare the measured pressure with the pressure of the TCM read via diagnosis; The dynamic behaviour of both signals must be similar; Check also for the correct release gradient; If not OK, replace the TCM; 1,5 [s] after brake pedal is released, pressure must be lower than 0,5 [bar]. If not OK, replace the TCM; Delete fault codes and check for the fault code; Check pressure on yellow coupling head with all brakes released (SECURE VEHICLE). Pressure on yellow coupling must be zero and the pressure read by diagnostics must also be zero; Measure on the parking brake input on the TCM (port 43), should be higher than 6 [bar] [KB to verify] all the time; Press brakepedal fully and release it fast, then the pressure on the yellow coupling must release fast (1 s); Activate parking brake and release it fast and the the pressure must release fast (1..1.5 s); After that the pressure on the PB input (port 43) has to be higher 6 [bar]; If the input pressure is not ok then the problem is between APM and TCM. Else, check parking brake and stretch brake signal. if no problem found on the signals, replace the TCM	Trailer Control Module Pressure Control
xBS	C108873	- Difference between pressure measured by TCM and demanded pressure < 1.5 bar - Implausible inlet valve actuation (if inlet valve protection is active, and actual pressure < 4bar and supply pressure of trailer circuit > 5bar.)	Actuator Stuck Closed	Driver informed by YELLOW warning and brake malfunction indication. Disabled functions are: - Trailer Brake - ESP - HSA - Coupling Force Control	- Damaged piping or part between TCM and trailer - Damaged piping or part between TCM and APM - TCM damaged - Damaged park brake(pressure output or signal from APM)	Diagnostic Service 14. No additional action is needed	Disconnect trailer; Do the revers test to exclude temporary faults; Check pneumatic piping to the trailer including coupling head (high leakage); Check pneumatic piping to the APM (if piping is not according to the specification /i.e. wrong dimention, bend on the piping/, supply pressure information can be wrong); Compare supply pressure of the TCM measured by external pressure gauge (with a maximum tolerance of 2 %) with the supply pressure information transmitted by the APM; If the pressure difference is higher than 1 [bar] + 5 %, then check the APM; Check the wiring (pressure sensor ground and signal) between the TCM and the central ECU; Repair if faulty; Set ECU into Aftermarket Diagnostic Mode; Press the brake pedal fully; Check pressure of the TCM (yellow coupling) with an external pressure gauge with a maximum tolerance of 2 %; Compare the pressure with the pressure of the TCM read via diagnosis; If difference is higher than 0,5 [bar] + 5 % then replace the TCM; Otherwise delete fault codes and check for the fault code	Trailer Control Module Pressure Control
xBS	C200063	- ABS activity is implausibly long - Too low brake pressures is required by ABS for an implausibly long time.	Circuit / Component Protection Time-Out	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Fault of at least one wheel speed sensor or its mounting - Damages of at least one tone wheel, - Noisy wheel speed signal, - Differences in tyre sizes, - Puncture of a tyre, - Continuous brake force on a wheel (without brake pressure).	Diagnostic Service 14. No additional action is needed	Check wheel speeds in snapshot to see, which wheel can cause the fault. Check tyre for size and correct air pressure, check pneumatic and mechanical parts of wheel brakes. Check wheel speed signal components: check cable, sensor and tone wheel. Check if pneumatic and electronic installation to wheels correspond to each other (if pneumatic pipes are mixed up). Check if configuration stored in the ECU corresponds to vehicle brake system layout: number and location of sensed wheels, axle type such as driven, liftable. Check if configuration stored in the central ECU corresponds to wheel rolling circumference and tone-wheel teeth number. If those do not conform, then correct the configuration values or change the part of the vehicle.	Anti-Lock Brake System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U300007	ECU's internal moisture indicator indicates too high moisture content.	Mechanical Failures	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damage	Diagnostic Service 14. No additional action is needed	Replace central ECU and check connector	Electronic control unit
xBS	U300041	Checksum error in internal memory	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- ECU damaged - Previously interrupted flash download	Diagnostic Service 14. No additional action is needed	Download update of application software, if the error is still active, replace central ECU.	Electronic control unit
xBS	U300044	Error in internal RAM memory	Data Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damage	Diagnostic Service 14. No additional action is needed	Replace central ECU.	Electronic control unit
xBS	U300045	Error in internal Flash memory	Program Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damage	Diagnostic Service 14. No additional action is needed	Replace central ECU.	Electronic control unit
xBS	U300046	- Error in internal EEPROM memory - Incompatibilities between EEPROM dataset and drive software, hardware or bootloader versions - Implausible data in the dataset	Calibration / Parameter Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damaged	Diagnostic Service 14. No additional action is needed	Download update of application software, if the error is still active, replace central ECU.	Electronic control unit
xBS	U300049	Internal fault in the ECU	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damage	Diagnostic Service 14. No additional action is needed	Replace central ECU.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U30004B	Temperature to high in the ECU	Over Temperature	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Ambient temperature too high - ECU damaged	Diagnostic Service 14. No additional action is needed	Check ECU location and suroundings for high temperatures. - If fault is active, replace it.	Electronic control unit
xBS	U300094	ECU internal plausibility monitoring function detected an ECU hardware failure	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	ECU damagedamage of central ECU	Diagnostic Service 14. No additional action is needed	Replace central ECU	Electronic control unit
xBS	U300096	Internal non saftey critical fault in the ECU	Component Internal Failure	Driver informed by YELLOW warning and brake malfunction indication	ECU damage	Diagnostic Service 14. No additional action is needed	- Check wiring to TCM - Replace central ECU	Electronic control unit
xBS	U300316	Supply voltage <= 16 V	Circuit Voltage Below Threshold	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - Battery is not charged enough - Damage of the vehicle's power supply system - ECU damaged	Diagnostic Service 14. No additional action is needed	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
xBS	U300317	Supply voltage >= 36 V	Circuit Voltage Above Threshold	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	Damage of the vehicle's power supply system	Diagnostic Service 14. No additional action is needed	If the state is active continuously, check vehicle's power supply system.	Battery Power

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U30031C	Supply voltage out of range. Overrange voltage levels: >36 V if RASec not configured (URSTS) >32 V if RASec configured (RSTS-EHY) Underrange voltage levels: <19.5 V if RASec configured (RSTS-EHY) <18.75 V if RASec not configured (URSTS) and PCVs (ABS valves) on front axle. <18 V if RASec not configured (URSTS), no PCVs (ABS valves)	Circuit Voltage Out of Range	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Battery is not charged enough - Damage of the vehicle's power supply system - ECU damaged	Diagnostic Service 14. No additional action is needed	If the state is active continuously, check vehicle's power supply system and voltage supply wiring to ECU.	Battery Power
xBS	U114655	Vehicle layout and configuration do not match.	Not Configured	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Wrong dataset	Diagnostic Service 14. No additional action is needed	Download vehicle layout matching dataset, if still active contact technical service	ESP Function Mode
xBS	C108D07	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 3 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108D38	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 3 Braking Monitoring
xBS	C108D62	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significantly different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 3 Braking Monitoring
xBS	C108C07	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 2 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108C38	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	"Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters."	Wheel 2 Braking Monitoring
xBS	C108C62	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 2 Braking Monitoring
xBS	C108707	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 1 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108738	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)vehicle one-side/asymmetric loaded for longer time, damaged vehicle suspension,	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 1 Braking Monitoring
xBS	C108762	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 1 Braking Monitoring
xBS	C108E07	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 4 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108E38	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 4 Braking Monitoring
xBS	C108E62	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensorr)	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 4 Braking Monitoring
xBS	C108F07	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 5 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C108F38	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)er time, damaged vehicle suspension, incorrect adjusted track of vehicle, damaged EPM	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 5 Braking Monitoring
xBS	C108F62	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 5 Braking Monitoring
xBS	C109007	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 6 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109038	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 6 Braking Monitoring
xBS	C109062	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 6 Braking Monitoring
xBS	C109107	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 7 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109138	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 7 Braking Monitoring
xBS	C109162	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 7 Braking Monitoring
xBS	C109207	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 8 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109238	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 8 Braking Monitoring
xBS	C109262	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 8 Braking Monitoring
xBS	C109307	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 9 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109338	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 9 Braking Monitoring
xBS	C109362	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 9 Braking Monitoring
xBS	C109407	Difference between left and right pads of an axle > 20%	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Mechanical fault in the brake calipers - Damaged disc or brake pads. - Damaged Lining wear sensor	Diagnostic Service 14. No additional action is needed	Measure actual wear values externally. Compare them with the wear values measured by the EBS system (read out via diagnostics). If they match, check brake calipers, disc and pads. calculate wear values, Check lining wear sensor Otherwise check connection resistance between lining wear sensor and EPM, including increased resitance of connectors due to e.g. corrosion.	Wheel 10 Braking Monitoring

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109438	Frequent non symmetric axle internal slip is detected.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged brake or brake cylinder, - Dirtened moving brake parts such as brake pads/discs/drums (oil, grease) - Leaky piping between EPM and brake cylinder - Different tyre at axle (wear, tyre type, season type, aging) - Different brake pads/discs/drums at axle (wear, type, aging) - Vehicle one-side/asymmetric loaded for longer time - Damaged vehicle suspension - Incorrect adjusted track of vehicle - EPM damaged (different pressure left/right in case of 2CH EPM)	Diagnostic Service 14. No additional action is needed	Check wheel brakes including actuator device and tyres according service instruction. Measure pressures in both brake chambers of the corresponding axle externally with 2% precision. Apply brake pedal until ca. 4 bar is reached. If difference between the two externally measured pressure is greater than 1 bar + 5% or pressure gradinets of the two sides are significally different then pneumatic piping between EPM and brake chamber should be checked. If there is no failure with the pneumatic piping, the EPM should be checked as follows: Ensure that reservoirs are fully charged (until cut-off pressure). Switch to Aftermarket mode with diagnostics. Control pressure at EPM via diagnostics increased from 0 to 8 bar with 1 bar steps. Read pressure values via diagnostics. Measure pressure in brake chamber externally with 2% precision. If difference between read and externally measured pressure is greater than 0.5 bar + 5%, then EPM should be replaced. In case of changed components run diagnostic routine reinitializeSelfLearnedParameters.	Wheel 10 Braking Monitoring
xBS	C109462	Current wheel speed differs (>25%) from the average of the other wheels.	Signal Compare Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted - Puncture of the tyre - Continuous brake force on a wheel without brake activated by EBS. - Wrong tone wheel - Damaged wire(s) or connector(s) - Undetected fault of the wheel speed sensor	Diagnostic Service 14. No additional action is needed	Check tire and check the brakes. If error still active, check Wheel speed sensors by driving and monitoring. Wheel speed shall not differ more than 25 %. check tone wheel	Wheel 10 Braking Monitoring
xBS	C109695	Wheel speeds on one axle during curve driving is not plausible.	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (Right and left hand side sensors are switched)	Diagnostic Service 14. No additional action is needed	Check the installation order of the wheel speed sensors of the affected axle; If installation is correct check dataset for correct configuration	Wheel Speed Sensor Configuration
xBS	U115E55	Configuration is not correct for the inteligent slip control	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	False parameters for ISC in configuration	Diagnostic Service 14. No additional action is needed	Download correct dataset, if still active contect technical service	Intelligent Slip Control Configuration
xBS	U116107	Difference between demanded deceleration and actual deceleration is too high.	Mechanical Failures	Driver informed by YELLOW warning and brake malfunction indication	- Not sufficient supply pressure - Fading or damaged brakes - overload	Diagnostic Service 14. No additional action is needed	Check for overloaded vehicle/trailer.	Vehicle Braking State

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U116168	Calculated disc temperature > 550 degrees	Event Information	Driver informed by YELLOW warning and brakes overheated indication. - Driver may feel that the brakes are fading.	- The brakes were overstressed (too frequent and long activation no time for cooling). - Wrong Ambient temperature or time/date information - Vehicle construction changed in a way that estimation parameters are no longer valid	Diagnostic Service 14. No additional action is needed	Driver should use a less agressive driving style, check tires for right size (cooling condition), download correct dataset	Vehicle Braking State
xBS	U116177	Vehicle deceleration does not match to the demaned deceleration within a tolerance in normal vehicle condition.	Commanded Position Not Reachable	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Software error - ECU damage	Diagnostic Service 14. No additional action is needed	Check for sufficient supply pressure, check for fading brakes, overloaded vehicle/trailer, check the piping and the brakes, if fault becomes still active and the whole mechanical brake system is ok. for info: Wrong vehicle weight do not cause this error	Vehicle Braking State
xBS	C107211	FBM supply soft short to ground	Circuit Short To Ground	No impact on functionality	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check cabling => acceptable minimum voltage on the power supply line (FBM_SPL) from ECU to DFBM min: 10 [V] (during ignition is on and off); If the problem still persists replace the central ECU; If the problem still persists replace the FBM	Foot Brake Module Electrical Link
xBS	C107212	- Communication from the ECU to the dFBM (S2) is disturbed. - Power down process of FBM does not work - FBM send continuous status message to the ECU because of missing task trigger. is working properly.	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - FBM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the wiring and connectors between the central ECU and the foot brake module (especially S2 signal) Measure insulation resitance of foot brake module wires to vehicle GND and all X1 of central ECU wires against S2 signal with disconnected Central ECU and disconnected FBM. Resistance must be higher than 200kOhm If resitance is lower repair wiring. Mesure resitance of foot brake module wires between foot brake module and Central ECU (especially S2 signal). Resistance must be lower than 5Ohm If resistance is higher repair wiring. If wiring is ok and error is still avtive measure signal voltage on S2 line. If oltage is lower than 1,5V replace foot brake moduleM otherwise replace Central ECU	Foot Brake Module Electrical Link
xBS	C107282	FBM message counter is inconsistent with timestamp of reception	Alive / Sequence Counter Incorrect / Not Updated	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	FBM damaged	Diagnostic Service 14. No additional action is needed	Replace foot brake module.	Foot Brake Module Electrical Link
xBS	C107287	Pedal position data missing or no valid data is received from FBM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - FBM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the wiring and connectors between the central ECU and the foot brake module Measure insulation resistance of all foot brake module wires to vehicle ground and to all other wires on the ECU-X1 connector with disconnected Central ECU and disconnected FBM. Resistance must be higher than 200kOhm If resitance is lower repair wiring Mesure resitance of all foot brake module wires between foot brake module and central ECU. Resistance must be lower than 5Ohm If resistance is higher repair wiring. If error is still active and no error found measure signal S1 with scope. If signal is toggling between 0,5 and 4,5V replace ECU otherwise replace foot brake module or measure voltage on S1 line with voltage meter. If voltage is between 1.5 and 3.5V replace ECU otherwise replace foot brake module.	Foot Brake Module Electrical Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C107294	- The stroke sensed by the two stroke sensors is not identical, but the offset is still within a tolerable range - The wakeup switch position does not match the measured stroke value	Unexpected Operation	Driver informed by YELLOW warning and brake malfunction indication	- Rod is not according to the specification - FBM damaged	Diagnostic Service 14. No additional action is needed	Check rod from pedal for foot brake module activation for correct part If rod is according to the specification replace the foot brake module.	Foot Brake Module Electrical Link
xBS	C107128	Too high offset internaly in FBM.	Signal Bias Level Out of Range / Zero Adjustment Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- FBM is blocked (stuck in actuated position) - Incorrect mounting/installation - FBM damaged	Diagnostic Service 14. No additional action is needed	Check if rod for activation of foot brake module has enough clearence (installtion manual). Adjust if needed. Do revers test and if the error do not disapear replace foot brake module.	Foot Brake Module
xBS	C107141	Checksum error of the drive software is detected in the A or B controller by the bootloader of the DFBM	General Checksum Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	FBM damaged	Diagnostic Service 14. No additional action is needed	Download again the digitial foot brake module software, if the fault is still active Replace foot brake module	Foot Brake Module
xBS	C107142	Flash, EEPROM or RAM memory fault	General Memory Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	FBM damaged	Diagnostic Service 14. No additional action is needed	Replace foot brake module	Foot Brake Module
xBS	C107149	Internal fault in the foot brake module	Internal Electronic Failure	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	FBM damaged	Diagnostic Service 14. No additional action is needed	Download latest software version to the foot brake module, if fault is still active, Replace foot brake Module.	Foot Brake Module
xBS	C10714A	- Invalid FBM hardware type - Incompatibility is detected by the central ECU between the FBM and the central ECU	Incorrect Component Installed	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	Wrong FBM hardware is mounted	Diagnostic Service 14. No additional action is needed	- Check part number on foot brake module, replace foot brake module with correct part number - Check that the MSW is the latest SW. If not, download new Central ECU SW.	Foot Brake Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C10714B	One of the two dFBM temperature sensors delivers an implausible value.	Over Temperature	No impact on functionality	FBM damaged	Diagnostic Service 14. No additional action is needed	Replace the foot brake module	Foot Brake Module
xBS	C107194	Incompatible SW parts in dFBM	Unexpected Operation	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	FBM damaged	Diagnostic Service 14. No additional action is needed	Update the foot brake module with compatible boot/driver SW for the used hardware, if the fault is still active check Replace foot brake module.	Foot Brake Module
xBS	C107195	Wakeup time of the system when ignition is off > 1h	Incorrect Assembly	Driver informed by YELLOW warning and brake malfunction indication	- FBM is blocked (stuck in actuated position) - Incorrect mounting/installation - FBM damaged	Diagnostic Service 14. No additional action is needed	Check if rod for activation of foot brake module has enough clearence (installtion manual). Adjust if needed, if adjustment not successful then replace foot brake module.	Foot Brake Module
xBS	U116368	CRC or messsage counter error in external brake demand messages	Event Information	Driver informed by YELLOW warning and brake malfunction indication - Safety functions most likely switched off. - Transmission brake requests not exectuted (for example truck not braking automaticaly when changing to revers)	Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting BBM, DACU or TECU signals.	External Brake Request Message Checksum
xBS	U116968	No of DTCs > 30.	Event Information	No impact on functionality	Flush of faults set because of any central ECU problem	Diagnostic Service 14. No additional action is needed	Check the active faults and correct them	DTC Notification Max Limit Reached
xBS	C109768	Roller test bench mode activated.	Event Information	ESP and traction control disabled	Roller test bench mode was activated by diagnostic routine or instrument cluster activatation.	Diagnostic Service 14. No additional action is needed	No action required, it is not a failure, only an event information (Roller Test Bench Mode is active)	Roller Test Bench Mode
xBS	C100811	Sensor supply voltage < 4.75V	Circuit Short To Ground	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between EBS ECU and rear axle steering angle sensor (insulation resistance should be >200 kO). - Disconnect the sensor - Run control routine „Trigger Aftermarket Mode“ (0x0414). Actual state of the after maket mode can be read with identifier „Aftermarket Mode“ (0x1249) - Measure the voltage of the sensor supply. If it's lower than 4.8V then replace the central ECU else replace rear axle steering angle sensor.	Rear Active Steering Angle Sensor
xBS	C100816	Any of the sensor signals < 0.2V	Circuit Voltage Below Threshold	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and steering angle sensor signal for short-circuits and open-circuits.The insulation resistance should be higher than 200 kO and the wire resistance should be less than 5 O. - Run control routine „Trigger Aftermarket Mode“ (0x0414). Actual state of the after maket mode can be read with identifier „Aftermarket Mode“ (0x1249) - Measure the voltage of the sensor signal. If it's lower than 0.2V, then replace rear axle steering angle sensor else replace the central ECU.	Rear Active Steering Angle Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100817	Any of the two sensor signals > 4.8V	Circuit Voltage Above Threshold	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the wires and connectors between central ECU and steering angle sensor signal for short-circuits.The insulation resistance should be higher than 200 kO. - Run control routine „Trigger Aftermarket Mode“ (0x0414). Actual state of the after maket mode can be read with identifier „Aftermarket Mode“ (0x1249) - Measure the voltage of the sensor signal. If it's higher than 4.8V, then replace rear axle steering angle sensor else replace the central ECU.	Rear Active Steering Angle Sensor
xBS	C10081C	Sum of sensor signals < 4.7V or > 5.3V.	Circuit Voltage Out of Range	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and steering angle sensor signal for short-circuits and open-circuits. The insulation resistance should be higher than 200 kO and the wire resistance should be less than 5 O. - If the error is not active, then the axle must turn between the 2 end positions (by diagnostics) while the sensor voltages are measured. - If the error is continuously active then run control routine „Trigger Aftermarket Mode“ (0x0414). Actual state of the after maket mode can be read with identifier „Aftermarket Mode“ (0x1249) - Measure the voltage of both sensor signals. If sum of the signals is lower than 4.8V (end of line parameter) or higher than 5.2 then replace rear axle steering angle sensor else replace the central ECU (only if the error is continuously present and the sum of the sensor voltages measured by external tool is correct).	Rear Active Steering Angle Sensor
xBS	C100854	Sensor not calibrated	Missing Calibration	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Sensor not calibrated.	Diagnostic Service 14. No additional action is needed	- Calibrate sensor - Read RAS status information (P1DE9) to ensure that the axle is ready for calibration. If not, check mechanical parts, specs and other active failures. If error still active replace the sensor.	Rear Active Steering Angle Sensor
xBS	C100911	Feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kO. - If the resistance of the proportional valve is lower than 20 O replace the proportional valve. - If error is still active replace the central ECU.	Rear Active Steering Proportional Valve A
xBS	C100915	Feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kO. - If the resistance of the proportional valve is lower than 20 O replace the proportional valve. - If error is still active replace the central ECU.	Rear Active Steering Proportional Valve A
xBS	C100A11	Feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kO. - If the resistance of the proportional valve is lower than 20 O replace the proportional valve. - If error is still active replace the central ECU.	Rear Active Steering Propotional Valve B
xBS	C100A15	Feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and proportional valve for short-circuits. The insulation resistance must be higher than 200 kO. - If the resistance of the proportional valve is lower than 20 O replace the proportional valve. - If error is still active replace the central ECU.	Rear Active Steering Propotional Valve B
xBS	C100B11	Feedback does not correspond to the actual valve high state.	Circuit Short To Ground	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and cut-off valve for short-circuit. The insulation resistance must be higher than 200 kO. - If the resistance of the cut-off valve is lower than 18 O replace the cut-off valve. - If error is still active replace the central ECU.	Rear Active Steering Cutoff Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100B15	Feedback does not correspond to the actual valve low state.	Circuit Short To Battery or Open	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged wire(s) or connector(s) - Valve damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	- Check the wires and connectors between central ECU and cutoff valve for short-circuits and for open circuit. The insulation resistance must be higher than 200 kOhm and the wire resistance should be less than 3 Ohm. - If the resistance of the cutoff valve is higher than 28 O replace the cutoff valve. - If error is still active replace the central ECU.	Rear Active Steering Cutoff Valve
xBS	C100B71	Axle moves > 2 degrees when cutoff valve is opened and a target value is requested.	Actuator Stuck	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Valve damaged (Valve is stuck in closed position)	Diagnostic Service 14. No additional action is needed	Replace hydraulic valve	Rear Active Steering Cutoff Valve
xBS	C109B68	External CAN signals not available	Event Information	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	- Damaged wire(s) or connector(s) - Sending ECU or component	Diagnostic Service 14. No additional action is needed	Check other ECUs for related fault codes	Rear Active Steering Function
xBS	C108B02	Difference between the calibrated vehicle speed and mean drive axle wheelspeed > 20 km/h	General Signal Failure	Normal function. Highest speed of mean front axle wheel speed and vehicle speed is used	- Tachograph signal not calibrated (signal not plausible) - Wheel speed sensor failure	Diagnostic Service 14. No additional action is needed	Check other related fault codes. Check tyre sizes of the driven wheels. If tyre sizes are correct, check tachosignal (according to regulations). If tachosignal is not correct, check components related to tachometer.	Rear Active Steering Inter Axle Vehicle Speed
xBS	C108A01	Offset compensation > 2 degrees	General Electrical Failure	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Incorrect wheel alignment - Rear axle angle sensor value drift	Diagnostic Service 14. No additional action is needed	Align wheels and recalibrate	Rear Active Steering Rear Angle
xBS	C108A64	Rear steering angle is too high	Signal Plausibility Failure	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Damaged ECU	Diagnostic Service 14. No additional action is needed	Replace ECU	Rear Active Steering Rear Angle
xBS	C108A77	Difference between actual rear wheel angle and the target angle is to high	Commanded Position Not Reachable	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	- Damaged hydraulics (low hydraulic oil level, hydraulic pump) - Damaged hydraulic valve/cylinder	Diagnostic Service 14. No additional action is needed	Check hydraulic: - Oil level - Pump - Valve/cylinder	Rear Active Steering Rear Angle
xBS	C10894A	Rear axle steering circuits is not populated in ECU but configured.	Incorrect Component Installed	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Wrong ECU type	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Download correct dataset.	Rear Active Steering Control Valves
xBS	C108955	Valves or rear angle sensor is detected, but not configured.	Not Configured	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Installed but not configured components.	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Download correct dataset.	Rear Active Steering Control Valves
xBS	C108964	Both proportional valves are activated at the same time	Signal Plausibility Failure	Rear axle steering will be in passive failuremode i.e. the axle will be self steered and hydraulically damped	Damaged ECU	Diagnostic Service 14. No additional action is needed	Replace ECU	Rear Active Steering Control Valves

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C109C62	Sensor signal is implausible compared to wheel speeds signals	Signal Compare Failure	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	- Sensor is incorrectly calibrated - Drive of the sensor defect - Incorrect mounting - Steering system does not correspond to the vehicle specification (wrong steering gearbox or steering gear arm) - Sensor damaged - Wheelbase or wheeltrack do not correspond to the vehicle specification - Calibrated vehicle speed (C3) does not correspond to the wheel speeds	Diagnostic Service 14. No additional action is needed	- Check steering wheel angle sensor calibration - Check steering wheel angle sensor installation - Check Steering system (Steering gear etc.)	Rear Active Steering Internal Signal
xBS	C109C68	Internal EBS signals not available i.e. steering wheel angle or wheel speed is not available.	Event Information	Rear axle steering will be in active failuremode i.e. the axle will be actively steered towards target angle zero (straight ahead)	- Damaged wire(s) or connector(s) - Damaged sensor	Diagnostic Service 14. No additional action is needed	See other steering angle sensor or wheel speed sensor related faults	Rear Active Steering Internal Signal
xBS	C101721	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements (- lining wear sensor ground, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101722	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements (- lining wear sensor ground, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for	Lining Wear Sensor 1
xBS	C101821	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of [""to be defined by V3P""] Ohm) V3P to check for possible measurements (- lining wear sensor ground, it should be in range of [""to be defined by V3P""] Ohm) V3P to check for possible measurements	Lining Wear Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101822	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements (- lining wear sensor ground, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 2
xBS	C101921	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101922	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 3
xBS	C101A21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101A22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 4
xBS	C101B21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101B22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 5
xBS	C101C21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101C22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 6
xBS	C101D21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 7

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101D22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 7
xBS	C101E21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 8

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101E22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 8
xBS	C101F21	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 9

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101F22	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 9
xBS	C102021	Sensor signal < 0.25 V	Signal Amplitude Less Than Minimum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible measurements	Lining Wear Sensor 10

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C102022	Sensor signal > 4.75 V	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and Lining wear sensor. Switch on lining wear sensor supply with diagnostics and read recorded voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement externally (between LWS GND and LWS signal) during the time period, when lining wear sensor supply is switch on. If measured value is not within range 0.25-4,75V (EOL values) the problem is in the sensor or wiring. Compare values recorded by diagnostics and measured externally. If the values do not match (within 10% tolerance), EPM should be changed. If the values match disconnect lining wear sensor connector from EPM. Measure resistance of sensor to - vehicle ground, it should be larger than 200kOhm. - Then with wiring disconnected from both EPM and lining wear sensor: Resistance of each wire should be less than 5 Ohm and resistance between two pins of the wiring should be larger than 200 kOhm If all values are correct then wiring is OK, and sensor should be replaced. (- lining wear sensor supply, it should be in range of ["to be defined by V3P"] Ohm) V3P to check for possible	Lining Wear Sensor 10
xBS	C103F22	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 1 Axle Load Sensor
xBS	C103F55	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 1 Axle Load Sensor
xBS	C104022	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 2 Axle Load Sensor
xBS	C104055	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 2 Axle Load Sensor
xBS	C104122	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 3 Axle Load Sensor
xBS	C104155	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 3 Axle Load Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104222	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 4 Axle Load Sensor
xBS	C104255	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 4 Axle Load Sensor
xBS	C104322	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 5 Axle Load Sensor
xBS	C104355	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 5 Axle Load Sensor
xBS	C104422	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 6 Axle Load Sensor
xBS	C104455	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 6 Axle Load Sensor
xBS	C104522	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 7 Axle Load Sensor
xBS	C104555	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 7 Axle Load Sensor
xBS	C104622	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 8 Axle Load Sensor
xBS	C104655	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 8 Axle Load Sensor
xBS	C104722	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 9 Axle Load Sensor
xBS	C104755	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 9 Axle Load Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104822	Sensor signal > range of the sensor	Signal Amplitude Greater Than Maximum	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM. If error still active, check with other central ECU.	Wheel 10 Axle Load Sensor
xBS	C104855	ALS is configured but no signal or sensor supply detected.	Not Configured	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from the ALS to the corresponding EPM (including check for chafing damage). If no error found, check the EoL parameters. If no error found, check with other ALS. If error still active, check with other EPM (check that the EPM is prepared for ALS connected) . If error still active, check with other central ECU.	Wheel 10 Axle Load Sensor
xBS	C104911	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 1 Brake Lining Wear Supply Circuit
xBS	C104912	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 1 Brake Lining Wear Supply Circuit
xBS	C104924	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 1 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104A11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 2 Brake Lining Wear Supply Circuit
xBS	C104A12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 2 Brake Lining Wear Supply Circuit
xBS	C104A24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 2 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104B11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 3 Brake Lining Wear Supply Circuit
xBS	C104B12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 3 Brake Lining Wear Supply Circuit
xBS	C104B24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 3 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104C11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 4 Brake Lining Wear Supply Circuit
xBS	C104C12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 4 Brake Lining Wear Supply Circuit
xBS	C104C24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 4 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104D11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 5 Brake Lining Wear Supply Circuit
xBS	C104D12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 5 Brake Lining Wear Supply Circuit
xBS	C104D24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 5 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104E11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 6 Brake Lining Wear Supply Circuit
xBS	C104E12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 6 Brake Lining Wear Supply Circuit
xBS	C104E24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 6 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C104F11	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 7 Brake Lining Wear Supply Circuit
xBS	C104F12	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 7 Brake Lining Wear Supply Circuit
xBS	C104F24	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 7 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105011	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 8 Brake Lining Wear Supply Circuit
xBS	C105012	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 8 Brake Lining Wear Supply Circuit
xBS	C105024	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 8 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105111	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 9 Brake Lining Wear Supply Circuit
xBS	C105112	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 9 Brake Lining Wear Supply Circuit
xBS	C105124	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 9 Brake Lining Wear Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C105211	Supply voltage < 4.5V	Circuit Short To Ground	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors of all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Switch on lining wear sensor supply with diagnostics and read recorded supply voltage values (readLiningWearSensorValues_Start, readLiningWearSensorValues_Stop, readLiningWearSensorValues_Results). Perform voltage measurement of lining wear sensor output of EPM externally during the time period, when lining wear sensor supply is switch on. Compare values recorded by diagnostics and measured externally. Lining wear supply will not remain on as long as a defective Sensor connected to the corresponding EPM. In this case reconnect tested sensor, disconnect the next one and repeat process. Repeat until all Sensors of corresponding EPM have been disconnected once and readLiningWearSensorValues_Results delivers a plausible result (see below). If the values do not match (tolerance 10%), or they are out of range (4.5V...5.5V), EPM should be changed. With wiring disconnected from both EPM and lining wear sensor: resistance of each wire should be less than 3 Ohm, and resistance between two pins of the wiring should be larger than 200 kOhm. If all values are correct then wiring is OK, and sensor should be replaced	Wheel 10 Brake Lining Wear Supply Circuit
xBS	C105212	Supply voltage > 6V	Circuit Short To Battery	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS, which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated. The voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 10 Brake Lining Wear Supply Circuit
xBS	C105224	Sensor signal > 0.5V	Signal Stuck High	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors from all LWS that are connected to the EPM that supplies the LWS which is indicated by DTC (including check for chafing damage) for short circuits. Disconnect first the lining wear sensor from EPM where the DTC points on. Measure the lining wear sensor supply externally with lining wear sensor supply deactivated and the voltage must be below 0.5V. Repeat for each LWS that is connected to the corresponding EPM. If not replace EPM. Activate the lining wear supply and the value shall be within 4.5-5.5V. If not replace EPM. Disconnect also EPM from central ECU. Check resistance between voltage supply pin of wire to EPM and wire to lining wear sensor supply. If resistance is smaller than 200 kOhm then error is in the wiring.	Wheel 10 Brake Lining Wear Supply Circuit
xBS	U001088	ECU CAN controller in bus off.	Bus off	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - ECU damaged - Other ECU damaged connected to Backbone 1 network	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between other backbone ECU's and central ECU. Check for sufficient power supply. If no fault found, check if no disturbing improper devices are mounted. If error still active check with other backbone ECU's. If error still active check with other central ECU.	CAN Communication Backbone 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U200768	EBS on Trailer request red warning	Event Information	TRAILER EBS System in pneumatic backup. - Driver informed by RED warning sign and TRAILER brake malfunction indication.	- Trailer ECU damaged - Low pressure on trailer - Low voltage on trailer	Diagnostic Service 14. No additional action is needed	Check the fault memory of trailer EBS ECU(s).	CAN Communication Brake Proprietary Net
xBS	U200788	CAN controller in bus off, and change to one wire mode was not successful.	Bus off	Driver informed by YELLOW warning and brake malfunction indication	Damaged wire(s) or connector(s) (incl. connection between truck and trailer)	Diagnostic Service 14. No additional action is needed	Check if cables and connectors of ISO 11992 network are ok. Short circuit of both CAN wires, dirt or water in the connectors.	CAN Communication Brake Proprietary Net
xBS	U115387	Lost communication with ECU HMIIOM.	Missing Message	Driver informed by YELLOW warning and brake malfunction indication - Fading warning will be inaccurate.	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting HMIIOM signals and connected components.	Lost Communication With HMIIOM
xBS	U115487	Lost communication with ECU BBM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Check for correct EOL configuration regarding BBM signals. If configuration is correct, then failure is out of EBS system. Check ECU:s transmitting BBM signals and connected components.	Lost Communication With BBM
xBS	U115587	Lost communication with ECU CIOM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU due to missing switch information.	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting CIOM signals and connected components.	Lost Communication With CIOM
xBS	U115687	Lost communication with ECU RCIOM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting RCIOM signals and connected components.	Lost Communication With RCIOM
xBS	U115787	Lost communication with ECU CCIOM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting CCIOM signals and connected components.	Lost Communication With CCIOM
xBS	U115287	Lost communication with ECU ACM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication - Driveline retarder blending can not be active even if the lever is in "A" position.	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting RECU signals (ACM or TECU) and connected components.	Lost Communication With RECU
xBS	U115887	Lost communication with ECU DACU	Missing Message	Safety functions most likely switched off.	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Check for correct EOL configuration regarding DACU signals. If configuration is correct, then failure is out of EBS system. Check ECU:s transmitting DACU signals and connected components.	Lost Communication With DACU
xBS	U114887	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 1 Brake CAN Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U114888	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 1 Brake CAN Net
xBS	U114987	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 2 brake CAN Net
xBS	U114988	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 2 brake CAN Net
xBS	U010087	Lost communication with ECU EMS	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting EMS signals and connected components.	Lost Communication with EMS
xBS	U010187	Lost communication with ECU TECU	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Check for correct EOL configuration regarding signals of TECU. If configuration is correct, then failure is out of EBS system. Check ECU:s transmitting TECU signals and connected components.	Lost Communication with TECU
xBS	U014087	Lost communication with ECU VMCU	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting VMCU signals and connected components.	Lost Communication With VMCU
xBS	U114087	Lost communication with ECU APM	Missing Message	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Failure is out of EBS system. Check ECU:s transmitting APM signals and connected components.	Lost Communication With APM
xBS	U114A87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 3 brake CAN Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U114A88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 3 brake CAN Net
xBS	U114B87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 4 brake CAN Net
xBS	U114B88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 4 brake CAN Net
xBS	U114C87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 5 brake CAN Net
xBS	U114C88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 5 brake CAN Net
xBS	U114D87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 6 brake CAN Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U114D88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 6 brake CAN Net
xBS	U114E87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 7 brake CAN Net
xBS	U114E88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 7 brake CAN Net
xBS	U114F87	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 8 brake CAN Net
xBS	U114F88	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 8 brake CAN Net
xBS	U115087	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 9 brake CAN Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U115088	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 9 brake CAN Net
xBS	U115187	Lost communication with the EPM	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Check for sufficient (UB+-3V) power supply on the EPM. If same communication problem on other EPM. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU.	Wheel 10 brake CAN Net
xBS	U115188	EPM CAN controller in bus off	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	- Damaged wire(s) or connector(s) - EPM damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Disconnect the EPM and check if faultcode is still active. - If active check shortcut between EPM supply and CAN High and Low. - If inactive replace EPM.	Wheel 10 brake CAN Net
xBS	U002888	ECU CAN controller in bus off	Bus off	Driver informed by YELLOW warning and brake malfunction indication General reduced functionality of the ECU	- Damaged wire(s) or connector(s) - ECU damaged - Other ECU damaged connected to Chassis subnet network	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between other Chassis CAN ECU's and central ECU. Check for sufficient power supply. If no fault found, check if no disturbing improper devices are mounted. If error still active check with other chassis CAN ECU's. If error still active check with other central ECU.	CAN Communication Chassis Net
xBS	C005122	Sensor signal is to high (level depends on parameter "Maximum steering angle", P1BL0)	Signal Amplitude Greater Than Maximum	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Sensor is incorrectly calibrated - Incorrect mounting - Steering system does not correspond to the vehicle specification (wrong steering gearbox or steering gear arm) - Sensor damaged.	Diagnostic Service 14. No additional action is needed	Check if sensor is calibrated correctly: with engine running, set steered wheels in straight position, read steering angle via diagnosis, check if the value of steering angle is 0 +- 10 degrees. If value is not OK, recalibrate the sensor. While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle values via diagnosis (+-360 degrees). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005127	Sensor signal changes too fast.	Signal Rate of Change Above Threshold	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Drive of the sensor defect - Sensor damaged.	Diagnostic Service 14. No additional action is needed	Check the mounting and the drive of the sensor. If fault still active, replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005128	Sensor signal offset > 90 degrees	Signal Bias Level Out of Range / Zero Adjustment Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Sensor is incorrectly calibrated - Incorrect mounting - Sensor damaged.	Diagnostic Service 14. No additional action is needed	Check the mounting of the sensor and check if sensor is calibrated correctly (+-10 degree in straight line position). If necessary recalibrate the sensor. If fault is still active check steering system for specified parts. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C005149	Internal fault received from sensor via Sensor CAN	Internal Electronic Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Sensor damaged	Diagnostic Service 14. No additional action is needed	Replace the steering angle sensor.	Steering Wheel Position Sensor
xBS	C005154	Control routine "Enable Fast Steering Angle Sensor Adjustment" (0x0406) has not yet been started.	Missing Calibration	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Sensor not calibrated	Diagnostic Service 14. No additional action is needed	Recalibrate the steering angle sensor. (Start fast adjustment routine)	Steering Wheel Position Sensor
xBS	C005155	Sensor is detected, but not configured.	Not Configured	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Installed but not configured component	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Download correct dataset.	Steering Wheel Position Sensor
xBS	C005161	Sensor not calibrated	Signal Calculation Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	New steering angle sensor was mounted	Diagnostic Service 14. No additional action is needed	Calibrate the sensor again. If calibration cannot be performed successfully, check with an other sensor. If it still not possible to calibrate the sensor, check with other ECU.	Steering Wheel Position Sensor
xBS	C005162	Sensor signal is implausible compared to yaw rate sensor signal and reference yaw rate.	Signal Compare Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Sensor is incorrectly calibrated - Drive of the sensor defect - Incorrect mounting - Steering system does not correspond to the vehicle specification (wrong steering gearbox or steering gear arm) - Sensor damaged - Wheelbase or wheeltrack do not correspond to the vehicle specification - Calibrated vehicle speed does not correspond to the wheel speeds	Diagnostic Service 14. No additional action is needed	Check if sensor is calibrated correctly: with engine running, set steered wheels in straight position, read steering angle via diagnosis, check if the value of steering angle is 0 +- 10 degrees. If value is not OK, recalibrate the sensor. While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle values via diagnosis (+-360 +-10 degrees). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the steering angle sensor. Check whether wheelbase and wheeltrack correspond to the vehicle specification in the dataset and download correct dataset. Check for correct tacho signal calibration. If fault is still active replace the yaw rate sensor.	Steering Wheel Position Sensor
xBS	C005176	Sign of the Sensor signal is incorrect.	Wrong Mounting Position	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle sign values via diagnosis (+-360 degrees, sign should be negative while turning left and positive while turning right). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the steering angle sensor.	Steering Wheel Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C005193	The failure is reported if after a vehicle standstill a left and a right curve has been detected, but the steering angle sensor signal has not changed.	No Operation	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check correct mounting or damages of the sensor. If mounting is OK, then replce the sensor.	Steering Wheel Position Sensor
xBS	C006322	Sensor signal > range of the yaw-rate sensor.	Signal Amplitude Greater Than Maximum	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If mounting of the yaw rate sensor/central ECU is OK and the fault is still active, If the fault is still active, replace the yaw rate sensor/central ECU.	Yaw Rate Sensor
xBS	C006327	Sensor signal changed too fast.	Signal Rate of Change Above Threshold	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If mounting of the sensor is OK and the fault is still active, replace the yaw rate sensor/central ECU.	Yaw Rate Sensor
xBS	C00632F	Noise, vibrations or implausible frequencies on the yaw rate sensor signal are detected by the central ECU.	Signal Erratic	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If mounting of the sensor is OK and the fault is still active, replace the yaw rate sensor/central ECU.	Yaw Rate Sensor
xBS	C006349	Internal fault received from sensor via Sensor CAN	Internal Electronic Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Sensor damaged	Diagnostic Service 14. No additional action is needed	Change the sensor	Yaw Rate Sensor
xBS	C00634A	The failure is detected if either iYRS is configured but not populated (according HW coding), or ext YRS is configured but ESP sensor CAN or ESP sensor supply is not populated.	Incorrect Component Installed	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- ECU hardware type. - Wrong dataset	Diagnostic Service 14. No additional action is needed	Check if configuration matches the installation/central ECU type. Download correct dataset to Central ECU. Use correct part number of central ECU.	Yaw Rate Sensor
xBS	C006355	Sensor is detected, but not configured.	Not Configured	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Installed but not configured component	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Download correct dataset.	Yaw Rate Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C006364	Sensor signal is implausible compared to steering angle sensor signal and/or reference yaw rate.	Signal Plausibility Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged - Wheelbase or wheeltrack do not correspond to the vehicle specification - Calibrated vehicle speed does not correspond to the wheel speeds	Diagnostic Service 14. No additional action is needed	Check the mounting of the sensor. If the fault is still active check replace YRS/ECU.	Yaw Rate Sensor
xBS	C006376	Sign of the Sensor signal is incorrect.	Wrong Mounting Position	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If mounting of the yaw rate sensor/central ECU is OK and the fault is still active, calibrate the yaw rate sensor. If the fault is still active, replace the yaw rate sensor/central ECU.	Yaw Rate Sensor
xBS	C006392	- Difference between measured and reference lateral acceleration is > 2 m/s^2. - Temporary error Flag is read from sensor CAN sent by yaw rate sensor	Performance or Incorrect Operation	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	See other reported ESP related faults	Diagnostic Service 14. No additional action is needed	Check for other ESP-related faults.	Yaw Rate Sensor
xBS	C006396	Offset of sensor signal is too high.	Component Internal Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Conditions of calibration are not met - Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check and correct the mounting of the yaw rate sensor/central ECU. If fault is still active, replace the yaw rate sensor/central ECU.	Yaw Rate Sensor
xBS	C006122	Sensor signal > 15 m/s^2	Signal Amplitude Greater Than Maximum	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If the DTC counter is below 3, recalibrate the sensor. If the DTC counter is higher then 2, replace the yaw rate sensor/central ECU.	Lateral Acceleration Sensor
xBS	C006128	Sensor signal offset > 2.5 m/s^2	Signal Bias Level Out of Range / Zero Adjustment Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Incorrect mounting - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. If mounting of the yaw rate sensor/central ECU is OK and the fault is still active, calibrate the yaw rate sensor/central ECU. If fault is still active, replace the yaw rate sensor/central ECU.	Lateral Acceleration Sensor
xBS	C006154	Sensor not calibrated	Missing Calibration	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	Sensor not calibrated	Diagnostic Service 14. No additional action is needed	Recalibrate the lateral acceleration sensor	Lateral Acceleration Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C006164	Sensor signal is implausible compared to other ESP sensor signals and the reference yaw rate.	Signal Plausibility Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Sensor is incorrectly calibrated - Incorrect mounting - Steering system does not correspond to the vehicle specification (wrong steering gearbox or steering gear arm) - Sensor damaged - Wheelbase or wheeltrack do not correspond to the vehicle specification - Calibrated vehicle speed does not correspond to the wheel speeds	Diagnostic Service 14. No additional action is needed	Check the mounting of the yaw rate sensor/central ECU. Perform standstill calibration. Check whether wheelbase and wheeltrack correspond to the vehicle specification and are correct programmed in the dataset, if necessary. Check calibration of vehicle speed. Check if sensor is calibrated correctly: with engine running, set steered wheels in straight position, read steering angle via diagnosis, check if the value of steering angle is 0 +- 10 degrees. If value is not OK, recalibrate the sensor. While engine still running, mark zero position on the steering wheel and turn it in both directions by one turn (360 degrees). Check the steering angle values via diagnosis (+-360 degrees). If values are not OK, check the mounting and the drive of the sensor. If fault is still active check steering system for specified parts (e.g. steering lever). With engine running turn the steering wheel to its end positions, read the steering angle values via diagnosis and check if the read values are within the allowed range defined in the dataset. If fault is still active, replace the yaw rate sensor/central ECU.	Lateral Acceleration Sensor
xBS	U012687	No CAN messages recevied from the sensor.	Missing Message	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the wiring and connectors between steering angle sensor and central ECU and external yaw rate sensor and central ECU for open circuit and physical damage (including check for chafing damage), check the power supply of the ESP sensors 8,3V - 14,5V, if error still active, replace steering angle sensor, if error still active, replace central ECU.	Lost Communication With Steering Angle Sensor Module
xBS	U012787	No CAN messages recevied from the sensor.	Missing Message	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the wiring and connectors between external yaw rate sensor and central ECU and between steering angle sensor and central ECU for open circuit and physical damage (including check for chafing damage), check the power supply of the ESP sensors 8,3V - 14,5V, if error still active, replace external yaw rate sensor, if error still active, replace central ECU.	Lost Communication With External Yaw Rate Sensor Module
xBS	C100C01	- Voltage > 4.5V before switching ON sensor supply - Current > 200-600 mA. - Voltage error information received from sensor via Sensor CAN.	General Electrical Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and ESP sensors for short-circuits. 1. Remove sensor(s) and check for faultcodes again. If only communication fault codes apear the problem is in one of the sensors. Connect one by one to identify the faulty sensor. 2. If still a problem with sensors removed the problem is in the harness or ECU. 3. Check wiring for shortcuts to ground or other signals and CNT (below 3 ohm). Repair if faulty 4. If harness is ok and there is still active faultcode, replace ECU. 5. If harness is ok and no active fault try to move harness to reproduce the fault.	EBS ESP Sensor Supply
xBS	C100C16	Voltage < 8.3 V	Circuit Voltage Below Threshold	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and ESP sensor(s) for short-circuits. If error is still active replace central ECU.	EBS ESP Sensor Supply
xBS	C100C17	Voltage > 14.5 V	Circuit Voltage Above Threshold	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - ECU damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between central ECU and ESP sensor(s) for short-circuits to other signals higher than 12 V. If error is still active replace central ECU.	EBS ESP Sensor Supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100C49	ECU Internal test indicating error	Internal Electronic Failure	ESP function disabled (Also other safety systems may be disabled). Driver informed by YELLOW warning and ESP malfunction indication	ECU damaged	Diagnostic Service 14. No additional action is needed	Replace central ECU	EBS ESP Sensor Supply
xBS	C100D01	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 1
xBS	C100D31	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100D38	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 1
xBS	C100D3A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 1
xBS	C100D54	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100D55	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 1
xBS	C100D95	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 1
xBS	C100E01	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100E31	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 2
xBS	C100E38	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 2
xBS	C100E3A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100E54	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 2
xBS	C100E55	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 2
xBS	C100E95	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 2
xBS	C100F01	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100F31	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 3
xBS	C100F38	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 3
xBS	C100F3A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 3

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C100F54	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 3
xBS	C100F55	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 3
xBS	C100F95	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Shortcut between sensors connected to the same EPM	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 3
xBS	C101001	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring.	Wheel Speed Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101031	Implausible drop of signal (signal is frequently jumping between above 11km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 4
xBS	C101038	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 4
xBS	C10103A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 4

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101054	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 4
xBS	C101055	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 4
xBS	C101095	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 4
xBS	C101101	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101131	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 5
xBS	C101138	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 5
xBS	C10113A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101154	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 5
xBS	C101155	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 5
xBS	C101195	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 5
xBS	C101201	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101231	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 6
xBS	C101238	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 6
xBS	C10123A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101254	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 6
xBS	C101255	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 6
xBS	C101295	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 6
xBS	C101301	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 7

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101331	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 7
xBS	C101338	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 7
xBS	C10133A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 7

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101354	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 7
xBS	C101355	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 7
xBS	C101395	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 7
xBS	C101401	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 8

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101431	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 8
xBS	C101438	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 8
xBS	C10143A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 8

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101454	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 8
xBS	C101455	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 8
xBS	C101495	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 8
xBS	C101501	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 9

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101531	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 9
xBS	C101538	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 9
xBS	C10153A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 9

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101554	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 9
xBS	C101555	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 9
xBS	C101595	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 9
xBS	C101601	Voltage between EPM GND and wheel speed sensor (any pin) < 0,5V or > 4.3V.	General Electrical Failure	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor damaged - EPM damaged	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Disconnect wheel speed sensor from EPM. 1. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. -> Change Sensor or wiring if out of range 2. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. -> Change Sensor or wiring if out of range 3. Connect the WSS, Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must be between 0,5 - 4.3V. -> If it is in this range and error is still active, replace the EPM. 4. If it is out of range (Below 0,5 or above 4.3 V) , change sensor and or wiring. 5. If the faultcode is still active, change the EPM.	Wheel Speed Sensor 10

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101631	Implausible drop of signal (signal is frequently jumping between above 11 km/h and zero)	No Signal	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) - Sensor or mounting damaged - EPM damaged - Too large airgap between tone wheel and sensor	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits (between H and L) and open-circuits. Measure Voltage between EPM GND and wheel speed sensor (any pin) Voltage must in range 0.5-4.3V. Disconnect wheel speed sensor from EPM. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Switch left/right WSS (with brakeout box) and see if the problem remain on the same side (vehicle has to be driven) =>replace EPM. If problem move to other side change defect sensor	Wheel Speed Sensor 10
xBS	C101638	- Plausibility test for the wheel speed sensor signal frequency - Rate of the ABS active/inactive status change is too high.	Signal Frequency Incorrect	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Sensor or mounting damaged - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. Do waveform check with oscilloscope. Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-5km/h). Check wheel speed sensor mounting (Check 'Klämhylsa'). Check the tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth.	Wheel Speed Sensor 10
xBS	C10163A	Minimum measurable wheel-speed >10 km/h	Signal Has Too Many Pulses	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Large airgap - Dirty or damaged tone wheel - Toggling tone wheel, production-failures of tone wheel	Diagnostic Service 14. No additional action is needed	Check and correct air gap between wheel speed sensor and tone wheel by measuring the wheel speed signal. (Wheel speed sensor signal must be higher than 100mV peak to peak at wheel speed of about 3-4km/h). Check the cables and connectors between EPM and wheel-speed sensor for short-circuits and open-circuits. Measure resistance of Sensor and cabling. Resistance must be between 800 …. 2000 Ohm. Measure insulation resistance of sensor and cabling to vehicle GND. Resistance must be > 200kOhm. If possible : Check tone wheel by checking wheel speed sensor signal for correct wave(Not possible to define the correct waveform) form according the specification for wheel speed sensor wave form or check tone wheel for dirt and metallic particles, for wobble, for pitch error, for the correct number of teeth. Check wheel speed sensor mounting.	Wheel Speed Sensor 10

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	C101654	Difference between the learned wheel circumferences and programmed values > 25%	Missing Calibration	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Tyre with wrong circumference is mounted on one of the axels - Tachograph not calibrated - Wrong number of teeth of tone wheel.	Diagnostic Service 14. No additional action is needed	Check if tire size of all tires is according vehicle specification. Check C3 Signal calibration, check tachograp for additional faultcodes. Check number of teeth on tone wheel on all wheels. (Check tire size of all tires in data set.) V3P to check	Wheel Speed Sensor 10
xBS	C101655	Sensor is detected, but not configured.	Not Configured	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	- Damaged wire(s) or connector(s) (dirt in the connector ,low resistance detected)) - Connected but not correctly configured sensor - Sensor connected to incorrect EPM connector	Diagnostic Service 14. No additional action is needed	Check if configuration matches installation. Check if sensor connected according to wiring diagram. If configuration matches installation, check wheel speed sensor's connector of EPM	Wheel Speed Sensor 10
xBS	C101695	Shortcut between sensors connected to the same EPM	Incorrect Assembly	- ABS function not fully operational, possibility for wheel locking when braking. Driver informed by YELLOW warning and ABS malfunction indication. - ESP switched off. Driver informed by YELLOW warning and ESP malfunction indication	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Disconnect wheel speed sensor from EPM. Measure resistance between all connected sensor in pairs. Resistance must be > 200kOhm. Replace sensors or harness	Wheel Speed Sensor 10
xBS	U116649	Short circuit or open circuit on CAN high line.	Internal Electronic Failure	Driver informed by YELLOW warning and brake malfunction indication	Damaged wire(s) or connector(s) (incl. connection between truck and trailer)	Diagnostic Service 14. No additional action is needed	Check if cable and connectors of the high line (CAN H) are ok (interruption, short cut, dirt or water in the connectors) at tractor and trailer.	CAN Communication Brake Proprietary Net High Line
xBS	U116749	Short circuit or open circuit on CAN low line.	Internal Electronic Failure	Driver informed by YELLOW warning and brake malfunction indication	Short circuit or open circuit on CAN low line.	Diagnostic Service 14. No additional action is needed	Check if cable and connectors of the low line (CAN L) are ok (interruption, short cut, dirt or water in the connectors) at tractor and trailer.	CAN Communication Brake Proprietary Net Low Line
xBS	U116887	Communication is interrupted with all configured EPMs.	Missing Message	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. Measure CAN wiring resistances of the wires and between the CAN wires and between CAN wires and EPM voltage supply lines and between CAN wires and vehicle ground with disconnected wires from both central ECU and EPM:s. Wiring resistances should be below 3 Ohm and resistances between CAN wires, between CAN wires EPM voltage supply lines and between wires and vehicle ground should be above 200 kOhm.	CAN Communication EBS Private Brake Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
xBS	U116888	ECU CAN controller in bus off.	Bus off	EBS System in pneumatic backup. - Driver informed by RED warning sign and brake malfunction indication. - Changed Pedal feeling (need to brake harder). - System pressure limited to 10 bar.	Damaged wire(s) or connector(s)	Diagnostic Service 14. No additional action is needed	Check the cables and connectors and wiring nodes between EPM's and central ECU. - Measure CAN wiring resistances of each wire (less than 3 ohm) - Measure between the CAN wires (depends on No of connected EPMs) - Between CAN wires EPM voltage supply lines and between wires and vehicle ground should be above 200 kOhm. Try to remove EPMs one by one to detect the a faulty EPM. If the problem remains the problem is in the harness or ECU. If the error is still active replace ECU	CAN Communication EBS Private Brake Net
xBS	U114287	Lost communication with EBS trailer CAN	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Trailer ECU damaged	Diagnostic Service 14. No additional action is needed	Failure is out of tractor EBS system. Check Trailer ECU#1 and connected components.	Lost Communication with Trailer1 ECU
xBS	U115987	Lost communication with EBS trailer CAN	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Trailer ECU damaged	Diagnostic Service 14. No additional action is needed	Failure is out of tractor EBS system. Check Trailer ECU#2 and connected components.	Lost Communication with Trailer2 ECU
xBS	U114387	Lost communication with EBS trailer CAN	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Trailer ECU damaged	Diagnostic Service 14. No additional action is needed	Failure is out of tractor EBS system. Check Trailer ECU#4 and connected components.	Lost Communication with Trailer4 ECU
xBS	U114487	Lost communication with EBS trailer CAN	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Trailer ECU damaged	Diagnostic Service 14. No additional action is needed	Failure is out of tractor EBS system. Check Trailer ECU#5 and connected components.	Lost Communication with Trailer5 ECU
xBS	U114586	Axle load is sent as NOT AVAILABLE or ERROR	Signal Invalid	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Damaged axle load sensor - Damaged sending ECU	Diagnostic Service 14. No additional action is needed	Check for correct EOL configuration regarding vehicle weight signals. If configuration is correct, then failure is out of EBS system. Check ECU transmitting axle load signals and connected components.	Vehicle Weight Signal
xBS	U114781	- Too high accelerations at the signal - Swinging signal - Difference betwwen received vehicle speed and driven wheel speeds > 30%	Invalid Serial Data Received	Driver informed by YELLOW warning and brake malfunction indication	- Tachograph signal not calibrated (signal not plausible) - Wrong tyre size (compared either to the configuration or to the speed signal or to both)	Diagnostic Service 14. No additional action is needed	Check tyre sizes of the driven wheels. If tyre sizes are correct, check tachosignal (according to regulations). If tachosignal is not correct, check components related to tachometer.	Vehicle Speed Electronic Data Link
xBS	U115B87	Lost communication with EBS trailer CAN	Missing Message	Driver informed by YELLOW warning and brake malfunction indication	- Damaged wire(s) or connector(s) - Trailer ECU damaged	Diagnostic Service 14. No additional action is needed	Failure is out of tractor EBS system. Check Trailer ECU#3 and connected components.	Lost Communication with Trailer3 ECU
xBS	C107E68	Sensor value in procentage < 0.4%	Event Information	Driver informed by YELLOW warning and Brake Worn indication	- Brake pads worn out - Sensor damaged	Diagnostic Service 14. No additional action is needed	Check brake pads, if necessary exchange them.	Brake Lining Wear
FAS	U300004	Main algorithm has detected an error in an internal signal	System Internal Failures	N.A. - System behaviour determined by other DTC	Internal torque and angle sensor error	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
FAS	U300041	Faulty SW and/or Data Checksum Failure (invalid signature or checksum).	General Checksum Failure	VDS electrical motor disengaged. Power steering via hydraulics.	Invalid SW and/or Data downloaded	Diagnostic Service 14. No additional action is needed.	1. Download new SW 2. Replace VDS	Electronic control unit
FAS	U300044	RAM error	Data Memory Failure	VDS electrical motor disengaged. Power steering via hydraulics.	RAM has been corrupted	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Electronic control unit
FAS	U300045	NVRAM/ROM Error	Program Memory Failure	VDS electrical motor disengaged. Power steering via hydraulics.	1.CRC check mismatch occurs 2.Write to NVRAM fails 3. ECU detects a ROM memory failure	Diagnostic Service 14. No additional action is needed.	1. Download new SW 2. Replace VDS	Electronic control unit
FAS	U300046	EEPROM failure	Calibration / Parameter Memory Failure	VDS electrical motor disengaged. Power steering via hydraulics.	1.Read and write error 2. CRC check error	Diagnostic Service 14. No additional action is needed.	1. Download new SW 2. Replace VDS	Electronic control unit
FAS	U300049	All types of internal faults that are not memory related (this includes also slave processors)	Internal Electronic Failure	VDS electrical motor disengaged. Power steering via hydraulics.	- Torque/Angle Sensor short circuit battery or open circuit Torque/Angle Sensor short circuit ground FAS internal CAN communication Alive/sequence counter incorrect Error in A/D HW	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Electronic control unit
FAS	U30004B	PCB over temperature > 110 deg C	Over Temperature	VDS electrical motor disengaged. Power steering via hydraulics.	- Steering gear malfunction. Steering gear high oil temp	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. 2. Check steering gear malfunction. 3. Replace VDS.	Electronic control unit
FAS	U300054	Processor calibration not performed	Missing Calibration	VDS electrical motor disengaged. Power steering via hydraulics.	Necessary information missing on NC or DSP processor	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Electronic control unit
FAS	U300061	Main algorithm has detected a calculation error.	Signal Calculation Failure	N.A. - System behaviour determined by other DTC.	Could be caused by either an ECU internal error or an ECU external error	Diagnostic Service 14. No additional action is needed.	1. Check chassi CAN for fault codes on other nodes that could explain the behaviour. 2. Perform front wheel alignment (incl. calibration of VDS motor angle offset). 3. Replace VDS	Electronic control unit
FAS	U300064	Actual motor torque supervision - Discrepancy between requested motor torque and actual motor torque.	Signal Plausibility Failure	VDS electrical motor disengaged. Power steering via hydraulics.	Electrical motor malfunction.	Diagnostic Service 14. No additional action is needed.	Replace VDS	Electronic control unit
FAS	U300068	Failsafe mode is entered due to a serious error. DTC readout is needed to get the serious error causing failsafe mode.	Event Information	VDS electrical motor disengaged. Power steering via hydraulics.	DTC readout is needed to get the error causing failsafe mode	Diagnostic Service 14. No additional action is needed.	1. If no other DTC: Calibrate torque offset compensation on existing VDS. 2. N.A. - Service action determined by other DTC.	Electronic control unit
FAS	U300092	Main algorithm has detected a condition leading to an abnormal torque to the electrical motor.	Performance or Incorrect Operation	Possibly unnormal steering feel (other than motor disengaged or degraded mode)	- Wrong front wheel alignment Flatted tyre Wrong calibration of electrical motor or other external error	Diagnostic Service 14. No additional action is needed.	N/A - Service action determined by other DTC	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
FAS	U300094	Faulty operation (this includes also slave processors) Unexpected operation of type: 1. unexpected reset due to null pointer reference 2. division by zero 3. watchdog failure	Unexpected Operation	VDS electrical motor disengaged. Power steering via hydraulics.	Software error (division by zero, null pointer reference, etc)	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Electronic control unit
FAS	U300098	Hydraulic servo probably overheating	Component or System Over Temperature	Degraded Mode (Max torque N/A)	Failure in hydraulic servo or extensive steering at low speeds.	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. 2. Check steering gear malfunction. 3. Replace VDS.	Electronic control unit
FAS	U002868	A critical signal on Chassis CAN have been timed-out or have its error value set.	Event Information	N.A. - System behaviour determined by other DTC.	Probably failure in another ECU.	Diagnostic Service 14. No additional action is needed.	1. Check chassi CAN for fault codes on other nodes that could explain the behaviour. 2. Replace VDS.	CAN Communication Chassis Net
FAS	U002888	Bus off error occurred on Chassis CAN.	Bus off	VDS electrical motor disengaged. Power steering via hydraulics.	Short circuit on CAN cables (CAN High and/or CAN Low).	Diagnostic Service 14. No additional action is needed.	1. Check chassi subnet 2. Replace VDS.	CAN Communication Chassis Net
FAS	U114106	Main algorithm has detected a critical fault causing failsafe mode.	Algorithm Based Failures	VDS electrical motor disengaged. Power steering via hydraulics.	- Either an internal ECU error or ECU external error. Check other failure codes from this unit to determine the root cause.	Diagnostic Service 14. No additional action is needed.	N.A. - Service action determined by other DTC	Electronic Control Unit Degraded Mode
FAS	U114168	Main algorithm has set unit to degraded mode. N.B. This DTC is no longer used!	Event Information	N.B. This DTC is no longer used! Possibly unnormal steering feel (other than motor disengaged or degraded mode).	N.B. This DTC is no longer used!	Diagnostic Service 14. No additional action is needed.	N/A	Electronic Control Unit Degraded Mode
FAS	U1FFF68	Factory Mode is active.	Event Information	VDS in factory mode (for example VDS electrical motor disengaged).	Parameter FactoryModeActive is set to True.	Diagnostic Service 14. No additional action is needed.	Shut off factory mode.	Factory Mode Active
FAS	U300316	ECU voltage below 14 V	Circuit Voltage Below Threshold	VDS electrical motor disengaged. Power steering via hydraulics.	Broken power lines.	Diagnostic Service 14. No additional action is needed.	1. Check voltage supply to VDS unit. 2. Replace VDS.	Battery Power
FAS	U300317	ECU voltage above 37 V	Circuit Voltage Above Threshold	VDS electrical motor disengaged. Power steering via hydraulics.	BatteryPower, lines, Fuse (FAS generates power)	Diagnostic Service 14. No additional action is needed.	1. Check voltage supply to VDS unit. 2. Replace VDS.	Battery Power
FAS	C100016	Electrical Motor supply voltage too low i.e below 13.5 V	Circuit Voltage Below Threshold	VDS electrical motor disengaged. Power steering via hydraulics.	Broken power lines. Blown fuse.	Diagnostic Service 14. No additional action is needed.	1. Check voltage supply to VDS unit. 2. Replace VDS.	Front Active Steering Electrical Motor
FAS	C100017	ECU voltage above 37 V	Circuit Voltage Above Threshold	VDS electrical motor disengaged. Power steering via hydraulics.	- Battery Power cables FAS generates power	Diagnostic Service 14. No additional action is needed.	1. Check voltage supply to VDS unit. 2. Replace VDS.	Front Active Steering Electrical Motor
FAS	C100019	Motor overload - the average motor current consumption too high (> 130 deg C, calculated value).	Circuit Current Above Threshold	VDS electrical motor disengaged. Power steering via hydraulics.	- Steering Gear malfunction. Steering column (for example excessive friction).	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. 2. Check steering gear malfunction. 3. Replace VDS.	Front Active Steering Electrical Motor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
FAS	C10001C	Electrical motor supply voltage out of range that do not constitute disengagement of FAS motor (below 16 V or above 35 V).	Circuit Voltage Out of Range	Degraded Mode (Max torque N/A).	- Battery Power cables	Diagnostic Service 14. No additional action is needed.	1. Check voltage supply to VDS unit. . Replace VDS.	Front Active Steering Electrical Motor
FAS	C10001D	Motor minor overload - the average motor current consumption too high (> 120 deg C, calculated value).	Circuit Current Out of Range	Degraded Mode (Max torque N/A)	- Steering Gear malfunction. Steering column (for example excessive friction)	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. . Check steering gear malfunction. . Replace VDS.	Front Active Steering Electrical Motor
FAS	C10004B	Electrical motor critical over temperature (> 110 deg C).	Over Temperature	VDS electrical motor disengaged. Power steering via hydraulics.	Steering Gear malfunction. teering gear high oil temp. teering column (for example excessive friction)	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. . Check steering gear malfunction. . Replace VDS.	Front Active Steering Electrical Motor
FAS	C100098	Electrical motor minor over temperature (> 100 deg C) or temp sensor broken.	Component or System Over Temperature	Degraded Mode (Max torque N/A)	Steering Gear malfunction. teering gear high oil temp. teering column (for example excessive friction)	Diagnostic Service 14. No additional action is needed.	1. Clean VDS externally if dirty. . Check steering gear malfunction. . Replace VDS.	Front Active Steering Electrical Motor
FAS	C100364	Comparison of the two "redundant" torque sensor inputs failed.	Signal Plausibility Failure	VDS electrical motor disengaged. Power steering via hydraulics.	Torque sensor error.	Diagnostic Service 14. No additional action is needed.	Replace VDS	Front Active Steering Failsafe Signal Circuit
FAS	C100254	Angle offset calibration not performed.	Missing Calibration	Possibly unnormal steering feel (other than motor disengaged or degraded mode)	Angle offset calibration not performed	Diagnostic Service 14. No additional action is needed.	1. Calibrate motor angle offset on existing VDS. . Replace VDS	Front Active Steering Angle Sensor
FAS	C100262	ECU internal angle sensor sector mismatch against chassis CAN steering wheel angle signal.	Signal Compare Failure	VDS electrical motor disengaged. Power steering via hydraulics.	- ECU internal angle sensor error External steering wheel angle signal missing External steering wheel angle signal error	Diagnostic Service 14. No additional action is needed.	1. Check SAS steering wheel angle available. . Perform front wheel alignment (incl. calibration of VDS motor angle offset). . Replace VDS.	Front Active Steering Angle Sensor
FAS	C100264	Comparison of the two "redundant" angle inputs failed.	Signal Plausibility Failure	VDS electrical motor disengaged. Power steering via hydraulics.	Angle sensor error.	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Front Active Steering Angle Sensor
FAS	C100154	TOC (Torque Offset Compensation) calibration missing or faulty.	Missing Calibration	Possibly unnormal steering feel (other than motor disengaged or degraded mode)	TOC (Torque Offset Compensation) calibration missing or faulty.	Diagnostic Service 14. No additional action is needed.	1. Calibrate torque offset compensation on existing VDS. . Replace VDS.	Front Active Steering Torque Sensor
FAS	C100164	Comparison of the two "redundant" torque inputs failed.	Signal Plausibility Failure	VDS electrical motor disengaged. Power steering via hydraulics.	Torque sensor error.	Diagnostic Service 14. No additional action is needed.	Replace VDS.	Front Active Steering Torque Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	P007011	U < 0,187V	Circuit Short To Ground	*Ambient Air Temperature Sensor cannot be activated *AmbientAirTemperature = Error	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Ambient Air Temperature Sensor	Ambient Air Temperature Sensor "A"
XCIOM	P007015	4,702V <= U <= 5,000V	Circuit Short To Battery or Open	*Open circuit: Ambient Air Temperature Sensor cannot be activated *Short circuit to battery: Ambient Air Temperature Sensor cannot be deactivated *AmbientAirTemperature = Error	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Ambient Air Temperature Sensor	Ambient Air Temperature Sensor "A"
XCIOM	P007016	0,187V <= U < 0,3V	Circuit Voltage Below Threshold	*General reduced functionality of the Ambient Air Temperature Sensor *AmbientAirTemperature = Error	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	*Check Ambient Air Temperature Sensor *Check wires and connectors between ECU and Ambient Air Temperature Sensor	Ambient Air Temperature Sensor "A"
XCIOM	C109411	I > 3,022A..4,083A	Circuit Short To Ground	*Auxiliary Horn cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Auxiliary Horn	Auxiliary Horn Circuit
XCIOM	C109415	No Load detected (Offstate Diagnostic)	Circuit Short To Battery or Open	*Open circuit: Auxiliary Horn cannot be activated *Short circuit to battery: Auxiliary Horn cannot be deactivated	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Auxiliary Horn	Auxiliary Horn Circuit
XCIOM	C109611	I < 4,199A..5,672A	Circuit Short To Ground	*Left Front Fog Light cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *AuxFogLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Fog lights	Left Front Fog Light Circuit
XCIOM	C109615	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Left Front Fog lights cannot be activated. *Short circuit to battery: Left Front Fog lights cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *AuxFogLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Fog lights	Left Front Fog Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109711	I < 4,199A..5,672A	Circuit Short To Ground	*Right Front Fog Light cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *AuxFogLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Fog lights	Right Front Fog Light Circuit
XCIOM	C109715	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Right Front Fog lights cannot be activated. *Short circuit to battery: Right Front Fog lights cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *AuxFogLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Fog lights	Right Front Fog Light Circuit
XCIOM	C109811	I < 4,199A..5,672A	Circuit Short To Ground	*Left Extra Main Beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *ExtraMainBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Extra Main Beam	Left Extra Main Beam Circuit
XCIOM	C109815	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Left Extra Main Beam cannot be activated. *Short circuit to battery: Left Extra Main Beam cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *ExtraMainBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Extra Main Beam	Left Extra Main Beam Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109911	I < 4,199A..5,672A	Circuit Short To Ground	* Right Extra Main Beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *ExtraMainBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Extra Main Beam	Right Extra Main Beam Circuit
XCIOM	C109915	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Right Extra Main Beam cannot be activated. *Short circuit to battery: Right Extra Main Beam cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *ExtraMainBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Extra Main Beam	Right Extra Main Beam Circuit
XCIOM	C109A11	I < 4,199A..5,672A	Circuit Short To Ground	*Left Static Corner Light cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *StaticCornerLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Static Corner Light	Left Static Corner Light Circuit
XCIOM	C109A15	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Left Static Corner Light cannot be activated. *Short circuit to battery: Left Static Corner Light cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *StaticCornerLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Static Corner Light	Left Static Corner Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109B11	I < 4,199A..5,672A	Circuit Short To Ground	*Right Static Corner Light cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *StaticCornerLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Static Corner Light	Right Static Corner Light Circuit
XCIOM	C109B15	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Right Static Corner Light cannot be activated. *Short circuit to battery: Right Static Corner Light cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *StaticCornerLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Static Corner Light	Right Static Corner Light Circuit
XCIOM	C10FB11	I > 4,199A..5,672A	Circuit Short To Ground	*Left Front Fog and Static Corner Light cannot de activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *AuxFogLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Fog and Static Corner Light	Left Front Fog and Static Corner Light Circuit
XCIOM	C10FB15	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Left Front Fog and Static Corner Light cannot be activated. *Short circuit to battery: Left Front Fog and Static Corner Light cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *AuxFogLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Fog and Static Corner Light	Left Front Fog and Static Corner Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10FC11	I > 4,199A..5,672A	Circuit Short To Ground	*Right Front Fog and Static Corner Light cannot de activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *AuxLamp_Front_Status = 0 (Inactive) *AuxFogLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Fog and Static Corner Light	Right Front Fog and Static Corner Light Circuit
XCIOM	C10FC15	I < 0.7A (worst case: I < 450mA...760mA)	Circuit Short To Battery or Open	*Open circuit: Right Front Fog and Static Corner Light cannot be activated. *Short circuit to battery: Right Front Fog and Static Corner Light cannot be deactivated. *AuxLamp_Front_Status = 2 (Error) *AuxFogLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Fog and Static Corner Light	Right Front Fog and Static Corner Light Circuit
XCIOM	C103F11	I > 3,513A..4,745A I > 9.43A..12,737 (limphome)	Circuit Short To Ground	*Bending lights not functional *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *Bending lights are stuck	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and BLECU	Bending light ECU supply
XCIOM	C103F12	U > Ubat - 1V (Offstate)	Circuit Short To Battery	*The output cannot be deactivated. *The BLECU will always be awake and the trucks quiescient current will consume the battery.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and BLECU	Bending light ECU supply
XCIOM	C10CF11	I > 1,177A..1,589A	Circuit Short To Ground	*Instep Light cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *ChassiInstepLightsLamp_stat = 2 (Tested, Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Instep Light	Instep Light Circuit
XCIOM	C10CF15	I < 0.1A. (open circuit or two (VT) / one (RT) bulbs broken. (nominal:100mA worst case: I < 75mA...130mA)	Circuit Short To Battery or Open	*Open circuit: Instep light cannot be activated. *Short circuit to battery: Instep light cannot be deactivated. *ChassiInstepLightsLamp_stat = 2 (Tested, Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Instep Light	Instep Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10CF18	0.1A< I <0.63A (1 bulb OK) I > 0.63A --> 0 bulbs broken, everything ok. (nominal:630mA worst case: I < 590mA...685mA)	Circuit Current Below Threshold	*Indicates that one of the lamps are broken. *In case of open circuit, the output cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *ChassiInstepLightsLamp_stat = 2 (Tested, Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Instep Light	Instep Light Circuit
XCIOM	U30004A	- Incorrect Component Installed (Incorrect ECU variant OR incorrect Kola variant configuration used)	Incorrect Component Installed	Some of the I/Os may not work or behave unexpectedly.	- Incorrect Component Installed - Incorrect Kola variant configuration used	Diagnostic Service 14. No additional action is needed.	Re-configure correct Kola variant configuration OR replace with correct ECU variant.	Electronic control unit
XCIOM	U30004B	The temperature has exceeded the threshold on one of the outputs.	Over Temperature	*General reduced functionality of the ECU *The output that is overheated will be turned OFF.	The ambient temperature is too high.	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Electronic control unit
XCIOM	C104411	I > 3,807A..5,142A	Circuit Short To Ground	*City horn cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and City horn	City Horn Circuit
XCIOM	C104415	No Load detected (Offstate Diagnostic)	Circuit Short To Battery or Open	*Open circuit: City horn cannot be activated *Short circuit to battery: City horn cannot be deactivated	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and City horn	City Horn Circuit
XCIOM	U200488	Bus off detected	Bus off	*No communication on CAN *General reduced system functionality *ECU to sleep mode	- CAN_H and CAN_L short - CAN_H to Gnd - CAN_L to Battery	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check ECUs	CAN Communication Advanced Lightning Net
XCIOM	U300041	ECU checksum error	General Checksum Failure	*General reduced functionality of the ECU *HW Limp Home	- Memory hardware error - Software error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
XCIOM	U300044	Program memory corruption	Data Memory Failure	*General reduced functionality of the ECU *HW Limp Home	- Memory hardware error - Software error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
XCIOM	U300045	Program memory corruption	Program Memory Failure	*General reduced functionality of the ECU *HW Limp Home	- Memory hardware error - Software error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
XCIOM	U300049	ECU internal failure	Internal Electronic Failure	*General reduced functionality of the ECU *HW Limp Home	- Hardware error - Software error	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
XCIOM	U300093	Software failsafe	No Operation	HW Limp Home	- Hardware error - Software error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
XCIOM	U300094	Watchdog failure	Unexpected Operation	*General reduced functionality of the ECU *HW Limp Home	- Hardware error - Software error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	U002888	Bus off detected	Bus off	*No communication on CAN *General reduced system functionality *ECU to sleep mode	- CAN_H and CAN_L short - CAN_H to Gnd - CAN_L to Battery	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check ECUs	CAN Communication Chassis Net
XCIOM	U300716	U<16V	Circuit Voltage Below Threshold	General reduced functionality of the ECU	alternator problem or wrong battery	Diagnostic Service 14. No additional action is needed.	Check power supply system	Control Module Input Power "B"
XCIOM	U300717	U>32V	Circuit Voltage Above Threshold	*General reduced functionality of the ECU *HW limp home	1. Wrong or two batteries has been connected 2. Alternator problem	Diagnostic Service 14. No additional action is needed.	Check power supply system	Control Module Input Power "B"
XCIOM	U300616	U<16V	Circuit Voltage Below Threshold	General reduced functionality of the ECU	alternator problem or wrong battery	Diagnostic Service 14. No additional action is needed.	Check power supply system	Control Module Input Power "A"
XCIOM	U300617	U>32V	Circuit Voltage Above Threshold	*General reduced functionality of the ECU *HW limp home	1. Wrong or two batteries has been connected 2. Alternator problem	Diagnostic Service 14. No additional action is needed.	Check power supply system	Control Module Input Power "A"
XCIOM	U1FFF68	Factory mode parameter P1C54 is active	Event Information	A truck delivered with Factory mode active will have unspecified behavior that might influence customer safety.	The parameter P1C54 is set to TRUE	Diagnostic Service 14. No additional action is needed.	Correct the Factoty Mode parameter setting	Factory Mode Active
XCIOM	U114268	max limit of DTC's has been set	Event Information	*Normal functionality *Max. number of DTCs reached. No more DTCs will be stored.	- System broken - ECU broken - Harness broken	Diagnostic Service 14. No additional action is needed.	None	DTC Notification Max Limit Reached
XCIOM	C109511	I > 6,184A..8,352A	Circuit Short To Ground	*Front Fog lights cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *FogLamp_Front_Status = 0 (Inactive) *FogLightFrontLamp_stat = 2 (Tested, Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Fog lights	Front Fog Lights Circuit
XCIOM	C109515	I < 1A (worst case: I < 780mA...1360mA)	Circuit Short To Battery or Open	*Open circuit: Front Fog lights cannot be activated. *Short circuit to battery: Front Fog lights cannot be deactivated. *FogLamp_Front_Status = 2 (Error) *FogLightFrontLamp_stat = 2 (Tested, Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front Fog lights	Front Fog Lights Circuit
XCIOM	C109518	1A< I <3,6A (1 bulb OK) I > 3,6A --> 0 bulbs broken, everything ok. (worst case: I < 2,8A...4,2A)	Circuit Current Below Threshold	*Right or left fog lamp is broken. *FogLamp_Front_Status = 2 (Error) *FogLightFrontLamp_stat = 2 (Tested, Lamp NOK)	1. One bulb broken 2. Connected components broken. 3. Wires broken. Insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	*Check Front Fog lights *Check wires and connectors between ECU and Front Fog lights	Front Fog Lights Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108711	I > 1,27A..1,715A	Circuit Short To Ground	*Position lamp cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *PositionLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Position Lamp	Left Front Position Light Circuit
XCIOM	C108715	LED No Load detected (Offstate Diagnostic) BULB I < 100mA (worst case bulbs: I < 35mA…110mA)	Circuit Short To Battery or Open	*Open circuit: position light cannot be activated *Short circuit to battery: position light cannot be deactivated *PositionLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Position Lamp	Left Front Position Light Circuit
XCIOM	C108811	I > 1,27A..1,715A	Circuit Short To Ground	*Position lamp cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *PositionLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Position Lamp	Right Front Position Light Circuit
XCIOM	C108815	LED No Load detected (Offstate Diagnostic) BULB I < 100mA (worst case bulbs: I < 35mA…110mA)	Circuit Short To Battery or Open	*Open circuit: position light cannot be activated *Short circuit to battery: position light cannot be deactivated *PositionLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Position Lamp	Right Front Position Light Circuit
XCIOM	C102711	I > 2,942A..3,974A	Circuit Short To Ground	*Daytime Running Lights cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DRLLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Daytime Running Lights	Left Daytime Running Lights Circuit
XCIOM	C102715	I < 50mA (worst case bulbs: I < 17mA…115mA) (worst case LEDs: I < 28mA…77mA)	Circuit Short To Battery or Open	*Open circuit: Daytime running light cannot be activated *Short circuit to battery: Daytime running light cannot be deactivated *DRLLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Daytime Running Lights	Left Daytime Running Lights Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108911	I > 2,942A..3,974A	Circuit Short To Ground	*Daytime Running Lights cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DRLLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Daytime Running Lights	Right Daytime Running Lights Circuit
XCIOM	C108915	I < 50mA (worst case bulbs: I < 17mA…115mA) (worst case LEDs: I < 28mA…77mA)	Circuit Short To Battery or Open	*Open circuit: Daytime running light cannot be activated *Short circuit to battery: Daytime running light cannot be deactivated *DRLLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Daytime Running Lights	Right Daytime Running Lights Circuit
XCIOM	C108A11	I > 1,316A..1,777A	Circuit Short To Ground	*Bi-Xenon Lights cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Bi-Xenon Lights	Bi-Xenon Light Shutter Circuit
XCIOM	C108A15	I < 100mA (worst case: I < 72mA…140mA)	Circuit Short To Battery or Open	*Open circuit: Bi-Xenon Lights cannot be activated *Short circuit to battery: Bi-Xenon Lights cannot be deactivated	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Bi-Xenon Lights	Bi-Xenon Light Shutter Circuit
XCIOM	C108B11	I > 1,219A..1,647 (bulb & LED) I > 20A for 7s @-40 (Limphome) I > 20A for 1s @+85 (Limphome)	Circuit Short To Ground	*Side Direction Indicator cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Front_Status = 1 (Left direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Side Direction Indicator	Left Side Direction Indicator Circuit
XCIOM	C108B15	I < 40mA (bulb and LED) (worst case: I < 12mA…60mA)	Circuit Short To Battery or Open	*Open circuit: Side Direction Indicator cannot be activated *Short circuit to battery: Side Direction Indicator cannot be deactivated *DirInd_Front_Status = 1 (Left direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Side Direction Indicator	Left Side Direction Indicator Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108C11	I > 1,219A..1,647 (bulb & LED) I > 20A for 7s @-40 (Limphome) I > 20A for 1s @+85 (Limphome)	Circuit Short To Ground	*Side Direction Indicator cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Front_Status = 2 (Right direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Side Direction Indicator	Right Side Direction Indicator Circuit
XCIOM	C108C15	I < 40mA (bulb and LED) (worst case: I < 12mA...60mA)	Circuit Short To Battery or Open	*Open circuit: Side Direction Indicator cannot be activated *Short circuit to battery: Side Direction Indicator cannot be deactivated *DirInd_Front_Status = 2 (Right direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Side Direction Indicator	Right Side Direction Indicator Circuit
XCIOM	C108D11	"I > 7,198A..9,724A (xenon start) I > 3,922A..5,298A (xenon nom) I > 3,484A..4,706A (bulb) I > 11,767A..15,895 (limphome)"	Circuit Short To Ground	*Dipped beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DippedBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Dipped Beam	Left Head Light Dipped Beam Circuit
XCIOM	C108D15	I < 0.7A (worst case: I < 530mA...860mA)	Circuit Short To Battery or Open	*Open circuit: Dipped Beam cannot be activated *Short circuit to battery: Dipped Beam cannot be deactivated *DippedBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Dipped Beam	Left Head Light Dipped Beam Circuit
XCIOM	C108E11	"I > 7,198A..9,724A (xenon start) I > 3,922A..5,298A (xenon nom) I > 3,484A..4,706A (bulb) I > 11,767A..15,895 (limphome)"	Circuit Short To Ground	*Dipped beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DippedBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Dipped Beam	Right Head Light Dipped Beam Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108E15	I < 0.7A (worst case: I < 530mA...860mA)	Circuit Short To Battery or Open	*Open circuit: Dipped Beam cannot be activated *Short circuit to battery: Dipped Beam cannot be deactivated *DippedBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Dipped Beam	Right Head Light Dipped Beam Circuit
XCIOM	C108F11	I > 3,484A..4,706A	Circuit Short To Ground	*Main Beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *MainBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Main Beam	Left Head Light Main Beam Circuit
XCIOM	C108F15	I < 1A (worst case: I < 820mA...1240mA)	Circuit Short To Battery or Open	*Open circuit: Main Beam cannot be activated *Short circuit to battery: Main Beam cannot be deactivated *MainBeamLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Main Beam	Left Head Light Main Beam Circuit
XCIOM	C109011	I > 3,484A..4,706A	Circuit Short To Ground	*Main Beam cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *MainBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Main Beam	Right Head Light Main Beam Circuit
XCIOM	C109015	I < 1A (worst case: I < 820mA...1240mA)	Circuit Short To Battery or Open	*Open circuit: Main Beam cannot be activated *Short circuit to battery: Main Beam cannot be deactivated *MainBeamLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Main Beam	Right Head Light Main Beam Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109111	I > 1,219A..1,647 (bulb & LED) I > 20A for 7s @-40 (Limphome) I > 20A for 1s @+85 (Limphome)	Circuit Short To Ground	*Front Direction Indicator cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Front_Status = 1 (Left direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Direction Indicators	Left Front Direction Indicator Circuit
XCIOM	C109115	I < 50mA (bulb and LED) (worst case: I < 31mA...81mA)	Circuit Short To Battery or Open	*Open circuit: Front Direction Indicator cannot be activated *Short circuit to battery: Front Direction Indicator cannot be deactivated *DirInd_Front_Status = 1 (Left direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left Front Direction Indicators	Left Front Direction Indicator Circuit
XCIOM	C109211	I > 1,219A..1,647 (bulb & LED) I > 20A for 7s @-40 (Limphome) I > 20A for 1s @+85 (Limphome)	Circuit Short To Ground	*Front Direction Indicator cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Front_Status = 2 (Right direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Direction Indicators	Right Front Direction Indicator Circuit
XCIOM	C109215	I < 50mA (bulb and LED) (worst case: I < 31mA...81mA)	Circuit Short To Battery or Open	*Open circuit: Front Direction Indicator cannot be activated *Short circuit to battery: Front Direction Indicator cannot be deactivated *DirInd_Front_Status = 2 (Right direction indicator lamp fault) or 3 = (Both left and right direction indicators fault)	"1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right Front Direction Indicators	Right Front Direction Indicator Circuit
XCIOM	C104011	I > 2.515A..3,396A	Circuit Short To Ground	*Headlight level motor cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Headlight level motor	Headlight Level motor 1&2 Power supply
XCIOM	C104012	Short to bat	Circuit Short To Battery	*The Headlight level motor cannot be deactivated *Headlights are stuck in same position as when they were last active.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Headlight level motor	Headlight Level motor 1&2 Power supply

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C104111	U < 0.5V	Circuit Short To Ground	*Headlight level motor cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed *Headlights are stuck	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Headlight level motor	Headlight Level motor 1/2
XCIOM	C104112	U > Ubat - 1V	Circuit Short To Battery	*The Headlight level motor cannot be deactivated *Headlights are stuck in end position	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Headlight level motor	Headlight Level motor 1/2
XCIOM	C109C11	<0,15V	Circuit Short To Ground	*Windscreen Washer Fluid Level Sensor cannot be activated *WasherFluidLevel = 6 (Error)	" 1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Windscreen Washer Fluid Level Sensor	Windscreen Washer Fluid Level Sensor Circuit
XCIOM	C109C12	U = 5V	Circuit Short To Battery	*Windscreen Washer Fluid Level Sensor cannot be deactivated *WasherFluidLevel = 6 (Error)	"1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Windscreen Washer Fluid Level Sensor	Windscreen Washer Fluid Level Sensor Circuit
XCIOM	C109C13	4,208V <= U < 5V	Circuit Open	*Windscreen Washer Fluid Level Sensor cannot be activated *WasherFluidLevel = 6 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Windscreen Washer Fluid Level Sensor	Windscreen Washer Fluid Level Sensor Circuit
XCIOM	C109C16	0,15V <= U <= 0,8V	Circuit Voltage Below Threshold	*General reduced functionality of the Windscreen Washer Fluid Level Sensor *WasherFluidLevel = 6 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check power supply system	Windscreen Washer Fluid Level Sensor Circuit
XCIOM	C109C17	3,87V < U < 4,208V	Circuit Voltage Above Threshold	*Risk for component damage *WasherFluidLevel = 6 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check power supply system	Windscreen Washer Fluid Level Sensor Circuit
XCIOM	C109311	I > 7,164A..9,677A	Circuit Short To Ground	*Windscrenn Washer Fluid Pump cannot be activated *Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Washer pump motor frozen. 2. Connected component broken. 3. Wires broken. Insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Windscreen Washer Fluid Pump	Windscreen Washer Fluid Pump Circuit
XCIOM	C109315	I < 100mA (nominal: 102mA worst case: I < 15mA...200mA)	Circuit Short To Battery or Open	*Open circuit: Windscrenn Washer Fluid Pump cannot be activated *Short circuit to battery: Windscrenn Washer Fluid Pump cannot be deactivated, constant flow of washer fluid	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Windscreen Washer Fluid Pump	Windscreen Washer Fluid Pump Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U000188	CAN bus off	Bus off	CAN communication interrupted.	Communication bus is short circuited to ground or battery, or there are connector or harness problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on Backbone 2.	CAN Communication Backbone 2 Net
HMIIOM	U001088	CAN bus off	Bus off	CAN communication interrupted.	Communication bus is short circuited to ground or battery, or there are connector or harness problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on Backbone 1.	CAN Communication Backbone 1 Net
HMIIOM	U004688	CAN bus off	Bus off	CAN communication interrupted.	Communication bus is short circuited to ground or battery, or there are connector or harness problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on Infotainment Net.	CAN Communication Infotainment Net
HMIIOM	U200A88	CAN bus off	Bus off	CAN communication interrupted.	Communication bus is short circuited to ground or battery, or there are connector or harness problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on Display Net.	CAN Communication Display Net
HMIIOM	U300041	CRC checksum occured	General Checksum Failure	Reset of Electronic Control Unit, HMIIOM	Multiple possible root causes e.g. flash error, EMC problem	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U300044	RAM check failed	Data Memory Failure	Reset of Electronic Control Unit, HMIIOM	ECU hardware error	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U300045	NVRAM/ROM check failed	Program Memory Failure	Reset of Electronic Control Unit, HMIIOM	ECU hardware error	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U300046	EEPROM check failed	Calibration / Parameter Memory Failure	Reset of Electronic Control Unit, HMIIOM	ECU hardware error	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U300049	ECU Internal failure	Internal Electronic Failure	Reset of Electronic Control Unit, HMIIOM	ECU hardware error	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U30004A	Slave Module on wrong LIN channel	Incorrect Component Installed	Reset of Electronic Control Unit, HMIIOM	Wrong LIN slave Module or not correct connected	Diagnostic Service 14. No additional action is needed.	Check that the correct LIN-slaves are connected and that they are connected on the correct channels.	Electronic control unit
HMIIOM	U30004B	ECU temperature is above 85ºC	Over Temperature	Reset of Electronic Control Unit, HMIIOM	1. Environment temperature above normal operating range (-40 °C - 85 °C) 2. ECU damaged	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Electronic control unit
HMIIOM	U300094	Unexpected operation detected	Unexpected Operation	Reset of Electronic Control Unit, HMIIOM	SW Application error (Watch dog failure et.c)	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU. 2. Replace ECU.	Electronic control unit
HMIIOM	U300316	Voltage < 8 V +- 10%	Circuit Voltage Below Threshold	None	1. Drained battery 2. Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check battery status.	Battery Power
HMIIOM	U300317	Voltage > 32V +- 10%	Circuit Voltage Above Threshold	None	1. Drained battery 2. Power supply system failure	Diagnostic Service 14. No additional action is needed.	Check battery status.	Battery Power
HMIIOM	U1FFF68	Truck is in Factory Mode	Event Information	None	Truck is in Factory Mode	Diagnostic Service 14. No additional action is needed.	Set P1C54 to FALSE	Factory Mode Active
HMIIOM	U1FE049	LIN related HW fault detected.	Internal Electronic Failure	LIN communication interrupted.	ECU hardware error.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	ECU LIN Hardware

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U114300	Signal from BBM missing	No Sub Type Information	BBM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and BBM. If this is the only lost communication DTC then fault is more probable to be close to the BBM. This fault can also appear by incorrect calibration.	Lost Communication With BBM
HMIIOM	U114400	Signal from RCIOM missing	No Sub Type Information	RCIOM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and RCIOM. If this is the only lost communication DTC then fault is more probable to be close to the RCIOM. This fault can also appear by incorrect calibration.	Lost Communication With RCIOM
HMIIOM	U114500	Signal from CCIOM missing	No Sub Type Information	CCIOM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U300316 and 300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and CCIOM. If this is the only lost communication DTC then fault is more probable to be close to the CCIOM. This fault can also appear by incorrect calibration.	Lost Communication With CCIOM
HMIIOM	U114600	Signal from DACU missing	No Sub Type Information	DACU diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and DACU. If this is the only lost communication DTC then fault is more probable to be close to the DACU. This fault can also appear by incorrect calibration.	Lost Communication With DACU
HMIIOM	U010000	Signal from EMS missing	No Sub Type Information	EMS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and EMS. If this is the only lost communication DTC then fault is more probable to be close to the EMS. This fault can also appear by incorrect calibration.	Lost Communication with EMS

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U010100	Signal from TECU missing	No Sub Type Information	TECU diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TECU. If this is the only lost communication DTC then fault is more probable to be close to the TECU. This fault can also appear by incorrect calibration.	Lost Communication with TECU
HMIIOM	U014000	Signal from VMCU missing	No Sub Type Information	VMCU diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and VMCU. If this is the only lost communication DTC then fault is more probable to be close to the VMCU. This fault can also appear by incorrect calibration.	Lost Communication With VMCU
HMIIOM	U114000	Signal from APM missing	No Sub Type Information	APM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and APM. If this is the only lost communication DTC then fault is more probable to be close to the APM. This fault can also appear by incorrect calibration.	Lost Communication With APM
HMIIOM	U109F00	Signal from IC missing	No Sub Type Information	IC diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and IC. If this is the only lost communication DTC then fault is more probable to be close to the IC. This fault can also appear by incorrect calibration.	Lost Communication With IC
HMIIOM	U10A500	Signal from SID missing	No Sub Type Information	SIDH diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and SIDH. If this is the only lost communication DTC then fault is more probable to be close to the SIDH. This fault can also appear by incorrect calibration.	Lost Communication With SID

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U10A700	Signal from FCIOM missing	No Sub Type Information	FCIOM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and FCIOM. If this is the only lost communication DTC then fault is more probable to be close to the FCIOM. This fault can also appear by incorrect calibration.	Lost Communication With FCIOM
HMIIOM	U10AB00	Signal from CCM missing	No Sub Type Information	CCM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and CCM. If this is the only lost communication DTC then fault is more probable to be close to the CCM. This fault can also appear by incorrect calibration.	Lost Communication With CCM
HMIIOM	U10AC00	Signal from DDM missing	No Sub Type Information	DDM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and DDM. If this is the only lost communication DTC then fault is more probable to be close to the DDM. This fault can also appear by incorrect calibration.	Lost Communication With DDM
HMIIOM	U10AD00	Signal from FAS missing	No Sub Type Information	FAS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U300316 and 300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and FAS. If this is the only lost communication DTC then fault is more probable to be close to the FAS. This fault can also appear by incorrect calibration.	Lost Communication With FAS
HMIIOM	U10AE00	Signal from Alarm missing	No Sub Type Information	Alarm diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Alarm. If this is the only lost communication DTC then fault is more probable to be close to the Alarm. This fault can also appear by incorrect calibration.	Lost Communication With Alarm

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U10AF00	Signal from PDM missing	No Sub Type Information	PDM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and PDM. If this is the only lost communication DTC then fault is more probable to be close to the PDM. This fault can also appear by incorrect calibration.	Lost Communication With PDM
HMIIOM	U10B000	Signal from SRS missing	No Sub Type Information	SRS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and SRS. If this is the only lost communication DTC then fault is more probable to be close to the SRS. This fault can also appear by incorrect calibration.	Lost Communication With SRS
HMIIOM	U10B100	Signal from TPM missing	No Sub Type Information	TPM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TPM. If this is the only lost communication DTC then fault is more probable to be close to the TPM. This fault can also appear by incorrect calibration.	Lost Communication With TPM
HMIIOM	U10B200	Signal from CIOM missing	No Sub Type Information	CIOM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and CIOM. If this is the only lost communication DTC then fault is more probable to be close to the CIOM. This fault can also appear by incorrect calibration.	Lost Communication With CIOM
HMIIOM	U10B400	Signal from LECM missing	No Sub Type Information	LECM diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LECM. If this is the only lost communication DTC then fault is more probable to be close to the LECM. This fault can also appear by incorrect calibration.	Lost Communication With LECM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U10B600	Signal from Audio missing	No Sub Type Information	Audio diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Audio. If this is the only lost communication DTC then fault is more probable to be close to the Audio. This fault can also appear by incorrect calibration.	Lost Communication With Audio
HMIIOM	U10B800	Signal from VS missing	No Sub Type Information	VS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and VS. If this is the only lost communication DTC then fault is more probable to be close to the VS. This fault can also appear by incorrect calibration.	Lost Communication With VS
HMIIOM	U114200	Signal from TGW2 missing	No Sub Type Information	TGW2 diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and TGW2. If this is the only lost communication DTC then fault is more probable to be close to the TGW2. This fault can also appear by incorrect calibration.	Lost Communication With TGW2
HMIIOM	U115900	Signal from EAS missing	No Sub Type Information	EAS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and EAS. If this is the only lost communication DTC then fault is more probable to be close to the EAS. This fault can also appear by incorrect calibration.	Lost Communication With EAS - Extra Axle Steering
HMIIOM	U010E00	Signal from ACM missing	No Sub Type Information	Aftertreatment Control Module diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and ACM. If this is the only lost communication DTC then fault is more probable to be close to the ACM. This fault can also appear by incorrect calibration.	Lost Communication With Reductant Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U114E00	Signal from Tachograph missing	No Sub Type Information	Tachograph diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Tachograph. If this is the only lost communication DTC then fault is more probable to be close to the Tachograph. This fault can also appear by incorrect calibration.	Lost Communication With Tachograph
HMIIOM	U012900	Signal from Brake system control module missing	No Sub Type Information	Wire Brake System diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake system. If this is the only lost communication DTC then fault is more probable to be close to the Brake system. This fault can also appear by incorrect calibration.	Lost Communication With Brake System Control Module
HMIIOM	U114F00	Signal from BLECU missing	No Sub Type Information	Bending Light diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U014000, U10A700, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and BLECU. If this is the only lost communication DTC then fault is more probable to be close to the BLECU. This fault can also appear by incorrect calibration.	Lost Communication With Bending Light ECU
HMIIOM	U115000	Signal from FLS missing	No Sub Type Information	Forward Looking Sensor diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U114600, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and FLS. If this is the only lost communication DTC then fault is more probable to be close to the FLS. This fault can also appear by incorrect calibration.	Lost Communication With Forward Looking Sensor
HMIIOM	U115100	Signal from LPOS missing	No Sub Type Information	Lane Position diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U114600, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LPOS. If this is the only lost communication DTC then fault is more probable to be close to the LPOS. This fault can also appear by incorrect calibration.	Lost Communication With Lane Position Object Control Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U115200	Signal from WRC missing	No Sub Type Information	Wireless Remote Control diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-911 and D-950. It can be triggered at the same time as: U10B200, U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and WRC. If this is the only lost communication DTC then fault is more probable to be close to the WRC. This fault can also appear by incorrect calibration.	Lost Communication With Wireless Remote Control System
HMIIOM	U115500	Max number of DTC:s within the system reached.	No Sub Type Information	No more DTC can be set after this DTC is set.	Reception of 29 DTCs will trigger the setting of this DTC.	Diagnostic Service 14. No additional action is needed.	Control other set DTCs. When they are erased this DTC should also be erased.	DTC Notification Max Limit Reached
HMIIOM	U115600	TIPSI (The Astonishing Tribe (TAT) Inter Process Software Interface) hearbeat signalling is missing or other TIPSI related transport errors occur.	No Sub Type Information	Driver Information Display will stop working. Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Software error, software mismatch between HMIIOM and IC or communication problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between HMIIOM and IC. Also check that the software versions match.	Driver Information Display
HMIIOM	U115694	TIPSI (The Astonishing Tribe (TAT) Inter Process Software Interface) memory allocation error occur.	Unexpected Operation	Driver Information Display resets	Software Error	Diagnostic Service 14. No additional action is needed.	Reprogram HMIIOM or IC.	Driver Information Display
HMIIOM	U115700	TIPSI (The Astonishing Tribe (TAT) Inter Process Software Interface) hearbeat signalling is missing or other TIPSI related transport errors occur.	No Sub Type Information	Secondary Information Display will stop working. Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Software error, software mismatch between HMIIOM and IC or communication problems.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between HMIIOM and IC. Also check that the software versions match.	Secondary Information Display
HMIIOM	U115794	TIPSI (The Astonishing Tribe (TAT) Inter Process Software Interface) memory allocation error occur.	Unexpected Operation	Secondary Information Display resets	Software Error	Diagnostic Service 14. No additional action is needed.	Reprogram HMIIOM or IC.	Secondary Information Display
HMIIOM	U114D11	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to ground.	Circuit Short To Ground	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 15.	LIN15 Communication Bus

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U114D12	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to battery.	Circuit Short To Battery	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 15.	LIN15 Communication Bus
HMIIOM	U114D88	LIN transceiver has detected too many transmit or receive errors.	Bus off	LIN communication interrupted. Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-854. It can be triggered at the same time as: U114D11 and U114D12.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 15.	LIN15 Communication Bus
HMIIOM	U114786	Gear Lever LIN slave Module Invalid Signal	Signal Invalid	GearShift function will be effected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-798. It can be triggered at the same time as: U114B11, U114B12 and U114B88.	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and GLU/DashGSU. 2. Check GLU/DashGSU functionality. If faulty then replace GLU/DashGSU.	Gear Lever LIN Module
HMIIOM	U114787	Gear Lever LIN Module slave not responding	Missing Message	GearShift function will be effected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-798. It can be triggered at the same time as: U114B11, U114B12 and U114B88.	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and GLU/DashGSU. 2. Check GLU/DashGSU functionality. If faulty then replace GLU/DashGSU.	Gear Lever LIN Module
HMIIOM	U114886	Stalk Module LIN slave Module Invalid Signal	Signal Invalid	Functions controlled by the Stalk module will be effected	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SM. 2. Check SM functionality. If faulty then replace SM.	Stalk Module LIN Module
HMIIOM	U114887	Stalk Module LIN slave Module not responding	Missing Message	Functions controlled by the Stalk module will be effected	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SM. 2. Check SM functionality. If faulty then replace SM.	Stalk Module LIN Module
HMIIOM	U114986	Steering Wheel Switch LIN slave Module Invalid Signal	Signal Invalid	Functions controlled by the Steering Wheel Switch module will be effected	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SWS. 2. Check SWS functionality. If faulty then replace SWS.	Steering Wheel Switch LIN Module
HMIIOM	U114987	Steering Wheel Switch LIN slave Module not responding	Missing Message	Functions controlled by the Steering Wheel Switch module will be effected.	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SWS. 2. Check SWS functionality. If faulty then replace SWS.	Steering Wheel Switch LIN Module
HMIIOM	U114A86	Secondary Information Display Control Panel LIN slave Module Invalid Signal	Signal Invalid	Functions related to Display Control Panel will be effected.	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SIDCP. 2. Check SIDCP functionality. If faulty then replace SIDCP.	Secondary Information Display Control Panel LIN Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U114A87	Secondary Information Display Control Panel LIN slave Module not responding	Missing Message	Functions related to Display Control Panel will be effected	LIN Slave Module error	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors on between ECU and SIDCP. 2. Check SIDCP functionality. If faulty then replace SIDCP.	Secondary Information Display Control Panel LIN Module
HMIIOM	U114B11	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to ground.	Circuit Short To Ground	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 13.	LIN13 Communication Bus
HMIIOM	U114B12	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to battery.	Circuit Short To Battery	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 13.	LIN13 Communication Bus
HMIIOM	U114B88	LIN transceiver has detected too many transmit or receive errors.	Bus off	LIN communication interrupted. Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-854. It can be triggered at the same time as: U114B11 and U114B12.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and LIN nodes on LIN 13. 2. Replace HMIIOM ECU.	LIN13 Communication Bus
HMIIOM	U114C11	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to ground.	Circuit Short To Ground	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 14.	LIN14 Communication Bus
HMIIOM	U114C12	LIN transceiver has detected too many transmit or receive errors and pin is short circuited to battery.	Circuit Short To Battery	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 14.	LIN14 Communication Bus
HMIIOM	U114C88	LIN transceiver has detected too many transmit or receive errors.	Bus off	LIN communication interrupted. Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-854. It can be triggered at the same time as: U114C11 and U114C12.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and LIN nodes on LIN 14.	LIN14 Communication Bus
HMIIOM	B150216	Voltage < 8V	Circuit Voltage Below Threshold	Gear shifting functionality stops to work.	1. Connecter or harness problems. 2. GLU or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and GLU. 2. Check output voltage from ECU to GLU.	Gear Lever Unit
HMIIOM	B150217	Voltage > 18V	Circuit Voltage Above Threshold	Gear shifting functionality stops to work.	1. Connecter or harness problems. 2. GLU or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and GLU. 2. Check output voltage from ECU to GLU.	Gear Lever Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	B150249	Sensor fault detected.	Internal Electronic Failure	Gear position will be set to last reliable position.	Faulty sensors or lever stuck	Diagnostic Service 14. No additional action is needed.	Replace GLU.	Gear Lever Unit
HMIIOM	B150294	Both up and down shift button activation simultaneously	Unexpected Operation	Gear position will be set to last reliable position.	Unexpected Operation	Diagnostic Service 14. No additional action is needed.	Replace GLU.	Gear Lever Unit
HMIIOM	B150716	Voltage < 7 V	Circuit Voltage Below Threshold	LIN is switched off. SIDCP will not work.	1. Connecter or harness problems. 2. SIDCP or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SIDCP. 2. Check output voltage from ECU to SIDCP.	Secondary Information Display Control Panel
HMIIOM	B150717	Voltage > 20 V	Circuit Voltage Above Threshold	LIN is switched off. SIDCP will not work.	1. Connecter or harness problems. 2. SIDCP or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SIDCP. 2. Check output voltage from ECU to SIDCP.	Secondary Information Display Control Panel
HMIIOM	B150744	RAM error, read compare failure after write operation.	Data Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SIDCP	Secondary Information Display Control Panel
HMIIOM	B150745	ROM error, 16 bit checksum incorrect	Program Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SIDCP	Secondary Information Display Control Panel
HMIIOM	B150746	EEPROM checksum error	Calibration / Parameter Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SIDCP	Secondary Information Display Control Panel
HMIIOM	B150794	Watchdog error read from reset register	Unexpected Operation	ECU resets continuously. DTC will never be set.	Watchdog failure	Diagnostic Service 14. No additional action is needed.	Replace SIDCP	Secondary Information Display Control Panel
HMIIOM	B150311	Input Signals of Brake Program Button <= 2,44 V OR Rain Sensor Button <= 0,88 V OR Scroll Wheel <= 0,88 V	Circuit Short To Ground	It will not be possible to activate on or more of the following functions: Brake Program, Rain Sensor and/or use wipers.	Brake program button, rain sensor button or scroll wheel short circuit to ground.	Diagnostic Service 14. No additional action is needed.	Manual Check of Brake Program Button, Rainsensor Button and Scroll Wheel. Replace Stalk with erroneous function.	Stalk Module Variant 1
HMIIOM	B150312	Input Signals of Brake Program Button >= 4,5 V OR Rain Sensor Button >= 4,78 V OR Scroll Wheel >= 4,88 V	Circuit Short To Battery	It will not be possible to activate on or more of the following functions: Brake Program, Rain Sensor and/or use wipers.	Brake program button, rain sensor button or scroll wheel short circuit to battery.	Diagnostic Service 14. No additional action is needed.	Manual Check of Brake Program Button, Rainsensor Button and Scroll Wheel. Replace Stalk with erroneous function.	Stalk Module Variant 1
HMIIOM	B150313	Open circuit conditions detected on Rain Sensor button OR Scroll Wheel.	Circuit Open	It will not be possible to activate on or more of the following functions: Rain Sensor and/or use wipers.	Rain sensor button or scroll wheel has an open circuit.	Diagnostic Service 14. No additional action is needed.	Manual Check of Rainsensor Button and Scroll Wheel. Replace Stalk with erroneous function.	Stalk Module Variant 1
HMIIOM	B150316	Voltage < 8V	Circuit Voltage Below Threshold	LIN communicastion Interface shut down.	1. Connecter or harness problems. 2. SM1 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SM1. 2. Check output voltage from ECU to SM1.	Stalk Module Variant 1
HMIIOM	B150317	Voltage > 18V	Circuit Voltage Above Threshold	LIN communicastion Interface shut down.	1. Connecter or harness problems. 2. SM1 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SM1. 2. Check output voltage from ECU to SM1.	Stalk Module Variant 1
HMIIOM	B15031C	Voltage < 2.1 V	Circuit Voltage Out of Range	None	1. Connecter or harness problems. 2. SM1 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	No action required.	Stalk Module Variant 1
HMIIOM	B150344	RAM Failure	Data Memory Failure	LIN communication Interface shut down.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace base plate of SM1.	Stalk Module Variant 1
HMIIOM	B150345	ROM Failure	Program Memory Failure	LIN communication Interface shut down.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace base plate of SM1.	Stalk Module Variant 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	B150346	EEPROM failure	Calibration / Parameter Memory Failure	LIN communication Interface shut down.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace base plate of SM1.	Stalk Module Variant 1
HMIIOM	B15034A	Incorrect Component detected	Incorrect Component Installed	It will not be possible to activate on or more of the following functions: Direction Indicators, Main Beam, Washers and Wipers and/or Retarder.	Direction Indicator stalk, washer wiper stalk or auxiliary stalk incorrectly mounted or damaged.	Diagnostic Service 14. No additional action is needed.	Check correct mountig of each Stalk.	Stalk Module Variant 1
HMIIOM	B150364	Scroll Wheel Unplausible Signal	Signal Plausibility Failure	It will not be possible to activate the wipers manually.	Scroll Wheel damaged.	Diagnostic Service 14. No additional action is needed.	Replace washer wiper stalk.	Stalk Module Variant 1
HMIIOM	B150394	Watchdog Reset	Unexpected Operation	None	Watchdog failure	Diagnostic Service 14. No additional action is needed.	Replace base plate of SM1.	Stalk Module Variant 1
HMIIOM	B150395	Detected to low input voltage from direction indicator stalk, washer wiper stalk or auxiliary stalk.	Incorrect Assembly	It will not be possible to activate on or more of the following functions: Direction Indicators, Main Beam, Washers and Wipers and/or Retarder.	Direction Indicator stalk, washer wiper stalk or auxiliary stalk dismounted.	Diagnostic Service 14. No additional action is needed.	Check correct mountig of each Stalk. Replace uncorrect mounted or damaged Stalk.	Stalk Module Variant 1
HMIIOM	B150416	Voltage < 7 volts +- 8%	Circuit Voltage Below Threshold	None	1. Connecter or harness problems. 2. SM2 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SM2. 2. Check output voltage from ECU to SM2.	Stalk Module Variant 2
HMIIOM	B150417	Voltage > 18 volts +- 8%	Circuit Voltage Above Threshold	None	1. Connecter or harness problems. 2. SM2 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SM2. 2. Check output voltage from ECU to SM2.	Stalk Module Variant 2
HMIIOM	B150444	RAM Error. read compare failure after write operation.	Data Memory Failure	RAM DATA is untrustable, may affect functionality related to SM2. What it affects depends on where in RAM error occured.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SM2.	Stalk Module Variant 2
HMIIOM	B150446	EEPROM checksum error.	Calibration / Parameter Memory Failure	EEPROM DATA is untrustable, may affect functionality related to SM2. What it affects depends on where in EEPROM error occured.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SM2.	Stalk Module Variant 2
HMIIOM	B150494	Watchdog error read from reset register	Unexpected Operation	Functionality related to SM2 will not work.	Watchdog failure	Diagnostic Service 14. No additional action is needed.	Replace SM2.	Stalk Module Variant 2
HMIIOM	B150516	Voltage < 7 volts	Circuit Voltage Below Threshold	LIN is switched off	1. Connecter or harness problems. 2. SWS1 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SWS1. 2. Check output voltage from ECU to SWS1.	Steering Wheel Switch Variant 1
HMIIOM	B150517	Voltage > 20 volts	Circuit Voltage Above Threshold	LIN is switched off LEDs are switched off	1. Connecter or harness problems. 2. SWS1 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors between ECU and SWS1. 2. Check output voltage from ECU to SWS1.	Steering Wheel Switch Variant 1
HMIIOM	B150544	RAM error, read compare failure after write operation.	Data Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SWS1.	Steering Wheel Switch Variant 1
HMIIOM	B150545	ROM error, 16 bit checksum incorrect.	Program Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SWS1.	Steering Wheel Switch Variant 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	B150546	EEPROM checksum error.	Calibration / Parameter Memory Failure	ECU resets continuously. DTC will never be set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SWS1.	Steering Wheel Switch Variant 1
HMIIOM	B150584	One or several SWS buttons or scroll wheels short circuit to ground.	Signal Below Allowable Range	Reduced functionality for the right steering wheel switch. Possible that switches are not useable.	Short to ground	Diagnostic Service 14. No additional action is needed.	1. Check connection to left SWS1 2. Replace SWS1 if fault is still present	Steering Wheel Switch Variant 1
HMIIOM	B150585	One or several SWS buttons or scroll wheels short circuit to battery or open circuit.	Signal Above Allowable Range	Reduced functionality for the right steering wheel switch. Possible that switches are not useable.	Short to battery or open circuit	Diagnostic Service 14. No additional action is needed.	1. Check connection to left SWS1 2. Replace SWS1 if fault is still present	Steering Wheel Switch Variant 1
HMIIOM	B150594	Watchdog error read from reset register	Unexpected Operation	ECU resets continuously. DTC will never be set.	Watchdog failure	Diagnostic Service 14. No additional action is needed.	Replace SWS1.	Steering Wheel Switch Variant 1
HMIIOM	B150611	Voltage = 0V	Circuit Short To Ground	Cruise Control functionality will be effected	Connection between right SWS buttons and left SWS buttons short circuited to ground.	Diagnostic Service 14. No additional action is needed.	1. Check connection between left hand side and right hand side of SWS2. 2. Replace SWS2.	Steering Wheel Switch Variant 2
HMIIOM	B150616	Voltage < 7V +- 10%	Circuit Voltage Below Threshold	SWS buttons may not work.	1. Connecter or harness problems. 2. SWS2 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check output voltage from HMIIOM to SWS2. 2. Replace SWS2	Steering Wheel Switch Variant 2
HMIIOM	B150617	Voltage > 19V +- 10%	Circuit Voltage Above Threshold	SWS buttons may not work.	1. Connecter or harness problems. 2. SWS2 or HMIIOM damaged.	Diagnostic Service 14. No additional action is needed.	1. Check output voltage from HMIIOM to SWS2. 2. Replace SWS2	Steering Wheel Switch Variant 2
HMIIOM	B150694	MCU watchdog reset	Unexpected Operation	SWS2 resets	Watchdog failure	Diagnostic Service 14. No additional action is needed.	Replace SWS2.	Steering Wheel Switch Variant 2
HMIIOM	B15084A	Incorrect Gearbox Stalk detected	Incorrect Component Installed	Gearshifting functionality will be effected	Incorrect Gearbox Stalk installed or wrong configuration	Diagnostic Service 14. No additional action is needed.	Control that the correct gearbox stalk is installed if the truck should have a gearbox stalk.	Gearbox Stalk
HMIIOM	B15094A	Incorrect Radio Satelite Stalk detected	Incorrect Component Installed	Audio functionality will be effected	Incorrect Radio Satelite Stalk installed or wrong configuration	Diagnostic Service 14. No additional action is needed.	Control that the correct radio satellite stalk is installed if the truck should have a radio satellite stalk.	Radio Satelite Stalk
HMIIOM	B150A4A	Incorrect Retarder Stalk detected	Incorrect Component Installed	Auxiliary brakes will be effected	Incorrect Retarder Stalk installed or wrong configuration	Diagnostic Service 14. No additional action is needed.	Control that the correct retarder stalk is installed if the truck should have a retarder stalk.	Retarder Stalk
HMIIOM	B150B4A	Incorrect cruise control switch detected	Incorrect Component Installed	Cruise Control functionality will be effected	Incorrect cruise control switch installed or wrong configuration	Diagnostic Service 14. No additional action is needed.	Control that the correct speed control switches are installed if the truck should have speed control switches.	Cruise Control Switches
HMIIOM	B150C4A	Incorrect phone control switch detected	Incorrect Component Installed	Will not be possible to naviage in menus and phone functionality will be effected	Incorrect phone control switch installed or wrong configuration	Diagnostic Service 14. No additional action is needed.	Control that the correct phone control switches are installed if the truck should have phone control switches.	Phone Control Switches
HMIIOM	B150D11	Voltage = 0V	Circuit Short To Ground	LIN communication interrupted.	-Main connector pin 8 short circuit to ground.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 13.	LIN13 Power Supply Circuit
HMIIOM	B150D17	VP_LINA > 14V (VP_LINA_MON < 2,54)	Circuit Voltage Above Threshold	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 13.	LIN13 Power Supply Circuit
HMIIOM	B150E11	Voltage = 0V	Circuit Short To Ground	LIN communication interrupted.	-Main connector pin 9 short circuit to ground.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 14.	LIN14 Power Supply Circuit
HMIIOM	B150E17	VP_LINB > 14V (VP_LINB_MON < 2,54)	Circuit Voltage Above Threshold	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 14.	LIN14 Power Supply Circuit
HMIIOM	B150F11	Voltage = 0V	Circuit Short To Ground	LIN communication interrupted.	-Main connector pin 10 short circuit to ground.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 15.	LIN15 Power Supply Circuit
HMIIOM	B150F17	VP_LINC > 14V (VP_LINC_MON < 2,54)	Circuit Voltage Above Threshold	LIN communication interrupted.	Connector or harness problems	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and other ECUs on LIN 15.	LIN15 Power Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
HMIIOM	U115371	Detection that there is a button that is stuck. Button is stuck after 1-4095s, depending on which button is stuck.	Actuator Stuck	Functions will be effected	Button stuck or driver has held button pressed longer than the allowed time	Diagnostic Service 14. No additional action is needed.	Control that GLU, SM, SWS and SIDCP buttons are working as they should.	Softkey and Rocker Buttons
HMIIOM	C100054	Input parameter is missing.	Missing Calibration	Will not be possible to calibrate load for front axle.	Wrong calibration.	Diagnostic Service 14. No additional action is needed.	Correct calibration parameters.	Leaf Calculation Failure
HMIIOM	U200B93	No response from Real Time Clock	No Operation	Clock may show incorrect time. Alarm clock, parking heater and refrigerator timer will not work.	Internal HW fault	Diagnostic Service 14. No additional action is needed.	Replace HMIIOM.	Real Time Clock (RTC)
HMIIOM	U115A00	Signal from EMS missing	No Sub Type Information	Engine Malfunction Indication telltale will be activated "On steady" at least during current drive cycle Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	"Connector or harness problems. Data link EMC problems. Problem with sending ECU."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and EMS. This fault can also appear by incorrect calibration.	Lost Communication with EMS
HMIIOM	U115B00	Signal from EMS missing	No Sub Type Information	Engine Malfunction Indication telltale will be activated "On steady" at least during current drive cycle Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	"Connector or harness problems. Data link EMC problems. Problem with sending ECU."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and EMS. This fault can also appear by incorrect calibration.	Lost Communication with EMS
LPOS	U200288	Bus off detection	Bus off	No communication on CAN General reduced system functionality	Short circuit to ground/battery Broken cables	Diagnostic Service 14. No additional action is needed.	Check CAN wires Battery reset	CAN Communication Active Safety Proprietary 2 Net
LPOS	U300041	Checksum failure detected	General Checksum Failure	ECU not alive	No valid application SW found	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with correct SW and calibrate 2. Replace ECU	Electronic control unit
LPOS	U300044	RAM check failed	Data Memory Failure	LPOS is set to error	RAM cells damaged	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
LPOS	U300046	"EEPROM record CRC error EEPROM access failed"	Calibration / Parameter Memory Failure	LPOS is set to error	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with correct SW 2. Perform new calibration 3. Replace ECU	Electronic control unit
LPOS	U300049	HW malfunction detected inside the ECU	Internal Electronic Failure	LPOS is set to error	ECU damaged internally	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
LPOS	U30004B	ECU temperature > 85 Celcius	Over Temperature	LPOS is set to error	Environment temperature above normal operating range ECU damaged	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in normal range	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
LPOS	U300054	Missing calibration	Missing Calibration	LPOS is set to error	LPOS not calibrated	Diagnostic Service 14. No additional action is needed.	Perform Sensor Calibration	Electronic control unit
LPOS	U300055	VIN not set ECU VIN and Camera VIN mismatch	Not Configured	LPOS is set to error	Camera or ECU has been replaced with incorrect VIN	Diagnostic Service 14. No additional action is needed.	Reload software and calibrate	Electronic control unit
LPOS	U300068	Message missing or invalid data received	Event Information	LPOS is set to error	Message missing or invalid data received	Diagnostic Service 14. No additional action is needed.	Check that input CAN signals are correct Yaw RateCompensated VehicleSpeedCompensated LPOSPowerMode_rqst WiperParkStatus_Global	Electronic control unit
LPOS	U300094	Unexpected operation failure (eg. Watchdog)	Unexpected Operation	LPOS is set to errror	Unexpected reset due to internal ECU failure	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Check that input CAN signals are correct 3. Replace ECU	Electronic control unit
LPOS	U300316	Power supply voltage < 16 V	Circuit Voltage Below Threshold	LPOS is set to errror	1. Drained battery 2. Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
LPOS	U300317	Power supply voltage > 32 V	Circuit Voltage Above Threshold	LPOS is set to error	1. Power supply system failure 2. Overload alternator	Diagnostic Service 14. No additional action is needed.	Check power supply system	Battery Power
LPOS	U114113	Camera signals missing	Circuit Open	LPOS is set to error	Connection between ECU and camera broken	Diagnostic Service 14. No additional action is needed.	1. Check connection between camera and ECU 2. Replace LPOS-kit	LVDS Video Data Circuit
LPOS	B150249	Internal HW failure detection	Internal Electronic Failure	LPOS is set to error	Camera damaged	Diagnostic Service 14. No additional action is needed.	Replace camera and reload software and calibrate	Lane Position Camera
LPOS	B15024B	Camera temperature > 85 Celsius	Over Temperature	LPOS is set to error	Environment temperature above normal operating range Camera damaged	Diagnostic Service 14. No additional action is needed.	1. Check camera environment 2. Replace camera if fault is still present when environment temperature is in normal range and calibrate	Lane Position Camera
LPOS	B150278	Camera misalignment detected	Alignment or Adjustment Incorrect	LPOS is set to error	Camera misalignment detected by the algorithm	Diagnostic Service 14. No additional action is needed.	1. Check that camera is correctly mounted in bracket and also check bracket for damage and position 2. Recalibrate 3. If not correct in pitch, follow calibration routine for adjusting the screw in the camera bracket 4. If not correct in roll or yaw, replace bracket	Lane Position Camera
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P054415	Measured voltage outside the limits specified by the sensor	Circuit Short To Battery or Open		Fault code set by ACM: IDTC_ENGEXHTEMP_OC	Diagnostic Service 14. No additional action is needed	Check: Fault code set by ACM: IDTC_ENGEXHTEMP_OC	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260013	Measured voltage outside the limits specified by the sensor	Circuit Open	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit/Open
EMS	P260200	Measured voltage below limit	No Sub Type Information	Pump always runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit Low
EMS	P260300	Measured voltage above limit	No Sub Type Information	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit High
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P054500	Measured voltage below limit	No Sub Type Information		Fault code set by ACM: IDTC_ENGEXHTEMP_SCL	Diagnostic Service 14. No additional action is needed	Check: Fault code set by ACM: IDTC_ENGEXHTEMP_SCL	Exhaust Gas Temperature Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P108700		No Sub Type Information	None	No fault None	Diagnostic Service 14. No additional action is needed	Check: No fault	Injector 1 Missing Glitch Signature
EMS	P107B00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 2 Missing Glitch Signature
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	P108900		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 3 Missing Glitch Signature
EMS	P108B00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 4 Missing Glitch Signature
EMS	P108D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 5 Missing Glitch Signature
EMS	P108F00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 6 Missing Glitch Signature
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	P220A1C		Circuit Voltage Out of Range		Too high voltage to NOx-sensor ECU Battery voltage	Diagnostic Service 14. No additional action is needed	Check: Battery voltage	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1)
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	Measured voltage below limit	Circuit Short To Ground	Oil thermostat alwaysclosed.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113712	Measured voltage above limit	Circuit Short To Battery	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113713	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P020A25		Signal Shape / Waveform Failure	None	No fault None	Diagnostic Service 14. No additional action is needed	Check: No fault	Cylinder 1 Injection Timing

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P020D25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 4 Injection Timing
EMS	P020E25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 5 Injection Timing
EMS	P020F25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 6 Injection Timing

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020B25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 2 Injection Timing
EMS	P020C25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 3 Injection Timing
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P25AB00	Measured voltage above limit	No Sub Type Information	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit High
EMS	P25AA00	Measured voltage below limit	No Sub Type Information	No piston cooling=>major engine damage if high loads.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit Low
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P25AE00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Too Low
EMS	P25A913	Measured voltage outside the limits specified by the sensor	Circuit Open	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P055A13	Measured voltage outside the limits specified by the sensor	Circuit Open		.	Diagnostic Service 14. No additional action is needed	.	Engine Oil Pressure Sensor/Switch "B"
EMS	P055B64		Signal Plausibility Failure	The control function closes the PCJ valve. When the fault is set, the PCJ oil pressure is forced to max (if calibrated so in the pcj_ function).	The sensor is stuck, has an offset or another kind of rationality fault. It could also be the valve that cannot be closed. Sensor/Valve	Diagnostic Service 14. No additional action is needed	Check: Sensor/Valve	Piston Cooling Oil Pressure Sensor Range/Performance
EMS	P055D00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Sensor Circuit High
EMS	P112800		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Moderately Low
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P054415	Measured voltage outside the limits specified by the sensor	Circuit Short To Battery or Open		Fault code set by ACM: IDTC_ENGEXHTEMP_OC	Diagnostic Service 14. No additional action is needed	Check: Fault code set by ACM: IDTC_ENGEXHTEMP_OC	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P260013	Measured voltage outside the limits specified by the sensor	Circuit Open	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit/Open
EMS	P260200	Measured voltage below limit	No Sub Type Information	Pump always runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit Low
EMS	P260300	Measured voltage above limit	No Sub Type Information	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit High
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P054500	Measured voltage below limit	No Sub Type Information		Fault code set by ACM: IDTC_ENGEXHTEMP_SCL	Diagnostic Service 14. No additional action is needed	Check: Fault code set by ACM: IDTC_ENGEXHTEMP_SCL	Exhaust Gas Temperature Short Circuit Low
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P108700		No Sub Type Information	None	No fault None	Diagnostic Service 14. No additional action is needed	Check: No fault	Injector 1 Missing Glitch Signature
EMS	P107B00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 2 Missing Glitch Signature
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	P108900		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 3 Missing Glitch Signature
EMS	P108B00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 4 Missing Glitch Signature
EMS	P108D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 5 Missing Glitch Signature

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P108F00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Injector 6 Missing Glitch Signature
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	P220A1C		Circuit Voltage Out of Range		Too high voltage to NOx-sensor ECU Battery voltage	Diagnostic Service 14. No additional action is needed	Check: Battery voltage	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1)
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	Measured voltage below limit	Circuit Short To Ground	Oil thermostat alwaysclosed.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113712	Measured voltage above limit	Circuit Short To Battery	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113713	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P020A25		Signal Shape / Waveform Failure	None	No fault None	Diagnostic Service 14. No additional action is needed	Check: No fault	Cylinder 1 Injection Timing
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P020D25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 4 Injection Timing
EMS	P020E25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 5 Injection Timing
EMS	P020F25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 6 Injection Timing
EMS	P020B25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 2 Injection Timing
EMS	P020C25		Signal Shape / Waveform Failure		.	Diagnostic Service 14. No additional action is needed	.	Cylinder 3 Injection Timing
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P25AB00	Measured voltage above limit	No Sub Type Information	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit High
EMS	P25AA00	Measured voltage below limit	No Sub Type Information	No piston cooling=>major engine damage if high loads.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit Low
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P25AE00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Too Low
EMS	P25A913	Measured voltage outside the limits specified by the sensor	Circuit Open	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P055A13	Measured voltage outside the limits specified by the sensor	Circuit Open		.	Diagnostic Service 14. No additional action is needed	.	Engine Oil Pressure Sensor/Switch "B"
EMS	P055B64		Signal Plausibility Failure	The control function closes the PCJ valve. When the fault is set, the PCJ oil pressure is forced to max (if calibrated so in the pcj_ function).	The sensor is stuck, has an offset or another kind of rationality fault. It could also be the valve that cannot be closed. Sensor/Valve	Diagnostic Service 14. No additional action is needed	Check: Sensor/Valve	Piston Cooling Oil Pressure Sensor Range/Performance
EMS	P055D00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Sensor Circuit High
EMS	P112800		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Moderately Low
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P054415	Measured voltage above limit	Circuit Short To Battery or Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P054500	Measured voltage below limit	No Sub Type Information	None	Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Exhaust Gas Temperature Short Circuit Low
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P054415	Measured voltage above limit	Circuit Short To Battery or Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P054500	Measured voltage below limit	No Sub Type Information	None	Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Exhaust Gas Temperature Short Circuit Low
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260013	Measured voltage outside the limits specified by the sensor	Circuit Open	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit/Open
EMS	P260200	Measured voltage below limit	No Sub Type Information	Pump always runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit Low
EMS	P260300	Measured voltage above limit	No Sub Type Information	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit High
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P047013	Measured voltage outside the limits specified by the sensor	Circuit Open	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Exhaust Pressure Sensor "A"
EMS	P047164		Signal Plausibility Failure	Open loop control enabled. Decreased brake power.	sensor, Closed loop butterfly, AVU sensor, butterfly, AVU	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly, AVU	Engine Exhaust Back Pressure Circuit Range/Performance
EMS	P047300	Measured voltage above limit	No Sub Type Information	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Engine Exhaust Back Pressure Circuit High
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	P220A1C		Circuit Voltage Out of Range		Too high voltage to NOx-sensor ECU Battery voltage	Diagnostic Service 14. No additional action is needed	Check: Battery voltage	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1)
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113700	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Oil Thermostat
EMS	P113711	Measured voltage below limit	Circuit Short To Ground	Oil thermostat alwaysclosed.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113712	Measured voltage above limit	Circuit Short To Battery	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113713	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P25AB00	Measured voltage above limit	No Sub Type Information	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit High
EMS	P25AA00	Measured voltage below limit	No Sub Type Information	No piston cooling=>major engine damage if high loads.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit Low
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P25AE00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Too Low
EMS	P25A900	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Piston Cooling Oil Control
EMS	P25A913	Measured voltage outside the limits specified by the sensor	Circuit Open	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P055A00	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A13	Measured voltage outside the limits specified by the sensor	Circuit Open		.	Diagnostic Service 14. No additional action is needed	.	Engine Oil Pressure Sensor/Switch "B"
EMS	P055B64		Signal Plausibility Failure	The control function closes the PCJ valve. When the fault is set, the PCJ oil pressure is forced to max (if calibrated so in the pcj_ function).	The sensor is stuck, has an offset or another kind of rationality fault. It could also be the valve that cannot be closed. Sensor/Valve	Diagnostic Service 14. No additional action is needed	Check: Sensor/Valve	Piston Cooling Oil Pressure Sensor Range/Performance
EMS	P055D00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Sensor Circuit High
EMS	P112800		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Moderately Low
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P047F71		Actuator Stuck	Decreased brake power.	sensor, Closed loop butterfly sensor, butterfly	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly	Exhaust Pressure Control Valve "A" Stuck Open
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260013	Measured voltage outside the limits specified by the sensor	Circuit Open	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit/Open
EMS	P260200	Measured voltage below limit	No Sub Type Information	Pump always runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit Low
EMS	P260300	Measured voltage above limit	No Sub Type Information	Pump always at half speed. High coolant temperatures may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Coolant Pump Control Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P047013	Measured voltage outside the limits specified by the sensor	Circuit Open	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Exhaust Pressure Sensor "A"
EMS	P047164		Signal Plausibility Failure	Open loop control enabled. Decreased brake power.	sensor, Closed loop butterfly, AVU sensor, butterfly, AVU	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly, AVU	Engine Exhaust Back Pressure Circuit Range/Performance
EMS	P047300	Measured voltage above limit	No Sub Type Information	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Engine Exhaust Back Pressure Circuit High
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P109000		No Sub Type Information	Display of the message, without bip, simultaneously with the gear engaged pictogram	Driver started at too high gear at start Protective alert for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective alert for clutch/transmission	Transmission High Gear At Start
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	P220A1C		Circuit Voltage Out of Range		Too high voltage to NOx-sensor ECU Battery voltage	Diagnostic Service 14. No additional action is needed	Check: Battery voltage	NOx Sensor Supply Voltage Circuit (Bank 1 Sensor 1)
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113700	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Oil Thermostat
EMS	P113711	Measured voltage below limit	Circuit Short To Ground	Oil thermostat alwaysclosed.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113712	Measured voltage above limit	Circuit Short To Battery	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P113713	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil thermostat always open. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Oil Thermostat
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error
EMS	P116500		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at low speed but still shifts to high gear, range high. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too High Gear
EMS	P116600		No Sub Type Information	Buzzer signal (alert), Display of the message, with bip, simultaneously with the gear engaged pictogram.	Driver is at high speed but still down-shift to low gear, range low. Protective measure for clutch/transmission	Diagnostic Service 14. No additional action is needed	Check: Protective measure for clutch/transmission	Driver Selected Too Low Gear

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P25AB00	Measured voltage above limit	No Sub Type Information	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit High
EMS	P25AA00	Measured voltage below limit	No Sub Type Information	No piston cooling=>major engine damage if high loads.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P25AE00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Too Low
EMS	P25A900	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Piston Cooling Oil Control
EMS	P25A913	Measured voltage outside the limits specified by the sensor	Circuit Open	Possible black smoke during start.	Harness/Broken actuator Harness/Valve	Diagnostic Service 14. No additional action is needed	Check: Harness/Valve	Piston Cooling Oil Control
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P110F90		Stuck On	Stop switch can not be activated	Switch Engine stop switch	Diagnostic Service 14. No additional action is needed	Check: Engine stop switch	Engine Stop Switch Stuck
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P055A00	DUMMY fault. Not intented for fault tracing	No Sub Type Information	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	DUMMY fault. Not intented for fault tracing	Check: DUMMY fault. Not intented for fault tracing	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A13	Measured voltage outside the limits specified by the sensor	Circuit Open		.	Diagnostic Service 14. No additional action is needed	.	Engine Oil Pressure Sensor/Switch "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P055B64		Signal Plausibility Failure	The control function closes the PCJ valve. When the fault is set, the PCJ oil pressure is forced to max (if calibrated so in the pcj_ function).	The sensor is stuck, has an offset or another kind of rationality fault. It could also be the valve that cannot be closed. Sensor/Valve	Diagnostic Service 14. No additional action is needed	Check: Sensor/Valve	Piston Cooling Oil Pressure Sensor Range/Performance
EMS	P055D00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Sensor Circuit High
EMS	P112800		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Piston Cooling Oil Pressure Moderately Low
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P047F71		Actuator Stuck	Decreased brake power.	sensor, Closed loop butterfly sensor, butterfly	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly	Exhaust Pressure Control Valve "A" Stuck Open
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P047013	Measured voltage outside the limits specified by the sensor	Circuit Open	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Exhaust Pressure Sensor "A"
EMS	P047164		Signal Plausibility Failure	Open loop control enabled. Decreased brake power.	sensor, Closed loop butterfly, AVU sensor, butterfly, AVU	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly, AVU	Engine Exhaust Back Pressure Circuit Range/Performance
EMS	P047300	Measured voltage above limit	No Sub Type Information	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Engine Exhaust Back Pressure Circuit High
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115211	Measured voltage below limit	Circuit Short To Ground	Priming pump permanently on. Possible damage of priming pump.	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115212	Measured voltage above limit	Circuit Short To Battery	Priming or drain aborted and inhibited	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115213	Measured voltage outside the limits specified by the sensor	Circuit Open	Priming or drain inhibited	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P047F71		Actuator Stuck	Decreased brake power.	sensor, Closed loop butterfly sensor, butterfly	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly	Exhaust Pressure Control Valve "A" Stuck Open
EMS	U000188	It has not been possible to send frames on High Speed CAN Communication Bus and It has not been possible to receive frames on High Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 2 Net
EMS	U001088	It has not been possible to send frames on Medium Speed CAN Communication Bus and It has not been possible to receive frames on Medium Speed CAN Communication Bus	Bus off		.	Diagnostic Service 14. No additional action is needed	.	CAN Communication Backbone 1 Net
EMS	U010187		Missing Message		ETC2 not responding .	Diagnostic Service 14. No additional action is needed	Check: ETC2 not responding	Lost Communication with TECU
EMS	P034031		No Signal	The engine start time will increase. Possible loss of power	Sensor/Harness Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138		Signal Frequency Incorrect	The engine start time will increase. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Camshaft Position Sensor A Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052612	Measured voltage above limit	Circuit Short To Battery	Either 1. Fan will work as on/off fan. 100% fan speed if cooling is needed. Higher fuel consumption. Or 2. Fan will work in open loop mode. Deteriorated fan control, but might not be noticable to driver.	Broken sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Fan Speed Sensor Circuit
EMS	P054300	Measured voltage outside the limits specified by the sensor	No Sub Type Information	Might get start problems in cold climate.	Heating element or wire broken. Preheat sense 1	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1	Intake Air Heater A Circuit Open
EMS	P001676		Wrong Mounting Position	Fuel consumption increased	Bad correlation between cam and crank, engine transmission not correct Engine transmission	Diagnostic Service 14. No additional action is needed	Check: Engine transmission	Crankshaft Position - Camshaft Position Correlation
EMS	P007100		No Sub Type Information		Ambient air temp sensor or boost temp sensor faulty Ambient air temp sensor or Boost temp sensor	Diagnostic Service 14. No additional action is needed	Check: Ambient air temp sensor or Boost temp sensor	Ambient Air Temperature Sensor Range/Performance
EMS	P02CD00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700		No Sub Type Information	Larger errors can be noticeable at engine idle speed	- PTO is connected incorrectly so that ECM is not informed if it is engaged. - Low or high flow injector. - Uneven spaced tooth on flywheel. - Bad compression. Injector, see Probable errors	Diagnostic Service 14. No additional action is needed	Check: Injector, see Probable errors	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531		No Signal	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033638		Signal Frequency Incorrect	Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Sensor monting Sensor/Lack of shims	Diagnostic Service 14. No additional action is needed	Check: Sensor/Lack of shims	Crankshaft Position Sensor A Circuit Range/Performance
EMS	P033900		No Sub Type Information	The engine start time will increase.Engine is running with imprecise timing causing risk of high fuel consumption and smoke. Cylinder balancing not working with risk of uneven running. Possible loss of power	Harness/sensor monting Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankshaft Position Sensor A Circuit Intermittent
EMS	P203F00		No Sub Type Information	No urea dosing Urea symbol in cluster	Tank almost empty, stop dosing to maintain cooling ECM interpretation of raw value. Urea level. DST problem.	Diagnostic Service 14. No additional action is needed	Check: ECM interpretation of raw value. Urea level. DST problem.	Reductant Level Low
EMS	P205A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank temp sensor Cables Connectors Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor
EMS	P208064		Signal Plausibility Failure	None	The sensor is not plausible Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1
EMS	P220013	Measured voltage outside the limits specified by the sensor	Circuit Open		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220164		Signal Plausibility Failure		Excessive urea dosing (only for SCR system) Exhaust system leakage Air intake leakage Nox-sensor Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Dosing system (only for SCR system) Exhaust system Air intake system Nox-sensor	NOx Sensor Circuit Range/Performance
EMS	P060100		No Sub Type Information	Engine derate	signature failure .	Diagnostic Service 14. No additional action is needed	Check: signature failure	Internal Control Module Memory Check Sum Error
EMS	P060700		No Sub Type Information	Not possible to reprogram ECU	.	Diagnostic Service 14. No additional action is needed	.	Control Module Performance
EMS	P06151C	Measured voltage outside the limits specified by the sensor	Circuit Voltage Out of Range	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P061600	Measured voltage below limit	No Sub Type Information	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit Low
EMS	P061700	Measured voltage above limit	No Sub Type Information	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Circuit High
EMS	P066800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit Low
EMS	P06B200		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply A Circuit High
EMS	P06B400		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit Low
EMS	P06B500		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Sensor Power Supply B Circuit High
EMS	P20EE00		No Sub Type Information	NOx-conversion reduced by appr. 50% Possible ammonia smell	Regent out of specification, Dosing performance low Urea specification Dosing system	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P261087		Missing Message	Timestamps from Cluster not available	J1708 time data failure and J1939 message does not exist. (Cluster error). Cluster	Diagnostic Service 14. No additional action is needed	Check: Cluster	ECM/PCM Internal Engine Off Timer Performance
EMS	P2BA700		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Empty Reagent Tank
EMS	P023400		No Sub Type Information		Too high pressure Turbine , Compressor	Diagnostic Service 14. No additional action is needed	Check: Turbine , Compressor	Turbocharger/Supercharger Overboost Condition
EMS	P029900		No Sub Type Information		Too low pressure Boost leakage, wastegate, VGT, compressor.	Diagnostic Service 14. No additional action is needed	Check: Boost leakage, wastegate, VGT, compressor.	Turbocharger/Supercharger Underboost
EMS	U300001	Measured voltage outside the limits specified by the sensor	General Electrical Failure	Possible damage on EMS system components depending on the high voltage (>36V)	Generator, external charger	Diagnostic Service 14. No additional action is needed	Check: Generator, external charger	Electronic control unit
EMS	U300041	Program memory corruption	General Checksum Failure	The engine can not be started	No software Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Electronic control unit
EMS	P007087		Missing Message	None	J1708 and CAN ambient air temperature message does not exist. Harness. Cluster	Diagnostic Service 14. No additional action is needed	Check: Harness. Cluster	Ambient Air Temperature Sensor "A"
EMS	P011013	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Measured voltage below limit	No Sub Type Information	Engine is running but could be difficult to start in cold climate.	Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Measured voltage below limit	No Sub Type Information	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P021700		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Over Temperature Condition
EMS	P047013	Measured voltage outside the limits specified by the sensor	Circuit Open	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Exhaust Pressure Sensor "A"
EMS	P047164		Signal Plausibility Failure	Open loop control enabled. Decreased brake power.	sensor, Closed loop butterfly, AVU sensor, butterfly, AVU	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly, AVU	Engine Exhaust Back Pressure Circuit Range/Performance
EMS	P047300	Measured voltage above limit	No Sub Type Information	Open loop control enabled. Decreased brake power.	.	Diagnostic Service 14. No additional action is needed	.	Engine Exhaust Back Pressure Circuit High
EMS	P048013	Measured voltage outside the limits specified by the sensor	Circuit Open	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator or wire. Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit
EMS	P051B64		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The crank case pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Crankcase Pressure Sensor
EMS	P051D00	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine High Resolution Crankcase Pressure
EMS	P052300	Measured voltage above limit	No Sub Type Information	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Circuit High
EMS	P052400		No Sub Type Information	Low pressure, powerloss	Oil leakage, broken oil pump, clogged oil system Oil system	Diagnostic Service 14. No additional action is needed	Check: Oil system	Engine Oil Pressure Too Low
EMS	P222800	Measured voltage below limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit Low
EMS	P222900	Measured voltage above limit	No Sub Type Information	None	Internal fault in ECM Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Circuit High
EMS	P226400		No Sub Type Information	None	No error: Used for indication when priming pump is active None	Diagnostic Service 14. No additional action is needed	Check: No error: Used for indication when priming pump is active	Water in Fuel Sensor
EMS	P226600	Measured voltage below limit	No Sub Type Information	Will give water in fuel indication.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit Low
EMS	P226700	Measured voltage above limit	No Sub Type Information	Undetected water in fuel can lead to engine stop	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Water In Fuel Indicator Circuit High
EMS	P250C00	Measured voltage below limit	No Sub Type Information	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250F00		No Sub Type Information	None	Critically low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Engine Oil Level Too Low
EMS	P255613	Measured voltage outside the limits specified by the sensor	Circuit Open	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Sensor/Switch
EMS	P255900	Measured voltage above limit	No Sub Type Information	Coolant level can not be detected.	Harness Harness connector sensor	Diagnostic Service 14. No additional action is needed	Check: Harness connector sensor	Engine Coolant Level Circuit High
EMS	P256000		No Sub Type Information	Powerloss	Coolant level below range if not Harness Coolant system	Diagnostic Service 14. No additional action is needed	Check: Coolant system	Engine Coolant Level Low
EMS	P108600		No Sub Type Information	None	Low oil level Leakage Leakage Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage Sensor	Oil Level Moderately Low
EMS	P20BF00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 2 Short Circuit Low
EMS	U014100	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With VMCU
EMS	U012A87	Lost communication with the node	Missing Message		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With CCIOM
EMS	P208E00		No Sub Type Information	No urea dosing	Sensor/Harness dosing valve injector, clogged hoses	Diagnostic Service 14. No additional action is needed	Check: dosing valve injector, clogged hoses	Aftertreatment Reagent Dosing Valve Clogged
EMS	P109541		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Control Unit
EMS	P109641		General Checksum Failure	Engine derate	.	Diagnostic Service 14. No additional action is needed	.	Transmission Control Unit
EMS	P069200	Measured voltage above limit	No Sub Type Information	Fan runs at full speed. Increased fuel consumption may be seen.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit High
EMS	P069100	Measured voltage below limit	No Sub Type Information	Fan always deactivated or always activated if fault is intermittent.	Broken actuator, wire or ECM driver output Actuator/Harness	Diagnostic Service 14. No additional action is needed	Check: Actuator/Harness	Fan 1 Control Circuit Low
EMS	U015500	Lost communication with the node on Medium Speed CAN Communication Bus	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil pressure shows 0 in the cluster. Engine is running.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Pressure Sensor/Switch "A"
EMS	P066613	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Internal fault in ECM ECM	Diagnostic Service 14. No additional action is needed	Check: ECM	Control Module Internal Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020113	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 1 Injector "A"
EMS	P020313	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 3 Injector "A"
EMS	P020413	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 4 Injector "A"
EMS	P020513	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 5 Injector "A"
EMS	P020613	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 6 Injector "A"
EMS	P220E93		No Operation		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Heater Control Circuit Range/Performance (Bank 1 Sensor 1)
EMS	P250A13	Measured voltage outside the limits specified by the sensor	Circuit Open	Oil level can not be measured.	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Level Sensor
EMS	P020213	Measured voltage outside the limits specified by the sensor	Circuit Open	Loss of power and uneven running. Run on three to five cylinders	Harness/Broken solenoid Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Cylinder 2 Injector "A"
EMS	P100E18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 2
EMS	P100F18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 3
EMS	P101018		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 4
EMS	P101518		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 5
EMS	P101618		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 6
EMS	P101711	Measured voltage below limit	Circuit Short To Ground	High fuel consumption due to fuel leakage.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101712	Measured voltage above limit	Circuit Short To Battery	Valve constantly shut.	Broken solenoid, wire or ECM driver output Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve
EMS	P101713	Measured voltage outside the limits specified by the sensor	Circuit Open	Valve constantly shut.	Broken solenoid or wire. Water-in-fuel valve driver	Diagnostic Service 14. No additional action is needed	Check: Water-in-fuel valve driver	Water In Fuel Drainage Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220300		No Sub Type Information		Cabling between Nox-sensor and Nox-sensor ECU Nox-sensor Nox-sensor cabling Nox-sensor	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor cabling Nox-sensor	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P2BA800		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Interruption of Reagent Dosing Activity
EMS	P2BA900		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality
EMS	P2BAA00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Low Reagent Consumption
EMS	P2BAD00		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown
EMS	P2BAE96		Component Internal Failure		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System
EMS	P100211	Measured voltage below limit	Circuit Short To Ground	Induction air is hot. Preheat relay is impossible to turn off.	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Preheat Relay
EMS	P100212	Measured voltage above limit	Circuit Short To Battery	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Preheat relay solenoid shorted Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Preheat relay.	Preheat Relay
EMS	P100213	Measured voltage outside the limits specified by the sensor	Circuit Open	Preheat relay not activated. White smoke for cold start. Start problems in cold climate.	Harness or Preheat relay solenoid Harness. Preheat relay.	Diagnostic Service 14. No additional action is needed	Check: Harness. Preheat relay.	Preheat Relay
EMS	P100629		Signal invalid	None	The boost temperature sensor output is showing constant value. Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Intake Air Temperature Sensor 1 Stuck Bank 1
EMS	P100713	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator
EMS	P100D18		Circuit Current Below Threshold	None	Injector/Harness resistance too high. Injector/Harness	Diagnostic Service 14. No additional action is needed	Check: Injector/Harness	Injector 1
EMS	P102111	Measured voltage below limit	Circuit Short To Ground	VCB can not be turned off. Causes engine to stop. Engine impossible to restart.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P102112	Measured voltage above limit	Circuit Short To Battery	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	Measured voltage outside the limits specified by the sensor	Circuit Open	VCB can not be turned on. Engine break function derated. Gear shift performance derated for some automatic transmission boxes.	Broken actuator/Harness VCB actuator	Diagnostic Service 14. No additional action is needed	Check: VCB actuator	Volvo Compression Brake (VCB) Control Circuit
EMS	U010E00	Lost communication with the node	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Lost Communication With Reductant Control Module
EMS	P019513	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Engine Oil Temperature Sensor "A"
EMS	P204A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Reductant Pressure Sensor
EMS	P203D00	Measured voltage above limit	No Sub Type Information	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Aftertreatment Reagent Level Short Circuit High
EMS	P208A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Reductant Pump "A" Control
EMS	P026500	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 2 Short Circuit High
EMS	P026800	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 3 Short Circuit High
EMS	P027100	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 4 Short Circuit High
EMS	P027400	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 5 Short Circuit High
EMS	P027700	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 6 Short Circuit High
EMS	P20B300	Measured voltage below limit	No Sub Type Information	None	.	Diagnostic Service 14. No additional action is needed	.	Reductant Heater Coolant Control Valve Circuit Low
EMS	P204800	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: DCU ACM Pump module: internal cabling ACM Pump module: DCU, ACM Pump module: internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: DCU, ACM Pump module: internal cabling	Aftertreatment Reagent Dosing Valve Short Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P204713	Measured voltage outside the limits specified by the sensor	Circuit Open	no urea dosing	Cable / Connectors ACM Dosing module: Dosing valve Cable / Connectors ACM Dosing module: Dosing valve	Diagnostic Service 14. No additional action is needed	Check: Cable / Connectors ACM Dosing module: Dosing valve	Reductant Injection Valve Control Bank 1 Unit 1
EMS	P20C300	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
EMS	P20BB00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
EMS	P20A200	Measured voltage below limit	No Sub Type Information	Not possible to perform afterrun	Internal electrical failure in valve within the ACM	Diagnostic Service 14. No additional action is needed	Check: Internal electrical failure in valve within the ACM	Aftertreatment Reagent Direction Valve Short Circuit Low
EMS	P065900	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit High
EMS	P065800	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	ACM Actuator High Supply 1 Short Circuit Low
EMS	P208D00	Measured voltage above limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circut High
EMS	P208C00	Measured voltage below limit	No Sub Type Information	No urea dosing	Wiring harness internal error in pump	Diagnostic Service 14. No additional action is needed	Check: Wiring harness internal error in pump	Aftertreatment Reagent Pump Control Short Circuit Low
EMS	P203A13	Measured voltage outside the limits specified by the sensor	Circuit Open	None	AdBlue tank level sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: AdBlue tank level sensor, Cables, Connectors	Reductant Level Sensor "A"
EMS	P019700	Measured voltage below limit	No Sub Type Information	None	Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Oil Temperature Short Circuit Low
EMS	P024313	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Broken actuator or wire. Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024471		Actuator Stuck	Possible turbo damage	Wategate mechanically damaged Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Stuck Close
EMS	P024500	Measured voltage below limit	No Sub Type Information		Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit Low
EMS	P024600	Measured voltage above limit	No Sub Type Information	Powerloss	Broken actuator, wire or ECM driver output Wastegate	Diagnostic Service 14. No additional action is needed	Check: Wastegate	Wastegate Short Circuit High
EMS	P026200	Measured voltage above limit	No Sub Type Information	Loss of power and uneven running. Run on three to five cylinders	Harness shorted Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Injector 1 Short Circuit High
EMS	P204300		No Sub Type Information	None	Urea tank temp sensor stuck high Urea tank temp sensor	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor	Aftertreatment Reagent Tank Temperature Sensor Stuck
EMS	P209F73		Actuator Stuck Closed	None	Urea tank temp sensor or Urea tank heating system Urea tank temp sensor, Urea tank heating system	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Urea tank heating system	Aftertreatment Reagent Tank Temperature

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20F400		No Sub Type Information	Too low urea consumption	Urea level sensor stuck Clogged urea system Wrong urea tank Wrong dosing module Nozzle or ACM Dosing module: dosing valve	Diagnostic Service 14. No additional action is needed	Check: Nozzle or ACM Dosing module: dosing valve	AdBlue Consumption Too Low
EMS	U029D00	Lost communication with the node	No Sub Type Information		MS from Nox-sensor ECU Nox-sensor power supply CAN communication between ECM and Nox-EUC	Diagnostic Service 14. No additional action is needed	Check: Nox-sensor power supply CAN communication between ECM and Nox-EUC	Lost Communication With NOx Sensor "A"
EMS	P204C00	Measured voltage below limit	No Sub Type Information	No urea dosing	ACM Pump module: AdBlue pressure sensor ACM Pump module: Internal cabling ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Diagnostic Service 14. No additional action is needed	Check: ACM Pump module: Urea pressure sensor, ACM Pump module: Internal cabling	Aftertreatment Reagent Pressure Sensor Circuit Low
EMS	U016700		No Sub Type Information	The engine can not be started	Immobilizer unit not installed. (When Immo in Cluster is used, the Cluster maybe did not wake up properly) Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Lost Communication With Vehicle Immobilizer Control Module
EMS	U042600		No Sub Type Information	The engine can not be started	ECM and Immobilizer(or Cluster) unit security codes do not match. Immo system	Diagnostic Service 14. No additional action is needed	Check: Immo system	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000		No Sub Type Information	No VCB	low engine oil temperature .	Diagnostic Service 14. No additional action is needed	Check: low engine oil temperature	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P011164		Signal Plausibility Failure	None	The boost temperature sensor output is showing either too high or to low value. (abnormal value) Harness Sensor Ambient air temperature sensor.	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor Ambient air temperature sensor.	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1
EMS	P20E892		Performance or Incorrect Operation	No urea dosing	ACM dosing module: Dosing valve leakage. Bleed orifice leakage, pressure hose lekage, pump internal leakage, empty ureatank Tank frozen. if pump/tank-lines are mixed-up when connecting them to the pump unit there will be no pressure built (check color mar Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Diagnostic Service 14. No additional action is needed	Check: Empty Urea tank, Urea filter clogged, Urea inlet pipe leakage/blocked, Urea tank ventilation, ACM Pump module: Pump	Reductant Pressure Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P21CA00		No Sub Type Information	No dozing	Voltage supply to ACM. Voltage supply to Urea pump.	Diagnostic Service 14. No additional action is needed	Check: Voltage supply to Urea pump.	Reductant Control Module Supply Voltage
EMS	P114411	Measured voltage below limit	Circuit Short To Ground	EMS can not control starter	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114412	Measured voltage above limit	Circuit Short To Battery	Impossible to activate starter motor.	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P114413	Measured voltage outside the limits specified by the sensor	Circuit Open	Impossible to activate starter motor	Start relay/harness	Diagnostic Service 14. No additional action is needed	Check: Start relay/Harness	Starter Relay Highside
EMS	P116A13	Measured voltage outside the limits specified by the sensor	Circuit Open	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side
EMS	P116B00	Measured voltage below limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit Low
EMS	P116C00	Measured voltage above limit	No Sub Type Information	No urea dosing	.	Diagnostic Service 14. No additional action is needed	.	Reductant Pump "A" Control Low Side Circuit High
EMS	P021900		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPD activated to lower espd below deactivation limit (commonly 1000rpm).	Engine is/was overspeeding. Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Diagnostic Service 14. No additional action is needed	Check: Engine could be running on oil, due to e.g. faulty crankcase oil filter.	Engine Overspeed Condition
EMS	P115211	Measured voltage below limit	Circuit Short To Ground	Priming pump permanently on. Possible damage of priming pump.	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115212	Measured voltage above limit	Circuit Short To Battery	Priming or drain aborted and inhibited	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115213	Measured voltage outside the limits specified by the sensor	Circuit Open	Priming or drain inhibited	Actuator/harness Priming pump	Diagnostic Service 14. No additional action is needed	Check: Priming pump	Fuel Priming Pump
EMS	P115C00	Measured voltage above limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit High
EMS	P115D00	Measured voltage below limit	No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "C" Circuit Low
EMS	P06E970		Actuator Stuck On	The starter motor is deactivated	The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the cogwheels of the starter and engine are stuck	Engine Starter Performance
EMS	P06E998		Component or System Over Temperature	The starter motor is deactivated	The starter motor is deactivated due to overheat None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to overheat	Engine Starter Performance
EMS	P115E97		Component or System Operation Obstructed or Blocked	No dozing	Bleed orifice clogged Bleed orifice, Urea pressure sensor,	Diagnostic Service 14. No additional action is needed	Check: Bleed orifice, Urea pressure sensor,	Reductant Return No Flow Detected
EMS	P11609B		High/Excessive Flow		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Root Cause Unknown (First Level)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P116192		Performance or Incorrect Operation		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - Insufficient Reagent Quality (First Level)
EMS	P116300		No Sub Type Information		.	Not possible to erase, will be healed after 400 days. In production, please re-flash the control unit and make sure to enter factory mode next time	.	NOx Exceedence - NOx Control Monitoring System (First Level)
EMS	P26047F	Measured voltage below limit	Actuator Stuck Off	Preheat relay problem or sense SC-.	Shorting wire may break when attempt Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Circuit Range/Performance
EMS	P115600		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit High
EMS	P115700		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Reductant Control Module Sensor Power Supply "A" Circuit Low
EMS	P115800		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit High
EMS	P115900		No Sub Type Information	Incorrect sensor readings	Harness/Sensor power supply Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Reductant Control Module Sensor Power Supply "B" Circuit Low
EMS	P115A00		No Sub Type Information	Possible incread risk of crystalization	Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Exhaust Gas Temperature Sensor Circuit Signal Compare Failure Bank 1 Sensor 1
EMS	P115B00		No Sub Type Information	None	The sensor is stuck Leakage in exhaust system Leakage in intake system Sensor	Diagnostic Service 14. No additional action is needed	Check: Leakage in exhaust system Leakage in intake system Sensor	Exhaust Gas Temperature Sensor Circuit Stuck Bank 1 Sensor 1
EMS	P116441		General Checksum Failure	All ACM functionality disabled, no urea dosing	No software in ACM .	Diagnostic Service 14. No additional action is needed	Check: No software in ACM	Reductant Control Module Internal Control Module Memory Checksum Error
EMS	P116700		No Sub Type Information	The starter motor is deactivated	The starter motor is deactivated due to that the PTO switch is active. None	Diagnostic Service 14. No additional action is needed	Check: The starter motor is deactivated due to that the PTO switch is active.	Starter Activated When PTO Is Active
EMS	P000041		General Checksum Failure	The engine can not be started	.	Diagnostic Service 14. No additional action is needed	.	Control Module
EMS	P24FF00		No Sub Type Information	If tank temp is too high, no urea dosing	Urea tank temp is too high Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	Reductant Temperature Too High
EMS	U1FFF68		Event Information	None	None	Diagnostic Service 14. No additional action is needed	.	Factory Mode Active
EMS	P117A00		No Sub Type Information	Possible ammonia smell	SCR catalyst removed, Regent out of specification, Clogged dosing system Urea specification Dosing system Catalyst deactivated	Diagnostic Service 14. No additional action is needed	Check: Urea specification Dosing system Catalyst deactivated	SCR NOx Catalyst Efficiency Critically Below Threshold Bank 1
EMS	P047800	Measured voltage above limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" High
EMS	P047700	Measured voltage below limit	No Sub Type Information	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A" Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss, engine break is reduced	Broken actuator/Harness AVU	Diagnostic Service 14. No additional action is needed	Check: AVU	Exhaust Pressure Control Valve "A"
EMS	P222764		Signal Plausibility Failure	Powerloss (minor) P3524: Black smoke (minor)	The ambient air pressure is showing either too high or to low value. (abnormal value) Sensor ECM	Diagnostic Service 14. No additional action is needed	Check: Sensor ECM	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P011513	Measured voltage outside the limits specified by the sensor	Circuit Open	Engine is difficult to start. Idle run regulation is detoriated.	Harness Sensor Sensor/Harness	Diagnostic Service 14. No additional action is needed	Check: Sensor/Harness	Engine Coolant Temperature Sensor 1
EMS	P205E1F		Circuit Intermittent		Urea tank temp sensor, Cables, Connectors	Diagnostic Service 14. No additional action is needed	Check: Urea tank temp sensor, Cables, Connectors	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic
EMS	P053E00		No Sub Type Information	Forced idle Engine shut down	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Too High
EMS	P010513	Measured voltage outside the limits specified by the sensor	Circuit Open	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Measured voltage above limit	No Sub Type Information	Powerloss	Connectors. Harness. Sensor Connectors. Harness. Sensor	Diagnostic Service 14. No additional action is needed	Check: Connectors. Harness. Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P010662		Signal Compare Failure	Powerloss Black smoke	The boost pressure sensor output is showing either too high or to low value. (abnormal value) Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Range/Performance
EMS	P054070	Measured voltage above limit	Actuator Stuck On	Fuse for shorting wire breaks.	Sense wire SC+ or preheat relay broken Preheat sense 1 Preheat relay	Diagnostic Service 14. No additional action is needed	Check: Preheat sense 1 Preheat relay	Intake Air Heater "A" Control
EMS	P225D00		No Sub Type Information	None except fault code and lamp	Sensor removed (measures surround air) Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10F200	Measured voltage below limit	No Sub Type Information	None	Pressure switch / Harness Pressure switch/Harness	Diagnostic Service 14. No additional action is needed	Check: Pressure switch/Harness	Air Filter Indicator Circuit Low
EMS	P10F300	Measured voltage above limit	No Sub Type Information	None	Harness	Diagnostic Service 14. No additional action is needed	Check: Harness	Air Filter Indicator Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P22FB00		No Sub Type Information		Sensor	Diagnostic Service 14. No additional action is needed	Check: Sensor	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P110400		No Sub Type Information	The engine can not be started	Bad software configuration Software download application	Diagnostic Service 14. No additional action is needed	Check: Software download application	Factory Mode
EMS	P10F824		Signal Stuck High		.	Diagnostic Service 14. No additional action is needed	.	Air Filter Clogged
EMS	P029800		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Over Temperature
EMS	P111600		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Moderately Too High
EMS	P111800		No Sub Type Information	Intermittent Powerloss	VGT stuck, Wastegate stuck VGT Turbo, Wastegate	Diagnostic Service 14. No additional action is needed	Check: VGT Turbo, Wastegate	Manifold Absolute Pressure Too High
EMS	P111D00		No Sub Type Information		.	Diagnostic Service 14. No additional action is needed	.	Engine Coolant Level Moderately Low
EMS	P111E00		No Sub Type Information	Powerloss	Extreme Driving conditions Extreme driving conditions Coolant thermostat Fan Blocked radiator	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Coolant thermostat Fan Blocked radiator	Engine Coolant Temperature Moderately High
EMS	P112500		No Sub Type Information	Powerloss	Extreme driving conditions Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Diagnostic Service 14. No additional action is needed	Check: Extreme driving conditions Fan Oil thermostat Coolant system Clogged oil cooler	Engine Oil Temperature Moderately High
EMS	U117888	It has not been possible to send frames on Vehicle Communication Powertrain CAN and It has not been possible to receive frames on Vehicle Communication Powertrain CAN	Bus off		.	Diagnostic Service 14. No additional action is needed	.	Vehicle Communication Powertrain CAN
EMS	P253913	Measured voltage outside the limits specified by the sensor	Circuit Open	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Measured voltage above limit	No Sub Type Information	None	Sensor/Harness Harness Sensor	Diagnostic Service 14. No additional action is needed	Check: Harness Sensor	Low Pressure Fuel System Sensor Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00		No Sub Type Information	Fuel injection can be affected.	High differential pressure over filter. Leakage Filter Sensor	Diagnostic Service 14. No additional action is needed	Check: Filter Sensor	Low Pressure Fuel System Pressure - Too Low
EMS	P113500		No Sub Type Information	Engine speed is limited (to commonly 1200rpm). VCB and EPG activated to lower espd below deactivation limit (commonly 1000rpm).	The non-filtered pressure difference (between crankcase pressure and ambient air pressure) is/was above limit. (The fault code will remain during the entire driving cycle (unless reset).) E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Diagnostic Service 14. No additional action is needed	Check: E.g. Blow-by at piston ring. NOT Crankcase pressure sensor or ambient air pressure sensor!!!	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P047F71		Actuator Stuck	Decreased brake power.	sensor, Closed loop butterfly sensor, butterfly	Diagnostic Service 14. No additional action is needed	Check: sensor, butterfly	Exhaust Pressure Control Valve "A" Stuck Open
APM	C100601	"Failure is set if the sensor shorted to ground. Failure is set if the sensor shorted to power supply."	General Electrical Failure	Circuit #1 pressure information will not available.	"- damage of the sensor wiring - damage of the sensor or - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Check pressure in Air supply circuit 1 by manometer. Pressure should be <12.6b. - If overpressure is measured, replace APM.	Air supply circuit 1 pressure sensor
APM	C100701	"Failure is set if the sensor shorted to ground. Failure is set if the sensor shorted to power supply."	General Electrical Failure	Circuit #2 pressure information will not available.	"- damage of the sensor wiring - damage of the sensor or - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Check pressure in Air supply circuit 2 by manometer. Pressure should be <12.6b. - If overpressure is measured, replace APM.	Air supply circuit 2 pressure sensor
APM	U300041	"Drive software checksum is wrong Flash, NVM checksum is wrong."	General Checksum Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	"- Demage of the flash and/or flash content - Last drive software download faild - Damage of the eeprom of the ECU"	Diagnostic Service 14. No additional action is needed.	- download SW and datasets - Replace APM	Electronic control unit
APM	U300044	"At least one dataset parameter out of range or relation error among parameters.	Data Memory Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- Incorrect/Unrealistic data written into APM - Dataset damage	Diagnostic Service 14. No additional action is needed.	- Download dataset - Execute Restore supplier data memory area routine - Check the improper parameters and their writing times. If it happened after parameter set up then do the programming again, if not then replace the sample. - Replace APM	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	U300045	Boot strap loader (BSL) invalid or incompatible with Main software (MSW)	Program Memory Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	"Main SW download is interrupted OR wrong compatibility number between BSL and Main SW.	Diagnostic Service 14. No additional action is needed.	- download SW - Replace APM	Electronic control unit
APM	U300046	Dataset compatibility or block checksum calculation fails	Calibration / Parameter Memory Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	Damage of the EEPROM of the ECU Damage of the flash and/or flash content Compatibility number of datasets and MSW differs. SM function for updating failures was called with a parameter which is out of range.	Diagnostic Service 14. No additional action is needed.	- download dataset - execute restore system supplier data - download MSW - replace APM	Electronic control unit
APM	U300049	EEPROM invalid or unpartitioned or protected EEPROM invalid or CSWC internal failure	Internal Electronic Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- No valid eeprom downloaded - Damage of the eeprom of the ECU - CSWC internal failure	Diagnostic Service 14. No additional action is needed.	- Replace APM	Electronic control unit
APM	U30004A	HW stepping / MV / heater / sensor configuration is incompatible with the Main SW	Incorrect Component Installed	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- Software is incompatible with the hardware or number of solenoids or sensors	Diagnostic Service 14. No additional action is needed.	- download Main software (MSW) and download dataset - Replace APM	Electronic control unit
APM	U30004B	Ecu temperature is too high. Normal operating range: -40 to +85°C	Over Temperature	High risk of unintended parking brake actuation	Environement temperature is above normal operating range or ECU damaged	- Diagnostic Service 14. No additional action is needed.	- Check the temperature arround APM : High temperature can be caused by (gaz Exhaust leakage, wrong installation of cab heating system...).	Electronic control unit
APM	U300054	End-of-line (vehicle related) parameter setting (dataset) is missing	Missing Calibration	Parking brake function related plausibility monitoring disbaled.	Dataset is missing.	Diagnostic Service 14. No additional action is needed.	- Download vehicle configuration dataset - Check parking brake chamber configuration of the axles	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	U300055	DFLASH dataset invalid	Not Configured	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkingBrakeActuation Status	No valid Dflash downloaded	Diagnostic Service 14. No additional action is needed.	- Flash the dataset. - Replace APM if not successful.	Electronic control unit
APM	U300094	Microcontroller halted unexpectedly	Unexpected Operation	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- Unexpected microcontroller error caused by software or hardware	Diagnostic Service 14. No additional action is needed.	- dowload Main software (MSW) - Replace APM	Electronic control unit
APM	U002849	CAN driver hardware / software error	Internal Electronic Failure	CAN communication restricted, worst case: Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	internal ECU Can driver failure	Diagnostic Service 14. No additional action is needed.	- Replace APM	CAN Communication Chassis Net
APM	U002886	At least one of the follwing signals has error value: WheelBasedVehicleSpeed, DriverEngineShutoff_rqst, Vehicle mode WheelBasedVehicleSpeed - updated only in high VM DriverEngineShutoff_rqst - updated in all VM	Signal Invalid	Vehicle mode invalid: No parking brake release is possible, automatic functions and trailer test is not available / WheelBasedVehicleSpeed, DriverEngineShutoff_rqst is invalid: If redundant signals are also missing then Autorelease not available XPR not available	Damage of other ECUs	Diagnostic Service 14. No additional action is needed.	- Read out failure codes of the Vehicle ECU (VMCU) - Check communication of the Chassis Net - Check ECUs connected to the Chassis Net	CAN Communication Chassis Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	U002887	Timeout of Vehicle mode signal is over Vehicle mode or WheelBasedVehicleSpeed or DriverEngineShutoff_rqst is not received in time (6x the missingignal peridoicity)	Missing Message	Vehicle mode missing: No parking brake release is possible, automatic functions and trailer test is not available / WheelBasedVehicleSpeed, DriverEngineShutoff_rqst is missing: If redundant signals are also missing then Autorelease not available XPR not available	Demage onVMCU. damage on CAN bus Damage on CAN wiring.	Diagnostic Service 14. No additional action is needed.	- Read out failure codes of the Vehicle ECU - Check APM connector and CAN wires and contacts - Check the Chassis Net wires and communication.	CAN Communication Chassis Net
APM	U002888	CAN electronic data link bus off detected if there is no difference between the CAN wires	Bus off	Wake up by LIN. Just manual parking is possible. Governing is working. No ParkingBrakeActuation Status	- CAN cabling errors (e.g. short circuit	Diagnostic Service 14. No additional action is needed.	- Check CAN wires and connector pins. Measure bus resistance - Replace APM	CAN Communication Chassis Net
APM	U300313	Direct battery supply line (pin 5) voltage level below 16 Volt threshold.	Circuit Open	Power supply redundancy is lost, possible loss of learned parameters. Shut down process is not guaranteed (Autopark, final reg, Compressor pipe exhaust). Parking brake actuation is not guaranteed at ignition off.	- wrong supply voltage level - damage of wiring - internal electric failure	Diagnostic Service 14. No additional action is needed.	- Check wiring, connectors, fuse. - Check battery voltage - Replace APM	Battery Power
APM	U300316	Maximum voltage of battery supply lines (pin 5 and pin14) below threshold longer than a timeframe 110ms If voltage level below 16V	Circuit Voltage Below Threshold	Further voltage decrease might cause Pneumatic backup of the air management control (idle cut-off, no regeneration) and No EPB functions Pneumatic backup release and apply of PB is possible.	- wrong supply voltage level - internal electric failure - damage of the ECU - damage of wiring	Diagnostic Service 14. No additional action is needed.	- Check wiring, connectors, fuse. - Check battery voltage - Replace APM	Battery Power
APM	U300317	Maximum voltage of battery supply lines (pin 5 and pin14) above threshold (32V) longer than a timeframe	Circuit Voltage Above Threshold	Further voltage increase might cause Pneumatic backup of the air management control (idle cut-off, no regeneration) and No EPB functions Pneumatic backup release and apply of PB is possible.	- wrong supply voltage level - internal electric failure - damage of the ECU - damage of wiring	Diagnostic Service 14. No additional action is needed.	- Check wiring, connectors, fuse. - Check battery voltage - Replace APM	Battery Power
APM	C100101	Failure detected in the electrical circuit of the regeneration magnet valve	General Electrical Failure	Pneumatic backup air management control (idle cut-off, no regeneration)	"- damage of the wiring - damage of the Magnet Valve - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Replace APM	Air Management Regeneration Solenoid Valve
APM	C100201	"Failure detected in the electrical circuit of the governor magnet valve	General Electrical Failure	Pneumatic backup air management control (idle cut-off, no regeneration)	"- damage of the wiring - damage of the Magnet Valve - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Replace APM	Air Management Governor Solenoid Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C100294	Failure is set in case of external excitation on the MV	Unexpected Operation	Compressor is continuously in off-load, vehicle will loose the stored compressed air.	- damage of the ECU - unintended actuation - shortcut in the magnet valve circuit (due to humidity)	Diagnostic Service 14. No additional action is needed.	- Replace APM	Air Management Governor Solenoid Valve
APM	C102A01	Failure is set if the sensor shorted to ground. Failure is set if the sensor shorted to ubat.	General Electrical Failure	EPB: No functional effect, redundant information is lost. Backup Level 1. PPB: Parking brake actuation status is not available.	damage of the sensor wiring damage of the sensor or damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Replace APM	Parking Brake Chamber Pressure Sensor
APM	C100A7A	System pressure level increases over cut off level more than 0.5 bar staying in off load phase *setting the DTC indirectly depends on vehicle modes when the engine is running	Fluid Leak or Seal Failure	Compressor cannot be switched to off-load. High pressure. Limited circuits are not affected.	"1. ESS line damaged 2. Unloader valve failure 3. Magnet valve failure 4. Sensor failure"	Diagnostic Service 14. No additional action is needed.	1 - Check if compressor control pipe line has no leakage when compressor is off load . 2 - Unloader - Open the cartridge, Check area below the cartridge in the sieve area (and below the sieve) against dirt or ice and blockings (sieve blocked), check main exhaust silencer if not blocked 3- If no problem on compressor pipe and no dirt, replace unloader valve. (Kit). 4 - If still problem , replace APM	Governor valve pressure
APM	C100C64	No more failure free pressure sensor available in circuits without pneumatic pressure limiter without back stream or check valve.System pressure cannot be estimated.	Signal Plausibility Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No electronic parking brake release is not possible.	See the root causes of failures reported simultaneously	Diagnostic Service 14. No additional action is needed.	- Check if power supply voltage is less than 8 Volt (in this case also the DTC U300316 is active) - Deflate the air tanks - fill-up and follow if pressures are increasing normally - Check if failure reoccurs . - if yes, exchange APM	Air Pressure System
APM	C100C7A	"Circuit 1 pressure level reached above the maximal cut off plus 1.5 bar Circuit 2 pressure level reached above the maximal cut off plus 1.5 bar"	Fluid Leak or Seal Failure	Pneumatic backup air management control (idle cut-off, no regeneration)	"1. ESS line damaged 2. Unloader valve failure 3. Magnet valve failure 4. Sensor failure"	Diagnostic Service 14. No additional action is needed.	1 - Check if compressor control pipe line has no leakage when compressor is off load . 2 - Unloader - Open the cartridge, Check area below the cartridge in the sieve area (and below the sieve) against dirt or ice and blockings (sieve blocked), check main exhaust silencer if not blocked 3- If no problem on compressor pipe and no dirt, replace unloader valve. (Kit). 4 - If still problem , replace APM	Air Pressure System
APM	C100C7B	At least one circuit pressure is under low limit when the vehicle is moving (v>10 km/h)	Low Fluid Level	Service brake performance is not sufficient	"1. High air consumption 2. Leakage or defect in air circuits"	Diagnostic Service 14. No additional action is needed.	- When compressor is loading, check leakage at APM main exhaust . No leakage expected until cut off pressure. - If leakage, exchange unloader valve.	Air Pressure System
APM	C100D7B	Circuit 1 pressure level reached minimal pressure level (parameter) after reaching the nominal pressure (paramater) when the vehicle is moving (v>10 km/h)	Low Fluid Level	Service brake performance lost	"1. High air consumption 2. Leakage or defect in air circuits"	Diagnostic Service 14. No additional action is needed.	- When compressor is loading, check leakage at APM main exhaust . No leakage expected until cut off pressure. - If leakage, exchange unloader valve.	Air supply pressure front brake circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C100E7B	Circuit 2 pressure level reached minimal pressure level (parameter) after reaching the nominal pressure (paramater) when the vehicle is moving (v>10 km/h)	Low Fluid Level	Service brake performance lost	"1. High air consumption 2. Leakage or defect in air circuits"	Diagnostic Service 14. No additional action is needed.	- When compressor is loading, check leakage at APM main exhaust . No leakage expected until cut off pressure. - If leakage, exchange unloader valve.	Air Supply Pressure Rear Brake Circuit
APM	C100F62	Pressure difference between Circuit 1 and 2 is above 1.5 bar	Signal Compare Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No electronic parking brake release is not possible.	"1. External fill up through Circuit 1 or Circuit 2 air tanks 2. Sensor failure 3. Protection valve failure"	Diagnostic Service 14. No additional action is needed.	- Deflation of system down to pressure level of 5 bar and then fill up the system up to cut off pressure and let APM switch to off load. Wait for 30s. - Check the service brake circuits for leakage - Replace APM	Front and Rear Brake Circuit
APM	C101024	The cartridge is over worn	Signal Stuck High	No APM functional effect, high risk to freeze brake parts.	"1. Cartridge is overworn 2. High air consumption 3. Leakage 4. Compressor failure 5. Wrong compressor parameter configured	Diagnostic Service 14. No additional action is needed.	-- Replace cartridge - Reset the maintenance data of APM cartridge (by techtool or menu in cluster); - Check of reservoirs: If there is water in reservoirs, drain water from reservoirs, actuate and release 30 times the park brake to eliminate water in park brake circuit (can cause a stop in the future). - If the periodicidy of maintenance is short (abnormal cartridge life time (under 6 months))following additional actions must be done: - Check vehicle for air leakage while system is filled up with PB released (engine running not needed) - Check compressor performance by measuring the inflating air system time (techtool action).	Air Filter Cartridge Wear
APM	C101068	The cartridge is 100% worn, and must be changed.	Event Information	No functional effect, water content in the compressed air reservoirs can be too high, freeze possible	"1. Cartridge is worn 2. High air consumption 3. Leakage 4. Compressor failure 5. Wrong compressor parameter configured	Diagnostic Service 14. No additional action is needed.	- Replace cartridge - Reset the maintenance data of APM cartridge (by techtool or menu in cluster); - Check of reservoirs: If there is water in reservoirs, drain water from reservoirs, actuate and release 30 times the park brake to eliminate water in park brake circuit (can cause a stop in the future). - If the periodicidy of maintenance is short (abnormal cartridge life time (under 6 months))following additional actions must be done: - Check vehicle for air leakage while system is filled up with PB released (engine running not needed) - Check compressor performance by measuring the inflating air system time (techtool action).	Air Filter Cartridge Wear
APM	C101164	Calculated duty factor of the compressor is above 85%	Signal Plausibility Failure	No sufficient regeneration	"1: High air consumption 2. air Leakage 3. Vehicle usage not corresponds to the designed compressor purpose 4. Compressor failure	Diagnostic Service 14. No additional action is needed.	- Check Air system for possible air leakage on the vehicle. - Check vehicle usage relative to Air compressor size. - Check of reservoirs: If there is water in reservoirs, drain water from reservoirs, actuate and release 30 times the park brake to eliminate water in park brake circuit (can cause a stop in the future). - if the fault is set in normal usage (like high air consumption on specific application) and there is no air leakage, the function" Engine Speed Control" can be enabled by parameter P1F08 to limit the occurence. - If the fault occurence is high , contact the after market product specialist for specific solution.	Air Compressor
APM	C101512	Safety relay of Air management short circuit to supply voltage or short between relays or BuC can not switch off relays	Circuit Short To Battery	No effect on the functionality Safety feature of the relay is not available.	- internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Replace APM	Regeneration and Governor Valve High Side Solenoid
APM	C101513	Safety relay of Air management open circuit to supply voltage.	Circuit Open	Pneumatic backup air management control (idle cut-off, no regeneration)	- internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Replace APM	Regeneration and Governor Valve High Side Solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C101268	The cartridge is nearly fully worn	Event Information	No functional effect.	"1. Cartridge is getting worn 2. High air consumption 3. Leakage 4. Compressor failure 5. Wrong engine speed signal 6. Wrong compressor parameter configured 7. Sensor failure"	Diagnostic Service 14. No additional action is needed	-- The replacement of cartridge can be done in advance (avoid to go go back soon for this in workshop) - Reset the maintenance data of APM cartridge (by techtool or menu in cluster); - Check of reservoirs: If there is water in reservoirs, drain water from reservoirs, actuate and release 30 times the park brake to eliminate water in park brake circuit (can cause a stop in the future). - If the periodicidy of maintenance is short (abnormal cartridge life time (under 6 months))following additional actions must be done: - Check vehicle for air leakage while system is filled up with PB released (engine running not needed) - Check compressor performance by measuring the inflating air system time (techtool action).	Air Filter Cartridge
APM	C102172	Parking brake pressure is above than the Circuit 3 and Circuit 4 limitation pressure plus 1500 mbar	Actuator Stuck Open	Parking brake status is not available, possible damage of the trailer	1. Pressure limiter failure 2. Sensor failure	Diagnostic Service 14. No additional action is needed	- Deflate fully air tanks and Fill up the system to the cut off pressure, apply and release the parking brake; check if problem reoccurs. - Check parking brake connection to the springbrake chambers - Check the APM exhaust ports - Compare value of DID P1E0K to value of manometer connected to port 29.2 - Compare value of DID P1E0J to value of manometer connected to port 28.1/28.2 - Replace APM	Pressure Limiter
APM	U114A47	Intercontroller communication error detected or software of secondary controller is incompatible with the software of primary controller	Watchdog / Safety µC Failure	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- Wrong compatibility number of BuC. - AuC and BuC communication lost	Diagnostic Service 14. No additional action is needed.	- dowload Main software (MSW) - Replace APM	Secondary Control Unit
APM	U114A68	Internal function failure or intercontroller communication error .	Event Information	Learned parameters are not stored. Lost wetness information. Frequent occurrence can lead to wetness in the system. Secondary controller can not prevent to send incorrect information by the primary one	- internal Electric failure - failure between AuC and BuC communication - power supply is increased very slow from low value, the CAN transceiver goes into an internal error state which possible is not handled by CSWC.	Diagnostic Service 14. No additional action is needed.	- perform a hard reset - download MSW - Replace APM	Secondary Control Unit
APM	U114913	CCIOM power supply line (pin 14) voltage level below threshold. 8V	Circuit Open	Power supply redundancy is lost.	- wrong supply voltage level - damage of wiring - internal electric failure	Diagnostic Service 14. No additional action is needed.	- Check wiring, connectors, fuse - Check battery voltage - check CCIOM failure - Replace APM	Backup Power Supply
APM	U114C68	Number of failed DTCs reached its maximal number (29 DTCs)	Event Information	No functional effect.	Too many errors in the system	Diagnostic Service 14. No additional action is needed.	- Check the workshop action for other active DTCs	DTC Notification Max Limit Reached
APM	C102668	estimated leakage > Level 1 leakage threshold parameter	Event Information	No functional effect	1. Pneumatic channel connection failure 2. Unexpected air flow out from a mechanical component	Diagnostic Service 14. No additional action is needed.	- Check air circuits for possible leakage. - Check the piping on the vehicle. And also possible leakage on trailer - when leakage repaired , Reset leakage monitoring by Routine service R1AEK	Vehicle Air Pressure System
APM	C10267A	estimated leakage > Level 2 leakge threshold parameter	Fluid Leak or Seal Failure	No functional effect on the air management, fuel consumption of the vehicle might be higher	1 Vehicle usage not corresponds to the designed purpose 2. Pneumatic channel connection failure 3. Unexpected air flow out from a mechanical component	Diagnostic Service 14. No additional action is needed.	- Check air circuits for possible leakage. - Check the piping on the vehicle. And also possible leakage on trailer - when leakage repaired , Reset leakage monitoring by Routine service R1AEK.	Vehicle Air Pressure System

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	U1FFF68	The ECU is in active Factory Mode. P1C54 set to TRUE.	Event Information	Auto-release, auto-park and park-brake alert are unavailable	Factory Mode not disabled.	Diagnostic Service 14. No additional action is needed.	Set the Factory Mode flag to false in NVM, then execute ECU reset.	Factory Mode Active
APM	C102B01	Failure detected in the electrical circuit of the heater or the temperature sensor	General Electrical Failure	Heater functionality is not available or restricted	- damage of the ECU internal wiring - damage of the heater - damage of the ECU	Diagnostic Service 14. No additional action is needed.	Replace APM	Heating Resistor
APM	C102B55	Heater functionality is enabled but heater is not present	Not Configured	Heater functionality is not available or restricted	- heater is not installed, - invalid configuration of EEPROM, - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- download dataset - download MSW - Replace ECU	Heating Resistor
APM	C102B68	Heater is inhibited in case of secondary braking.	Event Information	Heater functionality is not available or restricted	Secondary braking was activated.	Diagnostic Service 14. No additional action is needed.	No action	Heating Resistor
APM	C100401	Failure detected in the electrical circuit of the state manager magnet valve open circuit, short circuit, short circuit between 2 magnet valves, external excitation.	General Electrical Failure	Electronic Parking Brake functions are not avaiable.	- damage of the Magnet Valve wiring - damage of the Magnet Valve - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Replace APM	Electronic Parking Brake chassis state manager valve solenoid
APM	C102472	Parking brake cannot be locked in park position	Actuator Stuck Open	Last safe state restored depending on safety conditons. After 3 unsuccessful actuation attempt parking brake is no more operational	"1. TCM or its control line is: - damaged or - installed on wrong port or - removed or - replaced or - other trailer control port is not plugged 2. Sensor failure. 3. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- TCM: disconnect the TCM pipe on APM port; close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking. - Additional action: Check compressor pipe, area below the cartridge and scilencer for freezing. - Relay valve: exchange relay valve if other actions are not effective.	Pressure Control Valve
APM	C102473	Parking brake cannot be locked in released position OR initialisation state cannot be determined	Actuator Stuck Closed	Last safe state restored depending on safety conditons. After 3 unsuccessful release attempt parking brake is no more operational	1. Trailer control module or its control line is: - damaged or - installed on wrong port or - removed or - replaced or - other trailer control port is not plugged 2. Sensor failure 3. APM internal valve failure"	Diagnostic Service 14. No additional action is needed.	- TCM: disconnect the TCM pipe on APM port ; close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking - Relay valve: exchange relay valve if other actions are not effective	Pressure Control Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C102474	Parking brake initialisation state cannot be determined	Actuator Slipping	Parking brake cannot be released as long as parking brake is not requested to locked actuation position whatever the (P) lamp state is. Blinking HCU LED	"1. Trailer contriol module or its control line is: - damaged or - installed on wrong port or - removed or - replaced or - other trailer control port is not plugged 2. Sensor failure 3. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- TCM: disconnect the TCM pipe on APM port ; close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking. - Relay valve: exchange relay valve if other actions are not effective	Pressure Control Valve
APM	C100801	"Failure is set if the sensor shorted to ground. Failure is set if the sensor shorted to ubat."	General Electrical Failure	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	"- damage of the sensor wiring - damage of the sensor or - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Replace APM	Parking Brake Control Pressure Sensor
APM	C100864	Parking brake control pressure deviates from the reference pressure more than the threshold, threshold is based on multicriteria formula.	Signal Plausibility Failure	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	1. Sensor failure 2. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- TCM check : disconnect the Trailer control module pipe on APM port - close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust check: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking - APM change - If the other actions are not effective exchange APM	Parking Brake Control Pressure Sensor
APM	C100001	Failure detected in the electrical circuit of the load magnet valve	General Electrical Failure	Electronic Parking Brake functions are not avaiable.	"- damage of the wiring - damage of the Magnet Valve - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Replace APM	Electronic Parking Brake chassis load valve solenoid
APM	C100301	Failure detected in the electrical circuit of the exhasut magnet valvee is set	General Electrical Failure	Electronic Parking Brake functions are not avaiable.	"- damage of the wiring - damage of the Magnet Valve - damage of the ECU"	Diagnostic Service 14. No additional action is needed.	- Replace APM	Electronic Parking Brake chassis exhaust valve solenoid
APM	C102A64	Parking brake chamber pressure deviates the reference pressure more than the threshold, this threshold is based on multicriteria formula.	Signal Plausibility Failure	No functional effect, redundant information is lost. Backup Level 1.	1. Sensor failure 2. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- TCM: disconnect the TCM pipe on APM port ; close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking. -Relay valve: exchange relay valve if other actions are not effective.	Parking Brake Chamber Pressure Sensor
APM	C101612	Safety relay of Electronic Parking brake short circuit to Ubat or short between relays or BuC can not switch off relays	Circuit Short To Battery	No effect on the driver. Safety feature of the relay is not available. Backup level 1.	- internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Replace APM	Load, exhaust and state manager valve high side solenoid
APM	C101613	Safety relay of Electronic Parking Brake open circuit to supply voltage.	Circuit Open	Electronic Parking Brake functions are not avaiable.	- internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Wait 10mn ignition off park applied, to cool the ecu. - afterward if still active replace APM.	Load, exhaust and state manager valve high side solenoid

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C101862	Trailer control pressure provided by parking brake deviates from the reference pressure minus threshold, this threshold is based on multicriteria formula.	Signal Compare Failure	Trailer test is not successful and no more available, Autorelease, XPR and autopark functions are no more available.	1. Trailer control module or its pipe connection is damaged or incorrect trailer control port is not plugged 2. Trailer control module supply line is damaged. 3. Spring brake lines are damaged. 4. Parking brake relay valve failure 5. Sensor failure 6. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- TCM : disconnect the Trailer Control Module pipe on APM port - close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust : check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking - Relay valve: exchange relay valve if other actions are not effective	Trailer Control Circuit
APM	C101D62	Parking brake sensors are not correlating, faulty sensor cannot be identified	Signal Compare Failure	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	1. Sensor failure 2. APM internal valve failure 3. Wrong supply voltage level	Diagnostic Service 14. No additional action is needed	- APM exhausht in bottom: check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking - Relay valve: exchange relay valve if other actions are not effective.	Parking Brake Circuit
APM	C101D7A	Parking brake control and chamber pressure higher than the reference pressure minus threshold, this threshold is based on multicriteria formula.	Fluid Leak or Seal Failure	No functional effect. Risk of unintended release due to high pressure in parking brake	1. Trailer control module or its pipe connection is damaged or incorrect trailer control port is not plugged 2. Trailer control module supply line is damaged. 3. Spring brake lines are damaged. 4. Parking brake relay valve failure 5. Sensor failure 6. APM internal valve failure	Diagnostic Service 14. No additional action is needed	- TCM: disconnect the TCM pipe on APM port - close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking. - Relay valve: exchange relay valve if other actions are not effective.	Parking Brake Circuit
APM	C101D7B	Parking brake control and chamber pressure lower than the reference pressure plus threshold, this threshold is based on multicriteria formula.	Low Fluid Level	No functional effect. Risk of unintended braking due to low pressure in parking brake	1. Trailer control module or its pipe connection is damaged or incorrect trailer control port is not plugged 2. Trailer control module supply line is damaged. 3. Spring brake lines are damaged. 4. Parking brake relay valve failure 5. Sensor failure 6. APM internal valve failure	Diagnostic Service 14. No additional action is needed	-'TCM: disconnect the TCM pipe on APM port; close it's port with plug. check park brake application and release 5 times. if it's working failure free, replace TCM. - Exhaust: check Park brake relay valve silencers for blocking and dirt, check the plastic cover (plastic part can be removed for the check and put back after) of the secondary exhausts on the bottom against dirt and blocking - Additional action: Check compressor pipe, area below the cartridge and siclencer for freezing. - Relay valve: exchange relay valve if other actions are not effective.	Parking Brake Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C101F11	LIN signal received : HCU sensor shorted to ground	Circuit Short To Ground	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU sensor shorted to ground	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit
APM	C101F12	LIN signal received : HCU sensor shorted to supply	Circuit Short To Battery	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU sensor shorted to power supply	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit
APM	C101F1C	LIN signal received : HCU supply voltage failure	Circuit Voltage Out of Range	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU power supply out of range	Diagnostic Service 14. No additional action is needed.	- check wiring between APM and HCU - replace HCU - replace APM	Hand Control Unit
APM	C101F28	the HCU sensor angle failure set error code was received.	Signal Bias Level Out of Range / Zero Adjustment Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU sensor angle out of range failure	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C101F45	the HCU RAM/ROM failure set error code was received	Program Memory Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	RAM/ROM failure in HCU	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit
APM	C101F49	LIN signal received : HCU sensor angle jump check failure OR amplifier failure OR amplitude ratio is not correct	Internal Electronic Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU sensor amplifier failure	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit
APM	C101F54	LIN signal received : HCU sensor calibration failure	Missing Calibration	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	Sensor calibration failure in the HCU	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU (electronic Park brake lever)	Hand Control Unit
APM	C101F64	LIN signal received : HCU lever stays more than 2 minutes in a position different than neutral or release position	Signal Plausibility Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test.	Lever stays is any position more than 2 minutes, except Neutral and Release position.	Diagnostic Service 14. No additional action is needed.	- Check whether HCU movement is hindered or blocked - Replace HCU	Hand Control Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C101F86	Hand control unit CRC or signal invalid or UARTcommunication failure or HCU alive counter freeze or HCU area signal is not available	Signal Invalid	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	Electric disturbance on the LIN line - SCI (UART) controller is damaged - wiring damaged - HCU damaged	Diagnostic Service 14. No additional action is needed.	- Check the wiring between HCU and APM - Check whether HCU movement is hindered or blocked - Check HCU against foreign particule below lever - Check LIN communication (use another HCU to Check) - change HCU - change APM	Hand Control Unit
APM	C101F87	LIN communication failure detected by HCU or APM	Missing Message	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU is not connected or HCU cables are broken or electric disturbance	Diagnostic Service 14. No additional action is needed.	- Check the wiring between HCU and APM - Check LIN communication (use another HCU to Check) - change HCU - change APM	Hand Control Unit
APM	U114708	Backlight signal is undefined or error for HCU	Bus Signal / Message Failures	HCU backlight is not controlled	Backlight signal is undefined or error failure in HCU	Diagnostic Service 14. No additional action is needed.	- check the dimming function in VMCU - Check if backlight (white light) on HCU is working well.- (use another HCU to Check) - If not change the HCU (electronic Park brake lever)	Hand Control Unit Backlight Data Link
APM	U114808	Illumination signal is undefined or error for HCU	Bus Signal / Message Failures	PB indication on HCU lost	Illumination level signal is undefined or error failure in HCU	Diagnostic Service 14. No additional action is needed.	- check the dimming function in VMCU - with park applied, Check if red LED on HCU is working well. (use another HCU to Check) - If not change the HCU (electronic Park brake lever)	Hand Control Unit Indicator LInk
APM	U114A94	Failure intercontroller communication or unwanted parking brake actuation	Unexpected Operation	Pneumatic backup air management control (idle cut-off, no regeneration) No EPB functions Pneumatic backup release and apply of PB is possible No CAN and LIN communication. No ParkBrakeActuationStatus.	- Unexpected parking brake actuation. - AuC and BuC communication lost	Diagnostic Service 14. No additional action is needed.	- dowload Main software (MSW) - Replace APM	Secondary Control Unit
APM	C102316	HCU power supply (pin 18) voltage level below threshold 8V	Circuit Voltage Below Threshold	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are available. No secondary brake function, no trailer test. No indication on the HCU.	- wrong supply voltage level - internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Check the wiring - Check LIN communication (use another HCU to Check) - change HCU - change APM	Hand Control Unit Power Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C102317	HCU power supply (pin18) voltage level above threshold 16V - measured inside the APM. External supply of HCU does not activate the DTC.	Circuit Voltage Above Threshold	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are available. No secondary brake function, no trailer test. No indication on the HCU.	- APM internal wrong supply voltage to HCU - internal electric failure - damage of the ECU	Diagnostic Service 14. No additional action is needed.	- Check the wiring - Check LIN communication (use another HCU to Check) - change HCU - change APM	Hand Control Unit Power Supply Circuit
APM	U114B01	short to GND or Power supply or floating line	General Electrical Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are available. No secondary brake function, no trailer test. No indication on the HCU.	HCU wiring error, ECU, HCU HW damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring - Check LIN communication (use another HCU to Check) - change HCU - change APM	LIN12 Communication Net
APM	U114B49	CSWC internal failure	Internal Electronic Failure	Manual actuation, secondary braking and trailer test is not possible (due to hand control unit availability as it is disabled if any failure is indicated by the HCU).Backup autorelease and autopark are avaiable. No secondary brake function, no trailer test. No indication on the HCU.	HCU damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring - Check LIN communication (use another HCU to Check) - change HCU - change APM	LIN12 Communication Net
APM	C102503	ADR signal is floating	FM (Frequency Modulated) / PWM (Pulse Width Modulated) Failures	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	ADR circuit/wiring error, ECU HW damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring between ADR main switch and APM - Check ADR signal on Pin 9 of APM connector/compare with P1CAJ "Air Production Modulator Input Main switch signal status" - Compare the status in P1CAJ and what measured on Pin 9: if it is the same, Check/Replace ADR switch. - if satus in P1CAJ is different than what measured on Pin 9 : replace APM.	ADR Input Signal Circuit
APM	C102511	ADR signal is constantly low on GND voltage level	Circuit Short To Ground	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	ADR circuit/wiring error, ECU HW damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring between ADR main switch and APM - Check ADR signal on Pin 9 of APM connector/compare with P1CAJ "Air Production Modulator Input Main switch signal status" - Compare the status in P1CAJ and what measured on Pin 9: if it is the same, Check/Replace ADR switch. - if satus in P1CAJ is different than what measured on Pin 9 : replace APM.	ADR Input Signal Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
APM	C102512	ADR signal is constantly high on battery voltage level	Circuit Short To Battery	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available.	ADR circuit/wiring error, ECU HW damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring between ADR main switch and APM - Check ADR signal on Pin 9 of APM connector/compare with P1CAJ "Air Production Modulator Input Main switch signal status" - Compare the status in P1CAJ and what measured on Pin 9: if it is the same, Check/Replace ADR switch. - if satus in P1CAJ is different than what measured on Pin 9 : replace APM.	ADR Input Signal Circuit
APM	C102527	ADR signal frequency deviates from pre-defined frequency values	Signal Rate of Change Above Threshold	Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed. Trailer test not available.	ADR circuit/wiring error, ECU HW damage	Diagnostic Service 14. No additional action is needed.	- Check the wiring between ADR main switch and APM - Check ADR signal on Pin 9 of APM connector/compare with P1CAJ "Air Production Modulator Input Main switch signal status" - Compare the status in P1CAJ and what measured on Pin 9: if it is the same, Check/Replace ADR switch. - if status in P1CAJ is different than what measured on Pin 9 : replace APM.	ADR Input Signal Circuit
APM	C102768	At least one of the incoming CAN signals needed for automatic parking brake release or XPR function is missing or invalid.	Event Information	Autorelease not available XPR not available. Backup level 2	1. EBS failure 2. TECU failure 3. accelerator pedal, door switches, gear lever failure 4. BBM failure 5. DACU failure 4. CAN commnunication failure	Diagnostic Service 14. No additional action is needed.	- Resolve failures reported by other ECUs (EBS, TECU,accelerator pedal, door switches,gear lever, BBM, DACU) - Check the CAN communication and wiring	Parking Brake Function
TACHO	U001081	Invalid data received on the CAN bus.	Invalid Serial Data Received	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "13".	"1. CAN messages received by the DTCO are not formatted or addressed correctly. 2. Timinig errors within CAN protocol."	Diagnostic Service 14. No additional action is needed.	Check the CAN bus	CAN Communication Backbone 1 Net
TACHO	U001087	No response from a instrument connected to the DTCO.	Missing Message	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "15".	"1. No instrument cluster connected to the DTCO. 2. The connected instrument cluster does not respond to the DTCO."	Diagnostic Service 14. No additional action is needed.	Check the CAN bus	CAN Communication Backbone 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	U001088	Bus off state	Bus off	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "14".	"1. there are too many errorneous CAN messages send on the CAN bus. 2. There are too many collisions on the CAN bus."	Diagnostic Service 14. No additional action is needed.	Check the CAN bus	CAN Communication Backbone 1 Net
TACHO	U30001C	power interruption on the DTCO.	Circuit Voltage Out of Range	The DTCO continues to work normally (after the voltage is back). A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "power interruption". The memory code is "31".	"1. The power supply to the DTCO was below the specified range. 2. The power supply to the DTCO was above the specified range."	Diagnostic Service 14. No additional action is needed.	Check the voltage supply of the DTCO	Electronic control unit
TACHO	U300041	The ECU SW is corrupted.	General Checksum Failure	The DTCO can stop working properly. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "4".	The program memory is partly changed or deleted.	Diagnostic Service 14. No additional action is needed.	Replace the DTCO	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	U300044	Error in data memory detected.	Data Memory Failure	The DTCO can stop working properly. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "25".	"1. Writing to flash memory failed 2. CRC error in the recorded data."	Diagnostic Service 14. No additional action is needed.	If the error message appears repetitively, replace the DTCO	Electronic control unit
TACHO	U300046	Checksum check of calibration parameters failed.	Calibration / Parameter Memory Failure	The DTCO continues to work normally, but the data sent and stored can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "calibration fault". The memory code is "9".	1. Calibration parameters have been altered in the ECU memory.	Diagnostic Service 14. No additional action is needed.	If the error message appears despite repetitive acknowledgement, replace the DTCO	Electronic control unit
TACHO	U300049	An ECU internal component detected an error.	Internal Electronic Failure	The DTCO continues to work normally (except the faulty component). A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is: 1. "internal fault" 2. "time fault" 3. "internal fault" The memory code is: 1. "5" 2. "6" 3. "8"	1. The ECU internal communication with the RTC module failed. 2. The RTC does not deliver a time or the time is not plausible. 3. A malfunction of the ECU keyboard was detected.	Diagnostic Service 14. No additional action is needed.	If the error message appears despite repetitive acknowledgement, replace the DTCO	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	U300094	There was an unexpected situation that caused a reset of the Control Module	Unexpected Operation	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is: 1., 2., 3. "internal fault" 4. "security breach". The memory code is: 1. "1" 2. "2" 3. "3" 4. "15"	1. A runtime error occurred in the ECU SW leading to a reset of the ECU to solve the issue. 2. Power supply of the ECU is not delivering a constant power signal. 3. The ECU SW encountered an unexpected situation leading to a reset to solve the issue. 4. Security breach detected by the Security Controller.	Diagnostic Service 14. No additional action is needed.	If the error message appears despite repetitive acknowledgement, replace the DTCO	Electronic control unit
TACHO	P050004	The motion sensor connected to the DTCO is reporting an internal fault.	System Internal Failures	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "sensorfault". The memory code is "16".	"1. Motion sensor reports error on controller RAM 2. Motion sensor reports error on controller instruction 3. Motion sensor reports error on communication"	Diagnostic Service 14. No additional action is needed.	Replace Motion sensor	Wheel-Based Vehicle Speed
TACHO	P050006	Motion sensor authentication error.	Algorithm Based Failures	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "18" or "23".	"1. Motion sensor reports error on authentication 2. The motion sensor connected to the DTCO is not the one paired with the DTCO. 3. The motion sensor pairing failed"	Diagnostic Service 14. No additional action is needed.	Replace motion sensor	Wheel-Based Vehicle Speed

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	P050016	power interruption of the motion sensor.	Circuit Voltage Below Threshold	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "power interruption". The memory code is "19".	1. The motion sensor was cut off from the power supply.	Diagnostic Service 14. No additional action is needed.	Check power supply of DTCO and motion sensor	Wheel-Based Vehicle Speed
TACHO	P050042	The motion sensor connected to the DTCO is reporting a memory fault.	General Memory Failure	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "17".	Motion sensor reports error on non-volatile memory	Diagnostic Service 14. No additional action is needed.	Replace motion sensor	Wheel-Based Vehicle Speed
TACHO	P05004A	Wrong motion sensor connected.	Incorrect Component Installed	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "24".	1. The motion sensor connected has a different serial number than the motion sensor paired with the DTCO.	Diagnostic Service 14. No additional action is needed.	Check motion sensor and replace it if defective Check DTCO and replace it if defective	Wheel-Based Vehicle Speed

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	P050062	There is a difference (> 20 pulses) between the number of pulses delivered by the real time signal of the motion sensor and the number of pulses given in the motion sensor data signal or from independent vehicle motion signal (IMS).	Signal Compare Failure	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is: 1. "sensor fault" 2. "motion conflict" The memory code is: 1. "22" 2. "80".	1. The line for the real time signal from the motion sensor to the DTCO is not connected correctly. 2. The plausibility check of the speed information via motion sensor pulses and from addtional external vehicle motion information (IMS) fails.	Diagnostic Service 14. No additional action is needed.	Check motion sensor and replace it if defective Check DTCO and replace it if defective	Wheel-Based Vehicle Speed
TACHO	P050064	The real time signal from the motion sensor is missing.	Signal Plausibility Failure	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "sensor fault". The memory code is "21".	The motion sensor power supply and/or real time signal line to the DTCO is not connected or broken.	Diagnostic Service 14. No additional action is needed.	Check motion sensor and replace it if defective Check DTCO and replace it if defective	Wheel-Based Vehicle Speed
TACHO	P050068	The independent vehicle motion signal is missing.	Event Information	The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "IMS fault". The memory code is "81".	The independent vehicle motion signal (IMS) is missing / not valid.	Diagnostic Service 14. No additional action is needed.	Check the second speed source	Wheel-Based Vehicle Speed

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	P050087	The communication with the motion sensor failed.	Missing Message	The DTCO continues to work normally, but the data can be erroneous. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "sensor fault". The memory code is "20".	1. The motion sensor does not respond.	Diagnostic Service 14. No additional action is needed.	Check motion sensor and replace it if defective Check DTCO and replace it if defective	Wheel-Based Vehicle Speed
TACHO	B130100	Printing attempt without ignition, during driving, at excess temperature, at over- or undervoltage (>32V or <20V), at busy printer interface.	No Sub Type Information	The DTCO continues to work normally, except the printer. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "printout not possible". The memory code is "36".	"1. A printout was requested while the ignition was off on a device dedicated to carriage of dangerous goods. 2. Ignition was switched off while a printout was ongoing on a device dedicated to carriage of dangerous goods. 3. A printout was started while the vehicle was moving. 4. A printout was started or ongoing while the power supply of the DTCO was outside the specified range. 5. A printout was started while the temperature of the printer head was too high."	Diagnostic Service 14. No additional action is needed.	If the error message continues to appear despite these measures, replace the DTCO	Vehicle Unit Printout Not Possible
TACHO	B122C00	Card inserted during an unallowed system status.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "71".	"1. A driver or company card was inserted while the time of the DTCO was not reliable. 2. A chip card was inserted when the DTCO program code was corrupted. 3. A motion sensor pairing was triggered using a invalid workshop card."	Diagnostic Service 14. No additional action is needed.	Check UTC time on the DTCO. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 2 Insertion Not Allowed

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B121C00	Card inserted during an unallowed system status.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "53".	"1. A driver or company card was inserted while the time of the DTCO was not reliable. 2. A chip card was inserted when the DTCO program code was corrupted. 3. A motion sensor pairing was triggered using a invalid workshop card."	Diagnostic Service 14. No additional action is needed.	Check UTC time on the DTCO. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 1 Insertion Not Allowed
TACHO	B122B00	The chip card is locked.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card locked". The memory code is "70".	1. The chip card that was inserted was locked due to too many wrong PIN entries.	Diagnostic Service 14. No additional action is needed.	Check the workshop card	Chip Card Slot 2 Card Locked
TACHO	B121B00	The chip card is locked.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card locked". The memory code is "52".	1. The chip card that was inserted was locked due to too many wrong PIN entries.	Diagnostic Service 14. No additional action is needed.	Check the workshop card	Chip Card Slot 1 Card Locked

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B122A93	The card inserted does not react to any commands. the inserted tachograph card can not be read or written to.	No Operation	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card error". The memory code is "68".	"1. The chip card was inserted with the contact pads facing in the wrong direction. 2. The chip card contact pads are dirty. 3. The chip card is broken. 4. The card inserted is not a chip card."	Diagnostic Service 14. No additional action is needed.	Check card	Chip Card Slot 2 Card Insertion
TACHO	B121A93	The card inserted does not react to any commands. the inserted tachograph card can not be read or written to	No Operation	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card error". The memory code is "50".	"1. The chip card was inserted with the contact pads facing in the wrong direction. 2. The chip card contact pads are dirty. 3. The chip card is broken. 4. The card inserted is not a chip card."	Diagnostic Service 14. No additional action is needed.	Check card	Chip Card Slot 1 Card Insertion
TACHO	B13004B	The printout was paused due to too high temperature of the printer head. It is +95degreesC to stop, +85degreesC to resume the printout.	Over Temperature	The DTCO continues to work normally, except the printer. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "printout delayed". The memory code is "37".	The printer head got too hot from continuous printing.	Diagnostic Service 14. No additional action is needed.	Wait several minutes and repeat the process.	Vehicle Unit Printer

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B130200	There is no paper in the printer.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "no paper". The memory code is "39".	"1. A printout was requested while there is no paper in the printer. 2. The printer ran out of paper during a printout."	Diagnostic Service 14. No additional action is needed.	insert a new paper roll	Vehicle Unit Printer no Paper
TACHO	B121071	The chip card mechanics for card slot 1 are blocked.	Actuator Stuck	The DTCO continues to work normally, except the mechanical part. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "40".	1. Chip card movement is blocked	Diagnostic Service 14. No additional action is needed.	If the error message appears repetitively, replace the DTCO.	Chip Card Slot 1
TACHO	B110129	The voltage level at the B7 output does not match the expected value: - Low voltage <1,5V - High voltage >5,5V.	Signal invalid	The DTCO continues to work normally, except B7 pin. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "10".	"1. There is a short circuit between the line connected to B7 and any other line. 2. Signals are sent to the B7 output from a device connected to it."	Diagnostic Service 14. No additional action is needed.	Check the pin B7. If the error message appears repetitively, replace the DTCO.	Vehicle Speed Impulse

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B150102	A download communication was initiated but not successful.	General Signal Failure	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "download fault". The memory code is "12".	1. The download tool is not properly connected to the DTCO.	Diagnostic Service 14. No additional action is needed.	Check download interface on the DTCO. If the error message appears repetitively, replace the DTCO.	Initiate Download
TACHO	B100100	The DTCO housing was opened.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "26".	1. The DTCO case was opened.	Diagnostic Service 14. No additional action is needed.	If the error message appears repetitively, replace the DTCO.	Vehicle Unit Case Open
TACHO	B120100	The DTCO detected driving but there was no driver card inserted in slot 1.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "driving without card". The memory code is "28".	"1. Driving was detected, but there was no driver card in the chip card slot 1. 2. Driving was detected, but the chip card in slot 1 is not a valid driver card 3. Driving was detected but the currently inserted combination of chip cards does not allow driving."	Diagnostic Service 14. No additional action is needed.	Insert driver card	Driving without valid driver card

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B120200	Two cards which are not allowed to be used together are inserted in the DTCO.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card conflict". The memory code is "29".	1. There are two chip cards inserted in the DTCOs chip card slot which are not allowed to be inserted at the same time.	Diagnostic Service 14. No additional action is needed.	Check the combination of the inserted cards	Vehicle Unit Invalid Card Combination
TACHO	B122071	The chip card mechanics for card slot 2 are blocked.	Actuator Stuck	The DTCO continues to work normally, except the mechanical part. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "internal fault". The memory code is "58".	1. Chip card movement is blocked	Diagnostic Service 14. No additional action is needed.	If the error message appears repetitively, replace the DTCO.	Chip Card Slot 2
TACHO	B121100	Card communication failed in slot 1. A communication fault has appeared during reading or writting of card data.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card fault". The memory code is "41".	1. Dirt on chip card contacts. 2. Chip card was disconnected during communication. 3. Chip card is operating outside of the specified temperature range. 4. The chip card is broken. 5. Unknown card response / unknown card structure / unknown card data.	Diagnostic Service 14. No additional action is needed.	Check the card. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 1 Card Communication Error

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B122100	Card communication failed in slot 2.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card fault". The memory code is "59".	1. Dirt on chip card contacts. 2. Chip card was disconnected during communication. 3. Chip card is operating outside of the specified temperature range. 4. The chip card is broken. 5. Unknown card response / unknown card structure / unknown card data.	Diagnostic Service 14. No additional action is needed.	Check the card. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 2 Card Communication Error
TACHO	B121200	The chip card data could not be authenticated in slot 1.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "42".	1. The smart card data has been manipulated.	Diagnostic Service 14. No additional action is needed.	check the card	Chip Card Slot 1 Card Data Authenticity Error
TACHO	B122200	The chip card data could not be authenticated in slot 2.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "42".	1. The chip card data has been manipulated.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card Slot 2 Card Data Authenticity Error

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B121300	A different card was detected or the card is missing or the card lock is open in slot 1.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "43".	"1. The chip card was exchanged in an improper way. 2. The chip card was removed in an improper way. 3. The chip card lock is opened while a card is inserted."	Diagnostic Service 14. No additional action is needed.	Check the card. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 1 Card Missing or Changed
TACHO	B122300	A different card was detected or the card is missing or the card lock is open in slot 2.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "61".	"1. The chip card was exchanged in an improper way. 2. The chip card was removed in an improper way. 3. The chip card lock is opened while a card is inserted."	Diagnostic Service 14. No additional action is needed.	Check the card. If the error message appears repetitively, replace the DTCO.	Chip Card Slot 2 Card Missing or Changed
TACHO	B121400	The chip card was not properly ejected during its previous usage.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card not closed". The memory code is "44".	1. The chip card in slot 1 was not correctly closed during its last usage.	Diagnostic Service 14. No additional action is needed.	Check card and check the previous DTCO. The source of the error is not located in the current DTCO .	Chip Card Slot 1 Card not Closed in The Last Vehicle

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B122400	The chip card was not properly ejected during its previous usage.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card not closed". The memory code is "62".	1. The chip card in slot 2 was not correctly closed during its last usage.	Diagnostic Service 14. No additional action is needed.	Check card and check the previous DTCO. The source of the error is not located in the current DTCO .	Chip Card Slot 2 Card Not Closed in The Last Vehicle
TACHO	B121500	Negative time difference to last vehicle.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "time overlap". The memory code is "45".	1. The VehicleLastUse time - which contains the time of the previous ejecting of the card in slot 1 before the current usage - is more than 1 minute ahead of the current time in the current DTCO.	Diagnostic Service 14. No additional action is needed.	Check UTC time in the current and previous DTCO.	Chip Card slot 1 Negative Time Difference
TACHO	B122500	Negative time difference to last vehicle.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "time overlap". The memory code is "63".	1. The VehicleLastUse time - which contains the time of the previous ejecting of the card in slot 2 before the current usage - is more than 1 minute ahead of the current time in the current DTCO.	Diagnostic Service 14. No additional action is needed.	Check UTC time in the current and previous DTCO.	Chip Card Slot 2 Negative Time Difference

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B121600	A chip card was inserted while driving.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "insertion while driving". The memory code is "46".	1. A chip card was inserted in the chip card slot 1 of the DTCO while the DTCO was detecting driving.	Diagnostic Service 14. No additional action is needed.	No action.	Chip Card Slot 1 Card Insertion While Driving
TACHO	B122600	A chip card was inserted while driving.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "insertion while driving". The memory code is "64".	1. A chip card was inserted in the chip card slot 2 of the DTCO while the DTCO was detecting driving.	Diagnostic Service 14. No additional action is needed.	No action.	Chip Card Slot 2 Card Insertion While Driving
TACHO	B121768	Chip card could not be authenticated.	Event Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "47".	1. The chip card certificate is not valid in slot 1.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card slot 1 Authentication

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B122768	Chip card could not be authenticated.	Event Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "security breach". The memory code is "65".	1. The chip card certificate is not valid in slot 2.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card Slot 2 Authentication
TACHO	B121800	The chip card is not valid.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card not valid". The memory code is "48".	1. The validity of the chip card in slot 1 expired. 2. The chip card in slot 1 is not yet valid.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card Slot 1 Invalid Date
TACHO	B122800	The chip card is not valid.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "card not valid". The memory code is "66".	1. The validity of the chip card in slot 2 expired. 2. The chip card in slot 2 is not yet valid.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card Slot 2 Invalid Date

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B121900	The activity recording on the chip card contains a consistency error.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "recording inconsistent". The memory code is "49".	1. The activity recording on the chip card in slot 1 contains an error that does not allow adding new data without corrupting the whole file.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card slot 1 Activity Recording Inconsistent
TACHO	B122900	The activity recording on the chip card contains a consistency error.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "recording inconsistent". The memory code is "67".	1. The activity recording on the chip card in slot 2 contains an error that does not allow adding new data without corrupting the whole file.	Diagnostic Service 14. No additional action is needed.	Check the card	Chip Card Slot 2 Activity Recording Inconsistent

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TACHO	B120300	A chip card ejection was requested but can not be handled by the system at this time.	No Sub Type Information	The DTCO continues to work normally. A warning message appears on the DTCO display. The backlighting of the display is additionally flashing for about 30 seconds or until the warning message will be acknowledged. The warning message contains one error text and one memory code: The error text is "ejection not possible". The memory code is "34".	"1. Ejecting of a chip card was requested while the vehicle is driving. 2. Ejecting of a chip card was requested while a card download for that card is currently ongoing. 3. Ejecting of a chip card was requested while. 4. Ejecting of a chip card was requested while ignition was off on an ADR device. 5. Ejecting of a chip card was requested while the DTCO was in an exeptional system mode. 6. Ejecting of a chip card was requested within the same minute in which the card was inserted. 7. Ejecting of the workshop card was requested in the same minute as the reset of the DTCO during activation took place. 8. Ejecting of a chip card	Diagnostic Service 14. No additional action is needed.	Wait several minutes and repeat the process. If the error message continues to appear despite these measures, replace the DTCO	Card request during driving, download or switched-off ignition
FLS	U200288	Bussoff detection	Bus off	No communication on CAN General reduced system functionality	Short circuit to ground/battery Broken cables	Diagnostic Service 14. No additional action is needed.	Check CAN wires Battery reset	CAN Communication Active Safety Proprietary 2 Net
FLS	U300041	Invalid SW or data	General Checksum Failure	ECU not alive	Faulty or wrong data downloaded	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with correct SW 2. Replace ECU	Electronic control unit
FLS	U300044	RAM check failed	Data Memory Failure	ECU is set to ERROR; System is restarted; After 3 restarts without success the system is kept in permanent Error.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
FLS	U300046	"EEPROM record CRC error EEPROM access failed"	Calibration / Parameter Memory Failure	FLS is set to ERROR.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with correct SW 2. Perform new calibration	Electronic control unit
FLS	U300049	HW malfunction detected inside the ECU	Internal Electronic Failure	FLS is set to ERROR; System is restarted; After 3 restarts without success the system is kept in permanent Error.	ECU internal damaged	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
FLS	U30004B	ECU temperature detected to be above 105 degrees	Over Temperature	Over temperature will generate out of range/invalid values; FLS is set to ERROR.	Environment temperature above normal operating range ECU damaged	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Electronic control unit
FLS	U300054	Calibration has not been performed	Missing Calibration	Un-calibrated sensor will generate invalid data; FLS is set to ERROR.	A sensor calibration has not been performed	Diagnostic Service 14. No additional action is needed.	Perform Sensor Calibration	Electronic control unit
FLS	U300094	Unexpected operation failure(eg.Watchdog)	Unexpected Operation	FLS is set to error	Unexpected reset due to internal ECU failure	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Check that input CAN signals is correct 3. Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
FLS	U300097	Sensor blocked	Component or System Operation Obstructed or Blocked	FLS is set to ERROR.	Sensor is obstructed Radar or secondary surface is covered with ex snow or dirt.	Diagnostic Service 14. No additional action is needed.	Check that the sensor is not blocked by eg dirt, snow, obstacle	Electronic control unit
FLS	U300316	Power supply ECU Low voltage <= 10 V	Circuit Voltage Below Threshold	FLS is set to ERROR.	1. Drained battery 2. Power supply system failure	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
FLS	U300317	Power supply ECU High voltage > 32 V	Circuit Voltage Above Threshold	FLS is set to ERROR.	1. Power supply system failure 2. Overload alternator	Diagnostic Service 14. No additional action is needed.	Check power supply system Overload alternator	Battery Power
FLS	U300078	Sensor misaligned	Alignment or Adjustment Incorrect	FLS is set to ERROR.	"If the Forward Looking Sensor HW is misaligned then the ForwardLookingSensorObjectData shall be set to error or similar within 100ms from the time the misalignment was discovered. Sensor misaligned: e.g. Faulty installation / re-installation, Mechanical damages, FLS reports 'Sensor misalignment'. ""Alignment or adjustment incorrect"" (according with ISO15031-6)"	Diagnostic Service 14. No additional action is needed.	1. Look for external damage eg bracket, sensor, bumper 2. Recalibrate sensor. 3. Use Volvo orginal washers to adjust FLS assembly to be in valid range	Electronic control unit
FLS	U300093	"FLS azimuth motor failure, FLS elevation motor failure, FLS RF-module failure"	No Operation	FLS is set to ERROR; System is restarted; After 3 restarts without success the system is kept in permanent error.	Mechanical/electrical damage of the motor e.g. FLS azimuth motor failure, FLS elevation motor failure, FLS RF-module failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034038	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P054464	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or Exhaust Gas Temperature Sensor 1 below 100°C)))	Signal Plausibility Failure	Activation failure, regeneration	Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P03351F	(Intermittent crank speed signal)	Circuit Intermittent	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033538	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P203F00	(((Reductant level) below 2% for 15s))	No Sub Type Information	Engine derate	Reductant tank almost empty	Diagnostic Service 14. No additional action is needed.	Check: Reductant tank almost empty If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Low
EMS	P205A64	(Difference during Precrank between mean value of (Boost Temperature, EGR Temperature, Engine Coolant Temperature, Compressor Temperature) and Urea Tank Temperature) above 100°C	Signal Plausibility Failure	None	Malfunctioning DEF Tank Temperature Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning DEF Tank Temperature Sensor	Reductant Tank Temperature Sensor
EMS	P220012	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P22001C	(Internal Supply Voltage signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220064	(The Upstream NOx Difference (upstream NOx sensor measured NOx flow – modeled NOx flow) is compared to a low and high fault threshold map (NOx flow threshold as a function of modeled NOx flow). The difference must be below Low Limit or above High Limit repeatedly 3No Unit times. Evaluation time for each repeated evaluation is 4s)	Signal Plausibility Failure	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220093	(Internal status and error byte recieved from Upstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 1
EMS	P060741	(Internal Control Module Memory Check Sum Error)	General Checksum Failure	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Control Module Performance
EMS	P061511	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay
EMS	P061512	(Short circuit to battery is detected.)	Circuit Short To Battery	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20EE00	((Filtered NOx conversion) below 58% and (A filter is applied to the NOx conversion that may require the evaluation to be repeated a few times to set the fault. Time required depends on engine load and the NOx flow. With all enable conditions fulfilled it normally takes no more than a few minutes.))	No Sub Type Information	Engine derate	Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 45kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -30kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
EMS	P007064	(Difference during Precrank between mean value of (Engine Exhaust Temperature, HC Heat Temperature, DPF Temperature) and Ambient Air Temperature) above 100°C	Signal Plausibility Failure	None	Malfunctioning Ambient Air Temperature Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Ambient Air Temperature Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Ambient Air Temperature Sensor "A"
EMS	P008700	No information available.	No Sub Type Information	Engine cranks but does not start	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail/System Pressure - Too Low Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P009A00	((Difference between Sensed Boost Temperature and Modeled Boost Temperature) below -50°C or (Difference between Sensed Boost Temperature and Modeled Boost Temperature) above 50°C)	No Sub Type Information	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Air Intake Temperature - Correlation
EMS	P011001	(Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011011	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011013	Voltage on A:47 above 4.910889V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011064	(Difference during Precrank between Boost Temperature and mean value of(EGR Temperature, Compressor Temperature, Engine Coolant Temperature)) above 100°C	Signal Plausibility Failure	None	Malfunctioning Intake Manifold Air Temperature Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Manifold Air Temperature Sensor	Intake Air Temperature Sensor 1 Bank 1
EMS	P021700	se_CoolantTemp above (3.20000004768371 6No Unit + 107No unit + (Output from look-up tables with inputs se_AmbAirPres and EngineCoolantPressu re))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition
EMS	P047001	((Voltage on B:12 above 4.787598V and Voltage on B:12 below 4.852295V) or (Voltage on B:12 above 0.147705V and Voltage on B:12 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047012	Voltage on B:12 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047013	Voltage on B:12 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047064	((Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) below -45kPa or (Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) above 45kPa)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Sensor "A"
EMS	P048011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit
EMS	P048012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit
EMS	P051A01	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or (Voltage on B:28 above 0.147705V and Voltage on B:28 below 0.307617V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A12	Voltage on B:28 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A13	Voltage on B:28 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Crankcase Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052400	se_OilPres below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low
EMS	P203164	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 2 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
EMS	P222601	(Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222612	Voltage on N/A:ECU internal above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222613	Voltage on N/A:ECU internal below 0.117188V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222664	(se_AmbAirPres below 55kPa or se_AmbAirPres above 114kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Circuit
EMS	P226411	Voltage on B:8 below 0.211182V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P244A00	(Measured differential pressure as percentage of modeled differential pressure) below (55 %)	No Sub Type Information	None	DPF removed or severely damaged Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF removed or severely damaged Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too Low (Bank 1)
EMS	P244B00	(Measured differential pressure as percentage of modeled differential pressure) above (350 %)	No Sub Type Information	None	DPF melted DPF filled with soot Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF melted DPF filled with soot Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too High (Bank 1)
EMS	P245264	(((Failure event for diagnosis in AfterRun) and Particulate Filter Differential Pressure above 1.200000047683716 kPa))	Signal Plausibility Failure	None	Sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1
EMS	P250F00	((If calculated value for oil level) below -4.980500221252441 % and (Engine must be in prerunning during a specific time, defined by output from map with input se_OilTemp))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255601	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255611	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255612	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255613	(Voltage on B:23 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P256000	se_CoolantLvl below (100No unit - 75No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P207F9A	(((Estimated reductant concentration) below 18% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input se_OilTemp))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU
EMS	P229E12	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E64	(((The sensor shall respond to tip-in events (sudden power increase). Each time it does not respond correctly a timer is started. When this accumulated time has reached its time limit 60s a second phase is started with the purpose to confirm the suspected fault. This second phase is using the same method, but this time the reductant delivery is stopped in order to improve chances for the outlet NOx sensor to respond to the tip-in events. A second cumulative timer is started every time	Signal Plausibility Failure	None	Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor	NOx Sensor Bank 1 Sensor 2
EMS	P229E93	(Internal status and error byte recieved from Downstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 2
EMS	P009011	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine derate	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Pressure Regulator 1 Control
EMS	P009012	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine derate	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Pressure Regulator 1 Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P009013	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Engine derate	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Fuel Pressure Regulator 1 Control
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052001	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or (Voltage on B:11 above 0.147705V and Voltage on B:11 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052012	Voltage on B:11 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052013	Voltage on B:11 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052064	(se_OilPres above 100kPa or se_OilPres above 600kPa or se_OilPres below 300kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "A"
EMS	P019001	(Voltage on B:16 above 0.147705V and Voltage on B:16 below 0.208740V)	General Electrical Failure	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019012	Voltage on B:16 above 4.858398V	Circuit Short To Battery	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019013	Voltage on B:16 below 0.147705V	Circuit Open	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019064	No information available.	Signal Plausibility Failure	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1
EMS	P066601	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P009B73	No information available.	Actuator Stuck Closed	Low power and Engine shuts off unexpectedly - Only at low speed	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Pressure Relief Control
EMS	P018A01	(Voltage on A:23 above 0.147705V and Voltage on A:23 below 0.208740V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Pressure Sensor "B" Circuit
EMS	P018A12	Voltage on A:23 above 4.858398V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Pressure Sensor "B" Circuit
EMS	P018A13	Voltage on A:23 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Pressure Sensor "B" Circuit
EMS	P018A64	No information available.	Signal Plausibility Failure	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Pressure Sensor "B" Circuit
EMS	P020112	No information available.	Circuit Short To Battery	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 1 Injector "A"
EMS	P020113	No information available.	Circuit Open	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 1 Injector "A"
EMS	P020312	No information available.	Circuit Short To Battery	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 3 Injector "A"
EMS	P020313	No information available.	Circuit Open	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 3 Injector "A"
EMS	P020412	No information available.	Circuit Short To Battery	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 4 Injector "A"
EMS	P020413	No information available.	Circuit Open	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 4 Injector "A"
EMS	P020512	No information available.	Circuit Short To Battery	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 5 Injector "A"
EMS	P020513	No information available.	Circuit Open	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 5 Injector "A"
EMS	P020612	No information available.	Circuit Short To Battery	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 6 Injector "A"
EMS	P020613	No information available.	Circuit Open	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 6 Injector "A"
EMS	P250A11	Voltage on B:4 below 0.05V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P249F92	(Exhaust Aftertreatment Fuel Injecton Control Forced in Open Loop 0.5ratio of the time.)	Performance or Incorrect Operation	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Quantity Wrong.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Quantity Wrong. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control
EMS	P244C00	Exhaust Gas Temperature Sensor 1 below (240 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature From Engine During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature From Engine During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P24A000	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Maximum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low
EMS	P020212	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 2 Injector "A"
EMS	P020213	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 2 Injector "A"
EMS	P24A100	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Minimum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too High
EMS	P24A200	No information available.	No Sub Type Information	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Diesel Particulate Filter Regeneration Incomplete (Bank 1)
EMS	P00C600	No information available.	No Sub Type Information	Engine cranks but does not start	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Too Low - Engine Cranking
EMS	P016E00	No information available.	No Sub Type Information	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too High
EMS	P016F00	No information available.	No Sub Type Information	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too Low
EMS	P228D72	No information available.	Actuator Stuck Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too High
EMS	P228F00	No information available.	No Sub Type Information	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High
EMS	P228E00	No information available.	No Sub Type Information	Engine derate	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P101712	(Short circuit to battery is detected.)	Circuit Short To Battery	Uneven Torque or The engine misfires or Engine is difficult to start or High fuel consumption	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Water In Fuel Drainage Valve
EMS	P101713	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Water In Fuel Drainage Valve
EMS	P100190	((Stop button is active during start-up) or (Stop button becomes active when vehicle speed > 0))	Stuck On	None	Problems with plausibility check of the stop button, stop button stuck.	Diagnostic Service 14. No additional action is needed.	Check: Problems with plausibility check of the stop button, stop button stuck.	Engine stop switch
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100701	(Voltage on A:29 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:29 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (12.194354V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (4.148043V *	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100712	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019511	Voltage on A:31 below 0.100098V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019564	((se_OilTemp below -39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Temperature Sensor "A"
EMS	P056084	No information available.	Signal Below Allowable Range	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check: If fault not detected: Fault trace according to Tech Tool fault tree.	System Voltage
EMS	P203A86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A"
EMS	P20F400	(Calculated reductant consumption) below -50%	No Sub Type Information	None	Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	Diagnostic Service 14. No additional action is needed.	Check: Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	AdBlue Consumption Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114123	((Calculated pedal position percentage) above 200% and se_BufferedIvs equals to 0)	Signal Stuck Low	None	Signal stuck high/low	Diagnostic Service 14. No additional action is needed.	Check: Signal stuck high/low	Buffered Idle Validation Switch Wire - ECM Side
EMS	P114124	((Calculated pedal position percentage) below -100% and se_BufferedIvs equals to 1)	Signal Stuck High	None	Signal stuck high/low	Diagnostic Service 14. No additional action is needed.	Check: Signal stuck high/low	Buffered Idle Validation Switch Wire - ECM Side
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Oil Thermostat
EMS	P113712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Oil Thermostat
EMS	P113713	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Oil Thermostat
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 14% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115100	(Temperature in the aftertreatment system) above (999No Unit + 540No unit)	No Sub Type Information	Engine derate	Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Aftertreatment System Over Temperature
EMS	P008B00	((se_AhiFuelPres above (700kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	P229A00	No information available.	No Sub Type Information	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Pressure Regulator 2 Exceeded Control Limits - Pressure Too Low
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P118411	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Rail Drain Valve Control
EMS	P118412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Rail Drain Valve Control
EMS	P118413	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Rail Drain Valve Control
EMS	P118811	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Gas Vent System - Diesel Evacuation Pump Control
EMS	P118812	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Gas Vent System - Diesel Evacuation Pump Control
EMS	P118813	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Gas Vent System - Diesel Evacuation Pump Control
EMS	P118873	No information available.	Actuator Stuck Closed	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Gas Vent System - Diesel Evacuation Pump Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047511	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Pressure Control Valve "A"
EMS	P047512	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Exhaust Pressure Control Valve "A"
EMS	P047513	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Exhaust Pressure Control Valve "A"
EMS	P047572	((Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure without Engine Brake) below (20kPa + (Offset dependent on se_AmbAirPres and Engine Speed)) or (Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure with VCB/EPG activated) below (20kPa + (Offset dependent on se_AmbAirPres and Engine Speed)))	Actuator Stuck Open	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Control Valve "A"
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)
EMS	P011501	(Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P011511	Voltage on B:27 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011513	Voltage on B:27 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P01152A	(Difference during Precrank between Engine Coolant Temperature and mean value of(EGR Temperature, Compressor Temperature, Boost Temperature)) above 100°C	Signal Stuck In Range	Engine cooling fan does not shut off or Engine overheating or Engine cooling fan inoperative or Engine does not reach normal operating temperature	Malfunctioning Engine Coolant Temperature Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Engine Coolant Temperature Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Coolant Temperature Sensor 1
EMS	P012800	(se_CoolantTemp) below 65°C	No Sub Type Information	Engine does not reach normal operating temperature	Coolant Thermostat leaking or blocked open	Diagnostic Service 14. No additional action is needed.	Check: Coolant Thermostat leaking or blocked open If fault not detected: Fault trace according to Tech Tool fault tree.	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)
EMS	P04009C	(Difference between Sensed Boost Temperature and boost temperature stored during load/engine speed/vehicle speed operational conditions with closed EGR Valve) below (12C - (Offset dependent on Ambient Air Temperature))	Low/Insufficient Flow	None	EGR Valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P026A00	(Ratio between Modeled CaC Temp and difference between Modeled Deteriored and Modeled Nominal CaC Temp) below 0.100000001490116 ratio	No Sub Type Information	None	Deteriorated Charge Air Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated Charge Air Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Charge Air Cooler Efficiency Below Threshold
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229B00	No information available.	No Sub Type Information	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Pressure Regulator 2 Exceeded Control Limits - Pressure Too High
EMS	P269711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P26977A	se_AhiFuelPres below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD01	((Voltage on A:50 above 4.787598V and Voltage on A:50 below 4.852295V) or (Voltage on A:50 above 0.147705V and Voltage on A:50 below 0.212402V))	General Electrical Failure	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD12	Voltage on A:50 above 4.852295V	Circuit Short To Battery	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD13	Voltage on A:50 below 0.147705V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	P110573	(The starter motor is stopped due problems with rear start switch.)	Actuator Stuck Closed	None	Rear start switch problems.	Diagnostic Service 14. No additional action is needed.	Check: Rear start switch problems.	Engine Rear Start Switch "A" Circuit
EMS	P01B901	(Voltage on B:32 above 0.147705V and Voltage on B:32 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B911	Voltage on B:32 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B913	Voltage on B:32 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P242A64	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 3 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
EMS	P01A301	(Voltage on A:21 above 0.147705V and Voltage on A:21 below 0.208740V)	General Electrical Failure	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Rail Pressure Sensor
EMS	P01A312	Voltage on A:21 above 4.858398V	Circuit Short To Battery	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Rail Pressure Sensor
EMS	P01A313	Voltage on A:21 below 0.147705V	Circuit Open	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Rail Pressure Sensor
EMS	P01A364	No information available.	Signal Plausibility Failure	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Rail Pressure Sensor
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P040311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control
EMS	P266511	No information available.	Circuit Short To Ground	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Shutoff Valve "B" Control
EMS	P266512	No information available.	Circuit Short To Battery	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Shutoff Valve "B" Control
EMS	P266513	No information available.	Circuit Open	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Shutoff Valve "B" Control
EMS	P26657F	No information available.	Actuator Stuck Off	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Fuel Shutoff Valve "B" Control
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P06B016	Voltage on A:7 below 4.5V	Circuit Voltage Below Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B017	Voltage on A:7 above 5.5V	Circuit Voltage Above Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B316	Voltage on B:17 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06B317	Voltage on B:17 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06E616	Voltage on A:1 below 4.5V	Circuit Voltage Below Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P06E617	Voltage on A:1 above 5.5V	Circuit Voltage Above Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation
EMS	P20D711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D773	(se_AhiFuelPres below 70kPa and not (((se_AhiFuelPres below -10kPa and se_AhiFuelPres below -10kPa) or (se_AhiFuelPres above 710kPa and se_AhiFuelPres above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control
EMS	P010501	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or (Voltage on A:22 above 0.147705V and Voltage on A:22 below 0.297852V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010512	Voltage on A:22 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010513	Voltage on A:22 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P010562	ExhaustManifoldPressure above 50kPa	Signal Compare Failure	None	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P25AE00	(The signal se_OilPresPcj is above a critical limit)	No Sub Type Information	Engine derate	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Too Low
EMS	P25A911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Piston Cooling Oil Control
EMS	P25A912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Piston Cooling Oil Control
EMS	P25A913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Piston Cooling Oil Control
EMS	P25A971	(When piston cooling valve is forced open and closed the oil pressure change) below 20kPa	Actuator Stuck	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control
EMS	P054001	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (19.616229V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (20.775897V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054011	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P054013	(Voltage on B:7 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (19.616229V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P05407F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P225C00	((Upstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	No Sub Type Information	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 1
EMS	P225E67	((Downstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	Signal Incorrect After Event	None	Downstream NOx sensor Excessive reductant delivery causing NH3 slip	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor Excessive reductant delivery causing NH3 slip	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P225F00	(The Downstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Downstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 2
EMS	P225D00	(The Upstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Upstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10E100	Particulate Filter Differential Pressure above (30No unit + 0No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (30No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High
EMS	P22D912	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 1 Alternative Fuel Injector Control
EMS	P22D913	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 1 Alternative Fuel Injector Control
EMS	P22DD12	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 2 Alternative Fuel Injector Control
EMS	P22DD13	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 2 Alternative Fuel Injector Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P22FB92	(Internal status and error byte recieved from Upstream NOx sensor evaluated as bad sensor quality)	Performance or Incorrect Operation	None	Faulty Upstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P22FE00	(Internal status and error byte recieved from Downstream NOx sensor evaluated as bad sensor quality)	No Sub Type Information	None	Faulty Downstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2
EMS	P22E112	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 3 Alternative Fuel Injector Control
EMS	P22E113	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 3 Alternative Fuel Injector Control
EMS	P22E512	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 4 Alternative Fuel Injector Control
EMS	P22E513	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 4 Alternative Fuel Injector Control
EMS	P22E912	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 5 Alternative Fuel Injector Control
EMS	P22E913	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 5 Alternative Fuel Injector Control
EMS	P22ED12	No information available.	Circuit Short To Battery	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 6 Alternative Fuel Injector Control
EMS	P22ED13	No information available.	Circuit Open	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Cylinder 6 Alternative Fuel Injector Control
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P055A01	((Voltage on B:19 above 4.787598V and Voltage on B:19 below 4.852295V) or (Voltage on B:19 above 0.147705V and Voltage on B:19 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P055A12	Voltage on B:19 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A13	Voltage on B:19 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A64	((Difference between demanded piston cooling pressure and measured piston cooling pressure) between -15kPa and 15kPa or se_OilPresPcj above 30kPa)	Signal Plausibility Failure	None	Piston cooling pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "B"
EMS	P02E011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control
EMS	P02E012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control
EMS	P02E07C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09B	(se_ThrottlePos1 - Demanded Throttle Position) above 10%	High/Excessive Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09C	(Demanded Throttle Position - se_ThrottlePos1) above 10%	Low/Insufficient Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P112800	(The signal se_OilPresPcj is above a critical limit)	No Sub Type Information	None	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Moderately Low
EMS	P112900	PFTransmissionOilTemp above (999No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High
EMS	P029800	se_OilTemp above (125No unit + 3No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E601	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or (Voltage on B:24 above 0.147705V and Voltage on B:24 below 0.247803V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E612	Voltage on B:24 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E613	Voltage on B:24 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E697	(not ((Maximum Opened Throttle Position) between 73deg and 86deg) and not ((Minimum Closed Throttle Position) between -5deg and 7deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor
EMS	P02E699	(Calibration requested and no calibration performed)	Exceeded Learning Limit	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (999No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	P111600	BoostPressure above (Output from look-up tables with input EngineSpeed and se_AmbAirPres)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Moderately Too High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus
EMS	P111800	BoostPressure above (Output from look-up tables with inputs EngineSpeed and se_AmbAirPres)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Too High
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P111D00	se_CoolantLvl below (100No unit - 30No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low
EMS	P111E00	se_CoolantTemp above (2No Unit + 107No unit + (Output from look-up tables with inputs se_AmbAirPres and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High
EMS	P112500	se_OilTemp above (125No unit + 1No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00	((se_AhiFuelPres below (200kPa + (Offset dependent on fuel value and Engine Speed)) or se_AhiFuelPres below (100kPa + (Offset dependent on fuel value and Engine Speed))) or se_FuelPres below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P113600	((not ((Upstream NOx Sensor available in EMS))) or (not ((Downstream NOx Sensor available in EMS))))	No Sub Type Information	None	Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	Diagnostic Service 14. No additional action is needed.	Check: Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	NOx Sensor 1/2 Correlation Calibration Error Bank 1
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below -150kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range
EMS	P06DA71	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 180kPa	Actuator Stuck	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control
EMS	P118A01	(Voltage on A:54 above 0.598145V and Voltage on A:54 below 3.901367V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Gas Vent System - Diesel Level Sensor
EMS	P2A3E00	No information available.	No Sub Type Information	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Delivery Error
EMS	P2A3F00	No information available.	No Sub Type Information	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Excessive Time to Enter Alternative Fuel Control
EMS	P2A4000	No information available.	No Sub Type Information	Diesel only mode	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel System - Injector Leakage to Rail Detected

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P2A6E00	No information available.	No Sub Type Information	Low power	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Rail Pressure Regulator Inlet Pressure Too Low
EMS	P24F811	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve Control Circuit
EMS	P24F813	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve Control Circuit
EMS	P2A9E11	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Alternative Fuel Return Line Valve Control
EMS	P2A9E12	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Alternative Fuel Return Line Valve Control
EMS	P2A9E13	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Alternative Fuel Return Line Valve Control
EMS	P2A9E73	No information available.	Actuator Stuck Closed	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Return Line Valve Control
EMS	P2AA200	No information available.	No Sub Type Information	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Return Line Insufficient Flow
EMS	P11A101	((Voltage on A:9 above 1.632080V and Voltage on A:9 below 2.393799V) or (Voltage on A:9 above 0.391846V and Voltage on A:9 below 0.727539V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Return to Tank Diesel Presence Sensor Circuit
EMS	P11A111	Voltage on A:9 below 0.391846V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Return to Tank Diesel Presence Sensor Circuit
EMS	P11A113	Voltage on A:9 above 4.726563V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Alternative Fuel Return to Tank Diesel Presence Sensor Circuit
EMS	P11A300	No information available.	No Sub Type Information	None	No information available.	Diagnostic Service 14. No additional action is needed.	Check:	Alternative Fuel Return to Tank Diesel Presence Detected
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034038	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P052612	CoolingFanSpeedSense equals to (0 rpm)	Circuit Short To Battery	Engine cooling fan does not shut off	Fan speed sensor malfunctioning	Diagnostic Service 14. No additional action is needed.	Check: Fan speed sensor malfunctioning	Fan Speed Sensor Circuit
EMS	P054464	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or Exhaust Gas Temperature Sensor 1 below 100°C)))	Signal Plausibility Failure	Activation failure, regeneration	Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P03351F	(Intermittent crank speed signal)	Circuit Intermittent	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033538	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P203F00	(((Reductant level) below 2% for 15s))	No Sub Type Information	Engine derate	Reductant tank almost empty	Diagnostic Service 14. No additional action is needed.	Check: Reductant tank almost empty If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Low
EMS	P220012	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P22001C	(Internal Supply Voltage signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220064	(The Upstream NOx Difference (upstream NOx sensor measured NOx flow – modeled NOx flow) is compared to a low and high fault threshold map (NOx flow threshold as a function of modeled NOx flow). The difference must be below Low Limit or above High Limit repeatedly 3No Unit times. Evaluation time for each repeated evaluation is 4s)	Signal Plausibility Failure	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220093	(Internal status and error byte recieved from Upstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 1
EMS	P060741	(Internal Control Module Memory Check Sum Error)	General Checksum Failure	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Control Module Performance
EMS	P061511	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay
EMS	P061512	(Short circuit to battery is detected.)	Circuit Short To Battery	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay
EMS	P20EE00	((Filtered NOx conversion) below 58% and (A filter is applied to the NOx conversion that may require the evaluation to be repeated a few times to set the fault. Time required depends on engine load and the NOx flow. With all enable conditions fulfilled it normally takes no more than a few minutes.))	No Sub Type Information	Engine derate	Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Coolant Pump Control Circuit/Open
EMS	P260012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Coolant Pump Control Circuit/Open
EMS	P260013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Coolant Pump Control Circuit/Open
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 45kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -30kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
EMS	P009A00	((Difference between Sensed Boost Temperature and Modeled Boost Temperature) below -50°C or (Difference between Sensed Boost Temperature and Modeled Boost Temperature) above 50°C)	No Sub Type Information	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Air Intake Temperature - Correlation
EMS	P011001	(Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011011	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011013	Voltage on A:47 above 4.910889V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P021700	EngineCoolantTemperature above (3.200000047683716No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047001	((Voltage on B:12 above 4.787598V and Voltage on B:12 below 4.852295V) or (Voltage on B:12 above 0.147705V and Voltage on B:12 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047012	Voltage on B:12 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047013	Voltage on B:12 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047064	((Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) below -45kPa or (Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) above 45kPa)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Sensor "A"
EMS	P048011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit
EMS	P048012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit
EMS	P051A01	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or (Voltage on B:28 above 0.147705V and Voltage on B:28 below 0.307617V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A12	Voltage on B:28 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A13	Voltage on B:28 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P051A64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Crankcase Pressure Sensor
EMS	P052400	EngineOilPressure below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low
EMS	P203164	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 2 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
EMS	P222601	(Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222612	Voltage on N/A:ECU internal above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222613	Voltage on N/A:ECU internal below 0.117188V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P222664	(ECUAmbientPressure below 55kPa or ECUAmbientPressure above 114kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Circuit
EMS	P226411	Voltage on B:8 below 0.211182V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor
EMS	P244A00	(Measured differential pressure as percentage of modeled differential pressure) below (55 %)	No Sub Type Information	None	DPF removed or severely damaged Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF removed or severely damaged Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too Low (Bank 1)
EMS	P244B00	(Measured differential pressure as percentage of modeled differential pressure) above (350 %)	No Sub Type Information	None	DPF melted DPF filled with soot Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF melted DPF filled with soot Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too High (Bank 1)
EMS	P245264	(((Failure event for diagnosis in AfterRun) and Particulate Filter Differential Pressure above 1.200000047683716 kPa))	Signal Plausibility Failure	None	Sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250F00	((If calculated value for oil level) below -4.980500221252441 % and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low
EMS	P255601	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255611	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255612	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255613	(Voltage on B:23 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P256000	EngineCoolantLevel below (100No unit - 75No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low
EMS	P207F9A	(((Estimated reductant concentration) below 18% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229E12	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E64	(((The sensor shall respond to tip-in events (sudden power increase). Each time it does not respond correctly a timer is started. When this accumulated time has reached its time limit 60s a second phase is started with the purpose to confirm the suspected fault. This second phase is using the same method, but this time the reductant delivery is stopped in order to improve chances for the outlet NOx sensor to respond to the tip-in events. A second cumulative timer is started every time	Signal Plausibility Failure	None	Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor	NOx Sensor Bank 1 Sensor 2
EMS	P229E93	(Internal status and error byte recieved from Downstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052001	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or (Voltage on B:11 above 0.147705V and Voltage on B:11 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052012	Voltage on B:11 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052013	Voltage on B:11 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052064	(EngineOilPressure above 100kPa or EngineOilPressure above 600kPa or EngineOilPressure below 300kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "A"
EMS	P066601	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P020112	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 1 Injector "A"
EMS	P020113	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 1 Injector "A"
EMS	P020312	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 3 Injector "A"
EMS	P020313	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 3 Injector "A"
EMS	P020412	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 4 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020413	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 4 Injector "A"
EMS	P020512	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 5 Injector "A"
EMS	P020513	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 5 Injector "A"
EMS	P020612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 6 Injector "A"
EMS	P020613	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 6 Injector "A"
EMS	P250A11	Voltage on B:4 below 0.05V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P249F92	Exhaust Gas Temperature Sensor 1 below (240 C)	Performance or Incorrect Operation	None	Exhaust Temperature Too Low For Particulate Filter Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Temperature Too Low For Particulate Filter Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control
EMS	P244C00	Average Particulate Filter Temperature below (250 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)
EMS	P24A000	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Maximum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low
EMS	P020212	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 2 Injector "A"
EMS	P020213	(Current driving the injector rises to slow)	Circuit Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 2 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P24A100	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Minimum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too High
EMS	P24A200	((Particulate Filter Differential Pressure above 2h))	No Sub Type Information	None	Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor. If fault not detected: Fault trace according to Tech Tool fault tree.	Diesel Particulate Filter Regeneration Incomplete (Bank 1)
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P100101	(Voltage on A:27 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:27 above (20.303109V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:27 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (18.135422V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100111	(Voltage on A:27 above (0.731483V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100112	(Voltage on A:27 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100190	((Stop button is active during start-up) or (Stop button becomes active when vehicle speed > 0))	Stuck On	None	Problems with plausibility check of the stop button, stop button stuck.	Diagnostic Service 14. No additional action is needed.	Check: Problems with plausibility check of the stop button, stop button stuck.	Engine stop switch
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100701	(Voltage on A:29 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:29 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (12.194354V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (4.148043V *	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100711	Voltage on A:29 below (1.587853V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100712	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100713	(Voltage on A:29 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019511	Voltage on A:31 below 0.100098V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019564	((EngineOilTemperature below -39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Temperature Sensor "A"
EMS	P203A86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A"
EMS	P024311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Turbocharger/Supercharger Wastegate Solenoid "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P024312	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P20F400	(Calculated reductant consumption) below -50%	No Sub Type Information	None	Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	Diagnostic Service 14. No additional action is needed.	Check: Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	AdBlue Consumption Too Low
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Oil Thermostat
EMS	P113712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Oil Thermostat
EMS	P113713	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Oil Thermostat
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 14% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Priming Pump
EMS	P115212	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Priming Pump
EMS	P115213	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fuel Priming Pump
EMS	P115100	(Temperature in the aftertreatment system) above (999No Unit + 540No unit)	No Sub Type Information	Engine derate	Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Aftertreatment System Over Temperature
EMS	P008B00	((AftertreatmentFuel Pressure above (700kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P117C00	(The selected gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Range Gear Too High
EMS	P117D00	(The selected gear is too low.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Gear Too Low
EMS	P117E00	(The selected starting gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Launch Gear Too High
EMS	P047511	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Pressure Control Valve "A"
EMS	P047512	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Exhaust Pressure Control Valve "A"
EMS	P047513	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Exhaust Pressure Control Valve "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047572	((Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure without Engine Brake) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)) or (Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure with VCB/EPG activated) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)))	Actuator Stuck Open	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Control Valve "A"
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)
EMS	P011501	(Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011511	Voltage on B:27 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011513	Voltage on B:27 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P01152A	(Difference between and maximum and minimum coolant temperature) below 3°C	Signal Stuck In Range	None	Coolant Temperature sensor signal stuck in operating range	Diagnostic Service 14. No additional action is needed.	Check: Coolant Temperature sensor signal stuck in operating range If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Coolant Temperature Sensor 1
EMS	P012800	(EngineCoolantTemperature) below 65°C	No Sub Type Information	Engine does not reach normal operating temperature	Coolant Thermostat leaking or blocked open	Diagnostic Service 14. No additional action is needed.	Check: Coolant Thermostat leaking or blocked open If fault not detected: Fault trace according to Tech Tool fault tree.	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)
EMS	P04009C	(Difference between Sensed Boost Temperature and boost temperature stored during load/engine speed/vehicle speed operational conditions with closed EGR Valve) below (12C - (Offset dependent on Ambient Air Temperature))	Low/Insufficient Flow	None	EGR Valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P026A00	(Ratio between Modeled CaC Temp and difference between Modeled Deteriored and Modeled Nominal CaC Temp) below 0.100000001490116 ratio	No Sub Type Information	None	Deteriorated Charge Air Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated Charge Air Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Charge Air Cooler Efficiency Below Threshold
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit
EMS	P269711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P269773	(AftertreatmentFuel Pressure above 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P26977A	AftertreatmentFuelPressure below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD01	((Voltage on A:50 above 4.787598V and Voltage on A:50 below 4.852295V) or (Voltage on A:50 above 0.147705V and Voltage on A:50 below 0.212402V))	General Electrical Failure	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD12	Voltage on A:50 above 4.852295V	Circuit Short To Battery	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD13	Voltage on A:50 below 0.147705V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	P01B901	(Voltage on B:32 above 0.147705V and Voltage on B:32 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B911	Voltage on B:32 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B913	Voltage on B:32 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P242A64	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 3 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P040311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P06B016	Voltage on A:7 below 4.5V	Circuit Voltage Below Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B017	Voltage on A:7 above 5.5V	Circuit Voltage Above Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06B316	Voltage on B:17 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06B317	Voltage on B:17 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06E616	Voltage on A:1 below 4.5V	Circuit Voltage Below Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P06E617	Voltage on A:1 above 5.5V	Circuit Voltage Above Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20D711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D773	(AftertreatmentFuel Pressure below 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control
EMS	P010501	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or (Voltage on A:22 above 0.147705V and Voltage on A:22 below 0.297852V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010512	Voltage on A:22 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010513	Voltage on A:22 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010562	ExhaustManifoldPressure above 50kPa	Signal Compare Failure	None	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P25AE00	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	Engine derate	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Too Low
EMS	P25A911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Piston Cooling Oil Control
EMS	P25A912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Piston Cooling Oil Control
EMS	P25A913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Piston Cooling Oil Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P25A971	(When piston cooling valve is forced open and closed the oil pressure change) below 20kPa	Actuator Stuck	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control
EMS	P054001	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (19.616229V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (22.747332V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054011	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054013	(Voltage on B:7 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (19.616229V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P05407F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P225C00	((Upstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	No Sub Type Information	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 1
EMS	P225E67	((Downstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	Signal Incorrect After Event	None	Downstream NOx sensor Excessive reductant delivery causing NH3 slip	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor Excessive reductant delivery causing NH3 slip	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 2
EMS	P225F00	(The Downstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Downstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P225D00	(The Upstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Upstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10E100	Particulate Filter Differential Pressure above (30No unit + 0No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (30No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High
EMS	P22FB92	(Internal status and error byte recieved from Upstream NOx sensor evaluated as bad sensor quality)	Performance or Incorrect Operation	None	Faulty Upstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P22FE00	(Internal status and error byte recieved from Downstream NOx sensor evaluated as bad sensor quality)	No Sub Type Information	None	Faulty Downstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P055A01	((Voltage on B:19 above 4.787598V and Voltage on B:19 below 4.852295V) or (Voltage on B:19 above 0.147705V and Voltage on B:19 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A12	Voltage on B:19 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A13	Voltage on B:19 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A64	((Difference between demanded piston cooling pressure and measured piston cooling pressure) between -15kPa and 15kPa or PistonCoolPressure above 30kPa)	Signal Plausibility Failure	None	Piston cooling pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "B"
EMS	P02E011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control
EMS	P02E012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E07C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09B	(ThrottlePosition - Demanded Throttle Position) above 10%	High/Excessive Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09C	(Demanded Throttle Position - ThrottlePosition) above 10%	Low/Insufficient Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P112800	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	None	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Moderately Low
EMS	P112900	PFTransmissionOilTemp above (1000No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High
EMS	P029800	EngineOilTemperature above (125No unit + 3No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature
EMS	P02E601	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or (Voltage on B:24 above 0.147705V and Voltage on B:24 below 0.247803V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E612	Voltage on B:24 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E613	Voltage on B:24 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E697	(not ((Maximum Opened Throttle Position) between 73deg and 86deg) and not ((Minimum Closed Throttle Position) between -5deg and 7deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor
EMS	P02E699	(Calibration requested and no calibration performed)	Exceeded Learning Limit	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (1000No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	P111600	BoostPressure above (Output from look-up tables with input EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Moderately Too High
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus
EMS	P111800	BoostPressure above (Output from look-up tables with inputs EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Too High
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P111D00	EngineCoolantLevel below (100No unit - 30No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low
EMS	P111E00	EngineCoolantTemperature above (2No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High
EMS	P112500	EngineOilTemperature above (125No unit + 1No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN
EMS	P253901	((Voltage on B:16 above 4.787598V and Voltage on B:16 below 4.852295V) or (Voltage on B:16 above 0.147705V and Voltage on B:16 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Low Pressure Fuel System Sensor Circuit
EMS	P253912	Voltage on B:16 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Low Pressure Fuel System Sensor Circuit
EMS	P253913	Voltage on B:16 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Low Pressure Fuel System Sensor Circuit
EMS	P008A00	((AftertreatmentFuel Pressure below (200kPa + (Offset dependent on fuel value and Engine Speed)) or AftertreatmentFuelP ressure below (100kPa + (Offset dependent on fuel value and Engine Speed))) or FuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P113600	((not ((Upstream NOx Sensor available in EMS))) or (not ((Downstream NOx Sensor available in EMS))))	No Sub Type Information	None	Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	Diagnostic Service 14. No additional action is needed.	Check: Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	NOx Sensor 1/2 Correlation Calibration Error Bank 1
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below -150kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range
EMS	P06DA71	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 180kPa	Actuator Stuck	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control
EMS	P113411	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113413	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113472	(AftertreatmentFuel Pressure above 25kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Open	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113473	(AftertreatmentFuel Pressure below 30kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening	Diagnostic Service 14. No additional action is needed.	Check: No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034038	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P052612	CoolingFanSpeedSense equals to (0 rpm)	Circuit Short To Battery	Engine cooling fan does not shut off	Fan speed sensor malfunctioning	Diagnostic Service 14. No additional action is needed.	Check: Fan speed sensor malfunctioning	Fan Speed Sensor Circuit
EMS	P054464	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or Exhaust Gas Temperature Sensor 1 below 100°C)))	Signal Plausibility Failure	Activation failure, regeneration	Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P03351F	(Intermittent crank speed signal)	Circuit Intermittent	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"
EMS	P033538	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P203F00	(((Reductant level) below 0.100000001490116 % for 15s))	No Sub Type Information	Engine derate	Reductant tank almost empty	Diagnostic Service 14. No additional action is needed.	Check: Reductant tank almost empty If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Low
EMS	P220012	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P22001C	(Internal Supply Voltage signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220064	(The Upstream NOx Difference (upstream NOx sensor measured NOx flow – modeled NOx flow) is compared to a low and high fault threshold map (NOx flow threshold as a function of modeled NOx flow). The difference must be below Low Limit or above High Limit repeatedly 5No Unit times. Evaluation time for each repeated evaluation is 2.5s)	Signal Plausibility Failure	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220093	(Internal status and error byte recieved from Upstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 1
EMS	P060741	(Internal Control Module Memory Check Sum Error)	General Checksum Failure	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Control Module Performance
EMS	P061511	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay
EMS	P061512	(Short circuit to battery is detected.)	Circuit Short To Battery	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20EE00	((Filtered NOx conversion) below 50% and (A filter is applied to the NOx conversion that may require the evaluation to be repeated a few times to set the fault. Time required depends on engine load and the NOx flow. With all enable conditions fulfilled it normally takes no more than a few minutes.))	No Sub Type Information	Engine derate	Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 60kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -30kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
EMS	P008700	(The rail pressure below 285bar for 3s	No Sub Type Information	Engine cranks but does not start	Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail/System Pressure - Too Low Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P009A00	((Difference between Sensed Boost Temperature and Modeled Boost Temperature) below -40°C or (Difference between Sensed Boost Temperature and Modeled Boost Temperature) above 50°C)	No Sub Type Information	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Air Intake Temperature - Correlation
EMS	P011001	(Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011011	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011013	Voltage on A:47 above 4.910889V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P021700	EngineCoolantTemperature above (3.200000047683716No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition
EMS	P047001	((Voltage on B:12 above 4.787598V and Voltage on B:12 below 4.852295V) or (Voltage on B:12 above 0.147705V and Voltage on B:12 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047012	Voltage on B:12 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047013	Voltage on B:12 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047064	((Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) below -50kPa or (Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) above 50kPa)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Sensor "A"
EMS	P048011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit
EMS	P048012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit
EMS	P051A01	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or (Voltage on B:28 above 0.147705V and Voltage on B:28 below 0.307617V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A12	Voltage on B:28 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A13	Voltage on B:28 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Crankcase Pressure Sensor
EMS	P052400	EngineOilPressure below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P203164	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 2 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
EMS	P222601	(Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222612	Voltage on N/A:ECU internal above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222613	Voltage on N/A:ECU internal below 0.117188V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222664	(ECUAmbientPressure below 50kPa or ECUAmbientPressure above 118kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Circuit
EMS	P226411	Voltage on B:8 below 0.211182V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor
EMS	P244A00	(Measured differential pressure as percentage of modeled differential pressure) below (55 %)	No Sub Type Information	None	DPF removed or severely damaged Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF removed or severely damaged Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too Low (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P244B00	(Measured differential pressure as percentage of modeled differential pressure) above (350 %)	No Sub Type Information	None	DPF melted DPF filled with soot Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF melted DPF filled with soot Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too High (Bank 1)
EMS	P245264	(((Failure event for diagnosis in AfterRun) and Particulate Filter Differential Pressure above 1.200000047683716 kPa))	Signal Plausibility Failure	None	Sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1
EMS	P250F00	((If calculated value for oil level) below -4.980500221252441 % and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255601	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255611	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255612	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255613	(Voltage on B:23 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P256000	EngineCoolantLevel below (100No unit - 75No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low
EMS	P257831	(Turbo Speed) below 60000rpm	No Signal	None	Malfunctioning VGT Actuator	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning VGT Actuator	Turbocharger Speed Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P25783A	(Ratio between Fault Timer (incremented when Turbo Overspeed Flag is TRUE) and Evaluation Timer) above 0.899999976158142 ratio	Signal Has Too Many Pulses	None	Malfunction in VGT Control	Diagnostic Service 14. No additional action is needed.	Check: Malfunction in VGT Control If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger Speed Sensor
EMS	P207F9A	(((Estimated reductant concentration) below 20% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229E12	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E64	(((The sensor shall respond to tip-in events (sudden power increase). Each time it does not respond correctly a timer is started. When this accumulated time has reached its time limit 40s a second phase is started with the purpose to confirm the suspected fault. This second phase is using the same method, but this time the reductant delivery is stopped in order to improve chances for the outlet NOx sensor to respond to the tip-in events. A second cumulative timer is started every time	Signal Plausibility Failure	None	Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor	NOx Sensor Bank 1 Sensor 2
EMS	P229E93	(Internal status and error byte recieved from Downstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052001	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or (Voltage on B:11 above 0.147705V and Voltage on B:11 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052012	Voltage on B:11 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052013	Voltage on B:11 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052064	(EngineOilPressure above 50kPa or EngineOilPressure above 600kPa or EngineOilPressure below 350kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "A"
EMS	P019001	(Voltage on A:19 above 0.147705V and Voltage on A:19 below 0.447998V)	General Electrical Failure	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019011	Voltage on A:19 below 0.147705V	Circuit Short To Ground	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019013	Voltage on A:19 above 4.877930V	Circuit Open	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019062	(Rail fuel pressure fluctuations less than expected.)	Signal Compare Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1
EMS	P019064	(Rail pressure before engine start. above 200bar or Rail pressure before engine start. below -100bar)	Signal Plausibility Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P066601	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P009B11	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine derate	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Pressure Relief Control
EMS	P009B12	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine derate	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Pressure Relief Control
EMS	P009B13	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Engine derate	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Fuel Pressure Relief Control
EMS	P020112	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 1 Injector "A"
EMS	P020113	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 1 Injector "A"
EMS	P020312	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 3 Injector "A"
EMS	P020313	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 3 Injector "A"
EMS	P020412	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 4 Injector "A"
EMS	P020413	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 4 Injector "A"
EMS	P020512	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 5 Injector "A"
EMS	P020513	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 5 Injector "A"
EMS	P020612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 6 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020613	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 6 Injector "A"
EMS	P250A11	Voltage on B:4 below 0.05V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P249F92	Exhaust Gas Temperature Sensor 1 below (240 C)	Performance or Incorrect Operation	None	Exhaust Temperature Too Low For Particulate Filter Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Temperature Too Low For Particulate Filter Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control
EMS	P244C00	Average Particulate Filter Temperature below (250 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)
EMS	P24A000	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Maximum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low
EMS	P025112	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "A" (Cam/Rotor/Injector)
EMS	P025114	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "A" (Cam/Rotor/Injector)
EMS	P020212	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 2 Injector "A"
EMS	P020213	(Current driving the injector rises to slow)	Circuit Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 2 Injector "A"
EMS	P24A100	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Minimum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P24A200	((Particulate Filter Differential Pressure above 2h))	No Sub Type Information	None	Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor. If fault not detected: Fault trace according to Tech Tool fault tree.	Diesel Particulate Filter Regeneration Incomplete (Bank 1)
EMS	P00C600	(The rail pressure during cranking phase below 285bar for 15s	No Sub Type Information	Engine cranks but does not start	Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Too Low - Engine Cranking
EMS	P009400	(During engine shutdown, an abnormal fuel pressure drop n the rail is observed during evaluation period. 	No Sub Type Information	None	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel System Leak Detected - Small Leak
EMS	P016E00	(Average rail pressure deviation above 150bar for 5s	No Sub Type Information	Engine derate	Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed	Diagnostic Service 14. No additional action is needed.	Check: Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too High
EMS	P016F00	(Average rail pressure deviation below - 150bar for 5s	No Sub Type Information	Engine derate	Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too Low
EMS	P025612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control B
EMS	P025614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control B
EMS	P025692	(Loss of high fuel pressure pumping capacity 	Performance or Incorrect Operation	Low power	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control B

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P228F00	(Fuel pressure controller adjustment value saturated high	No Sub Type Information	Engine derate	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High
EMS	P228E00	(Fuel pressure controller adjustment value saturated low	No Sub Type Information	Engine derate	Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100711	Voltage on A:29 below (1.587853V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100712	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100713	(Voltage on A:29 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019511	Voltage on A:31 below 0.100098V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019564	((EngineOilTemperature below 39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Temperature Sensor "A"
EMS	P203A86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A"
EMS	P20F400	(Calculated reductant consumption) below -50%	No Sub Type Information	None	Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	Diagnostic Service 14. No additional action is needed.	Check: Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	AdBlue Consumption Too Low
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U010C00	(Lost communication with VGTM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	VGTM not powered Software incompatibility with VGTM	Diagnostic Service 14. No additional action is needed.	Check: VGTM not powered Software incompatibility with VGTM	Lost Communication With Turbocharger/Supercharger Control Module A

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 12% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition
EMS	P115211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Priming Pump
EMS	P115212	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Priming Pump
EMS	P115213	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fuel Priming Pump
EMS	P115100	(Temperature in the aftertreatment system) above (999No Unit + 540No unit)	No Sub Type Information	Engine derate	Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Aftertreatment System Over Temperature
EMS	P2A1212	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "C" (Cam/Rotor/Injector)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P2A1214	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "C" (Cam/Rotor/Injector)
EMS	P2A1612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1692	(Loss of high fuel pressure pumping capacity P	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1A12	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "E" (Cam/Rotor/Injector)
EMS	P2A1A14	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "E" (Cam/Rotor/Injector)
EMS	P2A1E12	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E14	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E92	(Loss of high fuel pressure pumping capacity P	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P008B00	((AftertreatmentFuel Pressure above (700kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P117C00	(The selected gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Range Gear Too High
EMS	P117D00	(The selected gear is too low.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Gear Too Low
EMS	P117E00	(The selected starting gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Launch Gear Too High
EMS	P040A01	(Voltage on B:48 above 0.023671V and Voltage on B:48 below 0.181149V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P040A11	Voltage on B:48 below 0.023671V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P040A13	Voltage on B:48 above 0.772330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P040A64	((((EGRTemperature - EngineCoolantTemperature) below -40°C or (EGRTemperature - EngineCoolantTemperature) above 40°C)))	Signal Plausibility Failure	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P047511	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Pressure Control Valve "A"
EMS	P047512	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Exhaust Pressure Control Valve "A"
EMS	P047513	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Exhaust Pressure Control Valve "A"
EMS	P047572	((Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure without Engine Brake) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)) or (Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure with VCB/EPG activated) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)))	Actuator Stuck Open	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Control Valve "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)
EMS	P011501	(Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011511	Voltage on B:27 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011513	Voltage on B:27 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P01152A	(Difference between and maximum and minimum coolant temperature) below 3°C	Signal Stuck In Range	None	Coolant Temperature sensor signal stuck in operating range	Diagnostic Service 14. No additional action is needed.	Check: Coolant Temperature sensor signal stuck in operating range If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Coolant Temperature Sensor 1
EMS	P012800	(EngineCoolantTemperature) below 65°C	No Sub Type Information	Engine does not reach normal operating temperature	Coolant Thermostat leaking or blocked open	Diagnostic Service 14. No additional action is needed.	Check: Coolant Thermostat leaking or blocked open If fault not detected: Fault trace according to Tech Tool fault tree.	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)
EMS	P245700	(Ratio (Modeled Exhaust Manifold Temp - EGR Temp) / (Modeled Exhaust Manifold Temp - Engine Coolant Temp)) below 0.800000011920929 ratio	No Sub Type Information	None	Deteriorated EGR Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated EGR Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation Cooler Efficiency Below Threshold
EMS	P040901	(Voltage on A:21 above 0.147705V and Voltage on A:21 below 0.207520V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor
EMS	P040912	Voltage on A:21 above 4.831543V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P040913	Voltage on A:21 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor
EMS	P040964	((((35kPa - (Offset dependent on EGR valve Control PWM) - DifferentialPressureFeedbackEGR) below 0kPa or (DifferentialPressure FeedbackEGR - 0kPa - (Offset dependent on EGR valve Control PWM)) below 0kPa)) or DifferentialPressureFeedbackEGR above 1.5kPa)	Signal Plausibility Failure	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P040984	DifferentialPressureFeedbackEGR below -2.200000047683716 kPa	Signal Below Allowable Range	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P040985	DifferentialPressureFeedbackEGR above 2.200000047683716 kPa	Signal Above Allowable Range	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P04007C	(EGR valve fails to follow command)	Slow Response	None	EGR valve stuck	Diagnostic Service 14. No additional action is needed.	Check: EGR valve stuck If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P04009B	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) above (130% + (Offset dependent on ECUAmbientPressure))	High/Excessive Flow	None	Malfunctioning EGR Valve Malfunctioning VGT	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning EGR Valve Malfunctioning VGT If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P04009C	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) below (70% + (Offset dependent on ECUAmbientPressure))	Low/Insufficient Flow	None	EGR system blocked/restricted Malfunctioning EGR Valve	Diagnostic Service 14. No additional action is needed.	Check: EGR system blocked/restricted Malfunctioning EGR Valve If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P026A00	(Ratio between Modeled CaC Temp and difference between Modeled Deteriored and Modeled Nominal CaC Temp) below 0.200000002980232 ratio	No Sub Type Information	None	Deteriorated Charge Air Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated Charge Air Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Charge Air Cooler Efficiency Below Threshold
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit
EMS	P269711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269773	(AftertreatmentFuel Pressure above 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P26977A	AftertreatmentFuelPressure below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD01	((Voltage on A:50 above 4.787598V and Voltage on A:50 below 4.852295V) or (Voltage on A:50 above 0.147705V and Voltage on A:50 below 0.212402V))	General Electrical Failure	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD12	Voltage on A:50 above 4.852295V	Circuit Short To Battery	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20DD13	Voltage on A:50 below 0.147705V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	P242A64	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 3 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P04D900	(Difference between Demanded and Actual Burned Air Fraction) above (Output MAP Value dependant on Engine Speed and Engine Torque)	No Sub Type Information	None	EGR Valve Stuck Open VGT Control Saturated Low	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve Stuck Open VGT Control Saturated Low If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop EGR Control At Limit - Flow Too Low
EMS	P04DA00	(Difference between Demanded and Actual Burned Air Fraction) below (Output MAP Value dependant on Engine Speed and Engine Torque)	No Sub Type Information	None	EGR Valve Stuck Closed EGR System restricted VGT Control Saturated High	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve Stuck Closed EGR System restricted VGT Control Saturated High If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop EGR Control At Limit - Flow Too High
EMS	P040311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P06B016	Voltage on A:7 below 4.5V	Circuit Voltage Below Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B017	Voltage on A:7 above 5.5V	Circuit Voltage Above Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B316	Voltage on B:17 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06B317	Voltage on B:17 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06E616	Voltage on A:1 below 4.5V	Circuit Voltage Below Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P06E617	Voltage on A:1 above 5.5V	Circuit Voltage Above Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation
EMS	P20D711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D773	(AftertreatmentFuel Pressure below 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P010501	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or (Voltage on A:22 above 0.147705V and Voltage on A:22 below 0.297852V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010512	Voltage on A:22 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010513	Voltage on A:22 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010562	ExhaustManifoldPressure above 60kPa	Signal Compare Failure	None	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P25A971	(When piston cooling valve is forced open and closed the oil pressure change) below 30kPa	Actuator Stuck	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control
EMS	P054001	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (19.616229V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (22.747332V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P054011	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054013	(Voltage on B:7 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (19.616229V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P05407F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P225C00	((Upstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	No Sub Type Information	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 1
EMS	P225E67	((Downstream NOx level) above 300ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	Signal Incorrect After Event	None	Downstream NOx sensor Excessive reductant delivery causing NH3 slip	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor Excessive reductant delivery causing NH3 slip	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P225F00	(The Downstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Downstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 2
EMS	P225D00	(The Upstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Upstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10E100	Particulate Filter Differential Pressure above (30No unit + 0No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (30No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High
EMS	P22FB92	(Internal status and error byte recieved from Upstream NOx sensor evaluated as bad sensor quality)	Performance or Incorrect Operation	None	Faulty Upstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P22FE00	(Internal status and error byte recieved from Downstream NOx sensor evaluated as bad sensor quality)	No Sub Type Information	None	Faulty Downstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P02E011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control
EMS	P02E012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control
EMS	P02E07C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09B	(ThrottlePosition - Demanded Throttle Position) above 10%	High/Excessive Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09C	(Demanded Throttle Position - ThrottlePosition) above 10%	Low/Insufficient Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P112900	PFTransmissionOilTemp above (1000No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P112B00	(High temperature failure detected by Turbocharger/Supercharger A.)	No Sub Type Information	None	High temperature in Turbocharger/Supercharger A	Diagnostic Service 14. No additional action is needed.	Check: High temperature in Turbocharger/Supercharger A	Turbocharger/Supercharger Actuator "A" Temperature High
EMS	P029800	EngineOilTemperature above (125No unit + 3No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature
EMS	P02E601	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or (Voltage on B:24 above 0.147705V and Voltage on B:24 below 0.247803V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E612	Voltage on B:24 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E613	Voltage on B:24 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E697	(not ((Maximum Opened Throttle Position) between 78.3000030517578deg and 84.6999969482422deg) and not ((Minimum Closed Throttle Position) between -2.799999952316284deg and 2.799999952316284deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor
EMS	P02E699	(Calibration requested and no calibration performed)	Exceeded Learning Limit	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (1000No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	P111600	BoostPressure above (Output from look-up tables with input EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Moderately Too High
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus
EMS	P111800	BoostPressure above (Output from look-up tables with inputs EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Too High
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P111D00	EngineCoolantLevel below (100No unit - 30No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low
EMS	P111E00	EngineCoolantTemperature above (2No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High
EMS	P112100	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112198	EGRTemperature above (220No unit + 0No unit)	Component or System Over Temperature	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112200	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P112500	EngineOilTemperature above (125No unit + 1No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN
EMS	P008A00	((AftertreatmentFuel Pressure below (200kPa + (Offset dependent on fuel value and Engine Speed)) or AftertreatmentFuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed))) or FuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P113600	((not ((Upstream NOx Sensor available in EMS))) or (not ((Downstream NOx Sensor available in EMS))))	No Sub Type Information	None	Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	Diagnostic Service 14. No additional action is needed.	Check: Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	NOx Sensor 1/2 Correlation Calibration Error Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below 50kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range
EMS	P06DA71	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 100kPa	Actuator Stuck	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control
EMS	P113411	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113413	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113472	(AftertreatmentFuel Pressure above 25kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Open	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113473	(AftertreatmentFuel Pressure below 30kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening	Diagnostic Service 14. No additional action is needed.	Check: No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034038	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P052612	CoolingFanSpeedSense equals to (0 rpm)	Circuit Short To Battery	Engine cooling fan does not shut off	Fan speed sensor malfunctioning	Diagnostic Service 14. No additional action is needed.	Check: Fan speed sensor malfunctioning	Fan Speed Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P054464	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or Exhaust Gas Temperature Sensor 1 below 100°C)))	Signal Plausibility Failure	Activation failure, regeneration	Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P03351F	(Intermittent crank speed signal)	Circuit Intermittent	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"
EMS	P033538	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P203F00	(((Reductant level) below 2% for 15s))	No Sub Type Information	Engine derate	Reductant tank almost empty	Diagnostic Service 14. No additional action is needed.	Check: Reductant tank almost empty If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Low
EMS	P220012	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P22001C	(Internal Supply Voltage signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220064	(The Upstream NOx Difference (upstream NOx sensor measured NOx flow – modeled NOx flow) is compared to a low and high fault threshold map (NOx flow threshold as a function of modeled NOx flow). The difference must be below Low Limit or above High Limit repeatedly 5No Unit times. Evaluation time for each repeated evaluation is 2.5s)	Signal Plausibility Failure	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Bank 1 Sensor 1
EMS	P220093	(Internal status and error byte recieved from Upstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 1
EMS	P060741	(Internal Control Module Memory Check Sum Error)	General Checksum Failure	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Control Module Performance
EMS	P061511	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay
EMS	P061512	(Short circuit to battery is detected.)	Circuit Short To Battery	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20EE00	((Filtered NOx conversion) below 50% and (A filter is applied to the NOx conversion that may require the evaluation to be repeated a few times to set the fault. Time required depends on engine load and the NOx flow. With all enable conditions fulfilled it normally takes no more than a few minutes.))	No Sub Type Information	Engine derate	Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Coolant Pump Control Circuit/Open
EMS	P260012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Coolant Pump Control Circuit/Open
EMS	P260013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Coolant Pump Control Circuit/Open
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 65kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -38kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008700	(The rail pressure below 285bar for 4s	No Sub Type Information	Engine cranks but does not start	Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail/System Pressure - Too Low Bank 1
EMS	P009A00	((Difference between Sensed Boost Temperature and Modeled Boost Temperature) below -50°C or (Difference between Sensed Boost Temperature and Modeled Boost Temperature) above 50°C)	No Sub Type Information	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Air Intake Temperature - Correlation
EMS	P011001	(Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011011	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011013	Voltage on A:47 above 4.910889V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P021700	EngineCoolantTemperature above (3.20000004768371 6No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P047001	((Voltage on B:12 above 4.787598V and Voltage on B:12 below 4.852295V) or (Voltage on B:12 above 0.147705V and Voltage on B:12 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047012	Voltage on B:12 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047013	Voltage on B:12 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047064	((Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) below -50kPa or (Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) above 50kPa)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Sensor "A"
EMS	P048011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit
EMS	P048012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit
EMS	P051A01	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or (Voltage on B:28 above 0.147705V and Voltage on B:28 below 0.307617V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A12	Voltage on B:28 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A13	Voltage on B:28 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P051A64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Crankcase Pressure Sensor
EMS	P052400	EngineOilPressure below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low
EMS	P203164	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 2 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
EMS	P222601	(Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222612	Voltage on N/A:ECU internal above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222613	Voltage on N/A:ECU internal below 0.117188V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P222664	(ECUAmbientPressure below 50kPa or ECUAmbientPressure above 118kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Circuit
EMS	P226411	Voltage on B:8 below 0.211182V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor
EMS	P244A00	(Measured differential pressure as percentage of modeled differential pressure) below (55 %)	No Sub Type Information	None	DPF removed or severely damaged Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF removed or severely damaged Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too Low (Bank 1)
EMS	P244B00	(Measured differential pressure as percentage of modeled differential pressure) above (350 %)	No Sub Type Information	None	DPF melted DPF filled with soot Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF melted DPF filled with soot Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too High (Bank 1)
EMS	P245264	(((Failure event for diagnosis in AfterRun) and Particulate Filter Differential Pressure above 1.200000047683716 kPa))	Signal Plausibility Failure	None	Sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P250F00	((If calculated value for oil level) below -4.980500221252441 % and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low
EMS	P255601	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255611	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255612	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255613	(Voltage on B:23 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P256000	EngineCoolantLevel below (100No unit - 75No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low
EMS	P207F9A	(((Estimated reductant concentration) below 18% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229E12	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E64	(((The sensor shall respond to tip-in events (sudden power increase). Each time it does not respond correctly a timer is started. When this accumulated time has reached its time limit 40s a second phase is started with the purpose to confirm the suspected fault. This second phase is using the same method, but this time the reductant delivery is stopped in order to improve chances for the outlet NOx sensor to respond to the tip-in events. A second cumulative timer is started every time	Signal Plausibility Failure	None	Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor	NOx Sensor Bank 1 Sensor 2
EMS	P229E93	(Internal status and error byte recieved from Downstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052001	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or (Voltage on B:11 above 0.147705V and Voltage on B:11 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052012	Voltage on B:11 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052013	Voltage on B:11 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052064	(EngineOilPressure above 100kPa or EngineOilPressure above 600kPa or EngineOilPressure below 300kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "A"
EMS	P019001	(Voltage on A:19 above 0.147705V and Voltage on A:19 below 0.447998V)	General Electrical Failure	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019011	Voltage on A:19 below 0.147705V	Circuit Short To Ground	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019013	Voltage on A:19 above 4.877930V	Circuit Open	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019062	(Rail fuel pressure fluctuations less than expected.)	Signal Compare Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1
EMS	P019064	(Rail pressure before engine start. above 200bar or Rail pressure before engine start. below -100bar)	Signal Plausibility Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P066601	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P009B11	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine derate	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Pressure Relief Control
EMS	P009B12	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine derate	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Pressure Relief Control
EMS	P009B13	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Engine derate	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Fuel Pressure Relief Control
EMS	P020112	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 1 Injector "A"
EMS	P020113	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 1 Injector "A"
EMS	P020312	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 3 Injector "A"
EMS	P020313	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 3 Injector "A"
EMS	P020412	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 4 Injector "A"
EMS	P020413	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 4 Injector "A"
EMS	P020512	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 5 Injector "A"
EMS	P020513	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 5 Injector "A"
EMS	P020612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 6 Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P020613	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 6 Injector "A"
EMS	P250A11	Voltage on B:4 below 0.05V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P249F92	Exhaust Gas Temperature Sensor 1 below (240 C)	Performance or Incorrect Operation	None	Exhaust Temperature Too Low For Particulate Filter Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Temperature Too Low For Particulate Filter Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control
EMS	P244C00	Average Particulate Filter Temperature below (250 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)
EMS	P24A000	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Maximum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low
EMS	P020212	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 2 Injector "A"
EMS	P020213	(Current driving the injector rises to slow)	Circuit Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 2 Injector "A"
EMS	P24A100	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Minimum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too High
EMS	P24A200	((Particulate Filter Differential Pressure above 2h))	No Sub Type Information	None	Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor. If fault not detected: Fault trace according to Tech Tool fault tree.	Diesel Particulate Filter Regeneration Incomplete (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P00C600	(The rail pressure during cranking phase below 285bar for 15s	No Sub Type Information	Engine cranks but does not start	Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Too Low - Engine Cranking
EMS	P009400	(During engine shutdown, an abnormal fuel pressure drop in the rail is observed during evaluation period. 	No Sub Type Information	None	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel System Leak Detected - Small Leak
EMS	P016E00	(Average rail pressure deviation above 150bar for 5s	No Sub Type Information	Engine derate	Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed	Diagnostic Service 14. No additional action is needed.	Check: Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too High
EMS	P016F00	(Average rail pressure deviation below - 150bar for 5s	No Sub Type Information	Engine derate	Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too Low
EMS	P025612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control B
EMS	P025614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control B
EMS	P025692	(Loss of high fuel pressure pumping capacity 	Performance or Incorrect Operation	Low power	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control B

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P228F00	(Fuel pressure controller adjustment value saturated high	No Sub Type Information	Engine derate	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High
EMS	P228E00	(Fuel pressure controller adjustment value saturated low	No Sub Type Information	None	Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P100101	(Voltage on A:27 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:27 above (20.303109V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:27 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (18.135422V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100111	(Voltage on A:27 above (0.731483V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100112	(Voltage on A:27 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100190	((Stop button is active during start-up) or (Stop button becomes active when vehicle speed > 0))	Stuck On	None	Problems with plausibility check of the stop button, stop button stuck.	Diagnostic Service 14. No additional action is needed.	Check: Problems with plausibility check of the stop button, stop button stuck.	Engine stop switch
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100701	(Voltage on A:29 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:29 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (12.194354V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (4.148043V *	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100711	Voltage on A:29 below (1.587853V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100712	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100713	(Voltage on A:29 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019511	Voltage on A:31 below 0.100098V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019564	((EngineOilTemperature below -39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Temperature Sensor "A"
EMS	P203A86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20F400	(Calculated reductant consumption) below -50%	No Sub Type Information	None	Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	Diagnostic Service 14. No additional action is needed.	Check: Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	AdBlue Consumption Too Low
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Oil Thermostat
EMS	P113712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Oil Thermostat
EMS	P113713	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Oil Thermostat
EMS	P113B01	((Voltage on A:23 above 4.787598V and Voltage on A:23 below 4.852295V) or (Voltage on A:23 above 0.147705V and Voltage on A:23 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Turbo Compound Buffer Air Pressure Sensor
EMS	P113B12	Voltage on A:23 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Turbo Compound Buffer Air Pressure Sensor
EMS	P113B13	Voltage on A:23 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Turbo Compound Buffer Air Pressure Sensor
EMS	P113B64	(Averaged TC Buffer Air Pressure) below 7.5kPa	Signal Plausibility Failure	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	Turbo Compound Buffer Air Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113E11	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Turbo Compound Buffer Air Valve Control
EMS	P113E12	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Turbo Compound Buffer Air Valve Control
EMS	P113E13	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Turbo Compound Buffer Air Valve Control
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 14% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition
EMS	P115211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Priming Pump
EMS	P115212	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Priming Pump
EMS	P115213	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fuel Priming Pump
EMS	P115100	(Temperature in the aftertreatment system) above (999No Unit + 500No unit)	No Sub Type Information	Engine derate	Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Aftertreatment System Over Temperature

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P2A1612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1692	(Loss of high fuel pressure pumping capacity)	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1E12	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E14	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E92	(Loss of high fuel pressure pumping capacity)	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P008B00	((AftertreatmentFuel Pressure above (700kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P117C00	(The selected gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Range Gear Too High
EMS	P117D00	(The selected gear is too low.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Gear Too Low
EMS	P117E00	(The selected starting gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Launch Gear Too High
EMS	P040A01	(Voltage on B:48 above 0.023671V and Voltage on B:48 below 0.181149V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P040A11	Voltage on B:48 below 0.023671V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P040A13	Voltage on B:48 above 0.772330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P040A64	((((EGRTemperature - EngineCoolantTemperature) below -46°C or (EGRTemperature - EngineCoolantTemperature) above 40°C)))	Signal Plausibility Failure	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P047511	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Pressure Control Valve "A"
EMS	P047512	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Exhaust Pressure Control Valve "A"
EMS	P047513	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Exhaust Pressure Control Valve "A"
EMS	P047572	((Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure without Engine Brake) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)) or (Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure with VCB/EPG activated) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)))	Actuator Stuck Open	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Control Valve "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)
EMS	P011501	(Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011511	Voltage on B:27 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011513	Voltage on B:27 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P01152A	(Difference between and maximum and minimum coolant temperature) below 3°C	Signal Stuck In Range	None	Coolant Temperature sensor signal stuck in operating range	Diagnostic Service 14. No additional action is needed.	Check: Coolant Temperature sensor signal stuck in operating range If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Coolant Temperature Sensor 1
EMS	P012800	(EngineCoolantTemperature) below 65°C	No Sub Type Information	Engine does not reach normal operating temperature	Coolant Thermostat leaking or blocked open	Diagnostic Service 14. No additional action is needed.	Check: Coolant Thermostat leaking or blocked open If fault not detected: Fault trace according to Tech Tool fault tree.	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)
EMS	P245700	(Ratio (Modeled Exhaust Manifold Temp - EGR Temp) / (Modeled Exhaust Manifold Temp - Engine Coolant Temp)) below 0.769999980926514 ratio	No Sub Type Information	None	Deteriorated EGR Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated EGR Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation Cooler Efficiency Below Threshold
EMS	P040901	(Voltage on A:21 above 0.147705V and Voltage on A:21 below 0.207520V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor
EMS	P040912	Voltage on A:21 above 4.831543V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P040913	Voltage on A:21 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor
EMS	P040964	((((35kPa - (Offset dependent on EGR valve Control PWM) - DifferentialPressureFeedbackEGR) below 0kPa or (DifferentialPressure FeedbackEGR - 0kPa - (Offset dependent on EGR valve Control PWM)) below 0kPa)) or DifferentialPressureFeedbackEGR above 1.5kPa)	Signal Plausibility Failure	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P040984	DifferentialPressureFeedbackEGR below -2.200000047683716 kPa	Signal Below Allowable Range	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P040985	DifferentialPressureFeedbackEGR above 2.200000047683716 kPa	Signal Above Allowable Range	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor
EMS	P04007C	(EGR valve fails to follow command)	Slow Response	None	EGR valve stuck	Diagnostic Service 14. No additional action is needed.	Check: EGR valve stuck If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P04009B	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) above (130% + (Offset dependent on ECUAmbientPressure))	High/Excessive Flow	None	Malfunctioning EGR Valve Malfunctioning VGT	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning EGR Valve Malfunctioning VGT If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P04009C	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) below (70% + (Offset dependent on ECUAmbientPressure))	Low/Insufficient Flow	None	EGR system blocked/restricted Malfunctioning EGR Valve	Diagnostic Service 14. No additional action is needed.	Check: EGR system blocked/restricted Malfunctioning EGR Valve If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P026A00	(Ratio between Modeled CaC Temp and difference between Modeled Deteriored and Modeled Nominal CaC Temp) below 0.200000002980232 ratio	No Sub Type Information	None	Deteriorated Charge Air Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated Charge Air Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Charge Air Cooler Efficiency Below Threshold
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit
EMS	P269711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269773	(AftertreatmentFuel Pressure above 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P26977A	AftertreatmentFuelPressure below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD01	((Voltage on A:50 above 4.787598V and Voltage on A:50 below 4.852295V) or (Voltage on A:50 above 0.147705V and Voltage on A:50 below 0.212402V))	General Electrical Failure	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD12	Voltage on A:50 above 4.852295V	Circuit Short To Battery	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20DD13	Voltage on A:50 below 0.147705V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	P01B901	(Voltage on B:32 above 0.147705V and Voltage on B:32 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B911	Voltage on B:32 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B913	Voltage on B:32 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P242A64	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 3 below 100°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P04D900	(Difference between Demanded and Actual Burned Air Fraction) above (Output MAP Value dependant on Engine Speed and Engine Torque)	No Sub Type Information	None	EGR Valve Stuck Open VGT Control Saturated Low	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve Stuck Open VGT Control Saturated Low If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop EGR Control At Limit - Flow Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P04DA00	(Difference between Demanded and Actual Burned Air Fraction) below (Output MAP Value dependant on Engine Speed and Engine Torque)	No Sub Type Information	None	EGR Valve Stuck Closed EGR System restricted VGT Control Saturated High	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve Stuck Closed EGR System restricted VGT Control Saturated High If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop EGR Control At Limit - Flow Too High
EMS	P040311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P06B016	Voltage on A:7 below 4.5V	Circuit Voltage Below Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B017	Voltage on A:7 above 5.5V	Circuit Voltage Above Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B316	Voltage on B:17 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06B317	Voltage on B:17 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06E616	Voltage on A:1 below 4.5V	Circuit Voltage Below Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P06E617	Voltage on A:1 above 5.5V	Circuit Voltage Above Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation
EMS	P20D711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P20D773	(AftertreatmentFuel Pressure below 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control
EMS	P010501	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or (Voltage on A:22 above 0.147705V and Voltage on A:22 below 0.297852V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010512	Voltage on A:22 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010513	Voltage on A:22 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010562	ExhaustManifoldPressure above 60kPa	Signal Compare Failure	None	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P25AE00	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	Engine derate	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Too Low
EMS	P25A911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Piston Cooling Oil Control
EMS	P25A912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Piston Cooling Oil Control
EMS	P25A913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Piston Cooling Oil Control
EMS	P25A971	(When piston cooling valve is forced open and closed the oil pressure change) below 20kPa	Actuator Stuck	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P054001	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (19.616229V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (22.747332V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054011	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054013	(Voltage on B:7 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (19.616229V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P05407F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P225C00	((Upstream NOx level) above 200ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	No Sub Type Information	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 1
EMS	P225E67	((Downstream NOx level) above 200ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	Signal Incorrect After Event	None	Downstream NOx sensor Excessive reductant delivery causing NH3 slip	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor Excessive reductant delivery causing NH3 slip	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 2
EMS	P225F00	(The Downstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Downstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 2
EMS	P225D00	(The Upstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Upstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10E100	Particulate Filter Differential Pressure above (30No unit + 0No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (30No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High
EMS	P22FB92	(Internal status and error byte recieved from Upstream NOx sensor evaluated as bad sensor quality)	Performance or Incorrect Operation	None	Faulty Upstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P22FE00	(Internal status and error byte recieved from Downstream NOx sensor evaluated as bad sensor quality)	No Sub Type Information	None	Faulty Downstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P055A01	((Voltage on B:19 above 4.787598V and Voltage on B:19 below 4.852295V) or (Voltage on B:19 above 0.147705V and Voltage on B:19 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A12	Voltage on B:19 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P055A13	Voltage on B:19 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A64	((Difference between demanded piston cooling pressure and measured piston cooling pressure) between -15kPa and 15kPa or PistonCoolPressure above 30kPa)	Signal Plausibility Failure	None	Piston cooling pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "B"
EMS	P02E011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control
EMS	P02E012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control
EMS	P02E07C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09B	(ThrottlePosition - Demanded Throttle Position) above 10%	High/Excessive Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09C	(Demanded Throttle Position - ThrottlePosition) above 10%	Low/Insufficient Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P112800	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	None	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Moderately Low
EMS	P112900	PFTransmissionOilTemp above (1000No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High
EMS	P029800	EngineOilTemperature above (125No unit + 3No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature
EMS	P02E601	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or (Voltage on B:24 above 0.147705V and Voltage on B:24 below 0.247803V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E612	Voltage on B:24 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E613	Voltage on B:24 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E697	(not ((Maximum Opened Throttle Position) between 73deg and 86deg) and not ((Minimum Closed Throttle Position) between -5deg and 7deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor
EMS	P02E699	(Calibration requested and no calibration performed)	Exceeded Learning Limit	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (1000No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	P111600	BoostPressure above (Output from look-up tables with input EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Moderately Too High
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P111800	BoostPressure above (Output from look-up tables with inputs EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Too High
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P111D00	EngineCoolantLevel below (100No unit - 30No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low
EMS	P111E00	EngineCoolantTemperature above (2No Unit + 107No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High
EMS	P112100	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112198	EGRTemperature above (220No unit + 0No unit)	Component or System Over Temperature	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112200	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Moderately High
EMS	P112500	EngineOilTemperature above (125No unit + 1No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00	((AftertreatmentFuel Pressure below (200kPa + (Offset dependent on fuel value and Engine Speed)) or AftertreatmentFuelP ressure below (100kPa + (Offset dependent on fuel value and Engine Speed))) or FuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P113600	((not ((Upstream NOx Sensor available in EMS))) or (not ((Downstream NOx Sensor available in EMS))))	No Sub Type Information	None	Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	Diagnostic Service 14. No additional action is needed.	Check: Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	NOx Sensor 1/2 Correlation Calibration Error Bank 1
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below -150kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range
EMS	P06DA71	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 180kPa	Actuator Stuck	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control
EMS	P113411	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113413	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113472	(AftertreatmentFuel Pressure above 25kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Open	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113473	(AftertreatmentFuel Pressure below 30kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening	Diagnostic Service 14. No additional action is needed.	Check: No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve
ACM	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Auxiliary brakes, low performance (Reduced retarder braking)	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
ACM	U010000	(Lost communication with ECM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	ECM not powered Software incompatibility with ECM	Diagnostic Service 14. No additional action is needed.	Check: ECM not powered Software incompatibility with ECM	Lost Communication with EMS
ACM	P054401	((Voltage on A:42 above 2.260742V and Voltage on A:42 below 4.852295V) or (Voltage on A:42 above 0.147705V and Voltage on A:42 below 0.677490V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Temperature Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P054411	Voltage on A:42 below 0.147705V	Circuit Short To Ground	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
ACM	P054415	Voltage on A:42 above 4.852295V	Circuit Short To Battery or Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
ACM	P205A01	(Voltage on A:18 above 4.794922V and Voltage on A:18 below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Tank Temperature Sensor
ACM	P205A11	Voltage on A:18 below 0.037842V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Tank Temperature Sensor
ACM	P205A13	Voltage on A:18 above 4.852295V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Tank Temperature Sensor
ACM	P205A64	(Temperature rise not above 3°C)	Signal Plausibility Failure	None	Coolant level low Extremely cold weather conditions Coolant valve damaged	Diagnostic Service 14. No additional action is needed.	Check: Coolant level low Extremely cold weather conditions Coolant valve damaged	Reductant Tank Temperature Sensor
ACM	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
ACM	P203101	((Voltage on A:34 above 2.260742V and Voltage on A:34 below 4.852295V) or (Voltage on A:34 above 0.147705V and Voltage on A:34 below 0.677490V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
ACM	P203111	Voltage on A:34 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P203115	Voltage on A:34 above 4.852295V	Circuit Short To Battery or Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
ACM	P245211	Voltage on A:26 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
ACM	P245213	Voltage on A:26 above 4.901123V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
ACM	P20BD11	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Reductant Heater "B" Control
ACM	P20BD12	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Reductant Heater "B" Control
ACM	P20BD13	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater "B" Control
ACM	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU
ACM	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
ACM	P066601	((Voltage on N/A:ECU internal above 2.874756V and Voltage on N/A:ECU internal below 4.852295V) or (Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.237793V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
ACM	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P066613	Voltage on N/A:ECU internal above 4.852295V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
ACM	P107B01	((Voltage on A:11 above 2.214355V and Voltage on A:11 below 4.852295V) or (Voltage on A:11 above 0.147705V and Voltage on A:11 below 1.165771V))	General Electrical Failure	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Water Temperature Sensor
ACM	P107B11	Voltage on A:11 below 0.147705V	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Water Temperature Sensor
ACM	P107B13	Voltage on A:11 above 4.852295V	Circuit Open	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Water Temperature Sensor
ACM	P107C01	((Voltage on A:15 above 2.214355V and Voltage on A:15 below 4.852295V) or (Voltage on A:15 above 0.147705V and Voltage on A:15 below 1.165771V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Oil Temperature Sensor
ACM	P107C11	Voltage on A:15 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Oil Temperature Sensor
ACM	P107C13	Voltage on A:15 above 4.852295V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Oil Temperature Sensor
ACM	P107C98	(The retarder oil temperature is above critical limit.)	Component or System Over Temperature	Auxiliary brakes, low performance (Reduced retarder braking)	Bad cooling Sensor malfunction	Diagnostic Service 14. No additional action is needed.	Check: Bad cooling Sensor malfunction	Retarder Oil Temperature Sensor
ACM	P107D11	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced retarder braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Retarder Valve
ACM	P107D12	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced retarder braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Retarder Valve
ACM	P107D13	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced retarder braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Retarder Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P107E01	((Voltage on A:19 above 4.797363V and Voltage on A:19 below 4.852295V) or (Voltage on A:19 above 0.147705V and Voltage on A:19 below 0.202637V))	General Electrical Failure	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Air Pressure Sensor
ACM	P107E12	Voltage on A:19 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Air Pressure Sensor
ACM	P107E13	Voltage on A:19 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced retarder braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Retarder Air Pressure Sensor
ACM	P107E64	(Too big difference between requested retarder pressure and actual retarder pressure.)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced retarder braking)	Retarder valve malfunction Air pressure sensor malfunction	Diagnostic Service 14. No additional action is needed.	Check: Retarder valve malfunction Air pressure sensor malfunction	Retarder Air Pressure Sensor
ACM	P107E85	(The retarder air pressure is above critical limit.)	Signal Above Allowable Range	Auxiliary brakes, low performance (Reduced retarder braking)	Retarder valve malfunction Air pressure sensor malfunction	Diagnostic Service 14. No additional action is needed.	Check: Retarder valve malfunction Air pressure sensor malfunction	Retarder Air Pressure Sensor
ACM	P204A01	((Voltage on A:27 above 4.65V and Voltage on A:27 below 4.85V) or (Voltage on A:27 above 0.147V and Voltage on A:27 below 0.3V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Pressure Sensor
ACM	P204A11	Voltage on A:27 below 0.147V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pressure Sensor
ACM	P204A13	Voltage on A:27 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P204A64	(ACM_AirPressure above 250kPa or ((Difference between inital reductant pressure and final reductant pressure during startup) above 30kPa and not ((Difference between inital air pressure and final air pressure during startup) above 20kPa)))	Signal Plausibility Failure	None	Faulty urea pressure sensor Blocked urea line or nozzle	Diagnostic Service 14. No additional action is needed.	Check: Faulty urea pressure sensor Blocked urea line or nozzle	Reductant Pressure Sensor
ACM	P208A11	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Reductant Pump "A" Control
ACM	P208A12	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector short circuited to battery.	Reductant Pump "A" Control
ACM	P208A13	(Open circuit is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Pump "A" Control
ACM	P056084	not (Control Module below 8V)	Signal Below Allowable Range	None	Bad pins or connection. Cabling issues. Generator issues.	Diagnostic Service 14. No additional action is needed.	Check: Bad pins or connection. Cabling issues. Generator issues.	System Voltage
ACM	P056085	not (Control Module above 36V)	Signal Above Allowable Range	None	Bad pins or connection. Cabling issues. Generator issues.	Diagnostic Service 14. No additional action is needed.	Check: Bad pins or connection. Cabling issues. Generator issues.	System Voltage
ACM	P20B111	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Heater Coolant Control Valve Control
ACM	P20B112	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Reductant Heater Coolant Control Valve Control
ACM	P20B113	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater Coolant Control Valve Control
ACM	P20B172	(Reductant Tank Temperature above 70°C or Reductant Pump Temperature above 120°C)	Actuator Stuck Open	None	Tankheatvalve stuck open Inappropriate location of the pump or tank	Diagnostic Service 14. No additional action is needed.	Check: Tankheatvalve stuck open Inappropriate location of the pump or tank	Reductant Heater Coolant Control Valve Control
ACM	P20B173	((Temperature rise not above 3°C) and (Temperature rise not above 5°C))	Actuator Stuck Closed	None	Coolan valve damaged Coolat valve stuck	Diagnostic Service 14. No additional action is needed.	Check: Coolan valve damaged Coolat valve stuck	Reductant Heater Coolant Control Valve Control
ACM	P20B911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Reductant Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P20B912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Reductant Heater "A" Control
ACM	P20B913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater "A" Control
ACM	P20B992	(Start the engine and drive the truck during cold weather conditions)	Performance or Incorrect Operation	None	Heating valve stuck close Electrical heaters failure	Diagnostic Service 14. No additional action is needed.	Check: Heating valve stuck close Electrical heaters failure	Reductant Heater "A" Control
ACM	P20A072	Reagent Pressure AlbonAir above 30kPa	Actuator Stuck Open	None	Damaged checkvalve	Diagnostic Service 14. No additional action is needed.	Check: Damaged checkvalve	Reductant Purge Control Valve
ACM	P20A073	(not ((Difference between inital reductant pressure and final reductant pressure during startup) above 30kPa) and (Difference between inital air pressure and final air pressure during startup) above 20kPa and (Pump priming has been completed))	Actuator Stuck Closed	None	Damaged checkvalve	Diagnostic Service 14. No additional action is needed.	Check: Damaged checkvalve	Reductant Purge Control Valve
ACM	P203A01	((Voltage on A:46 above 4.582520V and Voltage on A:46 below 4.852295V) or (Voltage on A:46 above 0.147705V and Voltage on A:46 below 0.457764V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Level Sensor "A"
ACM	P203A12	Voltage on A:46 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Level Sensor "A"
ACM	P203A13	Voltage on A:46 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Level Sensor "A"
ACM	P21CAA1	not (Control Module between 20V and 32V)	System Voltage	None	Connector/pins issues Cabling issues	Diagnostic Service 14. No additional action is needed.	Check: Connector/pins issues Cabling issues	Reductant Control Module Supply Voltage
ACM	P218F92	not ((Turn engine on and run truck with medium to high load during at least 10 minutes))	Performance or Incorrect Operation	None	Damaged pump Empty Urea tank Leak in suction line	Diagnostic Service 14. No additional action is needed.	Check: Damaged pump Empty Urea tank Leak in suction line	Reductant No Flow Detected

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P26E711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage D" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage D" is short circuited. If fault not detected: Fault trace according to Tech Tool fault tree.	Actuator Supply Voltage "D"
ACM	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
ACM	P204211	Voltage on A:35 below 0.080566V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Temperature Sensor
ACM	P204213	Voltage on A:35 above 4.882813V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Temperature Sensor
ACM	P204264	(Temperature rise not above 5°C)	Signal Plausibility Failure	None	Coolant level low Extremely cold weather conditions Coolant valve damaged	Diagnostic Service 14. No additional action is needed.	Check: Coolant level low Extremely cold weather conditions Coolant valve damaged	Reductant Temperature Sensor
ACM	P20A611	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Reductant Injection Air Pressure Control Valve Control
ACM	P20A612	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Reductant Injection Air Pressure Control Valve Control
ACM	P20A613	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Injection Air Pressure Control Valve Control
ACM	P20A672	(Reagent Pressure AlbonAir above 200kPa and ACM_AirPressure above 250kPa)	Actuator Stuck Open	None	Faulty or damaged airvalve	Diagnostic Service 14. No additional action is needed.	Check: Faulty or damaged airvalve	Reductant Injection Air Pressure Control Valve Control
ACM	P20A673	not (((Difference between inital reductant pressure and final reductant pressure during startup) above 30kPa and (Difference between inital air pressure and final air pressure during startup) above 20kPa))	Actuator Stuck Closed	None	Damaged airvalve	Diagnostic Service 14. No additional action is needed.	Check: Damaged airvalve	Reductant Injection Air Pressure Control Valve Control
ACM	P203701	((Voltage on A:47 above 4.696045V and Voltage on A:47 below 4.852295V) or (Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.302734V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Injection Air Pressure Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P203711	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Injection Air Pressure Sensor "A"
ACM	P203713	Voltage on A:47 above 4.852295V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Reductant Injection Air Pressure Sensor "A"
ACM	P203764	((not (ACM_AirPressure above 200kPa) and (Difference between inital reductant pressure and final reductant pressure during startup) above 30kPa) or ACM_AirPressure above 250kPa)	Signal Plausibility Failure	None	Faulty air pressure sensor	Diagnostic Service 14. No additional action is needed.	Check: Faulty air pressure sensor	Reductant Injection Air Pressure Sensor "A"
ACM	P20CA00	((Turn engine on and run truck with medium to high load during at least 10 minutes) and not ((The difference between controlled airvalve dutycycle and the predicted airvalve dutycycle) above -25%))	No Sub Type Information	None	Damaged air lines Leaking air lines	Diagnostic Service 14. No additional action is needed.	Check: Damaged air lines Leaking air lines	Reductant Injection Air Pressure Leakage
ACM	P20E797	((Turn engine on and run truck with medium to high load during at least 10 minutes) and not ((The difference between controlled airvalve dutycycle and the predicted airvalve dutycycle) below -25%))	Component or System Operation Obstructed or Blocked	None	Dirt in the air lines/nozzle Squished air line	Diagnostic Service 14. No additional action is needed.	Check: Dirt in the air lines/nozzle Squished air line	Reductant Injection Air Pressure Too High
ACM	P065711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage A" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage A" is short circuited. If fault not detected: Fault trace according to Tech Tool fault tree.	Actuator Supply Voltage "A"
ACM	P065712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to battery voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Actuator Supply Voltage "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P266911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage B" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage B" is short circuited.	Actuator Supply Voltage "B"
ACM	P266912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to battery voltage.	Actuator Supply Voltage "B"
ACM	P20E997	(One evaluation is made each pump stroke. This condition is TRUE If 75% of the evaluations fails after 1200No Unitstrokes.)	Component or System Operation Obstructed or Blocked	None	Frozen or clogged urea in urea pathway/nozzle	Diagnostic Service 14. No additional action is needed.	Check: Frozen or clogged urea in urea pathway/nozzle	Reductant Pressure Too High
ACM	P242A01	((Voltage on A:39 above 2.260742V and Voltage on A:39 below 4.852295V) or (Voltage on A:39 above 0.147705V and Voltage on A:39 below 0.677490V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
ACM	P242A11	Voltage on A:39 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
ACM	P242A15	Voltage on A:39 above 4.852295V	Circuit Short To Battery or Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
ACM	P06B016	Voltage on A:50 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
ACM	P06B017	Voltage on A:50 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P06B316	Voltage on A:51 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
ACM	P06B317	Voltage on A:51 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
ACM	U114600	(Lost communication with ECM on Vehicle Communication Engine Subnet)	No Sub Type Information	None	ECM not powered Software incompatibility with ECM	Diagnostic Service 14. No additional action is needed.	Check: ECM not powered Software incompatibility with ECM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With EECU on Engine Subnet
ACM	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	None	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
ACM	P109A00	(The filtered water temperature of the retarder is above limit.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	Bad cooling Sensor malfunction	Diagnostic Service 14. No additional action is needed.	Check: Bad cooling Sensor malfunction	Retarder Water Temperature Moderately High
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034038	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P052612	CoolingFanSpeedSense equals to (0 rpm)	Circuit Short To Battery	Engine cooling fan does not shut off	Fan speed sensor malfunctioning	Diagnostic Service 14. No additional action is needed.	Check: Fan speed sensor malfunctioning	Fan Speed Sensor Circuit
EMS	P054464	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or Exhaust Gas Temperature Sensor 1 below 200°C)))	Signal Plausibility Failure	Activation failure, regeneration	Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor Sensor 1 Not Mounted Exhaust Gas Temperature Sensor Sensor 1 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P03351F	(Intermittent crank speed signal)	Circuit Intermittent	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033538	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor "A"
EMS	P203F00	(((Reductant level) below 2% for 15s))	No Sub Type Information	Engine derate	Reductant tank almost empty	Diagnostic Service 14. No additional action is needed.	Check: Reductant tank almost empty If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Low
EMS	P220012	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P22001C	(Internal Supply Voltage signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P220064	(The Upstream NOx Difference (upstream NOx sensor measured NOx flow – modeled NOx flow) is compared to a low and high fault threshold map (NOx flow threshold as a function of modeled NOx flow). The difference must be below Low Limit or above High Limit repeatedly 3No Unit times. Evaluation time for each repeated evaluation is 17s)	Signal Plausibility Failure	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Bank 1 Sensor 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P220093	(Internal status and error byte recieved from Upstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 1
EMS	P060741	(Internal Control Module Memory Check Sum Error)	General Checksum Failure	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Control Module Performance
EMS	P061511	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay
EMS	P061512	(Short circuit to battery is detected.)	Circuit Short To Battery	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay
EMS	P20EE00	((Filtered NOx conversion) below 71% and (A filter is applied to the NOx conversion that may require the evaluation to be repeated a few times to set the fault. Time required depends on engine load and the NOx flow. With all enable conditions fulfilled it normally takes no more than a few minutes.))	No Sub Type Information	Engine derate	Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	SCR NOx Catalyst Efficiency Below Threshold (Bank 1)
EMS	P260011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Coolant Pump Control Circuit/Open
EMS	P260012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Coolant Pump Control Circuit/Open
EMS	P260013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Coolant Pump Control Circuit/Open
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 40kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -40kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
EMS	P008700	(The rail pressure below 225bar for 1.5s	No Sub Type Information	Engine cranks but does not start	Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Any type of leakage Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail/System Pressure - Too Low Bank 1
EMS	P009A00	((Difference between Sensed Boost Temperature and Modeled Boost Temperature) below -40°C or (Difference between Sensed Boost Temperature and Modeled Boost Temperature) above 40°C)	No Sub Type Information	None	Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Temperature sensor drifting Temperature sensor biased high or low Temperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Air Intake Temperature - Correlation
EMS	P011001	(Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011011	Voltage on A:47 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011013	Voltage on A:47 above 4.910889V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P021700	EngineCoolantTemperature above (4.199999809265137No Unit + 105No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition
EMS	P047001	((Voltage on B:12 above 4.787598V and Voltage on B:12 below 4.852295V) or (Voltage on B:12 above 0.147705V and Voltage on B:12 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047012	Voltage on B:12 above 4.852295V	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047013	Voltage on B:12 below 0.147705V	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Pressure Sensor "A"
EMS	P047064	((Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) below -45kPa or (Difference beteween Sensed Exhaust Back Pressure and Modeled Exhaust Back Pressure) above 45kPa)	Signal Plausibility Failure	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Sensor "A"
EMS	P048011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit
EMS	P048012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P051A01	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or (Voltage on B:28 above 0.147705V and Voltage on B:28 below 0.307617V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A12	Voltage on B:28 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A13	Voltage on B:28 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051A64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Crankcase Pressure Sensor
EMS	P052400	EngineOilPressure below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low
EMS	P203164	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 2) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 2 below 150°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 2 Not Mounted Exhaust Gas Temperature Sensor 2 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
EMS	P222601	(Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P222612	Voltage on N/A:ECU internal above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222613	Voltage on N/A:ECU internal below 0.117188V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222664	(ECUAmbientPressure below 55kPa or ECUAmbientPressure above 114kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Circuit
EMS	P226411	Voltage on B:8 below 0.211182V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor
EMS	P244A00	(Measured differential pressure as percentage of modeled differential pressure) below (55 %)	No Sub Type Information	None	DPF removed or severely damaged Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF removed or severely damaged Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too Low (Bank 1)
EMS	P244B00	(Measured differential pressure as percentage of modeled differential pressure) above (350 %)	No Sub Type Information	None	DPF melted DPF filled with soot Hose to differential pressure sensor not mounted	Diagnostic Service 14. No additional action is needed.	Check: DPF melted DPF filled with soot Hose to differential pressure sensor not mounted If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Too High (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P245264	(((Failure event for diagnosis in AfterRun) and Particulate Filter Differential Pressure above 1kPa) or ((Failure event for diagnosis during low load) and Particulate Filter Differential Pressure above 1kPa) or ((Failure event for diagnosis during high load) and Particulate Filter Differential Pressure below 1.299999952316284 kPa))	Signal Plausibility Failure	None	Sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1
EMS	P250F00	((If calculated value for oil level) below 0% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255601	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255611	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255612	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255613	(Voltage on B:23 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P256000	EngineCoolantLevel below (100No unit - 75No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P207F9A	(((Estimated reductant concentration) below 28% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU
EMS	P229E12	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Short To Battery	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P229E64	(((The sensor shall respond to tip-in events (sudden power increase). Each time it does not respond correctly a timer is started. When this accumulated time has reached its time limit 300s a second phase is started with the purpose to confirm the suspected fault. This second phase is using the same method, but this time the reductant delivery is stopped in order to improve chances for the outlet NOx sensor to respond to the tip-in events. A second cumulative timer is started every time	Signal Plausibility Failure	None	Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor	NOx Sensor Bank 1 Sensor 2
EMS	P229E93	(Internal status and error byte recieved from Downstream NOx sensor not evaluated as GOOD within a reasonable time)	No Operation	None	Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. Incorrect Start Up Behaviour	NOx Sensor Bank 1 Sensor 2
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P052001	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or (Voltage on B:11 above 0.147705V and Voltage on B:11 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052012	Voltage on B:11 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052013	Voltage on B:11 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"
EMS	P052064	(EngineOilPressure above 100kPa or EngineOilPressure above 550kPa or EngineOilPressure below 350kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "A"
EMS	P019001	(Voltage on A:19 above 0.147705V and Voltage on A:19 below 0.447998V)	General Electrical Failure	Low power	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019011	Voltage on A:19 below 0.147705V	Circuit Short To Ground	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019013	Voltage on A:19 above 4.877930V	Circuit Open	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Fuel Rail Pressure Sensor Bank 1
EMS	P019062	(Rail fuel pressure fluctuations less than expected.)	Signal Compare Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the sensor wiring harness Pressure sensor Malfunctioning Mprop if applicable If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1
EMS	P019064	(Rail pressure before engine start. above 100bar or Rail pressure before engine start. below -50bar)	Signal Plausibility Failure	Engine derate	Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor connector and/or pins corrosive Damaged insulation on the wiring harness Pressure sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Sensor Bank 1
EMS	P066601	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P066611	Voltage on N/A:ECU internal below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P009B11	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine derate	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Pressure Relief Control
EMS	P009B12	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine derate	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Pressure Relief Control
EMS	P009B13	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Engine derate	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Fuel Pressure Relief Control
EMS	P020112	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 1 Injector "A"
EMS	P020113	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 1 Injector "A"
EMS	P020312	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 3 Injector "A"
EMS	P020313	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 3 Injector "A"
EMS	P020412	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 4 Injector "A"
EMS	P020413	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 4 Injector "A"
EMS	P020512	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 5 Injector "A"
EMS	P020513	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 5 Injector "A"
EMS	P020612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 6 Injector "A"
EMS	P020613	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 6 Injector "A"
EMS	P250A11	Voltage on B:4 below 0.05V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P249F92	Exhaust Gas Temperature Sensor 1 below (240 C)	Performance or Incorrect Operation	None	Exhaust Temperature Too Low For Particulate Filter Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Temperature Too Low For Particulate Filter Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control
EMS	P244C00	Average Particulate Filter Temperature below (250 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)
EMS	P24A000	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Maximum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low.	Diagnostic Service 14. No additional action is needed.	Check: Regeneration Target Temperature Cannot be Reached. Exhaust Aftertreatment Fuel Injecton Too Low. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low
EMS	P020212	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Cylinder 2 Injector "A"
EMS	P020213	(Current driving the injector rises to slow)	Circuit Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 2 Injector "A"
EMS	P24A100	(Closed Loop Exhaust Aftertreatment Fuel Injecton Control At Minimum Limit 0.899999976158142 ratio of the time. Regeneration Temperature Too Low.)	No Sub Type Information	None	Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too High During Particulate Filter Regeneration. Exhaust Aftertreatment Fuel Injecton Too High. If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop DPF Regeneration Control At Limit - Temperature Too High
EMS	P24A200	((Particulate Filter Differential Pressure above 5h))	No Sub Type Information	None	Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor.	Diagnostic Service 14. No additional action is needed.	Check: Temperature Too Low During Particulate Filter Regeneration. Particulate Filter Differential Pressure Sensor. If fault not detected: Fault trace according to Tech Tool fault tree.	Diesel Particulate Filter Regeneration Incomplete (Bank 1)
EMS	P00C600	(The rail pressure during cranking phase below 300bar for 15s	No Sub Type Information	Engine cranks but does not start	Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit	Diagnostic Service 14. No additional action is needed.	Check: Injector leakage High pressure pipe leakage Inlet metering valve with connector/harness damaged Bad fuel quality Air leakage in low pressure system Empty fuel tank High pressure pump damaged Low fuel pressure circuit If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Rail Pressure Too Low - Engine Cranking

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P009400	(During engine shutdown, an abnormal fuel pressure drop in the rail is observed during evaluation period.	No Sub Type Information	None	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System Rail Fuel Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel System Leak Detected - Small Leak
EMS	P016E00	(Average rail pressure deviation above 100bar for 5s	No Sub Type Information	Engine derate	Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed	Diagnostic Service 14. No additional action is needed.	Check: Rotated Cam profile Leaking Fuel Pressure Release Valve Fuel Pressure Regulator Low Fuel Pressure System Bad fuel quality Pressure release valve stuck closed If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too High
EMS	P016F00	(Average rail pressure deviation below -100bar for 5s	No Sub Type Information	Engine derate	Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Inlet metering valve with connector/harness damaged Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Closed Loop Fuel Pressure Control At Limit - Pressure Too Low
EMS	P025612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control B
EMS	P025614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control B
EMS	P025692	(Loss of high fuel pressure pumping capacity	Performance or Incorrect Operation	Low power	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control B
EMS	P228F00	(Fuel pressure controller adjustment value saturated high	No Sub Type Information	Engine derate	Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve Leaking High Pressure Fuel System High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System, fuel supply If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High
EMS	P228E00	(Fuel pressure controller adjustment value saturated low	No Sub Type Information	None	Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: Leaking Fuel Pressure Release Valve High Fuel Pressure Pump Fuel Pressure Regulator Low Fuel Pressure System Rotated Cam profile If fault not detected: Fault trace according to Tech Tool fault tree.	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P100101	(Voltage on A:27 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:27 above (20.303109V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:27 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (18.135422V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100111	(Voltage on A:27 above (0.731483V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100112	(Voltage on A:27 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:27 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine stop switch
EMS	P100190	((Stop button is active during start-up) or (Stop button becomes active when vehicle speed > 0))	Stuck On	None	Problems with plausibility check of the stop button, stop button stuck.	Diagnostic Service 14. No additional action is needed.	Check: Problems with plausibility check of the stop button, stop button stuck.	Engine stop switch
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100701	(Voltage on A:29 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:29 above (19.589467V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (12.194354V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (4.148043V *	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100711	Voltage on A:29 below (1.587853V * (UnfilteredBatteryVoltage / 24.0V))	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100712	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P100713	(Voltage on A:29 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (19.589467V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019511	Voltage on A:31 below 0.100098V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P019564	((EngineOilTemperature below -39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Temperature Sensor "A"
EMS	P203A86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A"
EMS	P024311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024312	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P024313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Turbocharger/Supercharger Wastegate Solenoid "A"
EMS	P20F400	(Calculated reductant consumption) below -50%	No Sub Type Information	None	Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	Diagnostic Service 14. No additional action is needed.	Check: Reductant level sensor stuck Clogged reductant delivery system Incompatible reductant tank installed	AdBlue Consumption Too Low
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P113711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Oil Thermostat
EMS	P113712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Oil Thermostat
EMS	P113713	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Oil Thermostat
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 14% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition
EMS	P115211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fuel Priming Pump
EMS	P115212	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fuel Priming Pump
EMS	P115213	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fuel Priming Pump

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P115100	(Temperature in the aftertreatment system) above (999No Unit + 540No unit)	No Sub Type Information	Engine derate	Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions may result in high temperatures. High soot load in aftertreatment diesel particulate filter may result in high temperatures. Exhaust Gas Temperature Sensor Bank 1 Sensor 1 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 2 may be faulty. Exhaust Gas Temperature Sensor Bank 1 Sensor 3 may be faulty.	Aftertreatment System Over Temperature
EMS	P2A1612	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1614	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1692	(Loss of high fuel pressure pumping capacity P	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "D" (Cam/Rotor/Injector)
EMS	P2A1E12	(Current driving the injector rises to quickly)	Circuit Short To Battery	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E14	(Current driving the injector rises to slow)	Circuit Short To Ground or Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P2A1E92	(Loss of high fuel pressure pumping capacity P	Performance or Incorrect Operation	Engine derate	High Fuel Pressure Pump Rotated Cam profile	Diagnostic Service 14. No additional action is needed.	Check: High Fuel Pressure Pump Rotated Cam profile	Injection Pump Fuel Metering Control "F" (Cam/Rotor/Injector)
EMS	P008B00	((AftertreatmentFuel Pressure above (700kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P117C00	(The selected gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Range Gear Too High
EMS	P117D00	(The selected gear is too low.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Gear Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P117E00	(The selected starting gear is too high.)	No Sub Type Information	None	Improper gear selection	Diagnostic Service 14. No additional action is needed.	Check: Improper gear selection	Driver Requested Launch Gear Too High
EMS	P047511	(Short circuit to ground is detected.)	Circuit Short To Ground	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Pressure Control Valve "A"
EMS	P047512	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Exhaust Pressure Control Valve "A"
EMS	P047513	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Exhaust Pressure Control Valve "A"
EMS	P047572	((Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure without Engine Brake) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)) or (Difference between Sensed Exhaust Manifold Pressure and Modeled Exhaust Manifold Pressure with VCB/EPG activated) below (20kPa + (Offset dependent on ECUAmbientPressure and Engine Speed)))	Actuator Stuck Open	Auxiliary brakes, low performance (Reduced engine braking)	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Pressure Control Valve "A"
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P011501	(Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011511	Voltage on B:27 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P011513	Voltage on B:27 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P01152A	(Difference between and maximum and minimum coolant temperature) below 3°C	Signal Stuck In Range	None	Coolant Temperature sensor signal stuck in operating range	Diagnostic Service 14. No additional action is needed.	Check: Coolant Temperature sensor signal stuck in operating range If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Coolant Temperature Sensor 1
EMS	P012800	(EngineCoolantTemperature) below 70°C	No Sub Type Information	Engine does not reach normal operating temperature	Coolant Thermostat leaking or blocked open	Diagnostic Service 14. No additional action is needed.	Check: Coolant Thermostat leaking or blocked open If fault not detected: Fault trace according to Tech Tool fault tree.	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)
EMS	P04009C	(Difference between Sensed Boost Temperature and boost temperature stored during load/engine speed/vehicle speed operational conditions with closed EGR Valve) below (12C - (Offset dependent on Ambient Air Temperature))	Low/Insufficient Flow	None	EGR Valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: EGR Valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Recirculation "A" Flow
EMS	P026A00	(Ratio between Modeled CaC Temp and difference between Modeled Deteriored and Modeled Nominal CaC Temp) below 0.25ratio	No Sub Type Information	None	Deteriorated Charge Air Cooler	Diagnostic Service 14. No additional action is needed.	Check: Deteriorated Charge Air Cooler If fault not detected: Fault trace according to Tech Tool fault tree.	Charge Air Cooler Efficiency Below Threshold
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit
EMS	P269711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P269773	(AftertreatmentFuel Pressure above 150kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P26977A	AftertreatmentFuelPressure below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD01	((Voltage on A:50 above 4.787598V and Voltage on A:50 below 4.852295V) or (Voltage on A:50 above 0.147705V and Voltage on A:50 below 0.212402V))	General Electrical Failure	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD12	Voltage on A:50 above 4.852295V	Circuit Short To Battery	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20DD13	Voltage on A:50 below 0.147705V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	P01B901	(Voltage on B:32 above 0.147705V and Voltage on B:32 below 0.231934V)	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P01B911	Voltage on B:32 below 0.147705V	Circuit Short To Ground	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P01B913	Voltage on B:32 above 4.913330V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "B"
EMS	P242A64	(((Evaluation Performed During Engine Running:) and ((Difference between Exhaust Gas Temperature Sensor 1 and Sensor 3) above 70°C or (Difference between Exhaust Gas Temperature Sensor 2 and Sensor 3) above 65°C or Exhaust Gas Temperature Sensor 3 below 150°C)))	Signal Plausibility Failure	None	Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Gas Temperature Sensor 3 Not Mounted Exhaust Gas Temperature Sensor 3 Broken If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P040311	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P06B016	Voltage on A:7 below 4.5V	Circuit Voltage Below Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06B017	Voltage on A:7 above 5.5V	Circuit Voltage Above Threshold	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "A"
EMS	P06B316	Voltage on B:17 below 4.5V	Circuit Voltage Below Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06B317	Voltage on B:17 above 5.5V	Circuit Voltage Above Threshold	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "B"
EMS	P06E616	Voltage on A:1 below 4.5V	Circuit Voltage Below Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P06E617	Voltage on A:1 above 5.5V	Circuit Voltage Above Threshold	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C"
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation
EMS	P20D711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D773	(AftertreatmentFuel Pressure below 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control
EMS	P010501	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or (Voltage on A:22 above 0.147705V and Voltage on A:22 below 0.297852V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010512	Voltage on A:22 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010513	Voltage on A:22 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010562	ExhaustManifoldPressure above 50kPa	Signal Compare Failure	None	Malfunctioning Exhaust Manifold Pressure Sensor	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Exhaust Manifold Pressure Sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P25AE00	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	Engine derate	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P25A911	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Piston Cooling Oil Control
EMS	P25A912	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Piston Cooling Oil Control
EMS	P25A913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Piston Cooling Oil Control
EMS	P25A971	(When piston cooling valve is forced open and closed the oil pressure change) below -100kPa	Actuator Stuck	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control
EMS	P054001	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (19.616229V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (22.747332V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (18.893666V * (UnfilteredBatteryVoltage / 24.0V))))	General Electrical Failure	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054011	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Short To Ground	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control
EMS	P054013	(Voltage on B:7 above (18.893666V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (19.616229V * (UnfilteredBatteryVoltage / 24.0V)))	Circuit Open	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P05407F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P225C00	((Upstream NOx level) above 200ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	No Sub Type Information	None	Upstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx sensor	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 1
EMS	P225E67	((Downstream NOx level) above 200ppm and (A filter is applied to the NOx level that may require the evaluation to be repeated a few times to set the fault.))	Signal Incorrect After Event	None	Downstream NOx sensor Excessive reductant delivery causing NH3 slip	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx sensor Excessive reductant delivery causing NH3 slip	NOx Sensor Performance - Signal Stuck High Bank 1 Sensor 2
EMS	P225F00	(The Downstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Downstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P225D00	(The Upstream NOx Sensor considered removed from the exhaust system when the sensor measure too high lambda value or too low NOx value during high engine speed and high load. The fault evaluation need to be repeated for a number of times to take a decission, and report the IDTC)	No Sub Type Information	None	Upstream NOx Sensor has been removed from exhaust flow	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor has been removed from exhaust flow	NOx Sensor Performance - Signal Stuck Low Bank 1 Sensor 1
EMS	P10E100	Particulate Filter Differential Pressure above (24No unit + 0No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (24No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High
EMS	P22FB92	(Internal status and error byte recieved from Upstream NOx sensor evaluated as bad sensor quality)	Performance or Incorrect Operation	None	Faulty Upstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Upstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1
EMS	P22FE00	(Internal status and error byte recieved from Downstream NOx sensor evaluated as bad sensor quality)	No Sub Type Information	None	Faulty Downstream NOx Sensor. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream NOx Sensor. (Internal sensor fault)	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P055A01	((Voltage on B:19 above 4.787598V and Voltage on B:19 below 4.852295V) or (Voltage on B:19 above 0.147705V and Voltage on B:19 below 0.212402V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A12	Voltage on B:19 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A13	Voltage on B:19 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055A64	((Difference between demanded piston cooling pressure and measured piston cooling pressure) between -15kPa and 15kPa or PistonCoolPressure above 50kPa)	Signal Plausibility Failure	None	Piston cooling pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch "B"
EMS	P02E011	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control
EMS	P02E012	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E07C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09B	(ThrottlePosition - Demanded Throttle Position) above 5%	High/Excessive Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P02E09C	(Demanded Throttle Position - ThrottlePosition) above 5%	Low/Insufficient Flow	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control
EMS	P112800	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	None	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Moderately Low
EMS	P112900	PFTransmissionOilTemp above (1000No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High
EMS	P029800	EngineOilTemperature above (125No unit + 2No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature
EMS	P02E601	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or (Voltage on B:24 above 0.147705V and Voltage on B:24 below 0.247803V))	General Electrical Failure	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E612	Voltage on B:24 above 4.852295V	Circuit Short To Battery	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E613	Voltage on B:24 below 0.147705V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P02E697	(not ((Maximum Opened Throttle Position) between 78.3000030517578deg and 84.6999969482422deg) and not ((Minimum Closed Throttle Position) between -2.799999952316284deg and 2.799999952316284deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E699	(Calibration requested and no calibration performed)	Exceeded Learning Limit	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (1000No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	P111600	BoostPressure above (Output from look-up tables with input EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Moderately Too High
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus
EMS	P111800	BoostPressure above (Output from look-up tables with inputs EngineSpeed and ECUAmbientPressure)	No Sub Type Information	Engine derate	VGT is stuck Wastegate is stuck	Diagnostic Service 14. No additional action is needed.	Check: VGT is stuck Wastegate is stuck	Manifold Absolute Pressure Too High
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P111D00	EngineCoolantLevel below (100No unit - 30No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P111E00	EngineCoolantTemperature above (2No Unit + 105No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High
EMS	P112500	EngineOilTemperature above (125No unit + 0No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN
EMS	P008A00	((AftertreatmentFuel Pressure below (200kPa + (Offset dependent on fuel value and Engine Speed)) or AftertreatmentFuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed))) or FuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113600	((not ((Upstream NOx Sensor available in EMS))) or (not ((Downstream NOx Sensor available in EMS))))	No Sub Type Information	None	Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	Diagnostic Service 14. No additional action is needed.	Check: Wrong configuration in EMS. Upstream/Downstream NOx Sensors configurated in (control part) do not match setup in (diagnostic part)	NOx Sensor 1/2 Correlation Calibration Error Bank 1
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below -250kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range
EMS	P06DA71	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 120kPa	Actuator Stuck	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control
EMS	P113411	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113413	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113472	(AftertreatmentFuel Pressure above 50kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Open	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113473	(AftertreatmentFuel Pressure below 35kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	Actuator Stuck Closed	Activation failure, regeneration	No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening	Diagnostic Service 14. No additional action is needed.	Check: No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve
DM	U005588	CAN Security failure detected.	Bus off	Passenger Door Module does not communicate on CAN bus. One or more functions of Passenger Door Module are not working as expected.	1. Harness issue: - Short circuit between CAN High & CAN low; - Short circuit between CAN High & GND; - Short circuit between CAN Low & BAT; 2. Issue on one of the CAN nodes connected to Security subnet.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	CAN Communication Security Net
DM	U300005	Not supported SW configuration; Invalid parameters settings;	System Programming Failures	One or more functions of Passenger Door Module are not working as expected.	1. SW release issue; 2. Not supported truck configuration build; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U300041	Flash memory HW failure.	General Checksum Failure	MSW invalidation : DTC reported by Primary Boot Loader. Passenger Door Module are not working as expected.	SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U300044	RAM corruption detected.	Data Memory Failure	ECU reset.	1. Invalid SW processing; 2. RAM corruption (intermittent fault due to working environment); 3. RAM physical issue (continuous defect); SW or HW complex system issue, to be rised to back-office .	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U300045	Program memory corruption.	Program Memory Failure	MSW invalidation after ECU reset. Passenger Door Module are not working as expected.	1. SWDL invalid process (cut of power during SWDL, etc) 2. Flash memory physical defect; 3. Invalid SW operation; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U300046	Dataset memory corruption.	Calibration / Parameter Memory Failure	MSW invalidation after ECU reset.	1. Incorrect dataset download/change (e.g. cut of power during change). 2. Invalid SW operation (mainly in case of parameters change processing); 3. Flash memory physical defect; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U300049	Internal electronic failure.	Internal Electronic Failure	One or more functions of Passenger Door Module are not working as expected.	Internal electronic failure: - µC peripheric circuit invalid state; - LIN tranceiver circuit broken; - CAN tranceiver circuit broken; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U30004A	Power window function with antipinch activated on not valid HW. Electrical Power window function activated on not valid HW.	Incorrect Component Installed	Power window function is not working as expected.	Not correct component installed or not valid SW configuration downloaded. SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Contact back office.	Electronic control unit
DM	U300055	No specific SW dataset downloaded.	Not Configured	One or more functions of Passenger Door Module are not working as expected.	No valid SW configuration downloaded: - no valid LIN schedule table selected through dataset - no valid latch motor type selected through dataset - no valid window motor type selected through dataset - no valid mirror motor type selected through dataset SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Contact back office.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U300094	SW unexpected behaviour;	Unexpected Operation	ECU reset.	1. SW bug; - unexpected SW operation executed; - watchdog reset performed; - overrun situation detected; - insynchronized communication circuit state detected; 2. RAM corruption; 3. Flash corruption; SW or HW complex system issue, to be rised to back-office.	Diagnostic Service 14. No additional action is needed.	- Replace the Passenger Door Module Unit.	Electronic control unit
DM	U300316	Power supply voltage is lower than 16V (configurable).	Circuit Voltage Below Threshold	One or more functions of Passsenger Door Module are not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the Passenger Door Module Unit. - Inspect the harness and connectors.	Battery Power
DM	U300317	Power supply voltage is higher than 32V (configurable).	Circuit Voltage Above Threshold	One or more functions of Passsenger Door Module are not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. ECU HW issue;	Diagnostic Service 14. No additional action is needed.	- Inspect the battery - Inspect the Passenger Door Module Unit. - Inspect the harness and connectors.	Battery Power
DM	U1FFF68	Factory Mode is active in the ECU.	Event Information	No functional impact.	P1C54 parameter is set to TRUE.	Diagnostic Service 14. No additional action is needed.	Exit Factory mode.	Factory Mode Active
DM	U114668	Maximum number of DTC reached : 29 DTC.	Event Information	No new DTC set until the failure is removed or DTC list cleared.	1. Unexpected environment conditions; 2. ECU failure	Diagnostic Service 14. No additional action is needed.	- Clear All DTC.	DTC Notification Max Limit Reached
DM	B100D13	Both lock & unlock signals stay in Open Circuit state longer than allowed (1500 ms configurable) after electrical lock or unlock request.	Circuit Open	if Open Circuit is on lock sensor : -for a lock command, the latch will be lock and unlock 1s later. -for an unlock command,the latch will be unlock. if Open Circuit is on unlock sensor : -for a lock command, the latch will be lock. -for an unlock command, the latch will be unlock.	1. Harness issue; 2. Latch failure;	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch lock unlock sensors
DM	B100411	Output pin short circuit to GND.	Circuit Short To Ground	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, the latch will be unlock.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 1
DM	B100412	Output pin short circuit to BAT+.	Circuit Short To Battery	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, the latch will be unlock.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 1
DM	B100511	Output pin short circuit to GND.	Circuit Short To Ground	Locking function is not working as expected : - for Lock command, the latch will be lock - for Unlock command, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 2

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100512	Output pin short circuit to BAT+.	Circuit Short To Battery	Locking function is not working as expected : - for Lock command, the latch will be lock - for Unlock command, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch actuator circuit 2
DM	B150513	Open circuit conditions detected on output pin.	Circuit Open	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Actuator Circuit
DM	B100D11	Both lock & unlock signals stay in SC to ground state.	Circuit Short To Ground	Locking function is not working as expected : - for Lock command, it will not be executed. - for Unlock command, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door latch lock unlock sensors
DM	B150712	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is open. If sensor has blown, door will always be considered as closed; - if short circuit with protected output, this output will be affected when door is open (short to ground from concerned output perspective);	1. Harness issue; 2. Passenger Door Module failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Open Sensor
DM	B150812	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is locked. If sensor has blown, door will always be considered as unlocked. - if short circuit with protected output, this output will be affected when door is locked (short to ground from concerned output perspective);	1. Harness issue. 2. Passenger Door Module failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Lock Sensor
DM	B150912	Resident voltage level present when not expected by function.	Circuit Short To Battery	- if direct 24V short circuit with fuse, fuse or sensor will blow when door is unlocked. If sensor has blown, door will always be considered as locked. - if short circuit with protected output, this output will be affected when door is unlocked (short to ground from concerned output perspective);	1. Harness issue; 2. Passenger Door Module failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Latch. - Inspect the harness and connectors.	Door Latch Unlock Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150214	24V Power supply lost (Fault Code Indication received from Window Antipinch).	Circuit Short To Ground or Open	Power window function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors.	Window Antipinch Module
DM	B150216	Window Antipinch measures 24V power supply voltage lower than 18,5 V (Fault Code Indication reception from Window Antipinch).	Circuit Voltage Below Threshold	Power window function is not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors. - Inspect the Door seal ageing.	Window Antipinch Module
DM	B150217	Window Antipinch measures power supply voltage higher than 32V (Fault Code Indication received from Window Antipinch).	Circuit Voltage Above Threshold	Power window function is not working as expected.	1. Harness issue; 2. Power supply circuit issue; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors. - Inspect the Door seal ageing.	Window Antipinch Module
DM	B15021C	Window Antipinch measures 12V power supply voltage lower than 7.5V or higher than 18,5V (Fault Code Indication received from Window Antipinch).	Circuit Voltage Out of Range	Power window function is not working as expected.	1. Harness issue; 2. PDM 12V output failure; 3. Faulty Window Antipinch hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness and connectors.	Window Antipinch Module
DM	B150249	Window Antipinch internal fault (Fault Code Indication received from Window Antipinch)	Internal Electronic Failure	Power window function is not working as expected.	1. Internal uC failure detected; 2. Temperature sensor defective; 3. Hall sensor defective	Diagnostic Service 14. No additional action is needed.	- Replace the Window Antipinch Module.	Window Antipinch Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150254	Window position information is lost (Fault Code Indication received from Window Antipinch).	Missing Calibration	Power window function is not working as expected. Manual Up is working. Manual Down is working. Auto-Down is working. Auto-Up is not working.	1. During a window motor movement the 12V or 24V power supply was suddenly interrupted; 2. Internal Window Antipinch failure (Hall sensor error, Window moving while position counting not possible etc..); 3. Block event occured during reversal; 4. Window positon is unplausible; 5. Sealing ageing (that generated decalibration) 6. Change of door and glass type information sent over LIN during Antipinch run time.	Diagnostic Service 14. No additional action is needed.	- Perform the calibration (Press window up button until the window raise to the top and maintain it press for 3 seconds after that). - Inspect the Door seal ageing.	Window Antipinch Module
DM	B150255	No valid configuration information received from Passenger Door Module over LIN: - Glass Type - Door Type (Fault Code Indication received from Window Antipinch).	Not Configured	Power window function is not working as expected. Manual Up is working. Manual Down is working. Auto-Down is working. Auto-Up is not working.	1. Harness issue; 2. Glass and door configuration information sent over LIN not available or error; 3. Change of door and glass type information sent over LIN during Window Antipinch run time.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness and connectors. - Redownload the Door Module Software. - Contact back office.	Window Antipinch Module
DM	B150271	Internal Window Antipinch relay is stuck (Fault Code Indication received from Window Antipinch).	Actuator Stuck	Power window function is not working as expected.	1. Mechanical defect of the relay contacts.	Diagnostic Service 14. No additional action is needed.	- Replace the Window Antipinch Module. - Inspect the Door seal ageing.	Window Antipinch Module
DM	B150298	Window Motor is too hot according Window Antipinch overheating SW algorithm (LIN Fault Code Indication received from Window Antipinch).	Component or System Over Temperature	Window motor is inhibited by Window Antipinch. Power window function is not working as expected, but return automatically to normal operation once the motor has cooled down.	1. Motor thermal protection is activated because motor was actived during too long time.	Diagnostic Service 14. No additional action is needed.	- Cool down the motor during 20 minutes minimum (Do not use it). - Inspect the Door seal ageing.	Window Antipinch Module
DM	U114286	Invalid frame received from LIN slave.	Signal Invalid	Passenger Door Module Invalidates Window Antipinch signals.	LIN slave issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	Window Antipinch Link
DM	U114287	LIN slave not responding.	Missing Message	Passenger Door Module Invalidates Window Antipinch signals.	LIN slave issue. Harness issue. Invalid parameters setting.	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	Window Antipinch Link

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	U114488	Communication fault detected on the LIN bus.	Bus off	Passenger Door Module Invalidates Window Antipinch signals. Passenger Door Module invalidates the Passenger Control Panel signals.	1. Harness issue: short circuit to GND/BAT/12V Power Supply or LIN bus is out of operational range: [8V..18V]; 2. Issue on one of the LIN nodes;	Diagnostic Service 14. No additional action is needed.	- Inspect the harness or connectors.	LIN9 Communication Net
DM	U114586	Invalid frame received from LIN slave.	Signal Invalid	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	LIN slave issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Passenger Door Control Panel Link
DM	U114587	LIN slave not responding.	Missing Message	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	LIN slave issue. Harness issue. Invalid parameters setting.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Passenger Door Control Panel Link
DM	B150416	Door Control Panel measures power supply voltage lower than 7.5V(Fault Code Indication received from Door Control Panel).	Circuit Voltage Below Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue; 2. Passenger Door Module 12V output failure; 3. Faulty Door Control Panel hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness and connectors.	Passenger Door Control Panel
DM	B150417	Door Control Panel measures power supply voltage higher than 18.5V (Fault Code Indication received from Door Control Panel).	Circuit Voltage Above Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue; 2. Passenger Door Module 12V output failure; 3. Faulty Door Control Panel hardware that misdetect the supply level.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness and connectors.	Passenger Door Control Panel
DM	B150444	Door Control Panel RAM check fail (Fault Code Indication received from Door Control Panel).	Data Memory Failure	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	1. Door Control Panel slave SW issue. 2. Door Control Panel RAM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Passenger Door Control Panel
DM	B150445	Door Control Panel ROM check fail (Fault Code Indication received from Door Control Panel).	Program Memory Failure	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	1. Door Control Panel ROM physical corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Passenger Door Control Panel

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150446	Door Control Panel flash corruption detected during writing of calibration data (Fault Code Indication received from Door Control Panel).	Calibration / Parameter Memory Failure	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	1. Door Control Panel flash memory corruption.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Passenger Door Control Panel
DM	B150449	Door Control Panel Buttons configuration failure (Fault Code Indication received from Door Control Panel).	Internal Electronic Failure	All functions handles by Door Control Panel are not working as expected.	1. Door Control Panel End Of Line configuration not correct.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Passenger Door Control Panel
DM	B150494	Door Control Panel unexpected SW operation executed; Door Control Panel slave watchdog reset performed; Door Control Panel slave overrun situation detected; (Fault Code Indication received from Door Control Panel).	Unexpected Operation	Passenger Door Module invalidates the Passenger Control Panel signals. All functions handled by Passenger Control Panel are not working as expected.	1. Door Control Panel SW issue; 2. Door Control Panel MCU not correct.	Diagnostic Service 14. No additional action is needed.	- Replace the Door Control Panel.	Passenger Door Control Panel
DM	B100011	Output pin short circuit to GND.	Circuit Short To Ground	Turns off Door Control Panel 12V power supply; All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. Door Control Panel internal short circuit.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100012	Output pin short circuit to BAT+.	Circuit Short To Battery	Turns off Door Control Panel 12V power supply; All functions handled by Door Control Panel are not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100013	Open circuit conditions detected on output pin.	Circuit Open	All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. Door Contrl Panel internal failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100016	Feedback voltage measured on the output is lower than 11,4V threshold (configurable).	Circuit Voltage Below Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. Passenger Door Module internal HW issue. 3. Door Control Panel internal HW failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100017	Feedback voltage measured on the output is higher 14V threshold (configurable).	Circuit Voltage Above Threshold	All functions handled by Door Control Panel are not working as expected.	1. Harness issue. 2. Passenger Door Module internal HW issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Door Control Panel. - Inspect the harness or connectors.	Door Control Panel 12V power supply circuit
DM	B100111	Output pin short circuit to GND.	Circuit Short To Ground	Turns off Window Antipinch 12V power supply. Power window function is not working as expected.	1. Harness issue. 2.Antipinch internal short circuit.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100112	Output pin short circuit to BAT+.	Circuit Short To Battery	Turns off Window Antipinch 12V power supply. Power window function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100113	Open circuit conditions detected on output pin.	Circuit Open	Power window function is not working as expected.	1. Harness issue. 2. Antipinch internal failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100116	Feedback voltage measured on the output is lower than 11,4V threshold (configurable).	Circuit Voltage Below Threshold	Power window function is not working as expected.	1. Harness issue. 2. PDM internal HW issue. 3. Antipinch internal HW failure.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100117	Feedback voltage measured on the output is higher 14V threshold (configurable).	Circuit Voltage Above Threshold	Power window function is not working as expected.	1. Harness issue. 2. PDM internal HW issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Window Antipinch Module. - Inspect the harness or connectors.	Antipinch module 12V power supply circuit
DM	B100811	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Mirrors adjustment actuators common circuit
DM	B100812	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Mirrors adjustment actuators common circuit
DM	B100911	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100912	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100913	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit
DM	B100919	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror X adjustment actuator circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100A11	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A12	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A13	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100A19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Main Mirror Y adjustment actuator circuit
DM	B100B11	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B12	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B13	Open circuit conditions detected on output pin	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100B19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror X adjustment actuator circuit
DM	B100C11	Output pin short circuit to GND.	Circuit Short To Ground	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100C12	Output pin short circuit to BAT+.	Circuit Short To Battery	Mirror function is not working as expected.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100C13	Open circuit conditions detected on output pin.	Circuit Open	Mirror function is not working as expected.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100C19	Feedback load current measured on the output is higher than defined threshold. Threshold depends on multiple criteria (voltage, temperature and motor type).	Circuit Current Above Threshold	Mirror function is not working as expected. ECU switches off main and wide mirror motors outputs for 60 seconds.	1. Mirror actuator issue; 2. Mirror mechanical issue (anormal resistance).	Diagnostic Service 14. No additional action is needed.	- Inspect the mirrors. - Inspect the harness and connectors.	Wide Mirror Y adjustment actuator circuit
DM	B100211	Output pin short circuit to GND.	Circuit Short To Ground	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp mounting issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100212	Output pin short circuit to BAT+.	Circuit Short To Battery	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100213	Open circuit conditions detected on output pin.	Circuit Open	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door instep lamp circuit
DM	B100311	Output pin short circuit to GND.	Circuit Short To Ground	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp mounting issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit
DM	B100312	Output pin short circuit to BAT+.	Circuit Short To Battery	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit
DM	B100313	Open circuit conditions detected on output pin.	Circuit Open	Switch off output until : 1. next Vehicle Mode change from lower than Pre-Running to higher or equal to PreRunning 2. New activation request.	1. Harness issue. 2. Lamp blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the lamp. - Inspect the harness and connectors.	Door warning lamp circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B100611	Output pin short circuit to GND.	Circuit Short To Ground	Power window function is not working as expected. - For window down request, window will be driven down. - For window up request, it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 1
DM	B100612	Output pin short circuit to BAT+.	Circuit Short To Battery	Power window function is not working as expected. - For window down request, it will not be executed. - For window up request, window will driven up.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 1
DM	B100711	Output pin short circuit to GND.	Circuit Short To Ground	Power window function is not working as expected. - For window down request, it will not be executed. - For window up request, window will driven up.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 2
DM	B100712	Output pin short circuit to BAT+.	Circuit Short To Battery	Power window function is not working as expected. - For window down request, window will be driven down. - For window up request, it will not be executed.	1. Harness issue.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the harness and connectors.	Power window actuator circuit 2
DM	B150613	Open circuit conditions detected on output pin.	Circuit Open	Power window function is not working as expected. - For down command, it will not be executed. - For up command,it will not be executed.	1. Harness issue. 2. Actuator motor blown.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor
DM	B150618	Feedback load current measured on the output is lower than defined threshold. Threshold is dynamic and based on multicriteria formula (voltage, temperature and motor type).	Circuit Current Below Threshold	Power window function is not working as expected.	1. Anormal window mechanical resistance, 2. Power motor issue (Thermal switch open, etc), 3. Door sealing ageing.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
DM	B150619	Feedback load current measured on the output is higher than defined threshold. Threshold is dynamic and based on multicriteria formula (voltage, temperature and motor type).	Circuit Current Above Threshold	Power window function is not working as expected.	1. Anormal window mechanical resistance, 2. Power motor issue, 3. Door sealing ageing.	Diagnostic Service 14. No additional action is needed.	- Inspect the Power Window Motor. - Inspect the Door seal ageing. - Inspect the harness and connectors.	Power Window Motor
WRG	B150224	Button stuck in WRC for more than 3 mins	Signal Stuck High	* Not able to use the button that has been stuck * Not possible to control ECS functions	Button stuck in WRC	Diagnostic Service 14. No additional action is needed.	1. If all buttons working OK - Clear DTC 2. If not all buttons working - replace WRC	Wireless Remote Control
WRG	B150244	Bit fault in RAM	Data Memory Failure	WRC stays in boot, displays red screen	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150245	Bit fault in RAM	Program Memory Failure	WRC stays in boot, displays red screen	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150246	EEProm memory failure	Calibration / Parameter Memory Failure	WRC starts application and report error, possible loss of user data (e.g. backlight setting) WRG function as usual	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150249	Radio chip do not answer commands, MSP failure	Internal Electronic Failure	Application will run normally but without radio. WRG will not get any connection to the WRC	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B15024B	At start up	Over Temperature	WRC is shut off after displaying a message on the display. It is possible to start the WRC but it will be shut off if the temperature still is too high. Return to normal functionality when temperature is below threshold and no components are damaged	Battery temperature to high	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If the WRC is working normally - do nothing 3. If there is damages to battery - replace battery 4. If there is damages to display - replace WRC	Wireless Remote Control
WRG	B150294	Watchdog timeout, software assert	Unexpected Operation	Reset of WRC WRG not affected	Software error	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If it's set periodical - replace WRC	Wireless Remote Control
WRG	B150393	Battery supervising circuit reporting a faulty battery	No Operation	WRC will not function other than docked to WRG and in a power mode that support WRG 5V output to be enabled.	Battery failure	Diagnostic Service 14. No additional action is needed.	1. Make sure the battery is connected 2. Replace battery	WRC Battery
WRG	U003711	CAN short to GND	Circuit Short To Ground	No communication possible. Will report the DTC when communication comes back No affect on application	CAN short to GND	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
WRG	U003712	CAN short to BATT	Circuit Short To Battery	No communication possible. Will report the DTC when communication comes back No affect on application	CAN short to BATT	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net
WRG	U003788	CAN controller Bus Off	Bus off	No communication possible. Will report the DTC when communication comes back No affect on application	Bad wiring, bad connectors	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net
WRG	U300041	Checksum error	General Checksum Failure	Stay/Jump to BOOT	Missing software	Diagnostic Service 14. No additional action is needed.	1. Try to download complete SW 2. Replace WRG	Electronic control unit
WRG	U300044	Bit fault in RAM	Data Memory Failure	Application will start up and report error Could not guarante correct operation of WRG	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300046	Failing write to memory	Calibration / Parameter Memory Failure	* No functional effect * Possible loss of DTC information	Hardware fault, outside temperatue or voltage specification	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300049	Radio chip do not answer commands	Internal Electronic Failure	Application will run normally but without radio. WRC will not get any connection to the WRG	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300094	Watchdog timeout, software assert	Unexpected Operation	Reset of WRG	Software error	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If it's set periodical - replace WRG	Electronic control unit
WRG	U300316	U<8V	Circuit Voltage Below Threshold	Application running as normal 5V output (used for charging the WRC battery) is turned off when U<16V ECU will shut down if U<7V (All ratings are approximate)	Input voltage below 8V	Diagnostic Service 14. No additional action is needed.	1. Problem with wire harness? 2. Problem with power transformation?	Battery Power
WRG	U300317	U>32V	Circuit Voltage Above Threshold	Application running as normal 5V output (used for charging the WRC battery) is turned off when U>36V (All ratings are approximate)	Input voltage over 32V	Diagnostic Service 14. No additional action is needed.	1. Problem with wire harness? 2. Problem with power transformation?	Battery Power
WRG	U100068	More than 29 DTC with "warning indicator bit"	Event Information	None	More than 29 DTC with "warning indicator bit"	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. Investigate the other DTCs	DTC Notification Max Limit Reached

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Auxiliary brakes, low performance (Reduced retarder braking)	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
ACM	U010000	(Lost communication with ECM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	ECM not powered Software incompatibility with ECM	Diagnostic Service 14. No additional action is needed.	Check: ECM not powered Software incompatibility with ECM	Lost Communication with EMS
ACM	P054415	(Voltage on A:42 above 2.260742V or (Voltage on A:42 above 0.147705V and Voltage on A:42 below 0.677490V))	Circuit Short To Battery or Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 1
ACM	P205A13	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 5	Circuit Open	None	Open circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Tank Temperature Sensor
ACM	P205B64	((Reductant Tank Temperatureshall increase by at least 5°Cduring evaluation time) and ((Reductant Temperature below -4.5°C or UreaTankTemp below -9.5°C) and (EngineSpeed above 500rpm and Control Module between 16V and 32V and UreaTankTemp below 68°C)) and not ((Reductant Temperatureshall increase by at least 5°Cduring evaluation time)))	Signal Plausibility Failure	None	Tanktemperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Tanktemperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Tank Temperature Sensor Circuit Range/Performance
ACM	P20BA00	((Defrosting has been active for 10800s) or (Number of tries to start the pump 1000No Unithas been reached))	No Sub Type Information	None	Broken heaters Extremely cold weather conditions Low coolant flow	Diagnostic Service 14. No additional action is needed.	Check: Broken heaters Extremely cold weather conditions Low coolant flow If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Heater A Control Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P066800	Voltage on N/A:ECU internal below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
ACM	P06B100	Voltage on A:50 below 4.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply A Circuit Low
ACM	P06B200	Voltage on A:50 above 5.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply A Circuit High
ACM	P06B400	Voltage on A:51 below 4.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply B Circuit Low
ACM	P06B500	Voltage on A:51 above 5.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply B Circuit High
ACM	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
ACM	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
ACM	P203115	(Voltage on A:34 above 2.260742V or (Voltage on A:34 above 0.147705V and Voltage on A:34 below 0.677490V))	Circuit Short To Battery or Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 2
ACM	P203200	Voltage on A:34 below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Exhaust Gas Temperature Circuit Low (Bank 1 Sensor 2)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P245213	Voltage on A:26 above 4.901123V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Particulate Filter Pressure Sensor "A"
ACM	P245400	Voltage on A:26 below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment 1 Diesel Particulate Filter Differential Pressure Circuit Low
ACM	P204F94	(Start engine and wait 1140s)	Unexpected Operation	None	Pump malfunction Blocked suction line Leakage on sunction or pressure line	Diagnostic Service 14. No additional action is needed.	Check: Pump malfunction Blocked suction line Leakage on sunction or pressure line If fault not detected: Fault trace according to Tech Tool fault tree.	SCR System Performance Dosing Failure
ACM	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU
ACM	P208E00	(Pump motor speed has not increased with 1.5-in 80% of 80-)	No Sub Type Information	None	Clogging in reductant injector Pressure line blocked	Diagnostic Service 14. No additional action is needed.	Check: Clogging in reductant injector Pressure line blocked If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Dosing Valve Clogged
ACM	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
ACM	P066613	(Voltage on N/A:ECU internal above 2.874756V or (Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.237793V))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
ACM	P202D00	(Pump duty cycle) above (60% + (Offset dependent on Reductant Tank Temperature))	No Sub Type Information	None	Leakage on the pressure line Pump error	Diagnostic Service 14. No additional action is needed.	Check: Leakage on the pressure line Pump error If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Pressure Leakage

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P204A13	Voltage on A:30 above 4.887695V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pressure Sensor
ACM	P205C00	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 4	No Sub Type Information	None	Short circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Short circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Tank Temperature Short Circuit Low
ACM	P208A13	(Open circuit is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pump "A" Control
ACM	P204B64	ACM_UreaPressure below 100kPa	Signal Plausibility Failure	None	Faulty reductant pressure sensor	Diagnostic Service 14. No additional action is needed.	Check: Faulty reductant pressure sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent PumpOrifice Orifice Blocked
ACM	P054500	Voltage on A:42 below 0.147705V	No Sub Type Information	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Short Circuit Low
ACM	P20B297	(Reductant Tank Temperature below 10°C and ((Reductant Temperature below -4.5°C or UreaTankTemp below -9.5°C) and (EngineSpeed above 500rpm and Control Module between 16V and 32V and UreaTankTemp below 68°C)) and (Reductant Tank Temperatureshall increase by at least 5°Cduring evaluation time) and (Reductant Temperatureshall increase by at least 5°Cduring evaluation time))	Component or System Operation Obstructed or Blocked	None	Pumptemperature sensor broken	Diagnostic Service 14. No additional action is needed.	Check: Pumptemperature sensor broken If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Heater Coolant Control Valve Performance/Stuck Open
ACM	P20B400	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Reductant Heater Coolant Control Valve Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P20B300	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Heater Coolant Control Valve Circuit Low
ACM	P20B113	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater Coolant Control Valve Control
ACM	P208B00	((Error state indicates that a severe error has occurred which prohibits the system from running) and (Error state is confirmed when the pump has failed to run after 5No Unitwithin300s))	No Sub Type Information	None	Damaged pump Broken pump motor inside pumpunit	Diagnostic Service 14. No additional action is needed.	Check: Damaged pump Broken pump motor inside pumpunit If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pump "A" Control Performance/Stuck Off
ACM	P204900	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector short circuited to battery. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Dosing Valve Short Circuit High
ACM	P204800	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Dosing Valve Short Circuit Low
ACM	P204713	(Open circuit or short circuit to battery is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector. Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Harness, actuator or connector is short circuited to battery voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Injection Valve Control Bank 1 Unit 1
ACM	P20C400	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Aftertreatment Reagent Hose Heater 3 Short Circuit High
ACM	P20C300	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Aftertreatment Reagent Hose Heater 3 Short Circuit Low
ACM	P20C113	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater "C" Control
ACM	P20BC00	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Aftertreatment Reagent Hose Heater 1 Short Circuit High
ACM	P20BB00	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Aftertreatment Reagent Hose Heater 1 Short Circuit Low
ACM	P20B913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Reductant Heater "A" Control
ACM	P20A300	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Direction Valve Short Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P20A200	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Direction Valve Short Circuit Low
ACM	P20A013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Purge Control Valve
ACM	P20A107	((Turn key on) and (A fault has been set in previous driving cycle))	Mechanical Failures	None	Reverting valve stuck	Diagnostic Service 14. No additional action is needed.	Check: Reverting valve stuck If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Direction Valve Mechanical Problem
ACM	P065900	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to battery voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	ACM Actuator High Supply 1 Short Circuit High
ACM	P065800	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage A" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage A" is short circuited. If fault not detected: Fault trace according to Tech Tool fault tree.	ACM Actuator High Supply 1 Short Circuit Low
ACM	P208D00	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector short circuited to battery. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Pump Control Short Circut High
ACM	P208C00	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Pump Control Short Circuit Low
ACM	P203A13	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 5	Circuit Open	None	Open circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Sensor "A"
ACM	P056300	not (Control Module above 36V)	No Sub Type Information	None	Bad pins or connection. Cabling issues. Generator issues.	Diagnostic Service 14. No additional action is needed.	Check: Bad pins or connection. Cabling issues. Generator issues. If fault not detected: Fault trace according to Tech Tool fault tree.	ECU Battery Potential Above Range
ACM	P056200	not (Control Module below 8V)	No Sub Type Information	None	Bad pins or connection. Cabling issues. Generator issues.	Diagnostic Service 14. No additional action is needed.	Check: Bad pins or connection. Cabling issues. Generator issues. If fault not detected: Fault trace according to Tech Tool fault tree.	ECU Battery Potential Below Range

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P204C00	Voltage on A:30 below 0.112305V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Pressure Sensor Circuit Low
ACM	P203C00	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 4	No Sub Type Information	None	Short circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Short circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Aftertreatment Reagent Level Short Circuit Low
ACM	P20E892	ACM_UreaPressure above 650kPa	Performance or Incorrect Operation	None	Empty reagent tank Leakage in suction line Broken pump	Diagnostic Service 14. No additional action is needed.	Check: Empty reagent tank Leakage in suction line Broken pump If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pressure Too Low
ACM	P21CA00	not (Control Module between 16V and 32V)	No Sub Type Information	None	Cabling issues Alternator malfunction	Diagnostic Service 14. No additional action is needed.	Check: Cabling issues Alternator malfunction If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Control Module Supply Voltage
ACM	P10AD13	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pump "A" Control Low Side
ACM	P10AE00	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pump "A" Control Low Side Circuit Low
ACM	P10AF00	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage. If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Pump "A" Control Low Side Circuit High
ACM	P10CE97	ACM_UreaPressure above 1100kPa	Component or System Operation Obstructed or Blocked	None	Reductant return line blocked Orifice blocked	Diagnostic Service 14. No additional action is needed.	Check: Reductant return line blocked Orifice blocked If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Return No Flow Detected
ACM	P26E800	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage D" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage D" is short circuited. If fault not detected: Fault trace according to Tech Tool fault tree.	Actuator Supply Voltage "D" Circuit Low
ACM	P24FF00	Reductant Tank Temperature above 70°C	No Sub Type Information	None	Coolant valve stuck open Tank temperature sensor stuck high	Diagnostic Service 14. No additional action is needed.	Check: Coolant valve stuck open Tank temperature sensor stuck high If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Temperature Too High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
ACM	P10D100	(Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 2 or Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 9 or Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 12 or Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 13 or Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 19)	No Sub Type Information	None	UQS sensor faulty	Diagnostic Service 14. No additional action is needed.	Check: UQS sensor faulty	Reductant Quality Sensor Circuit - Component Internal Failure
ACM	P206C00	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 4	No Sub Type Information	None	Short circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Short circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Quality Sensor Low
ACM	P206A13	Failure mode identifier broadcasted by reductant quality sensor (UQS) equals to 5	Circuit Open	None	Open circuit in UQS sensor	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in UQS sensor If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Quality
ACM	P242C00	Voltage on A:39 below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Circuit Low (Bank 1 Sensor 3)
ACM	P267100	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to battery voltage.	Actuator Supply Voltage "B" Circuit High
ACM	P267000	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage B" is short circuited.	Diagnostic Service 14. No additional action is needed.	Check: Harness or connector is short circuited to ground voltage or High and Low side on any of the actuators connected to "Actuator supply voltage B" is short circuited.	Actuator Supply Voltage "B" Circuit Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
ACM	P242A15	(Voltage on A:39 above 2.260742V or (Voltage on A:39 above 0.147705V and Voltage on A:39 below 0.677490V))	Circuit Short To Battery or Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Gas Temperature Sensor Bank 1 Sensor 3
ACM	U02A200	(Lost communication with UQSM on Vehicle Communication Engine Subnet)	No Sub Type Information	None	UQSM not powered Software incompatibility with UQSM	Diagnostic Service 14. No additional action is needed.	Check: UQSM not powered Software incompatibility with UQSM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Quality Module
ACM	U114600	(Lost communication with ECM on Vehicle Communication Engine Subnet)	No Sub Type Information	None	ECM not powered Software incompatibility with ECM	Diagnostic Service 14. No additional action is needed.	Check: ECM not powered Software incompatibility with ECM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With EECU on Engine Subnet
ACM	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	None	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U000188	((It has not been possible to send frames on High Speed CAN Communication Bus) and (It has not been possible to receive frames on High Speed CAN Communication Bus))	Bus off	TEA2+: Engine does not start	Lost communication on High Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on High Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 2 Net
EMS	U001088	((It has not been possible to send frames on Medium Speed CAN Communication Bus) and (It has not been possible to receive frames on Medium Speed CAN Communication Bus))	Bus off	Starter motor does not run	Lost communication on Medium Speed CAN Communication Bus	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Medium Speed CAN Communication Bus If fault not detected: Fault trace according to Tech Tool fault tree.	CAN Communication Backbone 1 Net
EMS	U116600	(Lost communication with CHCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	CHCM not powered Software incompatibility with CHCM	Diagnostic Service 14. No additional action is needed.	Check: CHCM not powered Software incompatibility with CHCM	Lost Communication With CIOM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U116900	(Lost communication with CRCM on High Speed CAN Communication Bus)	No Sub Type Information	Activation failure, Cruise control	CRCM not powered Software incompatibility with CRCM	Diagnostic Service 14. No additional action is needed.	Check: CRCM not powered Software incompatibility with CRCM	Lost Communication With DACU
EMS	U010100	(Lost communication with TCM on Medium Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TECU
EMS	P034031	(No signal is detected on the cam wheel)	No Signal	None	Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Connector detached Sensor broken Sensor uncorrectly mounted (e.g. far away from the wheel)	Camshaft Position Sensor "A" Bank 1 or Single Sensor
EMS	P034138	(Noisy signal detected)	Signal Frequency Incorrect	None	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Camshaft Position Sensor A Circuit Range/Performance
EMS	P054100	(Voltage on B:7 above (0.722562V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V)))	No Sub Type Information	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater A Circuit Low
EMS	P054300	(Voltage on B:7 above (25.138032V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:7 above (2.863488V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:7 below (22.747332V * (UnfilteredBatteryVoltage / 24.0V))))	No Sub Type Information	Engine is difficult to start - Only cold conditions	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Heater A Circuit Open
EMS	P001676	(Phase error due to incorrect angle between cam and flywheel)	Wrong Mounting Position	None	Flywheel not correctly mounted	Diagnostic Service 14. No additional action is needed.	Check: Flywheel not correctly mounted	Crankshaft Position - Camshaft Position Correlation

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02CD00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Max Limit
EMS	P02CF00	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning at Max Limit
EMS	P02D100	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning at Max Limit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02D300	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning at Max Limit
EMS	P02D500	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning at Max Limit
EMS	P02D700	100Percentage above (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment VCB valve malfunction Worn injector cam lobe Unadjusted tappet clearance for valves or unit injectors Misaligned or broken valve springs or guides Cracked valves Piston/cylinder worn/damaged Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning at Max Limit
EMS	P033531	(No crank teeth detected between cam teeth by the control unit)	No Signal	High fuel consumption and Uneven Torque	Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Diagnostic Service 14. No additional action is needed.	Check: Sensor broken Connector detached Sensor uncorrectly mounted (e.g. far away from the wheel)	Crankshaft Position Sensor "A"
EMS	P033638	(More crank teeth than expected detected between cam teeth by the control unit)	Signal Frequency Incorrect	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor A Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P033900	(Intermittent crank speed signal)	No Sub Type Information	High fuel consumption and Uneven Torque	Noisy signal due to harness Sensor broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Noisy signal due to harness Sensor broken Connector detached	Crankshaft Position Sensor A Circuit Intermittent
EMS	P220013	(Internal Open Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Upstream NOx Sensor Open Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Open Circuit (Internal sensor fault)	NOx Sensor Bank 1 Sensor 1
EMS	P060100	(Internal Control Module Memory Check Sum Error)	No Sub Type Information	Engine derate	Corrupted dataset or software	Diagnostic Service 14. No additional action is needed.	Check: Corrupted dataset or software	Internal Control Module Memory Check Sum Error
EMS	P061600	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Circuit Low
EMS	P061700	(Short circuit to battery is detected.)	No Sub Type Information	Starter motor does not run	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken If fault not detected: Fault trace according to Tech Tool fault tree.	Starter Relay Circuit High
EMS	P066800	Voltage on N/A:ECU internal below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	PCM/ECM/TCM Internal Temperature Sensor Circuit Low
EMS	P06B100	Voltage on A:7 below 4.5V	No Sub Type Information	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply A Circuit Low
EMS	P06B200	Voltage on A:7 above 5.5V	No Sub Type Information	Engine derate	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply A Circuit High
EMS	P06B400	Voltage on B:17 below 4.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply B Circuit Low
EMS	P06B500	Voltage on B:17 above 5.5V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply B Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P023400	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) above 60kPa	No Sub Type Information	None	Blocked wastegate Boost Pressure Sensor biased high	Diagnostic Service 14. No additional action is needed.	Check: Blocked wastegate Boost Pressure Sensor biased high If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Overboost Condition
EMS	P029900	(Difference between Sensed Boost Pressure and Estimated Boost Pressure) below -60kPa	No Sub Type Information	Low power	Significant leakage in intake air system Boost Pressure Sensor biased low	Diagnostic Service 14. No additional action is needed.	Check: Significant leakage in intake air system Boost Pressure Sensor biased low If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Underboost
EMS	U300001	Voltage on N/A:ECU internal below 0.148926V	General Electrical Failure	Starter motor does not run	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Electronic control unit
EMS	U300041	(Program memory corruption)	General Checksum Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300044	(RAM corruption)	Data Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300045	(Program memory corruptionSoftware memory checksum error)	Program Memory Failure	Starter motor does not run	Download issue	Diagnostic Service 14. No additional action is needed.	Check: Download issue	Electronic control unit
EMS	U300049	(ECU internal error)	Internal Electronic Failure	Starter motor does not run	ECU Hardware issue	Diagnostic Service 14. No additional action is needed.	Check: ECU Hardware issue	Electronic control unit
EMS	P011013	(Voltage on A:47 above 4.910889V or (Voltage on A:47 above 0.147705V and Voltage on A:47 below 0.289307V))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Bank 1
EMS	P011200	Voltage on A:47 below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Intake Air Temperature Sensor 1 Circuit Low Bank 1
EMS	P011700	Voltage on B:27 below 0.147705V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Circuit Low
EMS	P021700	EngineCoolantTemperature above (1.2999999523162841No Unit + 106.7No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	Engine derate	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Over Temperature Condition
EMS	P040100	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) below (70% + (Offset dependent on ECUAmbientPressure))	No Sub Type Information	None	EGR system blocked/restricted Malfunctioning EGR Valve	Diagnostic Service 14. No additional action is needed.	Check: EGR system blocked/restricted Malfunctioning EGR Valve If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Exhaust Gas Recirculation 1 Valve Position Flow Insufficient Detected

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P040200	(Percentage of Ratio between Egr Mass Flow and Demanded Egr Mass Flow) above (130% + (Offset dependent on ECUAmbientPressure))	No Sub Type Information	None	Malfunctioning EGR Valve Malfunctioning VGT	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning EGR Valve Malfunctioning VGT If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Exhaust Gas Recirculation 1 Valve Position Flow Excessive Detected
EMS	P048013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Fan 1 Control Circuit
EMS	P051B64	((Crankcase pressure compared to reference pressure) below -15kPa or (Crankcase pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine High Resolution Crankcase Pressure Circuit Range/Performance
EMS	P051A13	((Voltage on B:28 above 4.757080V and Voltage on B:28 below 4.852295V) or Voltage on B:28 below 0.307617V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Crankcase Pressure Sensor
EMS	P051D00	Voltage on B:28 above 4.852295V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine High Resolution Crankcase Pressure
EMS	P052300	Voltage on B:11 above 4.852295V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Circuit High
EMS	P052400	EngineOilPressure below (Output from a look-up table with input EngineSpeed)	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Oil leakage The oil pump is broken The oil system is clogged	Diagnostic Service 14. No additional action is needed.	Check: Oil leakage The oil pump is broken The oil system is clogged	Engine Oil Pressure Too Low
EMS	P222613	((Voltage on N/A:ECU internal above 4.782715V and Voltage on N/A:ECU internal below 4.852295V) or Voltage on N/A:ECU internal below 0.117188V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit
EMS	P222900	Voltage on N/A:ECU internal above 4.852295V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Barometric Pressure Circuit High
EMS	P226413	(Voltage on B:8 above 3.234863V and Voltage on B:8 below 4.296875V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water in Fuel Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P226600	Voltage on B:8 below 0.211182V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Water In Fuel Indicator Circuit Low
EMS	P246300	(Moderately high soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Bank 1
EMS	P250C00	Voltage on B:4 below 0.05V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Circuit Low
EMS	P250F00	((If calculated value for oil level) below -4.980500221252441 % and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Critically low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Critically low oil level Oil leakage	Engine Oil Level Too Low

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P255613	(Voltage on B:23 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on B:23 above (16.378079V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (11.739407V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (12.943678V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on B:23 above (7.091815V *	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Sensor/Switch
EMS	P255800	Voltage on B:23 below (2.863488V * (UnfilteredBatteryVoltage / 24.0V))	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Circuit Low
EMS	P255900	(Voltage on B:23 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on B:23 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Level Circuit High
EMS	P256000	EngineCoolantLevel below (1No unit - 0.800000011920929 No Unit)	No Sub Type Information	None	Coolant level below range	Diagnostic Service 14. No additional action is needed.	Check: Coolant level below range	Engine Coolant Level Low
EMS	P257831	(Turbo Speed) below 60000rpm	No Signal	None	Malfunctioning VGT Actuator	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning VGT Actuator	Turbocharger Speed Sensor
EMS	P204F93	(Fault code in ACM causing stopped dosing)	No Operation	Engine derate	Fault code in ACM causing stopped dosing	Diagnostic Service 14. No additional action is needed.	Check: Fault code in ACM causing stopped dosing	SCR System Performance Dosing Failure

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P108600	((If calculated value for oil level) below 4% and (Engine must be in prerunning during a specific time, defined by output from map with input EngineOilTemperature))	No Sub Type Information	Engine shuts off unexpectedly - Only at low speed	Low oil level Oil leakage	Diagnostic Service 14. No additional action is needed.	Check: Low oil level Oil leakage	Oil Level Moderately Low
EMS	U014100	(Lost communication with BCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Low power	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication With VMCU
EMS	P229E13	(Internal Open Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Open	None	Downstream NOx Sensor Open Circuit. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Open Circuit. (Internal sensor fault)	NOx Sensor Bank 1 Sensor 2
EMS	P220B1C	(Internal Supply Voltage signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	Circuit Voltage Out of Range	None	Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Faulty Downstream Nox Sensor Supply. (Internal sensor fault)	NOx Sensor Supply Voltage Circuit Bank 1 Sensor 2
EMS	P069200	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Fan 1 Control Circuit High
EMS	P069100	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Fan 1 Control Circuit Low
EMS	U015500	(Lost communication with IPC on Medium Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication With Instrument Panel Cluster (IPC) Control Module
EMS	P052013	((Voltage on B:11 above 4.787598V and Voltage on B:11 below 4.852295V) or Voltage on B:11 below 0.212402V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P06E800	Voltage on A:1 above 5.5V	No Sub Type Information	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C" Circuit High
EMS	P06E700	Voltage on A:1 below 4.5V	No Sub Type Information	None or EUVI HDEP: Engine Cooling fan inoperative	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault If fault not detected: Fault trace according to Tech Tool fault tree.	Sensor Power Supply "C" Circuit Low
EMS	P00AF00	(Internal failure detected by Turbocharger/Supercharger A.)	No Sub Type Information	Low power	Internal actuator fault in Turbocharger/Supercharger A	Diagnostic Service 14. No additional action is needed.	Check: Internal actuator fault in Turbocharger/Supercharger A If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Boost Control "A" Module Performance
EMS	P066613	(Voltage on N/A:ECU internal above 0.147705V and Voltage on N/A:ECU internal below 1.119385V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Control Module Internal Temperature Sensor "A"
EMS	P040C00	Voltage on B:48 below 0.023671V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A" Circuit Low
EMS	P020113	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 1 Injector "A"
EMS	P020313	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 3 Injector "A"
EMS	P020413	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 4 Injector "A"
EMS	P020513	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 5 Injector "A"
EMS	P020613	(Current driving the injector rises to slow)	Circuit Open	Engine derate	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 6 Injector "A"
EMS	P250A13	Voltage on B:4 above 4.85V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Level Sensor
EMS	P249F00	Exhaust Gas Temperature Sensor 1 below (240 C)	No Sub Type Information	None	Exhaust Temperature Too Low For Particulate Filter Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Temperature Too Low For Particulate Filter Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Excessive Time To Enter Closed Loop DPF Regeneration Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P244C00	Average Particulate Filter Temperature below (250 C)	No Sub Type Information	Activation failure, regeneration	Too Low Exhaust Temperature During Regeneration.	Diagnostic Service 14. No additional action is needed.	Check: Too Low Exhaust Temperature During Regeneration. If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Temperature Too Low For Particulate Filter Regeneration (Bank 1)
EMS	P20DC00	(AftertreatmentFuel Pressure below 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	No Sub Type Information	Activation failure, regeneration	Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Supply Control Valve Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Supply Control Stuck Closed
EMS	P20CF7A	AftertreatmentFuelPressure below 70kPa	Fluid Leak or Seal Failure	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A" Stuck Open
EMS	P20D000	(AftertreatmentFuel Pressure above 70kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	No Sub Type Information	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injector Stuck Closed	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injector Stuck Closed If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Injector "A" Stuck Closed
EMS	P020213	(Current driving the injector rises to slow)	Circuit Open	Low power	Harness issue Actuator broken Connector detached	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Actuator broken Connector detached	Cylinder 2 Injector "A"
EMS	P052164	(EngineOilPressure above 100kPa or EngineOilPressure above 600kPa or EngineOilPressure below 300kPa)	Signal Plausibility Failure	None	Oil pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Sensor/Switch Range/Performance
EMS	P101711	(Short circuit to ground is detected.)	Circuit Short To Ground	Activation failure, Fuel Priming or Water Draining	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Water In Fuel Drainage Valve
EMS	P101712	(Short circuit to battery is detected.)	Circuit Short To Battery	Uneven Torque or The engine misfires or Engine is difficult to start or High fuel consumption	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Water In Fuel Drainage Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P101713	(Open circuit fault is detected.)	Circuit Open	Activation failure, Fuel Priming or Water Draining	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Water In Fuel Drainage Valve
EMS	P220300	(Internal Short Circuit signal containing status and error byte received from Upstream NOx sensor evaluated as NOT_RELIABLE)	No Sub Type Information	None	Upstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Upstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Circuit High Bank 1 Sensor 1
EMS	P242800	EngineExhaustGasTemperature above (530No unit + 999No Unit)	No Sub Type Information	Engine derate	High exhaust back pressure. Extreme driving conditions.	Diagnostic Service 14. No additional action is needed.	Check: High exhaust back pressure. Extreme driving conditions.	Exhaust Gas Temperature Too High
EMS	P100211	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Preheat Relay
EMS	P100212	(Short circuit to battery is detected.)	Circuit Short To Battery	Engine is difficult to start - Only cold conditions	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Preheat Relay
EMS	P100213	(Open circuit fault is detected.)	Circuit Open	Engine is difficult to start - Only cold conditions	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Preheat Relay
EMS	P100713	(Voltage on A:29 above (24.424390V * (UnfilteredBatteryVoltage / 24.0V)) or (Voltage on A:29 above (12.194354V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (23.559100V * (UnfilteredBatteryVoltage / 24.0V))) or (Voltage on A:29 above (4.148043V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (8.367450V * (UnfilteredBatteryVoltage / 24.0V))))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator
EMS	P102111	(Short circuit to ground is detected.)	Circuit Short To Ground	Engine starts but does not continue to run or Bridgeplan depending on installation: None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102112	(Short circuit to battery is detected.)	Circuit Short To Battery	Auxiliary brakes, low performance (Reduced engine braking)	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Volvo Compression Brake (VCB) Control Circuit
EMS	P102113	(Open circuit fault is detected.)	Circuit Open	Auxiliary brakes, low performance (Reduced engine braking)	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Volvo Compression Brake (VCB) Control Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U012900	(Lost communication with BSCM on Medium Speed CAN Communication Bus)	No Sub Type Information	Auxiliary brakes, low performance (Reduced retarder braking)	BSCM not powered Software incompatibility with BSCM	Diagnostic Service 14. No additional action is needed.	Check: BSCM not powered Software incompatibility with BSCM	Lost Communication With Brake System Control Module
EMS	P019513	Voltage on A:31 above 4.931641V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Temperature Sensor "A"
EMS	P026500	(Current driving the injector rises to quickly)	No Sub Type Information	Low power	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 2 Short Circuit High
EMS	P026800	(Current driving the injector rises to quickly)	No Sub Type Information	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 3 Short Circuit High
EMS	P027100	(Current driving the injector rises to quickly)	No Sub Type Information	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 4 Short Circuit High
EMS	P027400	(Current driving the injector rises to quickly)	No Sub Type Information	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 5 Short Circuit High
EMS	P027700	(Current driving the injector rises to quickly)	No Sub Type Information	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 6 Short Circuit High
EMS	P019664	((EngineOilTemperature below -39°C))	Signal Plausibility Failure	None	Oil temperature sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Oil temperature sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Oil Temperature
EMS	P019700	Voltage on A:31 below 0.100098V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Oil Temperature Short Circuit Low
EMS	P026200	(Current driving the injector rises to quickly)	No Sub Type Information	Engine derate	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Injector 1 Short Circuit High
EMS	U029E00	(Lost communication with NOX2 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX2 not powered Software incompatibility with NOX2	Diagnostic Service 14. No additional action is needed.	Check: NOX2 not powered Software incompatibility with NOX2 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "B"
EMS	U029D00	(Lost communication with NOX1 on Vehicle Communication Engine Subnet)	No Sub Type Information	None	NOX1 not powered Software incompatibility with NOX1	Diagnostic Service 14. No additional action is needed.	Check: NOX1 not powered Software incompatibility with NOX1 If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With NOx Sensor "A"
EMS	U010C00	(Lost communication with VGTM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	VGTM not powered Software incompatibility with VGTM	Diagnostic Service 14. No additional action is needed.	Check: VGTM not powered Software incompatibility with VGTM	Lost Communication With Turbocharger/Supercharger Control Module A

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	U042600	(Faulty immobilizer code)	No Sub Type Information	TEA2+: Starter motor does not run. Bridgeplan: Engine crank but does not start	EMS amd Immobilizer unit security codes do not match	Diagnostic Service 14. No additional action is needed.	Check: EMS amd Immobilizer unit security codes do not match	Invalid Data Received From Vehicle Immobilizer Control Module
EMS	P22A100	(Internal Short Circuit signal containing status and error byte received from Downstream NOx sensor evaluated as NOT_RELIABLE)	No Sub Type Information	None	Downstream NOx Sensor Short Circuit (Internal sensor fault)	Diagnostic Service 14. No additional action is needed.	Check: Downstream NOx Sensor Short Circuit (Internal sensor fault)	NOx Sensor Circuit High Bank 1 Sensor 2
EMS	P114000	(Engine oil temperature is too low when engine brake is demanded.)	No Sub Type Information	Auxiliary brakes, low performance (Reduced engine braking)	No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Diagnostic Service 14. No additional action is needed.	Check: No fault, only for driver information. Low engine oil temperature when engine brake is requested.	Volvo Compression Brake (VCB) Control Oil Temperature Too Low
EMS	P06DD00	(Difference in Oil Pressure (Piston Cooling Valve forced open) and Oil Pressure (Piston Cooling Valve forced closed)) above 180kPa	No Sub Type Information	None	Oil pump regulation valve failure	Diagnostic Service 14. No additional action is needed.	Check: Oil pump regulation valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Control Circuit Performance/Stuck Off
EMS	P113711	(Short circuit to ground is detected.)	Circuit Short To Ground	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Oil Thermostat
EMS	P113712	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Oil Thermostat
EMS	P113713	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Oil Thermostat
EMS	P114411	(Short circuit to ground is detected.)	Circuit Short To Ground	Starter motor does not run	Harness issue Short circuit to ground Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to ground Actuator broken	Starter Relay Highside
EMS	P114412	(Short circuit to battery is detected.)	Circuit Short To Battery	None	Harness issue Short circuit to battery Actuator broken	Diagnostic Service 14. No additional action is needed.	Check: Harness issue Short circuit to battery Actuator broken	Starter Relay Highside
EMS	P114500	(Reductant level) below 13% for 15s	No Sub Type Information	Engine derate	Reductant level moderately low	Diagnostic Service 14. No additional action is needed.	Check: Reductant level moderately low If fault not detected: Fault trace according to Tech Tool fault tree.	Reductant Level Moderately Low
EMS	P114800	(Communication failure detected by Turbocharger/Supercharger A.)	No Sub Type Information	Low power	Disturbance on CAN.	Diagnostic Service 14. No additional action is needed.	Check: Disturbance on CAN. If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication With ECM According To Turbocharger/Supercharger Control Module "A"

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P006E00	(Low voltage detected by Turbocharger/Supercharger A.)	No Sub Type Information	Low power	Low voltage detected by Turbocharger/Supercharger A	Diagnostic Service 14. No additional action is needed.	Check: Low voltage detected by Turbocharger/Supercharger A	Turbocharger/Supercharger Boost Control "A" Supply Voltage Circuit
EMS	P021900	(Too high engine speed detected.)	No Sub Type Information	The engine is shut off when the vehicle speed is less than {approx. 3 km/h}	Too high engine speed.	Diagnostic Service 14. No additional action is needed.	Check: Too high engine speed.	Engine Overspeed Condition
EMS	P26047F	((The preheat element is inactive when there is a preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck Off	Engine is difficult to start - Only cold conditions	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Circuit Range/Performance
EMS	P115300	(Calibration requested and no calibration performed)	No Sub Type Information	None	Intake Throttle Valve calibration not performed	Diagnostic Service 14. No additional action is needed.	Check: Intake Throttle Valve calibration not performed	Diesel Intake Air Flow Position Sensor Exceeded Learning Limit
EMS	P008B00	((FuelPressure above (100kPa + (Offset dependent on fuel value and Engine Speed))))	No Sub Type Information	None	Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used)	Diagnostic Service 14. No additional action is needed.	Check: Overflow valve stuck closed Low Pressure Fuel System Pressure Sensor malfunction (if this sensor is used) Exhaust Aftertreatment Fuel Pressure Sensor malfunction (if this sensor is used) If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too High
EMS	U1FFF68	(The factory mode is active through P1HCF)	Event Information	None	Factory Mode was not turned off by P1HCF	Diagnostic Service 14. No additional action is needed.	Check: Factory Mode was not turned off by P1HCF	Factory Mode Active
EMS	P040A13	(Voltage on B:48 above 0.772330V or (Voltage on B:48 above 0.023671V and Voltage on B:48 below 0.181149V))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Gas Recirculation Temperature Sensor "A"
EMS	P004607	(Failure detected by Turbocharger/Supercharger A.)	Mechanical Failures	Low power	Turbocharger/Supercharger A is not following command	Diagnostic Service 14. No additional action is needed.	Check: Turbocharger/Supercharger A is not following command If fault not detected: Fault trace according to Tech Tool fault tree.	Turbocharger/Supercharger Boost Control "A" Circuit Range/Performance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P222764	(ECUAmbientPressure below 55kPa or ECUAmbientPressure above 114kPa or (Ambient air pressure compared to reference pressure) below -15kPa or (Ambient air pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Barometric Pressure Sensor "A" Circuit Range/Performance
EMS	P24A400	(The aftertreatment diesel particulate filter load is considered critically high.)	No Sub Type Information	High fuel consumption	Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: Critically high soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty engine exhaust gas recirculation differential pressure sensor.	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1)
EMS	P011513	(Voltage on B:27 above 4.913330V or (Voltage on B:27 above 0.147705V and Voltage on B:27 below 0.231934V))	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Coolant Temperature Sensor 1
EMS	P040913	Voltage on A:21 below 0.207520V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P046C64	((((35kPa - (Offset dependent on EGR valve Control PWM) - DifferentialPressureFeedbackEGR) below 0kPa or (DifferentialPressure FeedbackEGR - 0kPa - (Offset dependent on EGR valve Control PWM)) below 0kPa)) or DifferentialPressureFeedbackEGR above 1.5kPa)	Signal Plausibility Failure	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor Circuit Range/Performance
EMS	P040600	Voltage on A:21 above 4.882813V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	EGR Differential Pressure Sensor Circuit High
EMS	P02CC00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 1 Fuel Injector Offset Learning at Min Limit
EMS	P269900	(Short circuit to ground is detected.)	No Sub Type Information	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Injector "A" Circuit Low
EMS	P269713	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Injector "A"
EMS	P20DD13	Voltage on A:50 below 0.112305V	Circuit Open	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor
EMS	P20E000	Voltage on A:50 above 4.887695V	No Sub Type Information	Activation failure, regeneration	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Exhaust Aftertreatment Fuel Pressure Sensor Circuit High
EMS	P053E00	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Diagnostic Service 14. No additional action is needed.	Check: The pressure difference between crankcase pressure and ambient air pressure is/was above limit	Crankcase Pressure Too High
EMS	U012100	(Lost communication with ABS on Medium Speed CAN Communication Bus)	No Sub Type Information	None	ABS not powered Software incompatibility with ABS	Diagnostic Service 14. No additional action is needed.	Check: ABS not powered Software incompatibility with ABS	Lost Communication With Anti-Lock Brake System (ABS) Control Module
EMS	P040313	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	EGR "A" Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P048900	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	EGR "A" Control Circuit Low
EMS	P02D400	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 5 Fuel Injector Offset Learning At Min Limit
EMS	P02D600	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 6 Fuel Injector Offset Learning At Min Limit
EMS	P02D000	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 3 Fuel Injector Offset Learning At Min Limit
EMS	P02CE00	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 2 Fuel Injector Offset Learning At Min Limit
EMS	P02D200	-100Percentage below (Cylinder balancing fueling offset ratios in percentage of the fault code limit for each injector.)	No Sub Type Information	Uneven Torque	Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Diagnostic Service 14. No additional action is needed.	Check: Losses via PTO or other power consuming equipment Worn injector Cracked injector tip Damaged spray holes in the injector	Cylinder 4 Fuel Injector Offset Learning At Min Limit
EMS	P25AB00	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Piston Cooling Oil Control Circuit High
EMS	P25AA00	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Piston Cooling Oil Control Circuit Low
EMS	P25AC00	(When piston cooling valve is forced open and closed the oil pressure change) below 20kPa	No Sub Type Information	None	Piston cooling valve failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling valve failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Control Circuit Performance/Stuck Off

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P006964	((Boost pressure compared to reference pressure) below -15kPa or (Boost pressure compared to reference pressure) above 15kPa)	Signal Plausibility Failure	None	Sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Manifold Absolute Pressure - Barometric Pressure Correlation
EMS	P20D713	(Open circuit fault is detected.)	Circuit Open	Activation failure, regeneration	Harness, actuator or connector is either disconnected or short circuited to battery.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is either disconnected or short circuited to battery.	Exhaust Aftertreatment Fuel Supply Control
EMS	P20D900	(Short circuit to ground is detected.)	No Sub Type Information	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Supply Control Circuit Low
EMS	P010513	((Voltage on A:22 above 4.702148V and Voltage on A:22 below 4.852295V) or Voltage on A:22 below 0.297852V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor
EMS	P010800	Voltage on A:22 above 4.852295V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High
EMS	P25AE00	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	Engine derate	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Too Low
EMS	P25A913	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Piston Cooling Oil Control
EMS	P054070	((The preheat element is active when there is no preheat activation request.) and (1No Unit equals to 1 or 1No Unit equals to 2))	Actuator Stuck On	None	Preheat relay problem Wiring harness defect	Diagnostic Service 14. No additional action is needed.	Check: Preheat relay problem Wiring harness defect	Intake Air Heater "A" Control
EMS	P003A00	(Self-calibration failure detected by Turbocharger/Supercharger A.)	No Sub Type Information	None	Calibration of Turbocharger/Supercharger A failed.	Diagnostic Service 14. No additional action is needed.	Check: Calibration of Turbocharger/Supercharger A failed.	Turbocharger/Supercharger Boost Control "A" Position Exceeded Learning Limit
EMS	P10E100	Particulate Filter Differential Pressure above (26No unit + 7No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Critically High
EMS	P10E200	Particulate Filter Differential Pressure above (26No unit + 999No Unit)	No Sub Type Information	Engine derate	Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Diagnostic Service 14. No additional action is needed.	Check: Chunk of ice is influencing measure values of the sensor Muck soot in the particle filter The filter is broken	Particulate Filter Differential Pressure Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P10EB00	DifferentialPressureFeedbackEGR below -2kPa	No Sub Type Information	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor "A" Circuit Below Range
EMS	P10EC00	DifferentialPressureFeedbackEGR above 2kPa	No Sub Type Information	None	Sensor	Diagnostic Service 14. No additional action is needed.	Check: Sensor	EGR Differential Pressure Sensor "A" Circuit Above Range
EMS	P10F200	Voltage on A:29 below (1.587853V * (UnfilteredBatteryVoltage / 24.0V))	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator Circuit Low
EMS	P10F300	(Voltage on A:29 above (23.559100V * (UnfilteredBatteryVoltage / 24.0V)) and Voltage on A:29 below (24.424390V * (UnfilteredBatteryVoltage / 24.0V)))	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Air Filter Indicator Circuit High
EMS	P10F800	AirFilterPressureLoss above 0	No Sub Type Information	None	Clogged air filter	Diagnostic Service 14. No additional action is needed.	Check: Clogged air filter	Air Filter Clogged
EMS	P10FE00	(High soot load in aftertreatment diesel particulate filter.)	No Sub Type Information	Engine derate	High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Diagnostic Service 14. No additional action is needed.	Check: High soot load in aftertreatment diesel particulate filter. High ash load in aftertreatment diesel particulate filter. Clogging of aftertreatment diesel particulate filter. Faulty aftertreatment diesel particulate filter differential pressure sensor. Faulty boost pressure sensor. Faulty exhaust gas recirculation differential pressure sensor.	Particulate Filter Restriction - Soot Accumulation Moderately High Bank 1
EMS	P055A13	Voltage on B:19 below 0.112305V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Engine Oil Pressure Sensor/Switch "B"
EMS	P055B64	((Difference between demanded piston cooling pressure and measured piston cooling pressure) between -15kPa and 15kPa or PistonCoolPressure above 30kPa)	Signal Plausibility Failure	None	Piston cooling pressure sensor failure	Diagnostic Service 14. No additional action is needed.	Check: Piston cooling pressure sensor failure If fault not detected: Fault trace according to Tech Tool fault tree.	Piston Cooling Oil Pressure Sensor Range/Performance
EMS	P055D00	Voltage on B:19 above 4.887695V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Piston Cooling Oil Pressure Sensor Circuit High
EMS	P02E013	(Open circuit fault is detected.)	Circuit Open	None	Open circuit in harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector.	Diesel Intake Air Flow Control

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02E17C	((Estimated Current) above 3.25A or (Estimated Current) above 5A)	Slow Response	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control Performance
EMS	P112800	(The signal PistonCoolPressure is above a critical limit)	No Sub Type Information	None	Low oil level	Diagnostic Service 14. No additional action is needed.	Check: Low oil level	Piston Cooling Oil Pressure Moderately Low
EMS	P02E200	(Short circuit to ground is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Diesel Intake Air Flow Control Circuit Low
EMS	P112900	PFTransmissionOilTemp above (154No unit + 0No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Temperature Moderately High
EMS	P02E300	(Short circuit to battery is detected.)	No Sub Type Information	None	Harness, actuator or connector is short circuited to battery voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to battery voltage.	Diesel Intake Air Flow Control Circuit High
EMS	P112B00	(High temperature failure detected by Turbocharger/Supercharger A.)	No Sub Type Information	None	High temperature in Turbocharger/Supercharger A	Diagnostic Service 14. No additional action is needed.	Check: High temperature in Turbocharger/Supercharger A	Turbocharger/Supercharger Actuator "A" Temperature High
EMS	P029800	EngineOilTemperature above (124.5No unit + 3.5No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Over Temperature
EMS	P02E613	((Voltage on B:24 above 4.757080V and Voltage on B:24 below 4.852295V) or Voltage on B:24 below 0.247803V)	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor
EMS	P021800	PFTransmissionOilTemp above (154No unit + 6No Unit)	No Sub Type Information	Engine derate	Transmission cooler failure Cooling system problem	Diagnostic Service 14. No additional action is needed.	Check: Transmission cooler failure Cooling system problem	Transmission Fluid Over Temperature Condition
EMS	P02E900	Voltage on B:24 above 4.852295V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Diesel Intake Air Flow Position Sensor Circuit High
EMS	U116B00	(Lost communication with TCM on High Speed CAN Communication Bus)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on High Speed CAN Communication Bus
EMS	U116C00	(Lost communication with TCM on Vehicle Communication Powertrain CAN)	No Sub Type Information	None	TCM not powered Software incompatibility with TCM	Diagnostic Service 14. No additional action is needed.	Check: TCM not powered Software incompatibility with TCM	Lost Communication with TCM on Powertrain CAN

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P02EC00	(ThrottlePosition - Demanded Throttle Position) above 10%	No Sub Type Information	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control System - High Air Flow Detected
EMS	P02ED00	(Demanded Throttle Position - ThrottlePosition) above 10%	No Sub Type Information	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Control System - Low Air Flow Detected
EMS	P02FA97	(not ((Maximum Opened Throttle Position) between 78.3000030517578deg and 84.6999969482422deg) and not ((Minimum Closed Throttle Position) between -2.799999952316284deg and 2.799999952316284deg))	Component or System Operation Obstructed or Blocked	None	Malfunctioning Intake Throttle Valve	Diagnostic Service 14. No additional action is needed.	Check: Malfunctioning Intake Throttle Valve	Diesel Intake Air Flow Position Sensor Minimum/Maximum Stop Performance
EMS	U116F00	(Lost communication with EACM on Vehicle Communication Engine Subnet)	No Sub Type Information	Engine derate	EACM not powered Software incompatibility with EACM	Diagnostic Service 14. No additional action is needed.	Check: EACM not powered Software incompatibility with EACM If fault not detected: Fault trace according to Tech Tool fault tree.	Lost Communication with Reductant Control Module on Engine Subnet
EMS	U117100	(Lost communication with BCM on High Speed CAN Communication Bus)	No Sub Type Information	Starter motor does not run	BCM not powered Software incompatibility with BCM	Diagnostic Service 14. No additional action is needed.	Check: BCM not powered Software incompatibility with BCM	Lost Communication with Body Control Module on High Speed CAN Communication Bus
EMS	U117300	(Lost communication with IPC on High Speed CAN Communication Bus)	No Sub Type Information	None	IPC not powered Software incompatibility with IPC	Diagnostic Service 14. No additional action is needed.	Check: IPC not powered Software incompatibility with IPC	Lost Communication with IPC Control Module on High Speed CAN Communication Bus
EMS	P10FF86	(An illegal value was received from the EACM on Vehicle Communication Engine Subnet)	Signal Invalid	Engine derate	A corresponding fault code is set in EACM	Diagnostic Service 14. No additional action is needed.	Check: A corresponding fault code is set in EACM	Reductant Level Sensor "A" Circuit Electrical
EMS	P111D00	EngineCoolantLevel below (1No unit - 2No Unit)	No Sub Type Information	None	Sensor out of range	Diagnostic Service 14. No additional action is needed.	Check: Sensor out of range	Engine Coolant Level Moderately Low
EMS	P111E00	EngineCoolantTemperature above (0.5No Unit + 106.7No unit + (Output from look-up tables with inputs ECUAmbientPressure and EngineCoolantPressure))	No Sub Type Information	None	Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions, resulting in too high temperature The coolant thermostat is broken Problems with the fan The radiator is blocked	Engine Coolant Temperature Moderately High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P112100	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112198	EGRTemperature above (220No unit + 0No unit)	Component or System Over Temperature	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Too High
EMS	P112200	EGRTemperature above (220No unit + 999No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions EGR cooler problems	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions EGR cooler problems	EGR Temperature Moderately High
EMS	P112E00	EngineExhaustGasTemperature above (530No unit + 999No Unit)	No Sub Type Information	Engine derate	High exhaust back pressure. Extreme driving conditions.	Diagnostic Service 14. No additional action is needed.	Check: High exhaust back pressure. Extreme driving conditions.	Exhaust Gas Temperature Moderately High Bank 1
EMS	P112500	EngineOilTemperature above (124.5No unit + 1No Unit)	No Sub Type Information	Engine derate	Extreme driving conditions	Diagnostic Service 14. No additional action is needed.	Check: Extreme driving conditions	Engine Oil Temperature Moderately High
EMS	U008088	((It has not been possible to send frames on Vehicle Communication Engine Subnet) and (It has not been possible to receive frames on Vehicle Communication Engine Subnet))	Bus off	Engine derate	Lost communication on Vehicle Communication Engine Subnet	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Engine Subnet If fault not detected: Fault trace according to Tech Tool fault tree.	Vehicle Communication Engine Subnet
EMS	U117888	((It has not been possible to send frames on Vehicle Communication Powertrain CAN) and (It has not been possible to receive frames on Vehicle Communication Powertrain CAN))	Bus off	None	Lost communication on Vehicle Communication Powertrain CAN	Diagnostic Service 14. No additional action is needed.	Check: Lost communication on Vehicle Communication Powertrain CAN	Vehicle Communication Powertrain CAN
EMS	P253913	Voltage on B:16 below 0.112305V	Circuit Open	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Low Pressure Fuel System Sensor Circuit
EMS	P254200	Voltage on B:16 above 4.887695V	No Sub Type Information	None	Wiring fault Sensor fault ECU fault	Diagnostic Service 14. No additional action is needed.	Check: Wiring fault Sensor fault ECU fault	Low Pressure Fuel System Sensor Circuit High

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P008A00	((AftertreatmentFuel Pressure below (200kPa + (Offset dependent on fuel value and Engine Speed)) or AftertreatmentFuelP ressure below (100kPa + (Offset dependent on fuel value and Engine Speed))) or FuelPressure below (100kPa + (Offset dependent on fuel value and Engine Speed)))	No Sub Type Information	None	Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open	Diagnostic Service 14. No additional action is needed.	Check: Fuel tank ventilation clogged or stuck close Fuel leakage, e.g. leaking fuel lines Air ingress in low pressure circquit LP system fuel pressure sensor malfunction (if this sensor is used) AHI fuel pressure sensor malfunction (if this sensor is used) Clogged fuel filter, if the pressure sensor is placed between the fuel pump and the fuel filter Fuel pump malfunction Overflow valve stuck open If fault not detected: Fault trace according to Tech Tool fault tree.	Low Pressure Fuel System Pressure - Too Low
EMS	P113500	(Too high difference between ambient air pressure and crank case pressure.)	No Sub Type Information	Engine derate	Too high difference between ambient air pressure and crank case pressure.	Diagnostic Service 14. No additional action is needed.	Check: Too high difference between ambient air pressure and crank case pressure.	Crankcase Pressure Relative To Ambient Pressure Too High
EMS	P21CE9A	(((Estimated reductant concentration) below 20% and (The time required to perform the estimation of reductant concentration depends on driving/operation condition.)) or ((Filtered measured reductant concentration) below 20% and (The evaluation of reductant concentration requires some time to set or reset the fault code due to signal filter constant. Normally within an hour depending on application.)))	Component or System Operating Conditions	None	Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Diagnostic Service 14. No additional action is needed.	Check: Reductant quality and concentration Reductant quality sensor Exhaust leakage Sulphur poisoning Reductant delivery SCR catalyst Upstream NOx sensor Downstream NOx sensor	Reductant Quality Module Performance
EMS	P055F00	(Difference between estimated oil pressure and real oil pressure) below -150kPa	No Sub Type Information	None	Low oil pressure	Diagnostic Service 14. No additional action is needed.	Check: Low oil pressure If fault not detected: Fault trace according to Tech Tool fault tree.	Engine Oil Pressure Out of Range

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
EMS	P113000	(AftertreatmentFuel Pressure below 30kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	No Sub Type Information	Activation failure, regeneration	No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening	Diagnostic Service 14. No additional action is needed.	Check: No Air Supply to the Exhaust Aftertreatment Fuel Injection System Pipe to the Exhaust Aftertreatment Fuel Injection Nozzle not Connected or Leakage Exhaust Aftertreatment Fuel Injection Air Purge Valve is not Opening If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve Stuck Closed
EMS	P113300	(AftertreatmentFuel Pressure above 25kPa and not (((AftertreatmentFuelPressure below -10kPa and AftertreatmentFuelPressure below -10kPa) or (AftertreatmentFuel Pressure above 710kPa and AftertreatmentFuelPressure above 710kPa))))	No Sub Type Information	Activation failure, regeneration	Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged	Diagnostic Service 14. No additional action is needed.	Check: Exhaust Aftertreatment Fuel Injection Air Purge Valve Stuck Open Exhaust Aftertreatment Fuel Supply Control Valve Stuck Open Exhaust Aftertreatment Fuel Injector Nozzel Clogged If fault not detected: Fault trace according to Tech Tool fault tree.	Exhaust Aftertreatment Fuel Air Purge Valve Stuck Open
EMS	P113413	(Either short circuit to battery, or open circuit is detected.)	Circuit Open	Activation failure, regeneration	Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Diagnostic Service 14. No additional action is needed.	Check: Open circuit in harness, actuator or connector. Short circuit to battery harness, actuator or connector.	Exhaust Aftertreatment Fuel Air Purge Valve
EMS	P113100	(Short circuit to ground is detected.)	No Sub Type Information	Activation failure, regeneration	Harness, actuator or connector is short circuited to ground voltage.	Diagnostic Service 14. No additional action is needed.	Check: Harness, actuator or connector is short circuited to ground voltage.	Exhaust Aftertreatment Fuel Air Purge Valve Circuit Low
TECU	P105816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105838		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	U000188		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 2 Net
TECU	U001088		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 1 Net
TECU	P10561C		Circuit Voltage Out of Range			Diagnostic Service 14. No additional action is needed.		ECU battery potential
TECU	P105621		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		ECU battery potential

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P105622		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		ECU battery potential
TECU	P060146		Calibration / Parameter Memory Failure			Diagnostic Service 14. No additional action is needed.		Internal Control Module Memory Check Sum Error
TECU	U007388		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet
TECU	P10B016		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B021		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B022		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B116		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Hydraulic Retarder Oil Temperature
TECU	P10B117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Hydraulic Retarder Oil Temperature
TECU	P10B816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B818		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B819		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B918		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B919		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10BA16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BB07		Mechanical Failures			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BB21		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BB22		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BC07		Mechanical Failures			Diagnostic Service 14. No additional action is needed.		Lockup Clutch
TECU	P10BC21		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Lockup Clutch
TECU	P10BC22		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Lockup Clutch
TECU	P10BD16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10BD17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BD18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BD19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BE16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BF18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		PTO Solenoid Valve
TECU	P10BF19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		PTO Solenoid Valve
TECU	P10C016		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C116		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Coolant Temperature Sensor
TECU	P10C117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Coolant Temperature Sensor
TECU	P10B202		General Signal Failure			Diagnostic Service 14. No additional action is needed.		Transmission Oil Filter
TECU	P10B297		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Transmission Oil Filter
TECU	P10B309		Component Failures			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B321		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B392		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B416		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B419		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B516		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10B517		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B518		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B519		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B616		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B618		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B619		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B716		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B719		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10C217		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Lubrication Oil Pressure Sensor
TECU	P10C221		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Lubrication Oil Pressure Sensor
TECU	P10C916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	P10C917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	P10C938		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	U114316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U114317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U114338		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U1FFF68		Event Information			Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	U114616		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	U114617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	U114638		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	P10C800		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Limphome Function
TECU	P10CB92		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Calibration Not Done
TECU	P105816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105838		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Countershaft Speed Sensor (SESC)
TECU	P105916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)
TECU	P105917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Gearbox Oil Temperature Sensor (SETeGO)

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	U000188		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 2 Net
TECU	U001088		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Backbone 1 Net
TECU	P10561C		Circuit Voltage Out of Range			Diagnostic Service 14. No additional action is needed.		ECU battery potential
TECU	P105621		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		ECU battery potential
TECU	P105622		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		ECU battery potential
TECU	P060146		Calibration / Parameter Memory Failure			Diagnostic Service 14. No additional action is needed.		Internal Control Module Memory Check Sum Error
TECU	U007388		Bus off			Diagnostic Service 14. No additional action is needed.		CAN Communication Powertrain Subnet
TECU	P10B016		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B021		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B022		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Transmission Oil Level
TECU	P10B116		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Hydraulic Retarder Oil Temperature
TECU	P10B117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Hydraulic Retarder Oil Temperature
TECU	P10B816		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B817		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B818		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B819		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B5 Pressure Control Valve
TECU	P10B916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B918		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10B919		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Lockup Clutch Pressure Control Valve
TECU	P10BA16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BA19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K1 Pressure Control Valve
TECU	P10BB07		Mechanical Failures			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BB21		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BB22		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Gearbox Clutch And Brake System
TECU	P10BC07		Mechanical Failures			Diagnostic Service 14. No additional action is needed.		Lockup Clutch

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10BC21		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Lockup Clutch
TECU	P10BC22		Signal Amplitude Greater Than Maximum			Diagnostic Service 14. No additional action is needed.		Lockup Clutch
TECU	P10BD16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BD17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BD18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BD19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Clutch K2 Pressure Control Valve
TECU	P10BE16		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE17		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BE19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Main Pressure Control Pilot Valve
TECU	P10BF18		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		PTO Solenoid Valve
TECU	P10BF19		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		PTO Solenoid Valve
TECU	P10C016		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C017		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C018		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C019		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Retarder Control Pilot Valve
TECU	P10C116		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Coolant Temperature Sensor
TECU	P10C117		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Coolant Temperature Sensor
TECU	P10B202		General Signal Failure			Diagnostic Service 14. No additional action is needed.		Transmission Oil Filter
TECU	P10B297		Component or System Operation Obstructed or Blocked			Diagnostic Service 14. No additional action is needed.		Transmission Oil Filter
TECU	P10B309		Component Failures			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B321		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B392		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Transmission Main Hydraulic Oil Pressure Sensor
TECU	P10B416		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B417		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TECU	P10B418		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B419		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B1 Pressure Control Valve
TECU	P10B516		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B517		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B518		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B519		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B2 Pressure Control Valve
TECU	P10B616		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B618		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B619		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B3 Pressure Control Valve
TECU	P10B716		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B717		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B718		Circuit Current Below Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10B719		Circuit Current Above Threshold			Diagnostic Service 14. No additional action is needed.		Brake B4 Pressure Control Valve
TECU	P10C217		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Lubrication Oil Pressure Sensor
TECU	P10C221		Signal Amplitude Less Than Minimum			Diagnostic Service 14. No additional action is needed.		Lubrication Oil Pressure Sensor
TECU	P10C916		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	P10C917		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	P10C938		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Engine Speed
TECU	U114316		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U114317		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U114338		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed
TECU	U1FFF68		Event Information			Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TECU	U114616		Circuit Voltage Below Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	U114617		Circuit Voltage Above Threshold			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	U114638		Signal Frequency Incorrect			Diagnostic Service 14. No additional action is needed.		Transmission Output Shaft Speed #2
TECU	P10C800		No Sub Type Information			Diagnostic Service 14. No additional action is needed.		Limphome Function
TECU	P10CB92		Performance or Incorrect Operation			Diagnostic Service 14. No additional action is needed.		Calibration Not Done

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109D11	I > 3,671A...4,96A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *BrakeLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake lights	Left Brake Light Circuit
XCIOM	C109D12	short to battery	Circuit Short To Battery	*The output cannot be deactivated	1. Intermittent failure due to poor connection 2. Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake lights	Left Brake Light Circuit
XCIOM	C109D15	I < 50 mA (worst case: I < 5mA...50mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *BrakeLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Brake lights 2. Check wires and connectors between ECU and Brake lights	Left Brake Light Circuit
XCIOM	C109D18	I < 450 mA (worst case: I < 370...570mA)	Circuit Current Below Threshold	*Reduced functionality of the lamp *BrakeLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Brake lights 2. Check wires and connectors between ECU and Brake lights	Left Brake Light Circuit
XCIOM	C109E11	I > 3,671A...4,96A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *BrakeLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake lights	Right Brake Light Circuit
XCIOM	C109E12	short to battery	Circuit Short To Battery	*The output cannot be deactivated	1. Intermittent failure due to poor connection 2. Short circuit to battery	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake lights	Right Brake Light Circuit
XCIOM	C109E15	I < 50 mA (worst case: I < 5mA...50mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *BrakeLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Brake lights 2. Check wires and connectors between ECU and Brake lights	Right Brake Light Circuit
XCIOM	C109E18	I < 450 mA (worst case: I < 370...570mA)	Circuit Current Below Threshold	*Reduced functionality of the lamp *BrakeLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Brake lights 2. Check wires and connectors between ECU and Brake lights	Right Brake Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C004412	U = 5V	Circuit Short To Battery	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake pressure sensor	Brake Pressure Sensor
XCIOM	C004414	U < 0,15V	Circuit Short To Ground or Open	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake pressure sensor	Brake Pressure Sensor
XCIOM	C004416	0,15V <= U < 0,3V	Circuit Voltage Below Threshold	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Brake pressure sensor	Brake Pressure Sensor
XCIOM	C004417	4,7V < U < 5V	Circuit Voltage Above Threshold	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Brake pressure sensor	Brake Pressure Sensor
XCIOM	C10A211	U < 0,172V	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake pressure sensor	Brake Pressure Sensor Power Supply
XCIOM	C10A212	5,578V <= U <= 5,750V	Circuit Short To Battery	*The component cannot be deactivated. *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Brake pressure sensor	Brake Pressure Sensor Power Supply
XCIOM	C10A216	0,172V <= U < 4,942V	Circuit Voltage Below Threshold	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Brake pressure sensor	Brake Pressure Sensor Power Supply
XCIOM	C10A217	5,081V <= U < 5,578V	Circuit Voltage Above Threshold	*Component function disabled *BrakePressureSensorDemand = 254 (Error) *SWBrakeSwitch = 2 (Error)	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	Check Brake pressure sensor	Brake Pressure Sensor Power Supply
XCIOM	U30004A	Incorrect Component Installed	Incorrect Component Installed	Not applicable	Incorrect Component Installed	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U30004B	The temperature has exceeded the threshold on one of the outputs.	Over Temperature	The output that is overheated will be turned OFF.	The ambient temperature is too high.	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in nomral range	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C107411	I > 1,2A... 1,63A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear inter axle diff lock valve	Rear inter axle diff lock valve
XCIOM	C107412	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	* The output cannot be deactivated	Possible Root Cause: 1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear inter axle diff lock valve	Rear inter axle diff lock valve
XCIOM	C107415	I < 100mA (worst case: I < 85,7mA...111,7mA)	Circuit Short To Battery or Open	*In case of open circuit, the output cannot be activated. *In case of short circuit to battery, the output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear inter axle diff lock valve	Rear inter axle diff lock valve
XCIOM	C107211	I > 1,2A... 1,63A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	Short circuit to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear wheel diff lock valve	Rear wheel diff lock Valve
XCIOM	C107212	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	* The output cannot be deactivated	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear wheel diff lock valve	Rear wheel diff lock Valve
XCIOM	C107215	I < 100mA (worst case: I < 74mA...130mA)	Circuit Short To Battery or Open	*In case of open circuit, the output cannot be activated. *In case of short circuit to battery, the output cannot be deactivated.	Short circuit to battery or open circuit	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear wheel diff lock valve	Rear wheel diff lock Valve
XCIOM	C109F11	I > 3,671A...4,96 A (Limphome I > 11,1A...15,0A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Rear_Status = 1 (Left direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Left rear direction indicator	Left Rear Direction Indicator Circuit
XCIOM	C109F15	I < 50mA (worst case: I < 5mA...50mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *DirInd_Rear_Status = 1 (Left direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Left rear direction indicator 2. Check wires and connectors between ECU and Left rear direction indicator	Left Rear Direction Indicator Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C109F18	I < 450 mA (worst case: I < 370...570mA)	Circuit Current Below Threshold	*Lamp functionality reduced *DirInd_Rear_Status = 1 (Left direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Left rear direction indicator 2. Check wires and connectors between ECU and Left rear direction indicator	Left Rear Direction Indicator Circuit
XCIOM	C10A011	I > 3,671A...4,96 A (Limphome I > 11,1A...15,0A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *DirInd_Rear_Status = 2 (Right direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Right rear direction indicator	Right Rear Direction Indicator Circuit
XCIOM	C10A015	I < 50mA (worst case: I < 5mA...50mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *DirInd_Rear_Status = 2 (Right direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Right rear direction indicator 2. Check wires and connectors between ECU and Right rear direction indicator	Right Rear Direction Indicator Circuit
XCIOM	C10A018	I < 450 mA (worst case: I < 370...570mA)	Circuit Current Below Threshold	*Lamp functionality reduced *DirInd_Rear_Status = 2 (Right direction indicator lamp fault) or 3 (Both left and right direction indicators fault)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Right rear direction indicator 2. Check wires and connectors between ECU and Right rear direction indicator	Right Rear Direction Indicator Circuit
XCIOM	C10F155	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 380]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Configuration
XCIOM	C10F255	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 381]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Levels
XCIOM	C10F355	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 382]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Wheelbase
XCIOM	C10F455	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 384]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Vehicle Component Characteristics

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10F555	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 385]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Axle Load Distribution
XCIOM	C10F655	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 386]: Inconsistent or erroneous parameter setting	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air suspension Parameter Check Controller Setup
XCIOM	C111955	Invalid parameter values.	Not Configured	Unpredictable ECS behaviour	Inconsistent ECS parameter settings in front and rear	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air Suspension Parameter Check Front/Rear Consistency
XCIOM	C111987	Timeout on CRC check message between front and rear ECS nodes.	Missing Message	Unpredictable ECS behaviour	CAN bus errors. Front ECS node not responding.	Diagnostic Service 14. No additional action is needed.	Check the parameters value which are in the error list.	Air Suspension Parameter Check Front/Rear Consistency
XCIOM	C10A311	I > 2,44A...3.3A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *FogLightRearLampCheck_stat = 2 (Tested, Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Fog lights	Rear Fog Lights
XCIOM	C10A315	Bulb: I < 0.5A (open circuit or 2 bulbs broken) (worst case: I < 455mA...560mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *FogLightRearLampCheck_stat = 2 (Tested, Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Fog lights 2. Check wires and connectors between ECU and Rear Fog lights	Rear Fog Lights
XCIOM	C10A318	Bulb only: I between 0,5A to 1.5A --> 1 bulb broken I>1,5A --> 0 bulb broken, everything ok. (worst case: I < 1430mA...1600mA)	Circuit Current Below Threshold	*One of the two bulbs is broken. *FogLightRearLampCheck_stat = 2 (Tested, Lamp NOK)	1. One bulb broken. 2. Connected component broken. 3. Wires broken. Insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Fog lights 2. Check wires and connectors between ECU and Rear Fog lights	Rear Fog Lights
XCIOM	C10D011	I > 2,219A...3,0A I > 9,43A...12,74A (Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *PositionLightRearLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Position lights	Left Rear Position Light and Registration Plate Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10D015	Bulb I < 100mA LED not possible	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *PositionLightRearLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Position lights 2. Check wires and connectors between ECU and Rear Position lights	Left Rear Position Light and Registration Plate Light Circuit
XCIOM	C10D018	I < 570mA bulb broken (5W bulb) (worst case: I < 395mA...620mA) I < 720mA bulb broken (10W bulb) (worst case: I < 500mA...790mA)	Circuit Current Below Threshold	*One or two of the three bulbs are broken. *PositionLightRearLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. One bulb broken 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Position lights 2. Check wires and connectors between ECU and Rear Position lights	Left Rear Position Light and Registration Plate Light Circuit
XCIOM	C10D111	I > 2,219A...3,0A I > 9,43A...12,74A (Limphome)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *PositionLightRearLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Position lights	Right Rear Position Light Circuit
XCIOM	C10D115	I < 100mA (worst case bulb: I < 40mA...130mA) LED not possible	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *PositionLightRearLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Position lights 2. Check wires and connectors between ECU and Rear Position lights	Right Rear Position Light Circuit
XCIOM	C10D118	I < 350mA bulb broken (worst case: I < 220mA...390mA)	Circuit Current Below Threshold	*One of the two bulbs is broken. *PositionLightRearLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. One bulb broken 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Rear Position lights 2. Check wires and connectors between ECU and Rear Position lights	Right Rear Position Light Circuit
XCIOM	U002888	Bus off detected, CAN communication not working	Bus off	Vehicle cannot be used.	1. Wiring harness 2. ECU failure (any ECU connected to Chassis network)	Diagnostic Service 14. No additional action is needed.	Check harness Replace ECU	CAN Communication Chassis Net
XCIOM	C10E07A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 150]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left Bellow Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10E17A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 151]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Right Bellow Circuit
XCIOM	C10E27A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 152]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left/Right Bellow Circuit
XCIOM	C10E37A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 153]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive Axle Left Bellow Circuit
XCIOM	C10E47A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 154]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive Axle Right Bellow Circuit
XCIOM	C10E57A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 166]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle Left/Right Bellow Circuit
XCIOM	C10E67A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 167]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle Lift Bellow Circuit
XCIOM	C10E77A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 170]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Left/Right Bellow Circuit
XCIOM	C10E87A	Air leakage detected by algorithm	Fluid Leak or Seal Failure	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 171]: Broken Bellows, Broken Pipes from valve to bellows	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Lift Bellow Circuit
XCIOM	C107F07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 216]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C107F16	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 215]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left Pressure Sensor Circuit
XCIOM	C107F17	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 214]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left Pressure Sensor Circuit
XCIOM	C108007	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 219]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Right Pressure Sensor Circuit
XCIOM	C108016	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 218]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Right Pressure Sensor Circuit
XCIOM	C108017	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 217]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Right Pressure Sensor Circuit
XCIOM	C108107	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 222]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left/Right Pressure Sensor Circuit
XCIOM	C108116	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 221]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left/Right Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C108117	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 220]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left/Right Pressure Sensor Circuit
XCIOM	C10A407	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 270]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A416	U_Normalized <= P1CBX.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 269]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A417	U_Normalized >= P1CBX.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 268]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A507	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 274]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A516	U_Normalized <= P1CBX.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 273]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A517	U_Normalized >= P1CBX.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 272]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10A607	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 278]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A616	U_Normalized <= P1CBX.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 277]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10A617	U_Normalized >= P1CBX.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 276]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C108216	U < P1CBZ.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 287]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Pressure Sensor Supply Voltage Circuit
XCIOM	C108217	U > P1CBZ.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 286]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Pressure Sensor Supply Voltage Circuit
XCIOM	C10C207	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 252]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Left/Right Pressure Sensor Circuit
XCIOM	C10C216	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 251]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Left/Right Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C217	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 250]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Left/Right Pressure Sensor Circuit
XCIOM	C10C307	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 255]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Lift Pressure Sensor Circuit
XCIOM	C10C316	U_Normalized <= P1CBW.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 254]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Lift Pressure Sensor Circuit
XCIOM	C10C317	U_Normalized >= P1CBW.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 253]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Lift Pressure Sensor Circuit
XCIOM	C10CA07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 376]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10CA16	U_Normalized <= P1CBX.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 375]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C10CA17	U_Normalized >= P1CBX.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 374]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Valve Pressure Sensor Circuit
XCIOM	C112707	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 243]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle Lift Pressure Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C112716	U_Normalized <= P1CBX.Sensor_Voltage_Error_Low (default 0.3 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 241]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle Lift Pressure Sensor Circuit
XCIOM	C112717	U_Normalized >= P1CBX.Sensor_Voltage_Error_High (default 4.7 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 242]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle Lift Pressure Sensor Circuit
XCIOM	C10A711	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 6]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10A712	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 5] Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10A715	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 5]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 7]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10A772	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 8B]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Load/Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10A773	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 8A]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10A811	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 46]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10A812	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 45]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10A815	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 45]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 47]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10A872	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 48]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 1- or 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10BE11	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 128]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Left Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BE12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 127]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Left Circuit
XCIOM	C10BE15	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 127]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 129]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Left Circuit
XCIOM	C10BE73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 130]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Left Circuit
XCIOM	C10BF11	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 132]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Right Circuit
XCIOM	C10BF12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 131]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BF15	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 131]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 133]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Right Circuit
XCIOM	C10BF73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 134]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Second Drive Axle Right Circuit
XCIOM	C10D572	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 135]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Circuit
XCIOM	C10C011	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 137]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C012	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 136]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C015	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 136]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 138]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C072	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 140]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C073	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 139]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C111	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 142]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C112	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 141]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Load Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C115	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 141]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 143]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C172	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 145]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C173	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 144]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10AA11	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 38]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C10AA12	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 37)]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Left Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10AA15	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 37]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 39]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C10AA72	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 40]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Left Circuit
XCIOM	C10AB11	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 42]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C10AB12	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 41]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C10AB15	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 41]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 43]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10AB72	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 44]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle 2-channel Solenoid Valve Right Circuit
XCIOM	C10B011	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 66]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B012	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 65]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B015	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 65]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 67]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B073	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 68]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B111	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 70]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B112	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 69]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B115	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 69]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 71]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B173	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 72]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B211	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 74]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Axle Left/Right Circuit
XCIOM	C10B212	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 73]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Axle Left/Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B215	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 73]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 75]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Axle Left/Right Circuit
XCIOM	C10B273	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 76]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Axle Left/Right Circuit
XCIOM	C10B311	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 78]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Lift Circuit
XCIOM	C10B312	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 77]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Lift Circuit
XCIOM	C10B315	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 77]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 79]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Lift Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B373	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 80]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Pusher Lift Circuit
XCIOM	C10D372	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 81]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Circuit
XCIOM	C10B411	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 83]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10B412	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 82]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10B415	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 82]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 84]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10B472	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 86]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B473	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 85]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10B511	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 88]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10B512	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 87]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10B515	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 87]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 89]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10B572	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 91]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10B573	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 90]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Pusher Axle 4-channel Solenoid Valve Load Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B611	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 93]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B612	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 92]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B615	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 92]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 94]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B673	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 95]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10B711	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 97]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B712	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 96]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B715	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 96]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 98]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B773	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 99]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10B811	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 101]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10B812	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 100]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10B815	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 100]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 102]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10B873	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 103]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10B911	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 105]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift
XCIOM	C10B912	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 104]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift
XCIOM	C10B915	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 104]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 106]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift
XCIOM	C10B973	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 107]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift
XCIOM	C10D472	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 108]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BA11	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 110]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10BA12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 109]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10BA15	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 109]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 111]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10BA72	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 113]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10BA73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 112]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10BB11	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 115]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Load Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BB12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 114]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10BB15	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 114]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 116]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10BB72	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 118]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10BB73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 117]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10BC11	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 120]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Left Circuit
XCIOM	C10BC12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 119]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Left Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BC15	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 119]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 121]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Left Circuit
XCIOM	C10BC73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 122]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Left Circuit
XCIOM	C10BD11	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 124]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Right Circuit
XCIOM	C10BD12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 123]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Right Circuit
XCIOM	C10BD15	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 123]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 125] Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10BD73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 126]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Second Drive Axle 4-channel Solenoid Valve First Drive Axle Right Circuit
XCIOM	C10C411	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 348]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10C412	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 347]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10C415	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 347]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 349]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10C473	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 350]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Left Circuit
XCIOM	C10C511	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 352]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C512	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 351]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10C515	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 351]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 353]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10C573	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 354]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Drive Axle Right Circuit
XCIOM	C10C611	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 356]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10C612	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 355]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C615	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 355]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 357]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10C673	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 358]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Left/Right Circuit
XCIOM	C10C911	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 370]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C912	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 369]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C915	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 369]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 371]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Load Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C972	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 373]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C973	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 372]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Load Circuit
XCIOM	C10C811	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 365]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C812	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 364]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C815	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 364]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 366]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C872	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 368]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10C873	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 367]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Exhaust Circuit
XCIOM	C10C711	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 360]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift Circuit
XCIOM	C10C712	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 359]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift Circuit
XCIOM	C10C715	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 359]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 361]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift Circuit
XCIOM	C10C773	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 362]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Tag Axle Lift Circuit
XCIOM	C10D772	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 363]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Second Drive and Tag Axle 4-channel Solenoid Valve Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10D811	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 14]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10D812	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 13]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10D815	I < 50mA	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 13]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 15]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10D872	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 16B]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10D873	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 16A]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Load/Exhaust Circuit
XCIOM	C10D911	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 340]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Left/Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10D912	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 339]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10D915	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 339]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 341]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10D972	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 342]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Left/Right Circuit
XCIOM	C10DA11	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 344]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Lift Circuit
XCIOM	C10DA12	U > U_bat - 2 V	Circuit Short To Battery	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 343]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay..	Air Suspension Pusher Axle 2-channel Solenoid Valve Lift Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10DA15	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 343]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 345]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Lift Circuit
XCIOM	C10DA72	Mechanical fault detected by algorithm	Actuator Stuck Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 346]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle 2-channel Solenoid Valve Lift Circuit
XCIOM	C10DB07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 40]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left Channel Circuit
XCIOM	C10DC07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 44]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Right Channel Circuit
XCIOM	C10DD07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 48]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes, mech fault on a sensor	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First Drive Axle Left/Right Channel Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10DE07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 342]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Left/Right Channel Circuit
XCIOM	C10DF07	Mechanical fault detected by algorithm	Mechanical Failures	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 346]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Pusher Axle Lift Channel Circuit
XCIOM	C112811	I > 4.0A	Circuit Short To Ground	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 393]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Left Right Circuit
XCIOM	C112812	U > U_bat - 2 V	Circuit Short To Battery	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 394]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Left Right Circuit
XCIOM	C112815	I > 4.0A	Circuit Short To Battery or Open	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 392]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 394]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Left Right Circuit
XCIOM	C112873	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 395]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Left Right Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C112911	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 398]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Lift Circuit
XCIOM	C112912	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 397]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong.	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Lift Circuit
XCIOM	C112915	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 397]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 399]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Lift Circuit
XCIOM	C112973	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve activation are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 400]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension First and Sec Drive Axle 4-channel Solenoid Valve Sec Drive Axle Lift Circuit
XCIOM	C112A11	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 320]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Load/Exhaust Circuit
XCIOM	C112A12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 319]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Load/Exhaust Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C112A15	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 321]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Load/Exhaust Circuit
XCIOM	C112A72	Mechanical fault detected by algorithm	Actuator Stuck Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 322]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Load/Exhaust Circuit
XCIOM	C112A73	Mechanical fault detected by algorithm	Actuator Stuck Closed	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 322]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Load/Exhaust Circuit
XCIOM	C112B11	I > 4.0A	Circuit Short To Ground	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 323]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Tag Lift Circuit
XCIOM	C112B12	U > U_bat - 2 V	Circuit Short To Battery	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 324]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Tag Lift Circuit
XCIOM	C112B15	I < 50mA	Circuit Short To Battery or Open	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 325]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Tag Lift Circuit
XCIOM	C112B71	Mechanical fault detected by algorithm	Actuator Stuck	Air suspension total limphome: No valve actuations are allowed. All functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 326]: Solenoid Mechanical stuck, pipes wrongly connected, pipes broken, dirt in valve, dirt in pneumatic connectors, dirt in pipes	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Tag Axle 1ch Solenoid Valve Tag Lift Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C112664	Rear1ClutchPosition Status is not received according to the command Rear1ClutchCommand.	Signal Plausibility Failure	Rear1ClutchStatus = 2 (Error)	1. Open circuit on one or two of the steering servo switches. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Inter Axle Dog Clutch Valve.	Rear Dog Clutch Feedback Sensor
XCIOM	C112511	I > 1,2A... 1,63A	Circuit Short To Ground	Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Inter Axle Dog Clutch Valve.	Rear Inter Axle Dog Clutch Valve
XCIOM	C112512	V > Ubat-1,80V (worst case: V > Ubat-2,10V... Ubat-1,50V)	Circuit Short To Battery	The output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Inter Axle Dog Clutch Valve.	Rear Inter Axle Dog Clutch Valve
XCIOM	C112515	I < 100mA (worst case: I < 85,7mA...111,7mA)	Circuit Short To Battery or Open	In case of open circuit, the output cannot be activated. In case of short circuit to battery, the output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear Inter Axle Dog Clutch Valve.	Rear Inter Axle Dog Clutch Valve
XCIOM	C107B07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 186]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left Level Sensor Circuit
XCIOM	C107B16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 185]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left Level Sensor Circuit
XCIOM	C107B17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 184]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left Level Sensor Circuit
XCIOM	C107B54	Level sensor is not calibrated	Missing Calibration	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 187]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left Level Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C107C07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 190]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Right Level Sensor Circuit
XCIOM	C107C16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 189]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Right Level Sensor Circuit
XCIOM	C107C17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 188]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Right Level Sensor Circuit
XCIOM	C107C54	Level sensor is not calibrated	Missing Calibration	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 191]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Right Level Sensor Circuit
XCIOM	C107D07	Mechanical fault detected by algorithm	Mechanical Failures	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 194]: Level Sensor Arm broken, Sensor Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left/Right Level Sensor Circuit
XCIOM	C107D16	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 193]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left/Right Level Sensor Circuit
XCIOM	C107D17	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 192]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong, Level Sensor Arm broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left/Right Level Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C107D54	Level sensor is not calibrated	Missing Calibration	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 195]: No level sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Left/Right Level Sensor Circuit
XCIOM	C107E16	U < P1B9W.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 285]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Level Sensor Supply Voltage Circuit
XCIOM	C107E17	U > P1B9W.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Air suspension full limphome: All sensor inputs are ignored and it is only possible to control the vehicle in manual mode. All bogie functionality is deactivated.	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 284]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Air Suspension Rear Level Sensor Supply Voltage Circuit
XCIOM	C10F916	U < P1B9W.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 391]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Rear Axle Load Sensor Supply Voltage Circuit
XCIOM	C10F917	U > P1B9W.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 390]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Rear Axle Load Sensor Supply Voltage Circuit
XCIOM	C10D616	U_Normalized <= P1E4H.SensorVoltageErrorLow (default 0.3 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 283]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Rear Axle Load Sensor Circuit
XCIOM	C10D617	U_Normalized >= P1E4H.SensorVoltageErrorHigh (default 4.7 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 282]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Rear Axle Load Sensor Circuit
XCIOM	C10D654	Load sensor is not calibrated.	Missing Calibration	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 379]: No load sensor calibration has been performed	Diagnostic Service 14. No additional action is needed.	Ensure the issues described in Possible Root Cause are okay.	Rear Axle Load Sensor Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C105B11	I > 4,674A...6,313A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *RevLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Reverse lights	Left Reverse Light Circuit
XCIOM	C105B15	Bulb: I < 300mA (worst case: I < 240mA...380mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *RevLightLamp_stat = 2 (Tested, left Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Reverse lights 2. Check wires and connectors between ECU and Reverse lights	Left Reverse Light Circuit
XCIOM	C105C11	I > 4,674A...6,313A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *RevLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Reverse lights	Right Reverse Light Circuit
XCIOM	C105C15	Bulb: I < 300mA (worst case: I < 240mA...380mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *RevLightLamp_stat = 3 (Tested, right Lamp NOK) or 4 (Tested left and right Lamp NOK)	1. Bulb broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Reverse lights 2. Check wires and connectors between ECU and Reverse lights	Right Reverse Light Circuit
XCIOM	C10A911	I > 1,487A...2,008A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	"1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection."	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Reverse back-up alarm	Reverse Warning Sound Circuit
XCIOM	C10A912	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated. Reverse beeper continues to beep.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Reverse back-up alarm	Reverse Warning Sound Circuit
XCIOM	C107511	I > 1,2A... 1,63A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Axle lock valve	axle lock valve

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C107515	I < 100mA (worst case: I < 85,7mA...111,7mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Axle lock valve	axle lock valve
XCIOM	C105811	I > 3,172A...4,285 A (Limphome I > 11,08A...14,97A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear side markers	Rear Side Markers Circuit
XCIOM	C105812	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Rear side markers	Rear Side Markers Circuit
XCIOM	C10A111	I > 3,162A...4,27 A (Limphome I > 9,43A...12,74A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front side markers	Front Side Markers Circuit
XCIOM	C10A112	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Front side markers	Front Side Markers Circuit
XCIOM	C106211	I > 14,65A...19,8A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer Brake lights	Trailer brake lights
XCIOM	C106215	I < 250mA (worst case: I < 100mA...345mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Trailer Brake lights 2. Check wires and connectors between ECU and Trailer Brake lights	Trailer brake lights
XCIOM	C10AF11	I > 7,32A...9,9A (Limphome I > 20,6...27,9A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed *TrailerDirInd_status = 2 (Trailer Right Direction Indicator Fault) or 3 (Both Left and Right Trailer Direction Indicators Fault).	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer Direction indicators	Trailer Right Direction Indicator Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10AF15	I < 50mA (worst case: I < 14mA...95mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *TrailerDirInd_status = 2 (Trailer Right Direction Indicator Fault) or 3 (Both Left and Right Trailer Direction Indicators Fault).	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Trailer Direction indicator lights 2. Check wires and connectors between ECU and Trailer Direction indicators	Trailer Right Direction Indicator Circuit
XCIOM	C10AE11	I > 7,32A...9,9A (Limphome I > 20,6...27,9A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed. *TrailerDirInd_status = 1 (Trailer Left Direction Indicator Fault) or 3 (Both Left and Right Trailer Direction Indicators Fault).	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer Direction indicators	Trailer Left Direction Indicator Light Circuit
XCIOM	C10AE15	I < 50mA (worst case: I < 14mA...95mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated. *TrailerDirInd_status = 1 (Trailer Left Direction Indicator Fault) or 3 (Both Left and Right Trailer Direction Indicators Fault).	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Trailer Direction indicator lights 2. Check wires and connectors between ECU and Trailer Direction indicators	Trailer Left Direction Indicator Light Circuit
XCIOM	C10AD11	I > 11,5A...15,56 A (Limphome I > 15,4A... 20,8A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer Position lights	Trailer Right Position Light Circuit
XCIOM	C10AD15	I < 125mA (worst case: I < 45mA...215mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Trailer Position lights 2. Check wires and connectors between ECU and Trailer Position lights	Trailer Right Position Light Circuit
XCIOM	C10AC11	I > 11,5A...15,56 A (Limphome I > 15,4A... 20,8A)	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer Position lights	Trailer Left Position Light Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10AC15	I < 125mA (worst case: I < 45mA...215mA)	Circuit Short To Battery or Open	*In case of open circuit, the component cannot be activated. *In case of short circuit to battery, the component cannot be deactivated.	1. Bulb/LED broken. 2. Connected component broken. 3. Wires broken. No continuity or insulation damage. 4. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check Trailer Position lights 2. Check wires and connectors between ECU and Trailer Position lights	Trailer Left Position Light Circuit
XCIOM	C10F711	I > 6,45A...8,72A	Circuit Short To Ground	*Output is deactivated until reactivation requirements are fulfilled and the fault is removed. *No trailer ABS functionality.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer ABS Power supply	Trailer ABS Power Supply
XCIOM	C10F712	No Load detected (Offstate Diagnostic)	Circuit Short To Battery	*The output cannot be deactivated. The trailer ABS ECU will always be awake and the trucks quiescient current will consume the battery.	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and Trailer ABS Power supply	Trailer ABS Power Supply
XCIOM	C10FA16	U < P1CBZ.Voltage_Low (default 4 V)	Circuit Voltage Below Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID TBD]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Pressure Switch Supply Voltage Circuit
XCIOM	C10FA17	U > P1CBZ.Voltage_High (default 5.5 V)	Circuit Voltage Above Threshold	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID TBD]: Broken Cable, Sensor broken, Connector Broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Pressure Switch Supply Voltage Circuit
XCIOM	C10E911	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 312]: Broken Cable, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Motor OnOff Relay Circuit
XCIOM	C10E915	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 311]: Broken Cable, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 313]: Broken Cable, Connector Not Connected, Connector Stift Wrong, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Motor OnOff Relay Circuit
XCIOM	C10EA11	I > 4.0A	Circuit Short To Ground	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 315]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Motor Direction Solenoid Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	C10EA15	I < 50mA	Circuit Short To Battery or Open	Normal software operation	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 314]: Broken Cable, Solenoid in valve block broken, Connector Broken, ECU Broken, Connector Stift Wrong [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 316]: Broken Cable, Connector Not Connected, Connector Stift Wrong, Solenoid in valve block broken, ECU Broken	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Motor Direction Solenoid Circuit
XCIOM	C10EB16	U <= P1D6P (default 0.3 V)	Circuit Voltage Below Threshold	ECS downgraded mode [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 318]:	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 318]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Pressure Switch Circuit
XCIOM	C10EB17	U >= P1D6Q (default 4.7 V)	Circuit Voltage Above Threshold	ECS downgraded mode [ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 317]:	[ECS4 Fault Codes.xls (Phoenix reg no 50084405), ID 317]: Broken Cable, Sensor broken, Connector Broken, ECU Broken, Connector Stift Wrong	Diagnostic Service 14. No additional action is needed.	Ensure the the issues described in Possible Root Cause are okay.	A-Ride Pressure Switch Circuit
XCIOM	U300041	ECU ROM checksum error.	General Checksum Failure	Check started at ECU startup	ECU SW has been tampered ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300044	ECU volatile memory error.	Data Memory Failure	RCIOM does not work properly.	1. ECU faulty (volatile memory)	Diagnostic Service 14. No additional action is needed.	If this DTC occurs frequently with similar DTC, this could indicate HW failure	Electronic control unit
XCIOM	U300045	ECU non volatile memory error.	Program Memory Failure	RCIOM does not work properly.	1. ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	If this DTC occurs frequently with similar DTC, this could indicate HW failure	Electronic control unit
XCIOM	U300049	ECU internal hardware fault.	Internal Electronic Failure	ECU does not work properly.	1. ECU faulty (BSW driver / transceiver). 2. Software error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300094	ECU watchdog fault.	Unexpected Operation	ECU does not work properly, potentially in FailSafe mode.	1. ECU faulty (watchdog fault) 2. Software error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U1FFF68	Factory mode parameter P1C54 is active	Event Information	A truck delivered with Factory mode active will have unspecified behavior that might influence customer safety.	Factory mode not disabled	Diagnostic Service 14. No additional action is needed.	Set P1C54 (Shared Factory mode parameter) for the ECU to FALSE before shipping from factory.	Factory Mode Active
XCIOM	U114268	Too many DTC for Diagnostic Warning Manger (DWM) to handle.	Event Information	Max. number of DTCs reached. No more DTCs will be stored.	1. Refer to the DTC's that are active.	Diagnostic Service 14. No additional action is needed.	None, this is a notification to the driver	DTC Notification Max Limit Reached

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
XCIOM	U250145	Internal Data Fragmentation detected by internal routines. The ECU cannot save internal status information like updated DTCs at resets.	Program Memory Failure	ECU cannot save internal data like new DTC	- Inproper shutdown - power cut - In rare situations HW failure due to memory degradation - Software error	Diagnostic Service 14. No additional action is needed.	Transfer data, run RID R1AHD, restore data may fix the problem	Internal Data Storage Fragmented
XCIOM	U250245	Internal Data Storage Discarded. The error is selfhealing if possible, but Required for fault tracing	Program Memory Failure	Fault tracing	- Software error - Memory degradation due to end-of-life	Diagnostic Service 14. No additional action is needed.	No action due to this DTC, the ECU has tried to recover. There are likely other DTCs that need to be handled as a consequence.	Internal Data Storage Discarded
XCIOM	U300046	- Program memory corruption - ROM Checksum error	Calibration / Parameter Memory Failure	Not applicable	- Memory hardware error	Diagnostic Service 14. No additional action is needed.	Reprogram SW Replace ECU	Electronic control unit
XCIOM	U300716	U < 16 V	Circuit Voltage Below Threshold	ECU only partly working - some outputs are not powered	- Power supply "B" broken - Battery voltage too low - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "B"
XCIOM	U300717	U > 32 V	Circuit Voltage Above Threshold	ECU potentially blown	- Battery voltage too high - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "B"
XCIOM	U300616	U < 16 V	Circuit Voltage Below Threshold	ECU only partly working - some outputs are not powered	- Power supply "A" broken - Battery voltage too low - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "A"
XCIOM	U300617	U > 32 V	Circuit Voltage Above Threshold	ECU potentially blown	- Battery voltage too high - ECU malfunctioning	Diagnostic Service 14. No additional action is needed.	Check cables, connectors and fuses	Control Module Input Power "A"
XCIOM	U300093	Unexpected ECU operation.	No Operation	ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors. Too many restarts or RAM/ROM errors.	ECU in failsafe (limited, limphome) mode due to unexpected HW or SW errors. Too many restarts or RAM/ROM errors.	Diagnostic Service 14. No additional action is needed.	Powercycle ECU Reprogram SW Replace ECU	Electronic control unit
IC	U115D92	Unable to open/play requested audio file.	Performance or Incorrect Operation	Instrument Cluster will send CAN signal "SoundStatus" to "Degraded function" on Display Subnet	1. No sound library downloaded 2. Requested sound file corrupt	Diagnostic Service 14. No additional action is needed.	Reflash Instrument Cluster.	Instrument Cluster Loadspeaker
IC	U200A88	Instrument Cluster communication error on Display Subnet	Bus off	No communication is possible to the Instrument Cluster. The instrument Cluster can not receive any data on the CAN-network. The instrument Cluster will keep the last recevied VehicleMode	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Investigate root-cause: - CAN-network error => Fix problem. - CAN-cable => Fix problem. Transceiver failure => Replace ECU	CAN Communication Display Net
IC	U300041	Internal EEPROM failure	General Checksum Failure	Enter limp-home mode. Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory where the patchable data is stored.	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U300044	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U300045	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Electronic control unit
IC	U300049	Internal electronic failure	Internal Electronic Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	- CAN-tranciever error	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U30004B	Temperature of TFT-Display >70 degrees Celsius	Over Temperature	ECU (IC) will set the background illumination level to Max 25% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300094	Application performs an unexpected operation	Unexpected Operation	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Watch dog reset (software / parameter bug)	Diagnostic Service 14. No additional action is needed.	Talk to driver 1. Can indicate improper shutdown - see possible root causes, no Instrument Cluster action required 2. If Cluster has been behaving strangely reflash Instrument Cluster	Electronic control unit
IC	U300098	Temperature of TFT-display >65 degrees Celsius	Component or System Over Temperature	ECU (IC) will set the background illumination level to Max 50% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300316	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	All functionalities are switched off (including the illumination of the IC). Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
IC	U300317	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	All functionalities are switched off (including the illumination of the IC). Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
IC	U116C05	Programming of Basic SID (from IC) has failed.	System Programming Failures	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Programming of SID-Basic intterupted due to: -Power Supply problem -Communication link failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Unit
IC	U116C44	Internal RAM failure	Data Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic SID Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Unit
IC	U116C45	Internal ROM failure	Program Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic SID Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U116C46	Internal EEPROM failure	Calibration / Parameter Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic SID Memory Failure	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Unit
IC	B150216	Circuit Voltage Below Threshold (UnderVoltage) (<= 8.25 Volts)	Circuit Voltage Below Threshold	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Voltage supply too low	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Power Supply Circuit
IC	B150217	Circuit Voltage Above Threshold (OverVoltage) (>= 36 Volts)	Circuit Voltage Above Threshold	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Voltage supply too high	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	DTC NOT SUPPORTED IN THE SW: CANNOT BE SET	Basic Secondary Infotainment Display Power Supply Circuit
IC	U116E68	The number of DTC's set in the ECU has reached 29	Event Information	No reduced functionality due to that the DTC is set.	Maximum limit of DTC's set in ECU	Diagnostic Service 14. No additional action is needed.	Clear DTCs	DTC Notification Max Limit Reached
IC	P000041	Program memory corruption.	General Checksum Failure	Application should not start (remains in boot-mode). Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Control Module
IC	U1FFF68	Factory Mode is active in the ECU.	Event Information	No reduced functionality due to that the DTC is set.	Factory Mode is active in the ECU. Whenever P1C54 parameter is set to TRUE, this DTC shall be set.	Diagnostic Service 14. No additional action is needed.	Write parameter P1C54 to false.	Factory Mode Active
IC	U116F98	Ambient Temperature > +85 degrees C during Vehicle Operation	Component or System Over Temperature	None	Ambient Temperature is High Ambient Temperature Sensor Fault	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Ambient Air Temperature
IC	U116B87	Missing heartbeat signal TIPSITP_DIDHeartbeat_Up	Missing Message	Instrument Cluster is missing a message from HMIIOM. Instrument Cluster will make a re-start of the Driver Interface Display (DID)	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check compatibility of HMIIOM and IC software. Check integrity of Display Subnet CAN cables	Lost Communication With HMIIOM
IC	U116A87	Engine Malfunction Indication CAN message is missing	Missing Message	Instrument Cluster will lit up the MalFunction Indication Lamp	The Engine Malfunction Indication CAN message is not recieved by the Instrument Cluster.	Diagnostic Service 14. No additional action is needed.	Check integrity of Display Subnet CAN cables	Malfunction Indication Lamp Fault Status Electronic Data Link
SID	U200A88	Communication error on Display Subnet	Bus off	No communication is possible to the Secondary Information Display (SID). The SID can not receive any data on the CAN-network. The SID keep the last recevied VehicleMode	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connector damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Investigate root-cause: - CAN-network error => Fix problem. - CAN-cable => Fix problem. Transceiver failure => Replace ECU	CAN Communication Display Net
SID	U300041	Internal EEPROM failure	General Checksum Failure	Enter limp-home mode. Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory where the patchable data is stored.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SID	U300044	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SID	U300045	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Electronic control unit
SID	U300049	Internal electronic failure	Internal Electronic Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	-CAN-tranciever error	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
SID	U30004B	Temperature of TFT Display > 70 degrees Celcius	Over Temperature	ECU (IC) will set the background illumination level to Max 25% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
SID	U300094	Application performs an unexpected operation	Unexpected Operation	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Watch dog reset (software / parameter bug) Sudden loss of power	Diagnostic Service 14. No additional action is needed.	Talk to driver 1. Can indicate improper shutdown - see possible root causes, no Instrument Cluster action required 2. If Cluster has been behaving strangely reflash Instrument Cluster	Electronic control unit
SID	U300098	Temperature of TFT display > 65 degrees Celsius	Component or System Over Temperature	ECU (IC) will set the background illumination level to Max 50% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
SID	U300316	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	All functionalities are switched off (including the illumination of the SID). Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
SID	U300317	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	All functionalities are switched off (including the illumination of the SID). Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
SID	U114863	Continuous touch screen press events for >= 10 seconds	Circuit / Component Protection Time-Out	SID will notify other ECU that touch screen information is not reliable anymore by sending "Error" coordinates	- Touch screen pressed / struck for a long time (for atleast 10 seconds continuously) - Touch screen faulty	Diagnostic Service 14. No additional action is needed.	Verify that no objects are put on the touch screen. If not, replace ECU	Touch Screen Display Module
SID	U114768	The number of DTC's set in the ECU has reached 29	Event Information	No reduced functionality due to that the DTC is set.	To many internal diagnostic trouble codes set in the ECU.	Diagnostic Service 14. No additional action is needed.	Clear DTCs	DTC Notification Max Limit Reached

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SID	P000041	Program memory corruption.	General Checksum Failure	Application should not start (remains in boot-mode). Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Control Module
SID	U1FFF68	Factory Mode is active in the ECU.	Event Information	No reduced functionality due to that the DTC is set.	Factory Mode is active in the ECU. Whenever P1C54 parameter is set to TRUE, this DTC shall be set.	Diagnostic Service 14. No additional action is needed.	Write parameter P1C54 to false	Factory Mode Active
SID	U114998	Ambient Temperature > +85 degrees C during Vehicle Operation	Component or System Over Temperature	None	Ambient Temperature is High Ambient Temperature Sensor Fault	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Ambient Air Temperature
SID	U114608	LVDS communication error with Secondary information Display	Bus Signal / Message Failures	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Low Level failure on LVDS Communication LVDS communication link error or LVDS internal hardware error	Diagnostic Service 14. No additional action is needed.	Check integrity of LVDS cable Ensure LVDS cable is connected properly both at SID and Video Switch Check Video Switch for DTCs	LVDS Video Data Circuit
SID	B150363	Push button error: -GPIO(General Purpose Input/Output) pin is read as 1 continuously for 30 seconds	Circuit / Component Protection Time-Out	One button can no longer be used.	One of the push buttons from SID is stucked. Internal SID hardware failure.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Secondary Infotainment Display Button Switches
LECM	U114107	Button in pressed state for 60s	Mechanical Failures	A release event will be sent when a softkey button is detected stuck. An error event will be sent when a rocker button is detected stuck	Mechanical - Button stuck	Diagnostic Service 14. No additional action is needed.	Check if any ECU button is stuck	Softkey and Rocker Buttons
LECM	U003788	Bus off state	Bus off	No communication on CAN General reduced system functionality	CAN bus short circuit	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors	CAN Communication Cab Net
LECM	U300041	CPU memory corrupted or un-validated SW downloaded	General Checksum Failure	Unit will do a reset and stay in boot mode. Re-programming will resolve.	EMC	Diagnostic Service 14. No additional action is needed.	Update SW	Electronic control unit
LECM	U300045	CPU memory read/write error	Program Memory Failure	DTC itself does not affect any functionality. Functional impact depends on each specific case - what actual (EEPROM) data failed to read/write.	Data retention	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
LECM	U300046	Calculated checksum/signature does not match stored	Calibration / Parameter Memory Failure	Calculated checksum/signature of CPU memory does not match stored	EMC or wrong handling	Diagnostic Service 14. No additional action is needed.	Update SW	Electronic control unit
LECM	U300049	Can tranceiver/driver failure	Internal Electronic Failure	DTC itself does not inhibit any functionality. Functional impact will be as result of can bus error.	CAN tranceiver error	Diagnostic Service 14. No additional action is needed.	Check cabling and connectors Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
LECM	U30004B	Measured temperature > 95 degrees C	Over Temperature	Display and button illumination reduced/switched off.	Ambient temperature too high	Diagnostic Service 14. No additional action is needed.	Check area around ECU for sources of heat. Replace ECU.	Electronic control unit
LECM	U300094	Critical internal SW error	Unexpected Operation	ECU will reset	Code error / assert	Diagnostic Service 14. No additional action is needed.	Update SW	Electronic control unit
LECM	U300316	Voltage < 15±1V	Circuit Voltage Below Threshold	If voltage TOO low, ECU operation will suffer. Normal functionality will deteriorate.	ECU incorrectly connected. Bad battery/generator.	Diagnostic Service 14. No additional action is needed.	Check supply cabling, battery and generator	Battery Power
LECM	U300317	Voltage > 34±2V	Circuit Voltage Above Threshold	If voltage TOO high for TOO long, ECU may be damaged. Normal functionality will deteriorate.	ECU incorrectly connected. Bad battery/generator.	Diagnostic Service 14. No additional action is needed.	Check battery and generator	Battery Power
LECM	U1FFF68	Truck is in Factory Mode. Factory mode address parameter is set to value "1"	Event Information	No functionality affected	Whenever P1C54 (Shared Factory mode parameter) is set to TRUE, this DTC shall be set. Unit not configured correctly when leaving factory.	Diagnostic Service 14. No additional action is needed.	Clear the factory mode parameter.	Factory Mode Active
LECM	U114368	Too many (>29) DTCs with warning indicator bit set are active at the same time.	Event Information	No more DTCs will be reported in diagnostic fault message on CAN.	Maximum limit of DTC's set in ECU	Diagnostic Service 14. No additional action is needed.	Analyze the other reported DTCs	DTC Notification Max Limit Reached
SRS	U003788	CAN lines problems	Bus off	Deployment of Airbag and Seatbelt retractors will occure	CAN_L line leakage to UBAT CAN_H leakage to GND or UBAT CAN_L in short with CAN_H	Diagnostic Service 14. No additional action is needed.	Check CAN wires Battery reset	CAN Communication Cab Net
SRS	U300041	cheksum calculation error	General Checksum Failure	ECU not alive	EEPROM checksum error	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300044	memory failure	Data Memory Failure	Deployment of Airbag and Seatbelt retractors is disabled.	RAM, stack failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300045	Flash memory failure	Program Memory Failure	Deployment of Airbag and Seatbelt retractors is disabled.	ROM failure	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300046	EEPROM CRC failure	Calibration / Parameter Memory Failure	Deployment of Airbag and Seatbelt retractors is disabled.	EEPROM corrupted data	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with correct SW 2. Replace ECU	Electronic control unit
SRS	U300049	Internal voltage failure	Internal Electronic Failure	Deployment of Airbag and Seatbelt retractors is disabled.	ASIC defect (power supply ASIC or firing logic ASIC)	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300054	EOL configuration error	Missing Calibration	Deployment of Airbag and Seatbelt retractors is disabled.	No EOL data, wrong platform selection, wrong calibration data	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with complete SW package 2. Replace ECU	Electronic control unit
SRS	U300064	Sensor faults	Signal Plausibility Failure	Deployment of Airbag and Seatbelt retractors is disabled.	drift, settling sensor faults	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300094	unexpected reset	Unexpected Operation	Deployment of Airbag and Seatbelt retractors will occure	watchdog, reset, safing controller faults	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
SRS	U300316	ECU Low Voltage < 9 V	Circuit Voltage Below Threshold	Deployment of Airbag and Seatbelt retractors will occure	broken power lines, voltage lower than 9 V	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
SRS	U300317	ECU High Voltage > 33V	Circuit Voltage Above Threshold	Deployment of Airbag and Seatbelt retractors will occure	problematic power lines, voltage hogher than 33V	Diagnostic Service 14. No additional action is needed.	Check power supply system Overload alternator	Battery Power

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SRS	U300286	Wrong response to VIN request	Signal Invalid	Deployment of Airbag and Seatbelt retractors is disabled. Error Status reporting on CAN	ECU installed in wrong cabin	Diagnostic Service 14. No additional action is needed.	1. Check instrument cluster for correct VIN 2. Reprogram ECU 3. Replace ECU	Vehicle Identification Number
SRS	U300287	No response to VIN request	Missing Message	Error Status reporting on CAN	Instrument cluster wrong behavior	Diagnostic Service 14. No additional action is needed.	Check instrument cluster	Vehicle Identification Number
SRS	B000101	Wrong connection of airbag squib	General Electrical Failure	Airbag not possible to be deployed Error status reporting on CAN	The squib is not connected to the designed pins	Diagnostic Service 14. No additional action is needed.	1. Check airbag and Seatbelt retractors connection 2. Check wiring harness for poor connection (short circuit)	Driver Frontal Stage 1 Ignition Loop
SRS	B000111	Short - circuit	Circuit Short To Ground	Airbag not possible to be deployed Error status reporting on CAN	short - circuit on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Frontal Stage 1 Ignition Loop
SRS	B000112	short to battery	Circuit Short To Battery	Airbag not possible to be deployed Error status reporting on CAN	short to battery on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Frontal Stage 1 Ignition Loop
SRS	B00011A	resistance of airbag squib is < 1.3 Ohm	Circuit Resistance Below Threshold	Airbag not possible to be deployed Error status reporting on CAN	value of the squib resistance is low -Short circuit in airbag wires -Airbag not correctly connected -Not correct airbag	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Frontal Stage 1 Ignition Loop
SRS	B00011B	resistance of airbag squib is > 5.8 Ohm	Circuit Resistance Above Threshold	Airbag not possible to be deployed Error status reporting on CAN	value of the squib resistance is high - Open circuit to airbag - Incorrect airbag installed - Airbag incorrectly installed	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Frontal Stage 1 Ignition Loop
SRS	B000155	Configuration squib mismatch	Not Configured	Airbag not possible to be deployed Error status reporting on CAN	EEPROM configuration for squib is not correct with the actual squib	Diagnostic Service 14. No additional action is needed.	Replace of ECU	Driver Frontal Stage 1 Ignition Loop
SRS	B007001	Wrong connection of squib	General Electrical Failure	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	The squib is not connected to the designed pins	Diagnostic Service 14. No additional action is needed.	1. Check airbag and Seatbelt retractors connection 2. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop
SRS	B007011	Short - circuit	Circuit Short To Ground	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	short - circuit on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop
SRS	B007012	short to battery	Circuit Short To Battery	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	short to battery on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SRS	B00701A	resistance of driver seatbelt pretentioner squib is <1.3 Ohm	Circuit Resistance Below Threshold	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	value of the squib resistance is low - Short circuit in wires to driver seatbelt pretensioner - Incorrect squib in driver seatbelt pretensioner - Driver seatbelt pretensioner squib incorrectly installed	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop
SRS	B00701B	resistance of driver seatbelt pretensioner squib is > 5.8 Ohm	Circuit Resistance Above Threshold	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	value of the squib resistance is high - Open circuit in wires to driver seatbelt pretensioner - Incorrect driver seatbelt pretensioner squib installed - Driver seatbelt pretensioner incorrectly installed	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop
SRS	B007055	Configuration squib mismatch	Not Configured	Driver Seatbelt tensioner not possible to be deployed Error status reporting on CAN	EEPROM configuration for squib is not correct with the actual squib	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with complete SW package 2. Reprogram new parameter data for seatbelt tensioner 3. Check wiring harness for poor connection (short circuit)	Driver Seatbelt Pretensioner Ignition Loop
SRS	B007201	Wrong connection of squib	General Electrical Failure	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	The squib is not connected to the designed pins	Diagnostic Service 14. No additional action is needed.	1. Check airbag and Seatbelt retractors connection 2. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop
SRS	B007211	Short - circuit	Circuit Short To Ground	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	short - circuit on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop
SRS	B007212	short to battery	Circuit Short To Battery	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	short to battery on the squib pins	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop
SRS	B00721A	resistance of passenger seatbelt pretensioner squib is < 1.3 Ohm	Circuit Resistance Below Threshold	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	value of the squib resistance is low - Short circuit in wires to passenger seatbelt pretensioner - Incorrect passenger seatbelt pretensioner squib installed - Passenger seatbelt pretensioner incorrectly installed	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SRS	B00721B	resistance of passenger seatbelt pretensioner squib is > 5.8 Ohm	Circuit Resistance Above Threshold	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	value of the squib resistance is high - Open circuit in wires to passenger seatbelt pretensioner - Incorrect passenger seatbelt pretensioner squib installed - Passenger seatbelt pretensioner incorrectly installed	Diagnostic Service 14. No additional action is needed.	1. Change airbag fuse 2. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop
SRS	B007255	Configuration squib mismatch	Not Configured	Passenger Seatbelt tensioner not possible to be deployed Error status reporting on CAN	EEPROM configuration for squib is not correct with the actual squib	Diagnostic Service 14. No additional action is needed.	1. Reprogram ECU with complete SW package 2. Reprogram new parameter data for seatbelt tensioner 3. Check wiring harness for poor connection (short circuit)	Passenger Seatbelt Pretensioner Ignition Loop
SRS	B150200	A crash has occured	No Sub Type Information	ECU has to be scrapped	The ECU deployed, i.e. the vehicle has been in a crash severe enough to trigger the airbag and pretensioners	Not possible	ECU needs to be replaced	Airbag Control Unit Front Crash Detected
TGW2	U004688	Bus off state	Bus off	No contact with TGWII, TGWII will act according to latest value received on CAN	1. CAN circuit open/CAN detached 2. CAN tranceiver faulty	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check Internal voltage for HMIIOM. 3. Check ECUs	CAN Communication Infotainment Net
TGW2	U300041	Erroneous checksum on data or application files	General Checksum Failure	No telematics functions	1. Faulty flash memory 2. Faulty data or application file	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
TGW2	U300049	Internal hardware failure on control unit	Internal Electronic Failure	No telematics functions	Faulty ECU	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
TGW2	U30004B	Temperature > 75degrees C	Over Temperature	No telematics functions	1. Internal error resulting in high temperature 2. High installation environment temperature	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in normal range	Electronic control unit
TGW2	U300316	Voltage < 11V+-0.5V	Circuit Voltage Below Threshold	No telematics functions	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
TGW2	U300317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Battery Power
TGW2	B100092	Incorrect operation detected	Performance or Incorrect Operation	Communication with Back office not possible.	1. Internal error in GSM modem 2. Internal circuit failure in connection between GSM modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GSM modem
TGW2	B100192	No response from GPS unit	Performance or Incorrect Operation	Positioning not possible.	1. Internal error in GPS unit 2. Internal circuit failure in connection between GPS modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GPS unit
TGW2	B100211	Short circuit detected	Circuit Short To Ground	Positioning not possible.	1. Antenna cabling shorted to ground	Diagnostic Service 14. No additional action is needed.	Check antenna and antenna cable	TGW2 GPS Antenna
TGW2	B100213	Open circuit detected	Circuit Open	Positioning not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GPS Antenna
TGW2	B100311	Short circuit detected	Circuit Short To Ground	Communication with Back office not possible.	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and GSM antenna	TGW2 GSM Antenna

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	B100313	Open circuit detected	Circuit Open	Communication with Back office not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GSM Antenna
TGW2	B100409	SIM blocked/invalid	Component Failures	Communication with Back office not possible.	1. GSM SIM card missing or incorrectly inserted 2. GSM SIM card faulty	Diagnostic Service 14. No additional action is needed.	1. Check SIM card is inserted correctly. 2. Replace SIM card.	TGW2 GSM SIM card
TGW2	B100592	No response from WLAN module	Performance or Incorrect Operation	WLAN not working	1. Internal error in WLAN module 2. Internal circuit failure in connection between WLAN module and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 WLAN module
TGW2	B100611	Short circuit detected	Circuit Short To Ground	WLAN not working	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and WLAN antenna	TGW2 WLAN Antenna
TGW2	B100613	Antenna disconnected	Circuit Open	WLAN not working	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 WLAN Antenna
TGW2	B150281	Invalid signal received	Invalid Serial Data Received	No connection with tachograph, driver card login not possible. Time management functionality limited.	1. Tachograph not properly installed 2. INFO IF cabling disconnected 3. Internal error in tachograph 4. INFO IF cabling damaged 5. Unsupported tachograph installed	Diagnostic Service 14. No additional action is needed.	1. Check tachograph cable 2. Check tachograph ECU	Tachograph Information Inteface Circuit
TGW2	B150317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Telematics Extended Service Platform Module
TGW2	B150349	TESP internal hardware error detected	Internal Electronic Failure	Unstable functionality, or no functions at all	1. Internal component error, unit must be replaced.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Telematics Extended Service Platform Module
TGW2	B150351	New software for TESP detected on TGWII at startup.	Not Programmed	TESP not running on latest software	Routine control to install TESP SW not performed.	Diagnostic Service 14. No additional action is needed.	Run routine (R1AD1 in Tech Tool) in order to complete SW upgrade on TESP	Telematics Extended Service Platform Module
TGW2	B150393	No response from TESP	No Operation	No telematics features or navigation will be available.	1. TESP not installed 2. Ethernet cable connecting TGW2 and TESP disconnected or damaged 3. TESP not powered 4. Internal TESP failure	Diagnostic Service 14. No additional action is needed.	1. Check TESP installed 2. Check ethernet cable connected between TESP and TGW 3. Check TESP has power	Telematics Extended Service Platform Module
TGW2	B150416	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150417	Voltage > 32V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150444	RAM error detected	Data Memory Failure	Keyboard can not be used.	Internal RAM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150445	ROM error detected	Program Memory Failure	Keyboard can not be used.	Internal ROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150446	EEPROM error detected	Calibration / Parameter Memory Failure	Keyboard can not be used.	Internal EEPROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150447	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	Internal hardware error	Diagnostic Service 14. No additional action is needed.	Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150486	Invalid signal received	Signal Invalid	Keyboard can not be used..	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	SID Control Panel Infrared Eye Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	B150516	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Infrared Eye Module
TGW2	B150517	Voltage > 20V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Infrared Eye Module
TGW2	B150547	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	1. internal HW error	Diagnostic Service 14. No additional action is needed.	Replace Infrared Eye Module	Infrared Eye Module
TGW2	B15054B	Operating Temperature > 85°C	Over Temperature	Keyboard can not be used.	1. Environnement T°C too high 2. Processor workload too high	Diagnostic Service 14. No additional action is needed.	1. Check Infrared Eye Module environment 2. Replace Infrared Eye Module if fault is still present when environment temperature is in normal range	Infrared Eye Module
TGW2	B150586	Invalid signal received	Signal Invalid	Keyboard can not be used.	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	Infrared Eye Module
TGW2	B150611	Short circuit detected	Circuit Short To Ground	No voice guidance from navigation available	1. Audio cable shorted to ground	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150612	Short circuit detected	Circuit Short To Battery	No voice guidance from navigation available	1. Audio cable faulty 2. Audio cable connected to battery supply	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150613	Open circuit detected	Circuit Open	No voice guidance from navigation available	1. Audio cable disconnected 2. Audio cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable connection 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150713	Open circuit detected	Circuit Open	No HMI from telematics available	1. Video cable disconnected 2. Video cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check video cable connection 2. Replace video cable	TESP LVDS Video Data Circuit
TGW2	U114111	Short circuit detected	Circuit Short To Ground	Keyboard not available	1. LIN connector faulty 2. LIN connector grounded	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus
TGW2	U114112	Short circuit detected	Circuit Short To Battery	Keyboard not available	1. LIN connector faulty 2. LIN connector short to battery	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus
TGW2	U114186	Invalid signal received	Signal Invalid	Keyboard not available	Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	U114188	Bus off state	Bus off	Keyboard not available	1. LIN Slave disconnected from BUS 2. Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	1. Check LIN connection 2. Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	B150813	Open circuit detected	Circuit Open	No traffic information data for navigation available	1. FM antenna disconnected 2. FM antenna faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna	TESP FM-Antenna Circuit
TGW2	B150818	No supply voltage for phantom feed to FM antenna	Circuit Current Below Threshold	No traffic information data for navigation available	1. FM antenna faulty 2. No phantom feed to FM antenna 3. Faulty FM-splitter	Diagnostic Service 14. No additional action is needed.	1. Replace FM antenna 2. Replace FM antenna splitter	TESP FM-Antenna Circuit
TGW2	B15094A	Faulty SD card	Incorrect Component Installed	No software upgrade from SD card possible.	1. Faulty SD card inserted	Diagnostic Service 14. No additional action is needed.	Replace SD card	TESP SD-Card
TGW2	U004688	Bus off state	Bus off	No contact with TGWII, TGWII will act according to latest value received on CAN	1. CAN circuit open/CAN detached 2. CAN tranceiver faulty	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check Internal voltage for HMIIOM. 3. Check ECUs	CAN Communication Infotainment Net
TGW2	U300041	Erroneous checksum on data or application files	General Checksum Failure	No telematics functions	1. Faulty flash memory 2. Faulty data or application file	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
TGW2	U300049	Internal hardware failure on control unit	Internal Electronic Failure	No telematics functions	Faulty ECU	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	U30004B	Temperature > 75degrees C	Over Temperature	No telematics functions	1. Internal error resulting in high temperature 2. High installation environment temperature	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in normal range	Electronic control unit
TGW2	U300316	Voltage < 11V+-0.5V	Circuit Voltage Below Threshold	No telematics functions	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
TGW2	U300317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Battery Power
TGW2	B100092	Incorrect operation detected	Performance or Incorrect Operation	Communication with Back office not possible.	1. Internal error in GSM modem 2. Internal circuit failure in connection between GSM modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GSM modem
TGW2	B100192	No response from GPS unit	Performance or Incorrect Operation	Positioning not possible.	1. Internal error in GPS unit 2. Internal circuit failure in connection between GPS modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GPS unit
TGW2	B100211	Short circuit detected	Circuit Short To Ground	Positioning not possible.	1. Antenna cabling shorted to ground	Diagnostic Service 14. No additional action is needed.	Check antenna and antenna cable	TGW2 GPS Antenna
TGW2	B100213	Open circuit detected	Circuit Open	Positioning not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GPS Antenna
TGW2	B100311	Short circuit detected	Circuit Short To Ground	Communication with Back office not possible.	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and GSM antenna	TGW2 GSM Antenna
TGW2	B100313	Open circuit detected	Circuit Open	Communication with Back office not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GSM Antenna
TGW2	B100409	SIM blocked/invalid	Component Failures	Communication with Back office not possible.	1. GSM SIM card missing or incorrectly inserted 2. GSM SIM card faulty	Diagnostic Service 14. No additional action is needed.	1. Check SIM card is inserted correctly. 2. Replace SIM card.	TGW2 GSM SIM card
TGW2	B100592	No response from WLAN module	Performance or Incorrect Operation	WLAN not working	1. Internal error in WLAN module 2. Internal circuit failure in connection between WLAN module and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 WLAN module
TGW2	B100611	Short circuit detected	Circuit Short To Ground	WLAN not working	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and WLAN antenna	TGW2 WLAN Antenna
TGW2	B100613	Antenna disconnected	Circuit Open	WLAN not working	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 WLAN Antenna
TGW2	B150281	Invalid signal received	Invalid Serial Data Received	No connection with tachograph, driver card login not possible. Time management functionality limited.	1. Tachograph not properly installed 2. INFO IF cabling disconnected 3. Internal error in tachograph 4. INFO IF cabling damaged 5. Unsupported tachograph installed	Diagnostic Service 14. No additional action is needed.	1. Check tachograph cable 2. Check tachograph ECU	Tachograph Information Inteface Circuit
TGW2	B150317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Telematics Extended Service Platform Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	B150349	TESP internal hardware error detected	Internal Electronic Failure	Unstable functionality, or no functions at all	1. Internal component error, unit must be replaced.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Telematics Extended Service Platform Module
TGW2	B150351	New software for TESP detected on TGWII at startup.	Not Programmed	TESP not running on latest software	Routine control to install TESP SW not performed.	Diagnostic Service 14. No additional action is needed.	Run routine (R1AD1 in Tech Tool) in order to complete SW upgrade on TESP	Telematics Extended Service Platform Module
TGW2	B150393	No response from TESP	No Operation	No telematics features or navigation will be available.	1. TESP not installed 2. Ethernet cable connecting TGW2 and TESP disconnected or damaged 3. TESP not powered 4. Internal TESP failure	Diagnostic Service 14. No additional action is needed.	1. Check TESP installed 2. Check ethernet cable connected between TESP and TGW 3. Check TESP has power	Telematics Extended Service Platform Module
TGW2	B150416	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150417	Voltage > 32V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150444	RAM error detected	Data Memory Failure	Keyboard can not be used.	Internal RAM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150445	ROM error detected	Program Memory Failure	Keyboard can not be used.	Internal ROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150446	EEPROM error detected	Calibration / Parameter Memory Failure	Keyboard can not be used.	Internal EEPROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150447	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	Internal hardware error	Diagnostic Service 14. No additional action is needed.	Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150486	Invalid signal received	Signal Invalid	Keyboard can not be used..	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	SID Control Panel Infrared Eye Module
TGW2	B150516	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Infrared Eye Module
TGW2	B150517	Voltage > 20V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Infrared Eye Module
TGW2	B150547	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	1. internal HW error	Diagnostic Service 14. No additional action is needed.	Replace Infrared Eye Module	Infrared Eye Module
TGW2	B15054B	Operating Temperature > 85°C	Over Temperature	Keyboard can not be used.	1. Environnement T°C too high 2. Processor workload too high	Diagnostic Service 14. No additional action is needed.	1. Check Infrared Eye Module environment 2. Replace Infrared Eye Module if fault is still present when environment temperature is in normal range	Infrared Eye Module
TGW2	B150586	Invalid signal received	Signal Invalid	Keyboard can not be used.	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	Infrared Eye Module
TGW2	B150611	Short circuit detected	Circuit Short To Ground	No voice guidance from navigation available	1. Audio cable shorted to ground	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150612	Short circuit detected	Circuit Short To Battery	No voice guidance from navigation available	1. Audio cable faulty 2. Audio cable connected to battery supply	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150613	Open circuit detected	Circuit Open	No voice guidance from navigation available	1. Audio cable disconnected 2. Audio cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable connection 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150713	Open circuit detected	Circuit Open	No HMI from telematics available	1. Video cable disconnected 2. Video cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check video cable connection 2. Replace video cable	TESP LVDS Video Data Circuit
TGW2	U114111	Short circuit detected	Circuit Short To Ground	Keyboard not available	1. LIN connector faulty 2. LIN connector grounded	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	U114112	Short circuit detected	Circuit Short To Battery	Keyboard not available	1. LIN connector faulty 2. LIN connector short to battery	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus
TGW2	U114186	Invalid signal received	Signal Invalid	Keyboard not available	Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	U114188	Bus off state	Bus off	Keyboard not available	1. LIN Slave disconnected from BUS 2. Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	1. Check LIN connection 2. Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	B150813	Open circuit detected	Circuit Open	No traffic information data for navigation available	1. FM antenna disconnected 2. FM antenna faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna	TESP FM-Antenna Circuit
TGW2	B150818	No supply voltage for phantom feed to FM antenna	Circuit Current Below Threshold	No traffic information data for navigation available	1. FM antenna faulty 2. No phantom feed to FM antenna 3. Faulty FM-splitter	Diagnostic Service 14. No additional action is needed.	1. Replace FM antenna 2. Replace FM antenna splitter	TESP FM-Antenna Circuit
TGW2	B15094A	Faulty SD card	Incorrect Component Installed	No software upgrade from SD card possible.	1. Faulty SD card inserted	Diagnostic Service 14. No additional action is needed.	Replace SD card	TESP SD-Card
TGW2	U004688	Bus off state	Bus off	No contact with TGWII, TGWII will act according to latest value received on CAN	1. CAN circuit open/CAN detached 2. CAN tranceiver faulty	Diagnostic Service 14. No additional action is needed.	1. Check CAN wires and connectors 2. Check Internal voltage for HMIIOM. 3. Check ECUs	CAN Communication Infotainment Net
TGW2	U300041	Erroneous checksum on data or application files	General Checksum Failure	No telematics functions	1. Faulty flash memory 2. Faulty data or application file	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	Electronic control unit
TGW2	U300049	Internal hardware failure on control unit	Internal Electronic Failure	No telematics functions	Faulty ECU	Diagnostic Service 14. No additional action is needed.	Replace ECU	Electronic control unit
TGW2	U30004B	Temperature > 75degrees C	Over Temperature	No telematics functions	1. Internal error resulting in high temperature 2. High installation environment temperature	Diagnostic Service 14. No additional action is needed.	1. Check ECU environment 2. Replace ECU if fault is still present when environment temperature is in normal range	Electronic control unit
TGW2	U300316	Voltage < 11V+-0.5V	Circuit Voltage Below Threshold	No telematics functions	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
TGW2	U300317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Battery Power
TGW2	B100092	Incorrect operation detected	Performance or Incorrect Operation	Communication with Back office not possible.	1. Internal error in GSM modem 2. Internal circuit failure in connection between GSM modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GSM modem
TGW2	B100192	No response from GPS unit	Performance or Incorrect Operation	Positioning not possible.	1. Internal error in GPS unit 2. Internal circuit failure in connection between GPS modem and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 GPS unit
TGW2	B100211	Short circuit detected	Circuit Short To Ground	Positioning not possible.	1. Antenna cabling shorted to ground	Diagnostic Service 14. No additional action is needed.	Check antenna and antenna cable	TGW2 GPS Antenna
TGW2	B100213	Open circuit detected	Circuit Open	Positioning not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GPS Antenna
TGW2	B100311	Short circuit detected	Circuit Short To Ground	Communication with Back office not possible.	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and GSM antenna	TGW2 GSM Antenna

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	B100313	Open circuit detected	Circuit Open	Communication with Back office not possible.	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 GSM Antenna
TGW2	B100409	SIM blocked/invalid	Component Failures	Communication with Back office not possible.	1. GSM SIM card missing or incorrectly inserted 2. GSM SIM card faulty	Diagnostic Service 14. No additional action is needed.	1. Check SIM card is inserted correctly. 2. Replace SIM card.	TGW2 GSM SIM card
TGW2	B100592	No response from WLAN module	Performance or Incorrect Operation	WLAN not working	1. Internal error in WLAN module 2. Internal circuit failure in connection between WLAN module and CPU	Diagnostic Service 14. No additional action is needed.	Replace ECU	TGW2 WLAN module
TGW2	B100611	Short circuit detected	Circuit Short To Ground	WLAN not working	1. Antenna cable shorted to ground	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and WLAN antenna	TGW2 WLAN Antenna
TGW2	B100613	Antenna disconnected	Circuit Open	WLAN not working	1. Antenna cable detached 2. Antenna or antenna cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna cable	TGW2 WLAN Antenna
TGW2	B150281	Invalid signal received	Invalid Serial Data Received	No connection with tachograph, driver card login not possible. Time management functionality limited.	1. Tachograph not properly installed 2. INFO IF cabling disconnected 3. Internal error in tachograph 4. INFO IF cabling damaged 5. Unsupported tachograph installed	Diagnostic Service 14. No additional action is needed.	1. Check tachograph cable 2. Check tachograph ECU	Tachograph Information Inteface Circuit
TGW2	B150317	Voltage > 32V	Circuit Voltage Above Threshold	Risk for component damage	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Telematics Extended Service Platform Module
TGW2	B150349	TESP internal hardware error detected	Internal Electronic Failure	Unstable functionality, or no functions at all	1. Internal component error, unit must be replaced.	Diagnostic Service 14. No additional action is needed.	Replace ECU	Telematics Extended Service Platform Module
TGW2	B150351	New software for TESP detected on TGWII at startup.	Not Programmed	TESP not running on latest software	Routine control to install TESP SW not performed.	Diagnostic Service 14. No additional action is needed.	Run routine (R1AD1 in Tech Tool) in order to complete SW upgrade on TESP	Telematics Extended Service Platform Module
TGW2	B150393	No response from TESP	No Operation	No telematics features or navigation will be available.	1. TESP not installed 2. Ethernet cable connecting TGW2 and TESP disconnected or damaged 3. TESP not powered 4. Internal TESP failure	Diagnostic Service 14. No additional action is needed.	1. Check TESP installed 2. Check ethernet cable connected between TESP and TGW 3. Check TESP has power	Telematics Extended Service Platform Module
TGW2	B150416	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150417	Voltage > 32V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	SID Control Panel Infrared Eye Module
TGW2	B150444	RAM error detected	Data Memory Failure	Keyboard can not be used.	Internal RAM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150445	ROM error detected	Program Memory Failure	Keyboard can not be used.	Internal ROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150446	EEPROM error detected	Calibration / Parameter Memory Failure	Keyboard can not be used.	Internal EEPROM error	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150447	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	Internal hardware error	Diagnostic Service 14. No additional action is needed.	Replace ECU	SID Control Panel Infrared Eye Module
TGW2	B150486	Invalid signal received	Signal Invalid	Keyboard can not be used..	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	SID Control Panel Infrared Eye Module

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TGW2	B150516	Voltage < 7V	Circuit Voltage Below Threshold	Keyboard can not be used.	1. Low battery output voltage 2. Power feed cabling shorted	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Infrared Eye Module
TGW2	B150517	Voltage > 20V	Circuit Voltage Above Threshold	Keyboard can not be used.	1. High battery output voltage	Diagnostic Service 14. No additional action is needed.	Check power supply system	Infrared Eye Module
TGW2	B150547	Watchdog reset occurs	Watchdog / Safety µC Failure	Keyboard can not be used.	1. internal HW error	Diagnostic Service 14. No additional action is needed.	Replace Infrared Eye Module	Infrared Eye Module
TGW2	B15054B	Operating Temperature > 85°C	Over Temperature	Keyboard can not be used.	1. Environnement T°C too high 2. Processor workload too high	Diagnostic Service 14. No additional action is needed.	1. Check Infrared Eye Module environment 2. Replace Infrared Eye Module if fault is still present when environment temperature is in normal range	Infrared Eye Module
TGW2	B150586	Invalid signal received	Signal Invalid	Keyboard can not be used.	1. low battery in Keyboard 2. Keyboard error	Diagnostic Service 14. No additional action is needed.	1. Check batteries in Keyboard 2. Replace Keyboard	Infrared Eye Module
TGW2	B150611	Short circuit detected	Circuit Short To Ground	No voice guidance from navigation available	1. Audio cable shorted to ground	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150612	Short circuit detected	Circuit Short To Battery	No voice guidance from navigation available	1. Audio cable faulty 2. Audio cable connected to battery supply	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150613	Open circuit detected	Circuit Open	No voice guidance from navigation available	1. Audio cable disconnected 2. Audio cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check Audio cable connection 2. Replace Audio cable	TESP Audio Circuit
TGW2	B150713	Open circuit detected	Circuit Open	No HMI from telematics available	1. Video cable disconnected 2. Video cable faulty	Diagnostic Service 14. No additional action is needed.	1. Check video cable connection 2. Replace video cable	TESP LVDS Video Data Circuit
TGW2	U114111	Short circuit detected	Circuit Short To Ground	Keyboard not available	1. LIN connector faulty 2. LIN connector grounded	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus
TGW2	U114112	Short circuit detected	Circuit Short To Battery	Keyboard not available	1. LIN connector faulty 2. LIN connector short to battery	Diagnostic Service 14. No additional action is needed.	1. Check LIN connector 2. Replace LIN connector	TESP LIN16 Communication Bus
TGW2	U114186	Invalid signal received	Signal Invalid	Keyboard not available	Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	U114188	Bus off state	Bus off	Keyboard not available	1. LIN Slave disconnected from BUS 2. Broken LIN slave component	Diagnostic Service 14. No additional action is needed.	1. Check LIN connection 2. Replace IR-CIDCP or IR-eye	TESP LIN16 Communication Bus
TGW2	B150813	Open circuit detected	Circuit Open	No traffic information data for navigation available	1. FM antenna disconnected 2. FM antenna faulty	Diagnostic Service 14. No additional action is needed.	1. Check antenna cable is connected 2. Replace antenna	TESP FM-Antenna Circuit
TGW2	B150818	No supply voltage for phantom feed to FM antenna	Circuit Current Below Threshold	No traffic information data for navigation available	1. FM antenna faulty 2. No phantom feed to FM antenna 3. Faulty FM-splitter	Diagnostic Service 14. No additional action is needed.	1. Replace FM antenna 2. Replace FM antenna splitter	TESP FM-Antenna Circuit
TGW2	B15094A	Faulty SD card	Incorrect Component Installed	No software upgrade from SD card possible.	1. Faulty SD card inserted	Diagnostic Service 14. No additional action is needed.	Replace SD card	TESP SD-Card
TPM	U300041	CRC check mismatch occurs.	General Checksum Failure		SW issues or broken ECU	Diagnostic Service 14. No additional action is needed.	Download update of application software, if the error is still active, replace TPM ECU	Electronic control unit
TPM	U300049	RAM error ROM error (EEPROM, NVRAM) Wireless comm faults Faulty operation Other HW related faults	Internal Electronic Failure		Replace ECU	Diagnostic Service 14. No additional action is needed.		Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
TPM	U30004B	Temperature over normal working values	Over Temperature		Broken wiring, broken ECU.	Diagnostic Service 14. No additional action is needed.		Electronic control unit
TPM	U300054	Vehicle is not completely configured.	Missing Calibration		Vehicle is not completely configured.	Diagnostic Service 14. No additional action is needed.		Electronic control unit
TPM	U002888	Chassis CAN off	Bus off		Fault is out of TPMS	Diagnostic Service 14. No additional action is needed.		CAN Communication Chassis Net
TPM	U300316	ECU low voltage < 12V	Circuit Voltage Below Threshold		Broken power lines	Diagnostic Service 14. No additional action is needed.		Battery Power
TPM	U300317	ECU low voltage > 24V	Circuit Voltage Above Threshold		Broken power lines	Diagnostic Service 14. No additional action is needed.		Battery Power
TPM	U1FFF68	Factory Mode Active	Event Information		Factory Mode Active	Diagnostic Service 14. No additional action is needed.		Factory Mode Active
TPM	U114100	Communication with TPM ECU on vehicle unit in front lost.	No Sub Type Information		Broken ECU, wireless environmental disturbances.	Diagnostic Service 14. No additional action is needed.		Lost Communication with TPM Unit in Front
TPM	U114200	Communication with TPM ECU on vehicle unit behind is lost.	No Sub Type Information		Broken ECU, wireless environmental disturbances.	Diagnostic Service 14. No additional action is needed.		Lost Communication with TPM Unit Behind
TPM	U114387		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle1, Left, Inner
TPM	U115287		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle4, Right, Outer
TPM	U115387		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle5, Left, Outer
TPM	U115487		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle5, Left, Inner
TPM	U115587		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle5, Right, Inner
TPM	U115687		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle5, Right, Outer
TPM	U114487		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle1, Left, Outer
TPM	U114587		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle2, Left, Outer
TPM	U114687		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle1, Right, Inner
TPM	U114787		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle1, Right, Outer
TPM	U114887		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle2, Left, Inner
TPM	U114987		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle2, Right, Inner
TPM	U114A87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle2, Right, Outer
TPM	U114B87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle3, Left, Outer
TPM	U114C87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle3, Left, Inner
TPM	U114D87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle3, Right, Inner
TPM	U114E87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle3, Right, Outer
TPM	U114F87		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle4, Left, Outer
TPM	U115087		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle4, Left, Inner
TPM	U115187		Missing Message			Diagnostic Service 14. No additional action is needed.		Wheel Module Axle4, Right, Inner

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B15AE34	Accelerator pedal sensor 1 signal is below operational range (Low fault range defined by parameter P1GF5).	Signal High Time Less Than Minimum	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 1 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Sensor Measuring Circuit 1
VMCU	B15AE35	Accelerator pedal sensor 1 signal is above operational range (High fault range defined by parameter P1GF6).	Signal High Time Greater Than Maximum	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 1 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Sensor Measuring Circuit 1
VMCU	B15AF34	Accelerator pedal sensor 2 signal is below operational range (Low fault range defined by parameter P1GF7).	Signal High Time Less Than Minimum	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 2 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Sensor Measuring Circuit 2
VMCU	B15AF35	Accelerator pedal sensor 2 signal is above operational range (High fault range defined by parameter P1GF8).	Signal High Time Greater Than Maximum	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 2 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Sensor Measuring Circuit 2
VMCU	B100111	Accelerator pedal power supply 1 has been shorted to ground.	Circuit Short To Ground	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Power Supply 1 Circuit
VMCU	B100115	Accelerator pedal power supply 1 has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Power Supply 1 Circuit
VMCU	B100211	Accelerator pedal power supply 2 has been shorted to ground.	Circuit Short To Ground	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Power Supply 2 Circuit
VMCU	B100215	Accelerator pedal power supply 2 has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Power Supply 2 Circuit
VMCU	B150C62	Deviation in accelerator pedal sensor values.	Signal Compare Failure	Accelerator Pedal Sensor (#1 or #2) with the lowest value is used to calculate the accelerator pedal position.	"1. Wiring harness from accelerator pedal sensor faulty 2. Accelerator pedal sensor faulty 3. VMCU faulty"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Accelerator Pedal Sensor
VMCU	B100411	Body builder flexible switch 1 power supply has been shorted to ground.	Circuit Short To Ground	No power is supplied to the device connected to the BBEC.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Flexible Switch 1 Output

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B100415	Body builder flexible switch 1 power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	No power is supplied to the device connected to the BBEC.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Flexible Switch 1 Output
VMCU	B100511	Body builder flexible switch 2 power supply has been shorted to ground.	Circuit Short To Ground	No power is supplied to the device connected to the BBEC.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Flexible Switch 2 Output Circuit
VMCU	B100515	Body builder flexible switch 2 power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	No power is supplied to the device connected to the BBEC.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Flexible Switch 2 Output Circuit
VMCU	B151962	Vehicle deceleration to zero speed without brake switch indicating a pressed brake pedal.	Signal Compare Failure	No functional impact.	"1. Wiring harness from brake switch faulty 2. Brake switch faulty 3. Vehicle speed faulty (wiring harness / sensor)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check brake switch (B03) 3. Check vehicle speed sensor	Brake Pedal Switch
VMCU	B100611	Power supply to the cab tilt motor pump has been shorted to ground.	Circuit Short To Ground	Not possible to operate the cab tilt.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Relay K28 Cab Tilt Motor Pump Control Circuit
VMCU	B100615	Power supply to the cab tilt motor pump has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to operate the cab tilt.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Relay K28 Cab Tilt Motor Pump Control Circuit
VMCU	B15AD34	Clutch pedal sensor signal is below operational range defined by P1HIR.	Signal High Time Less Than Minimum	Not possible to determine the clutch pedal position.	1. Connector or wiring harness faulty 2. Clutch pedal sensor faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check clutch pedal 3. Check VMCU	Clutch Pedal Sensor Measuring Circuit
VMCU	B15AD35	Clutch pedal sensor signal is above operational range defined by P1HIR.	Signal High Time Greater Than Maximum	Not possible to determine the clutch pedal position.	1. Connector or wiring harness faulty 2. Clutch pedal sensor faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check clutch pedal 3. Check VMCU	Clutch Pedal Sensor Measuring Circuit
VMCU	B152A11	Clutch pedal switch/sensor power supply has been shorted to ground.	Circuit Short To Ground	Not possible to determine the clutch pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Clutch Pedal Linear Sensor/Switch Power Supply Circuit
VMCU	B152A15	Clutch pedal switch/sensor power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to determine the clutch pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Clutch Pedal Linear Sensor/Switch Power Supply Circuit
VMCU	B15BC54	Clutch Pedal Linear Sensor not calibrated.	Missing Calibration	Not possible to determine the clutch pedal position.	Default values for pressed / released end positions.	Diagnostic Service 14. No additional action is needed.	Perform a calibration of the Clutch Pedal Linear Sensor using Routine Identifier R1AFT.	Clutch Pedal Linear Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B15C082	1. Input signal status is NOT OK 2. Input signal is received	Alive / Sequence Counter Incorrect / Not Updated	The status signal sets to error 1. Until the input status is OK 2. Until the input signal is received with correct expected signal value	1. Input signal status is NOT OK 2. Software error (signal publisher swc sends incorrect expected value)	Diagnostic Service 14. No additional action is needed.	Reprogram ECU	Collision Warning Emergency Braking System
VMCU	U210164	Unexpected change in transmission ratio.	Signal Plausibility Failure	Cruise Control is exited if it is active when the failure occurrs.	"1. Wiring harness from clutch switch faulty 2. Clutch switch faulty 3. Wiring harness from neutral switch faulty 4. Neutral switch faulty 5. Vehicle speed faulty (wiring harness / sensor) or engine speed (EMS) faulty"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check clutch switch 3. Check wires and connectors 4. Check neutral switch 5. Check vehicle speed sensor	Cruise control status
VMCU	B151812	Mirror heating relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to operate the mirror heater in the driver and passenger doors.	1. Mirror heating relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check mirror heating relay (K21) 2. Check VMCU	Relay K21 Mirror Heating Circuit
VMCU	B151814	Mirror heating relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to operate the mirror heater in the driver and passenger doors.	1. Mirror heating relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check mirror heating relay (K21) 2. Check VMCU	Relay K21 Mirror Heating Circuit
VMCU	B150211	Body Builder Tail Lift/Crane power supply has been shorted to ground.	Circuit Short To Ground	Not possible to operate the Tail Lift or Crane.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Tail Lift/Crane Enable Circuit
VMCU	B150215	Body Builder Tail Lift/Crane power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to operate the Tail Lift or Crane.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Body Builder Tail Lift/Crane Enable Circuit
VMCU	B150511	Cab body working lamps has been shorted to ground.	Circuit Short To Ground	Not possible to use the Cab/Body Working Light.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check Actuator 3. Check VMCU	Cab Body Working Lamps Circuit
VMCU	B150515	Cab body working lamps has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to use the Cab/Body Working Light.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check Actuator 3. Check VMCU	Cab Body Working Lamps Circuit
VMCU	B150611	Side reverse working lights has been shorted to ground.	Circuit Short To Ground	Not possible to use the Side/Reverse Working Light.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Side Reverse Working Lights
VMCU	B150615	Side reverse working lights has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to use the Side/Reverse Working Light.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Side Reverse Working Lights
VMCU	B150711	Front extra spot lamps have been shorted to ground.	Circuit Short To Ground	Not possible to use the Extra Spot Lamp Front.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Front Extra Spot Lamps Circuit
VMCU	B150715	Front extra spot lamps have been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to use the Extra Spot Lamp Front.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Front Extra Spot Lamps Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B150811	Extra parking lights have been shorted to ground.	Circuit Short To Ground	Not possible to use the Extra Parking Lights.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Extra Parking Lights
VMCU	B150815	Extra parking lights have been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to use the Extra Parking Lights.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check FSP 3. Check VMCU	Extra Parking Lights
VMCU	B152212	Roof light sign relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Roof Sign Light.	1. Roof light sign relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check roof light sign relay (K12) 2. Check VMCU	Relay K12 Roof Light Sign / Roof Triangle Circuit
VMCU	B152214	Roof light sign relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Roof Sign Light.	1. Roof light sign relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check roof light sign relay (K12) 2. Check VMCU	Relay K12 Roof Light Sign / Roof Triangle Circuit
VMCU	B152312	Extra spot lamps relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Extra Spot Lamps on Roof.	1. Extra spot lamps relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check extra spot lamps relay (K10) 2. Check VMCU	Relay K10 Extra Spot Lamps Roof / Roof Triangle Power Circuit
VMCU	B152314	Extra spot lamps relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Extra Spot Lamps on Roof.	1. Extra spot lamps relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check extra spot lamps relay (K10) 2. Check VMCU	Relay K10 Extra Spot Lamps Roof / Roof Triangle Power Circuit
VMCU	B151E12	Trailer body light relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Trailer Body Light.	1. Trailer body light relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check trailer body light relay (K19) 2. Check VMCU	Relay K19 Trailer Body Lighting Circuit
VMCU	B151E14	Trailer body light relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Trailer Body Light.	1. Trailer body light relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check trailer body light relay (K19) 2. Check VMCU	Relay K19 Trailer Body Lighting Circuit
VMCU	B151F12	Beacon warning lamps relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Beacon Warning Lamps.	1. Beacon warning lamps relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check beacon warning lamps relay (K13) 2. Check VMCU	Relay K13 Beacon Warning / Roof Spot Lamps Circuit
VMCU	B151F14	Beacon warning lamps relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Beacon Warning Lamps.	1. Beacon warning lamps relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check beacon warning lamps relay (K13) 2. Check VMCU	Relay K13 Beacon Warning / Roof Spot Lamps Circuit
VMCU	B15B868	Roller Bench Mode is enabled.	Event Information	Not possible to engage Front Wheel Drive.	Roller Bench Mode is enabled by parameter P1KM2.	Diagnostic Service 14. No additional action is needed.	Disable Roller Bench Mode by parameter P1KM2.	Roller Bench Mode Active
VMCU	U30004A	Incorrect data from FRC diagnostics.	Incorrect Component Installed	FRC diagnostics will be unavailable.	Wrong type of FRC mounted.	Diagnostic Service 14. No additional action is needed.	Replace FRC.	Electronic control unit
VMCU	B153B13	Fuse blown	Circuit Open	Loss of SID power supply	Fuse F21 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F21
VMCU	B155F13	Fuse blown	Circuit Open	None	Fuse F82 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F82
VMCU	B153C13	Fuse blown	Circuit Open	Loss of Trailer ABS/EBS power supply	Fuse F46 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F46
VMCU	B156013	Fuse blown	Circuit Open	None	Fuse F83 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F83
VMCU	B153D13	Fuse blown	Circuit Open	Loss of ABS/EBS power supply	Fuse F37 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F37
VMCU	B156113	Fuse blown	Circuit Open	None	Fuse F79 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F79

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B153E13	Fuse blown	Circuit Open	Loss of DACU power supply	Fuse F31 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F31
VMCU	B156213	Fuse blown	Circuit Open	Loss of FMS connector ignition supply	Fuse F85 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F85
VMCU	B153F13	Fuse blown	Circuit Open	Loss of Toll collect Battery supply	Fuse F25 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F25
VMCU	B156313	Fuse blown	Circuit Open	Loss of BB Electrical Centre ignition supply 1	Fuse F73 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F73
VMCU	B154013	Fuse blown	Circuit Open	Loss of OBDII diagnostic connector supply	Fuse F62 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F62
VMCU	B156413	Fuse blown	Circuit Open	Loss of BB Electrical Centre ignition supply 2	Fuse F74 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F74
VMCU	B154113	Fuse blown	Circuit Open	Loss of FMS connector power supply	Fuse F91 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F91
VMCU	B156513	Fuse blown	Circuit Open	Loss of EMS ECU power supply	Fuse F41 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F41
VMCU	B154213	Fuse blown	Circuit Open	None	Fuse F35 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F35
VMCU	B156613	Fuse blown	Circuit Open	Loss of EMS controlled loads 1	Fuse F42 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F42
VMCU	B154313	Fuse blown	Circuit Open	None	Fuse F47 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F47
VMCU	B156713	Fuse blown	Circuit Open	Loss of Fuel filter heater	Fuse F43 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F43
VMCU	B154413	Fuse blown	Circuit Open	Loss of Electrical front lid lock supply	Fuse F34 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F34
VMCU	B156813	Fuse blown	Circuit Open	Loss of EMS controlled loads 2	Fuse F44 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F44
VMCU	B154513	Fuse blown	Circuit Open	None	Fuse F89 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F89
VMCU	B154613	Fuse blown	Circuit Open	None	Fuse F09 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F09
VMCU	B154713	Fuse blown	Circuit Open	None	Fuse F72 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F72
VMCU	B154813	Fuse blown	Circuit Open	Loss of Sun visor supply	Fuse F22 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F22
VMCU	B154913	Fuse blown	Circuit Open	Loss of TIS&Telematics System (old TGW) power supply	Fuse F66 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F66
VMCU	B154A13	Fuse blown	Circuit Open	Loss of VS (Video Switch) power supply	Fuse F53 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F53
VMCU	B159D16	No power detected	Circuit Voltage Below Threshold	Voltage drop on 20A headersherlf DC/DC convertor outputs (12V Power Supply for Wired phone, Mobile phone, TV-DVD, CB Pwr)	1. Header shelf DC/DC converter is damaged. 2. 24 V power supply connection problem 3. 12 V output power supply connection problem	Diagnostic Service 14. No additional action is needed.	Check DC/DC converter.	Front Shelf 24V/12V Converter
VMCU	B154B13	Fuse blown	Circuit Open	Loss of power on Dashboard 12V-Socket	Fuse F01 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F01
VMCU	B159E16	No power detected	Circuit Voltage Below Threshold	Voltage drop on 12V power supply for Dashboard 12V-Socket, Sleeper 12V-Socket, Audio Control unit 12V-power supply, Audio Control unit 12V-Accessories	1. Dashboard DC/DC converter is damaged. 2. 24 V power supply connection problem 3. 12 V output power supply connection problem	Diagnostic Service 14. No additional action is needed.	Check DC/DC converter.	Dashboard 24V/12V Converter
VMCU	B154C13	Fuse blown	Circuit Open	Loss of power supply on TV-DVD, Wired phone, Mobile phone	Fuse F03 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F03

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B154D13	No power detected	Circuit Open	Loss of Power supply for Coffee maker	1. Fuse F50 is blown 2. Relay K08 is damaged.	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay	Fuse F50/Relay K08
VMCU	B154E13	Fuse blown	Circuit Open	None	1. Fuse F54 is blown 2. Relay K08 is damaged.	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F54
VMCU	B152B13	Fuse blown	Circuit Open	Loss of TACHO Power supply	Fuse F23 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F23
VMCU	B154F13	Fuse blown	Circuit Open	Loss of BB Electrical Centre main supply 1	Fuse F07 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F07
VMCU	B152C13	Fuse blown	Circuit Open	Loss of CIOM Power supply	Fuse F63 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F63
VMCU	B155013	Fuse blown	Circuit Open	Loss of BB Electrical Centre main supply 2	Fuse F08 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F08
VMCU	B152D13	Fuse blown	Circuit Open	Loss of ALARM Power supply	Fuse F55 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F55
VMCU	B155113	Fuse blown	Circuit Open	Loss of BB Electrical centre switches power supply	Fuse F06 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F06
VMCU	B152E13	Fuse blown	Circuit Open	Loss of HMIIOM Power supply	Fuse F18 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse	Fuse F18
VMCU	B155213	Fuse blown	Circuit Open	Loss of Trailer plug power supply	Fuse F04 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F04
VMCU	B152F13	Fuse blown	Circuit Open	Loss of DDM power supply	Fuse F61 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F61
VMCU	B155313	Fuse blown	Circuit Open	Loss of BB Chassis power supply	Fuse F05 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F05
VMCU	B153013	Fuse blown	Circuit Open	Loss of PDM power supply	Fuse F20 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F20
VMCU	B155413	Fuse blown	Circuit Open	Loss of 24V power outlet supply 1	Fuse F64 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F64
VMCU	B153113	Fuse blown	Circuit Open	Loss of Wireless remote control charger supply	Fuse F33 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F33
VMCU	B155513	Fuse blown	Circuit Open	Loss of 24V power outlet supply 2	Fuse F65 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F65
VMCU	B153213	Fuse blown	Circuit Open	None	Fuse F84 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F84
VMCU	B155613	Fuse blown	Circuit Open	Loss of 24V power outlet supply 3	Fuse F10 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F10
VMCU	B153313	Fuse blown	Circuit Open	Loss of Fuel line heater power supply	Fuse F39 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F39
VMCU	B155713	Fuse blown	Circuit Open	Loss of Cigarette lighter power supply	Fuse F67 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F67
VMCU	B153413	Fuse blown	Circuit Open	Loss of Cab parking heater power supply	Fuse F69 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F69
VMCU	B155813	Fuse blown	Circuit Open	Loss of AlcoLock power supply	Fuse F88 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F88
VMCU	B153513	Fuse blown	Circuit Open	Loss of APM power supply 1	Fuse F32 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F32
VMCU	B155913	Fuse blown	Circuit Open	Loss of BB Prep kit power supply	Fuse F56 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F56
VMCU	B153613	Fuse blown	Circuit Open	None	Fuse F48 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F48
VMCU	B153713	Fuse blown	Circuit Open	Loss of IC power supply	Fuse F24 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F24
VMCU	B155A13	Fuse blown	Circuit Open	Loss of Amplifier power supply	Fuse F58 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F58
VMCU	B155B13	Fuse blown	Circuit Open	Loss of Headershelf DC/DC Converter power supply	Fuse F59 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F59
VMCU	B153813	Fuse blown	Circuit Open	Loss of BBM power supply	Fuse F19 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F19
VMCU	B155C13	Fuse blown	Circuit Open	Loss of Dashboard DC/DC Converter power supply	Fuse F60 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F60

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B153913	Fuse blown	Circuit Open	Loss of LECM power supply	Fuse F80 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F80
VMCU	B155D13	Fuse blown	Circuit Open	Loss of Task lights power supply	Fuse F57 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F57
VMCU	B153A13	Fuse blown	Circuit Open	Loss of CCM power supply	Fuse F38 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F38
VMCU	B155E13	Fuse blown	Circuit Open	Loss of SRS power supply	Fuse F81 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F81
VMCU	B157772	Relay stuck in opened	Actuator Stuck Open	Loss of BB main power supply (Power Supply Relay 2)	Relay K07 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K07
VMCU	B157773	Relay stuck in closed	Actuator Stuck Closed	BB main power supply always ON	Relay K07 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K07
VMCU	B157872	Relay stuck in opened	Actuator Stuck Open	Loss of Power suppy 3	Relay K14 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K14
VMCU	B157873	Relay stuck in closed	Actuator Stuck Closed	Power supply 3 always ON	Relay K14 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K14
VMCU	B157972	Relay stuck in opened	Actuator Stuck Open	Loss of Power supply 4	Relay K11 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K11
VMCU	B157973	Relay stuck in closed	Actuator Stuck Closed	Power supply 4 always ON	Relay K11 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K11
VMCU	B157A72	Relay stuck in opened	Actuator Stuck Open	Loss of Ignition power supply	Relay K22 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K22
VMCU	B157A73	Relay stuck in closed	Actuator Stuck Closed	Ignition power supply always ON	Relay K22 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K22
VMCU	B157B72	Relay stuck in opened	Actuator Stuck Open	Loss of BB Ignition power supply	Relay K20 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K20
VMCU	B157B73	Relay stuck in closed	Actuator Stuck Closed	BB Ignition power supply always ON	Relay K20 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K20
VMCU	B157C73	Relay stuck in closed	Actuator Stuck Closed	Cab tilt pump power supply always ON	Relay K28 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K28
VMCU	B157D72	Relay stuck in opened	Actuator Stuck Open	Loss of EMS power supply	Relay K27 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K27
VMCU	B157D73	Relay stuck in closed	Actuator Stuck Closed	EMS power supply always ON	Relay K27 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K27
VMCU	B157E72	Relay stuck in opened	Actuator Stuck Open	Loss of Mirror Heating power supply	Relay K21 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K21
VMCU	B157E73	Relay stuck in closed	Actuator Stuck Closed	Mirror Heating power supply always ON	Relay K21 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K21
VMCU	B157F73	Relay stuck in closed	Actuator Stuck Closed	Seat movement power supply always ON	Relay K24 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K24
VMCU	B158073	Relay stuck in closed	Actuator Stuck Closed	Engine Running power supply always ON	Relay K09 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K09
VMCU	B156913	Fuse blown	Circuit Open	Loss of Tacho Ignition supply	Fuse F40 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F40
VMCU	B156A13	Fuse blown	Circuit Open	Loss of Mirror heating 1 power supply (Heater, rear-view mirror, right)	Fuse F29 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F29
VMCU	B156B13	Fuse blown	Circuit Open	Loss of Mirror heating 2 power supply (Heater, rear-view mirror, left)	Fuse F30 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F30
VMCU	B156C13	No power detected.	Circuit Open	Loss of power for El. adjustable seat, Hourmeter and Alcolock engine running signal	1. Fuse F13 is blown 2. Relay K09 is damaged.	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F13/Relay K09
VMCU	B156D13	Fuse blown	Circuit Open	None	Fuse F14 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F14
VMCU	B156E13	Fuse blown	Circuit Open	Lolss of power supply for Sun Roof motor	Fuse F52 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F52
VMCU	B156F13	Fuse blown	Circuit Open	Loss of power supply for Windscreen Wiper motor	Fuse F51 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F51

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B157013	Fuse blown	Circuit Open	None	Fuse F77 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F77
VMCU	B157113	Fuse blown	Circuit Open	None	Fuse F86 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F86
VMCU	B157213	Fuse blown	Circuit Open	None	Fuse F78 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F78
VMCU	B157313	Fuse blown	Circuit Open	Loss of power supply for Refrigerator and Electrical handle	Fuse F75 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F75
VMCU	B157413	Fuse blown	Circuit Open	None	Fuse F02 is blown	Diagnostic Service 14. No additional action is needed.	Replace fuse.	Fuse F02
VMCU	B159113	No power detected	Circuit Open	None	1. Fuse F87 is blown 2. Relay K25 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F87/Relay K25
VMCU	B159213	No power detected	Circuit Open	Loss of Cab tilt pump motor power supply	1. Fuse F45 is blown 2. Relay K28 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F45/Relay K28
VMCU	B159313	No power detected	Circuit Open	Loss of power supply for Seat motor	1. Fuse F90 is blown 2. Relay K24 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F90/Relay K24
VMCU	B159413	No power detected	Circuit Open	Loss of power supply for Headlight washer	1. Fuse F71 is blown 2. Relay K16 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F71/Relay K16
VMCU	B159513	No power detected	Circuit Open	Loss of power supply for Trailer Body Light EL8	1. Fuse F76 is blown 2. Relay K19 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F76/Relay K19
VMCU	B159613	No power detected	Circuit Open	Loss of power supply for Extra spot lamps roof/front	1. Fuse F11 is blown 2. Relay K10 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F11/Relay K10
VMCU	B159713	No power detected	Circuit Open	Loss of power supply for Beacon warning lamps	1. Fuse F12 is blown 2. Relay K13 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F12/Relay K13
VMCU	B159813	No power detected	Circuit Open	Loss of power supply for Roof light sign lamps	1. Fuse F15 is blown 2. Relay K12 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F15/Relay K12
VMCU	B159913	No power detected	Circuit Open	Loss of power supply for Full beam inside sign area lamps	1. Fuse F16 is blown 2. Relay K15 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F16/Relay K15
VMCU	B159A13	No power detected	Circuit Open	Loss of TECU power supply	1. Fuse F70 is blown 2. Relay K17 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F70/Relay K17
VMCU	B159B13	No power detected	Circuit Open	None	1. Fuse F36 is blown 2. Relay K26 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F36/Relay K26
VMCU	B159C13	No power detected	Circuit Open	None	1. Fuse F17 is blown 2. Relay K18 is damaged	Diagnostic Service 14. No additional action is needed.	Replace fuse or relay.	Fuse F17/Relay K18
VMCU	B157573	Relay stuck in closed	Actuator Stuck Closed	Coffee maker power supply always ON	Relay K08 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K08
VMCU	B157673	Relay stuck in closed	Actuator Stuck Closed	None	Relay K25 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K25
VMCU	B158172	Relay stuck in opened	Actuator Stuck Open	Loss of power supply for Sunroof motor1	Relay K04 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K04
VMCU	B158173	Relay stuck in closed	Actuator Stuck Closed	Sunroof motor1 power supply always ON	Relay K04 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K04
VMCU	B158272	Relay stuck in opened	Actuator Stuck Open	Loss of power supply for Sunroof motor2	Relay K03 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K03
VMCU	B158273	Relay stuck in closed	Actuator Stuck Closed	Sunroof motor2 power supply always ON	Relay K03 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K03
VMCU	B158373	Relay stuck in closed	Actuator Stuck Closed	Head Light Washer Pump power supply always ON	Relay K16 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K16
VMCU	B158472	Relay stuck in opened	Actuator Stuck Open	Intermittent mode of Windscreen wiper not available	Relay K05 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K05
VMCU	B158473	Relay stuck in closed	Actuator Stuck Closed	Intermittent mode of Windscreen wiper not available	Relay K05 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K05
VMCU	B158572	Relay stuck in opened (motor low speed position)	Actuator Stuck Open	High speed mode of Windscreen wiper not available	Relay K06 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K06

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B158573	Relay stuck in closed (motor high speed position)	Actuator Stuck Closed	Low speed mode of Windscreen wiper not available	Relay K06 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K06
VMCU	B158673	Relay stuck in closed	Actuator Stuck Closed	Trailer Body lights power supply always ON	Relay K19 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K19
VMCU	B158773	Relay stuck in closed	Actuator Stuck Closed	Extra Spot Lamps power supply always ON	Relay K10 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K10
VMCU	B158873	Relay stuck in closed	Actuator Stuck Closed	Beacon Warning Lamps power supply always ON	Relay K13 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K13
VMCU	B158973	Relay stuck in closed	Actuator Stuck Closed	Roof Light Sign power supply always ON	Relay K12 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K12
VMCU	B158A73	Relay stuck in closed	Actuator Stuck Closed	Full beam inside sign area power supply always ON	Relay K15 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K15
VMCU	B158B73	Relay stuck in closed	Actuator Stuck Closed	TECU power supply always ON	Relay K17 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K17
VMCU	B158C72	Relay stuck in opened	Actuator Stuck Open	Loss of Refrigerator power supply	Relay K02 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K02
VMCU	B158C73	Relay stuck in closed	Actuator Stuck Closed	Refrigerator power supply always ON	Relay K02 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K02
VMCU	B158D72	Relay stuck in opened	Actuator Stuck Open	None	Relay K01 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K01
VMCU	B158D73	Relay stuck in closed	Actuator Stuck Closed	None	Relay K01 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K01
VMCU	B158E73	Relay stuck in closed	Actuator Stuck Closed	None	Relay K26 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K26
VMCU	B158F73	Relay stuck in closed	Actuator Stuck Closed	None	Relay K18 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K18
VMCU	B159072	Relay stuck in opened	Actuator Stuck Open	None	Relay K23 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K23
VMCU	B159073	Relay stuck in closed	Actuator Stuck Closed	None	Relay K23 is damaged	Diagnostic Service 14. No additional action is needed.	Replace relay.	Relay K23
VMCU	B150B11	Cab component backlight has been shorted to ground.	Circuit Short To Ground	Not possible to use the Dash/Body Builder/Bunk/Headershelf backlight illumination.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check backlight illumination 3. Check VMCU	Cab/Body Builder Backlight Circuit
VMCU	B15B723	Hazard Button Signal Stuck Low for more than a configurable time. (Default 5 minutes)	Signal Stuck Low	Not possible to activate/deactivate hazard warning.	Hazard button mechanically stuck in pressed position or electrical fault in the button circuitry (short circuit to ground)	Diagnostic Service 14. No additional action is needed.	Repair/replace hazard warning button.	Hazard Button Circuit
VMCU	B150A11	Hazard button backlight has been shorted to ground.	Circuit Short To Ground	Not possible to use the Hazard Button backlight illumination.	"1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Hazard Button Backlight Circuit
VMCU	B151712	Headlight washer pump relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Head Lamp Washer.	1. Headlight washer pump relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check headlight washer pump relay (K16) 2. Check VMCU	Relay K16 Head Lamp Washer Pump Circuit
VMCU	B151714	Headlight washer pump relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Head Lamp Washer.	1. Headlight washer pump relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check headlight washer pump relay (K16) 2. Check VMCU	Relay K16 Head Lamp Washer Pump Circuit
VMCU	B100011	Redundant hazard output has been shorted to ground.	Circuit Short To Ground	Hazard indication on the hardwired output not possible.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check hazard switch and/or ELCP, if present 3. Check VMCU	Redundant Hazard Output Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B100015	Redundant hazard output has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Hazard indication on the hardwired output not possible.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check hazard switch and/or ELCP, if present 3. Check VMCU	Redundant Hazard Output Circuit
VMCU	U300A62	Invalid combination detected on the VMCU ignition key inputs.	Signal Compare Failure	Not possible to start the engine or use the vehicle.	1. Ignition key switch faulty. 2. Ignition key wiring harness to VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ignition key switch 2. Check wires and connectors	Key Position
VMCU	B152511	Cab max light 1 has been shorted to ground	Circuit Short To Ground	Not possible to use the left Roof light in the Headliner.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Max Light Circuit 1
VMCU	B152512	Cab max light 1 has been shorted to battery	Circuit Short To Battery	Not possible to use the left Roof light in the Headliner.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Max Light Circuit 1
VMCU	B152611	Cab max light 2 has been shorted to ground	Circuit Short To Ground	Not possible to use the right Roof light in the Headliner.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Max Light Circuit 2
VMCU	B152612	Cab max light 2 has been shorted to battery	Circuit Short To Battery	Not possible to use the right Roof light in the Headliner.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Max Light Circuit 2
VMCU	B152711	Cab resting light 1 has been shorted to ground	Circuit Short To Ground	Not possible to use the Front Shelf Light (white light).	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Resting Light Circuit 1
VMCU	B152712	Cab resting light 1 has been shorted to battery	Circuit Short To Battery	Not possible to use the Front Shelf Light (white light)	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Resting Light Circuit 1
VMCU	B152811	Cab resting light 2 has been shorted to ground	Circuit Short To Ground	Not possible to use the left/right Catwalk Light (white light).	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Resting Light Circuit 2
VMCU	B152911	Cab night driving light has been shorted to ground	Circuit Short To Ground	Not possible to use the left/right Catwalk Light (red light) and Front Shelf Light (red light).	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Night Driving Light Circuit
VMCU	B150E12	Main beam in roof sign relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Main Beam in the Roof Sign.	1. Main beam in roof sign relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check main beam in roof sign relay (K15) 2. Check VMCU	Relay K15 Main Beam in Roof Sign Circuit
VMCU	B150E14	Main beam in roof sign relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Main Beam in the Roof Sign.	1. Main beam in roof sign relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check main beam in roof sign relay (K15) 2. Check VMCU	Relay K15 Main Beam in Roof Sign Circuit
VMCU	U010000	Signal from EMS missing	No Sub Type Information	EMS diagnostic warnings and functional messages affected Note: This DTC can be set as a secondary DTC which has been agreed in deviation D-950. It can be triggered at the same time as: U300316 and U300317.	Connector or harness problems. Problem with sending ECU.	Diagnostic Service 14. No additional action is needed.	Check wires and connectors between ECU and EMS. If this is the only lost communication DTC then fault is more probable to be close to the EMS. This fault can also appear by incorrect calibration.	Lost Communication with EMS
VMCU	U114000	- No message received from APM	No Sub Type Information	No communication between APM and VMCU. No park brake information received.	- No power supply to APM - Broken CAN link	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2a. Check CAN wires and connectors 2b. Check ECUs	Lost Communication With APM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	P007011	<0,15V	Circuit Short To Ground	Not possible to determine ambient air temperature	1. Connected component broken. 2. Wires broken. Insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check connected sensors 3. Check ECUs	Ambient Air Temperature Sensor "A"
VMCU	P007015	4,622V < U < 5V OR U = 5V	Circuit Short To Battery or Open	Not possible to determine ambient air temperature	1. Connected component broken. 2. Wires broken. No continuity or insulation damage. 3. Intermittent failure due to poor connection.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check connected sensors 3. Check ECUs	Ambient Air Temperature Sensor "A"
VMCU	P007016	0,15V <= U < 0,3V	Circuit Voltage Below Threshold	Not possible to determine ambient air temperature	1. Connected component broken.	Diagnostic Service 14. No additional action is needed.	1. Check connected sensors 2. Check ECUs	Ambient Air Temperature Sensor "A"
VMCU	P012011	Accelerator pedal power supply 1 has been shorted to ground.	Circuit Short To Ground	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Throttle/Pedal Position Sensor "A"
VMCU	P012015	Accelerator pedal power supply 1 has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Throttle/Pedal Position Sensor "A"
VMCU	P012034	Accelerator pedal sensor 1 signal is below operational range (Low fault range defined by parameter P1GF5).	Signal High Time Less Than Minimum	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 1 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check for a loose connection 2b. Check accelerator pedal 3a. Check for a loose connection 3b. Check VMCU	Throttle/Pedal Position Sensor "A"
VMCU	P012035	Accelerator pedal sensor 1 signal is above operational range (High fault range defined by parameter P1GF6).	Signal High Time Greater Than Maximum	Accelerator Pedal Sensor #2 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 1 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check for a loose connection 2b. Check accelerator pedal 3a. Check for a loose connection 3b. Check VMCU	Throttle/Pedal Position Sensor "A"
VMCU	U015600	- No message received from HMIIOM	No Sub Type Information	No communication between HMIIOM and VMCU or between HMIIOM and IC.	- No power supply to HMIIOM - Broken CAN link	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2a. Check CAN wires and connectors 2b. Check ECUs	Lost Communication With HMIIOM
VMCU	P022011	Accelerator pedal power supply 2 has been shorted to ground.	Circuit Short To Ground	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Throttle/Pedal Position Sensor/Switch "B" Circuit
VMCU	P022015	Accelerator pedal power supply 2 has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check accelerator pedal 3. Check VMCU	Throttle/Pedal Position Sensor/Switch "B" Circuit
VMCU	P022034	Accelerator pedal sensor 2 signal is below operational range (Low fault range defined by parameter P1GF7).	Signal High Time Less Than Minimum	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 2 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check for a loose connection 2b. Check accelerator pedal 3a. Check for a loose connection 3b. Check VMCU	Throttle/Pedal Position Sensor/Switch "B" Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	P022035	Accelerator pedal sensor 2 signal is above operational range (High fault range defined by parameter P1GF8).	Signal High Time Greater Than Maximum	Accelerator Pedal Sensor #1 is used to calculate the accelerator pedal position.	1. Connector or wiring harness faulty 2. Accelerator pedal sensor 2 faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check for a loose connection 2b. Check accelerator pedal 3a. Check for a loose connection 3b. Check VMCU	Throttle/Pedal Position Sensor/Switch "B" Circuit
VMCU	P213562	Deviation in accelerator pedal sensor values.	Signal Compare Failure	Accelerator Pedal Sensor (#1 or #2) with the lowest value is used to calculate the accelerator pedal position.	1. Wiring harness from accelerator pedal sensor faulty 2. Accelerator pedal sensor faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check for a loose connection 2b. Check accelerator pedal 3a. Check for a loose connection 3b. Check VMCU	Throttle/Pedal Position Sensor/Switch "A"/"B" Voltage Correlation
VMCU	P215B61	Output shaft speed calculated by VMCU out of range (calculated value is more than 8000 rpm).	Signal Calculation Failure	Output Shaft Speed backup signal is set to Error (backup signal is always sent for vehicles without Tachograph. For vehicles with Tachograph the backup signal is sent when no output shaft speed is received from the Tachograph).	1.Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. Incorrect VMCU dataset or VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3a. Reload VMCU SW 3b. Check VMCU	Vehicle Speed - Output Shaft Speed Correlation
VMCU	P215A61	Vehicle speed calculated by VMCU out of range (calculated value is more than 250 km/h).	Signal Calculation Failure	Vehicle Speed backup signal is set to Error (backup signal is always sent for vehicles without Tachograph. For vehicles with Tachograph the backup signal is sent when no vehicle speed is received from the Tachograph).	1. Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. With TCO: TCO/VMCU K factor synchronization failure (VMCU/TCO faulty). 4: Without TCO: Incorrect K factor in VMCU.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3. Reload VMCU and TCO SW 4. Reload VMCU SW	Vehicle Speed - Wheel Speed Correlation
VMCU	P215A62	Vehicle speed calculated by VMCU deviates more than 10 km/h (parameter) from TCO vehicle speed.	Signal Compare Failure	Vehicle Speed signal is untrustable if the Tachograph vehicle speed is the cause for the deviation.	1. Wiring harness from TCO faulty. 2. TCO/VMCU K factor synchronization failure (VMCU/TCO faulty).	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Recalibrate K-factor 2b. Reload VMCU and TCO SW	Vehicle Speed - Wheel Speed Correlation
VMCU	P215A64	Vehicle speed deviates more than 10 km/h (parameter) from road speed (applicable for vehicles with ABS/EBS). Vehicle speed below or equal 0 km/h (parameter) while driveline is engaged and engine speed above 1200 rpm (parameter) (applicable for vehicles without ABS/EBS).	Signal Plausibility Failure	Vehicle Speed signal is set to Error.	1. Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. With TCO: Incorrect W/K factor in TCO. 4: Without TCO: Incorrect K factor in VMCU.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3a. Reload TCO SW 3b. Reload VMCU SW 4. Reload VMCU SW	Vehicle Speed - Wheel Speed Correlation

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	U031744	At least one NVM parameter is - in incorrect relation with one or more other parameters - out of the allowed minimal and maximal values from the NVM parameter descriptor table	Data Memory Failure	For APM, the following effect can be seen: No regeneration No compressor control Safety valve cut off. No EPB functions Pneumatic release of PB is possible by 29.2 port No TX messages on CAN	- Incorrect/Unrealistic data written into APM - APM HW damage	Diagnostic Service 14. No additional action is needed.	1. Reload APM SW 2. Check APM	Software Incompatibility With Park Brake Control Module
VMCU	U031746	If not permitted internal variable is used	Calibration / Parameter Memory Failure	For APM, the following effect can be seen: No regeneration No compressor control Safety valve cut off. No EPB functions Pneumatic release of PB is possible by 29.2 port No TX messages on CAN	Damaged dataset Dataset doesn't fit to the software Damaged software or damaged APM (microcontroller)	Diagnostic Service 14. No additional action is needed.	1 - 2. Reload APM SW 3. Check APM	Software Incompatibility With Park Brake Control Module
VMCU	U03174A	Drive software incompatible with the hardware	Incorrect Component Installed	For APM, the following effect can be seen: No regeneration No compressor control Safety valve cut off. No EPB functions Pneumatic release of PB is possible No TX messages on CAN	Wrong software version downloaded in APM	Diagnostic Service 14. No additional action is needed.	Reload APM SW	Software Incompatibility With Park Brake Control Module
VMCU	U012B00	- No message received from FCIOM	No Sub Type Information	No communication between FCIOM and VMCU. No information about ambient air temperature can be received.	- No power supply to FCIOM - Broken CAN link	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2a. Check CAN wires and connectors 2b. Check ECUs	Lost Communication With Chassis Control Module "B" (FCIOM)
VMCU	U015500	- No message received from IC	No Sub Type Information	No communication between IC and VMCU.	- No power supply to IC - Broken CAN link	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2a. Check CAN wires and connectors 2b. Check ECUs	Lost Communication With Instrument Panel Cluster (IPC) Control Module
VMCU	C100E01	Failure is set if both parking brake sensor shorted to ground or Ubat.	General Electrical Failure	Backup Level 3. Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available. Parking brake actuation status is not available.	1. Damage of the pressure sensor wiring 2. Damage of the pressure sensors 3. Damage of the APM	Diagnostic Service 14. No additional action is needed.	1. Replace APM	Parking Brake Circuit
VMCU	C100E62	Parking brake sensors are not correlating, faulty sensor cannot be identified or AD conversion failure.	Signal Compare Failure	Backup level 3. Autorelease not available XPR not available Autopark is not available Safe operation of the parking brake in driving is not guaranteed Trailer test not available. Parking brake actuation status is not available.	1. Leakage into the parking brake 2. Damage of the pressure sensor wiring 3. Damage of the pressure sensors 4. Damage of the APM 5. Incorrect supply voltage level	Diagnostic Service 14. No additional action is needed.	1. Check the Trailer control module; 2. Check leakages in the parking brake connection to the Trailer control module; 3. Check leakages in the parking brake connection to the springbrake chambers; 4. Check the APM exhaust ports; 5. Compare value of DID P1E0K to value of manometer connected to port 29.2; 6. Compare value of DID P1E0J to value of manometer connected to port 28.1/28.2; 7. Replace Parking brake relay valve; 8. Replace APM	Parking Brake Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	C100E64	Parking brake control pressure is lower than the reference pressure minus 1bar and Parking brake chamber pressure is lower than the reference pressure minus 1bar.	Signal Plausibility Failure	No functional effect. Risk of unintended braking due to low pressure in parking brake	1. Trailer Control Module or its control line is damaged or other trailer control port is not plugged 2. Trailer Control Module supply line is damaged. 3. Spring brake lines are damaged. 4. Parking brake relay valve failure 5. Sensor failure 6. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	1. Check the Trailer control module; 2. Check Spring brake lines; 3. Check Parking brake relay valve; 4. Check the APM exhaust ports; 5. Check Parking brake sensor; 6. Replace APM	Parking Brake Circuit
VMCU	C100E7A	Parking brake control and chamber pressure higher than the reference pressure minus threshold, this threshold is based on multicriteria formula.	Fluid Leak or Seal Failure	No functional effect. Risk of unintended release due to high pressure in parking brake.	1. Trailer control module or its pipe connection is damaged or incorrect trailer control port is not plugged 2. Trailer control module supply line is damaged. 3. Spring brake lines are damaged. 4. Parking brake relay valve failure 5. Sensor failure 6. APM internal valve failure	Diagnostic Service 14. No additional action is needed.	- Fill up the system to cut off, apply and release the parking brake manually - Check the Trailer control module - Check parking brake connection to the Trailer control module - Check parking brake connection to the springbrake chambers - Check the APM exhaust ports - Compare value of DID P1E0K to value of manometer connected to port 29.2 - Compare value of DID P1E0J to value of manometer connected to port 28.1/28.2 - Replace Parking brake relay valve - Replace APM	Parking Brake Circuit
VMCU	U114161	Total vehicle distance calculated by VMCU out of range (> 21055406 km).	Signal Calculation Failure	Incorrect odometer distance.	1. Without TCO: Incorrect distance offset configured in VMCU (if VMCU is replaced, a distance offset can be loaded in the replacement VMCU) 2. With TCO: TCO faulty (distance data from TCO is "in range" followed by "out of range", causes VMCU backup to be used and fault code to be set) 3. With/without TCO: VMCU faulty (incorrect stored distance (out of range value) or software fault) 4. With/without TCO: Actual distance which the vehicle has travelled is out of range	Diagnostic Service 14. No additional action is needed.	1. Reload SW 2. Check TCO 3a. Reload VMCU SW 3b. Check VMCU 4. No action	High Resolution Total Vehicle Distance
VMCU	U114261	Trip distance calculated by VMCU out of range (> 21055406 km).	Signal Calculation Failure	Incorrect trip distance.	"1. Power loss when trip counter is reset 2. VMCU faulty (incorrect stored distance when trip counter is reset)"	Diagnostic Service 14. No additional action is needed.	Reset trip meter, make a key Off/On reset and clear DTC. If the fault remains, replace VMCU.	High Resolution Trip Distance

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B159F64	Deviation in VMCU primary and secondary storage of vehicle distance data.	Signal Plausibility Failure	None. The largest stored odometer distance will be used.	"1. Power loss when writing data to primary and/or secondary storage 1. VMCU faulty (incorrect stored distance in primary and/or secondary storage)"	Diagnostic Service 14. No additional action is needed.	1. Check battery and power supply system 2. Check VMCU	Stored Total Vehicle Distance Data
VMCU	B150411	Position lights have been shorted to ground.	Circuit Short To Ground	Not possible to use the Position Lights or Identification Lights on Roof.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator and/or ELCP 3. Check VMCU	Position Light Circuit
VMCU	B150412	Position lights have been shorted to battery.	Circuit Short To Battery	Not possible to use the Position Lights or Identification Lights on Roof.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator and/or ELCP 3. Check VMCU	Position Light Circuit
VMCU	B100311	Accessories power supply has been shorted to ground.	Circuit Short To Ground	Not possible to operate the DC/DC converter (24V to 12V) in the header shelf.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Accessories Power Supply
VMCU	B100315	Accessories power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to operate the DC/DC converter (24V to 12V) in the header shelf.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check actuator 3. Check VMCU	Accessories Power Supply
VMCU	B100711	Ignition power supply has been shorted to ground.	Circuit Short To Ground	Not possible to operate the Toll Collect System, Fuel Line Heater among many other systems.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL and ignition key switch 3. Check VMCU	Ignition Signal Power Supply Circuit
VMCU	B100715	Ignition power supply has been shorted to battery or open circuit.	Circuit Short To Battery or Open	Not possible to operate the Toll Collect System, Fuel Line Heater among many other systems.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL and ignition key switch 3. Check VMCU	Ignition Signal Power Supply Circuit
VMCU	B150D12	EMS relay coil has been shorted to battery.	Circuit Short To Battery	No power supplied to the Engine Management System. Not possible to start or use the engine.	1. EMS relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check EMS relay (K27) 2. Check VMCU	Relay K27 Engine Management System Power Circuit
VMCU	B150D14	EMS relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	No power supplied to the Engine Management System. Not possible to start or use the engine.	1. EMS relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check EMS relay (K27) 2. Check VMCU	Relay K27 Engine Management System Power Circuit
VMCU	B150F12	Ignition relay coil has been shorted to battery.	Circuit Short To Battery	No power supplied to the Supplementary Restrain System (SRS).	1. Ignition relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check ignition relay (K22) 2. Check VMCU	Relay K22 Ignition Power Supply Circuit
VMCU	B150F14	Ignition relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	No power supplied to the Supplementary Restrain System (SRS).	1. Ignition relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check ignition relay (K22) 2. Check VMCU	Relay K22 Ignition Power Supply Circuit
VMCU	B151012	BB ignition relay coil has been shorted to battery.	Circuit Short To Battery	No power supplied to the BBEC Ignition terminals.	1. BB ignition relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check body builder ignition relay (K20) 2. Check VMCU	Relay K20 Body Builder Ignition Power Supply Circuit
VMCU	B151014	BB ignition relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	No power supplied to the BBEC Ignition terminals.	1. BB ignition relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check body builder ignition relay (K20) 2. Check VMCU	Relay K20 Body Builder Ignition Power Supply Circuit
VMCU	B151112	Power supply 2 relay coil has been shorted to battery.	Circuit Short To Battery	No power supplied to the BBEC Main and Switch terminals.	1. Power supply 2 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 2 (K07) 2. Check VMCU	Relay K07 Power Supply 2 Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B151114	Power supply 2 relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	No power supplied to the BBEC Main and Switch terminals.	1. Power supply 2 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 2 (K07) 2. Check VMCU	Relay K07 Power Supply 2 Circuit
VMCU	B151212	Power supply 3 relay coil has been shorted to battery.	Circuit Short To Battery	No power supplied to the Alcolock system, Cigarette Lighter or 24V Power Outlets.	1. Power supply 3 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 3 (K14) 2. Check VMCU	Relay K14 Power Supply 3 Circuit
VMCU	B151214	Power supply 3 relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	No power supplied to the Alcolock system, Cigarette Lighter or 24V Power Outlets.	1. Power supply 3 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 3 (K14) 2. Check VMCU	Relay K14 Power Supply 3 Circuit
VMCU	B151312	Power supply 4 relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Bed light side walls, luggage compartment switches and lamps or the DC/DC converter (24V to 12V) in the header shelf.	1. Power supply 4 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 4 (K11) 2. Check VMCU	Relay K11 Power Supply 4 Circuit
VMCU	B151314	Power supply 4 relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Bed light side walls, luggage compartment switches and lamps or the DC/DC converter (24V to 12V) in the header shelf.	1. Power supply 4 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check power supply circuit relay 4 (K11) 2. Check VMCU	Relay K11 Power Supply 4 Circuit
VMCU	B151412	Refrigerator relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Fridge in the Cab.	1. Refrigerator relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check refrigerator relay (K02) 2. Check VMCU	Relay K02 Refrigerator Power Supply Circuit
VMCU	B151414	Refrigerator relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Fridge in the Cab.	1. Fuse F75 blown 2. Refrigerator relay faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check fuse F75 2. Check refrigerator relay (K02) 3. Check VMCU	Relay K02 Refrigerator Power Supply Circuit
VMCU	B151612	Seat heating / Engine running relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the Seat Heater.	1. Seat heating / Engine running relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check engine running relay (K09) 2. Check VMCU	Relay K09 Seat Heating Power Supply Circuit
VMCU	B151614	Seat heating / Engine running relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the Seat Heater.	1. Seat heating / Engine running relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check engine running relay (K09) 2. Check VMCU	Relay K09 Seat Heating Power Supply Circuit
VMCU	B151D12	Accessory relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the coffee maker.	1. Accessory relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check accessories relay (K08) 2. Check VMCU	Relay K08 Accessory Power Supply Circuit
VMCU	B151D14	Accessory relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the coffee maker.	1. Accessory relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check accessories relay (K08) 2. Check VMCU	Relay K08 Accessory Power Supply Circuit
VMCU	B156516	EMS fuse blown.	Circuit Voltage Below Threshold	No power supplied to the Engine Management System. Not possible to start or use the engine.	Fuse F41 is blown.	Diagnostic Service 14. No additional action is needed.	Replace EMS fuse	Fuse F41

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B152012	Sunroof motor 1 relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to operate the Roof Hatch.	1. Sunroof motor 1 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check sunroof motor 1 relay (K04) 2. Check VMCU	Relay K04 Sunroof Motor 1 Circuit
VMCU	B152014	Sunroof motor 1 relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to operate the Roof Hatch.	1. Fuse F52 blown 2. Sunroof motor 1 relay faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check fuse F52 2. Check sunroof motor 1 relay (K04) 3. Check VMCU	Relay K04 Sunroof Motor 1 Circuit
VMCU	B152112	Sunroof motor 2 relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to operate the Roof Hatch.	1. Sunroof motor 2 relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check sunroof motor 2 relay (K03) 2. Check VMCU	Relay K03 Sunroof Motor 2 Circuit
VMCU	B152114	Sunroof motor 2 relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to operate the Roof Hatch.	1. Fuse F52 blown 2. Sunroof motor 2 relay faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check fuse F52 2. Check sunroof motor 2 relay (K03) 3. Check VMCU	Relay K03 Sunroof Motor 2 Circuit
VMCU	B151512	Seats movement relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to use the electrical seat adjustment.	1. Seats movement relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check seats movement relay (K24) 2. Check VMCU	Relay K24 Seat Adjustment Power Supply Circuit
VMCU	B151514	Seats movement relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to use the electrical seat adjustment.	1. Seats movement relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check seats movement relay (K24) 2. Check VMCU	Relay K24 Seat Adjustment Power Supply Circuit
VMCU	B15B016	Vehicle Speed Sensor Supply Voltage < 6 V	Circuit Voltage Below Threshold	For vehicles without Tachograph, vehicle speed information will not be available.	- Vehicle Speed Sensor or cable harness faulty. - Vehicle Speed Sensor Supply Circuitry In VMCU faulty	Diagnostic Service 14. No additional action is needed.	Check wires and connectors	Speed Pulse Sensor Power Supply Circuit
VMCU	B150929	Unexpected ESCL motor power supply detected.	Signal invalid	Not possible to lock or unlock the Electrical Steering Column Lock.	1. VMCU / ESCL motor power supply wiring harness faulty 2. ESCL faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL 3. Check VMCU	Electronic Steering Column Lock
VMCU	B150972	Maximum number of retries (P1CYX) reached for locking ESCL.	Actuator Stuck Open	Not possible to lock the Electrical Steering Column Lock.	ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock
VMCU	B150973	Maximum number of retries (P1CYX) reached for unlocking ESCL.	Actuator Stuck Closed	Not possible to unlock the Electrical Steering Column Lock.	1. Steering column jammed. 2. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1a. Turn steering wheel to a position that not jam the ESCL 1b. Check for other things that can jam the ESCL 2. Check ESCL	Electronic Steering Column Lock
VMCU	B150993	Operation timeout of the ESCL.	No Operation	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL wiring harness faulty. 2. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL	Electronic Steering Column Lock
VMCU	B15B968	Electrical Steering Column Lock locking/unlocking functionality not enabled.	Event Information	Not possible to lock/unlock the Electrical Steering Column Lock.	Electrical Steering Column Lock parameter P1G20 is inactive.	Diagnostic Service 14. No additional action is needed.	Enable Electrical Steering Column Lock by setting parameter P1G20 to TRUE (Enabled)	Electrical Steering Column Lock Locking/Unlocking Disabled
VMCU	B150311	ESCL motor power supply has been shorted to ground.	Circuit Short To Ground	Not possible to lock or unlock the Electrical Steering Column Lock.	1. Wiring harness faulty 2. ESCL faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL 3. Check VMCU	Electrical Steering Column Lock Supply Circuit
VMCU	B150312	ESCL motor power supply has been shorted to battery.	Circuit Short To Battery	Not possible to lock or unlock the Electrical Steering Column Lock.	1. Wiring harness faulty 2. ESCL faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL 3. Check VMCU	Electrical Steering Column Lock Supply Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B110129	MTCO fault translated: ""V-pulse output"	Signal invalid	Not possible to guarantee a legal use of the MTCO.	"1. B7 wiring harness faulty 2. VMCU faulty (to which B7 is connected) 3. MTCO faulty (Error at B7 output)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check VMCU 3. Check MTCO	Vehicle Speed Impulse
VMCU	B15A200	MTCO fault translated: "KBD Error"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Keyboard error)	Diagnostic Service 14. No additional action is needed.	Check MTCO	Tachograph Keyboard Error
VMCU	B15A308	MTCO fault translated: "PulseGen. COM"	Bus Signal / Message Failures	Not possible to guarantee a legal use of the MTCO.	"1. Speed sensor faulty 2. MTCO faulty (Error in sensor communcation))"	Diagnostic Service 14. No additional action is needed.	1. Check speed sensor 2. Check MTCO	Tachograph Speed Pulse Sensor
VMCU	B15A34A	MTCO fault translated: "PulseGen. COM"	Incorrect Component Installed	Not possible to guarantee a legal use of the MTCO.	"1. Speed sensor faulty 2. MTCO faulty (Error in temp. Sensorkey / serialnumber / realtime signal)"	Diagnostic Service 14. No additional action is needed.	1. Check speed sensor 2a. Reload MTCO SW 2b. Check MTCO	Tachograph Speed Pulse Sensor
VMCU	B15A400	MTCO fault translated: "DriveWithoutChart"	No Sub Type Information	No driver data can be registered since chart is missing.	"1. Chart not inserted in MTCO 2. MTCO faulty (Drive without chart)"	Diagnostic Service 14. No additional action is needed.	1. Insert Chart into MTCO 2a. Reload MTCO SW 2b. Check MTCO	Tachograph Trip Without Chart
VMCU	B15A500	MTCO fault translated: "Drawer Error"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	"1. MTCO drawer tray blocked 2. MTCO faulty (Error when opening the tray)"	Diagnostic Service 14. No additional action is needed.	Check MTCO	Tachograph Drawer Error
VMCU	B15A600	MTCO fault translated: "Missing Chart Driver1"	No Sub Type Information	No driver data can be registered since chart is missing.	"1. Chart for Driver 1 missing 2. MTCO faulty (Chart Driver 1 missing)"	Diagnostic Service 14. No additional action is needed.	1. Insert Chart for driver 1 into MTCO 2a. Reload MTCO SW 2b. Check MTCO	Tachograph Driver 1 Chart Missing
VMCU	B15A700	MTCO fault translated: "Missing Chart Driver2"	No Sub Type Information	No driver data can be registered since chart is missing.	"1. Chart for Driver 2 missing 2. MTCO faulty (Chart Driver 2 missing)"	Diagnostic Service 14. No additional action is needed.	1. Insert Chart for driver 2 into MTCO 2a. Reload MTCO SW 2b. Check MTCO	Tachograph Driver 2 Chart Missing
VMCU	B15A800	MTCO fault translated: "Chart Motion"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Chart motion)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph Chart Motion
VMCU	B15A900	MTCO fault translated: "V-recorder defective"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Error in recorder system)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph Recording System Defective
VMCU	B15AA00	MTCO fault translated: "MFS-recorder defective"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Error in duplicating recorder system)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph Duplicating Recording System Defective
VMCU	B15AB00	MTCO fault translated: "Recorder Liftsystem defective"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Error in recorder lifting system)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph Recording Lifting System Defective
VMCU	B15A001	MTCO fault translated: "Power supply"	General Electrical Failure	Not possible to guarantee a legal use of the MTCO.	"1. Power supply wiring harness. 2. MTCO faulty (power supply interruption)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check MTCO	Tachograph
VMCU	B15A008	MTCO fault translated: "Instrument COM"	Bus Signal / Message Failures	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (no instrument communication)	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check MTCO 3. Check VMCU	Tachograph
VMCU	B15A045	MTCO fault translated: "Program memory"	Program Memory Failure	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (error in FRAM, RTC, I2C, watchdog, program memory)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B15A087	"MTCO fault translated: ""CAN Other"" CAN message TCO1 timeout KWP2000 request timeout"	Missing Message	Not possible to guarantee a legal use of the MTCO.	"1. Backbone 1 wiring harness faulty 2. ECU failure (any ECU connected to Backbone 1 network) 3. MTCO faulty (other CAN problems)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check other ECUs 3. Check MTCO	Tachograph
VMCU	B15A100	MTCO fault translated: "LCD Error"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Display error)	Diagnostic Service 14. No additional action is needed.	Check MTCO	Tachograph Display Error
VMCU	B15AC00	MTCO fault translated: "Adjustment Clock request"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	MTCO faulty (Time difference between display and chart carrier)	Diagnostic Service 14. No additional action is needed.	1. Reload MTCO SW 2. Check MTCO	Tachograph Display And Chart Time Inconsistent
VMCU	B15BA00	MTCO fault translated: "Over Speed Warning"	No Sub Type Information	Not possible to guarantee a legal use of the MTCO.	1. Vehicle is actually travelling over speed limits. 2. Calibration problems. 3. Faulty vehicle speed sensors.	Diagnostic Service 14. No additional action is needed.	None. Speed limit has been exceeded.	Tachograph Over Speed Warning
VMCU	B15B412	TECU power supply has been shorted to battery.	Circuit Short To Battery	Not possible to deactivate the transmission ECU.	1. TECU relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Change TECU relay (K17) 2. Check VMCU	Relay K17 TECU Power Supply Circuit
VMCU	B15B414	TECU power supply has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to activate the transmission ECU.	1. TECU relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Change TECU relay (K17) 2. Check VMCU	Relay K17 TECU Power Supply Circuit
VMCU	U300041	- ECU ROM checksum error.	General Checksum Failure	Software is invalidated, reprogramming is required.	- ECU SW has been tampered - ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit
VMCU	U114461	Vehicle speed calculated by VMCU out of range (calculated value is more than 250 km/h).	Signal Calculation Failure	Calculated vehicle speed is set to Error (calculated vehicle speed is always used for vehicles without Tachograph. For vehicles with Tachograph the calculated vehicle speed is used in fallback mode. Fallback mode is entered when no vehicle speed is received from the Tachograph).	1. Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. With TCO: TCO/VMCU K factor synchronization failure (VMCU/TCO faulty). 4: Without TCO: Incorrect K factor in VMCU.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3. Reload VMCU and TCO SW 4. Reload VMCU SW	Wheel Based Vehicle Speed
VMCU	U114462	Vehicle speed calculated by VMCU deviates more than 10 km/h (parameter) from TCO vehicle speed.	Signal Compare Failure	Calculated vehicle speed is untrustable (for vehicles with Tachograph the calculated vehicle speed is used in fallback mode. Fallback mode is entered when no vehicle speed is received from the Tachograph).	1. Wiring harness from TCO faulty. 2. TCO/VMCU K factor synchronization failure (VMCU/TCO faulty).	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Reload VMCU and TCO SW	Wheel Based Vehicle Speed

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	U114464	Vehicle speed deviates more than 10 km/h (parameter) from road speed (applicable for vehicles with ABS/EBS). Vehicle speed below or equal 0 km/h (parameter) while driveline is engaged and engine speed above 1200 rpm (parameter) (applicable for vehicles without ABS/EBS).	Signal Plausibility Failure	Vehicle Speed signal is set to Error.	1. Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. With TCO: Incorrect W/K factor in TCO. 4: Without TCO: Incorrect K factor in VMCU.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3. Reload TCO SW 4. Reload VMCU SW	Wheel Based Vehicle Speed
VMCU	U114561	Output shaft speed calculated by VMCU out of range (calculated value is more than 8000 rpm).	Signal Calculation Failure	Calculated output shaft speed is set to Error (calculated output shaft speed is always used for vehicles without Tachograph. For vehicles with Tachograph the calculated output shaft speed is used in fallback mode. Fallback mode is entered when no output shaft speed is received from the Tachograph).	1.Wiring harness from speed sensor faulty. 2. Speed sensor faulty. 3. Incorrect VMCU dataset or VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check speed sensor 3a. Reload VMCU SW 3b. Check VMCU	Transmission Output Shaft Speed
VMCU	B151C11	Vehicle state for bodybuilder output has been shorted to ground.	Circuit Short To Ground	Not possible to operate power to the device connected to the BBEC. Output might be on always.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check connection/cable to CN11:3 on VMCU 2. Check VMCU	Vehicle State For Bodybuilder Circuit
VMCU	B151C15	Vehicle state for bodybuilder output has been shorted to battery or open circuit.	Circuit Short To Battery or Open	No power is supplied to the device connected to the BBEC. Output is probably always off.	1. Wiring harness faulty 2. Actuator faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check connection/cable to CN11:3 on VMCU 2. Check VMCU	Vehicle State For Bodybuilder Circuit
VMCU	U000188	CAN Backbone 2 network fault.	Bus off	Vehicle cannot be used.	1. Backbone 2 wiring harness faulty 2. ECU failure (any ECU connected to Backbone 2 network)	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ECUs	CAN Communication Backbone 2 Net
VMCU	U001088	CAN Backbone 1 network fault.	Bus off	Vehicle cannot be used.	"1. Backbone 1 wiring harness faulty 2. ECU failure (any ECU connected to Backbone 1 network)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ECUs	CAN Communication Backbone 1 Net
VMCU	U006488	CAN Diagnostic Link network fault.	Bus off	Not possible to access the vehicle from an external Diagnostic Tool.	"1. Diagnostic Link wiring harness faulty 2. External test tool / hardware faulty 3. VMCU faulty (only VMCU is connected to this network)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check test tool version 2b. Check HW 3. Check VMCU	CAN Communication Diagnostic Link Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	U300004	- Internal communication lost with high side and low side drivers. - Communication error indicated by hardware	System Internal Failures	Vehicle cannot be used. Not possible to control any power supplies or relays from the VMCU/FRC.	- Defect VMCU hardware - Defect VMCU software	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit
VMCU	U002888	CAN Chassis network fault.	Bus off	Vehicle cannot be used.	"1. Chassis wiring harness faulty 2. ECU failure (any ECU connected to Chassis network)"	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ECUs	CAN Communication Chassis Net
VMCU	U114649	VMCU LIN hardware fault.	Internal Electronic Failure	Rain Sensor cannot be used. Not possible to lock/unlock the Electrical Steering Column Lock.	1. VMCU faulty (BSW driver / transceiver).	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check VMCU	LIN7 Communication Net
VMCU	U300316	- ECU Voltage < 16V.	Circuit Voltage Below Threshold	Supply sensitive diagnostics are inhibited. Depending on how low the supply voltage is, the vehicle can be affected in various ways. Worst case it cannot be started or used. Best case it is not affected at all.	- Broken/bad power lines. - Battery voltage too low.	Diagnostic Service 14. No additional action is needed.	1. Check battery 2. Check power supply system	Battery Power
VMCU	U300317	- ECU Voltage > 32V.	Circuit Voltage Above Threshold	No functional impact.	- Alternator/charging circuit. - External reason (charging or incorrect power supply).	Diagnostic Service 14. No additional action is needed.	Check power supply system	Battery Power
VMCU	B15B116	- Voltage on VMCU Supply 1 Pin < 16V.	Circuit Voltage Below Threshold	Following functions cannot be used: - Max Light 1 - Max Light 2 - Resting Light 1 2DO: Improve the names of these functions and features.	- Fuse F28 blown. - Power supply or connector failure.	Diagnostic Service 14. No additional action is needed.	1. Replace fuse F28 2a. Check battery 2b. Check power supply system	Vehicle Master Control Unit Power Supply 1 Circuit
VMCU	B15B216	- Voltage on VMCU Supply 2 Pin < 16V.	Circuit Voltage Below Threshold	Following functions cannot be used: - Cab Body Working Light - Side Reverse Working Lights - Extra Spot Lamps Front - Extra Parking Lights - BB Flexible Switch 1 - BB Flexible Switch 2 - Tail Lift/Crane - Backlight - Resting Light 2 - Night Light - Vehicle State For Body Builder 2DO: Improve the names of these functions and features.	- Fuse F27 blown. - Power supply or connector failure.	Diagnostic Service 14. No additional action is needed.	1. Replace fuse F27 2a. Check battery 2b. Check power supply system	Vehicle Master Control Unit Power Supply 2 Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B15B316	- Voltage on VMCU Supply 3 Pin < 16V.	Circuit Voltage Below Threshold	Following functions cannot be used: - Hardwired Hazard Output - Accelerator Pedal 1 - Accelerator Pedal 2 - Accessories Signal - Ignition Signal - Cab Tilt - Position Light - Electrical Steering Column Lock - Hazard Switch Backlight - Clutch Pedal Sensor/Switch 2DO: Improve the names of these functions and features.	- Fuse F68 blown. - Power supply or connector failure.	Diagnostic Service 14. No additional action is needed.	1. Replace fuse F68 2a. Check battery 2b. Check power supply system	Vehicle Master Control Unit Power Supply 3 Circuit
VMCU	U114688	LIN7 network fault (invalid / corrupted headers sent on LIN7).	Bus off	Rain Sensor cannot be used. Not possible to lock/unlock the Electrical Steering Column Lock.	1. LIN7 wiring harness faulty. 2. RAINS or ESCL faulty. 3. VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2a. Check RAINS 2b. Check ESCL 3. Check VMCU	LIN7 Communication Net
VMCU	B152416	RAINS power supply below limit. Fault Code Information (FCI) sent from RAINS: 22	Circuit Voltage Below Threshold	Rain Sensor cannot be used.	1. Wiring harness faulty (short circuit to ground). 2. CIOM faulty. 3. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check CIOM 3. Check RAINS	Rain Sensor
VMCU	B152417	RAINS power supply voltage above limit. Fault Code Information (FCI) sent from RAINS: 21	Circuit Voltage Above Threshold	Rain Sensor cannot be used.	1. Wiring harness faulty (short circuit to supply). 2. CIOM faulty. 3. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check CIOM 3. Check RAINS	Rain Sensor
VMCU	B152446	RAINS EEPROM failure. Fault Code Information (FCI) sent from RAINS: 19	Calibration / Parameter Memory Failure	Rain Sensor cannot be used.	1. Calibration data lost/corrupted in RAINS (try calibration). 2. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. Recalibrate RAINS 2. Check RAINS	Rain Sensor
VMCU	B152449	RAINS hardware failure. Fault Code Information (FCI) sent (one or more) from RAINS: 5, 7, 8, 9, 11, 12, 13	Internal Electronic Failure	Rain Sensor cannot be used.	1. No EOL calibration done at supplier (RAINS faulty). 2. Twilight sensors defective (RAINS faulty). 3. Rain detection hardware faulty (RAINS faulty).	Diagnostic Service 14. No additional action is needed.	1. Recalibrate RAINS 2, 3. Check RAINS	Rain Sensor
VMCU	B15244B	RAINS over temperature. Fault Code Information (FCI) sent from RAINS: 20	Over Temperature	Rain Sensor cannot be used.	1. Ambient temperature too high. 2. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. No action 2. Check RAINS	Rain Sensor

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B152454	RAINS not calibrated at manufacturing. Fault Code Information (FCI) sent from RAINS: 10	Missing Calibration	Rains Sensor is not calibrated, degraded performance.	1. Missing 0km calibration data. 2. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. Perform a calibration of the Rain Sensor using Routine Identifier R1AEZ. 2. Check RAINS	Rain Sensor
VMCU	B152476	RAINS inproper installation to windscreen. Fault Code Information (FCI) sent (one or more) from RAINS: 1, 2, 3, 4, 6	Wrong Mounting Position	Rain Sensor cannot be used.	1. Inproper optical contact to windscreen. 2. Incorrect windscreen properties. 3. RAINS faulty.	Diagnostic Service 14. No additional action is needed.	1. Adjust RAINS position 2. Recalibrate RAINS 3. Check RAINS	Rain Sensor
VMCU	B152486	Faulty response from RAINS (invalid / corrupted response on RAINS headers sent on LIN7).	Signal Invalid	Rain Sensor cannot be used.	1. LIN7 wiring harness faulty. 2. RAINS faulty. 3. VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check RAINS 3. Check VMCU	Rain Sensor
VMCU	B152487	No response from RAINS (no response on RAINS headers sent on LIN7).	Missing Message	Rain Sensor cannot be used.	1. LIN7 wiring harness faulty. 2. RAINS faulty. 3. VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check RAINS 3. Check VMCU	Rain Sensor
VMCU	B152494	RAINS software failure. Fault Code Information (FCI) sent from RAINS: 23	Unexpected Operation	Rain Sensor cannot be used.	1. Watchog reset (RAINS faulty).	Diagnostic Service 14. No additional action is needed.	Check RAINS	Rain Sensor
VMCU	B150916	ESCL motor power supply below limit. Fault Code Information (FCI) sent (one or more) from ESCL: 7, 8, 11, 13	Circuit Voltage Below Threshold	Not possible to lock or unlock the Electrical Steering Column Lock.	1. System voltage too low. 2. Wiring harness faulty (short circuit to ground). 3. ESCL faulty. 4. VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2. Check wires and connectors 3. Check ESCL 4. Check VMCU	Electronic Steering Column Lock
VMCU	B150917	ESCL motor power supply above limit. Fault Code Information (FCI) sent (one or more) from ESCL: 1, 2, 3, 4, 9, 12, 14	Circuit Voltage Above Threshold	Not possible to lock or unlock the Electrical Steering Column Lock.	1. System voltage too high. 2. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1a. Check battery 1b. Check power supply system 2. Check ESCL	Electronic Steering Column Lock
VMCU	B150918	ESCL motor current below limit. Fault Code Information (FCI) sent from ESCL: 10	Circuit Current Below Threshold	Not possible to lock or unlock the Electrical Steering Column Lock.	1. Wiring harness faulty (too high impedance or bad connection) 2. VMCU faulty. 3. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL 3. Check VMCU	Electronic Steering Column Lock
VMCU	B150943	ESCL EEPROM (special memory) failure. Fault Code Information (FCI) sent from ESCL: 18	Special Memory Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	B150944	ESCL RAM (data memory) failure. Fault Code Information (FCI) sent from ESCL: 16	Data Memory Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock
VMCU	B150945	ESCL ROM (program memory) failure. Fault Code Information (FCI) sent from ESCL: 17	Program Memory Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock
VMCU	B150948	ESCL supervision software (flag check, lock sequence) failure. Fault Code Information (FCI) sent (one or more) from ESCL: 19, 20	Supervision Software Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock
VMCU	B150949	ESCL sensor defect. Fault Code Information (FCI) sent from ESCL: 21	Internal Electronic Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty.	Diagnostic Service 14. No additional action is needed.	1. Check ESCL	Electronic Steering Column Lock
VMCU	B150954	ESCL not mathced to the vehicle. Fault Code Information (FCI) sent from ESCL: 23	Missing Calibration	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL still in virgin mode (try matching).	Diagnostic Service 14. No additional action is needed.	Match the Electrical Steering Column Lock to the vehicle using Routine Identifier R1AEY.	Electronic Steering Column Lock
VMCU	B150961	ESCL detected incorrect authentication for unlock. Fault Code Information (FCI) sent from ESCL: 22	Signal Calculation Failure	Not possible to lock or unlock the Electrical Steering Column Lock.	1. ESCL faulty (encrypted communication key changed/corrupt). 2. VMCU faulty (encrypted communication key changed/corrupt).	Diagnostic Service 14. No additional action is needed.	1. Match ESCL 2. Reload VMCU SW 3. Check ESCL	Electronic Steering Column Lock
VMCU	B150986	Faulty response from ESCL (invalid / corrupted response on RAINS headers sent on LIN7).	Signal Invalid	Not possible to lock or unlock the Electrical Steering Column Lock.	1. LIN7 wiring harness faulty. 2. ESCL faulty. 3. VMCU faulty.	Diagnostic Service 14. No additional action is needed.	1. Check wires and connectors 2. Check ESCL 3. Check VMCU	Electronic Steering Column Lock
VMCU	B150987	No response from ESCL (no response on ESCL headers sent on LIN7).	Missing Message	Not possible to lock or unlock the Electrical Steering Column Lock.	1. LIN7 wiring harness faulty 2. CIOM / ESCL power supply wiring harness faulty 3. Backbone 2 wiring harness faulty (VMCU power supply command to CIOM) 4. ESCL faulty 5. CIOM faulty 6. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1 - 3. Check wires and connectors 4. Check ESCL 5. Check CIOM 6. Check VMCU	Electronic Steering Column Lock
VMCU	U300044	ECU volatile memory error.	Data Memory Failure	VMCU does not work properly.	1. ECU faulty (volatile memory)	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit
VMCU	U300045	ECU non volatile memory error.	Program Memory Failure	VMCU does not work properly.	1. ECU faulty (non volatile memory)	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit
VMCU	U300049	ECU internal hardware fault.	Internal Electronic Failure	VMCU does not work properly.	1. ECU faulty (BSW driver / transceiver). 2. Software error	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VMCU	U300094	ECU watchdog fault.	Unexpected Operation	VMCU does not work properly.	1. ECU faulty (watchdog fault) 2. Software error	Diagnostic Service 14. No additional action is needed.	1. Reload VMCU SW 2. Check VMCU HW	Electronic control unit
VMCU	U1FFF68	Factory mode parameter P1C54 is active	Event Information	A truck delivered with Factory mode active will have unspecified behavior that might influence customer safety.	Factory mode not disabled	Diagnostic Service 14. No additional action is needed.	Set P1C54 (Shared Factory mode parameter) for the ECU to FALSE before shipping from factory.	Factory Mode Active
VMCU	U114768	Too many DTC for Diagnostic Warning Manger (DWM) to handle.	Event Information	No functional impact.	1. Refer to the DTC's that are active.	Diagnostic Service 14. No additional action is needed.	According to other set DTCs.	DTC Notification Max Limit Reached
VMCU	U250145	ECU has detected Internal Data Fragmentation and cannot save internal status information like DTCs.	Program Memory Failure	ECU cannot save internal data like new DTC	- Inproper shutdown - power cut - In rare situations HW failure due to memory degradation - Software error	Diagnostic Service 14. No additional action is needed.	Transfer data, run RID R1AHD, restore data may fix the problem	Internal Data Storage Fragmented
VMCU	U250245	ECU detected Internal Data Storage Discarded, possibly discarding internal data but selfhealing attempted.	Program Memory Failure	No functional impact.	- Software error - Memory degradation due to end-of-life	Diagnostic Service 14. No additional action is needed.	No action due to this DTC, the ECU has tried to recover. There are likely other DTCs that need to be handled as a consequence.	Internal Data Storage Discarded
VMCU	B151A12	Windscreen wiper motor relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to operate the wipers.	1. Windscreen wiper motor relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check windscreen wiper motor On/Off relay (K05) 2. Check VMCU	Relay K05 Wiper Motor On/Off Circuit
VMCU	B151A14	Windscreen wiper motor relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to operate the wipers.	1. Fuse F51 blown 2. Windscreen wiper motor relay faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check fuse F51 2. Check windscreen wiper motor On/Off relay (K05) 3. Check VMCU	Relay K05 Wiper Motor On/Off Circuit
VMCU	B151B12	Windscreen wiper hi/lo relay coil has been shorted to battery.	Circuit Short To Battery	Not possible to switch between low/high wiper speed.	1. Windscreen wiper hi/lo relay faulty 2. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check windscreen wiper Hi/Lo relay (K06) 2. Check VMCU	Relay K06 Wiper Motor High/Low Speed Circuit
VMCU	B151B14	Windscreen wiper hi/lo relay coil has been shorted to ground or open circuit.	Circuit Short To Ground or Open	Not possible to switch between low/high wiper speed or not possible to operate the wipers.	1. Fuse F51 blown 2. Windscreen wiper hi/lo relay faulty 3. VMCU faulty	Diagnostic Service 14. No additional action is needed.	1. Check fuse F51 2. Check windscreen wiper Hi/Lo relay (K06) 3. Check VMCU	Relay K06 Wiper Motor High/Low Speed Circuit
VS	U004688	CAN bus short circuit	Bus off	No normal operation possible. All functionality is depending on incoming CAN data.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	CAN Communication Infotainment Net
VS	U300041	CPU memory corrupted or un-validated SW downloaded	General Checksum Failure	ECU will reset and stay in boot mode	CPU memory corrupted or un-validated SW downloaded	Clear DTC	Update SW	Electronic control unit
VS	U300045	CPU memory read/write error	Program Memory Failure	ECU will reset and stay in boot mode	Defect CPU	Clear DTC	Replace ECU	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VS	U300046	Calculated checksum/signature of CPU memory does not match stored	Calibration / Parameter Memory Failure	ECU will reset and stay in boot mode	CPU memory corrupted or un-validated SW downloaded	Clear DTC	Update SW	Electronic control unit
VS	U300049	CAN tranceiver/driver failure	Internal Electronic Failure	No normal operation possible. All functionality is depending on incoming CAN data.	Defect CPU or CAN tranceiver in ECU	Clear DTC	Check cabling and connectors Replace ECU	Electronic control unit
VS	U300094	Critical internal software error	Unexpected Operation	ECU will reset	Software "bug"	Clear DTC	Update SW	Electronic control unit
VS	U300316	Voltage under 15±1V	Circuit Voltage Below Threshold	If voltage TOO low, ECU operation will suffer. Normal functionality will deteriorate.	Defect battery, generator or supply cabling	Clear DTC	Check supply cabling, battery and generator	Battery Power
VS	U300317	Voltage over 34±2V	Circuit Voltage Above Threshold	If voltage TOO high for TOO long, ECU may be damaged. Normal functionality will deteriorate.	Defect battery, generator	Clear DTC	Check battery and generator	Battery Power
VS	B150211	Short circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Camera cannot be activated	Damaged cabling or connectors Camera cannot be activated	Clear DTC	Check cabling and connectors	Vision Camera1 Power Supply Circuit
VS	B150212	Short Circuit	Circuit Short To Battery	Camera cannot be de-activated	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Power Supply Circuit
VS	B150311	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Lens cover will be possible to open.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Activation Circuit
VS	B150312	Short Circuit	Circuit Short To Battery	Lens cover will be possible to close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Activation Circuit
VS	B150313	Short Circuit	Circuit Open	Lens cover will be possible to open or close.	Harness	Clear DTC	Check cabling and connectors	Vision Camera1 Activation Circuit
VS	B150411	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Lens heating cnanot be activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Heater Circuit
VS	B150412	Short Circuit	Circuit Short To Battery	Lens heating cnanot be de-activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Heater Circuit
VS	B150413	Open circuit	Circuit Open	Lens heating cannot be controlled.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1 Heater Circuit
VS	B150531	Video decoder status - detects no signal.	No Signal	No image displayed on SID.	Camera disconnected or damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera1
VS	B150631	Video decoder status - detects no signal.	No Signal	No image displayed on SID.	DVD disconnected or damaged cabling or connectors	Clear DTC	Check cabling and connectors	DVD

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VS	B150711	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Camera cannot be activated	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Power Supply Circuit
VS	B150712	Short Circuit	Circuit Short To Battery	Camera cannot be de-activated	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Power Supply Circuit
VS	B150811	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Lens cover will be possible to open.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Activation Circuit
VS	B150812	Short Circuit	Circuit Short To Battery	Lens cover will be possible to close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Activation Circuit
VS	B150813	Open circuit	Circuit Open	Lens cover will be possible to open or close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Activation Circuit
VS	B150911	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Lens heating cnanot be activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Heater Circuit
VS	B150912	Short Circuit	Circuit Short To Battery	Lens heating cnanot be de-activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Heater Circuit
VS	B150913	Open circuit	Circuit Open	Lens heating cannot be controlled.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2 Heater Circuit
VS	B150A31	Video decoder status - detects no signal.	No Signal	No image displayed on SID.	Camera disconnected or damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera2
VS	B150B11	Short Circuit	Circuit Short To Ground	"Output will be inhibited for 60s if output was active at the time of the SC. Camera cannot be activated"	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Power Supply Circuit
VS	B150B12	Short Circuit	Circuit Short To Battery	Camera cannot be de-activated	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Power Supply Circuit
VS	B150C11	Short Circuit	Circuit Short To Ground	Output will be inhibited for 60s if output was active at the time of the SC. Lens cover will be possible to open.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Activation Circuit
VS	B150C12	Short Circuit	Circuit Short To Battery	Lens heating cnanot be de-activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Activation Circuit
VS	B150C13	Open circuit	Circuit Open	Lens cover will be possible to open or close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Activation Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
VS	B150D11	Short Circuit	Circuit Short To Ground	"Output will be inhibited for 60s if output was active at the time of the SC. Lens heating cnanot be activated."	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Heater Circuit
VS	B150D12	Short Circuit	Circuit Short To Battery	Lens heating cnanot be de-activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Heater Circuit
VS	B150D13	Open circuit	Circuit Open	Lens heating cannot be controlled.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3 Heater Circuit
VS	B150E31	Video decoder status - detects no signal.	No Signal	No image displayed on SID.	Camera disconnected or damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera3
VS	B150F31	Video decoder status - detects no signal.	No Signal	No image displayed on SID.	Camera disconnected or damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4
VS	B151011	Short Circuit	Circuit Short To Ground	"Output will be inhibited for 60s if output was active at the time of the SC. Camera cannot be activated"	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Power Supply Circuit
VS	B151012	Short Circuit	Circuit Short To Battery	Camera cannot be de-activated	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Power Supply Circuit
VS	B151111	Short Circuit	Circuit Short To Ground	"Output will be inhibited for 60s if output was active at the time of the SC. Lens cover will be possible to open."	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Activation Circuit
VS	B151112	Short Circuit	Circuit Short To Battery	Lens cover will be possible to close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Activation Circuit
VS	B151113	Open circuit	Circuit Open	Lens cover will be possible to open or close.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Activation Circuit
VS	B151211	Short Circuit	Circuit Short To Ground	"Output will be inhibited for 60s if output was active at the time of the SC. Lens heating cnanot be activated."	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Heater Circuit
VS	B151212	Short Circuit	Circuit Short To Battery	Lens heating cnanot be de-activated.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Heater Circuit
VS	B151213	Open circuit	Circuit Open	Lens heating cannot be controlled.	Damaged cabling or connectors	Clear DTC	Check cabling and connectors	Vision Camera4 Heater Circuit
VS	U1FFF68	Factory mode address parameter P1C54 is active.	Event Information	No functionality affected	Factory mode not disabled.Unit not configured correctly when leaving factory	Clear DTC	Clear the factory mode parameter.	Factory Mode Active
VS	U114168	Too many (>29) DTCs with warning indicator bit set are active at the same time.	Event Information	No more DTCs will be reported in diagnostic fault message on CAN.	ECU has too many DTCs set	Clear DTC	Analyze the other reported DTCs	DTC Notification Max Limit Reached

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U115D92	Unable to open/play requested audio file.	Performance or Incorrect Operation	Instrument Cluster will send CAN signal "SoundStatus" to "Degraded function" on Display Subnet	1. No sound library downloaded 2. Requested sound file corrupt	Diagnostic Service 14. No additional action is needed.	Reflash Instrument Cluster.	Instrument Cluster Loadspeaker
IC	U200A88	Instrument Cluster communication error on Display Subnet	Bus off	No communication is possible to the Instrument Cluster. The instrument Cluster can not receive any data on the CAN-network. The instrument Cluster will keep the last recevied VehicleMode	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Investigate root-cause: - CAN-network error => Fix problem. - CAN-cable => Fix problem. Transceiver failure => Replace ECU	CAN Communication Display Net
IC	U300041	Internal EEPROM failure	General Checksum Failure	Enter limp-home mode. Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory where the patchable data is stored.	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U300044	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Proposed Workshop Action: Replace Instrument Cluster.	Electronic control unit
IC	U300045	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Electronic control unit
IC	U300049	Internal electronic failure	Internal Electronic Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	- CAN-tranciever error	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U30004B	Temperature of TFT-Display >70 degrees Celsius	Over Temperature	ECU (IC) will set the background illumination level to Max 25% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300094	Application performs an unexpected operation	Unexpected Operation	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Watch dog reset (software / parameter bug) Sudden loss of power	Diagnostic Service 14. No additional action is needed.	Talk to driver 1. Can indicate improper shutdown - see possible root causes, no Instrument Cluster action required 2. If Cluster has been behaving strangely reflash Instrument Cluster	Electronic control unit
IC	U300098	Temperature of TFT-display >65 degrees Celsius	Component or System Over Temperature	ECU (IC) will set the background illumination level to Max 50% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300316	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U300317	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
IC	U116C05	Programming of Basic SID (from IC) has failed.	System Programming Failures	Basic SID ECU will not be updated with new Main Software	Programming of SID-Basic intterupted due to: -Power Supply problem -Communication link failure	Diagnostic Service 14. No additional action is needed.	Check integrity of power supply lines to IC and BSID Check integrity of communications lines between IC and BSID	Basic Secondary Infotainment Display Unit
IC	U116C44	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	Replace BSID	Basic Secondary Infotainment Display Unit
IC	U116C45	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Basic Secondary Infotainment Display Unit
IC	U116C46	Internal EEPROM failure	Calibration / Parameter Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	Replace BSID	Basic Secondary Infotainment Display Unit
IC	B150216	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC and BSID.	Basic Secondary Infotainment Display Power Supply Circuit
IC	B150217	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC and BSID.	Basic Secondary Infotainment Display Power Supply Circuit
IC	U116E68	The number of DTC's set in the ECU has reached 29	Event Information	No reduced functionality due to that the DTC is set.	Maximum limit of DTC's set in ECU	Diagnostic Service 14. No additional action is needed.	Clear DTCs	DTC Notification Max Limit Reached

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	P000041	Program memory corruption.	General Checksum Failure	Application should not start (remains in boot-mode). Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Control Module
IC	U1FFF68	Factory Mode is active in the ECU.	Event Information	No reduced functionality due to that the DTC is set.	Factory Mode is active in the ECU. Whenever P1C54 parameter is set to TRUE, this DTC shall be set.	Diagnostic Service 14. No additional action is needed.	Write parameter P1C54 to false.	Factory Mode Active
IC	U116F98	Ambient Temperature > +85 degrees C during Vehicle Operation	Component or System Over Temperature	None	Ambient Temperature is High Ambient Temperature Sensor Fault	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Ambient Air Temperature
IC	U116B87	Missing heartbeat signal TIPSITP_DIDHeartbeat_Up	Missing Message	Instrument Cluster is missing a message from HMIIOM. Instrument Cluster will make a re-start of the Driver Interface Display (DID)	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check compatibility of HMIIOM and IC software. Check integrity of Display Subnet CAN cables.	Lost Communication With HMIIOM
IC	U200887	SID Basic communication error	Missing Message	No communication is possible between Instrument Cluster and SID Basic. The SID Basic can not anything on the display	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Check integrity of CAN cables between IC & BSID Check integrity of power supply cables to BSID Replace BSID	CAN Communication IC-SID Net
IC	U116A87	Engine Malfunction Indication CAN message is missing	Missing Message	Instrument Cluster will lit up the MalFunction Indication Lamp	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check integrity of Display Subnet CAN cables	Malfunction Indication Lamp Fault Status Electronic Data Link
IC	U114163	Trip button error (GPIO value is read as 1 continuously for 90s).	Circuit / Component Protection Time-Out	CAN signals TripmeterReset_rqst and Tripmetershift_rqst are sent as "Error Indicator" on Display Subnet	Trip button is stucked. Internal IC HW failure.	Diagnostic Service 14. No additional action is needed.	Replace Cluster	Instrument Cluster Tripmeter Status Button
IC	U115D92	Unable to open/play requested audio file.	Performance or Incorrect Operation	Instrument Cluster will send CAN signal "SoundStatus" to "Degraded function" on Display Subnet	1. No sound library downloaded 2. Requested sound file corrupt	Diagnostic Service 14. No additional action is needed.	Reflash Instrument Cluster.	Instrument Cluster Loadspeaker
IC	U200A88	Instrument Cluster communication error on Display Subnet	Bus off	No communication is possible to the Instrument Cluster. The instrument Cluster can not receive any data on the CAN-network. The instrument Cluster will keep the last recevied VehicleMode	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Investigate root-cause: - CAN-network error => Fix problem. - CAN-cable => Fix problem. Transceiver failure => Replace ECU	CAN Communication Display Net
IC	U300041	Internal EEPROM failure	General Checksum Failure	Enter limp-home mode. Cluster will be black except for STOP Lamp which will be lit	Memory Failure. Incorrect checksum of the memory where the patchable data is stored.	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U300044	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U300045	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Electronic control unit
IC	U300049	Internal electronic failure	Internal Electronic Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	- CAN-tranciever error	Diagnostic Service 14. No additional action is needed.	Replace Instrument Cluster.	Electronic control unit
IC	U30004B	Temperature of TFT-Display >70 degrees Celsius	Over Temperature	ECU (IC) will set the background illumination level to Max 25% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300094	Application performs an unexpected operation	Unexpected Operation	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Watch dog reset (software / parameter bug) Sudden loss of power	Diagnostic Service 14. No additional action is needed.	Talk to the driver 1. Can he indicate improper shutdown - see possible root causes, no Instrument Cluster action required 2. If Cluster has been behaving strangely reflash Instrument Cluster	Electronic control unit
IC	U300098	Temperature of TFT-display >65 degrees Celsius	Component or System Over Temperature	ECU (IC) will set the background illumination level to Max 50% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
IC	U300316	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
IC	U300317	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC.	Battery Power
IC	U116C05	Programming of Basic SID (from IC) has failed.	System Programming Failures	Basic SID ECU will not be updated with new Main Software	Programming of SID-Basic intterupted due to: -Power Supply problem -Communication link failure	Diagnostic Service 14. No additional action is needed.	Check integrity of power supply lines to IC and BSID Check integrity of communications lines between IC and BSID	Basic Secondary Infotainment Display Unit
IC	U116C44	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	Replace BSID	Basic Secondary Infotainment Display Unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U116C45	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If failure still present, replace unit	Basic Secondary Infotainment Display Unit
IC	U116C46	Internal EEPROM failure	Calibration / Parameter Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality in Basic SID ECU can not be guaranteed when this DTC is set.	Basic SID Memory Failure	Diagnostic Service 14. No additional action is needed.	Replace BSID	Basic Secondary Infotainment Display Unit
IC	B150216	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC and BSID.	Basic Secondary Infotainment Display Power Supply Circuit
IC	B150217	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to IC and BSID.	Basic Secondary Infotainment Display Power Supply Circuit
IC	U116E68	The number of DTC's set in the ECU has reached 29	Event Information	No reduced functionality due to that the DTC is set.	Maximum limit of DTC's set in ECU	Diagnostic Service 14. No additional action is needed.	Clear DTCs	DTC Notification Max Limit Reached
IC	P000041	Program memory corruption.	General Checksum Failure	Application should not start (stuck in boot-mode). Cluster will be black except for STOP Lamp which will be lit.	Memory Failure. Incorrect checksum of the memory.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Control Module
IC	U1FFF68	Factory Mode is active in the ECU.	Event Information	No reduced functionality due to that the DTC is set.	Factory Mode is active in the ECU. Whenever P1C54 parameter is set to TRUE, this DTC shall be set.	Diagnostic Service 14. No additional action is needed.	Write parameter P1C54 to false.	Factory Mode Active
IC	U116F98	Ambient Temperature > +85 degrees Celsius during Vehicle Operation	Component or System Over Temperature	None	Ambient Temperature is High Ambient Temperature Sensor Fault	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Ambient Air Temperature
IC	U116B87	Missing heartbeat signal TIPSITP_DIDHeartbeat_Up	Missing Message	Instrument Cluster is missing a message from HMIIOM. Instrument Cluster will make a re-start of the Driver Interface Display (DID)	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check compatibility of HMIIOM and IC software. Check integrity of Display Subnet CAN cables	Lost Communication With HMIIOM

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
IC	U200887	SID Basic communication error	Missing Message	No communication is possible between Instrument Cluster and SID Basic. The SID Basic can not anything on the display	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Check integrity of CAN cables between IC & BSID Check integrity of power supply cables to BSID Replace BSID	CAN Communication IC-SID Net
IC	U116A87	Engine Malfunction Indication CAN message is missing	Missing Message	Instrument Cluster will lit up the MalFunction Indication Lamp	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check integrity of Display Subnet CAN cables	Malfunction Indication Lamp Fault Status Electronic Data Link
IC	U114163	Trip button error (GPIO value is read as 1 continuously for 90s).	Circuit / Component Protection Time-Out	CAN signals TripmeterReset_rqst and Tripmetershift_rqst are sent as "Error Indicator" on Display Subnet	Trip button is stucked. Internal IC HW failure.	Diagnostic Service 14. No additional action is needed.	Replace Cluster	Instrument Cluster Tripmeter Status Button
SID	U200A88	Communication error on Display Subnet	Bus off	No communication is possible to the Secondary Infotainment Display (SID). The SID can not receive any data on the CAN-network. The SID will keep the last recevied VehicleMode	- CAN High Short Circuit to ground - CAN High short circuit to CAN Low - Open circuit CAN High - Open Circuit CAN Low - Connecotor damaged - Connector disconnected - Transciever damaged	Diagnostic Service 14. No additional action is needed.	Check CAN cables Check HMIIIOM DTCs	CAN Communication Display Net
SID	U300041	Internal EEPROM failure	General Checksum Failure	Enter limp-home mode. SID will be black.	Memory Failure. Incorrect checksum of the memory where the patchable data is stored.	Diagnostic Service 14. No additional action is needed.	Replace SID	Electronic control unit
SID	U300044	Internal RAM failure	Data Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	Replace SID	Electronic control unit
SID	U300045	Internal ROM failure	Program Memory Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Memory failure	Diagnostic Service 14. No additional action is needed.	1. Reflash software 2. If faliure still present, replace unit	Electronic control unit
SID	U300049	Internal electronic failure	Internal Electronic Failure	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	-CAN-tranciever error	Diagnostic Service 14. No additional action is needed.	1. Check CAN connector and Cable 2. If problem still present, replace SID	Electronic control unit
SID	U30004B	Temperature of TFT Display > 70 degrees Celsius	Over Temperature	ECU (SID) will set the background illumination level to Max25% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	1. Investigate driving environment - is truck being driven in very hot climate? 2. Investigate other nearby components - is another component running very hot? 3. Possible internal failure, replace SID	Electronic control unit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SID	U300094	Application performs an unexpected operation	Unexpected Operation	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Watch dog reset (software / parameter bug) Sudden loss of power	Diagnostic Service 14. No additional action is needed.	Talk to driver 1. Can indicate improper shutdown - see possible root causes, noSID action required 2. If SID has been behaving strangely reflash SID.	Electronic control unit
SID	U300098	Temperature of TFT display > 65 degrees Celsius	Component or System Over Temperature	ECU (SID) will set the background illumination level to Max50% to cool down.	High environmental temperature	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Electronic control unit
SID	U300316	Circuit Voltage Below Threshold. Considering component tolerances the fault will be detected between 7,7V....9,9V	Circuit Voltage Below Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too low	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to SID.	Battery Power
SID	U300317	Circuit Voltage Above Threshold. Considering component tolerances the fault will be detected between 32,1.....39,7V	Circuit Voltage Above Threshold	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Voltage supply too high	Diagnostic Service 14. No additional action is needed.	Check other ECUs for DTC Read battery voltage from other ECUs Check integrity of power and ground lines to SID.	Battery Power
SID	U114863	Continuous touch screen press events for >= 10 seconds	Circuit / Component Protection Time-Out	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	- Touch screen pressed / struck for a long time (for atleast 10 seconds continuously) - Touch screen faulty	Diagnostic Service 14. No additional action is needed.	1. Review that there is no material between the front frame and the touch panel that creates pessure 2. Remove objects and clean screen 3. Remove the front frame to clean the screen 4. Exhange the SID	Touch Screen Display Module
SID	U114768	The number of DTC's set in the ECU has reached 29	Event Information	No reduced functionality due to that the DTC is set.	To many internal diagnostic trouble codes set in the ECU.	Diagnostic Service 14. No additional action is needed.	Clear DTCs	DTC Notification Max Limit Reached
SID	P000041	Program memory corruption.	General Checksum Failure	Application should not start (stuck in boot-mode). SID will be black.	Memory Failure. Incorrect checksum of the memory.	Diagnostic Service 14. No additional action is needed.	Replace ECU.	Control Module
SID	U1FFF68	Factory Mode is active in the ECU.	Event Information	No reduced functionality due to that the DTC is set.	Factory Mode is active in the ECU. Whenever P1C54 parameter is set to TRUE, this DTC shall be set.	Diagnostic Service 14. No additional action is needed.	Write parameter P1C54 to false.	Factory Mode Active
SID	U114387	Missing heartbeat signal TIPSITP_DIDHeartbeat_Up	Missing Message	Secondary Infotainment Display (SID) is missing a message from HMIIOM. SID will make a re-start of the display.	SW Failure in the HMIIOM	Diagnostic Service 14. No additional action is needed.	Check compatibility of HMIIOM and IC software. Check integrity of Display Subnet CAN cables.	Lost Communication With HMIIOM
SID	U114608	LVDS communication error with Secondary information Display	Bus Signal / Message Failures	No controlled reduced functionality due to that the DTC is set. Normal functionality can not be guaranteed when this DTC is set.	Low Level failure on LVDS Communication LVDS communication link error or LVDS internal hardware error	Diagnostic Service 14. No additional action is needed.	Check integrity of LVDS cable Ensure LVDS cable is connected properly both at SID and Video Switch Check Video Switch for DTCs	LVDS Video Data Circuit

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
SID	U114998	Ambient Temperature > +85 degrees C during Vehicle Operation	Component or System Over Temperature	None	Ambient Temperature is High Ambient Temperature Sensor Fault	Diagnostic Service 14. No additional action is needed.	Investigate driving environment - is truck being driven in very hot climate? Investigate other nearby components - is another component running very hot?	Ambient Air Temperature
WRG	B150224	Button stuck in WRC for more than 3 mins	Signal Stuck High	* Not able to use the button that has been stuck * Not possible to control ECS functions	Button stuck in WRC	Diagnostic Service 14. No additional action is needed.	1. If all buttons working OK - Clear DTC 2. If not all buttons working - replace WRC	Wireless Remote Control
WRG	B150244	Bit fault in RAM	Data Memory Failure	WRC stays in boot, displays red screen	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150245	Bit fault in RAM	Program Memory Failure	WRC stays in boot, displays red screen	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150246	EEProm memory failure	Calibration / Parameter Memory Failure	WRC starts application and report error, possible loss of user data (e.g. backlight setting) WRG function as usual	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B150249	Radio chip do not answer commands, MSP failure	Internal Electronic Failure	Application will run normally but without radio. WRG will not get any connection to the WRC	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRC	Wireless Remote Control
WRG	B15024B	At start up	Over Temperature	WRC is shut off after displaying a message on the display. It is possible to start the WRC but it will be shut off if the temperature still is too high. Return to normal functionality when temperature is below threshold and no components are damaged	Battery temperature to high	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If the WRC is working normally - do nothing 3. If there is damages to battery - replace battery 4. If there is damages to display - replace WRC	Wireless Remote Control
WRG	B150294	Watchdog timeout, software assert	Unexpected Operation	Reset of WRC WRG not affected	Software error	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If it's set periodical - replace WRC	Wireless Remote Control
WRG	B150393	Battery supervising circuit reporting a faulty battery	No Operation	WRC will not function other than docked to WRG and in a power mode that support WRG 5V output to be enabled.	Battery failure	Diagnostic Service 14. No additional action is needed.	1. Make sure the battery is connected 2. Replace battery	WRC Battery
WRG	U003711	CAN short to GND	Circuit Short To Ground	No communication possible. Will report the DTC when communication comes back No affect on application	CAN short to GND	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net
WRG	U003712	CAN short to BATT	Circuit Short To Battery	No communication possible. Will report the DTC when communication comes back No affect on application	CAN short to BATT	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net

Node	DTC ISO	FailureEvent	FTB_Name	FunctBehaviorDesc	PossibleRootCause	ProcedureEraseConfirmedDtc	ProposedWorkshopAction	ProprietaryName
WRG	U003788	CAN controller Bus Off	Bus off	No communication possible. Will report the DTC when communication comes back No affect on application	Bad wiring, bad connectors	Diagnostic Service 14. No additional action is needed.	Search the error in wire harness	CAN Communication Cab Net
WRG	U300041	Checksum error	General Checksum Failure	Stay/Jump to BOOT	Missing software	Diagnostic Service 14. No additional action is needed.	1. Try to download complete SW 2. Replace WRG	Electronic control unit
WRG	U300044	Bit fault in RAM	Data Memory Failure	Application will start up and report error Could not guarante correct operation of WRG	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300046	Failing write to memory	Calibration / Parameter Memory Failure	* No functional effect * Possible loss of DTC information	Hardware fault, outside temperatue or voltage specification	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300049	Radio chip do not answer commands	Internal Electronic Failure	Application will run normally but without radio. WRC will not get any connection to the WRG	Hardware failure	Diagnostic Service 14. No additional action is needed.	Replace WRG	Electronic control unit
WRG	U300094	Watchdog timeout, software assert	Unexpected Operation	Reset of WRG	Software error	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. If it's set periodical - replace WRG	Electronic control unit
WRG	U300316	U<8V	Circuit Voltage Below Threshold	Application running as normal 5V output (used for charging the WRC battery) is turned off when U<16V ECU will shut down if U<7V (All ratings are approximate)	Input voltage below 8V	Diagnostic Service 14. No additional action is needed.	1. Problem with wire harness? 2. Problem with power transformation?	Battery Power
WRG	U300317	U>32V	Circuit Voltage Above Threshold	Application running as normal 5V output (used for charging the WRC battery) is turned off when U>36V (All ratings are approximate)	Input voltage over 32V	Diagnostic Service 14. No additional action is needed.	1. Problem with wire harness? 2. Problem with power transformation?	Battery Power
WRG	U100068	More than 29 DTC with "warning indicator bit"	Event Information	None	More than 29 DTC with "warning indicator bit"	Diagnostic Service 14. No additional action is needed.	1. Clear DTC 2. Investigate the other DTCs	DTC Notification Max Limit Reached